**Elite Mathematics**
Pearson Edexcel International GCSE

IGCSE MODULAR · UNIT 1 · 4WM1

# Modular Unit 1 Classified With Answers

Every question followed by Dr Eslam Ahmed's worked solution and final answer.

| 659 | 2020–2026 | Questions with worked solutions |
|---|---|---|
| topic placements | 50 past papers | checked by Dr Eslam |

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TOPIC

# Number Toolkit

Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 36

3 marks | Number Toolkit

**36** Here is a list of numbers.

2 9 15 17 21 25 27

From the list, write down

(a) the two prime numbers,

.....

**(1)**

(b) a square number that is also odd and greater than 10,

.....

**(1)**

(c) a cube number.

.....

**(1)**

**(Total for Question 36 is 3 marks)**

## WORKED SOLUTION

Question 36 · EP Specimen

**1** **(a) Exactly two factors**

$2$ and $17$ are prime ($15 = 3 \times 5$, $21 = 3 \times 7$).

**2** **(b) Square numbers in the list**

$9 = 3^2$ and $25 = 5^2$; the odd one greater than $10$ is $25$.

**3** **(c) Cube numbers**

$27 = 3^3$

✓ **FINAL ANSWER**

(a) $2, 17$ (b) $25$ (c) $27$

EP SPECIMEN

# Question 37

2 marks | Number Toolkit

**37** Use the information that $23 \times 47 = 1081$ to write down the value of

(a) $2.3 \times 4.7$

.....

**(1)**

(b) $1081 \div 4.7$

.....

**(1)**

**(Total for Question 37 is 2 marks)**

## WORKED SOLUTION

Question 37 · EP Specimen

**1** **(a) Each number is ten times smaller**

$2.3 \times 4.7 = 1081 \div 100 = 10.81$

**2** **(b) Reverse the multiplication**

$1081 \div 47 = 23$, and $4.7$ is ten times smaller than $47$, so $1081 \div 4.7 = 230$.

✓ **FINAL ANSWER**

(a) $10.81$ (b) $230$

4MA1/2HR JANUARY 2022

# Question 10

3 marks | Number Toolkit

**10** The table gives information about the population and the total amount of money, in dollars, spent on healthcare for two countries in 2016

| Country | Total population | Total spent on healthcare ($) |
|---|---|---|
| **Austria** | $8.7 \times 10^6$ | $4.2 \times 10^{10}$ |
| **Luxembourg** | $6.3 \times 10^5$ | $3.7 \times 10^9$ |

Work out how much more was spent **per person** on healthcare in Luxembourg than in Austria.
Give your answer correct to the nearest whole number.

..... dollars

**(Total for Question 10 is 3 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2HR January 2022

**1 Austria spending per person**

Austria spending per person:

$$\frac{4.2 \times 10^{10}}{8.7 \times 10^6} = 4827.586\ldots$$

**2 Luxembourg spending per person**

Luxembourg spending per person:

$$\frac{3.7 \times 10^9}{6.3 \times 10^5} = 5873.015\ldots$$

**3 Difference**

Difference:

$$5873.015\ldots - 4827.586\ldots = 1045.429\ldots$$

**4 Convert standard form**

To the nearest whole number, this is

$$1045$$

**FINAL ANSWER**

1045 dollars

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TOPIC

# Fractions

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 2

3 marks | Fractions

2 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

(Total for Question 2 is 3 marks)

## WORKED SOLUTION

Question 2 · 4WM1/01 Specimen

**1 Convert to improper fractions**

$$5\frac{1}{3} = \frac{16}{3}, \qquad 2\frac{6}{7} = \frac{20}{7}$$

**2 Use a common denominator**

$$\frac{16}{3} - \frac{20}{7} = \frac{112}{21} - \frac{60}{21} = \frac{52}{21}$$

**3 Convert back to a mixed number**

$$\frac{52}{21} = 2\frac{10}{21}$$

**FINAL ANSWER**

$2\frac{10}{21}$.

4MA1/1HR JANUARY 2020

# Question 2

3 marks | Fractions

**2** Show that $3\frac{1}{5} \times 2\frac{5}{8} = 8\frac{2}{5}$

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1HR January 2020

### 1 Simplify fraction

Convert the mixed numbers to improper fractions:

$$3\frac{1}{5} = \frac{16}{5}$$

$$2\frac{5}{8} = \frac{21}{8}$$

### 2 Simplify fraction

Now multiply:

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

$$= \frac{42}{5} = 8\frac{2}{5}$$

**FINAL ANSWER**

$8\frac{2}{5}$

4MA1/2H JANUARY 2020

# Question 3

3 marks | Fractions

3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2H January 2020

### 1 Convert to improper fractions

Convert to improper fractions:

$$4\frac{2}{3} = \frac{14}{3}$$

$$3\frac{4}{5} = \frac{19}{5}$$

### 2 Use a common denominator

Use a common denominator:

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

$$= \frac{127}{15} = 8\frac{7}{15}$$

**FINAL ANSWER**

$8\frac{7}{15}$

4MA1/1HR MAY/NOVEMBER 2020

# Question 1

3 marks | Fractions

**1** Show that $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1HR May/November 2020

**1** **Simplify fraction**

$$3\frac{3}{4} = \frac{15}{4}$$

**2** **Simplify fraction**

Now multiply:

$$\frac{15}{4} \times \frac{7}{9}$$

**3** **Simplify fraction**

Cancel $15$ and $9$:

$$\frac{5}{4} \times \frac{7}{3} = \frac{35}{12}$$

$$\frac{35}{12} = 2\frac{11}{12}$$

✓ **FINAL ANSWER**

$2\frac{11}{12}$

4MA1/1HR JANUARY 2021

# Question 1

3 marks | Fractions

**1** Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1HR January 2021

### 1 Simplify fraction

Convert the mixed numbers to improper fractions:

$$3\frac{1}{5} = \frac{16}{5}$$

$$1\frac{5}{6} = \frac{11}{6}$$

### 2 Simplify fraction

Now multiply:

$$\frac{16}{5} \times \frac{11}{6} = \frac{176}{30}$$

$$= \frac{88}{15} = 5\frac{13}{15}$$

**FINAL ANSWER**

$5\frac{13}{15}$

4MA1/2H MAY 2021

# Question 2

3 marks | Fractions

2 Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2H May 2021

**1 Simplify fraction**

$$2\frac{4}{7} = \frac{18}{7}$$

$$1\frac{1}{8} = \frac{9}{8}$$

**2 Divide by multiplying by the**

Divide by multiplying by the reciprocal:

$$\frac{18}{7} \div \frac{9}{8} = \frac{18}{7} \times \frac{8}{9}$$

$$= \frac{16}{7} = 2\frac{2}{7}$$

**FINAL ANSWER**

$2\frac{2}{7}$

4MA1/2H JANUARY 2022

# Question 2

3 marks | Fractions

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2H January 2022

### 1 Convert to improper fractions

Convert to improper fractions:

$$6\frac{3}{4} = \frac{27}{4}$$

$$2\frac{4}{7} = \frac{18}{7}$$

### 2 Divide by multiplying by the

Divide by multiplying by the reciprocal:

$$\frac{27}{4} \div \frac{18}{7} = \frac{27}{4} \times \frac{7}{18}$$

$$= \frac{21}{8} = 2\frac{5}{8}$$

**FINAL ANSWER**

$2\frac{5}{8}$

4MA1/2H JUNE 2022

# Question 4

3 marks | Fractions

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2H June 2022

### 1 Simplify fraction

$$5\frac{1}{3} = \frac{16}{3}$$

$$2\frac{6}{7} = \frac{20}{7}$$

### 2 Use a common denominator

Use a common denominator:

$$\frac{16}{3} - \frac{20}{7} = \frac{112}{21} - \frac{60}{21}$$

$$= \frac{52}{21} = 2\frac{10}{21}$$

**FINAL ANSWER**

$2\frac{10}{21}$

4MA1/1HR JUNE 2022

# Question 6

3 marks | Fractions

6 Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

**(Total for Question 6 is 3 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1HR June 2022

### 1 Simplify fraction

$$2\frac{2}{3} = \frac{8}{3}$$

$$3\frac{3}{4} = \frac{15}{4}$$

### 2 Use a common denominator

Use a common denominator:

$$\frac{8}{3} + \frac{15}{4} = \frac{32}{12} + \frac{45}{12}$$

$$= \frac{77}{12} = 6\frac{5}{12}$$

**FINAL ANSWER**

$6\frac{5}{12}$

4MA1/2H JUNE 2024

# Question 4

3 marks | Fractions

4 Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2H June 2024

**1 Simplify fraction**

$$2\frac{4}{7} = \frac{18}{7}$$

$$3\frac{1}{9} = \frac{28}{9}$$

**2 Simplify fraction**

Now multiply:

$$\frac{18}{7} \times \frac{28}{9}$$

**3 Cancel first**

Cancel first:

$$\frac{18}{9} \times \frac{28}{7} = 2 \times 4 = 8$$

**FINAL ANSWER**

8

4WM1/01 MAY 2025

# Question 1

3 marks | Fractions

**1** Without using a calculator, work out

$$3\frac{2}{3} \times 2\frac{4}{7}$$

Show each stage of your working.
Give your answer as a mixed number in its simplest form.

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4WM1/01 May 2025

**1** **Simplify fraction**

Convert the mixed numbers to improper fractions.

$$3\frac{2}{3} = \frac{11}{3}, \qquad 2\frac{4}{7} = \frac{18}{7}$$

**2** **Simplify fraction**

Then multiply.

$$\frac{11}{3} \times \frac{18}{7} = \frac{198}{21} = \frac{66}{7}$$

**3** **Convert to a mixed number**

Convert to a mixed number.

$$\frac{66}{7} = 9\frac{3}{7}$$

✓ **FINAL ANSWER**

$9\frac{3}{7}$

4WM1/01R MAY 2025

# Question 2

3 marks | Fractions

**2** Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4WM1/01R May 2025

**1** **Evaluate fraction**

$$2\frac{1}{4} = \frac{9}{4}, \qquad 1\frac{5}{7} = \frac{12}{7}$$

$$2\frac{1}{4} \times 1\frac{5}{7} = \frac{9}{4} \times \frac{12}{7}$$

$$= \frac{108}{28} = \frac{27}{7} = 3\frac{6}{7}$$

✓ **FINAL ANSWER**

$3\frac{6}{7}$.

4WM1/01 NOVEMBER 2025

# Question 3

3 marks | Fractions

**3** Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4WM1/01 November 2025

### 1 Evaluate fraction

$$\frac{3}{7} + \frac{1}{3}$$

### 2 Use denominator 21

Use denominator 21:

$$\frac{3}{7} = \frac{9}{21}, \qquad \frac{1}{3} = \frac{7}{21}$$

$$\frac{9}{21} + \frac{7}{21} = \frac{16}{21}$$

**FINAL ANSWER**

$\frac{16}{21}$.

4MA1/2H NOVEMBER 2025

# Question 3

4 marks | Fractions

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

**(Total for Question 3 is 4 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2H November 2025

**1 There are red and blue**

There are $9 + 43 = 52$ red and blue bricks.

**2 Simplify fraction**

Let the total number of bricks be $T$.

**3 Simplify fraction**

Since $\frac{7}{20}$ of the bricks are yellow, the remaining fraction is

$$1 - \frac{7}{20} = \frac{13}{20}$$

**4 Simplify fraction**

So

$$\frac{13}{20}T = 52$$

$$T = 52 \times \frac{20}{13} = 80$$

**5** **Yellow bricks**

Yellow bricks:

$$\frac{7}{20} \times 80 = 28$$

**6** **Simplify fraction**

Each brick weighs $35$ grams, so total yellow-brick weight is

$$28 \times 35 = 980$$

✓ **FINAL ANSWER**

$980$ grams

4MA1/1H JUNE 2026

# Question 1

3 marks | Fractions

**1** Show that $4\frac{1}{7} - 2\frac{3}{5} = 1\frac{19}{35}$

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H June 2026

**1** **Solution | Step 1**

$$4\frac{1}{7} = \frac{29}{7}, \qquad 2\frac{3}{5} = \frac{13}{5}.$$

Convert each mixed number to an improper fraction.

**2** **Solution | Step 2**

$$\frac{29}{7} - \frac{13}{5} = \frac{145 - 91}{35} = \frac{54}{35} = 1\frac{19}{35}.$$

Use the common denominator 35, subtract, then convert back to a mixed number.

✓ **FINAL ANSWER**

$1\frac{19}{35}$

4MA1/1HR JUNE 2026

# Question 1

3 marks | Fractions

1 Show that $2\frac{4}{5} \times 3\frac{3}{7} = 9\frac{3}{5}$

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1HR June 2026

**1 Solution | Step 1**

$$2\frac{4}{5} = \frac{14}{5}, \qquad 3\frac{3}{7} = \frac{24}{7}.$$

Convert each mixed number into an improper fraction.

**2 Solution | Step 2**

$$\frac{14}{5} \times \frac{24}{7} = \frac{2 \times 24}{5} = \frac{48}{5} = 9\frac{3}{5}.$$

Cancel the common factor 7, multiply, then convert back to a mixed number.

**✓ FINAL ANSWER**

$2\frac{4}{5} \times 3\frac{3}{7} = 9\frac{3}{5}$

4WM1/01 MAY 2026

# Question 1

3 marks | Fractions

**1** Show that $4\frac{1}{7} - 2\frac{3}{5} = 1\frac{19}{35}$

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4WM1/01 May 2026

**1 Improper fractions**

$$4\tfrac{1}{7} = \frac{29}{7}, \qquad 2\tfrac{3}{5} = \frac{13}{5}$$

**2 Common denominator 35**

$$\frac{29}{7} - \frac{13}{5} = \frac{145}{35} - \frac{91}{35} = \frac{54}{35}$$

**3 Back to a mixed number**

$$\frac{54}{35} = 1\tfrac{19}{35} \quad \text{as required.}$$

✓ **FINAL ANSWER**

$\frac{54}{35} = 1\frac{19}{35}$ (shown)

4MA1/2HR JUNE 2026

# Question 17

2 marks | Fractions

**17** Use algebra to show that $0.\dot{8}1\dot{4} = \frac{22}{27}$

**(Total for Question 17 is 2 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/2HR June 2026

**1** **Solution | Step 1**

Let $z = 0.\overline{814} = 0.814814814\ldots$. Then $1000z = 814.814814\ldots$.

Multiplication by 1000 aligns the repeating three-digit tails.

**2** **Solution | Step 2**

$$1000z - z = 814 \quad \Rightarrow \quad 999z = 814 \quad \Rightarrow \quad z = \frac{814}{999} = \frac{22}{27}.$$

Subtract to cancel the recurring tails, then cancel the common factor 37.

✓ **FINAL ANSWER**

$0.\overline{814} = \frac{22}{27}$.

TOPIC

# Fractions, Decimals & Percentages

Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 15

2 marks | Fractions, Decimals & Percentages

**15** Write these numbers in order of size. Start with the smallest number.

$0.62 \quad \frac{5}{8} \quad 61\% \quad \frac{2}{3} \quad \frac{3}{5}$

.....

**(2)**

**(Total for Question 15 is 2 marks)**

## WORKED SOLUTION

Question 15 · EP Specimen

**1 Change every number to a decimal**

$0.62,\ \frac{5}{8} = 0.625,\ 61\% = 0.61,\ \frac{2}{3} = 0.666\ldots,\ \frac{3}{5} = 0.6$

**2 Order them**

$0.6,\ 0.61,\ 0.62,\ 0.625,\ 0.666\ldots$

**FINAL ANSWER**

$\frac{3}{5},\ 61\%,\ 0.62,\ \frac{5}{8},\ \frac{2}{3}$

EP SPECIMEN

# Question 16

2 marks | Fractions, Decimals & Percentages

**16** (a) Write $\frac{7}{40}$ as a decimal.

.....

**(1)**

(b) Write $\frac{7}{40}$ as a percentage.

..... %

**(1)**

**(Total for Question 16 is 2 marks)**

## WORKED SOLUTION

Question 16 · EP Specimen

**1** **(a) Divide**

$$7 \div 40 = 0.175$$

**2** **(b) Multiply by 100**

$$0.175 \times 100 = 17.5\%$$

✓ **FINAL ANSWER**

(a) $0.175$ (b) $17.5\%$

EP SPECIMEN

# Question 17

3 marks | Fractions, Decimals & Percentages

**17** The students in a class travel to school by walking, by bicycle or by bus.

$\frac{3}{8}$ of the students walk. 0.35 of the students cycle. The rest travel by bus.

11 students travel by bus.

Work out the number of students in the class.

.....

**(3)**

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · EP Specimen

### 1 Proportion that walk or cycle

$$\frac{3}{8} + 0.35 = 0.375 + 0.35 = 0.725$$

### 2 Proportion that travel by bus

$$1 - 0.725 = 0.275$$

### 3 Number of students

$$0.275 \times n = 11 \Rightarrow n = \frac{11}{0.275} = 40$$

**FINAL ANSWER**

40

EP SPECIMEN

# Question 18

2 marks | Fractions, Decimals & Percentages

**18** Which is larger, $\frac{13}{20}$ or 0.66?

You must show your working.

.....

**(2)**

**(Total for Question 18 is 2 marks)**

## WORKED SOLUTION

Question 18 · EP Specimen

### 1 Write both as decimals

$$\frac{13}{20} = 13 \div 20 = 0.65$$

### 2 Compare

$0.66 > 0.65$, so $0.66$ is larger.

**FINAL ANSWER**

$0.66$

EP SPECIMEN

# Question 19

2 marks | Fractions, Decimals & Percentages

**19** Mia took two tests.
She scored 34 out of 40 in the first test.
She scored 87% in the second test.

In which test did Mia get the better score? You must show how you get your answer.

.....

**(2)**

**(Total for Question 19 is 2 marks)**

## WORKED SOLUTION

Question 19 · EP Specimen

### 1 Change the first score to a percentage

$$\frac{34}{40} \times 100 = 85\%$$

### 2 Compare

$87\% > 85\%$, so the second test was better.

**FINAL ANSWER**

The second test ($85\%$ against $87\%$)

EP SPECIMEN

# Question 20

2 marks | Fractions, Decimals & Percentages

**20** Write 0.0375 as a fraction in its simplest form.

.....

**(2)**

**(Total for Question 20 is 2 marks)**

## WORKED SOLUTION

Question 20 · EP Specimen

**1** **Write over a power of 10**

$$0.0375 = \frac{375}{10\,000}$$

**2** **Simplify (divide by 125)**

$$\frac{375}{10\,000} = \frac{3}{80}$$

✓ **FINAL ANSWER**

$$\frac{3}{80}$$

EP SPECIMEN

# Question 21

3 marks | Fractions, Decimals & Percentages

**21** A water tank is $\frac{2}{5}$ full.

120 litres of water are added to the tank.
The tank is now 65% full.

Work out the capacity of the tank.

..... litres

**(3)**

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · EP Specimen

**1 Change both amounts to percentages**

$\frac{2}{5} = 40\%$, so the $120$ litres fill $65\% - 40\% = 25\%$ of the tank.

**2 Find 100%**

$$25\% \to 120 \;\Rightarrow\; 100\% \to 120 \times 4 = 480 \text{ litres}$$

**FINAL ANSWER**

480 litres

TOPIC

# Recurring Decimals

Every past-paper question on this topic, 2020 to 2026.

4MA1/1H MAY 2021

# Question 16

2 marks | Recurring Decimals

**16** Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$

**(Total for Question 16 is 2 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H May 2021

**1 Let**

Let

$$x = 0.28131313\ldots$$

**2 Calculate value**

Then

$$100x = 28.131313\ldots$$

**3 Calculate value**

and

$$10000x = 2813.131313\ldots$$

**4 Subtract**

Subtract:

$$10000x - 100x = 2813.131313\ldots - 28.131313\ldots$$

$$9900x = 2785$$

$$x = \frac{2785}{9900} = \frac{557}{1980}$$

**FINAL ANSWER**

$0.28131313\ldots = \frac{557}{1980}$.

4MA1/1H NOVEMBER 2021

# Question 18

3 marks | Recurring Decimals

**18** $0.4\dot{x}$ is a recurring decimal.
$x$ is a whole number such that $1 \leqslant x \leqslant 9$

Find, in terms of $x$, the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1H November 2021

**1 Let**

Let

$$y = 0.4\dot{x}$$

**2 Simplify fraction**

So

$$y = 0.4xxx\ldots$$

**3 Simplify fraction**

Multiply by $10$ and $100$:

$$10y = 4.xxx\ldots$$

$$100y = 4x.xxx\ldots$$

**4 Subtract**

Subtract:

$$100y - 10y = (4x.xxx\ldots) - (4.xxx\ldots)$$

$$90y = x + 36$$

**5 Simplify fraction**

Therefore

$$y = \frac{x + 36}{90}$$

**FINAL ANSWER**

$$\frac{x+36}{90}$$

4WM1/01R MAY 2025

# Question 14

2 marks | Recurring Decimals

**14** Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \frac{101}{165}$

**(Total for Question 14 is 2 marks)**

## WORKED SOLUTION

Question 14 · 4WM1/01R May 2025

**1 Let**

Let

$$x = 0.6121212\ldots$$

**2 Calculate value**

Then

$$10x = 6.121212\ldots$$

**3 Calculate value**

and

$$1000x = 612.121212\ldots$$

**4 Subtract**

Subtract:

$$1000x - 10x = 612.121212\ldots - 6.121212\ldots$$

$$990x = 606$$

$$x = \frac{606}{990} = \frac{101}{165}$$

✓ **FINAL ANSWER**

$0.6121212\ldots = \frac{101}{165}$.

4MA1/1HR JUNE 2025

# Question 16

2 marks | Recurring Decimals

**16** Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \frac{101}{165}$

**(Total for Question 16 is 2 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1HR June 2025

**1 Let**

Let

$$x = 0.6121212\ldots$$

**2 Calculate value**

Then

$$10x = 6.121212\ldots$$

**3 Calculate value**

and

$$1000x = 612.121212\ldots$$

**4 Subtract**

Subtract:

$$1000x - 10x = 612.121212\ldots - 6.121212\ldots$$

$$990x = 606$$

$$x = \frac{606}{990} = \frac{101}{165}$$

✓ **FINAL ANSWER**

$0.6121212\ldots = \frac{101}{165}$.

TOPIC

# Rounding, Estimation & Bounds

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 1

1 marks | Rounding, Estimation & Bounds

1 The weight of a cake is 2.75 kg, correct to 2 decimal places.

Write down the lower bound of the weight of the cake.

..... kg

**(Total for Question 1 is 1 mark)**

## WORKED SOLUTION

Question 1 · 4WM1/01 Specimen

**1 Find half the rounding interval**

Correct to $2$ decimal places means the nearest $0.01$ kg.

$$\frac{0.01}{2} = 0.005$$

**2 Subtract from the rounded value**

$$2.75 - 0.005 = 2.745$$

**FINAL ANSWER**

$2.745$ kg.

# Question 15

4 marks | Rounding, Estimation & Bounds

**15** Martin and Lucia went on two different car journeys.

For Martin's journey

distance = 80 km correct to the nearest 5 km
time = 2.7 hours correct to 1 decimal place

For Lucia's journey

distance = 33 km correct to 2 significant figures
time = 1 hour correct to the nearest 0.1 hour

Martin says,

"My average speed could have been greater than Lucia's average speed."

By considering bounds, show that Martin is correct.
Show your working clearly.

**(Total for Question 15 is 4 marks)**

### 1 Use Martin's greatest possible speed

Martin's greatest possible distance is $82.5$ km and his least possible time is $2.65$ hours.

$$\frac{82.5}{2.65} = 31.132\ldots$$

### 2 Use Lucia's least possible speed

Lucia's least possible distance is $32.5$ km and her greatest possible time is $1.05$ hours.

$$\frac{32.5}{1.05} = 30.952\ldots$$

### 3 Compare the possible speeds

$$31.132\ldots > 30.952\ldots$$

So Martin's speed could have been greater than Lucia's speed.

**FINAL ANSWER**

Martin could have been faster because $31.132\ldots > 30.952\ldots$ $\text{km/h}$.

4MA1/1HR JANUARY 2020

# Question 11

4 marks | Rounding, Estimation & Bounds

**11** The weight of a cat is 4.3 kg correct to 2 significant figures.

(a) Write down the upper bound of the weight of the cat.

..... kg

(1)

(b) Write down the lower bound of the weight of the cat.

..... kg

(1)

$G = e - f$

$e$ = 17 correct to the nearest integer
$f$ = 9.4 correct to one decimal place

(c) Work out the upper bound for the value of $G$.

.....

(2)

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1HR January 2020

**1** **Estimate the value**

The weight $4.3$ kg correct to 2 significant figures has bounds

$$4.25 \leq w < 4.35$$

**2** **Find upper bound**

So the upper bound is $4.35$ kg and the lower bound is $4.25$ kg.

**3** **Estimate the value**

For

$$G = e - f$$

**4** **the upper bound uses the**

the upper bound uses the upper bound of $e$ and the lower bound of $f$.

$$e < 17.5, \quad f \geq 9.35$$

$$G_{\text{upper}} = 17.5 - 9.35 = 8.15$$

---

✓ **FINAL ANSWER**

(a) $4.35$ kg, (b) $4.25$ kg, (c) $8.15$

4MA1/1H JANUARY 2020

# Question 16

3 marks | Rounding, Estimation & Bounds

**16** $P = \dfrac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.
$c = 20$ correct to 2 significant figures.
$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of $P$.
Show your working clearly.
Give your answer correct to 2 decimal places.

.....

**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H January 2020

**1** **Estimate the value**

$$P = \frac{2a - c}{d}$$

**2** **Find upper bound**

For the upper bound of $P$, make the numerator as large as possible and the denominator as small as possible.

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

$$P_{\text{upper}} = \frac{2(58.45) - 19.5}{3.55} = 27.4366\ldots$$

---

✓ **FINAL ANSWER**

$27.44$

4MA1/1H MAY/NOVEMBER 2020

# Question 17

4 marks | Rounding, Estimation & Bounds

**17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3})\,m^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in $kg/m^3$ correct to 1 decimal place.
Show your working clearly.

..... $kg/m^3$

**(Total for Question 17 is 4 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1H May/November 2020

**1 Density is**

Density is

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

**2 To get the upper bound**

To get the upper bound for density, use the upper bound for mass and the lower bound for volume.

**3 The mass is kg correct**

The mass is $5$ kg correct to the nearest $50$ g.

$$50\text{ g} = 0.05\text{ kg}$$

**4 Find upper bound**

Half of this is $0.025$ kg, so

$$\text{upper bound for mass} = 5 + 0.025 = 5.025$$

**5 Find upper bound**

The volume is $1.84 \times 10^{-3} = 0.00184\ m^3$, correct to 3 significant figures.

**6 The lower bound is**

The lower bound is

$$0.00184 - 0.000005 = 0.001835$$

**7** **Find upper bound**

So the upper bound for the density is

$$\frac{5.025}{0.001835} = 2738.419\ldots$$

**8** **Find upper bound**

Correct to 1 decimal place,

$$2738.4$$

✓ **FINAL ANSWER**

$2738.4\ \mathrm{kg/m^3}$

4MA1/1HR MAY/NOVEMBER 2020

# Question 19

3 marks | Rounding, Estimation & Bounds

**19** The diagram shows rectangle $ABCD$ with rectangle $EFGH$ cut out to form the shaded region.

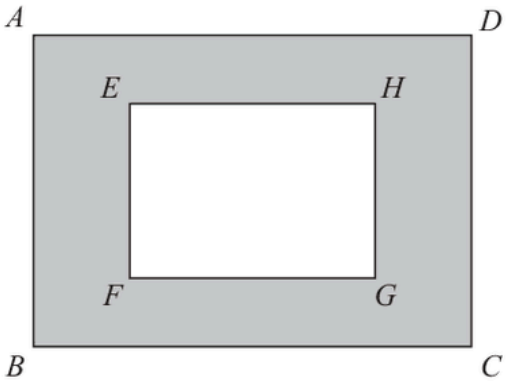



Diagram **NOT** accurately drawn

$AD = 8.3$ cm correct to one decimal place
$DC = 7.2$ cm correct to one decimal place
$EH = 6.2$ cm correct to one decimal place
$HG = 5.3$ cm correct to one decimal place

Work out the upper bound of the area of the shaded region.
Show your working clearly.

..... cm²

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1HR May/November 2020

**1** **Find upper bound**

For the upper bound of the shaded area, use the upper bound of the outer rectangle and subtract the lower bound of the cut-out rectangle.

**2** **Outer rectangle**

Outer rectangle:

$$AD < 8.35, \quad DC < 7.25$$

### 3 Cut out rectangle

Cut-out rectangle:

$$EH \geq 6.15, \quad HG \geq 5.25$$

$$\text{upper shaded area} = 8.35(7.25) - 6.15(5.25)$$

$$= 60.5375 - 32.2875 = 28.25$$

---

**FINAL ANSWER**

$28.25\ \text{cm}^2$

4MA1/1H JANUARY 2021

# Question 15

3 marks | Rounding, Estimation & Bounds

**15**

$$a = \frac{v - u}{t}$$

$v = 9.6$ correct to 1 decimal place
$u = 3.8$ correct to 1 decimal place
$t = 1.84$ correct to 2 decimal places

Calculate the upper bound for the value of $a$.
Give your answer as a decimal correct to 2 decimal places.
Show your working clearly.

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1H January 2021

### 1 Estimate the value

$$a = \frac{v - u}{t}$$

### 2 Find upper bound

For the upper bound, use the upper bound of $v$, the lower bound of $u$, and the lower bound of $t$.

$$v < 9.65, \quad u \geq 3.75, \quad t \geq 1.835$$

$$a_{\text{upper}} = \frac{9.65 - 3.75}{1.835} = 3.2152\ldots$$

**FINAL ANSWER**

$3.22$

4MA1/1HR JANUARY 2021

# Question 19

3 marks | Rounding, Estimation & Bounds

19 $k = \dfrac{t}{a-h}$

$t = 14$ correct to 2 significant figures
$a = 7.8$ correct to 2 significant figures
$h = 3.4$ correct to 2 significant figures

Work out the lower bound for the value of $k$.
Show your working clearly.

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1HR January 2021

### 1 Estimate the value

$$k = \frac{t}{a-h}$$

### 2 Find lower bound

For the lower bound of $k$, use the lower bound of $t$ and make the denominator as large as possible.

$$t \geq 13.5$$

$$a < 7.85, \quad h \geq 3.35$$

$$(a-h)_{\text{upper}} = 7.85 - 3.35 = 4.5$$

$$k_{\text{lower}} = \frac{13.5}{4.5} = 3$$

**FINAL ANSWER**

3

4MA1/1H MAY 2021

# Question 20

3 marks | Rounding, Estimation & Bounds

**20** $P = \dfrac{t - w}{y}$

$t$ = 9.7 correct to 1 decimal place

$w$ = 5.9 correct to 1 decimal place

$y$ = 3 correct to 1 significant figure

Calculate the upper bound for the value of $P$.
Show your working clearly.

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H May 2021

### 1 Estimate the value

$$P = \frac{t - w}{y}$$

### 2 Find upper bound

For the upper bound, make the numerator large and the denominator small.

$$t < 9.75, \quad w \geq 5.85, \quad y \geq 2.5$$

$$P_{\text{upper}} = \frac{9.75 - 5.85}{2.5} = 1.56$$

**FINAL ANSWER**

$1.56$

4MA1/1H NOVEMBER 2021

# Question 13

3 marks | Rounding, Estimation & Bounds

**13** Here is a triangle $XYZ$.

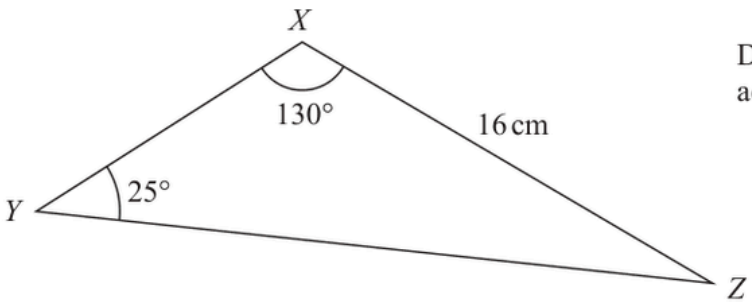



Diagram **NOT** accurately drawn

The length $XZ$ and the angles $YXZ$ and $XYZ$ are each given correct to 2 significant figures.

Calculate the upper bound for the length $YZ$.
Give your answer correct to one decimal place.
Show your working clearly.

..... cm

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H November 2021

**1 Use trigonometry**

Using the sine rule,

$$\frac{YZ}{\sin X} = \frac{XZ}{\sin Y}$$

**2 Find upper bound**

For the upper bound of $YZ$:

$$XZ < 16.5, \quad X < 25.5°, \quad Y < 135°$$

$$YZ_{\text{upper}} = \frac{16.5 \sin 25.5°}{\sin 135°} = 10.0457\ldots$$

**3 Use trigonometry**

Correct to one decimal place,

$$10.0457\ldots \approx 10.0$$

**FINAL ANSWER**

10.0 cm

4MA1/2H JANUARY 2022

# Question 4

2 marks | Rounding, Estimation & Bounds

4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

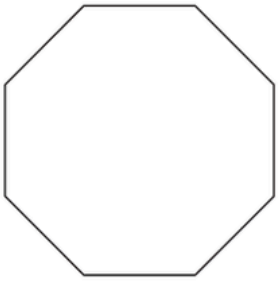


Diagram **NOT** accurately drawn

(a) Write down the lower bound of the length of each side of the octagon.

..... mm

(1)

(b) Write down the upper bound of the length of each side of the octagon.

..... mm

(1)

**(Total for Question 4 is 2 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2H January 2022

**1 The side length is mm**

The side length is $18$ mm correct to the nearest $0.5$ mm.

**2 Half of mm is mm**

Half of $0.5$ mm is $0.25$ mm.

$$18 - 0.25 = 17.75$$

$$18 + 0.25 = 18.25$$

**FINAL ANSWER**

(a) $17.75$ mm, (b) $18.25$ mm

4MA1/1H JANUARY 2022

# Question 18

3 marks | Rounding, Estimation & Bounds

**18** $A = w - \dfrac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.
$x = 1.9$ correct to 1 decimal place.
$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of $A$
Show your working clearly.

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1H January 2022

### 1 Estimate the value

$$A = w - \frac{x^2}{y}$$

### 2 Find lower bound

For the lower bound, use the lower bound of $w$, and make $\frac{x^2}{y}$ as large as possible.

$$w \geq 3.445, \quad x < 1.95, \quad y \geq 4.5$$

$$A_{\text{lower}} = 3.445 - \frac{1.95^2}{4.5}$$

$$A_{\text{lower}} = 2.6$$

**FINAL ANSWER**

$2.6$

4MA1/1HR JANUARY 2022

# Question 22

4 marks | Rounding, Estimation & Bounds

**22** The diagram shows triangle $ABC$

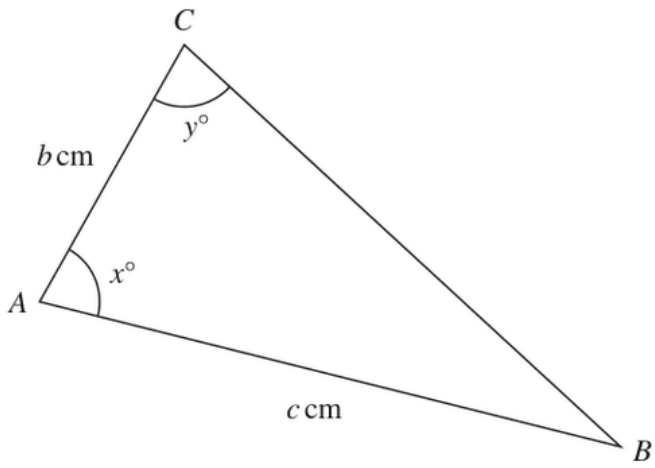



Diagram **NOT** accurately drawn

$c = 11.5$ correct to one decimal place
$x = 80$ correct to the nearest whole number
$y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of $b$
Show your working clearly.
Give your answer correct to 3 significant figures.

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1HR January 2022

### 1 Find upper bound

For an upper bound for $b$, use the upper bound of $c$:

$c = 11.55$

### 2 Use sine rule

Use the lower bounds of $x$ and $y$, because this makes angle $B$ as large as possible and makes $\sin y$ as small as possible.

$x = 79.5°, \quad y = 74.5°$

### 3 Use trigonometry

So

$\angle B = 180° - 79.5° - 74.5° = 26°$

**4** **Use trigonometry**

Using the sine rule,

$$\frac{b}{\sin 26^\circ} = \frac{11.55}{\sin 74.5^\circ}$$

$$b = \frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.254...$$

---

✓ **FINAL ANSWER**

$5.25$ cm to 3 significant figures.

4MA1/2H JUNE 2022

# Question 18

3 marks | Rounding, Estimation & Bounds

**18** $X = \dfrac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.
$b = 3.42$ correct to 2 decimal places.
$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of $X$
Show your working clearly.

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2H June 2022

**1 Estimate the value**

$$X = \frac{2a - b}{f}$$

**2 Find upper bound**

For the upper bound, make the numerator large and the denominator small.

$$a < 7.55, \quad b \geq 3.415, \quad f \geq 1.5$$

$$X_{\text{upper}} = \frac{2(7.55) - 3.415}{1.5} = 7.79$$

**FINAL ANSWER**

$7.79$

4MA1/2HR JUNE 2022

# Question 18

4 marks | Rounding, Estimation & Bounds

**18** Kaidan and Sonja went on two different car journeys.

For Kaidan's journey
distance = 80 km correct to the nearest 5 km
time = 2.7 hours correct to 1 decimal place

For Sonja's journey
distance = 33 km correct to 2 significant figures
time = 1 hour correct to the nearest 0.1 hour

Kaidan says,
"My average speed could have been greater than Sonja's average speed."

By considering bounds, show that Kaidan is correct.
Show your working clearly.

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2HR June 2022

### 1 To show Kaidan s average

To show Kaidan's average speed could be greater, compare Kaidan's greatest possible speed with Sonja's smallest possible speed.

### 2 Estimate the value

For Kaidan:

$$d < 82.5, \quad t \geq 2.65$$

$$\text{greatest speed} = \frac{82.5}{2.65} = 31.13\ldots \text{ km/h}$$

### 3 Estimate the value

For Sonja:

$$d \geq 32.5, \quad t < 1.05$$

$$\text{smallest speed} = \frac{32.5}{1.05} = 30.95\ldots \text{ km/h}$$

**4** **Estimate the value**

Since

$$31.13\ldots > 30.95\ldots$$

**5** **Kaidan s average speed could**

Kaidan's average speed could have been greater.

✓ **FINAL ANSWER**

Kaidan is correct.

4MA1/1HR MAY 2023

# Question 18

4 marks | Rounding, Estimation & Bounds

**18** Diego builds a fence using fence panels.

The total length of the fence is 50 metres, correct to the nearest 5 metres.
The length of each fence panel is 1.3 metres, correct to the nearest 10 cm.

The cost of each fence panel is £8.65

Diego may only buy complete fence panels.
Diego only pays for the number of panels he needs to build the fence.

Work out the greatest difference in the possible amounts that Diego could pay to build the fence.
Show your working clearly.

£.....

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1HR May 2023

### 1 The fence length m correct

The fence length $50$ m correct to the nearest $5$ m has bounds

$$47.5 \leq L < 52.5$$

### 2 The panel length m correct

The panel length $1.3$ m correct to the nearest $10$ cm has bounds

$$1.25 \leq p < 1.35$$

### 3 Smallest possible number of panels

Smallest possible number of panels:

$$\left\lceil \frac{47.5}{1.35} \right\rceil = 36$$

### 4 Greatest possible number of panels

Greatest possible number of panels:

$$\left\lceil \frac{52.5}{1.25} \right\rceil = 42$$

**5** **Difference in number of panels**

Difference in number of panels:

$$42 - 36 = 6$$

**6** **Greatest difference in cost**

Greatest difference in cost:

$$6 \times 8.65 = 51.90$$

✓ **FINAL ANSWER**

$51.90$ pounds

4MA1/2H JUNE 2023

# Question 25

4 marks | Rounding, Estimation & Bounds

**25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in $g/cm^3$ correct to 2 decimal places.
Show your working clearly.

..... $g/cm^3$

**(Total for Question 25 is 4 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/2H June 2023

**1 Find upper bound**

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

**2 The radius is cm correct**

The radius is $2.8$ cm correct to 1 decimal place, so the lower bound is
$r = 2.75$

**3 Find upper bound**

$M = 260$ correct to 2 significant figures, so the upper bound is
$M = 265$

**4 The mass is and the**

The mass is $M\pi$ and the volume of the sphere is $\frac{4}{3}\pi r^3$.

**5 Rearrange formula**

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3} = \frac{3M}{4r^3}$$

**6 Upper bound**

Upper bound:

$$\frac{3(265)}{4(2.75)^3} = 9.5567\ldots$$

**FINAL ANSWER**

$9.56\ \mathrm{g/cm^3}$ to 2 decimal places.

4MA1/2H NOVEMBER 2023

# Question 17

3 marks | Rounding, Estimation & Bounds

17 $P = a(c + y)$

$a =$ 8.3 correct to 2 significant figures
$c =$ 2 correct to 1 significant figure
$y =$ 15 correct to the nearest 5

Work out the upper bound for the value of $P$
Show your working clearly.

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/2H November 2023

### 1 Estimate the value

$$P = a(c + y)$$

### 2 Find upper bound

For the upper bound, use the upper bounds of all three values.

$$a < 8.35, \quad c < 2.5, \quad y < 17.5$$

$$P_{\text{upper}} = 8.35(2.5 + 17.5) = 167$$

**FINAL ANSWER**

167

4MA1/2H JUNE 2024

# Question 19

3 marks | Rounding, Estimation & Bounds

19 $G = \dfrac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number
$f = 6.62$ correct to 2 decimal places
$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of $G$
Give your answer correct to 3 decimal places.
Show your working clearly.

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2H June 2024

**1 Estimate the value**

$$G = \frac{c}{2f - 3h}$$

**2 Find lower bound**

For the lower bound, use the lower bound of $c$ and make the denominator as large as possible.

$$c \geq 7.5$$

$$f < 6.625, \quad h \geq 1.15$$

$$(2f - 3h)_{\text{upper}} = 2(6.625) - 3(1.15) = 9.8$$

$$G_{\text{lower}} = \frac{7.5}{9.8} = 0.765306\ldots$$

✓ **FINAL ANSWER**

$0.765$

4MA1/2HR JUNE 2024

# Question 19

3 marks | Rounding, Estimation & Bounds

**19** Aviv goes on a cycle journey.

For the cycle journey

average speed = 19 km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.
Give your answer correct to 3 significant figures.

..... km

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2HR June 2024

### 1 Distance is found using

Distance is found using

$$\text{distance} = \text{speed} \times \text{time}$$

### 2 Find upper bound

For the upper bound,

$$\text{speed} < 19.5, \quad \text{time} < 1.55$$

$$\text{distance}_{\text{upper}} = 19.5 \times 1.55 = 30.225$$

### 3 Estimate the value

Correct to 3 significant figures,

$$30.225 \approx 30.2$$

**FINAL ANSWER**

$30.2$ km

4MA1/2H NOVEMBER 2024

# Question 2

2 marks | Rounding, Estimation & Bounds

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres

(1)

(b) Write down the lower bound of the length of the table.

..... metres

(1)

**(Total for Question 2 is 2 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2H November 2024

**1 The length is m correct**

The length is $1.4$ m correct to one decimal place.

**2 Half of is**

Half of $0.1$ is $0.05$.

$$1.4 + 0.05 = 1.45$$

$$1.4 - 0.05 = 1.35$$

**FINAL ANSWER**

(a) $1.45$ m, (b) $1.35$ m

4MA1/1H NOVEMBER 2024

# Question 21

3 marks | Rounding, Estimation & Bounds

**21** $T = \dfrac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of $T$

Give your answer correct to 3 significant figures.
Show your working clearly.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H November 2024

**1 Estimate the value**

$$T = \frac{x^2 + y^2}{w}$$

**2 Find upper bound**

For the upper bound, use the upper bounds of $x$ and $y$, and the lower bound of $w$.

$$x < 28.45, \quad y < 17.5, \quad w \geq 87.5$$

$$T_{\text{upper}} = \frac{28.45^2 + 17.5^2}{87.5} = 12.7503\ldots$$

**3 Estimate the value**

Correct to 3 significant figures,

$$12.7503\ldots \approx 12.8$$

**FINAL ANSWER**

$12.8$

4WM1/01R MAY 2025

# Question 1

2 marks | Rounding, Estimation & Bounds

**1** The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

..... m

(1)

(ii) Write down the upper bound of the length of the ship.

..... m

(1)

**(Total for Question 1 is 2 marks)**

## WORKED SOLUTION

Question 1 · 4WM1/01R May 2025

**1** **Calculate value**

The value is correct to 1 decimal place, so the half-unit is $0.05$.

$$142.8 - 0.05 = 142.75, \qquad 142.8 + 0.05 = 142.85$$

✓ **FINAL ANSWER**

lower bound $142.75$ m, upper bound $142.85$ m.

4MA1/1HR JUNE 2025

# Question 2

2 marks | Rounding, Estimation & Bounds

2 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

..... m

(1)

(ii) Write down the upper bound of the length of the ship.

..... m

(1)

**(Total for Question 2 is 2 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1HR June 2025

**1 The length is m correct**

The length is $142.8$ m correct to 1 decimal place.

**2 Half of is**

Half of $0.1$ is $0.05$.

$$142.8 - 0.05 = 142.75$$

$$142.8 + 0.05 = 142.85$$

**FINAL ANSWER**

(i) $142.75$ m, (ii) $142.85$ m

4WM1/01 MAY 2025

# Question 2

4 marks | Rounding, Estimation & Bounds

**2** The height of a tree is 17.8 metres correct to one decimal place.

(a) Write down the lower bound for the height of the tree.

..... metres

(1)

The weight of an apple is 145 grams correct to the nearest 5 grams.

(b) Write down the upper bound for the weight of the apple.

..... grams

(1)

$$A = \frac{4.766 \times 815}{0.399}$$

(c) By rounding each number to one significant figure, work out an estimate for the value of $A$
Show your working clearly.

.....

(2)

**(Total for Question 2 is 4 marks)**

## WORKED SOLUTION

Question 2 · 4WM1/01 May 2025

**1** **Find lower bound**

For $17.8$ correct to 1 decimal place, the lower bound is

$$17.8 - 0.05 = 17.75$$

**2** **Find upper bound**

For $145$ correct to the nearest $5$ grams, the upper bound is

$$145 + 2.5 = 147.5$$

**3** **Estimate the value**

For the estimate,

$$A = \frac{4.766 \times 815}{0.399}$$

**4** **Round to 1 significant figure**

Round to 1 significant figure.

$$4.766 \approx 5, \quad 815 \approx 800, \quad 0.399 \approx 0.4$$

$$A \approx \frac{5 \times 800}{0.4} = 10000$$

---

✓ **FINAL ANSWER**

$17.75$ m, $147.5$ g, and $10000$.

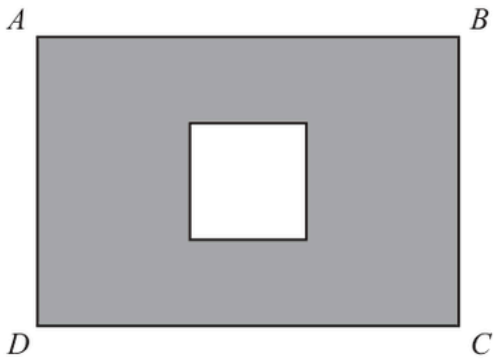
4MA1/2H JUNE 2025

# Question 22

4 marks | Rounding, Estimation & Bounds

**22** The diagram shows a square inside rectangle $ABCD$

Diagram **NOT** accurately drawn




The total area of the region shown shaded in the diagram is $X\text{cm}^2$

$AB = 11.5$ cm correct to the nearest 0.5 cm
$BC = 9.2$ cm correct to 2 significant figures
side of square $= 4.1$ cm correct to 2 significant figures

By considering bounds, work out the value of $X$ to a suitable degree of accuracy.
Show your working clearly.

$X =$ .....

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H June 2025

**1** **The shaded area is**

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

**2** **Bounds**

Bounds:

$$11.25 \leq AB < 11.75$$

$$9.15 \leq BC < 9.25$$

$$4.05 \leq \text{square side} < 4.15$$

**3** **Lower bound**

Lower bound:

$$11.25(9.15) - 4.15^2 = 85.715$$

**4** **Upper bound**

Upper bound:

$$11.75(9.25) - 4.05^2 = 92.285$$

**5** **Estimate the value**

So

$$85.715 < X < 92.285$$

**6** **Estimate the value**

Both bounds round to $90$ to the nearest $10$.

✓ **FINAL ANSWER**

$X = 90\ \text{cm}^2$ to a suitable degree of accuracy

4WM1/01 NOVEMBER 2025

# Question 2

2 marks | Rounding, Estimation & Bounds

2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

**(Total for Question 2 is 2 marks)**

## WORKED SOLUTION

Question 2 · 4WM1/01 November 2025

### 1 Round each number to one

Round each number to one significant figure:

$$2.11 \approx 2, \quad 58.9 \approx 60, \quad 8.859 \approx 9$$

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}} \approx \frac{2^2 \times 60}{\sqrt{9}}$$

$$= \frac{4 \times 60}{3} = 80$$

**FINAL ANSWER**

80

4MA1/1H NOVEMBER 2025

# Question 2

2 marks | Rounding, Estimation & Bounds

2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

**(Total for Question 2 is 2 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1H November 2025

### 1 Round each number to one

Round each number to one significant figure:

$$2.11 \approx 2, \quad 58.9 \approx 60, \quad 8.859 \approx 9$$

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}} \approx \frac{2^2 \times 60}{\sqrt{9}}$$

$$= \frac{4 \times 60}{3} = 80$$

**FINAL ANSWER**

80

4WM1/01 NOVEMBER 2025

# Question 6

2 marks | Rounding, Estimation & Bounds

**6** The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g

(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm

(1)

**(Total for Question 6 is 2 marks)**

## WORKED SOLUTION

Question 6 · 4WM1/01 November 2025

**1 Find lower bound**

The weight $475$ g correct to the nearest g has lower bound

$$475 - 0.5 = 474.5$$

**2 Find upper bound**

The height $120$ cm correct to the nearest $10$ cm has upper bound

$$120 + 5 = 125$$

**FINAL ANSWER**

(a) $474.5$ g, (b) $125$ cm

4MA1/1H NOVEMBER 2025

# Question 7

2 marks | Rounding, Estimation & Bounds

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g

(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm

(1)

**(Total for Question 7 is 2 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1H November 2025

**1 Find lower bound**

The weight $475$ g correct to the nearest g has lower bound

$$475 - 0.5 = 474.5$$

**2 Find upper bound**

The height $120$ cm correct to the nearest $10$ cm has upper bound

$$120 + 5 = 125$$

**FINAL ANSWER**

(a) $474.5$ g, (b) $125$ cm

4MA1/2H NOVEMBER 2025

# Question 24

5 marks | Rounding, Estimation & Bounds

**24** $D = \dfrac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place
$p = 7.24$ correct to 2 decimal places
$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of $D$ to a suitable degree of accuracy.
Show your working clearly.

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2H November 2025

**1 Estimate the value**

$$D = \frac{n}{p - q}$$

**2 Bounds**

Bounds:

$$10.25 \leq n < 10.35$$

$$7.235 \leq p < 7.245$$

$$4.385 \leq q < 4.395$$

**3 Lower bound**

Lower bound:

$$D_{\text{lower}} = \frac{10.25}{7.245 - 4.385} = 3.5839\ldots$$

**4 Upper bound**

Upper bound:

$$D_{\text{upper}} = \frac{10.35}{7.235 - 4.395} = 3.6443\ldots$$

**5** **Estimate the value**

So

$$3.5839\ldots < D < 3.6443\ldots$$

---

**6** **Estimate the value**

Both bounds round to $\mathbf{3.6}$ to 1 decimal place.

---

✓ **FINAL ANSWER**

$D = \mathbf{3.6}$ to a suitable degree of accuracy

4MA1/1HR JUNE 2026

# Question 4

2 marks | Rounding, Estimation & Bounds

**4** The weight of a giraffe is 1178 kg, correct to the nearest kg

(i) Write down the lower bound of the weight of the giraffe.

..... kg

**(1)**

(ii) Write down the upper bound of the weight of the giraffe.

..... kg

**(1)**

**(Total for Question 4 is 2 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1HR June 2026

### 1 Part (i) | Step 1

$$1178 - 0.5 = 1177.5 \text{ kg}.$$

Rounding to the nearest kilogram allows half a kilogram below the stated value.

### 2 Part (ii) | Step 1

$$1178 + 0.5 = 1178.5 \text{ kg}.$$

The rounding interval is $1177.5 \leq w < 1178.5$; the upper endpoint is the upper bound, not an included weight.

**FINAL ANSWER**

(i) $1177.5$ kg (ii) $1178.5$ kg

# Question 18

5 marks | Rounding, Estimation & Bounds

**18** The final velocity, $v$ m/s, of an object is given by

$$v = 2\left(\frac{s}{t}\right) - u$$

where

$s = 100$ correct to the nearest integer
$t = 1.68$ correct to 2 decimal places
$u = 0.5$ correct to 1 decimal place

By considering bounds, work out the value of $v$ to a suitable degree of accuracy.
Show your working clearly.

.....

**(Total for Question 18 is 5 marks)**

**1** **Bounds**

$99.5 \leq s < 100.5$, $1.675 \leq t < 1.685$, $0.45 \leq u < 0.55$

**2** **Upper bound of v**

Big $s$, small $t$, small $u$:

$$v_{\max} = 2\left(\frac{100.5}{1.675}\right) - 0.45 = 119.55$$

**3** **Lower bound of v**

Small $s$, big $t$, big $u$:

$$v_{\min} = 2\left(\frac{99.5}{1.685}\right) - 0.55 = 117.55\ldots$$

**4** **Suitable accuracy**

Both bounds round to $120$ to 2 s.f. (they differ to 3 s.f.), so $v = 120$.

✓ **FINAL ANSWER**

$v = 120$ m/s (2 s.f.)

4MA1/1H JUNE 2026

# Question 20

5 marks | Rounding, Estimation & Bounds

**20** The final velocity, $v$ m/s, of an object is given by

$$v = 2\left(\frac{s}{t}\right) - u$$

where

$s = 100$ correct to the nearest integer
$t = 1.68$ correct to 2 decimal places
$u = 0.5$ correct to 1 decimal place

By considering bounds, work out the value of $v$ to a suitable degree of accuracy.
Show your working clearly.

.....

**(Total for Question 20 is 5 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H June 2026

### 1 Solution | Step 1

$$99.5 \leq s < 100.5, \quad 1.675 \leq t < 1.685, \quad 0.45 \leq u < 0.55.$$

Use half a rounding unit on each side of the stated value.

### 2 Solution | Step 2

For $v = 2s/t - u$, the lower limiting value is

$$v_L = \frac{2(99.5)}{1.685} - 0.55 = 117.550890\ldots \text{ m/s}.$$

Decrease the numerator, increase the positive denominator and subtract the largest u.

### 3 Solution | Step 3

The upper limiting value is

$$v_U = \frac{2(100.5)}{1.675} - 0.45 = 119.55 \text{ m/s}.$$

Increase the numerator, decrease the denominator and subtract the smallest u.

4

### Solution | Step 4

Both limiting values round to $120\ \mathrm{m/s}$ to 2 significant figures. They do not round to the same integer, so

$$v = 120\ \mathrm{m/s} \quad (2\ \mathrm{s.f.}).$$

This is the greatest common significant-figure accuracy justified throughout the interval.

---

✓ **FINAL ANSWER**

$120\ \mathrm{m/s}$ to 2 significant figures.

4MA1/2HR JUNE 2026

# Question 21

3 marks | Rounding, Estimation & Bounds

**21** $p = \dfrac{d-e}{f}$

$d = 6.6$ correct to 1 decimal place
$e = 2.8$ correct to 1 decimal place
$f = 0.47$ correct to 2 decimal places

Calculate the lower bound for the value of $p$
Give your answer correct to 3 significant figures.
Show your working clearly.

.....

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2HR June 2026

**1 Solution | Step 1**

$$6.55 \le d < 6.65, \quad 2.75 \le e < 2.85, \quad 0.465 \le f < 0.475.$$

Use half of each rounding unit to obtain the bounds.

**2 Solution | Step 2**

$$p_{\text{lower}} = \frac{6.55 - 2.85}{0.475} = \frac{3.70}{0.475} = 7.789473684\ldots.$$

Minimize the positive numerator by using the lower bound of d and upper bound of e; maximize the positive denominator.

**3 Solution | Step 3**

$$p_{\text{lower}} = 7.79 \quad \text{(3 significant figures)}.$$

Round the boundary value to the requested accuracy.

✓ **FINAL ANSWER**

Lower bound $= 7.79$ (3 significant figures).

TOPIC

# Powers & Roots

Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 33

3 marks | Powers & Roots

**33** Find the value of

(a) $5^0$

.....

**(1)**

(b) $8^{2/3}$

.....

**(1)**

(c) $16^{-1/2}$

.....

**(1)**

**(Total for Question 33 is 3 marks)**

## WORKED SOLUTION

Question 33 · EP Specimen

**1** **(a) Any non-zero number to the power 0**

$5^0 = 1$

**2** **(b) Root first, then power**

$$8^{2/3} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$

**3** **(c) Negative power means reciprocal**

$$16^{-1/2} = \frac{1}{\sqrt{16}} = \frac{1}{4}$$

✓ **FINAL ANSWER**

(a) $1$ (b) $4$ (c) $\frac{1}{4}$

EP SPECIMEN

# Question 34

2 marks | Powers & Roots

**34** Write $\dfrac{3^5 \times 3^{-2}}{3^4}$ as a single power of 3

.....

**(2)**

**(Total for Question 34 is 2 marks)**

## WORKED SOLUTION

Question 34 · EP Specimen

**1 Multiply: add the powers**

$3^5 \times 3^{-2} = 3^3$

**2 Divide: subtract the powers**

$$\frac{3^3}{3^4} = 3^{-1}$$

✓ **FINAL ANSWER**

$3^{-1}$

EP SPECIMEN

# Question 35

3 marks | Powers & Roots

**35** $2^n = 8^4 \div 4^3$

Find the value of $n$.

$n =$ .....

**(3)**

**(Total for Question 35 is 3 marks)**

## WORKED SOLUTION

Question 35 · EP Specimen

**1 Write everything as a power of 2**

$8 = 2^3$, so $8^4 = 2^{12}$. $4 = 2^2$, so $4^3 = 2^6$.

**2 Divide**

$$\frac{2^{12}}{2^6} = 2^6 \Rightarrow n = 6$$

**FINAL ANSWER**

$n = 6$

# Question 3

5 marks | Powers & Roots

**3** $20^{70} \times 20^{8} = 20^{n}$

(a) Find the value of $n$

$n =$ .....

**(1)**

$36^{80} \div 36^{5} = 36^{m}$

(b) Find the value of $m$

$m =$ .....

**(1)**

$\left(52^{4}\right)^{16} = 52^{p}$

(c) Find the value of $p$

$p =$ .....

**(1)**

In a race, Dani ran a distance of 100 metres, correct to the nearest metre.

(d) Write down the lower bound for the distance.

..... metres

**(1)**

Dani ran this race in a time of 16.3 seconds, correct to one decimal place.

(e) Write down the upper bound for the time.

..... seconds

**(1)**

**(Total for Question 3 is 5 marks)**

**1** **(a) Multiply: add powers**

$$20^{70} \times 20^{8} = 20^{78} \Rightarrow n = 78$$

**2** **(b) Divide: subtract powers**

$$36^{80} \div 36^{5} = 36^{75} \Rightarrow m = 75$$

**3** **(c) Power of a power: multiply**

$$(52^{4})^{16} = 52^{64} \Rightarrow p = 64$$

**4** **(d) Lower bound**

Nearest metre means $\pm 0.5$ m: lower bound $= 99.5$ m.

**5** **(e) Upper bound**

One decimal place means $\pm 0.05$ s: upper bound $= 16.35$ s.

✓ **FINAL ANSWER**

(a) $78$ (b) $75$ (c) $64$ (d) $99.5$ m (e) $16.35$ s



TOPIC

# Surds

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 17

4 marks | Surds

17 Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where $p$ and $q$ are integers.

Show clearly each stage of your working.

.....

(Total for Question 17 is 4 marks)

## WORKED SOLUTION

Question 17 · 4WM1/01 Specimen

**1 Simplify the surds**

$$\sqrt{8}=2\sqrt{2}$$

$$(\sqrt{2}-1)^2=2-2\sqrt{2}+1=3-2\sqrt{2}$$

**2 Use the conjugate**

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}}\times\frac{3+2\sqrt{2}}{3+2\sqrt{2}}$$

The denominator is

$$9-(2\sqrt{2})^2=9-8=1$$

**3** **Expand the numerator**

$$(3+2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

**4** **Write in the requested form**

$$12\sqrt{2} = \sqrt{144 \times 2} = \sqrt{288}$$

✓ **FINAL ANSWER**

$17 + \sqrt{288}$.

4MA1/2H JANUARY 2020

# Question 20

3 marks | Surds

**20** The area of a rectangle is $18\,\text{cm}^2$

The length of the rectangle is $\left(\sqrt{7}+1\right)$ cm.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.
Give your answer in the form $a\sqrt{b}+c$ where $a$, $b$ and $c$ are integers.

..... cm

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H January 2020

### 1 Simplify surd

$$\text{width} = \frac{18}{\sqrt{7}+1}$$

### 2 Rationalise the denominator

Rationalise the denominator:

$$\frac{18}{\sqrt{7}+1} \times \frac{\sqrt{7}-1}{\sqrt{7}-1} = \frac{18(\sqrt{7}-1)}{7-1}$$

$$= \frac{18(\sqrt{7}-1)}{6} = 3(\sqrt{7}-1)$$

$$= 3\sqrt{7}-3$$

**FINAL ANSWER**

$3\sqrt{7}-3$ cm

4MA1/1H MAY/NOVEMBER 2020

# Question 19

3 marks | Surds

19 Without using a calculator, rationalise the denominator of $\frac{6}{3-\sqrt{7}}$

Simplify your answer.
You must show each stage of your working.

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1H May/November 2020

### 1 Simplify surd

$$\frac{6}{3-\sqrt{7}}$$

### 2 Rationalise the denominator

Rationalise the denominator:

$$\frac{6}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}} = \frac{6(3+\sqrt{7})}{9-7}$$

$$= \frac{6(3+\sqrt{7})}{2} = 3(3+\sqrt{7})$$

$$= 9+3\sqrt{7}$$

**FINAL ANSWER**

$9+3\sqrt{7}$

4MA1/1H JANUARY 2021

# Question 18

3 marks | Surds

**18** Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where $x$ is a prime number and $y$ is an integer,

find the value of $x$ and the value of $y$.
Show each stage of your working clearly.

$x =$ .....

$y =$ .....

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1H January 2021

**1 Simplify surd**

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

$$= 40 + 3\sqrt{x} - x$$

**2 Simplify surd**

This is equal to

$$y\sqrt{x} + 21$$

**3 Compare the rational parts**

Compare the rational parts:

$$40 - x = 21$$

$$x = 19$$

**4 Compare the surd coefficients**

Compare the surd coefficients:

$$y = 3$$

**FINAL ANSWER**

$x = 19, y = 3$

4MA1/2H NOVEMBER 2021

# Question 17

3 marks | Surds

**17** Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a}+b$ where $a$ and $b$ are integers.

Show each stage of your working clearly.

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/2H November 2021

### 1 Simplify surd

$$\frac{8}{\sqrt{5}-1}$$

### 2 Rationalise the denominator

Rationalise the denominator:

$$\frac{8}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{8(\sqrt{5}+1)}{5-1}$$

$$= 2(\sqrt{5}+1) = 2\sqrt{5}+2$$

### 3 Simplify surd

In the form $\sqrt{a}+b$,

$$2\sqrt{5}+2 = \sqrt{20}+2$$

**FINAL ANSWER**

$\sqrt{20}+2$

4MA1/1H JUNE 2022

# Question 16

3 marks | Surds

**16** Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - \left(\sqrt{2}\right)^5 = 2\sqrt{32} + 12$

Show your working clearly.

**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H June 2022

**1** **Proof**

Rationalise the denominator:

$$\frac{12}{\sqrt{2}-1} = \frac{12(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)}$$

$$= 12(\sqrt{2}+1) = 12\sqrt{2}+12$$

Also,

$$(\sqrt{2})^5 = (\sqrt{2})^4\sqrt{2} = 4\sqrt{2}$$

So the left side is

$$12\sqrt{2}+12-4\sqrt{2} = 8\sqrt{2}+12$$

And

$$2\sqrt{32}+12 = 2(4\sqrt{2})+12 = 8\sqrt{2}+12$$

So both sides are equal.

**FINAL ANSWER**

Shown.

4MA1/1HR JUNE 2022

# Question 21

4 marks | Surds

**21** Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where $p$ and $q$ are integers.

Show each stage of your working clearly.

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1HR June 2022

**1** **Simplify surd**

$$\frac{3+\sqrt{8}}{(\sqrt{2}-1)^2}$$

**2** **Simplify surd**

Since $\sqrt{8}=2\sqrt{2}$,

$$\frac{3+2\sqrt{2}}{(\sqrt{2}-1)^2}$$

**3** **Simplify surd**

Also

$$(\sqrt{2}-1)^2=3-2\sqrt{2}$$

**4** **Simplify surd**

So

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}}\times\frac{3+2\sqrt{2}}{3+2\sqrt{2}}=\frac{(3+2\sqrt{2})^2}{9-8}$$

$$=(3+2\sqrt{2})^2=9+12\sqrt{2}+8$$

$$=17+12\sqrt{2}=17+\sqrt{288}$$

**FINAL ANSWER**

$17+\sqrt{288}$

4MA1/2H JUNE 2023

# Question 23

4 marks | Surds

**23** The diagram shows a cuboid with a square cross section.

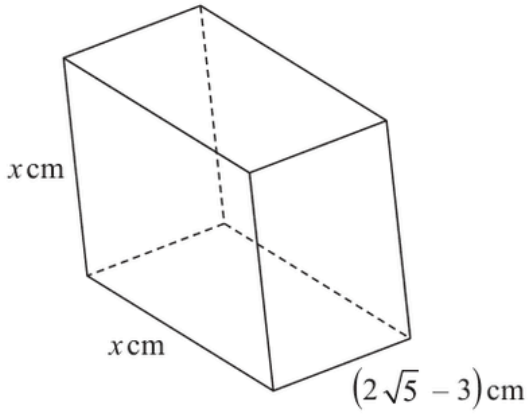



Diagram **NOT** accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of $x$
Give your answer in the form $a + \sqrt{b}$ where $a$ and $b$ are integers.
Show your working clearly.

$x =$ .....

**(Total for Question 23 is 4 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2H June 2023

**1 The volume is**

The volume is

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

**2 Simplify surd**

So

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

**3** **Rationalise**

Rationalise:

$$x^2 = \frac{(13+6\sqrt{5})(2\sqrt{5}+3)}{(2\sqrt{5})^2 - 3^2}$$

$$= \frac{26\sqrt{5}+39+60+18\sqrt{5}}{20-9}$$

$$= \frac{99+44\sqrt{5}}{11} = 9+4\sqrt{5}$$

---

**4** **Simplify surd**

Since

$$(\sqrt{5}+2)^2 = 5+4+4\sqrt{5} = 9+4\sqrt{5}$$

$$x = \sqrt{5}+2$$

---

✓ **FINAL ANSWER**

$2+\sqrt{5}$

4WM1/01R MAY 2025

# Question 16

4 marks | Surds

**16** $\sqrt{96} = 2\sqrt{k}$

(a) Find the value of $k$

$k =$ .....

**(1)**

(b) Show that $\dfrac{8}{3-\sqrt{5}}$ can be written in the form $a+\sqrt{b}$ where $a$ and $b$ are integers.

Show each stage of your working.

**(3)**

**(Total for Question 16 is 4 marks)**

## WORKED SOLUTION

Question 16 · 4WM1/01R May 2025

**1** **(a)**

(a)

$$\sqrt{96} = 4\sqrt{6} = 2\sqrt{24}$$

**2** **Calculate value**

So $k = 24$.

**3** **(b)**

(b)

$$\frac{8}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$$

$$= \frac{8(3+\sqrt{5})}{9-5} = 2(3+\sqrt{5})$$

$$= 6+2\sqrt{5}$$

✓ **FINAL ANSWER**

$k = 24, 6+2\sqrt{5}$.

4WM1/01 MAY 2025

# Question 17

4 marks | Surds

**17** $(\sqrt{3})^5 = k\sqrt{3}$ where $k$ is an integer.

(a) Find the value of $k$

$k =$ .....

(1)

(b) Show that $\dfrac{21}{3-\sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where $c$ and $d$ are integers.
Show each stage of your working clearly.

(3)

**(Total for Question 17 is 4 marks)**

## WORKED SOLUTION

Question 17 · 4WM1/01 May 2025

**1** **Simplify surd**

$$(\sqrt{3})^5 = (\sqrt{3})^4\sqrt{3} = 3^2\sqrt{3} = 9\sqrt{3}$$

**2** **Simplify surd**

So $k = 9$.

**3** **Split the ratio**

For part (b), rationalise the denominator.

$$\frac{21}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{21(3+\sqrt{2})}{9-2}$$

$$= 3(3+\sqrt{2}) = 9+3\sqrt{2}$$

**4** **Simplify surd**

Since $3\sqrt{2} = \sqrt{18}$,

$$9+3\sqrt{2} = 9+\sqrt{18}$$

**FINAL ANSWER**

$k = 9$, and $9 + \sqrt{18}$.

4WM1/01 NOVEMBER 2025

# Question 16

3 marks | Surds

**16** Show that $\dfrac{2\sqrt{7}+2}{\sqrt{7}-3}$ can be written in the form $a-\sqrt{b}$ where $a$ and $b$ are integers.

Show your working clearly.

**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4WM1/01 November 2025

**1** **Simplify surd**

$$\frac{2\sqrt{7}+2}{\sqrt{7}-3}\times\frac{\sqrt{7}+3}{\sqrt{7}+3}$$

$$=\frac{(2\sqrt{7}+2)(\sqrt{7}+3)}{7-9}$$

$$=\frac{20+8\sqrt{7}}{-2}$$

$$=-10-4\sqrt{7}=-10-\sqrt{112}$$

**FINAL ANSWER**

$-10-\sqrt{112}$.

# Question 16

6 marks | Surds

**16** (a) Solve $x^2 - \left(2\sqrt{3}\right)x - 13 = 0$

Give your solutions in the form $\sqrt{a} \pm b$ where $a$ and $b$ are integers.
Show your working clearly.

.....

**(3)**

(b) Show that $\dfrac{\sqrt{20}}{\sqrt{5}+2}$ can be written in the form $c - \sqrt{d}$ where $c$ and $d$ are integers.

**(3)**

**(Total for Question 16 is 6 marks)**

**1** **(a) Quadratic formula**

$$x = \frac{2\sqrt{3} \pm \sqrt{(2\sqrt{3})^2 + 4(13)}}{2} = \frac{2\sqrt{3} \pm \sqrt{64}}{2}$$

**2** **Simplify**

$$x = \frac{2\sqrt{3} \pm 8}{2} = \sqrt{3} \pm 4$$

**3** **(b) Rationalise**

$$\frac{\sqrt{20}}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2} = \frac{2\sqrt{5}(\sqrt{5}-2)}{5-4}$$

**4** **Simplify**

$$= 10 - 4\sqrt{5} = 10 - \sqrt{80} \quad (c = 10,\ d = 80)$$

✓ **FINAL ANSWER**

(a) $\sqrt{3} \pm 4$ (b) $10 - \sqrt{80}$

4MA1/1H JUNE 2026

# Question 17

3 marks | Surds

**17** Show that $\left(4-\sqrt{20}\right)\left(5-2\sqrt{5}\right)$ can be written in the form $a-\sqrt{b}$

where $a$ and $b$ are integers.

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1H June 2026

**1 Solution | Step 1**

$$\sqrt{20}=2\sqrt{5}, \qquad (4-\sqrt{20})(5-2\sqrt{5})=(4-2\sqrt{5})(5-2\sqrt{5}).$$

Simplify the surd first.

**2 Solution | Step 2**

$$20-8\sqrt{5}-10\sqrt{5}+20=40-18\sqrt{5}.$$

Expand; the product of the two surd terms is positive 20.

**3 Solution | Step 3**

$$18\sqrt{5}=\sqrt{18^2\times 5}=\sqrt{1620}, \quad (4-\sqrt{20})(5-2\sqrt{5})=40-\sqrt{1620}.$$

Thus the required integer values are a=40 and b=1620.

**FINAL ANSWER**

$40-\sqrt{1620}$, so $a=40,\ b=1620$.

TOPIC

# Using a Calculator

Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 1

2 marks | Using a Calculator

**1** Work out $\dfrac{5.7^2 - \sqrt{30.25}}{1.8 \times 0.65}$

Write down all the figures on your calculator display.

.....

**(2)**

**(Total for Question 1 is 2 marks)**

## WORKED SOLUTION

Question 1 · EP Specimen

**1** **Work out the top and the bottom**

$$5.7^2 - \sqrt{30.25} = 32.49 - 5.5 = 26.99, \qquad 1.8 \times 0.65 = 1.17$$

**2** **Divide**

$$\frac{26.99}{1.17} = 23.06837607\ldots$$

✓ **FINAL ANSWER**

$23.06837607$

EP SPECIMEN

# Question 2

3 marks | Using a Calculator

**2** (a) Work out $4.3^3 - \frac{1}{0.16}$

Write down all the figures on your calculator display.

.....

**(2)**

(b) Give your answer to part (a) correct to 3 significant figures.

.....

**(1)**

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · EP Specimen

**1** **(a) Use the calculator**

$$4.3^3 - \frac{1}{0.16} = 79.507 - 6.25 = 73.257$$

**2** **(b) Round to 3 significant figures**

The fourth figure is $5$, so round up: $73.3$.

**FINAL ANSWER**

(a) $73.257$ (b) $73.3$

EP SPECIMEN

# Question 3

2 marks | Using a Calculator

**3** Work out $\dfrac{8.4 \times \tan 35°}{\cos 62°}$

Give your answer correct to 3 significant figures.

.....

**(2)**

**(Total for Question 3 is 2 marks)**

## WORKED SOLUTION

Question 3 · EP Specimen

**1 Check the calculator is in degrees**

Then

$$\frac{8.4 \times \tan 35°}{\cos 62°} = \frac{5.8817\ldots}{0.46947\ldots} = 12.528\ldots$$

**2 Round**

$12.5$ (3 significant figures)

**FINAL ANSWER**

$12.5$

EP SPECIMEN

# Question 4

3 marks | Using a Calculator

**4** Work out $\dfrac{3.7 \times 10^5 + 2.9 \times 10^4}{8.1 \times 10^{-3}}$

Give your answer in standard form correct to 3 significant figures.

.....

**(3)**

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · EP Specimen

**1 Add the top**

$$3.7 \times 10^5 + 2.9 \times 10^4 = 370\,000 + 29\,000 = 399\,000$$

**2 Divide**

$$\frac{399\,000}{8.1 \times 10^{-3}} = 49\,259\,259.26\ldots$$

**3 Standard form, 3 significant figures**

$4.93 \times 10^7$

✓ **FINAL ANSWER**

$4.93 \times 10^7$

EP SPECIMEN

# Question 5

2 marks | Using a Calculator

**5** $p = 2.6 \quad q = -1.8$

Work out the value of $\dfrac{p^2 - 3q}{\sqrt{(p + q^2)}}$

Give your answer correct to 3 decimal places.

.....

**(2)**

**(Total for Question 5 is 2 marks)**

## WORKED SOLUTION

Question 5 · EP Specimen

**1** **Substitute (brackets round the negative)**

$$p^2 - 3q = 2.6^2 - 3(-1.8) = 6.76 + 5.4 = 12.16$$

$$\sqrt{p + q^2} = \sqrt{2.6 + (-1.8)^2} = \sqrt{5.84} = 2.4166\ldots$$

**2** **Divide and round**

$$\frac{12.16}{2.4166\ldots} = 5.03184\ldots \approx 5.032$$

✓ **FINAL ANSWER**

5.032

EP SPECIMEN

# Question 6

2 marks | Using a Calculator

6 Work out the value of $4.8^{-0.5} + 1.7^3$

Give your answer correct to 4 significant figures.

.....

(2)

**(Total for Question 6 is 2 marks)**

## WORKED SOLUTION

Question 6 · EP Specimen

### 1 Evaluate each power

$$4.8^{-0.5} = \frac{1}{\sqrt{4.8}} = 0.45643\ldots, \qquad 1.7^3 = 4.913$$

### 2 Add and round

$$0.45643\ldots + 4.913 = 5.36943\ldots \approx 5.369$$

✓ **FINAL ANSWER**

$5.369$

EP SPECIMEN

# Question 7

3 marks | Using a Calculator

7 Here is a calculation: $\frac{19.6 \times 0.48}{4.1}$

(a) By rounding each number to 1 significant figure, work out an estimate for the calculation.

.....

**(2)**

(b) Use your calculator to work out the calculation. Give your answer correct to 3 significant figures.

.....

**(1)**

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · EP Specimen

**1** **(a) Round each number to 1 significant figure**

$$\frac{20 \times 0.5}{4} = \frac{10}{4} = 2.5$$

**2** **(b) Calculator value**

$$\frac{19.6 \times 0.48}{4.1} = 2.2946\ldots \approx 2.29$$

✓ **FINAL ANSWER**

(a) $2.5$ (b) $2.29$

TOPIC

# Standard & Compound Units

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 5

3 marks | Standard & Compound Units

**5** A plane flies from New York City to Los Angeles.

The plane flies a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the plane.
Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

**(Total for Question 5 is 3 marks)**

## WORKED SOLUTION

Question 5 · 4WM1/01 Specimen

**1 Convert the time**

$$24 \text{ minutes} = \frac{24}{60} = 0.4 \text{ hours}$$

$$5 \text{ hours } 24 \text{ minutes} = 5.4 \text{ hours}$$

**2 Calculate the speed**

$$\text{speed} = \frac{3980}{5.4} = 737.037\ldots$$

**3 Round the answer**

$$737.037\ldots \approx 737$$

**FINAL ANSWER**

737 kilometres per hour.

4WM1/01 SPECIMEN

# Question 8

3 marks | Standard & Compound Units

(Total for Question 7 is 4 marks)

**8** Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

**(Total for Question 8 is 3 marks)**

## WORKED SOLUTION

Question 8 · 4WM1/01 Specimen

**1 Convert kilometres to metres**

$$81 \text{ km} = 81000 \text{ m}$$

**2 Convert hours to seconds**

$$1 \text{ hour} = 3600 \text{ seconds}$$

**3 Divide to find metres per second**

$$\frac{81000}{3600} = 22.5$$

**FINAL ANSWER**

22.5 metres per second.

4MA1/2H JANUARY 2020

# Question 2

2 marks | Standard & Compound Units

2 Change 32.4 $m^3$ into $cm^3$

..... $cm^3$

**(Total for Question 2 is 2 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2H January 2020

**1 Calculate value**

$$1\text{ m} = 100\text{ cm}$$

**2 Calculate value**

So

$$1\text{ m}^3 = 100^3\text{ cm}^3 = 1,000,000\text{ cm}^3$$

$$32.4\text{ m}^3 = 32.4 \times 1,000,000$$

$$= 32,400,000\text{ cm}^3$$

**FINAL ANSWER**

$32,400,000\text{ cm}^3$.

4MA1/2HR JANUARY 2020

# Question 5

3 marks | Standard & Compound Units

5 A plane takes 3 hours 36 minutes to fly from the Cayman Islands to New York.
The plane flies a distance of 2470 km.

Work out the average speed of the plane in km/h.
Give your answer correct to the nearest whole number.

..... km/h

**(Total for Question 5 is 3 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2HR January 2020

### 1 The time is

The time is

$$3\text{ h }36\text{ min} = 3.6\text{ h}$$

### 2 Average speed

Average speed:

$$2470 \div 3.6 = 686.111\ldots$$

**FINAL ANSWER**

$686\text{ km/h}$.

4MA1/1HR JANUARY 2020

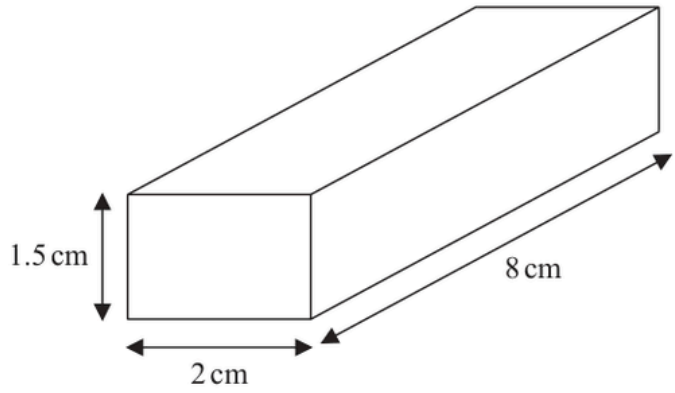
# Question 9

5 marks | Standard & Compound Units

9 Pablo made a solid gold statue.

He melted down some gold blocks and used the gold to make the statue.
Each block of gold was a cuboid, as shown below.

Diagram **NOT** accurately drawn




The mass of the statue is 5.73 kg.
The density of gold is 19.32 g/cm$^3$

Work out the least number of gold blocks Pablo melted down in order to make the statue.
Show your working clearly.

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1HR January 2020

### 1 Convert the mass to grams

Convert the mass to grams:

$$5.73 \text{ kg} = 5730 \text{ g}$$

### 2 Calculate volume

Using

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{volume of gold} = \frac{5730}{19.32} = 296.58\ldots \text{ cm}^3$$

**3** **Volume of one cuboid block**

Volume of one cuboid block is

$$1.5 \times 2 \times 8 = 24 \text{ cm}^3$$

**4** **Number of blocks needed is**

Number of blocks needed is

$$\frac{296.58\ldots}{24} = 12.35\ldots$$

**5** **Pablo must have a whole**

Pablo must have a whole number of blocks, so he needs at least 13 blocks.

✓ **FINAL ANSWER**

$13$ blocks.

4MA1/2H JANUARY 2020

# Question 9

2 marks | Standard & Compound Units

9 The density of gold is 19.3 g/cm$^3$
A gold bar has volume 150 cm$^3$

Work out the mass of the gold bar.

..... g

**(Total for Question 9 is 2 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2H January 2020

### 1 Calculate volume

$$\text{mass} = \text{density} \times \text{volume}$$

$$\text{mass} = 19.3 \times 150$$

$$\text{mass} = 2895$$

**FINAL ANSWER**

2895 g.

4MA1/2HR JANUARY 2020

# Question 9

3 marks | Standard & Compound Units

**9** The diagram shows a box in the shape of a cuboid.

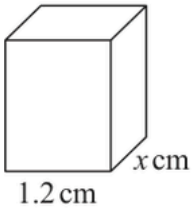



Diagram **NOT** accurately drawn

The box is put on a table.
The face of the box in contact with the table has length 1.2 metres and width $x$ metres.

The force exerted by the box on the table is 27 newtons.
The pressure on the table due to the box is 30 newtons/m$^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the value of $x$.

$x =$ .....

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2HR January 2020

### 1 Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$30 = \frac{27}{\text{area}}$$

$$\text{area} = \frac{27}{30} = 0.9 \text{ m}^2$$

**2** **The base area is**

The base area is

$$1.2x = 0.9$$

$$x = \frac{0.9}{1.2} = 0.75$$

---

✓ **FINAL ANSWER**

$x = 0.75\ \text{m}$.

4MA1/2H JANUARY 2020

# Question 10

3 marks | Standard & Compound Units

**10** Change a speed of 50 metres per second to a speed in kilometres per hour.

..... kilometres per hour

**(Total for Question 10 is 3 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2H January 2020

**1** **Convert m/s to km/h multiply**

To convert m/s to km/h, multiply by $\mathbf{3.6}$.

$$50 \times 3.6 = 180$$

**FINAL ANSWER**

$180\ \mathrm{km/h}$.

4MA1/1H JANUARY 2020

# Question 13

4 marks | Standard & Compound Units

**13** The diagram shows a solid cube.
The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

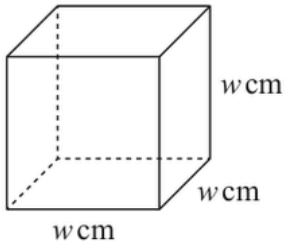



Diagram **NOT** accurately drawn

The cube exerts a force of 56 newtons on the table.
The pressure on the table due to the cube is 0.14 newtons/cm$^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

..... cm$^3$

**(Total for Question 13 is 4 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H January 2020

### 1 Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$0.14 = \frac{56}{\text{area}}$$

$$\text{area} = \frac{56}{0.14} = 400 \text{ cm}^2$$

**2** **The cube rests on one**

The cube rests on one square face, so

$$w^2 = 400$$

$$w = 20$$

**3** **Volume of the cube is**

Volume of the cube is

$$20^3 = 8000$$

**FINAL ANSWER**

$8000\ \text{cm}^3$.

4MA1/1H MAY/NOVEMBER 2020

# Question 6

3 marks | Standard & Compound Units

6 A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.
Give your answer in seconds, correct to the nearest second.

..... seconds

**(Total for Question 6 is 3 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1H May/November 2020

**1 Use**

Use

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$\text{time} = \frac{100}{28440} \text{ hours}$$

**2 Convert hours to seconds**

Convert hours to seconds:

$$\frac{100}{28440} \times 3600 = 12.658\ldots$$

**3 To the nearest second**

To the nearest second,

$$12.658\ldots \approx 13$$

**FINAL ANSWER**

13 seconds.

4MA1/1HR MAY/NOVEMBER 2020

# Question 7

4 marks | Standard & Compound Units

7 Pedro drove from Toulouse to Montpellier in 2 hours 42 minutes.
He drove at an average speed of 90 km/hour.

Janine drove from Toulouse to Montpellier along the same route as Pedro.
The journey took her 3 hours.

Work out Janine's average speed for the journey.

..... km/hour

**(Total for Question 7 is 4 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1HR May/November 2020

**1 Pedro s time was**

Pedro's time was

$$2\text{ h }42\text{ min} = 2.7\text{ h}$$

**2 Distance from Toulouse to Montpellier**

Distance from Toulouse to Montpellier:

$$90 \times 2.7 = 243\text{ km}$$

**3 Calculate value**

Janine took 3 hours, so her average speed was

$$243 \div 3 = 81$$

✓ **FINAL ANSWER**

$81\text{ km/hour}$.

4MA1/1HR MAY/NOVEMBER 2020

# Question 9

3 marks | Standard & Compound Units

9 A box is put on a horizontal table.

The face of the box in contact with the table is a square of side 1.5 metres.
The pressure on the table due to the box is 34.8 newtons/m$^2$

Work out the force exerted by the box on the table.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1HR May/November 2020

### 1 The area of the square

The area of the square face is

$$1.5 \times 1.5 = 2.25 \text{ m}^2$$

### 2 Use

Use

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

### 3 Calculate area

So

$$\text{force} = \text{pressure} \times \text{area}$$

$$34.8 \times 2.25 = 78.3$$

**FINAL ANSWER**

78.3 newtons.

4MA1/2H JANUARY 2021

# Question 1

3 marks | Standard & Compound Units

1 A train takes 6 hours 39 minutes to travel from New Delhi to Kanpur.
The train travels a distance of 429 km.

Work out the average speed of the train.
Give your answer in km/h correct to one decimal place.

..... km/h

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2H January 2021

**1 The time is**

The time is

$$6\text{ h }39\text{ min} = 6.65\text{ h}$$

**2 Average speed**

Average speed:

$$429 \div 6.65 = 64.511\ldots$$

✓ **FINAL ANSWER**

$64.5\text{ km/h}$.

4MA1/1HR JANUARY 2021

# Question 4

3 marks | Standard & Compound Units

4 A train journey from Paris to Amsterdam took 3 hours 24 minutes.
The total distance the train travelled was 433.5 km.

Work out the average speed of the train.
Give your answer in kilometres per hour.

..... km/h

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1HR January 2021

**1 The time is**

The time is

$$3\text{ h }24\text{ min} = 3.4\text{ h}$$

**2 Average speed**

Average speed:

$$433.5 \div 3.4 = 127.5$$

✓ **FINAL ANSWER**

$127.5\text{ km/h}$.

4MA1/1H MAY 2021

# Question 1

3 marks | Standard & Compound Units

1 A plane flew from Madrid to Dubai.

The distance the plane flew was 5658 km.
The flight time was 8 hours 12 minutes.

Work out the average speed of the plane.

..... km/h

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H May 2021

**1 Evaluate fraction**

$$8\text{ h }12\text{ min} = 8 + \frac{12}{60} = 8.2\text{ hours}$$

$$\text{average speed} = \frac{5658}{8.2} = 690$$

**FINAL ANSWER**

$690\text{ km/h}$.

4MA1/1H NOVEMBER 2021

# Question 10

5 marks | Standard & Compound Units

**10** A solid aluminium cylinder has radius 10 cm and height $h$ cm.

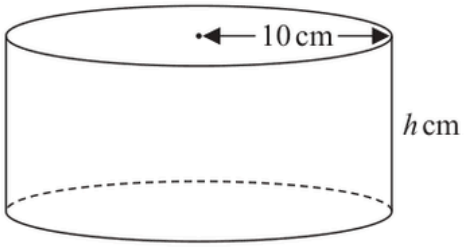



Diagram **NOT** accurately drawn

The mass of the cylinder is 5.4 kg.
The density of aluminium is 0.0027 kg/cm$^3$

Calculate the value of $h$.
Give your answer correct to one decimal place.

$h =$ .....

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1H November 2021

### 1 Calculate volume

The mass is $5.4\ \text{kg}$, and the density is $0.0027\ \text{kg/cm}^3$.

$$\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{5.4}{0.0027} = 2000\ \text{cm}^3$$

### 2 Evaluate fraction

For the cylinder,

$$V = \pi r^2 h$$

$$2000 = \pi(10)^2 h$$

$$h = \frac{2000}{100\pi} = 6.366\ldots$$

**FINAL ANSWER**

$6.4\ \text{cm}$.

4MA1/1HR JANUARY 2022

# Question 7

4 marks | Standard & Compound Units

7 Milly went on a car journey.
She travelled from Anesey to Breigh to Clando and then to Duckbridge.

For Anesey to Breigh, Milly drove the 245 km in 2.5 hours.
For Breigh to Clando, Milly drove the 220 km at an average speed of 80 km/h
For Clando to Duckbridge, Milly drove at an average speed of 72 km/h in 50 minutes.

Work out Milly's average speed, in km/h, for the journey from Anesey to Duckbridge.
Give your answer correct to one decimal place.

..... km/h

**(Total for Question 7 is 4 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1HR January 2022

### 1 Calculate value

First part:

$$\text{distance} = 245 \text{ km}, \qquad \text{time} = 2.5 \text{ h}$$

### 2 Second part

Second part:

$$\text{time} = \frac{220}{80} = 2.75 \text{ h}$$

### 3 Third part

Third part:

$$50 \text{ minutes} = \frac{5}{6} \text{ h}$$

$$\text{distance} = 72 \times \frac{5}{6} = 60 \text{ km}$$

### 4 Total distance

Total distance:

$$245 + 220 + 60 = 525 \text{ km}$$

**5** **Total time**

Total time:

$$2.5 + 2.75 + \frac{5}{6} = 6.0833\ldots\ \text{h}$$

**6** **Average speed**

Average speed:

$$\frac{525}{6.0833\ldots} = 86.301\ldots$$

✓ **FINAL ANSWER**

$86.3\ \text{km/h}$.

# Question 21

6 marks | Standard & Compound Units

**21** The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

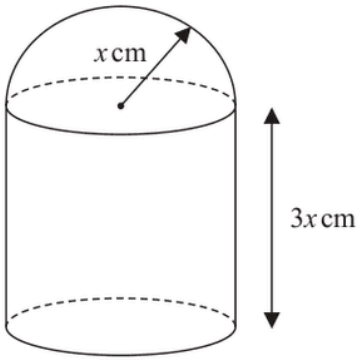



Diagram **NOT** accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both $x$ cm.
The height of the cylinder is $3x$ cm.

The total surface area of the solid is $81\pi$ cm$^2$
The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

| Metal | Density (g/cm$^3$) |
|---|---|
| Aluminium | 2.7 |
| Nickel | 8.9 |
| Gold | 19.3 |
| Silver | 10.5 |

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.
Show your working clearly.

**(Total for Question 21 is 6 marks)**

**1** **The outside surface area is**

The outside surface area is:

**2** **curved surface area of cylinder**

curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

**3** **bottom of cylinder**

bottom of cylinder:

$$\pi x^2$$

**4** **curved surface area of hemisphere**

curved surface area of hemisphere:

$$2\pi x^2$$

**5** **Calculate value**

So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

**6** **Volume of the cylinder**

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

**7** **Volume of the hemisphere**

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

**8** **Total volume**

Total volume:

$$3\pi(3)^3 + \frac{2}{3}\pi(3)^3$$

$$= 81\pi + 18\pi = 99\pi$$

**9** **Density**

Density:

$$\frac{840}{99\pi} = 2.70\ldots$$

**10** **Compare density**

This matches aluminium, whose density is $2.7\ \text{g/cm}^3$.

✓ **FINAL ANSWER**

Aluminium

4MA1/2H JUNE 2022

# Question 3

3 marks | Standard & Compound Units

3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.
Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2H June 2022

**1 The time is**

The time is

$$5\text{ h }24\text{ min} = 5.4\text{ h}$$

**2 Average speed**

Average speed:

$$3980 \div 5.4 = 737.037\ldots$$

**✓ FINAL ANSWER**

737 kilometres per hour.

4MA1/1H JUNE 2022

# Question 7

4 marks | Standard & Compound Units

7 A cylinder is placed on the ground.

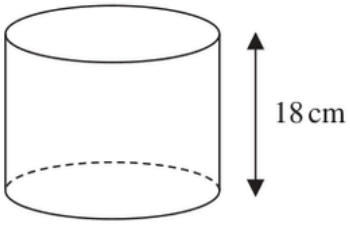



Diagram **NOT** accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.
The pressure on the ground due to the cylinder is 1.4 newtons/cm$^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.
Give your answer correct to 3 significant figures.

..... cm$^3$

**(Total for Question 7 is 4 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1H June 2022

### 1 Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$1.4 = \frac{72}{\text{area}}$$

$$\text{area} = \frac{72}{1.4} = 51.428\ldots \text{ cm}^2$$

### 2 The base area of the

The base area of the cylinder is $51.428\ldots$ cm$^2$.

**3** **Volume is**

Volume is

$$51.428\ldots \times 18 = 925.714\ldots$$

---

✓ **FINAL ANSWER**

$926\ \text{cm}^3$ to 3 significant figures.

4MA1/1HR JUNE 2022

# Question 7

3 marks | Standard & Compound Units

7 The diagram shows a solid cylinder made from iron.

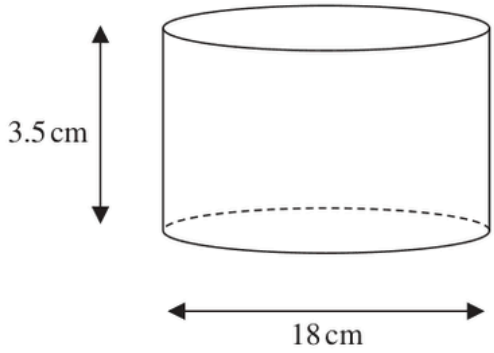



Diagram **NOT** accurately drawn

The cylinder has diameter 18 cm and height 3.5 cm
The mass of the cylinder is 7.04 kg

Work out the density of the iron.
Give your answer in g/cm$^3$ correct to 2 significant figures.

..... g/cm$^3$

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1HR June 2022

**1 Radius of the cylinder is**

Radius of the cylinder is

$$\frac{18}{2} = 9 \text{ cm}$$

**2 Volume is**

Volume is

$$\pi r^2 h = \pi(9^2)(3.5) = 283.5\pi$$

$$= 890.64\ldots \text{ cm}^3$$

**3 Mass is**

Mass is

$$7.04 \text{ kg} = 7040 \text{ g}$$

**4** **Density is**

Density is

$$\frac{7040}{890.64\ldots} = 7.904\ldots$$

---

✓ **FINAL ANSWER**

$7.9\ \mathrm{g/cm^3}$ to 2 significant figures.

4MA1/2H JUNE 2022

# Question 7

3 marks | Standard & Compound Units

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2H June 2022

**1 Convert km/h to m/s divide**

To convert km/h to m/s, divide by $3.6$.

$$81 \div 3.6 = 22.5$$

**FINAL ANSWER**

$22.5\ \text{m/s}$.

4MA1/2HR JUNE 2023

# Question 3

3 marks | Standard & Compound Units

3 Anjali travels on the Eurostar train from Paris to Amsterdam.

The distance the train travels between Paris and Amsterdam is 515 km.
The time taken by the train to travel between Paris and Amsterdam is 3 hours 18 minutes.

Work out the average speed of the train.
Give your answer in km/h correct to the nearest whole number.

..... km/h

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2HR June 2023

**1 Time hours minutes**

Time $= 3$ hours $18$ minutes:

$$3 + \frac{18}{60} = 3.3 \text{ hours}$$

$$\text{average speed} = \frac{515}{3.3} = 156.06\ldots$$

**FINAL ANSWER**

$156$ km/h.

4MA1/2H JUNE 2023

# Question 9

3 marks | Standard & Compound Units

9 Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2H June 2023

**1 Convert km/h to m/s divide**

To convert km/h to m/s, divide by $3.6$.

$$27 \div 3.6 = 7.5$$

**FINAL ANSWER**

$7.5$ m/s.

# Question 11

3 marks | Standard & Compound Units

**11** The diagram shows a block of iron in the shape of a cuboid.

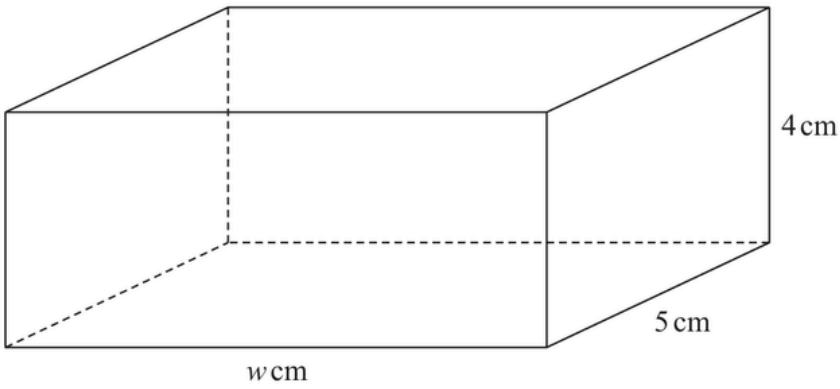



Diagram **NOT** accurately drawn

The block has length $w$ cm, width 5 cm and height 4 cm

The density of iron is 7.8 g/cm$^3$
The mass of the block is 1950 g

Work out the value of $w$

$w =$ .....

**(Total for Question 11 is 3 marks)**

## 1 Calculate volume

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$7.8 = \frac{1950}{\text{volume}}$$

$$\text{volume} = \frac{1950}{7.8} = 250 \text{ cm}^3$$

## 2 Calculate value

For the cuboid,

$$w \times 5 \times 4 = 250$$

$$20w = 250$$

$$w = 12.5$$

**FINAL ANSWER**

$w = 12.5$ cm.

4MA1/2H JUNE 2024

# Question 3

3 marks | Standard & Compound Units

3 A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.

The plane flies at an average speed of 820 km/h.

Work out the total distance the plane flies.

..... km

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2H June 2024

**1 Evaluate fraction**

$$9\text{ h }36\text{ min} = 9 + \frac{36}{60} = 9.6\text{ hours}$$

$$\text{distance} = 820 \times 9.6 = 7872$$

**FINAL ANSWER**

$7872$ km.

4MA1/2H JUNE 2024

# Question 6

3 marks | Standard & Compound Units

6 Change a speed of $w$ metres per second to a speed in kilometres per hour.
Give your answer in terms of $w$ in its simplest form.

..... kilometres per hour

**(Total for Question 6 is 3 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2H June 2024

**1 Convert m/s to km/h multiply**

To convert m/s to km/h, multiply by $3.6$.

$$w \text{ m/s} = 3.6w \text{ km/h}$$

$$3.6w = \frac{18w}{5}$$

**FINAL ANSWER**

$\frac{18w}{5}$ km/h.

4MA1/1HR MAY 2024

# Question 6

4 marks | Standard & Compound Units

**6** The diagram shows a solid wooden cylinder.

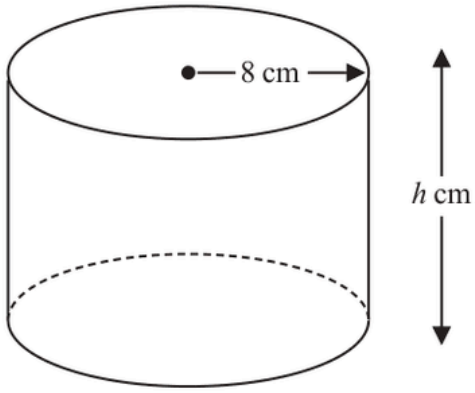



Diagram **NOT** accurately drawn

The cylinder has radius 8 cm and height $h$ cm.
The volume of the cylinder is 1208 $cm^3$

(a) Work out the value of $h$
Give your answer correct to the nearest whole number.

$h =$ .....

(2)

The density of the wood is 1.25 $g/cm^3$

(b) Work out the mass of the cylinder.
Give your answer in kilograms.

..... kilograms

(2)

**(Total for Question 6 is 4 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1HR May 2024

**1** **Evaluate fraction**

For the cylinder,

$$V = \pi r^2 h$$

$$1208 = \pi(8)^2 h$$

$$h = \frac{1208}{64\pi} = 6.006\ldots$$

## 2 Calculate value

So, to the nearest whole number,

$$h = 6$$

## 3 Calculate volume

For the mass, use the given volume:

$$\text{mass} = \text{density} \times \text{volume}$$

$$1.25 \times 1208 = 1510 \text{ g}$$

$$1510 \text{ g} = 1.51 \text{ kg}$$

**FINAL ANSWER**

$h = 6$ cm, and the mass is $1.51$ kg.

4MA1/1H MAY 2024

# Question 8

3 marks | Standard & Compound Units

**8** A cylinder is placed on a table.

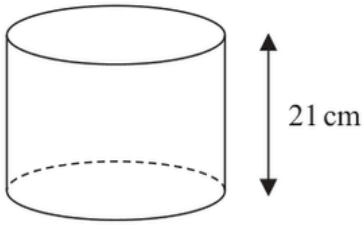



Diagram **NOT** accurately drawn

The volume of the cylinder is $1575\,\text{cm}^3$
The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

..... newtons/$\text{cm}^2$

**(Total for Question 8 is 3 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1H May 2024

### 1 The volume of a cylinder

The volume of a cylinder is

$$\text{volume} = \text{area of base} \times \text{height}$$

### 2 Calculate area

So the area in contact with the table is

$$1575 \div 21 = 75\ \text{cm}^2$$

### 3 Use

Use

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$\text{pressure} = \frac{84}{75} = 1.12$$

**FINAL ANSWER**

$1.12$ newtons/$\text{cm}^2$.

4WM1/01 MAY 2025

# Question 6

3 marks | Standard & Compound Units

**6** The diagram shows a block in the shape of a cuboid.

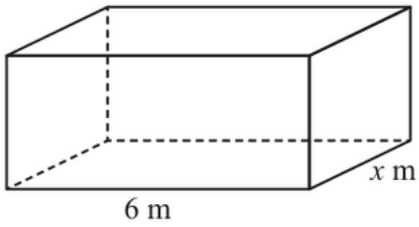



Diagram **NOT** accurately drawn

The block has a horizontal base with length 6 m and width $x$ m

The block is placed on a table so that the whole of its horizontal base is in contact with the table.

The block exerts a force of 702 newtons on the table.
The pressure on the table due to the block is 65 newtons/m²

Work out the value of $x$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$x =$ .....

**(Total for Question 6 is 3 marks)**

## WORKED SOLUTION

Question 6 · 4WM1/01 May 2025

### 1 Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$65 = \frac{702}{\text{area}}$$

$$\text{area} = \frac{702}{65} = 10.8$$

**2** **Calculate area**

The base area is $6x$, so

$$6x = 10.8$$

$$x = 1.8$$

---

✓ **FINAL ANSWER**

$x = 1.8$

4MA1/1HR JUNE 2025

# Question 7

3 marks | Standard & Compound Units

7 Change a speed of $50x$ metres per second to a speed in kilometres per hour.

..... km/h

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1HR June 2025

**1 Convert m/s to km/h multiply**

To convert m/s to km/h, multiply by $\mathbf{3.6}$.

$$50x \times 3.6 = 180x$$

**FINAL ANSWER**

$180x$ km/h.

4MA1/2HR JUNE 2025

# Question 7

2 marks | Standard & Compound Units

7 The mass of a silver coin is 48.3 g
The density of silver is 10.5 g/cm$^3$

Work out the volume of the silver coin.

..... cm$^3$

**(Total for Question 7 is 2 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2HR June 2025

**1 Use**

Use

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

**2 Calculate volume**

So

$$\text{volume} = \frac{\text{mass}}{\text{density}} = \frac{48.3}{10.5} = 4.6$$

**FINAL ANSWER**

$4.6\ \text{cm}^3$.

4WM1/01R MAY 2025

# Question 7

3 marks | Standard & Compound Units

7 Change a speed of 40 metres per second to a speed in kilometres per hour.
Show your working clearly.

..... kilometres per hour

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4WM1/01R May 2025

**1 Evaluate fraction**

$$40\ \text{m/s} = 40 \times \frac{3600}{1000}\ \text{km/h}$$

$$= 144\ \text{km/h}$$

**FINAL ANSWER**

144 kilometres per hour.

4WM1/01R MAY 2025

# Question 10

5 marks | Standard & Compound Units

**10** The diagram shows a rectangular sheet of metal.

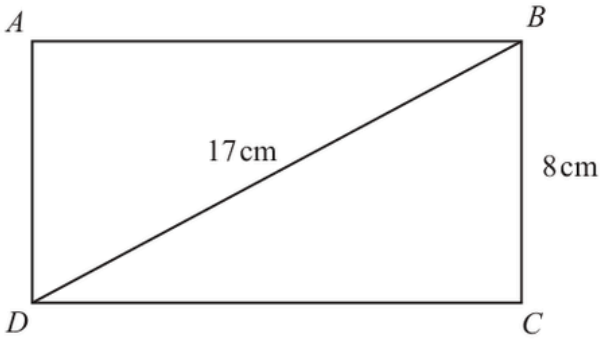



Diagram **NOT** accurately drawn

$BC = 8$ cm $\quad DB = 17$ cm

The sheet of metal lies flat on a table.
The force exerted by the sheet of metal on the table is 150 newtons.

Work out the pressure on the table due to the sheet of metal.
Show your working clearly.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons/cm$^2$

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4WM1/01R May 2025

**1** **Use Pythagoras to find the**

Use Pythagoras to find the other side of the rectangle.

$$DC = \sqrt{17^2 - 8^2} = 15$$

$$\text{area} = 15 \times 8 = 120 \text{ cm}^2$$

$$\text{pressure} = \frac{150}{120} = 1.25$$

**FINAL ANSWER**

$1.25$ newtons/cm$^2$.

4WM1/01 NOVEMBER 2025

# Question 10

3 marks | Standard & Compound Units

**10** A parcel in the shape of a cube is placed on a table so that the whole of one face of the parcel is in contact with the table.

The parcel has sides of length $7$ cm

The pressure on the table due to the parcel is $62$ newtons/cm$^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the force exerted by the parcel on the table.

..... newtons

**(Total for Question 10 is 3 marks)**

## WORKED SOLUTION

Question 10 · 4WM1/01 November 2025

**1** **The contact area is one**

The contact area is one square face:

$$7 \times 7 = 49 \text{ cm}^2$$

$$\text{force} = \text{pressure} \times \text{area} = 62 \times 49 = 3038$$

**FINAL ANSWER**

$3038$ newtons.

4MA1/2H NOVEMBER 2025

# Question 26

6 marks | Standard & Compound Units

**26** A solid cone is joined to a solid hemisphere to make the solid shown below.

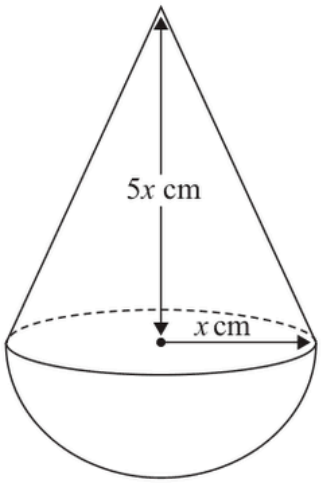



Diagram **NOT** accurately drawn

The cone is made from copper.
The density of copper is $9\,\text{g/cm}^3$

The hemisphere is made from a different metal.

The total mass of the solid is $4752\pi$ grams
The total volume of the solid is $504\pi\,\text{cm}^3$

Work out the density of the hemisphere.
Show your working clearly.

..... g/cm$^3$

**(Total for Question 26 is 6 marks)**

## WORKED SOLUTION

Question 26 · 4MA1/2H November 2025

**1** **Volume of cone**

Volume of cone:

$$\frac{1}{3}\pi x^2(5x) = \frac{5}{3}\pi x^3$$

**2** **Volume of hemisphere**

Volume of hemisphere:

$$\frac{2}{3}\pi x^3$$

**3** **Total volume**

Total volume:

$$\frac{5}{3}\pi x^3 + \frac{2}{3}\pi x^3 = 504\pi$$

$$\frac{7}{3}\pi x^3 = 504\pi$$

$$x^3 = 216$$

$$x = 6$$

**4** **Cone volume**

Cone volume:

$$\frac{5}{3}\pi (6)^3 = 360\pi$$

**5** **Mass of copper cone**

Mass of copper cone:

$$360\pi \times 9 = 3240\pi$$

**6** **Mass of hemisphere**

Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

**7** **Hemisphere volume**

Hemisphere volume:

$$\frac{2}{3}\pi (6)^3 = 144\pi$$

**8** **Density of the hemisphere**

Density of the hemisphere:

$$\frac{1512\pi}{144\pi} = 10.5$$

**FINAL ANSWER**

$10.5\ \text{g/cm}^3$

4MA1/1H JUNE 2026

# Question 4

4 marks | Standard & Compound Units

4 Henri competes in a race.

For the first part of the race Henri cycles 60 km at an average speed of 24 km/h
For the remaining part of the race Henri runs for 1 hour and 15 minutes at an average speed of 12 km/h

Work out Henri's average speed for the whole race.
Give your answer in km/h
Show your working clearly.

..... km/h

**(Total for Question 4 is 4 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1H June 2026

**1 Solution | Step 1**

Cycling time: $60 \div 24 = 2.5$ hours. Running time: $1 + 15/60 = 1.25$ hours.
Time equals distance divided by speed.

**2 Solution | Step 2**

Running distance: $12 \times 1.25 = 15$ km. Total distance: $60 + 15 = 75$ km. Total time: $2.5 + 1.25 = 3.75$ hours.
Combine the actual distances and times.

**3 Solution | Step 3**

$$\text{Average speed} = \frac{75}{3.75} = 20 \text{ km/h}.$$

Average speed is total distance divided by total time, not the mean of the two speeds.

**FINAL ANSWER**

$20 \text{ km/h}$

4MA1/1H JUNE 2026

# Question 8

3 marks | Standard & Compound Units

**8** Change a speed of 70 metres per second to a speed in kilometres per hour.
Show your working clearly.

..... km/h

**(Total for Question 8 is 3 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1H June 2026

### 1 Solution | Step 1

$$70\ \text{m/s} = 70 \times 3600 = 252000\ \text{m/h}.$$

There are 3600 seconds in one hour.

### 2 Solution | Step 2

$$252000 \div 1000 = 252\ \text{km/h}.$$

There are 1000 metres in one kilometre.

**FINAL ANSWER**

$252\ \text{km/h}$

4WM1/01 MAY 2026

# Question 8

3 marks | Standard & Compound Units

**8** A block of wood has a mass of $480$ kg
The wood has a density of $640$ kg/m$^3$

Work out the volume of the block of wood.

..... m$^3$

**(Total for Question 8 is 3 marks)**

## WORKED SOLUTION

Question 8 · 4WM1/01 May 2026

**1** **Volume = mass ÷ density**

$$V = \frac{480}{640} = 0.75 \text{ m}^3$$

**FINAL ANSWER**

$0.75$ m$^3$

4MA1/2HR JUNE 2026

# Question 9

3 marks | Standard & Compound Units

9 Change a speed of 198 kilometres per hour to a speed in metres per second.
Show your working clearly.

..... m/s

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2HR June 2026

**1 Solution | Step 1**

$198\ \text{km} = 198000\ \text{m}$ and $1\ \text{hour} = 3600\ \text{s}$.

Convert both the distance and the time in the rate.

**2 Solution | Step 2**

$$198\ \text{km/h} = \frac{198000}{3600} = 55\ \text{m/s}.$$

Metres per second means metres divided by seconds.

**✓ FINAL ANSWER**

$55\ \text{m/s}$.

4MA1/2HR JUNE 2026

# Question 10

3 marks | Standard & Compound Units

**10** The diagram shows a solid cylinder.

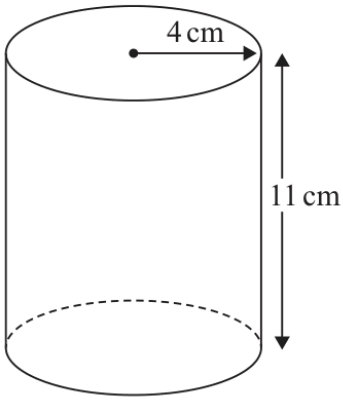



Diagram **NOT** accurately drawn

The cylinder has radius 4 cm and height 11 cm
The cylinder is made from wood with a density of 0.6 g/cm$^3$

Work out the mass of the cylinder.
Give your answer correct to 3 significant figures.

..... g

**(Total for Question 10 is 3 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2HR June 2026

**1** **Solution | Step 1**

$$V = \pi r^2 h = \pi(4)^2(11) = 176\pi \text{ cm}^3.$$

Use the cylinder volume formula.

**2** **Solution | Step 2**

$$m = \rho V = 0.6(176\pi) = 105.6\pi = 331.7521842\ldots \text{ g}.$$

Mass equals density times volume; keep full precision until the end.

**3** **Solution | Step 3**

$$m = 332 \text{ g} \quad (3 \text{ significant figures}).$$

Round the final mass as requested.

**FINAL ANSWER**

**332 g** (3 significant figures).



TOPIC

# Algebra Toolkit

Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 22

2 marks | Algebra Toolkit

**22** (a) Simplify $5a + 3b - 2a + 7b$

.....

**(1)**

(b) Simplify $4x \times 3y$

.....

**(1)**

**(Total for Question 22 is 2 marks)**

## WORKED SOLUTION

Question 22 · EP Specimen

**1** **(a) Collect like terms**

$$5a - 2a = 3a, \qquad 3b + 7b = 10b \Rightarrow 3a + 10b$$

**2** **(b) Multiply the numbers, then the letters**

$$4 \times 3 = 12 \Rightarrow 12xy$$

✓ **FINAL ANSWER**

(a) $3a + 10b$ (b) $12xy$

EP SPECIMEN

# Question 23

2 marks | Algebra Toolkit

**23** Work out the value of $3p^2 - 2q$ when $p = -4$ and $q = 5$

.....

**(2)**

**(Total for Question 23 is 2 marks)**

## WORKED SOLUTION

Question 23 · EP Specimen

**1** **Substitute, with brackets round the negative**

$$3(-4)^2 - 2(5) = 3 \times 16 - 10$$

**2** **Work it out**

$$48 - 10 = 38$$

✓ **FINAL ANSWER**

38

EP SPECIMEN

# Question 24

2 marks | Algebra Toolkit

**24** $v = u + at$

Work out the value of $v$ when $u = 12$, $a = -3.5$ and $t = 6$

$v =$ .....

**(2)**

**(Total for Question 24 is 2 marks)**

## WORKED SOLUTION

Question 24 · EP Specimen

**1 Substitute**

$$v = 12 + (-3.5) \times 6$$

**2 Work it out**

$$v = 12 - 21 = -9$$

**✓ FINAL ANSWER**

$v = -9$

EP SPECIMEN

# Question 25

2 marks | Algebra Toolkit

**25** Pens cost 35p each. Pencils cost 18p each.

Write down an expression, in terms of $x$ and $y$, for the total cost, in pence, of $x$ pens and $y$ pencils.

.....

**(2)**

**(Total for Question 25 is 2 marks)**

## WORKED SOLUTION

Question 25 · EP Specimen

**1** **Cost of each item × number bought**

$x$ pens cost $35x$ pence; $y$ pencils cost $18y$ pence.

**2** **Add**

$35x + 18y$

✓ **FINAL ANSWER**

$35x + 18y$

EP SPECIMEN

# Question 26

2 marks | Algebra Toolkit

**26** $T = \dfrac{a^2 + b}{2c}$

Work out the value of $T$ when $a = -3$, $b = 7$ and $c = 0.5$

$T =$ .....

**(2)**

**(Total for Question 26 is 2 marks)**

## WORKED SOLUTION

Question 26 · EP Specimen

**1 Substitute**

$$T = \frac{(-3)^2 + 7}{2 \times 0.5} = \frac{9 + 7}{1}$$

**2 Work it out**

$T = 16$

**FINAL ANSWER**

$T = 16$

EP SPECIMEN

# Question 27

2 marks | Algebra Toolkit

27 Simplify $6x^2y - 2xy^2 + 3x^2y + 5xy^2$

.....

(2)

**(Total for Question 27 is 2 marks)**

## WORKED SOLUTION

Question 27 · EP Specimen

**1 Like terms have exactly the same letters and powers**

$x^2y$ terms: $6 + 3 = 9$. $xy^2$ terms: $-2 + 5 = 3$.

**2 Write the answer**

$9x^2y + 3xy^2$

✓ **FINAL ANSWER**

$9x^2y + 3xy^2$

EP SPECIMEN

# Question 28

2 marks | Algebra Toolkit

**28** A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm.

Show that the perimeter, in cm, of the rectangle is $8x + 2$

.....

**(2)**

**(Total for Question 28 is 2 marks)**

## WORKED SOLUTION

Question 28 · EP Specimen

**1** **Perimeter = 2 lengths + 2 widths**

$$2(3x + 2) + 2(x - 1)$$

**2** **Expand and collect**

$$6x + 4 + 2x - 2 = 8x + 2$$

✓ **FINAL ANSWER**

Shown: $8x + 2$

EP SPECIMEN

# Question 29

2 marks | Algebra Toolkit

**29** Here are five expressions.

$3n$ $\quad n+3$ $\quad n^3$ $\quad 3n^2$ $\quad -n$

When $n = -2$, which expression has the greatest value?
You must show your working.

.....

**(2)**

**(Total for Question 29 is 2 marks)**

## WORKED SOLUTION

Question 29 · EP Specimen

**1 Substitute n = −2 into each expression**

$$3n = -6, \quad n+3 = 1, \quad n^3 = -8, \quad 3n^2 = 3 \times 4 = 12, \quad -n = 2$$

**2 Pick the greatest**

$3n^2 = 12$ is the greatest.

**FINAL ANSWER**

$3n^2$ (value $12$)

TOPIC

# Expanding Brackets

Every past-paper question on this topic, 2020 to 2026.

4MA1/1HR MAY/NOVEMBER 2020

# Question 14

3 marks | Expanding Brackets

**14** Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1HR May/November 2020

### 1 Expand brackets

$$(4x + 1)(x - 3)(5x + 6)$$

$$(x - 3)(5x + 6) = 5x^2 - 9x - 18$$

$$(4x + 1)(5x^2 - 9x - 18) = 20x^3 - 31x^2 - 81x - 18$$

**FINAL ANSWER**

$20x^3 - 31x^2 - 81x - 18$

4MA1/2HR JANUARY 2021

# Question 1

8 marks | Expanding Brackets

**1** $w = 5y^2 - y^3$

(a) Work out the value of $w$ when $y = -2$

$w =$ .....

**(2)**

(b) Factorise fully $8p^2 - 2p$

.....

**(2)**

(c) Expand $4t(3t - 2)$

.....

**(2)**

(d) Expand and simplify $(5x - 2)(x + 4)$

.....

**(2)**

**(Total for Question 1 is 8 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2HR January 2021

**1** **Expand brackets**

For $w = 5y^2 - y^3$ and $y = -2$:

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

$$8p^2 - 2p = 2p(4p - 1)$$

$$4t(3t - 2) = 12t^2 - 8t$$

$$(5x - 2)(x + 4) = 5x^2 + 18x - 8$$

✓ **FINAL ANSWER**

$28,\ 2p(4p - 1),\ 12t^2 - 8t,\ 5x^2 + 18x - 8$

4MA1/1H JANUARY 2021

# Question 12

6 marks | Expanding Brackets

**12** (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.
Give your answer in its simplest form.

.....

**(3)**

(b) Expand and simplify $2x(x-5)(x-3)$

.....

**(3)**

**(Total for Question 12 is 6 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H January 2021

**1** **(a)**

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

$$= \frac{4x + 4 - 3x + 6}{(x-2)(x+1)} = \frac{x+10}{(x-2)(x+1)}$$

**2** **(b)**

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

$$= 2x^3 - 16x^2 + 30x$$

✓ **FINAL ANSWER**

(a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

4MA1/1HR JANUARY 2021

# Question 16

5 marks | Expanding Brackets

**16** (a) Expand and simplify $(x + 4)(x - 2)(x + 1)$

.....

**(3)**

(b) Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where $a$ and $b$ are integers.

.....

**(2)**

**(Total for Question 16 is 5 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1HR January 2021

**1** **(a)**

(a)

$$(x+4)(x-2)(x+1)$$

$$(x-2)(x+1) = x^2 - x - 2$$

$$(x+4)(x^2 - x - 2) = x^3 + 3x^2 - 6x - 8$$

**2** **(b)**

(b)

$$x^2 - 10x + 40 = (x-5)^2 + 15$$

✓ **FINAL ANSWER**

(a) $x^3 + 3x^2 - 6x - 8$, (b) $(x-5)^2 + 15$

4MA1/1H MAY 2021

# Question 13

3 marks | Expanding Brackets

**13** Expand and simplify $4x(3x + 1)(2x - 3)$
Show your working clearly.

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H May 2021

### 1 Expand brackets

$$4x(3x+1)(2x-3) = 4x(6x^2 - 7x - 3)$$

$$= 24x^3 - 28x^2 - 12x$$

**FINAL ANSWER**

$24x^3 - 28x^2 - 12x$

4MA1/1H NOVEMBER 2021

# Question 1

3 marks | Expanding Brackets

**1** (a) Simplify $e^8 \div e^2$

.....

**(1)**

(b) Expand and simplify $(x - 3)(x + 1)$

.....

**(2)**

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H November 2021

**1** **(a)**

(a) $e^8 \div e^2 = e^6$.

**2** **(b)**

(b) $(x - 3)(x + 1) = x^2 - 2x - 3$.

✓ **FINAL ANSWER**

(a) $e^6$, (b) $x^2 - 2x - 3$

4MA1/2H NOVEMBER 2021

# Question 12

7 marks | Expanding Brackets

**12** (a) Simplify $(64p^9q^{12})^{\frac{2}{3}}$

.....

(2)

(b) Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

.....

(2)

(c) Expand and simplify $4x(x - 5)(2x + 3)$

Show your working clearly.

.....

(3)

**(Total for Question 12 is 7 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/2H November 2021

**1** **(a)**

(a)

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

**2** **(b)**

(b)

$$\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x} = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

**3** **(c)**

(c)

$$4x(x-5)(2x+3) = 4x(2x^2 - 7x - 15)$$

$$= 8x^3 - 28x^2 - 60x$$

---

✓ **FINAL ANSWER**

(a) $16p^6q^8$, (b) $\frac{17}{30x}$, (c) $8x^3 - 28x^2 - 60x$

4MA1/2H JANUARY 2022

# Question 1

4 marks | Expanding Brackets

**1** (a) Expand and simplify $(y + 4)(2 - y)$

.....

**(2)**

(b) Factorise fully $15b^5c - 35b^3c^9$

.....

**(2)**

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2H January 2022

**1** **(a)**

(a)

$$(y+4)(2-y) = 2y - y^2 + 8 - 4y = -y^2 - 2y + 8$$

**2** **(b)**

(b)

$$15b^5c - 35b^3c^9 = 5b^3c(3b^2 - 7c^8)$$

✓ **FINAL ANSWER**

(a) $-y^2 - 2y + 8$, (b) $5b^3c(3b^2 - 7c^8)$

4MA1/1H JANUARY 2022

# Question 13

6 marks | Expanding Brackets

**13** (a) Expand and simplify $5x(x + 2)(3x - 4)$

.....

(3)

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

.....

(3)

**(Total for Question 13 is 6 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H January 2022

**1** **(a)**

(a)

$$5x(x+2)(3x-4) = 5x(3x^2+2x-8)$$

$$= 15x^3 + 10x^2 - 40x$$

**2** **(b)**

(b)

$$\left(\frac{16w^8}{y^{20}}\right)^{-3/4} = 16^{-3/4}w^{-6}y^{15}$$

**3** **Expand brackets**

Since $16^{3/4} = 8$,

$$= \frac{y^{15}}{8w^6}$$

✓ **FINAL ANSWER**

(a) $15x^3 + 10x^2 - 40x$, (b) $\frac{y^{15}}{8w^6}$

4MA1/1HR JUNE 2022

# Question 16

6 marks | Expanding Brackets

**16** (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$

.....

(3)

(b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$

.....

(3)

**(Total for Question 16 is 6 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1HR June 2022

**1** **(a)**

(a)

$$(3x - 1)(x + 2)(3x + 1) = (9x^2 - 1)(x + 2)$$

$$= 9x^3 + 18x^2 - x - 2$$

**2** **(b)**

(b)

$$\left(\frac{2x^5}{8xy^2}\right)^{-2} = \left(\frac{x^4}{4y^2}\right)^{-2}$$

$$= \left(\frac{4y^2}{x^4}\right)^{2} = \frac{16y^4}{x^8}$$

✓ **FINAL ANSWER**

(a) $9x^3 + 18x^2 - x - 2$, (b) $\frac{16y^4}{x^8}$

4MA1/2HR JUNE 2023

# Question 1

4 marks | Expanding Brackets

**1** $P = m^2 - 4c$

(a) Work out the value of $P$ when $m = -5$ and $c = 3$

$P =$ .....

**(2)**

(b) Expand and simplify $(x + 5)(x - 7)$

.....

**(2)**

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2HR June 2023

**1** **Expand brackets**

$$P = m^2 - 4c = (-5)^2 - 4(3) = 25 - 12 = 13$$

$$(x + 5)(x - 7) = x^2 - 7x + 5x - 35 = x^2 - 2x - 35$$

**FINAL ANSWER**

$P = 13$, and $x^2 - 2x - 35$

4MA1/2H NOVEMBER 2023

# Question 16

3 marks | Expanding Brackets

**16** Expand and simplify $(2x + 3)(x - 5)(x + 4)$

**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/2H November 2023

### 1 Expand brackets

$$(x - 5)(x + 4) = x^2 - x - 20$$

$$(2x + 3)(x^2 - x - 20) = 2x^3 + x^2 - 43x - 60$$

**FINAL ANSWER**

$2x^3 + x^2 - 43x - 60$

4MA1/1HR MAY 2024

# Question 14

6 marks | Expanding Brackets

**14** (a) Expand and simplify $(3x+1)(2-x)(4+x)$

.....

(3)

(b) Simplify fully $\left(\frac{a^3b}{a^9b^5}\right)^{-\frac{1}{2}}$

.....

(3)

**(Total for Question 14 is 6 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1HR May 2024

**1** **(a)**

(a)

$$(2-x)(4+x) = 8 - 2x - x^2$$

$$(3x+1)(8-2x-x^2) = -3x^3 - 7x^2 + 22x + 8$$

**2** **(b)**

(b)

$$\left(\frac{a^3b}{a^9b^5}\right)^{-1/2} = \left(\frac{1}{a^6b^4}\right)^{-1/2} = a^3b^2$$

✓ **FINAL ANSWER**

(a) $-3x^3 - 7x^2 + 22x + 8$, (b) $a^3b^2$

4MA1/1H NOVEMBER 2024

# Question 15

3 marks | Expanding Brackets

**15** Write $3x(2x-1)(5x+4)$ in the form $ax^3+bx^2+cx$ where $a$, $b$ and $c$ are integers.

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1H November 2024

**1 Expand brackets**

$$3x(2x-1)(5x+4)$$

**2 Expand brackets**

First expand the two brackets:

$$(2x-1)(5x+4) = 10x^2+8x-5x-4$$

$$= 10x^2+3x-4$$

**3 Expand brackets**

Now multiply by $3x$:

$$3x(10x^2+3x-4) = 30x^3+9x^2-12x$$

**FINAL ANSWER**

$30x^3+9x^2-12x$

4WM1/01 MAY 2025

# Question 11

6 marks | Expanding Brackets

**11** (a) Expand and simplify $5x(5x + 4)(2x - 3)$

.....

**(3)**

(b) Express $\frac{9}{4} + \frac{y - 7}{5y}$ as a single fraction in its simplest form.

.....

**(3)**

**(Total for Question 11 is 6 marks)**

## WORKED SOLUTION

Question 11 · 4WM1/01 May 2025

**1** **Expand brackets**

$$5x(5x + 4)(2x - 3)$$

**2** **Expand brackets**

First,

$$(5x + 4)(2x - 3) = 10x^2 - 15x + 8x - 12 = 10x^2 - 7x - 12$$

**3** **Expand brackets**

Then multiply by $5x$.

$$5x(10x^2 - 7x - 12) = 50x^3 - 35x^2 - 60x$$

**4** **Expand brackets**

For part (b), use denominator $20y$.

$$\frac{9}{4} + \frac{y - 7}{5y} = \frac{45y}{20y} + \frac{4(y - 7)}{20y}$$

$$= \frac{49y - 28}{20y}$$

**FINAL ANSWER**

$50x^3 - 35x^2 - 60x$ and $\frac{49y - 28}{20y}$.

4MA1/2H JUNE 2025

# Question 12

3 marks | Expanding Brackets

**12** Expand and simplify $3x(2x+5)(7x-4)$

**(Total for Question 12 is 3 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/2H June 2025

### 1 Expand brackets

$$3x(2x+5)(7x-4) = 3x(14x^2+27x-20)$$

$$= 42x^3+81x^2-60x$$

**FINAL ANSWER**

$42x^3+81x^2-60x$

4MA1/2HR JUNE 2025

# Question 13

5 marks | Expanding Brackets

**13** (a) Factorise $4x^2 - 25y^2$

.....

**(2)**

(b) Show that $4x(x + 3)(2x - 5)$ can be written in the form $ax^3 + bx^2 + cx$ where $a$, $b$ and $c$ are integers to be found.

**(3)**

**(Total for Question 13 is 5 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/2HR June 2025

**1** **(a)**

(a)

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

**2** **(b)**

(b)

$$4x(x + 3)(2x - 5)$$

**3** **Factorise expression**

First,

$$(x + 3)(2x - 5) = 2x^2 + x - 15$$

**4** **Factorise expression**

Then

$$4x(2x^2 + x - 15) = 8x^3 + 4x^2 - 60x$$

**5** **Factorise expression**

So $a = 8$, $b = 4$, and $c = -60$.

**FINAL ANSWER**

(a) $(2x - 5y)(2x + 5y)$, (b) $8x^3 + 4x^2 - 60x$

4WM1/01R MAY 2025

# Question 15

6 marks | Expanding Brackets

**15** (a) Solve $3x^2 + 4x - 18 = 0$
Show your working clearly.
Give your solutions correct to 3 significant figures.

.....

**(3)**

(b) Expand and simplify $(2x - 3)(x + 4)(x - 1)$

.....

**(3)**

**(Total for Question 15 is 6 marks)**

## WORKED SOLUTION

Question 15 · 4WM1/01R May 2025

**1** **Simplify surd**

(a) For $3x^2 + 4x - 18 = 0$,

$$x = \frac{-4 \pm \sqrt{4^2 - 4(3)(-18)}}{2(3)}$$

$$x \approx 1.87 \quad \text{or} \quad x \approx -3.21$$

**2** **(b)**

(b)

$$(2x - 3)(x + 4)(x - 1)$$

$$= (2x^2 + 5x - 12)(x - 1)$$

$$= 2x^3 + 3x^2 - 17x + 12$$

**FINAL ANSWER**

$x = 1.87$ or $x = -3.21$; $2x^3 + 3x^2 - 17x + 12$.

4WM1/01 NOVEMBER 2025

# Question 13

6 marks | Expanding Brackets

**13** (a) Expand and simplify $7x(3x + 2)(2x - 5)$

.....

(3)

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$ .....

(3)

**(Total for Question 13 is 6 marks)**

## WORKED SOLUTION

Question 13 · 4WM1/01 November 2025

**1** **(a)**

(a)

$$7x(3x + 2)(2x - 5)$$

$$= 7x(6x^2 - 11x - 10)$$

$$= 42x^3 - 77x^2 - 70x$$

**2** **(b)**

(b)

$$\frac{9}{2y} + \frac{5}{7} = 5$$

$$\frac{9}{2y} = \frac{30}{7}$$

$$63 = 60y$$

$$y = \frac{21}{20}$$

**FINAL ANSWER**

$42x^3 - 77x^2 - 70x$, $y = \frac{21}{20}$.

4MA1/1H NOVEMBER 2025

# Question 14

6 marks | Expanding Brackets

**14** (a) Expand and simplify $7x(3x + 2)(2x - 5)$

.....

(3)

(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

$y =$ .....

(3)

**(Total for Question 14 is 6 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1H November 2025

**1** **(a)**

(a)

$$7x(3x+2)(2x-5) = 7x(6x^2 - 11x - 10)$$

$$= 42x^3 - 77x^2 - 70x$$

**2** **(b)**

(b)

$$\frac{9}{2y} + \frac{5}{7} = 5$$

$$\frac{9}{2y} = \frac{30}{7}$$

$$63 = 60y$$

**FINAL ANSWER**

(a) $42x^3 - 77x^2 - 70x$, (b) $y = \frac{21}{20}$

4MA1/1H JUNE 2026

# Question 5

3 marks | Expanding Brackets

**5** (a) Expand $4e(2e-3)$

.....

(1)

(b) Expand and simplify $(f+3)(f-6)$

.....

(2)

## WORKED SOLUTION

Question 5 · 4MA1/1H June 2026

**1** **Part (a) | Step 1**

$$4e(2e-3) = 8e^2 - 12e.$$

Multiply 4e by both terms in the bracket.

**2** **Part (b) | Step 1**

$$(f+3)(f-6) = f^2 - 6f + 3f - 18 = f^2 - 3f - 18.$$

Multiply each term in the first bracket by each term in the second and collect like terms.

✓ **FINAL ANSWER**

(a) $8e^2 - 12e$ (b) $f^2 - 3f - 18$

4MA1/1H JUNE 2026

# Question 12

6 marks | Expanding Brackets

**12** $\dfrac{2^3}{8^x} = 64$

(a) Find the value of $x$
Show clear algebraic working.

$x =$ .....

**(3)**

(b) Expand and simplify $4t(t + 5)(2t - 3)$

.....

**(3)**

**(Total for Question 12 is 6 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H June 2026

**1** **Part (a) | Step 1**

$$\frac{2^{x/3}}{8^x} = \frac{2^{x/3}}{2^{3x}} = 2^{x/3-3x} = 2^{-8x/3}.$$

Write 8 as a power of 2 and subtract exponents when dividing.

**2** **Part (a) | Step 2**

$$2^{-8x/3} = 64 = 2^6 \implies -\frac{8x}{3} = 6 \implies x = -\frac{9}{4}.$$

Equal powers of base 2 have equal exponents.

**3** **Part (b) | Step 1**

$$(t+5)(2t-3) = 2t^2 - 3t + 10t - 15 = 2t^2 + 7t - 15.$$

Expand the two brackets and collect like terms.

**4** **Part (b) | Step 2**

$$4t(t+5)(2t-3) = 4t(2t^2 + 7t - 15) = 8t^3 + 28t^2 - 60t.$$

Multiply every term by 4t.

**FINAL ANSWER**

(a) $x = -\frac{9}{4}$ (b) $8t^3 + 28t^2 - 60t$



TOPIC

# Factorising

Every past-paper question on this topic, 2020 to 2026.

4MA1/1H MAY/NOVEMBER 2020

# Question 7

8 marks | Factorising

7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$x =$ .....

(3)

(b) Factorise fully $16m^3g^3 + 24m^2g^5$

.....

(2)

(c) (i) Factorise $y^2 - 2y - 48$

.....

(2)

(ii) Hence, solve $y^2 - 2y - 48 = 0$

.....

(1)

**(Total for Question 7 is 8 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1H May/November 2020

**1** **(a)**

(a)

$$5(4 - x) = 7 - 3x$$

$$20 - 5x = 7 - 3x$$

$$13 = 2x$$

$$x = \frac{13}{2}$$

**2** **(b)**

(b)

$$16m^3g^3 + 24m^2g^5 = 8m^2g^3(2m + 3g^2)$$

3 **(c)(i)**

(c)(i)

$$y^2 - 2y - 48 = (y - 8)(y + 6)$$

4 **Factorise expression**

(c)(ii) $y = 8$ or $y = -6$.

**FINAL ANSWER**

$x = \frac{13}{2}$, $8m^2g^3(2m + 3g^2)$, $(y - 8)(y + 6)$, $y = 8$ or $y = -6$

4MA1/1H JANUARY 2021

# Question 9

5 marks | Factorising

**9** (a) Simplify $(2x^3y^5)^4$

.....

(2)

(b) (i) Factorise $x^2 + 5x - 36$

.....

(2)

(ii) Hence, solve $x^2 + 5x - 36 = 0$

.....

(1)

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1H January 2021

**1** **(a)**

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

**2** **(b)(i)**

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

**3** **Factorise expression**

(b)(ii) $x = -9$ or $x = 4$.

✓ **FINAL ANSWER**

(a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

4MA1/1H MAY 2021

# Question 9

5 marks | Factorising

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

.....

(2)

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

$x =$ .....

(3)

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1H May 2021

**1 (a)**

(a)

$$25a^4c^7d + 45a^9c^3h = 5a^4c^3(5c^4d + 9a^5h)$$

**2 (b)**

(b)

$$(2x + 5)^2 = (2x + 3)(2x - 1)$$

$$4x^2 + 20x + 25 = 4x^2 + 4x - 3$$

$$16x = -28$$

$$x = -\frac{7}{4}$$

**FINAL ANSWER**

(a) $5a^4c^3(5c^4d + 9a^5h)$, (b) $x = -\frac{7}{4}$

4MA1/2H MAY 2021

# Question 9

3 marks | Factorising

**9** (i) Factorise $x^2 + 2x - 24$

.....

**(2)**

(ii) Hence solve $x^2 + 2x - 24 = 0$

.....

**(1)**

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2H May 2021

**1** **Factorise expression**

$$x^2 + 2x - 24 = (x + 6)(x - 4)$$

**2** **Factorise expression**

So

$$(x + 6)(x - 4) = 0$$

✓ **FINAL ANSWER**

$x = -6$ or $x = 4$

4MA1/2H NOVEMBER 2021

# Question 8

4 marks | Factorising

8 (a) Simplify $(3x^2y)^0$

.....

(1)

(b) (i) Factorise $x^2 - 5x - 36$

.....

(2)

(ii) Hence solve $x^2 - 5x - 36 = 0$

.....

(1)

**(Total for Question 8 is 4 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/2H November 2021

**1 (a)**

(a) $(3x^2y)^0 = 1$.

**2 (b)(i)**

(b)(i)

$$x^2 - 5x - 36 = (x - 9)(x + 4)$$

**3 Factorise expression**

(b)(ii) $x = 9$ or $x = -4$.

**FINAL ANSWER**

$1$, $(x - 9)(x + 4)$, $x = 9$ or $x = -4$

4MA1/1HR JANUARY 2022

# Question 9

5 marks | Factorising

**9** (a) Simplify $x^4 \times x^5$

.....

(1)

(b) Simplify $(4y^2)^3$

.....

(2)

(c) Factorise $n^2 - 7n + 12$

.....

(2)

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1HR January 2022

**1** **(a)**

(a) $x^4 \times x^5 = x^9$.

**2** **(b)**

(b) $(4y^2)^3 = 64y^6$.

**3** **(c)**

(c)

$$n^2 - 7n + 12 = (n - 3)(n - 4)$$

✓ **FINAL ANSWER**

(a) $x^9$, (b) $64y^6$, (c) $(n - 3)(n - 4)$

4MA1/1HR JUNE 2022

# Question 5

5 marks | Factorising

**5** (a) Expand $3c^3(c + 4)$

.....

**(2)**

(b) (i) Factorise $x^2 + 8x - 9$

.....

**(2)**

(ii) Hence, solve $x^2 + 8x - 9 = 0$

.....

**(1)**

**(Total for Question 5 is 5 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1HR June 2022

**1** **(a)**

(a) $3c^3(c+4) = 3c^4 + 12c^3$.

**2** **(b)(i)**

(b)(i)

$$x^2 + 8x - 9 = (x+9)(x-1)$$

**3** **(b)(ii)**

(b)(ii)

$$(x+9)(x-1) = 0$$

✓ **FINAL ANSWER**

(a) $3c^4 + 12c^3$, (b)(i) $(x+9)(x-1)$, (b)(ii) $x = -9$ or $x = 1$

4MA1/1H JUNE 2022

# Question 11

5 marks | Factorising

**11** (a) Factorise $9x^2 - 4y^2$

.....

(2)

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

.....

(3)

**(Total for Question 11 is 5 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H June 2022

**1** **(a)**

(a)

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

**2** **(b)**

(b)

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

$$= \frac{7x - 2x - 6}{8x} = \frac{5x - 6}{8x}$$

**FINAL ANSWER**

(a) $(3x - 2y)(3x + 2y)$, (b) $\frac{5x-6}{8x}$

4MA1/2H JUNE 2022

# Question 11

3 marks | Factorising

**11** (i) Factorise $x^2 + 5x - 24$

.....

(2)

(ii) Hence, solve $x^2 + 5x - 24 = 0$

.....

(1)

**(Total for Question 11 is 3 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2H June 2022

**1** **Factorise expression**

$$x^2 + 5x - 24 = (x + 8)(x - 3)$$

**2** **Factorise expression**

So

$$(x + 8)(x - 3) = 0$$

✓ **FINAL ANSWER**

$x = -8$ or $x = 3$

4MA1/1H MAY 2023

# Question 6

8 marks | Factorising

**6** (a) Simplify $(2c^4d^7)^3$

.....

**(2)**

(b) Find the value of $5y^0$ where $y > 0$

.....

**(1)**

(c) Factorise fully $16a^2b^3 + 20a^3b$

.....

**(2)**

(d) (i) Factorise $x^2 + 9x - 22$

.....

**(2)**

(ii) Hence solve $x^2 + 9x - 22 = 0$

.....

**(1)**

**(Total for Question 6 is 8 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1H May 2023

**1** **(a)**

(a)

$$(2c^4d^7)^3 = 8c^{12}d^{21}$$

**2** **Factorise expression**

(b) $5y^0 = 5$, since $y^0 = 1$.

**3** **(c)**

(c)

$$16a^2b^3 + 20a^3b = 4a^2b(4b^2 + 5a)$$

4 **(d)(i)**

(d)(i)

$$x^2 + 9x - 22 = (x + 11)(x - 2)$$

5 **Factorise expression**

(d)(ii) $x = -11$ or $x = 2$.

✓ **FINAL ANSWER**

$8c^{12}d^{21}$, $5$, $4a^2b(4b^2 + 5a)$, $(x + 11)(x - 2)$, $x = -11$ or $x = 2$

4MA1/1HR MAY 2023

# Question 7

3 marks | Factorising

7 (a) Expand $4x(x - 5)$

.....

(1)

(b) Factorise $y^2 - 9y + 20$

.....

(2)

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1HR May 2023

**1 (a)**

(a)

$$4x(x-5) = 4x^2 - 20x$$

**2 (b)**

(b)

$$y^2 - 9y + 20 = (y-4)(y-5)$$

**FINAL ANSWER**

(a) $4x^2 - 20x$, (b) $(y-4)(y-5)$

4MA1/1H MAY 2023

# Question 14

3 marks | Factorising

**14** Factorise fully $50g^2 - 18$

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1H May 2023

### 1 Factorise expression

$$50g^2 - 18 = 2(25g^2 - 9)$$

$$= 2(5g - 3)(5g + 3)$$

**FINAL ANSWER**

$2(5g - 3)(5g + 3)$

4MA1/2H JUNE 2024

# Question 11

3 marks | Factorising

**11** (i) Factorise $x^2 + 9x - 22$

.....

(2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$

.....

(1)

**(Total for Question 11 is 3 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2H June 2024

**1** **Factorise expression**

$$x^2 + 9x - 22 = (x + 11)(x - 2)$$

**2** **Factorise expression**

So

$$(x + 11)(x - 2) = 0$$

**FINAL ANSWER**

$x = -11$ or $x = 2$

4MA1/1H JUNE 2025

# Question 6

6 marks | Factorising

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$x =$ .....

(3)

(b) (i) Factorise $y^2 - 11y + 30$

.....

(2)

(ii) Hence solve $y^2 - 11y + 30 = 0$

.....

(1)

**(Total for Question 6 is 6 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1H June 2025

**1** **(a)**

(a)

$$x - 4 = \frac{3 + 2x}{6}$$

$$6x - 24 = 3 + 2x$$

$$4x = 27$$

$$x = \frac{27}{4}$$

**2** **(b)(i)**

(b)(i)

$$y^2 - 11y + 30 = (y - 5)(y - 6)$$

3 **(b)(ii)**

(b)(ii)

$$y = 5 \quad \text{or} \quad y = 6$$

---

**FINAL ANSWER**

$x = \frac{27}{4}$, $(y-5)(y-6)$, $y = 5$ or $y = 6$

4WM1/01 MAY 2025

# Question 7

6 marks | Factorising

7 (a) Solve $x - 4 = \dfrac{3 + 2x}{6}$

Show clear algebraic working.

$x =$ .....

(3)

(b) (i) Factorise $y^2 - 11y + 30$

.....

(2)

(ii) Hence solve $y^2 - 11y + 30 = 0$

.....

(1)

**(Total for Question 7 is 6 marks)**

## WORKED SOLUTION

Question 7 · 4WM1/01 May 2025

**1 Factorise expression**

For part (a),

$$x - 4 = \frac{3 + 2x}{6}$$

**2 Multiply by**

Multiply by $6$.

$$6x - 24 = 3 + 2x$$

$$4x = 27$$

$$x = \frac{27}{4} = 6.75$$

**3 Factorise expression**

For part (b),

$$y^2 - 11y + 30 = (y - 5)(y - 6)$$

### 4 Factorise expression

Hence,

$$(y-5)(y-6)=0$$

$$y=5 \quad \text{or} \quad y=6$$

---

**FINAL ANSWER**

$x=6.75$, $(y-5)(y-6)$, and $y=5$ or $6$.

4MA1/1HR JUNE 2025

# Question 8

5 marks | Factorising

**8** (a) Simplify $a^6 \times a^{10}$

.....

(1)

(b) Simplify $c^{30} \div c^{12}$

.....

(1)

(c) (i) Factorise $y^2 - 10y + 21$

.....

(2)

(ii) Hence, solve $y^2 - 10y + 21 = 0$

.....

(1)

**(Total for Question 8 is 5 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1HR June 2025

**1** **(a)**

(a) $a^6 \times a^{10} = a^{16}$.

**2** **(b)**

(b) $c^{30} \div c^{12} = c^{18}$.

**3** **(c)(i)**

(c)(i)

$$y^2 - 10y + 21 = (y - 3)(y - 7)$$

**4** **(c)(ii)**

(c)(ii)

$$y = 3 \quad \text{or} \quad y = 7$$

✓ **FINAL ANSWER**

$a^{16}$, $c^{18}$, $(y - 3)(y - 7)$, $y = 3$ or $y = 7$

4WM1/01R MAY 2025

# Question 9

7 marks | Factorising

9 (a) Simplify $\left(5k^3n^4\right)^2$

.....

(2)

(b) Factorise fully $12m^2y^4 + 18m^3y$

.....

(2)

(c) (i) Factorise $x^2 - 13x + 42$

.....

(2)

(ii) Hence, solve $x^2 - 13x + 42 = 0$

.....

(1)

**(Total for Question 9 is 7 marks)**

## WORKED SOLUTION

Question 9 · 4WM1/01R May 2025

**1** **(a)**

(a)

$$(5k^3n^4)^2 = 25k^6n^8$$

**2** **(b)**

(b)

$$12m^2y^4 + 18m^3y^3 = 6m^2y^3(2y + 3m)$$

**3** **(c)**

(c)

$$x^2 - 13x + 42 = (x - 6)(x - 7)$$

**4** **Calculate value**

So

$$x = 6 \quad \text{or} \quad x = 7$$

**FINAL ANSWER**

$25k^6n^8$, $6m^2y^3(2y+3m)$, $(x-6)(x-7)$, $x=6,7$.

4MA1/2H NOVEMBER 2025

# Question 7

3 marks | Factorising

7 (i) Factorise $x^2 + 2x - 48$

.....

(2)

(ii) Hence, solve $x^2 + 2x - 48 = 0$

.....

(1)

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2H November 2025

**1 Factorise expression**

$$x^2 + 2x - 48 = (x + 8)(x - 6)$$

**2 Factorise expression**

So

$$(x + 8)(x - 6) = 0$$

**FINAL ANSWER**

$x = -8$ or $x = 6$

4WM1/01 NOVEMBER 2025

# Question 8

9 marks | Factorising

**8** (a) Expand $5w(3w - 4w^2)$

.....

(2)

(b) Simplify fully $\left(4a^3c^5\right)^3$

.....

(2)

(c) Factorise fully $14m^3p^2 + 21m^2p^4$

.....

(2)

(d) (i) Factorise $x^2 - 10x + 24$

.....

(2)

(ii) Hence solve $x^2 - 10x + 24 = 0$

.....

(1)

**(Total for Question 8 is 9 marks)**

## WORKED SOLUTION

Question 8 · 4WM1/01 November 2025

**1** **(a)**

(a)

$$5w(3w - 4w^2) = 15w^2 - 20w^3$$

**2** **(b)**

(b)

$$(4a^3c^5)^3 = 64a^9c^{15}$$

**3** **(c)**

(c)

$$14m^3p^2 + 21m^2p^4 = 7m^2p^2(2m + 3p^2)$$

**4** **(d)**

(d)

$$x^2 - 10x + 24 = (x - 4)(x - 6)$$

---

**5** **Calculate value**

So

$$x = 4 \quad \text{or} \quad x = 6$$

---

✓ **FINAL ANSWER**

$15w^2 - 20w^3$, $64a^9c^{15}$, $7m^2p^2(2m + 3p^2)$, $(x - 4)(x - 6)$, $x = 4, 6$.

# Question 10

7 marks | Factorising

**10** (a) Simplify $(6m)^0$

.....

**(1)**

(b) Solve $5y + 20 < 7y + 1$

.....

**(2)**

(c) Factorise fully $15w^2x^5 + 25w^3x^2$

.....

**(2)**

(d) Simplify fully $\dfrac{36a^5c^7}{12a^2c^3}$

.....

**(2)**

**(Total for Question 10 is 7 marks)**

**(a)**

(a) $(6m)^0 = 1$.

**(b)**

(b)

$$5y + 20 < 7y + 1$$

$$19 < 2y$$

$$y > \frac{19}{2}$$

**(c)**

(c)

$$15w^2x^5 + 25w^3x^2 = 5w^2x^2(3x^3 + 5w)$$

**(d)**

(d)

$$\frac{36a^5c^7}{12a^2c^3} = 3a^3c^4$$

**FINAL ANSWER**

$1$, $y > \frac{19}{2}$, $5w^2x^2(3x^3 + 5w)$, $3a^3c^4$

4MA1/1HR JUNE 2026

# Question 8

2 marks | Factorising

**8** Factorise $x^2 - 12x + 32$

.....

**(Total for Question 8 is 2 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1HR June 2026

### 1 Solution | Step 1

$$(-4)(-8) = 32, \qquad -4 + (-8) = -12.$$

Choose two numbers with product 32 and sum -12.

### 2 Solution | Step 2

$$x^2 - 12x + 32 = (x - 4)(x - 8).$$

The chosen numbers give the two constant terms in the factors.

**FINAL ANSWER**

$(x - 4)(x - 8)$

4MA1/2H JUNE 2026

# Question 9

7 marks | Factorising

**9** (a) Find the value of $7n^0$ where $n > 0$

.....

**(1)**

(b) Simplify fully $\left(2m^5p^4\right)^3$

.....

**(2)**

(c) Factorise fully $18c^5w^2 + 30cw^3$

.....

**(2)**

(d) Factorise $x^2 + 6x - 27$

.....

**(2)**

**(Total for Question 9 is 7 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2H June 2026

**1** **Part (a) | Step 1**

Since $n > 0$, $n^0 = 1$. Hence $7n^0 = 7(1) = 7$.

A nonzero number raised to power zero equals 1; only n is raised to zero.

**2** **Part (b) | Step 1**

$$(2m^5p^4)^3 = 2^3m^{5\times3}p^{4\times3} = 8m^{15}p^{12}.$$

Cube the coefficient and multiply each exponent by 3.

**3** **Part (c) | Step 1**

The highest common factor of $18c^5w^2$ and $30cw^3$ is $6cw^2$.

Take the numerical HCF and the smallest power of each shared variable.

**4** **Part (c) | Step 2**

$$18c^5w^2 + 30cw^3 = 6cw^2(3c^4 + 5w).$$

Divide each original term by the common factor.

5

### Part (d) | Step 1

Two numbers with product $-27$ and sum $6$ are $9$ and $-3$.

For a monic quadratic, find a factor pair with the required sum.

6

### Part (d) | Step 2

$$x^2 + 6x - 27 = (x + 9)(x - 3).$$

The cross terms are -3x+9x=6x.

✓ **FINAL ANSWER**

(a) $7$ (b) $8m^{15}p^{12}$ (c) $6cw^2(3c^4 + 5w)$ (d) $(x + 9)(x - 3)$

# Question 9

5 marks | Factorising

**9** (a) Factorise $15g^4h - 20gh^3$

.....

**(2)**

(b) (i) Factorise $x^2 + 3x - 28$

.....

**(2)**

(ii) Hence, solve $x^2 + 3x - 28 = 0$

.....

**(1)**

**(Total for Question 9 is 5 marks)**

**1** **(a) Take out the HCF**

HCF of $15g^4h$ and $20gh^3$ is $5gh$:

$$15g^4h - 20gh^3 = 5gh(3g^3 - 4h^2)$$

**2** **(b)(i) Two numbers: product −28, sum 3**

$7$ and $-4$:

$$x^2 + 3x - 28 = (x + 7)(x - 4)$$

**3** **(b)(ii) Each bracket = 0**

$$x = -7 \quad \text{or} \quad x = 4$$

✓ **FINAL ANSWER**

(a) $5gh(3g^3 - 4h^2)$ (b)(i) $(x + 7)(x - 4)$ (ii) $x = -7,\ 4$



TOPIC

# Algebraic Fractions

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 12

5 marks | Algebraic Fractions

**12** (a) Factorise $9x^2 - 4y^2$

.....

(2)

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

.....

(3)

**(Total for Question 12 is 5 marks)**

## WORKED SOLUTION

Question 12 · 4WM1/01 Specimen

**1** **Use difference of two squares**

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2$$

$$= (3x - 2y)(3x + 2y)$$

**2** **Use a common denominator**

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

**3** **Simplify the numerator**

$$\frac{7x - 2x - 6}{8x} = \frac{5x - 6}{8x}$$

✓ **FINAL ANSWER**

$(3x - 2y)(3x + 2y)$, and $\frac{5x-6}{8x}$.

# Question 21

3 marks | Algebraic Fractions

**21** Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

Show clear algebraic working.

.....

## 1 Factorise the denominator

$$x^2 - 36 = (x - 6)(x + 6)$$

## 2 Simplify the bracket

$$\frac{20}{(x-6)(x+6)} - \frac{2}{x-6} = \frac{20 - 2(x+6)}{(x-6)(x+6)}$$

$$= \frac{8 - 2x}{(x-6)(x+6)} = \frac{-2(x-4)}{(x-6)(x+6)}$$

## 3 Cancel the common factor

Since $4 - x = -(x - 4)$,

$$\frac{-2(x-4)}{(x-6)(x+6)} \times \frac{1}{4-x} = \frac{2}{(x-6)(x+6)}$$

**FINAL ANSWER**

$\frac{2}{x^2-36}$.

4MA1/2H JANUARY 2020

# Question 24

4 marks | Algebraic Fractions

24 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

**(Total for Question 24 is 4 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2H January 2020

**1 Use trigonometry**

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

**2 Use trigonometry**

First simplify the bracket:

$$\frac{4}{2x-5}-\frac{3}{2x-3}=\frac{4(2x-3)-3(2x-5)}{(2x-5)(2x-3)}$$

$$=\frac{8x-12-6x+15}{(2x-5)(2x-3)}$$

$$=\frac{2x+3}{(2x-5)(2x-3)}$$

**3 Use trigonometry**

Now factor the second fraction:

$$9x-4x^3=-x(2x-3)(2x+3)$$

$$6x^2-17x+5=(2x-5)(3x-1)$$

**4** **Dividing by a fraction means**

Dividing by a fraction means multiplying by its reciprocal:

$$\frac{2x+3}{(2x-5)(2x-3)} \times \frac{(2x-5)(3x-1)}{-x(2x-3)(2x+3)}$$

**5** **Cancel common factors**

Cancel common factors:

$$= -\frac{3x-1}{x(2x-3)^2}$$

$$= \frac{1-3x}{x(2x-3)^2}$$

✓ **FINAL ANSWER**

$$\boxed{\frac{1-3x}{x(2x-3)^2}}$$

4MA1/1H MAY/NOVEMBER 2020

# Question 21

5 marks | Algebraic Fractions

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H May/November 2020

**1 Use trigonometry**

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

**2 Factor**

Factor:

$$9x^2-4 = (3x-2)(3x+2)$$

$$3x^2-13x-10 = (3x+2)(x-5)$$

**3 Use trigonometry**

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

**4 Cancel common factors**

Cancel common factors:

$$\frac{1}{x-5}$$

**5 Use trigonometry**

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

### 6 Use the common denominator

Use the common denominator $(x-5)(x-1)$:

$$\frac{x-1-7(x-5)}{(x-5)(x-1)}$$

$$=\frac{x-1-7x+35}{(x-5)(x-1)}$$

$$=\frac{34-6x}{(x-5)(x-1)}$$

---

**FINAL ANSWER**

$$\boxed{\frac{34-6x}{(x-5)(x-1)}}$$

4MA1/2HR JANUARY 2021

# Question 18

5 marks | Algebraic Fractions

**18** Solve the equation

$$\frac{5}{x+2}+\frac{3}{x^2+2x}=2$$

Show clear algebraic working.

**(Total for Question 18 is 5 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2HR January 2021

**1 Solve equation**

$$\frac{5}{x+2}+\frac{3}{x^2+2x}=2$$

**2 Solve equation**

Since

$$x^2+2x=x(x+2)$$

**3 multiply by**

multiply by $x(x+2)$:

$$5x+3=2x(x+2)$$

$$5x+3=2x^2+4x$$

$$2x^2-x-3=0$$

**4 Factorise**

Factorise:

$$(2x-3)(x+1)=0$$

$$x=\frac{3}{2} \quad \text{or} \quad x=-1$$

**5** **Both values are allowed because**

Both values are allowed because $x \neq 0$ and $x \neq -2$.

---

✓ **FINAL ANSWER**

$x = \frac{3}{2}$ or $x = -1$.

4MA1/2H JANUARY 2021

# Question 21

4 marks | Algebraic Fractions

**21** Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2H January 2021

**1 Use trigonometry**

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

**2 Factor**

Factor:

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

$$5x^2 - 13x - 6 = (5x + 2)(x - 3)$$

$$x^2 - 8x + 15 = (x - 3)(x - 5)$$

**3 Use trigonometry**

So

$$\frac{(5x - 8)(5x + 8)}{(5x + 2)(x - 3)} \times \frac{(x - 3)(x - 5)}{5x + 8} - (x - 7)$$

**4 Cancel common factors**

Cancel common factors:

$$\frac{(5x - 8)(x - 5)}{5x + 2} - (x - 7)$$

**5** **Use the common denominator**

Use the common denominator $5x + 2$:

$$\frac{(5x - 8)(x - 5) - (x - 7)(5x + 2)}{5x + 2}$$

**6** **Expand**

Expand:

$$(5x - 8)(x - 5) = 5x^2 - 33x + 40$$

$$(x - 7)(5x + 2) = 5x^2 - 33x - 14$$

**7** **Use trigonometry**

So the numerator is

$$5x^2 - 33x + 40 - (5x^2 - 33x - 14) = 54$$

✓ **FINAL ANSWER**

$$\boxed{\frac{54}{5x + 2}}$$

4MA1/1H MAY 2021

# Question 21

4 marks | Algebraic Fractions

**21** Given that $x = \dfrac{5}{9y + 5}$ and that $y = \dfrac{5}{5a - 2}$

find an expression for $x$ in terms of $a$.
Give your expression as a single fraction in its simplest form.

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H May 2021

**1 Given**

Given

$$x = \frac{5}{9y + 5}$$

**2 Use trigonometry**

and

$$y = \frac{5}{5a - 2}$$

**3 Use cosine rule**

Substitute $y = \frac{5}{5a-2}$ into the expression for $x$:

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right) + 5}$$

$$x = \frac{5}{\frac{45}{5a-2} + 5}$$

**4** **Write the denominator as a**

Write the denominator as a single fraction:

$$\frac{45}{5a-2}+5=\frac{45+5(5a-2)}{5a-2}$$

$$=\frac{45+25a-10}{5a-2}$$

$$=\frac{25a+35}{5a-2}$$

---

**5** **Use trigonometry**

So

$$x=5\div\frac{25a+35}{5a-2}$$

$$x=5\times\frac{5a-2}{25a+35}$$

$$x=\frac{5(5a-2)}{5(5a+7)}$$

$$x=\frac{5a-2}{5a+7}$$

---

✓ **FINAL ANSWER**

$$\boxed{\frac{5a-2}{5a+7}}$$

4MA1/1HR JANUARY 2022

# Question 12

5 marks | Algebraic Fractions

**12** (a) Expand and simplify $n(n-4)(3n+5)$

.....

(2)

(b) Express

$$\frac{3}{x}+\frac{x+2}{2x}+\frac{1}{4}$$

as a single fraction in its simplest form.

.....

(3)

**(Total for Question 12 is 5 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1HR January 2022

**1** **(a)**

(a)

$$n(n-4)(3n+5) = n(3n^2-7n-20) = 3n^3-7n^2-20n$$

**2** **(b)**

(b)

$$\frac{3}{x}+\frac{x+2}{2x}+\frac{1}{4}$$

**3** **Use denominator**

Use denominator $4x$:

$$\frac{12}{4x}+\frac{2(x+2)}{4x}+\frac{x}{4x}=\frac{12+2x+4+x}{4x}$$

$$=\frac{3x+16}{4x}$$

**FINAL ANSWER**

(a) $3n^3-7n^2-20n$, (b) $\frac{3x+16}{4x}$

4MA1/2H JUNE 2022

# Question 19

3 marks | Algebraic Fractions

**19** $a = \dfrac{14}{3x - 7}$ $\quad x = \dfrac{7}{4y - 3}$

Express $a$ in the form $\dfrac{py + q}{ry + s}$ where $p$, $q$, $r$ and $s$ are integers.

Give your answer in its simplest form.

$a =$ .....

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2H June 2022

**1 Simplify fraction**

$$a = \frac{14}{3x - 7}, \qquad x = \frac{7}{4y - 3}$$

**2 Simplify fraction**

Substitute for $x$:

$$3x - 7 = \frac{21}{4y - 3} - 7$$

$$= \frac{21 - 7(4y - 3)}{4y - 3} = \frac{42 - 28y}{4y - 3}$$

**3 Simplify fraction**

So

$$a = \frac{14}{\frac{42 - 28y}{4y - 3}} = \frac{4y - 3}{3 - 2y}$$

**FINAL ANSWER**

$a = \frac{4y - 3}{3 - 2y}$

4MA1/2HR JUNE 2022

# Question 23

3 marks | Algebraic Fractions

**23** Express $\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$ as a single fraction in its simplest form.

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2HR June 2022

**1 Use trigonometry**

$$\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$$

**2 Factor**

Factor:

$$x^2-36=(x-6)(x+6)$$

**3 Use trigonometry**

So

$$\frac{20}{(x-6)(x+6)}-\frac{2}{x-6}$$

**4 Use the common denominator**

Use the common denominator $(x-6)(x+6)$:

$$\frac{20-2(x+6)}{(x-6)(x+6)}$$

$$=\frac{20-2x-12}{(x-6)(x+6)}$$

$$=\frac{8-2x}{(x-6)(x+6)}$$

$$=\frac{-2(x-4)}{(x-6)(x+6)}$$

**5** **Use trigonometry**

Now multiply by

$$\frac{1}{4-x} = -\frac{1}{x-4}$$

$$\frac{-2(x-4)}{(x-6)(x+6)} \times \left(-\frac{1}{x-4}\right)$$

---

**6** **Cancel**

Cancel $x-4$:

$$\frac{2}{(x-6)(x+6)}$$

$$= \frac{2}{x^2-36}$$

---

✓ **FINAL ANSWER**

$$\boxed{\frac{2}{x^2-36}}$$

4MA1/1HR MAY 2023

# Question 23

4 marks | Algebraic Fractions

23 Simplify $(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\frac{ax^2}{bx + c}$ where $a$, $b$ and $c$ are integers.

**(Total for Question 23 is 4 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/1HR May 2023

**1 Simplify surd**

$$(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x}\right) - 2x$$

**2 Change division into multiplication by**

Change division into multiplication by the reciprocal:

$$(x^2 - 4) \times \frac{x}{4x^2 - 7x - 2} - 2x$$

**3 Factorise**

Factorise:

$$x^2 - 4 = (x - 2)(x + 2)$$

$$4x^2 - 7x - 2 = (4x + 1)(x - 2)$$

**4 Simplify surd**

So

$$\frac{x(x - 2)(x + 2)}{(4x + 1)(x - 2)} - 2x = \frac{x(x + 2)}{4x + 1} - 2x$$

$$\frac{x(x + 2) - 2x(4x + 1)}{4x + 1} = \frac{x^2 + 2x - 8x^2 - 2x}{4x + 1} = \frac{-7x^2}{4x + 1}$$

**FINAL ANSWER**

$$\frac{-7x^2}{4x + 1}$$

4MA1/1H NOVEMBER 2023

# Question 11

3 marks | Algebraic Fractions

11 Express $\frac{3}{4}+\frac{5-x}{6x}$ as a single fraction in its simplest terms.

**(Total for Question 11 is 3 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H November 2023

### 1 Simplify fraction

$$\frac{3}{4}+\frac{5-x}{6x}$$

### 2 The common denominator is

The common denominator is $12x$:

$$\frac{3}{4}=\frac{9x}{12x}$$

$$\frac{5-x}{6x}=\frac{10-2x}{12x}$$

$$\frac{9x+10-2x}{12x}=\frac{7x+10}{12x}$$

**FINAL ANSWER**

$\frac{7x+10}{12x}$.

4MA1/2H NOVEMBER 2023

# Question 24

5 marks | Algebraic Fractions

24 Solve $\frac{45x^3-80x}{3x^2+x-4}\times\left(\frac{1}{3x-4}+\frac{1}{x}\right)=\frac{4(x+2)}{5x-8}$

Show clear algebraic working.

$x =$ .....

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2H November 2023

**1 Solve quadratic equation**

$$\frac{45x^3-80x}{3x^2+x-4}\left(\frac{1}{3x-4}+\frac{1}{x}\right)=\frac{4(x+2)}{5x-8}$$

**2 Factorise**

Factorise:

$$45x^3-80x=5x(3x-4)(3x+4)$$

$$3x^2+x-4=(3x+4)(x-1)$$

**3 Solve quadratic equation**

Also,

$$\frac{1}{3x-4}+\frac{1}{x}=\frac{x+3x-4}{x(3x-4)}=\frac{4(x-1)}{x(3x-4)}$$

**4 Solve quadratic equation**

So the left side becomes

$$\frac{5x(3x-4)(3x+4)}{(3x+4)(x-1)}\times\frac{4(x-1)}{x(3x-4)}=20$$

**5** **Solve quadratic equation**

Therefore,

$$20 = \frac{4(x+2)}{5x-8}$$

$$20(5x-8) = 4(x+2)$$

$$100x - 160 = 4x + 8$$

$$96x = 168$$

$$x = \frac{7}{4}$$

---

✓ **FINAL ANSWER**

$x = \frac{7}{4}$

4MA1/1H MAY 2024

# Question 12

5 marks | Algebraic Fractions

**12** (a) Factorise fully $6y^2 - 5y - 4$

.....

**(2)**

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

.....

**(3)**

**(Total for Question 12 is 5 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H May 2024

**1** **(a)**

(a)

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

**2** **(b)**

(b)

$$\frac{2x+1}{4x} + \frac{7-5x}{3x}$$

**3** **Use denominator**

Use denominator $12x$:

$$\frac{3(2x+1) + 4(7-5x)}{12x}$$

$$= \frac{6x + 3 + 28 - 20x}{12x} = \frac{31 - 14x}{12x}$$

**FINAL ANSWER**

(a) $(3y - 4)(2y + 1)$, (b) $\frac{31-14x}{12x}$

4MA1/2H JUNE 2024

# Question 20

3 marks | Algebraic Fractions

**20** Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where $a$, $b$, $c$ and $d$ are integers.

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H June 2024

**1** **Simplify fraction**

$$k = x - y, \quad x = \frac{1}{4y}$$

**2** **Simplify fraction**

So

$$k = \frac{1}{4y} - y = \frac{1 - 4y^2}{4y}$$

**3** **Simplify fraction**

Also

$$x + 2 = \frac{1}{4y} + 2 = \frac{1 + 8y}{4y}$$

**4** **Simplify fraction**

Therefore

$$\frac{5k}{x+2} = \frac{5\left(\frac{1-4y^2}{4y}\right)}{\frac{1+8y}{4y}}$$

**5** **Cancel**

Cancel $4y$:

$$\frac{5(1-4y^2)}{1+8y}$$

$$= \frac{5-20y^2}{1+8y}$$

---

✓ **FINAL ANSWER**

$$\frac{5-20y^2}{1+8y}$$

4MA1/2HR JUNE 2024

# Question 26

4 marks | Algebraic Fractions

**26** Write $4 - \left[(3x - 5) \div \dfrac{3x^2 + x - 10}{4x - 1}\right]$ as a single fraction in its simplest form.

**(Total for Question 26 is 4 marks)**

## WORKED SOLUTION

Question 26 · 4MA1/2HR June 2024

**1 Use trigonometry**

$$4 - \left[(3x - 5) \div \frac{3x^2 + x - 10}{4x - 1}\right]$$

**2 Factor the quadratic**

Factor the quadratic:

$$3x^2 + x - 10 = (3x - 5)(x + 2)$$

**3 Use trigonometry**

So the expression inside the square brackets is

$$(3x - 5) \div \frac{(3x - 5)(x + 2)}{4x - 1}$$

**4 Dividing by a fraction means**

Dividing by a fraction means multiplying by its reciprocal:

$$(3x - 5) \times \frac{4x - 1}{(3x - 5)(x + 2)}$$

**5 Cancel**

Cancel $3x - 5$:

$$\frac{4x - 1}{x + 2}$$

**6 Use trigonometry**

Therefore the full expression is

$$4 - \frac{4x - 1}{x + 2}$$

### 7 Use the common denominator

Use the common denominator $x + 2$:

$$\frac{4(x+2) - (4x-1)}{x+2}$$

$$= \frac{4x + 8 - 4x + 1}{x+2}$$

$$= \frac{9}{x+2}$$

---

**FINAL ANSWER**

$$\boxed{\frac{9}{x+2}}$$

4MA1/1H JUNE 2025

# Question 12

3 marks | Algebraic Fractions

**12** Express $\frac{5}{4}+\frac{x-3}{6x}$ as a single fraction in its simplest form.

**(Total for Question 12 is 3 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H June 2025

### 1 Simplify fraction

$$\frac{5}{4}+\frac{x-3}{6x}$$

### 2 The common denominator is

The common denominator is $12x$:

$$\frac{5}{4}=\frac{15x}{12x}$$

$$\frac{x-3}{6x}=\frac{2x-6}{12x}$$

$$\frac{15x}{12x}+\frac{2x-6}{12x}=\frac{17x-6}{12x}$$

**FINAL ANSWER**

$\frac{17x-6}{12x}$.

4WM1/01 MAY 2025

# Question 19

3 marks | Algebraic Fractions

**19** Simplify fully $\dfrac{3x^2 - 3y^2}{5x + 5y} \div \dfrac{xy^2 - x^2y}{10xy}$

Show clear algebraic working.

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4WM1/01 May 2025

**1** **Simplify fraction**

$$\frac{3x^2 - 3y^2}{5x + 5y} \div \frac{xy^2 - x^2y}{10xy}$$

**2** **Factorise**

Factorise.

$$= \frac{3(x - y)(x + y)}{5(x + y)} \div \frac{xy(y - x)}{10xy}$$

$$= \frac{3(x - y)}{5} \div \frac{y - x}{10}$$

$$= \frac{3(x - y)}{5} \times \frac{10}{y - x}$$

**3** **Simplify fraction**

Since $y - x = -(x - y)$, the expression is

$$-6$$

**FINAL ANSWER**

$-6$

4WM1/01 MAY 2025

# Question 20

3 marks | Algebraic Fractions

**20** Given that $k = m + n$ and $m = \dfrac{3}{4n}$

express $\dfrac{7k}{5-m}$ in the form $\dfrac{a+bn^2}{cn-d}$ where $a$, $b$ and $c$ are integers and $d$ is prime.

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4WM1/01 May 2025

**1 Simplify fraction**

$$k = m + n, \qquad m = \frac{3}{4n}$$

**2 Simplify fraction**

So

$$k = \frac{3}{4n} + n = \frac{3 + 4n^2}{4n}$$

**3 Simplify fraction**

Then

$$7k = \frac{21 + 28n^2}{4n}$$

**4 Simplify fraction**

and

$$5 - m = 5 - \frac{3}{4n} = \frac{20n - 3}{4n}$$

**5 Simplify fraction**

Therefore

$$\frac{7k}{5-m} = \frac{\frac{21+28n^2}{4n}}{\frac{20n-3}{4n}}$$

$$= \frac{21 + 28n^2}{20n - 3}$$

**FINAL ANSWER**

$$\frac{21+28n^2}{20n-3}$$

4MA1/2H JUNE 2025

# Question 23

4 marks | Algebraic Fractions

**23** $\dfrac{4x^2-4x-120}{5x^2-180} \div \dfrac{x^2+5x}{10x^2+60x} = p$ where $p$ is an integer.

Find the value of $p$

Show clear algebraic working.

$p =$ .....

**(Total for Question 23 is 4 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2H June 2025

**1 Split the ratio**

$$\frac{4x^2-4x-120}{5x^2-180} \div \frac{x^2+5x}{10x^2+60x}$$

**2 Factorise**

Factorise:

$$4x^2-4x-120 = 4(x-6)(x+5)$$

$$5x^2-180 = 5(x-6)(x+6)$$

$$x^2+5x = x(x+5)$$

$$10x^2+60x = 10x(x+6)$$

**3 Split the ratio**

So

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}$$

$$= \frac{4(x+5)}{5(x+6)} \times \frac{10(x+6)}{x+5} = 8$$

**FINAL ANSWER**

$p = 8$

4WM1/01R MAY 2025

# Question 24

4 marks | Algebraic Fractions

**24** Write $(2x-1)+\dfrac{2x^2-5x-12}{x^2-16}\times\dfrac{x^2+4x}{6x^2+11x+3}$ as a single fraction in its simplest form.

Show clear algebraic working.

**(Total for Question 24 is 4 marks)**

## WORKED SOLUTION

Question 24 · 4WM1/01R May 2025

### 1 Factor the fractional part first

Factor the fractional part first:

$$\frac{2x^2-5x-12}{x^2-16}\times\frac{x^2+4x}{6x^2+11x+3}$$

$$=\frac{(2x+3)(x-4)}{(x-4)(x+4)}\times\frac{x(x+4)}{(3x+1)(2x+3)}$$

$$=\frac{x}{3x+1}$$

### 2 Evaluate fraction

So

$$(2x-1)+\frac{x}{3x+1}=\frac{(2x-1)(3x+1)+x}{3x+1}$$

$$=\frac{6x^2-1}{3x+1}$$

**FINAL ANSWER**

$$\frac{6x^2-1}{3x+1}.$$

4MA1/1H NOVEMBER 2025

# Question 23

3 marks | Algebraic Fractions

23 $a = \dfrac{2x+5}{1-x}$ $x = \dfrac{5-2y}{3y}$

Write $a$ in the form $\dfrac{m+ny}{p(y-1)}$ where $m$, $n$ and $p$ are integers.

Show your working clearly.

$a =$ .....

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/1H November 2025

**1 Split the ratio**

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

**2 Split the ratio**

Substitute for $x$.

**3 Numerator**

Numerator:

$$2x+5 = 2\left(\frac{5-2y}{3y}\right)+5$$

$$= \frac{10-4y+15y}{3y} = \frac{10+11y}{3y}$$

**4 Denominator**

Denominator:

$$1-x = 1-\frac{5-2y}{3y}$$

$$= \frac{3y-5+2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

**5** **Split the ratio**

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

---

✓ **FINAL ANSWER**

$$\frac{10+11y}{5(y-1)}$$

4WM1/01 NOVEMBER 2025

# Question 25

3 marks | Algebraic Fractions

**25** $a = \dfrac{2x+5}{1-x}$ $x = \dfrac{5-2y}{3y}$

Write $a$ in the form $\dfrac{m+ny}{p(y-1)}$ where $m$, $n$ and $p$ are integers.

Show your working clearly.

$a =$ .....

**(Total for Question 25 is 3 marks)**

## WORKED SOLUTION

Question 25 · 4WM1/01 November 2025

**1 Split the ratio**

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

**2 Split the ratio**

Substitute for $x$.

**3 Numerator**

Numerator:

$$2x+5 = 2\left(\frac{5-2y}{3y}\right)+5$$

$$= \frac{10-4y+15y}{3y} = \frac{10+11y}{3y}$$

**4 Denominator**

Denominator:

$$1-x = 1-\frac{5-2y}{3y}$$

$$= \frac{3y-5+2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

**5** **Split the ratio**

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

---

✓ **FINAL ANSWER**

$$\frac{10+11y}{5(y-1)}$$

4WM1/01 NOVEMBER 2025

# Question 28

4 marks | Algebraic Fractions

**28** Solve $\dfrac{6x^3+5x^2+x}{4x^2-1} \div \dfrac{15x^2-x-2}{8x-4} = 2x$

Show clear algebraic working.

$x =$ .....

**(Total for Question 28 is 4 marks)**

## WORKED SOLUTION

Question 28 · 4WM1/01 November 2025

### 1 Factor each expression

Factor each expression:

$$\frac{x(3x+1)(2x+1)}{(2x-1)(2x+1)} \div \frac{(5x-2)(3x+1)}{4(2x-1)} = 2x$$

$$\frac{x(3x+1)}{2x-1} \times \frac{4(2x-1)}{(5x-2)(3x+1)} = 2x$$

$$\frac{4x}{5x-2} = 2x$$

$$4x = 2x(5x-2)$$

$$10x^2 - 8x = 0$$

$$2x(5x-4) = 0$$

### 2 The original fractions do not

The original fractions do not allow $x = 0$, so

$$x = \frac{4}{5}$$

**FINAL ANSWER**

$x = \frac{4}{5}$.

4MA1/1HR JUNE 2026

# Question 16

6 marks | Algebraic Fractions

**16** (a) Express $\frac{5}{8} - \frac{x+2}{4x}$ as a single fraction in its simplest form.

.....

**(3)**

(b) Expand and simplify $3x(2x - 1)(x + 4)$

.....

**(3)**

**(Total for Question 16 is 6 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1HR June 2026

**1** **Part (a) | Step 1**

$$\frac{5}{8} - \frac{x+2}{4x} = \frac{5x}{8x} - \frac{2(x+2)}{8x}, \qquad x \neq 0.$$

Use 8x as a common denominator.

**2** **Part (a) | Step 2**

$$\frac{5x - 2(x+2)}{8x} = \frac{5x - 2x - 4}{8x} = \frac{3x - 4}{8x}.$$

Subtract the entire numerator of the second fraction and collect like terms.

**3** **Part (b) | Step 1**

$$(2x - 1)(x + 4) = 2x^2 + 8x - x - 4 = 2x^2 + 7x - 4.$$

Multiply each term in the first bracket by each term in the second.

**4** **Part (b) | Step 2**

$$3x(2x^2 + 7x - 4) = 6x^3 + 21x^2 - 12x.$$

Multiply every term by 3x.

**FINAL ANSWER**

(a) $\frac{3x-4}{8x}$ (b) $6x^3 + 21x^2 - 12x$

4MA1/1H JUNE 2026

# Question 18

3 marks | Algebraic Fractions

**18** Simplify fully $\dfrac{4x^2 - 9}{2x^2 - 5x - 12}$

Show clear algebraic working.

.....

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1H June 2026

**1 Solution | Step 1**

$$4x^2 - 9 = (2x - 3)(2x + 3), \qquad 2x^2 - 5x - 12 = (2x + 3)(x - 4).$$

Use difference of squares for the numerator and factorise the denominator.

**2 Solution | Step 2**

$$\frac{4x^2 - 9}{2x^2 - 5x - 12} = \frac{(2x - 3)(2x + 3)}{(2x + 3)(x - 4)} = \frac{2x - 3}{x - 4}.$$

Cancel the common factor, not individual terms.

**3 Solution | Step 3**

The original fraction requires $x \neq -\frac{3}{2}$ and $x \neq 4$.
Cancellation does not restore excluded values.

✓ **FINAL ANSWER**

$\frac{2x-3}{x-4}$, $x \neq -\frac{3}{2}, 4$.

# Question 20

4 marks | Algebraic Fractions

**20** Express $\left(\frac{5}{4x-7}-\frac{3}{3x-5}\right)\div\frac{9x^2-16}{6x^2-7x-5}$ as a single fraction in its simplest form.

Show clear algebraic working.

.....

**(Total for Question 20 is 4 marks)**

**1** **Subtract the fractions**

$$\frac{5(3x-5)-3(4x-7)}{(4x-7)(3x-5)} = \frac{3x-4}{(4x-7)(3x-5)}$$

**2** **Factorise the second fraction**

$$9x^2-16=(3x-4)(3x+4), \qquad 6x^2-7x-5=(3x-5)(2x+1)$$

**3** **Divide = multiply by the reciprocal**

$$\frac{3x-4}{(4x-7)(3x-5)} \times \frac{(3x-5)(2x+1)}{(3x-4)(3x+4)}$$

**4** **Cancel**

$$= \frac{2x+1}{(4x-7)(3x+4)}$$

✓ **FINAL ANSWER**

$$\frac{2x+1}{(4x-7)(3x+4)}$$



TOPIC

# Algebraic Roots & Indices

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 7

4 marks | Algebraic Roots & Indices

7 (a) Simplify $8 \times (4t)^0$

.....

(1)

$x^6 \div x^{-5} = x^p$

(b) Find the value of $p$

$p =$ .....

(1)

(c) Simplify fully $(2k^2m^4)^3$

.....

(2)

## WORKED SOLUTION

Question 7 · 4WM1/01 Specimen

**1 Use the zero power**

$$(4t)^0 = 1$$

$$8(4t)^0 = 8$$

**2 Subtract the indices**

$$x^6 \div x^{-5} = x^{6-(-5)} = x^{11}$$

So $p = 11$.

**3** **Raise each factor to the power**

$$(2k^2m^4)^3 = 2^3k^{2\times3}m^{4\times3}$$

$$= 8k^6m^{12}$$

---

✓ **FINAL ANSWER**

$8$, $p = 11$, and $8k^6m^{12}$.

# Question 13

6 marks | Algebraic Roots & Indices

**13** (a) Expand and simplify $(3x-1)(x+2)(3x+1)$

.....

(3)

(b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$

.....

(3)

**(Total for Question 13 is 6 marks)**

**1** **Multiply the conjugate factors**

$$(3x-1)(3x+1) = 9x^2 - 1$$

**2** **Expand the remaining bracket**

$$(9x^2-1)(x+2) = 9x^3 + 18x^2 - x - 2$$

**3** **Simplify inside the bracket**

$$\frac{2x^5}{8xy^2} = \frac{x^4}{4y^2}$$

**4** **Apply the negative power**

$$\left(\frac{x^4}{4y^2}\right)^{-2} = \left(\frac{4y^2}{x^4}\right)^{2} = \frac{16y^4}{x^8}$$

✓ **FINAL ANSWER**

$9x^3 + 18x^2 - x - 2$, and $\frac{16y^4}{x^8}$.

# Question 22

3 marks | Algebraic Roots & Indices

**22** $\dfrac{18 \times (\sqrt{27})^{}}{6 \times 9^{2n+8}} = 3^x$

Express $x$ in terms of $n$

Show your working clearly and simplify your expression.

$x =$ .....

## 1 Write everything as powers of 3

$$\sqrt{27} = 3^{3/2}, \qquad 9 = 3^2, \qquad \frac{18}{6} = 3$$

## 2 Simplify the powers

$$\frac{18(\sqrt{27})^{4n+6}}{6(9)^{2n+8}} = 3 \times \frac{\left(3^{3/2}\right)^{4n+6}}{(3^2)^{2n+8}}$$

$$= 3^1 \times \frac{3^{6n+9}}{3^{4n+16}}$$

## 3 Subtract the indices

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

So

$$x = 2n - 6$$

**FINAL ANSWER**

$x = 2n - 6$.

4MA1/2H JANUARY 2020

# Question 1

3 marks | Algebraic Roots & Indices

**1** (a) Simplify $\dfrac{x^9}{x^2}$

.....

**(1)**

(b) Write $\dfrac{7^8 \times 7^4}{7^3}$ as a single power of 7

.....

**(2)**

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2H January 2020

**1** **(a)**

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

**2** **(b)**

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

✓ **FINAL ANSWER**

(a) $x^7$, (b) $7^9$.

4MA1/1H JANUARY 2020

# Question 17

6 marks | Algebraic Roots & Indices

**17** (a) Show that $\left(6+2\sqrt{12}\right)^2 = 12\left(7+4\sqrt{3}\right)$

Show each stage of your working.

(3)

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

.....

(3)

**(Total for Question 17 is 6 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1H January 2020

**1** **Simplify surd**

First,

$$\sqrt{12} = 2\sqrt{3}$$

**2** **Simplify surd**

So

$$6 + 2\sqrt{12} = 6 + 4\sqrt{3}$$

**3** **Simplify surd**

Now square:

$$(6 + 4\sqrt{3})^2 = 36 + 48\sqrt{3} + 48$$

$$= 84 + 48\sqrt{3}$$

**4** **Factor out**

Factor out $12$:

$$84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$$

**5** **Simplify surd**

For part (b),

$$\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}} = \left(\frac{t^{15}}{27a^{12}}\right)^{\frac{2}{3}}$$

$$= \frac{t^{10}}{9a^{8}}$$

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✓ **FINAL ANSWER**

(a) shown, (b) $\frac{t^{10}}{9a^{8}}$

4MA1/2HR JANUARY 2020

# Question 21

7 marks | Algebraic Roots & Indices

**21** (a) Simplify fully $\dfrac{10x^2 + 23x + 12}{4x^2 - 9}$

.....

(3)

$2^{2y} \times 2^{3y+2} = \dfrac{8^{5y}}{4^n}$

(b) Find an expression for $n$ in terms of $y$.
Show clear algebraic working and simplify your expression.

.....

(4)

**(Total for Question 21 is 7 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2HR January 2020

### 1 Factorise expression

For part (a), factorise the numerator and denominator:

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

$$4x^2 - 9 = (2x - 3)(2x + 3)$$

### 2 Use index laws

So

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

$$= \frac{5x + 4}{2x - 3}$$

### 3 Use index laws

For part (b),

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

**4** **Use index laws**

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

**5** **Equate powers of 2**

Equate powers of 2:

$$5y + 2 = 15y - 2n$$

$$2n = 10y - 2$$

$$n = 5y - 1$$

✓ **FINAL ANSWER**

(a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

4MA1/1HR MAY/NOVEMBER 2020

# Question 21

5 marks | Algebraic Roots & Indices

**21** Given that $M = \dfrac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of $n$ for which $M = 2$

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1HR May/November 2020

**1** **Use index laws**

$$18^{4n} = (2 \times 3^2)^{4n} = 2^{4n}3^{8n}$$

$$12^2 = (2^2 \times 3)^2 = 2^4 3^2$$

**2** **Use index laws**

So

$$M = \frac{2^{4n}3^{8n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{2^4 3^2}$$

**3** **Combine powers of 2**

Combine powers of 2:

$$2^{4n+3n^2-18n-4} = 2^{3n^2-14n-4}$$

**4** **Combine powers of 3**

Combine powers of 3:

$$3^{8n+2-8n-2} = 3^0 = 1$$

**5** **Use index laws**

So

$$M = 2^{3n^2-14n-4}$$

**6** **Use index laws**

Given $M = 2$,

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

$$(3n + 1)(n - 5) = 0$$

---

✓ **FINAL ANSWER**

$n = -\frac{1}{3}$ or $n = 5$

4MA1/2HR JANUARY 2021

# Question 4

4 marks | Algebraic Roots & Indices

4 (a) Simplify $(3k^2)^4$

.....

(2)

(b) Simplify $(21m^4n) \div (3n^{-5})$

.....

(2)

**(Total for Question 4 is 4 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2HR January 2021

**1 Use index laws**

For part (a),

$$(3k^2)^4 = 3^4(k^2)^4$$

$$= 81k^8$$

**2 Use index laws**

For part (b),

$$(21m^4n) \div (3n^{-5}) = \frac{21m^4n}{3n^{-5}}$$

$$= 7m^4n^{1-(-5)}$$

$$= 7m^4n^6$$

**FINAL ANSWER**

(a) $81k^8$, (b) $7m^4n^6$.

4MA1/2H JANUARY 2021

# Question 11

3 marks | Algebraic Roots & Indices

11 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

(Total for Question 11 is 3 marks)

## WORKED SOLUTION

Question 11 · 4MA1/2H January 2021

**1 Use index laws**

$$\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$$

**2 Simplify inside the brackets**

Simplify inside the brackets:

$$\frac{9t^4w^9}{18t^6w^{10}} = \frac{1}{2}t^{-2}w^{-1} = \frac{1}{2t^2w}$$

**3 Use index laws**

Now apply the power $-2$:

$$\left(\frac{1}{2t^2w}\right)^{-2} = (2t^2w)^2$$

$$= 4t^4w^2$$

✓ **FINAL ANSWER**

$4t^4w^2$.

4MA1/1HR JANUARY 2021

# Question 12

9 marks | Algebraic Roots & Indices

**12** (a) Simplify $(16e^{10}f^{6})^{\frac{1}{2}}$

.....

(2)

(b) Write $\frac{2x+1}{4}+\frac{x-2}{3}$ as a single fraction in its simplest form.

.....

(3)

Given that $4^{k+3}=16\times 2^{k}$

(c) find the value of $k$.
Show your working clearly.

$k =$ .....

(4)

**(Total for Question 12 is 9 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1HR January 2021

**1** **(a)**

(a)

$$(16e^{10}f^{6})^{\frac{1}{2}}=4e^{5}f^{3}$$

**2** **(b)**

(b)

$$\frac{2x+1}{4}+\frac{x-2}{3}=\frac{3(2x+1)+4(x-2)}{12}$$

$$=\frac{6x+3+4x-8}{12}=\frac{10x-5}{12}=\frac{5(2x-1)}{12}$$

**3** **(c)**

(c)

$$4^{k+3} = 16 \times 2^k$$

$$(2^2)^{k+3} = 2^4 \times 2^k$$

$$2^{2k+6} = 2^{k+4}$$

$$2k + 6 = k + 4$$

$$k = -2$$

---

**FINAL ANSWER**

(a) $4e^5f^3$, (b) $\frac{5(2x-1)}{12}$, (c) $k = -2$.

4MA1/2HR JANUARY 2021

# Question 19

7 marks | Algebraic Roots & Indices

**19** (a) Simplify $8^2 \times \sqrt[3]{4^6}$

Give your answer in the form $2^a$ where $a$ is an integer.

Show each stage of your working clearly.

.....

(3)

Given that $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$ where $n > 0$

(b) find the value of $n$.

$n =$ .....

(4)

**(Total for Question 19 is 7 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2HR January 2021

**1** **Simplify surd**

$$8^2 = (2^3)^2 = 2^6$$

$$\sqrt[3]{4^6} = \sqrt[3]{(2^2)^6} = \sqrt[3]{2^{12}} = 2^4$$

**2** **Simplify surd**

So

$$8^2 \times \sqrt[3]{4^6} = 2^6 \times 2^4 = 2^{10}$$

**3** **Use index laws**

For part (b),

$$n^{-4/5} = \left(\frac{1}{2}\right)^4 = 2^{-4}$$

**4** **Use index laws**

So

$$n^{4/5} = 2^4$$

**5** **Raise both sides to the**

Raise both sides to the power $\frac{5}{4}$:

$$n = (2^4)^{5/4} = 2^5 = 32$$

---

**FINAL ANSWER**

(a) $2^{10}$, (b) $n = 32$.

4MA1/1H MAY 2021

# Question 11

4 marks | Algebraic Roots & Indices

**11** $\sqrt{2} \times 16 = 2^x$

(a) Find the value of $x$.
Show your working clearly.

$x =$ .....

**(2)**

$$\frac{(11^{-6})^5}{11^4} = 11^n$$

(b) Find the value of $n$.
Show your working clearly.

$n =$ .....

**(2)**

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H May 2021

**1** **Simplify surd**

$$\sqrt{2} \times 16 = 2^x$$

**2** **Write both factors as powers**

Write both factors as powers of 2:

$$\sqrt{2} = 2^{1/2}, \quad 16 = 2^4$$

**3** **Use index laws**

So

$$2^{1/2} \times 2^4 = 2^{9/2}$$

**4** **Use index laws**

Therefore

$$x = \frac{9}{2}$$

**5** **Use index laws**

For part (b),

$$\frac{(11^{-6})^5}{11^4} = \frac{11^{-30}}{11^4} = 11^{-30-4} = 11^{-34}$$

**6** **Use index laws**

So

$$n = -34$$

**FINAL ANSWER**

(a) $x = \frac{9}{2}$, (b) $n = -34$.

4MA1/2H MAY 2021

# Question 14

3 marks | Algebraic Roots & Indices

**14** Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2H May 2021

### 1 Use index laws

$$\left(\frac{9x^4}{16y^{10}}\right)^{-1/2}$$

### 2 A negative power takes the

A negative power takes the reciprocal:

$$\left(\frac{16y^{10}}{9x^4}\right)^{1/2}$$

### 3 Simplify surd

Now take the square root of each factor:

$$\frac{\sqrt{16y^{10}}}{\sqrt{9x^4}} = \frac{4y^5}{3x^2}$$

**FINAL ANSWER**

$\frac{4y^5}{3x^2}$.

4MA1/2H NOVEMBER 2021

# Question 1

4 marks | Algebraic Roots & Indices

**1** (a) Write down the value of $m$, given that $3^4 \times 3^5 = 3^m$

$m =$ .....

**(1)**

(b) Write down the value of $n$, given that $(5^3)^7 = 5^n$

$n =$ .....

**(1)**

(c) Find the value of $p$, given that $\dfrac{7^8 \times 7^2}{7^p} = 7^6$

$p =$ .....

**(2)**

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2H November 2021

**1** **Use index laws**

(a) $3^4 \times 3^5 = 3^9$, so $m = 9$.

**2** **Use index laws**

(b) $(5^3)^7 = 5^{21}$, so $n = 21$.

**3** **(c)**

(c)

$$\frac{7^8 \times 7^2}{7^p} = 7^{10-p} = 7^6$$

**4** **Use index laws**

So $10 - p = 6$, hence $p = 4$.

✓ **FINAL ANSWER**

$m = 9$, $n = 21$, $p = 4$

4MA1/1H JANUARY 2022

# Question 1

6 marks | Algebraic Roots & Indices

**1** (a) Simplify $a^7 \times a^4$

.....

(1)

(b) Simplify $w^{15} \div w^3$

.....

(1)

(c) Simplify $(8x^5y^3)^2$

.....

(2)

(d) Make $t$ the subject of $c = t^3 - 8v$

.....

(2)

**(Total for Question 1 is 6 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H January 2022

**1** **(a)**

(a)

$$a^7 \times a^4 = a^{11}$$

**2** **(b)**

(b)

$$w^{15} \div w^3 = w^{12}$$

**3** **(c)**

(c)

$$(8x^5y^3)^2 = 64x^{10}y^6$$

**4** **(d)**

(d)

$$c = t^3 - 8v$$

$$t^3 = c + 8v$$

---

**FINAL ANSWER**

$a^{11}$, $w^{12}$, $64x^{10}y^6$, $t = \sqrt[3]{c + 8v}$

4MA1/2HR JANUARY 2022

# Question 6

4 marks | Algebraic Roots & Indices

**6** (a) Write down the value of $x^0$

.....

(1)

Given that $2^{-3} \times 2^9 = 2^n$

(b) find the value of $n$

$n =$ .....

(1)

Given that $\dfrac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

(c) find the value of $m$

$m =$ .....

(2)

**(Total for Question 6 is 4 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2HR January 2022

**1** **Use index laws**

$$x^0 = 1$$

**2** **Use index laws**

For part (b),

$$2^{-3} \times 2^9 = 2^{-3+9} = 2^6$$

**3** **Use index laws**

So

$$n = 6$$

**4** **Use index laws**

For part (c),

$$\frac{7^{206} \times 7^m}{7^{214}} = 7^{206+m-214}$$

**5** **Use index laws**

This equals $7^{-3}$, so

$$206 + m - 214 = -3$$

$$m = 5$$

---

✓ **FINAL ANSWER**

(a) $1$, (b) $n = 6$, (c) $m = 5$.

4MA1/2HR JUNE 2022

# Question 7

2 marks | Algebraic Roots & Indices

7 $\dfrac{2^k}{4^n} = 2^x$

Find an expression for $x$ in terms of $k$ and $n$

$x =$ .....

**(Total for Question 7 is 2 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2HR June 2022

**1 Use index laws**

$$\frac{2^k}{4^n} = 2^x$$

**2 Use index laws**

Since $4 = 2^2$,

$$4^n = (2^2)^n = 2^{2n}$$

**3 Use index laws**

So

$$\frac{2^k}{2^{2n}} = 2^{k-2n}$$

**4 Use index laws**

Therefore

$$x = k - 2n$$

✓ **FINAL ANSWER**

$x = k - 2n$.

4MA1/1H JUNE 2022

# Question 9

4 marks | Algebraic Roots & Indices

**9** (a) Simplify $8 \times (4t)^0$

.....

(1)

$x^6 \div x^{-5} = x^p$

(b) Find the value of $p$

$p =$ .....

(1)

(c) Simplify fully $(2k^2m^4)^3$

.....

(2)

**(Total for Question 9 is 4 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1H June 2022

**1** **Use index laws**

$$8 \times (4t)^0 = 8 \times 1 = 8$$

**2** **Use index laws**

For part (b),

$$x^6 \div x^{-5} = x^{6-(-5)} = x^{11}$$

**3** **Use index laws**

So

$$p = 11$$

**4** **Use index laws**

For part (c),

$$(2k^2m^4)^3 = 2^3k^{2\times3}m^{4\times3}$$

$$= 8k^6m^{12}$$

**FINAL ANSWER**

(a) $8$, (b) $p = 11$, (c) $8k^6m^{12}$.

4MA1/2H JUNE 2022

# Question 24

3 marks | Algebraic Roots & Indices

24 $$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express $x$ in terms of $n$
Show your working clearly and simplify your expression.

$x =$ .....

**(Total for Question 24 is 3 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2H June 2022

**1 Simplify surd**

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

**2 Use index laws**

First,

$$\frac{18}{6} = 3 = 3^1$$

**3 Simplify surd**

Also,

$$\sqrt{27} = \sqrt{3^3} = 3^{3/2}$$

**4 Simplify surd**

So

$$(\sqrt{27})^{4n+6} = 3^{\frac{3}{2}(4n+6)} = 3^{6n+9}$$

**5 Use index laws**

And

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

**6 Use index laws**

Therefore the whole expression is

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

**7** **Use index laws**

So

$$x = 2n - 6$$

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✓ **FINAL ANSWER**

$x = 2n - 6$.

4MA1/1HR MAY 2023

# Question 6

4 marks | Algebraic Roots & Indices

**6** (a) Simplify $m^{10} \div m^3$

.....

(1)

$k^n \times k^4 = k^{12}$

(b) Write down the value of $n$

$n =$ .....

(1)

(c) Simplify $(3x^6y^8)^2$

.....

(2)

**(Total for Question 6 is 4 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1HR May 2023

**1** **(a)**

(a) $m^{10} \div m^3 = m^{10-3} = m^7$.

**2** **Use index laws**

(b) $k^n \times k^4 = k^{n+4} = k^{12}$, so $n + 4 = 12$ and $n = 8$.

**3** **(c)**

(c) $(3x^6y^8)^2 = 3^2x^{12}y^{16} = 9x^{12}y^{16}$.

✓ **FINAL ANSWER**

(a) $m^7$, (b) $8$, (c) $9x^{12}y^{16}$

4MA1/2HR JUNE 2023

# Question 7

2 marks | Algebraic Roots & Indices

7 (a) Simplify $(4^{-2})^0$

.....

(1)

$3^{-14} \times 3^8 = 3^m$

(b) Find the value of $m$

$m =$ .....

(1)

**(Total for Question 7 is 2 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2HR June 2023

**1 Use index laws**

Any non-zero expression to the power 0 is 1:

$$(4^{-2})^0 = 1$$

**2 Use index laws**

For part (b),

$$3^{-14} \times 3^8 = 3^{-14+8} = 3^{-6}$$

**3 Use index laws**

So

$$m = -6$$

**FINAL ANSWER**

(a) $1$, (b) $m = -6$.

4MA1/2HR JUNE 2023

# Question 11

2 marks | Algebraic Roots & Indices

11 Express $\left(\frac{m^6k^{10}}{25}\right)^{\frac{3}{2}}$ in the form $\frac{m^ak^b}{c}$ where $a$, $b$ and $c$ are integers to be found.

**(Total for Question 11 is 2 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2HR June 2023

**1 Convert standard form**

$$\left(\frac{m^6k^{10}}{25}\right)^{\frac{3}{2}} = \frac{(m^6)^{\frac{3}{2}}(k^{10})^{\frac{3}{2}}}{25^{\frac{3}{2}}}$$

$$(m^6)^{\frac{3}{2}} = m^9$$

$$(k^{10})^{\frac{3}{2}} = k^{15}$$

**2 Convert standard form**

Also,

$$25^{\frac{3}{2}} = (\sqrt{25})^3 = 5^3 = 125$$

**3 Convert standard form**

So

$$\left(\frac{m^6k^{10}}{25}\right)^{\frac{3}{2}} = \frac{m^9k^{15}}{125}$$

✓ **FINAL ANSWER**

$$\frac{m^9k^{15}}{125}$$

4MA1/2H JUNE 2023

# Question 15

7 marks | Algebraic Roots & Indices

**15** (a) Simplify fully $\left(32a^{15}\right)^{\frac{3}{5}}$

.....

(2)

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form $px^n$ where $p$ and $n$ are integers.

.....

(2)

(c) Solve $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$

Show clear algebraic working.

$y =$ .....

(3)

**(Total for Question 15 is 7 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2H June 2023

**1** **Solve equation**

For part (a),

$$(32a^{15})^{3/5} = 32^{3/5}a^{15\times 3/5}$$

$$= 8a^9$$

**2** **Solve equation**

For part (b),

$$\left(\frac{1}{10x}\right)^{-3} = (10x)^3 = 1000x^3$$

**3** **Solve equation**

For part (c),

$$\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$$

**4** **Multiply by**

Multiply by $30$:

$$10(1-2y) = 24 - 15(2y-1)$$

$$10 - 20y = 39 - 30y$$

$$10y = 29$$

$$y = \frac{29}{10}$$

---

✓ **FINAL ANSWER**

(a) $8a^9$, (b) $1000x^3$, (c) $y = \frac{29}{10}$

4MA1/1HR MAY 2023

# Question 24

5 marks | Algebraic Roots & Indices

**24** Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for $x$ in terms of $n$
State any restrictions on $n$

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/1HR May 2023

**1** **Use index laws**

$$16^x = 2^{4x}, \qquad 8 = 2^3$$

**2** **Use index laws**

So

$$2^n = 2^{x^2} \times 2^{4x} \times 2^3 = 2^{x^2+4x+3}$$

**3** **Hence**

Hence

$$n = x^2 + 4x + 3 = (x+2)^2 - 1$$

$$(x+2)^2 = n + 1$$

**4** **Simplify surd**

Since $x > 0$,

$$x + 2 = \sqrt{n+1}$$

$$x = -2 + \sqrt{n+1}$$

**5** **Simplify surd**

For $x > 0$, $\sqrt{n+1} > 2$, so $n > 3$.

**FINAL ANSWER**

$x = -2 + \sqrt{n+1}, n > 3$

4MA1/2H NOVEMBER 2023

# Question 4

8 marks | Algebraic Roots & Indices

**4** (a) Solve $\frac{2x+5}{6} = 2x - 5$

Show clear algebraic working.

$x =$ .....

**(3)**

(b) Simplify $h^{15} \div h^3$

.....

**(1)**

(c) Simplify fully $\left(2g^3k^5\right)^4$

.....

**(2)**

(d) Given that $\frac{y^5 \times y^n}{y^7} = y^{12}$

work out the value of $n$

$n =$ .....

**(2)**

**(Total for Question 4 is 8 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2H November 2023

**1** **(a)**

(a)

$$\frac{2x+5}{6} = 2x - 5$$

$$2x + 5 = 12x - 30$$

$$35 = 10x$$

$$x = \frac{7}{2}$$

**2** **(b)**

(b)

$$h^{15} \div h^3 = h^{15-3} = h^{12}$$

---

**3** **(c)**

(c)

$$(2g^3k^5)^4 = 2^4 g^{12} k^{20} = 16g^{12}k^{20}$$

---

**4** **(d)**

(d)

$$\frac{y^5 \times y^n}{y^7} = y^{5+n-7} = y^{n-2}$$

---

**5** **Use index laws**

Given that this equals $y^{12}$,

$$n - 2 = 12$$

$$n = 14$$

---

✓ **FINAL ANSWER**

$x = \frac{7}{2}$, $h^{12}$, $16g^{12}k^{20}$, and $n = 14$.

4MA1/2H NOVEMBER 2023

# Question 12

4 marks | Algebraic Roots & Indices

**12** $2^7 \times 4^5 = 4^x$

(a) Calculate the value of $x$

$x =$ .....

**(2)**

(b) Simplify fully $\left(125p^6y^{24}\right)^{\frac{2}{3}}$

.....

**(2)**

**(Total for Question 12 is 4 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/2H November 2023

**1** **(a)**

(a)

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

**2** **Use index laws**

Also

$$4^x = (2^2)^x = 2^{2x}$$

**3** **Use index laws**

So

$$2x = 17$$

$$x = \frac{17}{2}$$

**4** **(b)**

(b)

$$(125p^6y^{24})^{2/3} = 125^{2/3}p^4y^{16}$$

$$= 25p^4y^{16}$$

**FINAL ANSWER**

(a) $x = \frac{17}{2}$, (b) $25p^4y^{16}$

4MA1/2HR JUNE 2024

# Question 7

2 marks | Algebraic Roots & Indices

7 $\dfrac{3^{-2}\times 3^{5}}{3^{10}} = 3^{n}$

Find the value of $n$

$n =$ .....

**(Total for Question 7 is 2 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2HR June 2024

**1 Use index laws**

$$\frac{3^{-2} \times 3^{5}}{3^{10}} = 3^{n}$$

**2 Use index laws**

Use index laws:

$$\frac{3^{-2+5}}{3^{10}} = 3^{3-10} = 3^{-7}$$

**3 Use index laws**

So

$$n = -7$$

**FINAL ANSWER**

$n = -7$.

4MA1/1H MAY 2024

# Question 10

4 marks | Algebraic Roots & Indices

**10** (a) Simplify $(2p)^0$ where $p > 0$

.....

(1)

$y^9 \times y^{-3} = y^n$

(b) Find the value of $n$

$n =$ .....

(1)

(c) Simplify fully $(5a^4c^2)^3$

.....

(2)

**(Total for Question 10 is 4 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1H May 2024

**1** **(a)**

(a) $(2p)^0 = 1$.

**2** **Use index laws**

(b) $y^9 \times y^{-3} = y^{9-3} = y^6$, so $n = 6$.

**3** **(c)**

(c) $(5a^4c^2)^3 = 5^3a^{12}c^6 = 125a^{12}c^6$.

✓ **FINAL ANSWER**

(a) $1$, (b) $6$, (c) $125a^{12}c^6$

4MA1/1HR MAY 2024

# Question 17

4 marks | Algebraic Roots & Indices

17 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of $n$

$n =$ .....

(1)

(b) Express $\dfrac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where $c$ and $d$ are integers.

Show your working clearly.

.....

(3)

**(Total for Question 17 is 4 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1HR May 2024

**1** **(a) so**

(a)

$$\left(\sqrt[4]{k^{12}}\right)^5 = (k^3)^5 = k^{15}$$

so $n = 15$.

**2** **(b)**

(b)

$$\frac{7}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{7(2+\sqrt{3})}{4-3} = 14 + 7\sqrt{3}$$

$$7\sqrt{3} = \sqrt{49 \times 3} = \sqrt{147}$$

**FINAL ANSWER**

(a) $15$, (b) $\sqrt{147} + 14$

4MA1/2H JUNE 2024

# Question 23

3 marks | Algebraic Roots & Indices

**23** Simplify $\dfrac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form $5^w$ where $w$ is an expression in terms of $x$
Show each stage of your working clearly.

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2H June 2024

**1 Simplify surd**

$$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$$

**2 Simplify the number part**

Simplify the number part:

$$\sqrt{180} = 6\sqrt{5}$$

$$\frac{30}{6\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5} = 5^{1/2}$$

**3 Simplify surd**

Now write the powers with base 5:

$$25^{2x+7} = (5^2)^{2x+7} = 5^{4x+14}$$

$$(\sqrt{5})^{4x+9} = 5^{(4x+9)/2} = 5^{2x+9/2}$$

**4 Use index laws**

So the full expression is

$$5^{1/2+4x+14-(2x+9/2)}$$

$$= 5^{2x+10}$$

**FINAL ANSWER**

$5^{2x+10}$.

4MA1/2H NOVEMBER 2024

# Question 14

3 marks | Algebraic Roots & Indices

**14** Given that $\dfrac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of $n$
Show your working clearly.

$n =$ .....

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2H November 2024

**1** **Use index laws**

$$\frac{3^{2n+3}}{3^4} = 3^{2n-1}$$

$$3^3 \times 3^{1-2n} = 3^{4-2n}$$

**2** **Use index laws**

So

$$2n - 1 = 4 - 2n$$

$$4n = 5$$

✓ **FINAL ANSWER**

$n = \frac{5}{4}$

4MA1/1H JUNE 2025

# Question 4

4 marks | Algebraic Roots & Indices

**4** $x^5 \times x^7 = x^m$

(a) Find the value of $m$

$m =$ .....

**(1)**

$y^8 \div y^3 = y^n$

(b) Find the value of $n$

$n =$ .....

**(1)**

(c) Simplify fully $(5a^4r^2)^3$

.....

**(2)**

**(Total for Question 4 is 4 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1H June 2025

**1** **Use index laws**

For part (a),

$$x^5 \times x^7 = x^{12}$$

**2** **Use index laws**

So $m = 12$.

**3** **Use index laws**

For part (b),

$$y^8 \div y^3 = y^{8-3} = y^5$$

**4** **Use index laws**

So $n = 5$.

**5** **Use index laws**

For part (c),

$$(5a^4r^2)^3 = 5^3a^{4\times3}r^{2\times3}$$

$$= 125a^{12}r^6$$

---

✓ **FINAL ANSWER**

(a) $m = 12$, (b) $n = 5$, (c) $125a^{12}r^6$.

4WM1/01 MAY 2025

# Question 5

4 marks | Algebraic Roots & Indices

**5** $x^5 \times x^7 = x^m$

(a) Find the value of $m$

$m =$ .....

**(1)**

$y^8 \div y^3 = y^n$

(b) Find the value of $n$

$n =$ .....

**(1)**

(c) Simplify fully $(5a^4r^2)^3$

.....

**(2)**

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4WM1/01 May 2025

**1** **Use index laws**

$$x^5 \times x^7 = x^{5+7} = x^{12}$$

**2** **Use index laws**

So $m = 12$.

$$y^8 \div y^3 = y^{8-3} = y^5$$

**3** **Use index laws**

So $n = 5$.

$$(5a^4r^2)^3 = 5^3(a^4)^3(r^2)^3 = 125a^{12}r^6$$

✓ **FINAL ANSWER**

$m = 12$, $n = 5$, and $125a^{12}r^6$.

# Question 5

4 marks | Algebraic Roots & Indices

**5** $3^8 \times 3^4 = 3^m$

(a) Find the value of $m$

$m =$ .....

**(1)**

$4^{12} \div 4^7 = 4^n$

(b) Find the value of $n$

$n =$ .....

**(1)**

$\left(6^2\right)^{10} = 6^p$

(c) Find the value of $p$

$p =$ .....

**(1)**

$5^r = 1$

(d) Write down the value of $r$

$r =$ .....

**(1)**

**(Total for Question 5 is 4 marks)**

## 1 Evaluate fraction

$$3^8 \times 3^4 = 3^{8+4} = 3^{12}$$

$$\frac{4^{12}}{4^7} = 4^{12-7} = 4^5$$

$$(6^2)^{10} = 6^{20}$$

$$5^0 = 1$$

**FINAL ANSWER**

$m = 12, n = 5, p = 20, r = 0.$

4MA1/2HR JUNE 2025

# Question 6

5 marks | Algebraic Roots & Indices

**6** (a) Write down the value of $5^0$

.....

(1)

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^k$$

(b) Find the value of $k$

$k =$ .....

(2)

(c) Simplify fully $\left(2d^4e^5\right)^3$

.....

(2)

**(Total for Question 6 is 5 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2HR June 2025

**1** **Use index laws**

$$5^0 = 1$$

**2** **Use index laws**

For part (b),

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^{9+(-3)-(-2)} = 5^8$$

**3** **Use index laws**

So

$$k = 8$$

**4** **Use index laws**

For part (c),

$$(2d^4e^5)^3 = 2^3 d^{4\times 3} e^{5\times 3}$$

$$= 8d^{12}e^{15}$$

**FINAL ANSWER**

(a) $\mathbf{1}$, (b) $\boldsymbol{k = 8}$, (c) $\boldsymbol{8d^{12}e^{15}}$.

4WM1/01 MAY 2025

# Question 9

3 marks | Algebraic Roots & Indices

9 $7^x = 1$

(a) Write down the value of $x$

$x =$ .....

(1)

$$\frac{8^{-4} \times 8^{11}}{8^{12}} = 8^n$$

(b) Work out the value of $n$

$n =$ .....

(2)

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4WM1/01 May 2025

**1 Use index laws**

$$7^x = 1$$

**2 Use index laws**

Since $7^0 = 1$,

$$x = 0$$

**3 Use index laws**

For part (b),

$$\frac{8^{-4} \times 8^{11}}{8^{12}} = 8^n$$

$$8^{-4} \times 8^{11} = 8^7$$

$$\frac{8^7}{8^{12}} = 8^{7-12} = 8^{-5}$$

**4 Use index laws**

So $n = -5$.

**FINAL ANSWER**

$x = 0, n = -5.$

4MA1/2H JUNE 2025

# Question 15

3 marks | Algebraic Roots & Indices

**15** Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2H June 2025

**1** **Use index laws**

$$\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$$

**2** **Use index laws**

First simplify inside the bracket:

$$\frac{2a^3}{10a^7c^2} = \frac{1}{5a^4c^2}$$

**3** **A negative power means take**

A negative power means take the reciprocal:

$$\left(\frac{1}{5a^4c^2}\right)^{-3} = (5a^4c^2)^3$$

$$= 125a^{12}c^6$$

✓ **FINAL ANSWER**

$125a^{12}c^6$.

4WM1/01 MAY 2025

# Question 16

3 marks | Algebraic Roots & Indices

16 Solve the equation $8^{3y+4} \times 4^{3y} = 2^{5y}$
Show your working clearly.

$y =$ .....

(Total for Question 16 is 3 marks)

## WORKED SOLUTION

Question 16 · 4WM1/01 May 2025

**1 Solve equation**

$$8^{3y+4} \times 4^{3y} = 2^{5y}$$

**2 Solve equation**

Write $8$ and $4$ as powers of $2$.

$$(2^3)^{3y+4}(2^2)^{3y} = 2^{5y}$$

$$2^{9y+12} \times 2^{6y} = 2^{5y}$$

$$2^{15y+12} = 2^{5y}$$

**3 Solve equation**

So

$$15y + 12 = 5y$$

$$y = -\frac{6}{5}$$

✓ **FINAL ANSWER**

$y = -\frac{6}{5}$

4WM1/01 NOVEMBER 2025

# Question 7

2 marks | Algebraic Roots & Indices

7 $\dfrac{8^{-2} \times 8^{9}}{8^{10}} = 8^{n}$

Work out the value of $n$

$n =$ .....

**(Total for Question 7 is 2 marks)**

## WORKED SOLUTION

Question 7 · 4WM1/01 November 2025

**1** **Use index laws**

$$\frac{8^{-2} \times 8^{9}}{8^{10}} = 8^{-2+9-10} = 8^{-3}$$

**FINAL ANSWER**

$n = -3$

4MA1/1H NOVEMBER 2025

# Question 8

2 marks | Algebraic Roots & Indices

**8** $\dfrac{8^{-2} \times 8^{9}}{8^{10}} = 8^{n}$

Work out the value of $n$

$n =$ .....

**(Total for Question 8 is 2 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1H November 2025

**1** **Use index laws**

$$\frac{8^{-2} \times 8^{9}}{8^{10}} = 8^{-2+9-10} = 8^{-3}$$

✓ **FINAL ANSWER**

$n = -3$

4MA1/2H NOVEMBER 2025

# Question 13

4 marks | Algebraic Roots & Indices

**13** (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^3$

.....

(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express $n$ in terms of $m$ and $p$

$n =$ .....

(2)

**(Total for Question 13 is 4 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/2H November 2025

**1** **(a)**

(a)

$$\left(\frac{125x^6}{y^{15}}\right)^{1/3} = \frac{5x^2}{y^5}$$

**2** **(b)**

(b)

$$64^p = (4^3)^p = 4^{3p}$$

$$4^n = \frac{4^m}{64^p} = \frac{4^m}{4^{3p}} = 4^{m-3p}$$

✓ **FINAL ANSWER**

(a) $\frac{5x^2}{y^5}$, (b) $n = m - 3p$

4WM1/01 NOVEMBER 2025

# Question 15

3 marks | Algebraic Roots & Indices

15 $\dfrac{16^{x-3} \times 8^{x+1}}{32^{x}} = 2^{n}$

Express $n$ in terms of $x$
Give your answer in its simplest form.

$n =$ .....

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4WM1/01 November 2025

**1 Evaluate fraction**

$$\frac{16^{x-3} \times 8^{x+1}}{32^{x}} = \frac{2^{4x-12} \times 2^{3x+3}}{2^{5x}}$$

$$= 2^{4x-12+3x+3-5x} = 2^{2x-9}$$

**FINAL ANSWER**

$n = 2x - 9$.

4MA1/1HR JUNE 2026

# Question 3

2 marks | Algebraic Roots & Indices

**3** (a) Simplify $e^7 \times e^3$

.....

**(1)**

(b) Simplify $g^5 \div g^2$

.....

**(1)**

**(Total for Question 3 is 2 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1HR June 2026

**1** **Part (a) | Step 1**

$$e^7 \times e^3 = e^{7+3} = e^{10}.$$

For equal bases, add the indices when multiplying.

**2** **Part (b) | Step 1**

$$g^5 \div g^2 = g^{5-2} = g^3 \quad (g \neq 0).$$

For equal bases, subtract the indices when dividing.

✓ **FINAL ANSWER**

(a) $e^{10}$ (b) $g^3$

4MA1/1H JUNE 2026

# Question 6

2 marks | Algebraic Roots & Indices

6 Simplify $\left(4x^5y^6\right)^3$

.....

**(Total for Question 6 is 2 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1H June 2026

### 1 Solution | Step 1

$$(4x^5y^6)^3 = 4^3x^{5\times3}y^{6\times3} = 64x^{15}y^{18}.$$

Raise each factor to the third power and multiply exponents for a power of a power.

**FINAL ANSWER**

$64x^{15}y^{18}$

4WM1/01 MAY 2026

# Question 6

2 marks | Algebraic Roots & Indices

6 Simplify $\left(4x^5y^6\right)^3$

.....

**(Total for Question 6 is 2 marks)**

## WORKED SOLUTION

Question 6 · 4WM1/01 May 2026

**1 Cube every part**

$$(4x^5y^6)^3 = 4^3x^{15}y^{18} = 64x^{15}y^{18}$$

✓ **FINAL ANSWER**

$64x^{15}y^{18}$

# Question 12

6 marks | Algebraic Roots & Indices

**12** $\dfrac{2^{\frac{x}{3}}}{8^x} = 64$

(a) Find the value of $x$
Show clear algebraic working.

$x =$ .....

**(3)**

(b) Expand and simplify $4t(t + 5)(2t - 3)$

.....

**(3)**

**(Total for Question 12 is 6 marks)**

**1** **(a) Write everything as powers of 2**

$$8^x = 2^{3x}, \quad 64 = 2^6 \Rightarrow \frac{2^{x/3}}{2^{3x}} = 2^6$$

**2** **Compare powers**

$$\frac{x}{3} - 3x = 6 \Rightarrow -\frac{8x}{3} = 6 \Rightarrow x = -\frac{9}{4}$$

**3** **(b) Expand two brackets first**

$$(t+5)(2t-3) = 2t^2 + 7t - 15$$

**4** **Multiply by 4t**

$$4t(2t^2 + 7t - 15) = 8t^3 + 28t^2 - 60t$$

✓ **FINAL ANSWER**

(a) $x = -\frac{9}{4}$ (b) $8t^3 + 28t^2 - 60t$

4MA1/2HR JUNE 2026

# Question 16

2 marks | Algebraic Roots & Indices

16 $\left(\frac{1}{\sqrt{a}}\right)^6 = a^n$

Work out the value of $n$

$n =$ .....

**(Total for Question 16 is 2 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/2HR June 2026

### 1 Solution | Step 1

$$\frac{1}{\sqrt{a}} = a^{-1/2}, \qquad \left(a^{-1/2}\right)^6 = a^{-3}.$$

A reciprocal changes the sign of the exponent and an outer power multiplies exponents.

**FINAL ANSWER**

$n = -3$.

4MA1/1HR JUNE 2026

# Question 26

3 marks | Algebraic Roots & Indices

**26**

$$\frac{27^x \times 3^{n+6}}{9^{4-n}} = \frac{6^x}{2^x}$$

Express $x$ in terms of $n$
Show your working clearly and simplify your expression.

$x =$ .....

**(Total for Question 26 is 3 marks)**

## WORKED SOLUTION

Question 26 · 4MA1/1HR June 2026

### 1 Solution | Step 1

$$27 = 3^3, \qquad 9 = 3^2, \qquad \frac{6^x}{2^x} = \left(\frac{6}{2}\right)^x = 3^x.$$

Write both sides in a common base.

### 2 Solution | Step 2

$$\frac{27^x \times 3^{n+6}}{9^{4-n}} = 3^{3x+n+6-2(4-n)} = 3^{3x+3n-2}.$$

Multiply powers by adding indices and divide powers by subtracting indices.

### 3 Solution | Step 3

$$3^{3x+3n-2} = 3^x \quad \Longrightarrow \quad 3x + 3n - 2 = x.$$

The base-3 exponential is one-to-one, so equal powers have equal indices.

### 4 Solution | Step 4

$$2x = 2 - 3n \quad \Longrightarrow \quad x = 1 - \frac{3n}{2}.$$

Collect the x terms and divide by 2.

**FINAL ANSWER**

$x = 1 - \frac{3n}{2}$

TOPIC

# Completing the Square

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 20

4 marks | Completing the Square

**20** Find the values of $a$, $b$ and $c$ so that

$$7 + 12x - 2x^2$$

is written as $a - b(x - c)^2$

$a =$ .....

$b =$ .....

$c =$ .....

**(Total for Question 20 is 4 marks)**

**1** **Factor out -2 from the quadratic terms**

$$7 + 12x - 2x^2 = -2(x^2 - 6x) + 7$$

**2** **Complete the square**

$$x^2 - 6x = (x - 3)^2 - 9$$

**3** **Simplify the expression**

$$-2((x - 3)^2 - 9) + 7 = -2(x - 3)^2 + 18 + 7$$

$$= 25 - 2(x - 3)^2$$

**4** **Read the values**

$$a = 25, \qquad b = 2, \qquad c = 3$$

✓ **FINAL ANSWER**

$a = 25,\ b = 2,\ c = 3.$

4MA1/2HR JANUARY 2020

# Question 23

3 marks | Completing the Square

**23** Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where $a$, $b$ and $c$ are integers.

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2HR January 2020

### 1 Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

$$= -2(x^2 + 6x) + 7$$

$$= -2\left((x + 3)^2 - 9\right) + 7$$

$$= -2(x + 3)^2 + 18 + 7$$

$$= 25 - 2(x + 3)^2$$

**FINAL ANSWER**

$25 - 2(x + 3)^2$

4MA1/2H MAY 2021

# Question 22

4 marks | Completing the Square

22 The curve **S** has equation $y = f(x)$ where $f(x) = x^2$
The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where $a$, $b$ and $c$ are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H May 2021

**1 Solve inequality**

$$g(x) = 2x^2 - 12x + 13$$

$$= 2(x^2 - 6x) + 13$$

$$= 2\left((x - 3)^2 - 9\right) + 13$$

$$= 2(x - 3)^2 - 18 + 13$$

$$= 2(x - 3)^2 - 5$$

**2 Calculate value**

Starting with $S : y = x^2$:

**3 Translate 3 units to the**

- Translate 3 units to the right to get $y = (x - 3)^2$.
- Stretch parallel to the $y$-axis by scale factor 2 to get $y = 2(x - 3)^2$.
- Translate 5 units down to get $y = 2(x - 3)^2 - 5$.

✓ **FINAL ANSWER**

$g(x) = 2(x - 3)^2 - 5$. Transformations: right 3, stretch parallel to the $y$-axis by scale factor 2, then down 5.

4MA1/1HR JANUARY 2022

# Question 19

3 marks | Completing the Square

**19** Given that $\left(\sqrt[3]{\frac{1}{x}}\right)^4 = x^m$

(a) find the value of $m$

$m =$ .....

(1)

Given that $a$, $b$ and $c$ are integers,

(b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

.....

(2)

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1HR January 2022

**1** **(a)**

(a)

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

**2** **Evaluate fraction**

So $m = -\frac{1}{3}$.

**3** **(b)**

(b)

$$3x^2 + 12x + 19 = 3(x^2 + 4x) + 19$$

$$= 3\left((x + 2)^2 - 4\right) + 19$$

$$= 3(x + 2)^2 + 7$$

**FINAL ANSWER**

$m = -\frac{1}{3}$, and $3(x + 2)^2 + 7$

4MA1/1H JANUARY 2022

# Question 20

4 marks | Completing the Square

**20** (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where $a$, $b$ and $c$ are integers.

.....

(3)

**C** is the curve with equation $y = 7 + 12x - 3x^2$
The point $A$ is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of $A$

(..... , .....)

(1)

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H January 2022

**1** **(a)**

(a)

$$7 + 12x - 3x^2 = -3x^2 + 12x + 7$$

$$= -3(x^2 - 4x) + 7$$

$$= -3\left((x - 2)^2 - 4\right) + 7$$

$$= 19 - 3(x - 2)^2$$

**2** **This is the same as**

This is the same as

$$19 - 3(x + (-2))^2$$

**3** **(b) The maximum point is**

(b) The maximum point is read from the completed square form.

$$y = 19 - 3(x - 2)^2$$

**4** **Calculate value**

The maximum occurs when $x = 2$, giving $y = 19$.

---

✓ **FINAL ANSWER**

(a) $19 - 3(x - 2)^2$, (b) $A = (2, 19)$

4MA1/1HR JUNE 2022

# Question 23

5 marks | Completing the Square

**23** (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where $a$, $b$ and $c$ are integers.

.....

(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point $M$ is the minimum point on **C**

(b) Find the coordinates of $M$

(..... , .....)

(2)

**(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/1HR June 2022

**1** **(a)**

(a)

$$2x^2 - 12x + 3 = 2(x^2 - 6x) + 3$$

$$= 2\left((x - 3)^2 - 9\right) + 3$$

$$= 2(x - 3)^2 - 18 + 3$$

$$= 2(x - 3)^2 - 15$$

**2** **(b)**

(b)

$$y = 2(x + 4)^2 - 12(x + 4) + 3$$

**3** **Let**

Let $u = x + 4$. From part (a),

$$2u^2 - 12u + 3 = 2(u - 3)^2 - 15$$

**4** **Calculate value**

So

$$y = 2((x + 4) - 3)^2 - 15$$

$$y = 2(x + 1)^2 - 15$$

**5** **The minimum point is when**

The minimum point is when $x = -1$, giving $y = -15$.

**FINAL ANSWER**

(a) $2(x - 3)^2 - 15$, (b) $M = (-1, -15)$

4MA1/1H JUNE 2022

# Question 24

4 marks | Completing the Square

**24** Express each of $a$, $b$ and $c$ in terms of $q$ so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ .....

$b =$ .....

$c =$ .....

**(Total for Question 24 is 4 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/1H June 2022

**1 Evaluate fraction**

$$q + 12x - qx^2 = -q\left(x^2 - \frac{12}{q}x\right) + q$$

**2 Complete the square**

Complete the square:

$$x^2 - \frac{12}{q}x = \left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}$$

**3 Evaluate fraction**

So

$$q + 12x - qx^2 = -q\left[\left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}\right] + q$$

$$= q + \frac{36}{q} - q\left(x - \frac{6}{q}\right)^2$$

**4 Compare coefficients**

Compare with

$$a - b(x - c)^2$$

**FINAL ANSWER**

$$a = q + \frac{36}{q},\ b = q,\ c = \frac{6}{q}$$

4MA1/2H JUNE 2023

# Question 19

3 marks | Completing the Square

**19** Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2H June 2023

**1** **Calculate value**

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5$$

$$= 3\left((x - 1)^2 - 1\right) + 5$$

$$= 3(x - 1)^2 + 2$$

**FINAL ANSWER**

$3(x - 1)^2 + 2$

# Question 20

5 marks | Completing the Square

**20** (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where $a$, $b$ and $c$ are numbers to be found.

.....

**(3)**

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point $P$ is the minimum point on **C**

(b) Find the coordinates of $P$

(..... , .....)

**(2)**

**(Total for Question 20 is 5 marks)**

**1** **Use Moved Coordinate Geometry Completing**

For part (a),

$$2x^2 - 11x + 9$$

**2** **Factor out 2 from the**

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

**3** **Complete the square**

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

**4** **Evaluate fraction**

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

**5** **Calculate value**

For part (b),

$$y = 2(x-3)^2 - 11(x-3) + 9$$

**6** **Let**

Let

$$u = x - 3$$

**7** **Evaluate fraction**

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

**8** **Evaluate fraction**

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

**9** **The minimum value of is**

The minimum value of $y$ is

$$-\frac{49}{8}$$

✓ **FINAL ANSWER**

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8},\ P = \left(\frac{23}{4}, -\frac{49}{8}\right)$$

4MA1/1H MAY 2024

# Question 25

4 marks | Completing the Square

**25** $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where $a$, $b$ and $c$ are constants.

$f(x) =$ .....

**(Total for Question 25 is 4 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/1H May 2024

**1 Solve inequality**

$$f(x) = 17 - 3x^2 + 12x$$

$$= -3x^2 + 12x + 17$$

$$= -3(x^2 - 4x) + 17$$

$$= -3\left((x - 2)^2 - 4\right) + 17$$

$$= -3(x - 2)^2 + 12 + 17$$

$$= 29 - 3(x - 2)^2$$

**FINAL ANSWER**

$f(x) = 29 - 3(x - 2)^2$

4WM1/01R MAY 2025

# Question 18

4 marks | Completing the Square

**18** (a) Express $2x^2 - 12x + 23$ in the form $a(x - b)^2 + c$ where $a$, $b$ and $c$ are integers.
Show clear algebraic working.

.....

(3)

(b) Hence, state the least value of $2x^2 - 12x + 23$

.....

(1)

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4WM1/01R May 2025

**1** **Calculate value**

$$2x^2 - 12x + 23 = 2(x^2 - 6x) + 23$$

$$= 2\left((x - 3)^2 - 9\right) + 23$$

$$= 2(x - 3)^2 + 5$$

**2** **The squared part is at**

The squared part is at least zero.

✓ **FINAL ANSWER**

$2(x - 3)^2 + 5$; least value $5$.

# Question 25

6 marks | Completing the Square

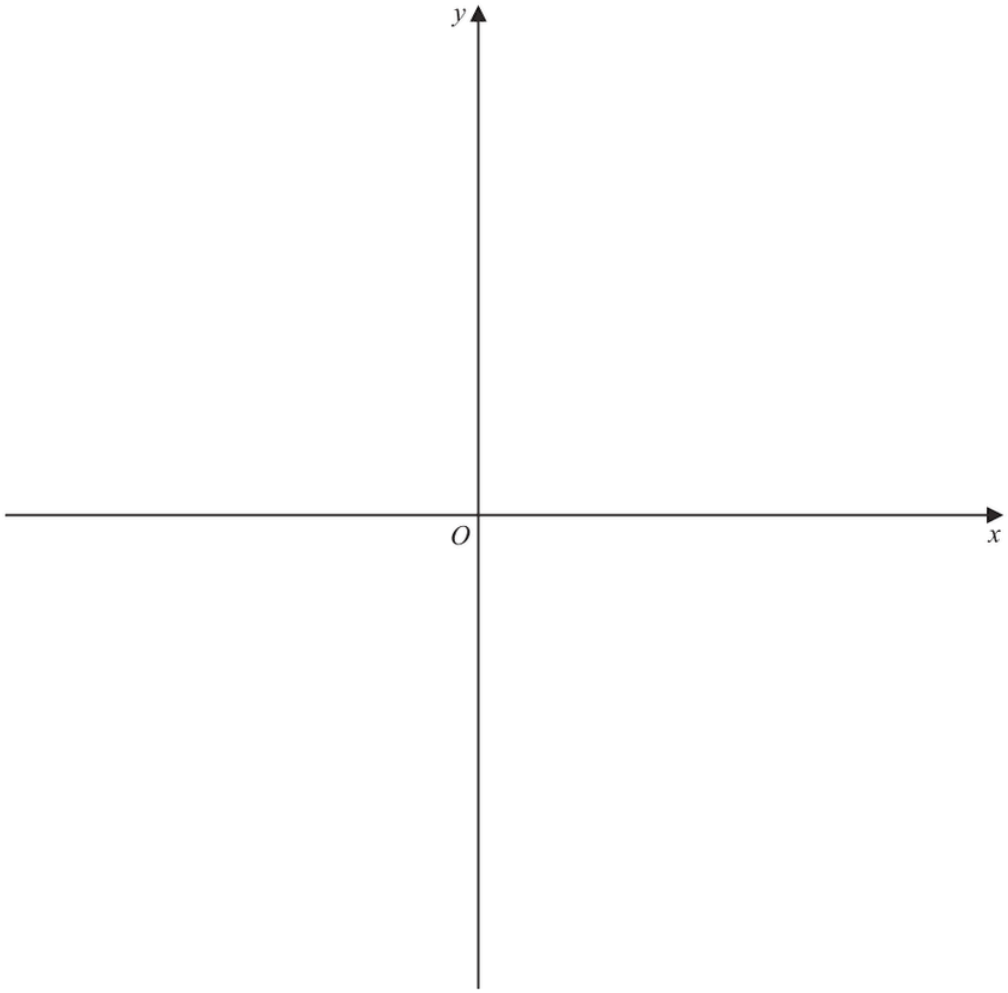
**25** (a) Write $28+24x-6x^2$ in the form $a-b(x-c)^2$ where $a$, $b$ and $c$ are integers.

.....

(3)

(b) On the axes below, sketch the graph of $y=28+24x-6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the $y$-axis.




(3)

**(Total for Question 25 is 6 marks)**

1 **(a)**

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

$$= -6(x^2 - 4x) + 28$$

$$= -6\left((x-2)^2 - 4\right) + 28$$

$$= 52 - 6(x-2)^2$$

2 **(b)**

(b)

$$y = 52 - 6(x-2)^2$$

3 **The graph is a downward**

The graph is a downward-opening parabola.

4 **The turning point is**

The turning point is

$$(2, 52)$$

5 **Calculate value**

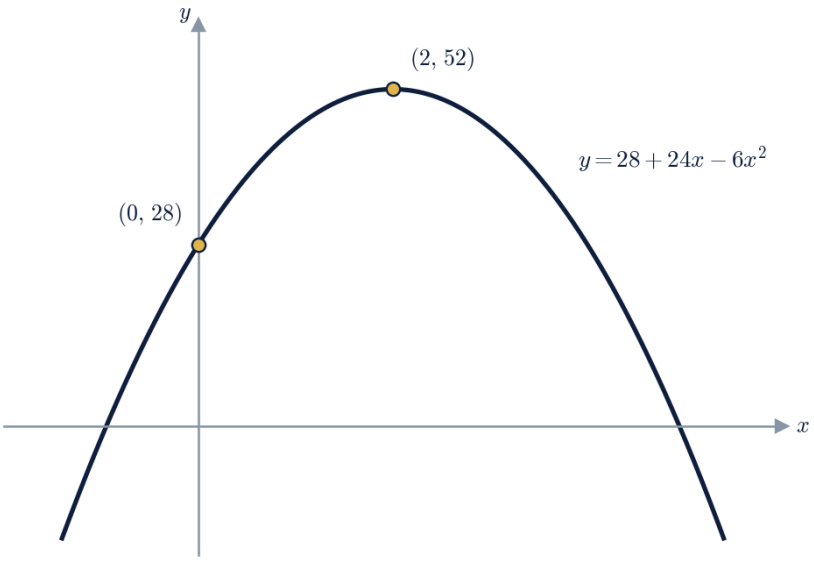
The point of intersection with the $y$-axis is found by putting $x = 0$:

$$y = 28$$

**6** **Find intercepts**

So the $y$-intercept is

$$(0, 28)$$




**FINAL ANSWER**

(a) $52 - 6(x - 2)^2$. For the sketch, show a downward parabola with turning point $(2, 52)$ and $y$-intercept $(0, 28)$.

4WM1/01 MAY 2025

# Question 25

5 marks | Completing the Square

**25** (a) Express $8 - 12x - 4x^2$ in the form $a - b(x + c)^2$
where $a$, $b$ and $c$ are constants.

.....

**(3)**

(b) Hence solve $8 - 12x - 4x^2 = 0$
Show your working clearly.
Give your solutions in surd form.

.....

**(2)**

**(Total for Question 25 is 5 marks)**

## WORKED SOLUTION

Question 25 · 4WM1/01 May 2025

**1** **Simplify surd**

$$8x^2 - 12x - 4 = 8\left(x^2 - \frac{3}{2}x\right) - 4$$

**2** **Complete the square**

Complete the square.

$$x^2 - \frac{3}{2}x = \left(x - \frac{3}{4}\right)^2 - \frac{9}{16}$$

**3** **Simplify surd**

So

$$8x^2 - 12x - 4 = 8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$$

**4** **Simplify surd**

Now solve

$$8x^2 - 12x - 4 = 0$$

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2} = 0$$

$$\left(x - \frac{3}{4}\right)^2 = \frac{17}{16}$$

$$x - \frac{3}{4} = \pm\frac{\sqrt{17}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

---

✓ **FINAL ANSWER**

$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$, and $x = \frac{3 \pm \sqrt{17}}{4}$.

4MA1/1H NOVEMBER 2025

# Question 21

3 marks | Completing the Square

**21** Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where $a$, $b$ and $c$ are integers.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H November 2025

### 1 Solve inequality

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5\left((x - 2)^2 - 4\right) + 23$$

$$= 5(x - 2)^2 - 20 + 23$$

$$= 5(x - 2)^2 + 3$$

**FINAL ANSWER**

$5(x - 2)^2 + 3$

4WM1/01 NOVEMBER 2025

# Question 22

2 marks | Completing the Square

**22** $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where $p$, $q$ and $r$ are integers.

Find the value of $q$

$q =$ .....

**(Total for Question 22 is 2 marks)**

## WORKED SOLUTION

Question 22 · 4WM1/01 November 2025

**1** **Calculate value**

$$9x^2 - 12x + q = (3x - 2)^2$$

$$(3x - 2)^2 = 9x^2 - 12x + 4$$

**FINAL ANSWER**

$q = 4$.

4WM1/01 NOVEMBER 2025

# Question 24

3 marks | Completing the Square

**24** Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where $a$, $b$ and $c$ are integers.

**(Total for Question 24 is 3 marks)**

## WORKED SOLUTION

Question 24 · 4WM1/01 November 2025

**1** **Calculate value**

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5\left((x - 2)^2 - 4\right) + 23$$

$$= 5(x - 2)^2 + 3$$

**FINAL ANSWER**

$5(x - 2)^2 + 3$.

4WM1/01 MAY 2026

# Question 24

3 marks | Completing the Square

**24** Express $22 - 3x^2 + 30x$ in the form $a - b(x - c)^2$ where $a$, $b$ and $c$ are integers.

.....

**(Total for Question 24 is 3 marks)**

## WORKED SOLUTION

Question 24 · 4WM1/01 May 2026

**1 Factor out −3 from the x terms**

$$22 - 3x^2 + 30x = 22 - 3(x^2 - 10x)$$

**2 Complete the square**

$$x^2 - 10x = (x - 5)^2 - 25$$

**3 Simplify**

$$22 - 3\big((x - 5)^2 - 25\big) = 97 - 3(x - 5)^2$$

**FINAL ANSWER**

$97 - 3(x - 5)^2$

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TOPIC

# Quadratic Formula

Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 30

3 marks | Quadratic Formula

**30** Solve $2x^2 - 5x - 4 = 0$

Show your working clearly.
Give your solutions correct to 3 significant figures.

.....

**(3)**

**(Total for Question 30 is 3 marks)**

## WORKED SOLUTION

Question 30 · EP Specimen

**1 Identify a, b and c**

$a = 2,\ b = -5,\ c = -4$

**2 Substitute into the formula**

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-4)}}{2(2)} = \frac{5 \pm \sqrt{57}}{4}$$

**3 Work out both solutions**

$$x = \frac{5 + 7.5498\ldots}{4} = 3.14, \qquad x = \frac{5 - 7.5498\ldots}{4} = -0.637$$

**FINAL ANSWER**

$x = 3.14$ or $x = -0.637$

EP SPECIMEN

# Question 31

3 marks | Quadratic Formula

**31** Solve $x^2 + 6x + 2 = 0$

Give your solutions in the form $a \pm \sqrt{b}$ where $a$ and $b$ are integers.

.....

**(3)**

**(Total for Question 31 is 3 marks)**

## WORKED SOLUTION

Question 31 · EP Specimen

**1** **Use the formula**

$$x = \frac{-6 \pm \sqrt{36 - 8}}{2} = \frac{-6 \pm \sqrt{28}}{2}$$

**2** **Simplify the surd**

$\sqrt{28} = 2\sqrt{7}$, so

$$x = \frac{-6 \pm 2\sqrt{7}}{2} = -3 \pm \sqrt{7}$$

✓ **FINAL ANSWER**

$x = -3 \pm \sqrt{7}$

EP SPECIMEN

# Question 32

5 marks | Quadratic Formula

**32** A rectangle has length $(x + 3)$ cm and width $(x - 1)$ cm.
The area of the rectangle is 20 $cm^2$.

(a) Show that $x^2 + 2x - 23 = 0$

.....

**(2)**

(b) Work out the length of the rectangle.
Give your answer correct to 3 significant figures.

..... cm

**(3)**

**(Total for Question 32 is 5 marks)**

## WORKED SOLUTION

Question 32 · EP Specimen

**1** **(a) Area = length × width**

$$(x+3)(x-1) = 20 \Rightarrow x^2 + 2x - 3 = 20 \Rightarrow x^2 + 2x - 23 = 0$$

**2** **(b) Use the formula**

$$x = \frac{-2 \pm \sqrt{4 + 92}}{2} = \frac{-2 \pm \sqrt{96}}{2} = -1 \pm \sqrt{24}$$

The width $x - 1$ must be positive, so $x = -1 + \sqrt{24} = 3.898\ldots$

**3** **(b) Length**

$$x + 3 = 6.898\ldots \approx 6.90 \text{ cm}$$

✓ **FINAL ANSWER**

(b) **6.90** cm

TOPIC

# Quadratic Equations

Every past-paper question on this topic, 2020 to 2026.

4MA1/1HR JANUARY 2020

# Question 6

3 marks | Quadratic Equations

6 Solve $x^2 - 5x - 36 = 0$
Show clear algebraic working.

**(Total for Question 6 is 3 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1HR January 2020

**1 Solve quadratic equation**

$$x^2 - 5x - 36 = 0$$

**2 Factorise**

Factorise:

$$(x - 9)(x + 4) = 0$$

**3 Solve quadratic equation**

So

$$x = 9 \quad \text{or} \quad x = -4$$

**FINAL ANSWER**

$x = 9$ or $x = -4$.

4MA1/2HR JANUARY 2021

# Question 9

3 marks | Quadratic Equations

9 Solve $x^2 - 21x + 20 = 0$
Show your working clearly.

(Total for Question 9 is 3 marks)

## WORKED SOLUTION

Question 9 · 4MA1/2HR January 2021

**1 Solve quadratic equation**

$$x^2 - 21x + 20 = 0$$

**2 Factorise**

Factorise:

$$(x - 20)(x - 1) = 0$$

**3 Solve quadratic equation**

So

$$x = 20 \quad \text{or} \quad x = 1$$

**FINAL ANSWER**

$x = 20$ or $x = 1$.

4WM1/01 MAY 2025

# Question 15

3 marks | Quadratic Equations

**15** Solve the equation $5x^2 + 9x - 17 = 0$
Show your working clearly.
Give your solutions correct to 3 significant figures.

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4WM1/01 May 2025

**1** **Solve quadratic equation**

$$5x^2 + 9x - 17 = 0$$

**2** **Use quadratic formula**

Use the quadratic formula with $a = 5$, $b = 9$, $c = -17$.

$$x = \frac{-9 \pm \sqrt{9^2 - 4(5)(-17)}}{2(5)}$$

$$x = \frac{-9 \pm \sqrt{421}}{10}$$

$$x = 1.152\ldots \quad \text{or} \quad x = -2.952\ldots$$

**3** **Solve quadratic equation**

To 3 significant figures,

$$x = 1.15 \quad \text{or} \quad x = -2.95$$

✓ **FINAL ANSWER**

$x = 1.15$ or $x = -2.95$.

TOPIC

# Linear Graphs

Every past-paper question on this topic, 2020 to 2026.

# Question 9

7 marks | Linear Graphs

**9** (a) Simplify $3a^4b^5 \times 4a^7b^2$

.....

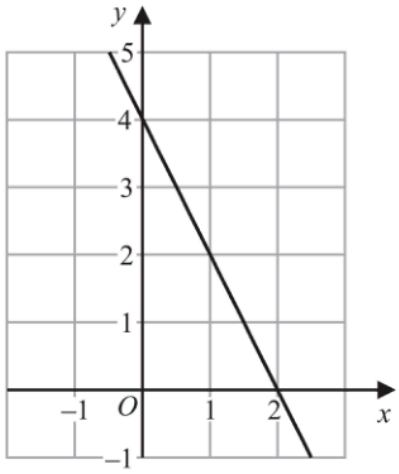
(2)

(b) Factorise fully $14x^2y^4 + 21x^3y^2$

.....

(2)

The diagram shows a straight line drawn on a grid.




(c) Find an equation of the line.

.....

(2)

A different straight line has equation $y = 3x - 5$

(d) Write down the coordinates of the point at which the line crosses the $y$-axis.

(....., .....)

(1)

**(Total for Question 9 is 7 marks)**

**1** **Simplify the product**

$$3a^4b^5 \times 4a^7b^2 = 12a^{11}b^7$$

**2** **Factorise fully**

$$14x^2y^4 + 21x^3y^2 = 7x^2y^2(2y^2 + 3x)$$

**3** **Find the gradient**

The line passes through $(0, 4)$ and $(2, 0)$.

$$m = \frac{0 - 4}{2 - 0} = -2$$

**4** **Write the equation**

The $y$-intercept is $4$, so

$$y = -2x + 4$$

**5** **Read the y-axis crossing**

For $y = 3x - 5$, put $x = 0$:

$$y = -5$$

So the crossing point is $(0, -5)$.

✓ **FINAL ANSWER**

$12a^{11}b^7$, $7x^2y^2(2y^2 + 3x)$, $y = -2x + 4$, and $(0, -5)$.

# Question 23

6 marks | Linear Graphs

**23** $ABCD$ is a kite.

$AB = AD$ and $CB = CD$

The point $B$ has coordinates $(k, 1)$ where $k$ is a negative constant.
The point $D$ has coordinates $(8, 7)$

The straight line **L** passes through the points $B$ and $D$ and has a gradient of $\frac{3}{5}$

Find an equation of $AC$
Give your answer in the form $px + qy = r$ where $p$, $q$ and $r$ are integers.
Show your working clearly.

.....

**(Total for Question 23 is 6 marks)**

## 1 Find k from the gradient

The gradient of $BD$ is

$$\frac{7-1}{8-k}=\frac{3}{5}$$

$$\frac{6}{8-k}=\frac{3}{5}$$

$$30=24-3k$$

$$k=-2$$

## 2 Find the midpoint of BD

So $B=(-2,1)$ and $D=(8,7)$.

$$\text{midpoint of } BD=\left(\frac{-2+8}{2},\frac{1+7}{2}\right)=(3,4)$$

## 3 Use the perpendicular gradient

In this kite, $AC$ is the perpendicular bisector of $BD$.

The gradient of $BD$ is $\frac{3}{5}$, so the gradient of $AC$ is

$$-\frac{5}{3}$$

## 4 Form the equation

Using point $(3,4)$:

$$y-4=-\frac{5}{3}(x-3)$$

$$3y-12=-5x+15$$

$$5x+3y=27$$

**FINAL ANSWER**

$5x+3y=27$.

4MA1/1H JANUARY 2020

# Question 1

4 marks | Linear Graphs

**1** The point $A$ has coordinates $(5, -4)$
The point $B$ has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of $AB$.

(.....  , .....)

**(2)**

Line **L** has equation $y = 2 - 3x$

(b) Write down the gradient of line **L**.

.....

**(1)**

Line **L** has equation $y = 2 - 3x$

(c) Does the point with coordinates $(100, -302)$ lie on line **L**?
You must give a reason for your answer.

.....

**(1)**

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H January 2020

**1** **Find the midpoint**

(a) The midpoint of $A(5, -4)$ and $B(13, 1)$ is

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

$$= (9, -1.5)$$

**2** **(b)**

(b)

$$y = 2 - 3x$$

**3** **The gradient is**

The gradient is $-3$.

**4** **Calculate value**

(c) For $x = 100$,

$$y = 2 - 3(100) = -298$$

**5** **Calculate value**

The point has $y = -302$, not $-298$, so it does not lie on the line.

**FINAL ANSWER**

$(9, -1.5)$, gradient $-3$, and no.

4MA1/2H JANUARY 2020

# Question 19

5 marks | Linear Graphs

**19** (a) Write down an equation of a line that is parallel to the line with equation $y = 7 - 4x$

.....

(1)

The line **L** passes through the points with coordinates (–3, 1) and (2, –2)

(b) Find an equation of the line that is perpendicular to **L** and passes through the point with coordinates (–6, 4)
Give your answer in the form $ax + by + c = 0$ where $a$, $b$ and $c$ are integers.

.....

(4)

**(Total for Question 19 is 5 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2H January 2020

**1** **(a)**

(a)

$$y = 7 - 4x$$

**2** **A parallel line has the**

A parallel line has the same gradient, $-4$. One possible answer is

$$y = -4x$$

**3** **(b) The gradient of line**

(b) The gradient of line $L$ through $(-3, 1)$ and $(2, -2)$ is

$$\frac{-2-1}{2-(-3)} = -\frac{3}{5}$$

**4** **A perpendicular line has gradient**

A perpendicular line has gradient $\frac{5}{3}$.

**5** **Evaluate fraction**

Through $(-6, 4)$:

$$y - 4 = \frac{5}{3}(x + 6)$$

**6** **Multiply by 3**

Multiply by 3:

$$3y - 12 = 5x + 30$$

$$5x - 3y + 42 = 0$$

---

✓ **FINAL ANSWER**

For example $y = -4x$, and $5x - 3y + 42 = 0$.

4MA1/1H JANUARY 2020

# Question 22

6 marks | Linear Graphs

**22** Triangle $HJK$ is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

$H$ is the point with coordinates $(-4, 1)$
$J$ is the point with coordinates $(j, 15)$ where $j < 0$
$K$ is the point with coordinates $(6, k)$

$M$ is the midpoint of $JK$.
The gradient of $HM$ is 2

Find the value of $j$ and the value of $k$.

$j =$ .....

$k =$ .....

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H January 2020

### 1 Find the midpoint

Since $M$ is the midpoint of $JK$,

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2}\right)$$

### 2 Find the gradient

The gradient of $HM$ is 2, and $H = (-4, 1)$.

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

$$\frac{k+13}{j+14} = 2$$

$$k = 2j + 15$$

**3** **Simplify surd**

Also,

$$JK = \sqrt{80}$$

**4** **Calculate value**

So

$$(6 - j)^2 + (k - 15)^2 = 80$$

**5** **Substitute**

Substitute $k = 2j + 15$:

$$(6 - j)^2 + (2j)^2 = 80$$

$$j^2 - 12j + 36 + 4j^2 = 80$$

$$5j^2 - 12j - 44 = 0$$

$$(5j + 10)(j - 4.4) = 0$$

**6** **Solve inequality**

Since $j < 0$,

$$j = -2$$

**7** **Calculate value**

Then

$$k = 2(-2) + 15 = 11$$

✓ **FINAL ANSWER**

$j = -2,\ k = 11$

4MA1/2H JANUARY 2020

# Question 22

5 marks | Linear Graphs

**22** The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points $A$ and $B$.

Find the coordinates of $A$ and $B$.
Show clear algebraic working.

(..... , .....)

(..... , .....)

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H January 2020

**1 The line is**

The line is

$$y = x + 2$$

**2 Substitute this into the circle**

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

$$x^2 + (x+2)^2 - 2(x+2) = 24$$

$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x+4)(x-3) = 0$$

**3** **Calculate value**

So

$$x = -4 \quad \text{or} \quad x = 3$$

---

**4** **Calculate value**

Using $y = x + 2$:

$$x = -4 \Rightarrow y = -2$$

$$x = 3 \Rightarrow y = 5$$

---

✓ **FINAL ANSWER**

$(-4, -2)$ and $(3, 5)$

4MA1/2HR JANUARY 2020

# Question 24

5 marks | Linear Graphs

24 $\mathbf{L}_1$ and $\mathbf{L}_2$ are two straight lines.
The origin of the coordinate axes is $O$.

$\mathbf{L}_1$ has equation $5x + 10y = 8$
$\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point with coordinates (8, 6)

$\mathbf{L}_2$ crosses the $x$-axis at the point $A$.
$\mathbf{L}_2$ intersects the straight line with equation $x = -3$ at the point $B$.

Find the area of triangle $AOB$.
Show your working clearly.

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2HR January 2020

**1 Evaluate fraction**

For $L_1$,

$$5x + 10y = 8$$

$$10y = -5x + 8$$

$$y = -\frac{1}{2}x + \frac{4}{5}$$

**2 Find the gradient**

So the gradient of $L_1$ is $-\frac{1}{2}$.

**3 Find the gradient**

Since $L_2$ is perpendicular to $L_1$, the gradient of $L_2$ is $2$.

**4 Calculate value**

Using the point $(8, 6)$:

$$y - 6 = 2(x - 8)$$

$$y = 2x - 10$$

**5** **Calculate value**

At $A$, the line crosses the $x$-axis, so $y = 0$:

$$0 = 2x - 10$$

$$x = 5$$

---

**6** **Calculate value**

So $A = (5, 0)$.

---

**7** **Calculate value**

At $B$, $x = -3$:

$$y = 2(-3) - 10 = -16$$

---

**8** **Calculate value**

So $B = (-3, -16)$.

---

**9** **Triangle has base and perpendicular**

Triangle $AOB$ has base $OA = 5$ and perpendicular height $16$.

$$\text{Area} = \frac{1}{2} \times 5 \times 16 = 40$$

---

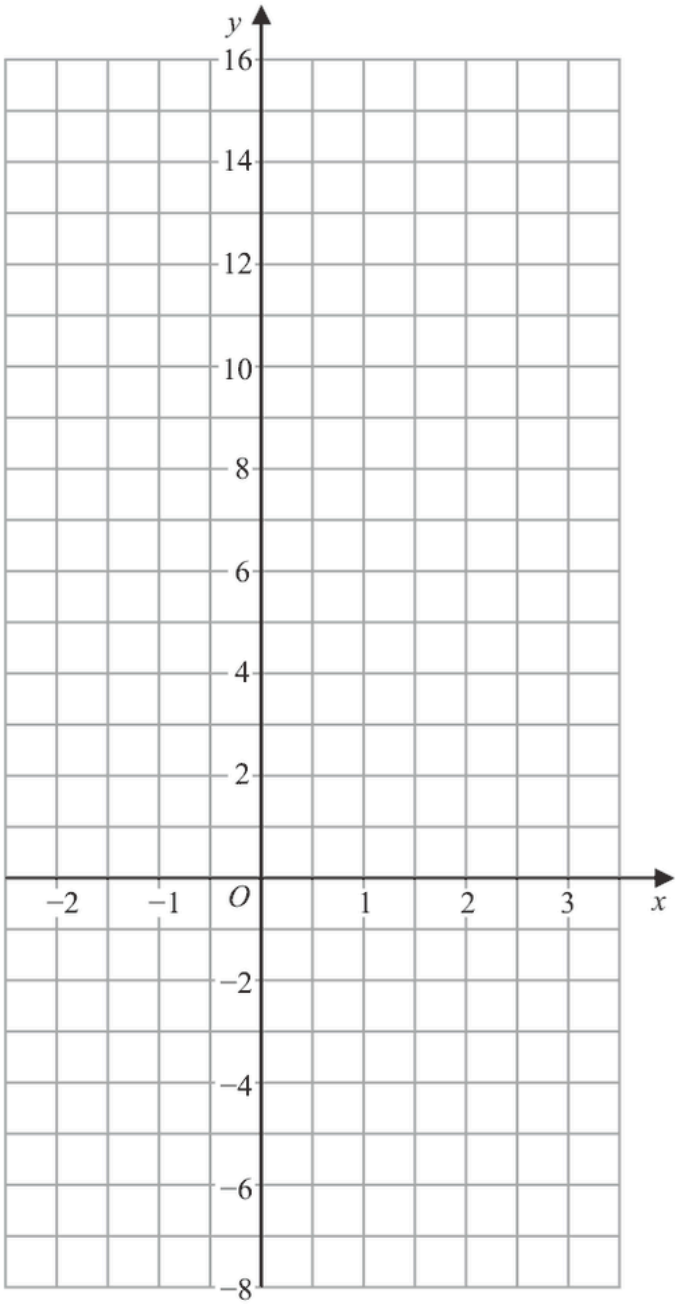
✓ **FINAL ANSWER**

$40$

# Question 3

3 marks | Linear Graphs

**3** On the grid, draw the graph of $y = 7 - 4x$ for values of $x$ from $-2$ to 3




**(Total for Question 3 is 3 marks)**

## 1 Read the graph

$$y = 7 - 4x$$

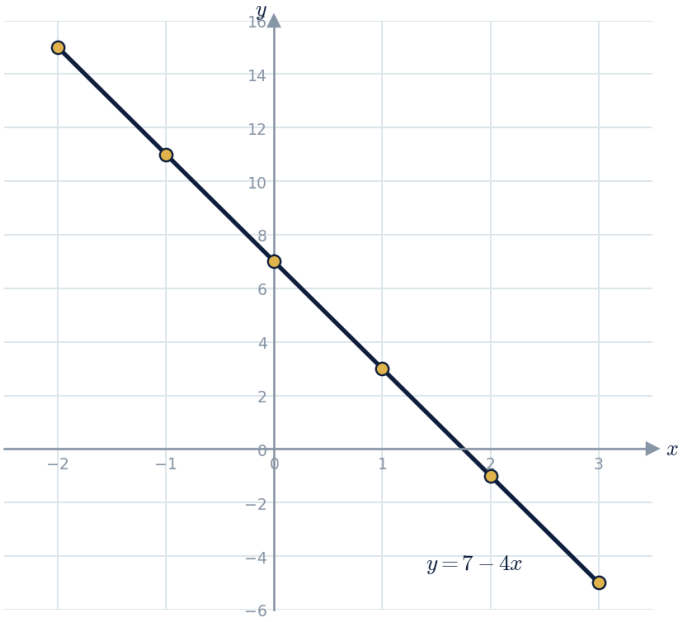
## 2 Useful points are

Useful points are:

$$(-2, 15), \quad (0, 7), \quad (1, 3), \quad (2, -1), \quad (3, -5)$$

## 3 Plot these points and join

Plot these points and join them with a straight line.




**FINAL ANSWER**

The graph is the straight line $y = 7 - 4x$.

4MA1/1HR MAY/NOVEMBER 2020

# Question 6

2 marks | Linear Graphs

6 Point $A$ has coordinates $(-3, 11)$
Point $B$ has coordinates $(47, b)$
The midpoint of $AB$ has coordinates $(a, -19)$

Find the value of $a$ and the value of $b$.

$a =$ .....

$b =$ .....

**(Total for Question 6 is 2 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1HR May/November 2020

**1 Calculate value**

$$A = (3, 11), \qquad B = (47, b)$$

**2 The midpoint is**

The midpoint is $(a, 19)$.

**3 Evaluate fraction**

For the x-coordinate:

$$a = \frac{3 + 47}{2} = 25$$

**4 Evaluate fraction**

For the y-coordinate:

$$19 = \frac{11 + b}{2}$$

$$38 = 11 + b$$

$$b = 27$$

**FINAL ANSWER**

$a = 25, b = 27$.

4MA1/1H MAY/NOVEMBER 2020

# Question 22

4 marks | Linear Graphs

**22** $ABCD$ is a rhombus.

The diagonals, $AC$ and $BD$, intersect at the point $M$.
The coordinates of $M$ are $(6, -11)$

The points $A$ and $C$ both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through $B$ and $D$ intersects the $y$-axis.

(..... , .....)

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H May/November 2020

**1 Find the gradient**

In a rhombus, the diagonals are perpendicular and bisect each other.

**2 The diagonal lies on**

The diagonal $AC$ lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

**3 Find the gradient**

So the gradient of $AC$ is

$$-\frac{7}{2}$$

**4 Find the gradient**

Therefore the gradient of $BD$ is

$$\frac{2}{7}$$

**5** **Calculate value**

The line $BD$ passes through

$$M = (6, -11)$$

**6** **Evaluate fraction**

So

$$y + 11 = \frac{2}{7}(x - 6)$$

**7** **To find where this line**

To find where this line intersects the $y$-axis, put $x = 0$:

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

✓ **FINAL ANSWER**

$$\left(0, -\frac{89}{7}\right)$$

4MA1/1H JANUARY 2021

# Question 13

3 marks | Linear Graphs

**13** Point $A$ has coordinates (5, 8)
Point $B$ has coordinates (9, –4)

(a) Work out the gradient of $AB$.

.....

(2)

The straight line **L** has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to **L**.

.....

(1)

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H January 2021

**1** **Find the gradient**

(a) The gradient of $AB$, where $A(5, 8)$ and $B(9, -4)$, is

$$\frac{-4-8}{9-5} = \frac{-12}{4} = -3$$

**2** **Find the gradient**

(b) A line perpendicular to a line with gradient $-3$ has gradient

$$\frac{1}{3}$$

**FINAL ANSWER**

Gradient of $AB = -3$, perpendicular gradient $\frac{1}{3}$.

4MA1/2H JANUARY 2021

# Question 18

7 marks | Linear Graphs

**18** A rectangle $ABCD$ is to be drawn on a centimetre grid such that

$A$ has coordinates $(-4, -2)$
$B$ has coordinates $(1, 10)$
$C$ has coordinates $(19, a)$
$D$ has coordinates $(b, c)$

(a) Work out the value of $a$, the value of $b$ and the value of $c$.

$a =$ .....

$b =$ .....

$c =$ .....

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

**(Total for Question 18 is 7 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2H January 2021

**1 Calculate value**

$$A = (-4, -2), \quad B = (1, 10), \quad C = (19, a), \quad D = (b, c)$$

**2 The vector is**

The vector $AB$ is

$$(1 - (-4), 10 - (-2)) = (5, 12)$$

**3 The vector is**

The vector $BC$ is

$$(19 - 1, a - 10) = (18, a - 10)$$

**4** **Adjacent sides of a rectangle**

Adjacent sides of a rectangle are perpendicular, so their dot product is zero:

$$5(18) + 12(a - 10) = 0$$

$$90 + 12a - 120 = 0$$

$$a = \frac{5}{2}$$

---

**5** **Calculate value**

Now

$$BC = (18, -7.5)$$

---

**6** **Calculate value**

So

$$D = A + BC = (-4, -2) + (18, -7.5) = (14, -9.5)$$

$$b = 14, \qquad c = -9.5$$

---

**7** **The side lengths are**

The side lengths are

$$AB = \sqrt{5^2 + 12^2} = 13$$

$$BC = \sqrt{18^2 + (-7.5)^2} = 19.5$$

---

**8** **Perimeter**

Perimeter:

$$2(13 + 19.5) = 65$$

---

✓ **FINAL ANSWER**

$a = \frac{5}{2}$, $b = 14$, $c = -9.5$, and perimeter $65\ \text{cm}$.

4MA1/1HR JANUARY 2021

# Question 22

5 marks | Linear Graphs

**22** $ABC$ is an isosceles triangle with $AB = AC$.

$B$ is the point with coordinates $(-1, 5)$
$C$ is the point with coordinates $(2, 10)$
$M$ is the midpoint of $BC$.

Find an equation of the line through the points $A$ and $M$.
Give your answer in the form $py + qx = r$ where $p$, $q$ and $r$ are integers.

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1HR January 2021

**1 Find the gradient**

Since $AB = AC$, point $A$ lies on the perpendicular bisector of $BC$.

**2 The midpoint of is**

The midpoint of $BC$ is

$$M = \left(\frac{-1+2}{2}, \frac{5+10}{2}\right)$$

$$M = \left(\frac{1}{2}, \frac{15}{2}\right)$$

**3 The gradient of is**

The gradient of $BC$ is

$$\frac{10-5}{2-(-1)} = \frac{5}{3}$$

**4 Find the gradient**

So the gradient of the perpendicular bisector is

$$-\frac{3}{5}$$

**5** **Find line equation**

Use the line through $M$:

$$y - \frac{15}{2} = -\frac{3}{5}\left(x - \frac{1}{2}\right)$$

**6** **Multiply by 10**

Multiply by 10:

$$10y - 75 = -6x + 3$$

$$10y + 6x = 78$$

**7** **Divide by 2**

Divide by 2:

$$5y + 3x = 39$$

✓ **FINAL ANSWER**

$5y + 3x = 39$

# Question 17

4 marks | Linear Graphs

**17** The straight line **L** passes through the points (4, −1) and (6, 4)

The straight line **M** is perpendicular to **L** and intersects the $y$-axis at the point (0, 8)

Find the coordinates of the point where **M** intersects the $x$-axis.

(..... , .....)

**(Total for Question 17 is 4 marks)**

**1** **Use Coordinate geometry tag correct**

Line $L$ passes through $(4, -1)$ and $(6, 4)$.

**2** **Its gradient is**

Its gradient is

$$\frac{4-(-1)}{6-4} = \frac{5}{2}$$

**3** **Find the gradient**

Line $M$ is perpendicular to $L$, so its gradient is

$$-\frac{2}{5}$$

**4** **Line intersects the y axis**

Line $M$ intersects the y-axis at $(0, 8)$, so

$$y = -\frac{2}{5}x + 8$$

**5** **Evaluate fraction**

At the x-axis, $y = 0$:

$$0 = -\frac{2}{5}x + 8$$

$$\frac{2}{5}x = 8$$

$$x = 20$$

✓ **FINAL ANSWER**

$(20, 0)$.

4MA1/2H NOVEMBER 2021

# Question 19

6 marks | Linear Graphs

**19** The straight line **L** has equation $x - y = 3$
The curve **C** has equation $3x^2 - y^2 + xy = 9$

**L** and **C** intersect at the points $P$ and $Q$.

Find the coordinates of the midpoint of $PQ$.
Show clear algebraic working.

(..... , .....)

**(Total for Question 19 is 6 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2H November 2021

**1 Line is**

Line $L$ is

$$x - y = 3$$

**2 Calculate value**

So

$$y = x - 3$$

**3 The curve is**

The curve is

$$3x - y + y^2 = 9$$

**4 Substitute**

Substitute $y = x - 3$:

$$3x - (x - 3) + (x - 3)^2 = 9$$

$$2x + 3 + x^2 - 6x + 9 = 9$$

$$x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0$$

**5** **Calculate value**

So $x = 1$ or $x = 3$.

**6** **Calculate value**

If $x = 1$, then $y = -2$. If $x = 3$, then $y = 0$.

**7** **The midpoint of and is**

The midpoint of $(1, -2)$ and $(3, 0)$ is

$$\left(\frac{1+3}{2}, \frac{-2+0}{2}\right) = (2, -1)$$

**FINAL ANSWER**

$(2, -1)$.

4MA1/1H NOVEMBER 2021

# Question 20

7 marks | Linear Graphs

**20** The straight line **L** passes through point $A$ $(-6, 2)$ and point $B$ $(5, 3)$
The straight line **M** is perpendicular to **L** and passes through the midpoint of $A$ and $B$.
The line **M** intersects the line $x = -1$ at point $C$.

Calculate the area of triangle $ABC$.

**(Total for Question 20 is 7 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H November 2021

**1** **Calculate value**

$$A = (-6, 2), \qquad B = (5, 3)$$

**2** **The midpoint of is**

The midpoint of $AB$ is

$$\left(\frac{-6+5}{2}, \frac{2+3}{2}\right) = \left(-\frac{1}{2}, \frac{5}{2}\right)$$

**3** **The gradient of is**

The gradient of $AB$ is

$$\frac{3-2}{5-(-6)} = \frac{1}{11}$$

**4** **Find the gradient**

So the perpendicular gradient is

$$-11$$

**5** **Evaluate fraction**

Line $M$ passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

$$y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$$

**6** **Evaluate fraction**

At point $C$, $x = -1$:

$$y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$$

$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

---

**7** **Calculate value**

So

$$C = (-1, 8)$$

---

**8** **Use vectors**

Now use vectors from $A$:

$$\vec{AB} = (11, 1), \qquad \vec{AC} = (5, 6)$$

---

**9** **Area of triangle is**

Area of triangle $ABC$ is

$$\frac{1}{2}|11(6) - 1(5)|$$

$$= \frac{1}{2}(61)$$

$$= 30.5$$

---

✓ **FINAL ANSWER**

$30.5$

4MA1/1H JANUARY 2022

# Question 10

3 marks | Linear Graphs

**10** The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

.....

**(2)**

(b) Find the coordinates of the point where **L** crosses the $y$-axis.

(..... , .....)

**(1)**

**(Total for Question 10 is 3 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1H January 2022

**1** **The line is**

The line is

$$2y + 7x = 10$$

$$2y = -7x + 10$$

$$y = -\frac{7}{2}x + 5$$

**2** **Find the gradient**

So the gradient is

$$-\frac{7}{2}$$

**3** **Evaluate fraction**

For the x-axis, $y = 0$:

$$2(0) + 7x = 10$$

$$x = \frac{10}{7}$$

**FINAL ANSWER**

Gradient $-\frac{7}{2}$, and x-axis intersection $\left(\frac{10}{7}, 0\right)$.

4MA1/2HR JANUARY 2022

# Question 18

5 marks | Linear Graphs

**18** The line with equation $2y = x + 1$ intersects the curve with equation $3y^2 + 7y + 16 = x^2 - x$ at the points $A$ and $B$

Find the coordinates of $A$ and the coordinates of $B$
Show clear algebraic working.

(..... , .....) and (..... , .....)

**(Total for Question 18 is 5 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2HR January 2022

### 1 The line is

The line is

$$2y = x + 1$$

### 2 Calculate value

So

$$x = 2y - 1$$

### 3 Substitute into the curve

Substitute into the curve

$$3y^2 + 7y + 16 = x^2 - x$$

$$3y^2 + 7y + 16 = (2y - 1)^2 - (2y - 1)$$

$$3y^2 + 7y + 16 = 4y^2 - 6y + 2$$

$$y^2 - 13y - 14 = 0$$

$$(y - 14)(y + 1) = 0$$

**4** **Calculate value**

So

$$y = 14 \quad \text{or} \quad y = -1$$

**5** **Calculate value**

If $y = 14$, then $x = 27$.

**6** **Calculate value**

If $y = -1$, then $x = -3$.

✓ **FINAL ANSWER**

$(27, 14)$ and $(-3, -1)$.

4MA1/2HR JANUARY 2022

# Question 22

5 marks | Linear Graphs

**22** $ABC$ is a triangle in which angle $ABC = 90°$

$p$ and $q$ are integers such that

the coordinates of $A$ are $(p, 10)$
the coordinates of $B$ are $(-1, -5)$
the coordinates of $C$ are $(8, q)$

Given that the gradient of $AC$ is $-\frac{6}{7}$

work out the value of $p$ and the value of $q$

$p =$ .....

$q =$ .....

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2HR January 2022

**1 The gradient of is**

The gradient of $AC$ is $-\frac{6}{7}$.

$$\frac{q-10}{8-p} = -\frac{6}{7}$$

$$7(q-10) = -6(8-p)$$

$$7q - 70 = -48 + 6p$$

$$6p - 7q = -22$$

**2** **Find the gradient**

Since $\angle ABC = 90°$, lines $AB$ and $BC$ are perpendicular.

$$\text{gradient of } AB = \frac{10-(-5)}{p-(-1)} = \frac{15}{p+1}$$

$$\text{gradient of } BC = \frac{q-(-5)}{8-(-1)} = \frac{q+5}{9}$$

---

**3** **The product of perpendicular gradients**

The product of perpendicular gradients is $-1$:

$$\frac{15}{p+1} \times \frac{q+5}{9} = -1$$

$$15(q+5) = -9(p+1)$$

$$5(q+5) = -3(p+1)$$

$$3p + 5q = -28$$

---

**4** **Solve**

Now solve

$$6p - 7q = -22$$

$$3p + 5q = -28$$

---

**5** **Double the second equation**

Double the second equation:

$$6p + 10q = -56$$

---

**6** **Subtract**

Subtract:

$$17q = -34$$

$$q = -2$$

---

**7** **Calculate value**

Then

$$3p + 5(-2) = -28$$

$$3p = -18$$

$$p = -6$$

---

✓ **FINAL ANSWER**

$p = -6,\ q = -2$

4MA1/1H JUNE 2022

# Question 10

4 marks | Linear Graphs

**10** Two circles, $C_1$ and $C_2$, are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle $C_1$ is at the point with coordinates $(-1, 3)$ and the radius of $C_1$ is 13 cm.

The centre of circle $C_2$ is at the point with coordinates $(7, 18)$ and the radius of $C_2$ is 6 cm.

(a) Work out the distance between the centre of $C_1$ and the centre of $C_2$

..... cm

(3)

(b) Explain why circle $C_1$ intersects circle $C_2$

.....

.....

(1)

**(Total for Question 10 is 4 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1H June 2022

### 1 The centres are

The centres are

$$C_1 = (-1, 3), \qquad C_2 = (7, 18)$$

### 2 The distance between the centres

The distance between the centres is

$$\sqrt{(7-(-1))^2 + (18-3)^2}$$

$$= \sqrt{8^2 + 15^2} = \sqrt{289} = 17$$

### 3 Calculate value

The radii are $3$ cm and $15$ cm.

$$3 + 15 = 18$$

4 **Use Coordinate geometry tag correct**

Since the distance between the centres is $17$ cm, which is less than the sum of the radii, the circles intersect.

---

✓ **FINAL ANSWER**

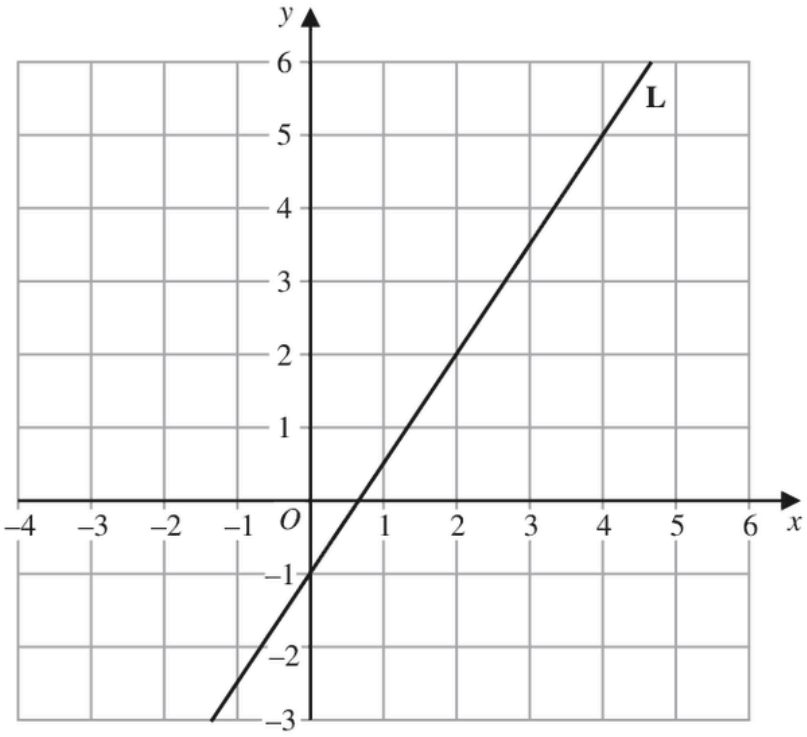
$17\ \text{cm}$, and the circles intersect because $17 < 18$.

4MA1/1HR JUNE 2022

# Question 10

3 marks | Linear Graphs

**10** Line **L** is drawn on the grid.




Find an equation for **L**
Give your answer in the form $y = mx + c$

**(Total for Question 10 is 3 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1HR June 2022

**1** **Solve equation**

From the graph, the line crosses the y-axis at $-1$, so

$$c = -1$$

**2** **Find the gradient**

Using two points on the line, for example $(0, -1)$ and $(2, 2)$, the gradient is

$$\frac{2-(-1)}{2-0} = \frac{3}{2}$$

**3** **Solve equation**

So the equation is

$$y = \frac{3}{2}x - 1$$

---

✓ **FINAL ANSWER**

$y = \frac{3}{2}x - 1$.

4MA1/2HR JUNE 2022

# Question 20

4 marks | Linear Graphs

**20** The centre $O$ of a circle has coordinates $(4, 7)$

The point $A$, on the circle, has coordinates $(6, 11)$ and $AOP$ is a diameter of the circle.

Find an equation of the tangent to the circle at the point $P$

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2HR June 2022

**1 The centre of the circle**

The centre of the circle is

$$O = (4, 7)$$

**2 The point is**

The point $A$ is

$$A = (6, 11)$$

**3 Find the midpoint**

Since $AOP$ is a diameter, $O$ is the midpoint of $AP$.

**4 Let**

Let $P = (x, y)$.

$$\left(\frac{6+x}{2}, \frac{11+y}{2}\right) = (4, 7)$$

$$6 + x = 8, \qquad 11 + y = 14$$

$$P = (2, 3)$$

**5 The gradient of is**

The gradient of $OP$ is

$$\frac{3-7}{2-4} = 2$$

**6** **Use circle theorem**

The tangent is perpendicular to the radius, so its gradient is

$$-\frac{1}{2}$$

---

**7** **Evaluate fraction**

Using point $P = (2, 3)$,

$$y - 3 = -\frac{1}{2}(x - 2)$$

$$y = -\frac{1}{2}x + 4$$

---

✓ **FINAL ANSWER**

$$y = -\frac{1}{2}x + 4$$

4MA1/2H JUNE 2022

# Question 22

7 marks | Linear Graphs

**22** $ABCD$ is a kite, with diagonals $AC$ and $BD$, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

$A$ is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point $M$ with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the $x$ coordinate of $B$ is positive,

find the coordinates of the points $B$ and $D$.

(..... , .....)

(..... , .....)

**(Total for Question 22 is 7 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H June 2022

**1 The midpoint of is**

The midpoint of $BD$ is $M = (0, 2)$.

**2 Find the gradient**

Since $A = (-3, 4)$, the gradient of $AM$ is

$$\frac{2-4}{0-(-3)} = -\frac{2}{3}$$

**3 Find the gradient**

For the kite, diagonal $AC$ is perpendicular to $BD$, so the gradient of $BD$ is

$$\frac{3}{2}$$

**4 Use direction vector along**

Use direction vector $(2, 3)$ along $BD$.

**5** **Let**

Let

$$B = (2t, 2 + 3t)$$

**6** **Calculate value**

and

$$D = (-2t, 2 - 3t)$$

**7** **Calculate value**

Since $AB = 6.5$,

$$(2t - (-3))^2 + (2 + 3t - 4)^2 = 6.5^2$$

$$(2t + 3)^2 + (3t - 2)^2 = 42.25$$

$$4t^2 + 12t + 9 + 9t^2 - 12t + 4 = 42.25$$

$$13t^2 + 13 = 42.25$$

$$13t^2 = 29.25$$

$$t^2 = 2.25$$

$$t = 1.5$$

**8** **Use**

The $x$-coordinate of $B$ is positive, so use $t = 1.5$.

$$B = (3, 6.5)$$

$$D = (-3, -2.5)$$

✓ **FINAL ANSWER**

$B = (3, 6.5),\ D = (-3, -2.5)$

4MA1/1H MAY 2023

# Question 13

3 marks | Linear Graphs

**13** $A$ is the point with coordinates $(-5, 12)$

$B$ is the point with coordinates $(19, -48)$

Find an equation of the straight line that passes through the points $A$ and $B$

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H May 2023

**1 Calculate value**

$$A = (-5, 12), \qquad B = (19, -48)$$

**2 The gradient is**

The gradient is

$$\frac{-48 - 12}{19 - (-5)} = \frac{-60}{24} = -\frac{5}{2}$$

**3 Evaluate fraction**

Using point $(-5, 12)$:

$$y - 12 = -\frac{5}{2}(x + 5)$$

**4 Multiply by 2**

Multiply by 2:

$$2y - 24 = -5x - 25$$

$$5x + 2y + 1 = 0$$

**FINAL ANSWER**

$5x + 2y + 1 = 0$.

4MA1/2HR JUNE 2023

# Question 21

5 marks | Linear Graphs

**21** Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(..... , .....)

(..... , .....)

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2HR June 2023

**1** **The line is**

The line is

$$y - 2x = 1$$

**2** **Calculate value**

So

$$y = 2x + 1$$

**3** **Calculate value**

Substitute into

$$y^2 + xy = 7$$

$$(2x + 1)^2 + x(2x + 1) = 7$$

$$4x^2 + 4x + 1 + 2x^2 + x = 7$$

$$6x^2 + 5x - 6 = 0$$

**4** **Factorise**

Factorise:

$$(3x - 2)(2x + 3) = 0$$

**5** **Evaluate fraction**

So

$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

---

**6** **Evaluate fraction**

Using $y = 2x + 1$:

$$x = \frac{2}{3} \Rightarrow y = \frac{7}{3}$$

$$x = -\frac{3}{2} \Rightarrow y = -2$$

---

✓ **FINAL ANSWER**

$\left(\frac{2}{3}, \frac{7}{3}\right)$ and $\left(-\frac{3}{2}, -2\right)$

4MA1/2H JUNE 2023

# Question 24

6 marks | Linear Graphs

**24** $ABCD$ is a kite with $AB = AD$ and $CB = CD$

$A$ is the point with coordinates $(-2, 10)$

$B$ is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

$C$ is the point with coordinates $(4, -5)$

Work out the coordinates of $D$

(..... , .....)

**(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2H June 2023

**1 Calculate value**

In the kite, $AB = AD$ and $CB = CD$, so $AC$ is the line of symmetry.

**2 Apply transformation**

Therefore $D$ is the reflection of $B$ in the line $AC$.

$$A = (-2, 10), \qquad B = \left(-\frac{27}{5}, 4\right), \qquad C = (4, -5)$$

**3 A direction vector for is**

A direction vector for $AC$ is

$$\begin{pmatrix} 4 - (-2) \\ -5 - 10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix}$$

**4 Use the simpler direction vector**

Use the simpler direction vector

$$\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

**5** **Find the gradient**

Let $H$ be the foot of the perpendicular from $B$ to $AC$.

$$\vec{AB} = \begin{pmatrix} -\frac{17}{5} \\ -6 \end{pmatrix}$$

**6** **The fraction along to is**

The fraction along $AC$ to $H$ is

$$\frac{\vec{AB} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix}}{2^2 + (-5)^2} = \frac{-\frac{34}{5} + 30}{29}$$

$$= \frac{\frac{116}{5}}{29} = \frac{4}{5}$$

**7** **Use matrices**

So

$$H = A + \frac{4}{5}\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

$$H = \left(-2 + \frac{8}{5}, 10 - 4\right)$$

$$H = \left(-\frac{2}{5}, 6\right)$$

**8** **Find the midpoint**

Since $H$ is the midpoint of $BD$,

$$D = 2H - B$$

$$D = \left(-\frac{4}{5}, 12\right) - \left(-\frac{27}{5}, 4\right)$$

$$D = \left(\frac{23}{5}, 8\right)$$

✓ **FINAL ANSWER**

$$D = \left(\frac{23}{5}, 8\right)$$

4MA1/2HR JUNE 2023

# Question 25

6 marks | Linear Graphs

**25** The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line $AB$ where $A$ is the point with coordinates $(j, 7)$ and $B$ is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line $AB$
Show clear algebraic working.

(..... , .....)

**(Total for Question 25 is 6 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/2HR June 2023

### 1 The perpendicular bisector is

The perpendicular bisector is

$$y - 2x = 7$$

### 2 Calculate value

so

$$y = 2x + 7$$

### 3 The gradient of the perpendicular

The gradient of the perpendicular bisector is $2$, so the gradient of $AB$ is

$$-\frac{1}{2}$$

**4** **Evaluate fraction**

Since $A = (j, 7)$ and $B = (6, k)$,

$$\frac{k-7}{6-j} = -\frac{1}{2}$$

$$2(k-7) = -(6-j)$$

$$j = 2k - 8$$

**5** **The midpoint of is**

The midpoint of $AB$ is

$$\left(\frac{j+6}{2}, \frac{7+k}{2}\right)$$

**6** **Substitute**

Substitute $j = 2k - 8$:

$$\left(k-1, \frac{k+7}{2}\right)$$

**7** **Find the midpoint**

The midpoint lies on $y = 2x + 7$, so

$$\frac{k+7}{2} = 2(k-1) + 7$$

$$k + 7 = 4k + 10$$

$$k = -1$$

**8** **Calculate value**

Then

$$j = 2(-1) - 8 = -10$$

**9** **Find the midpoint**

Therefore the midpoint is

$$\left(\frac{-10+6}{2}, \frac{7+(-1)}{2}\right) = (-2, 3)$$

**FINAL ANSWER**

$$\boxed{(-2, 3)}$$

4MA1/2H NOVEMBER 2023

# Question 9

2 marks | Linear Graphs

9 A straight line passes through the points with coordinates (0, –3) and (2, 0)

Find an equation of the line.

**(Total for Question 9 is 2 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2H November 2023

**1 Solve equation**

The line passes through $(0, -3)$ and $(2, 0)$.

**2 The gradient is**

The gradient is

$$\frac{0-(-3)}{2-0} = \frac{3}{2}$$

**3 Solve equation**

The y-intercept is $-3$, so

$$y = \frac{3}{2}x - 3$$

**FINAL ANSWER**

$y = \frac{3}{2}x - 3$.

4MA1/2H NOVEMBER 2023

# Question 21

5 marks | Linear Graphs

**21** The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points $A$ and $B$

Find the coordinates of $A$ and the coordinates of $B$
Show clear algebraic working.

(..... , .....)

(..... , .....)

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2H November 2023

**1 The line is**

The line is

$$x + 2y = 5$$

**2 Calculate value**

So

$$x = 5 - 2y$$

**3 Calculate value**

Substitute into

$$x^2 + 3y^2 = 13$$

$$(5 - 2y)^2 + 3y^2 = 13$$

$$25 - 20y + 4y^2 + 3y^2 = 13$$

$$7y^2 - 20y + 12 = 0$$

**4** **Factorise**

Factorise:

$$(7y - 6)(y - 2) = 0$$

**5** **Evaluate fraction**

So

$$y = \frac{6}{7} \quad \text{or} \quad y = 2$$

**6** **Evaluate fraction**

Using $x = 5 - 2y$:

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}$$

$$y = 2 \Rightarrow x = 1$$

✓ **FINAL ANSWER**

$\left(\frac{23}{7}, \frac{6}{7}\right)$ and $(1, 2)$

4MA1/1H NOVEMBER 2023

# Question 22

6 marks | Linear Graphs

22 [In this question 1 cm = 1 unit on the $x$-axis and 1 cm = 1 unit on the $y$-axis]

$P$ is a point on a circle with centre $(0, 0)$
The coordinates of $P$ are $(8, -10)$

The line **L** is the tangent to the circle at the point $P$
**L** crosses the $x$-axis at the point $Q$ and crosses the $y$-axis at the point $R$

Work out the length of $RQ$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H November 2023

**1 The centre of the circle**

The centre of the circle is $(0, 0)$, and

$$P = (8, -10)$$

**2 The gradient of is**

The gradient of $OP$ is

$$\frac{-10}{8} = -\frac{5}{4}$$

**3 Find the gradient**

So the gradient of the tangent is

$$\frac{4}{5}$$

**4 The tangent through is**

The tangent through $P$ is

$$y + 10 = \frac{4}{5}(x - 8)$$

$$y = \frac{4}{5}x - \frac{82}{5}$$

**5** **Evaluate fraction**

At the $x$-axis, $y = 0$:

$$0 = \frac{4}{5}x - \frac{82}{5}$$

$$x = \frac{41}{2}$$

---

**6** **Evaluate fraction**

So

$$Q = \left(\frac{41}{2}, 0\right)$$

---

**7** **Evaluate fraction**

At the $y$-axis, $x = 0$:

$$R = \left(0, -\frac{82}{5}\right)$$

---

**8** **Simplify surd**

Therefore

$$RQ = \sqrt{\left(\frac{41}{2}\right)^2 + \left(\frac{82}{5}\right)^2}$$

$$RQ = 26.252\ldots$$

---

✓ **FINAL ANSWER**

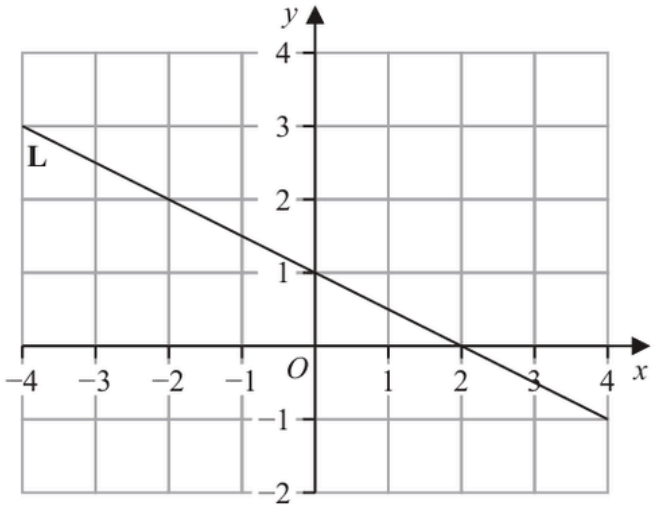
$26.3\ \text{cm}$

4MA1/1HR MAY 2024

# Question 9

3 marks | Linear Graphs

9 Line **L** is drawn on the grid.




Find an equation for **L**
Give your answer in the form $y = mx + c$

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1HR May 2024

**1 Solve equation**

From the graph, the line crosses the y-axis at $1$, so

$$c = 1$$

**2 It also crosses the x**

It also crosses the x-axis at $2$, so using $(0, 1)$ and $(2, 0)$,

$$m = \frac{0 - 1}{2 - 0} = -\frac{1}{2}$$

**3 Solve equation**

Therefore

$$y = -\frac{1}{2}x + 1$$

**FINAL ANSWER**

$y = -\frac{1}{2}x + 1$.

4MA1/2H JUNE 2024

# Question 18

2 marks | Linear Graphs

**18** The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

**(Total for Question 18 is 2 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2H June 2024

### 1 The line is

The line is

$$5y + 2x = 7$$

$$5y = -2x + 7$$

$$y = -\frac{2}{5}x + \frac{7}{5}$$

### 2 Find the gradient

So the gradient of line P is $-\frac{2}{5}$.

### 3 A line perpendicular to P

A line perpendicular to P has gradient

$$\frac{5}{2}$$

**FINAL ANSWER**

$\frac{5}{2}$.

4MA1/2HR JUNE 2024

# Question 21

4 marks | Linear Graphs

**21** $ABCD$ is a square.

The point $A$ has coordinates $(-5, 2)$
The point $B$ has coordinates $(3, 5)$

Find an equation of the line that passes through $B$ and $C$
Give your answer in the form $ax + by + c = 0$ where $a$, $b$ and $c$ are integers.

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2HR June 2024

**1 Calculate value**

$$A = (-5, 2), \qquad B = (3, 5)$$

**2 The gradient of is**

The gradient of $AB$ is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

**3 Find the gradient**

Since $ABCD$ is a square, $BC$ is perpendicular to $AB$.

**4 Find the gradient**

So the gradient of $BC$ is

$$-\frac{8}{3}$$

**5 The line through is**

The line through $B = (3, 5)$ is

$$y - 5 = -\frac{8}{3}(x - 3)$$

**6** **Multiply by 3**

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

---

✓ **FINAL ANSWER**

$8x + 3y - 39 = 0$

4MA1/2H JUNE 2024

# Question 22

5 marks | Linear Graphs

**22** The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(..... , .....) (..... , .....)

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H June 2024

### 1 The line is

The line is

$$x + y = 5$$

### 2 Calculate value

So

$$y = 5 - x$$

### 3 Substitute into the curve

Substitute into the curve:

$$2x^2 + 3y^2 = 210$$

$$2x^2 + 3(5 - x)^2 = 210$$

$$2x^2 + 3(x^2 - 10x + 25) = 210$$

$$5x^2 - 30x + 75 = 210$$

$$5x^2 - 30x - 135 = 0$$

**4** **Divide by 5**

Divide by 5:

$$x^2 - 6x - 27 = 0$$

**5** **Factorise**

Factorise:

$$(x - 9)(x + 3) = 0$$

**6** **Calculate value**

So

$$x = 9 \quad \text{or} \quad x = -3$$

**7** **Calculate value**

Using $y = 5 - x$:

$$x = 9 \Rightarrow y = -4$$

$$x = -3 \Rightarrow y = 8$$

✓ **FINAL ANSWER**

$(9, -4)$ and $(-3, 8)$

4MA1/2H NOVEMBER 2024

# Question 18

4 marks | Linear Graphs

**18** The straight line **L** is perpendicular to the straight line with equation $2x + y = 9$ and passes through the point with coordinates $(8, 11)$

Find an equation for **L**
Give your answer in the form $y = mx + c$

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2H November 2024

### 1 The given line is

The given line is

$$2x + y = 9$$

$$y = -2x + 9$$

### 2 Find the gradient

Its gradient is $-2$, so a perpendicular line has gradient $\frac{1}{2}$.

### 3 Find the gradient

Line $L$ passes through $(8, 11)$:

$$y - 11 = \frac{1}{2}(x - 8)$$

$$y = \frac{1}{2}x + 7$$

**FINAL ANSWER**

$y = \frac{1}{2}x + 7$.

4MA1/2HR JUNE 2025

# Question 10

2 marks | Linear Graphs

**10** A straight line, **L**, is parallel to the line with equation $y = 2 - 5x$
The line **L** passes through the point $(0, 6)$

Find an equation of the line **L**

**(Total for Question 10 is 2 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2HR June 2025

**1 The line has gradient**

The line

$$y = 2 - 5x$$

has gradient $-5$.

**2 A parallel line has the**

A parallel line has the same gradient, so line L has form

$$y = -5x + c$$

**3 Find the gradient**

Line L passes through $(0, 6)$, so

$$6 = c$$

**FINAL ANSWER**

$y = -5x + 6$.

4WM1/01R MAY 2025

# Question 23

5 marks | Linear Graphs

**23** A circle $C$ has centre $(3q, 4q)$ where $q$ is an integer.

The straight line with equation $y = -\frac{5}{4}x + \frac{21}{4}$ is the tangent to $C$ at the point $(p, p+3)$ where $p$ is an integer.

Work out the radius of $C$
Give your answer in the form $\sqrt{a}$ where $a$ is an integer.
Show your working clearly.

**(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Question 23 · 4WM1/01R May 2025

**1 The tangent point lies on**

The tangent point lies on the tangent line, so

$$p + 3 = -\frac{5}{4}p + \frac{21}{4}$$

$$p = 1$$

**2 Find the gradient**

The tangent gradient is $-\frac{5}{4}$, so the radius gradient is $\frac{4}{5}$.

**3 Use trigonometry**

Using centre $(3q, 4q)$ and tangent point $(1, 4)$,

$$\frac{4 - 4q}{1 - 3q} = \frac{4}{5}$$

$$q = 2$$

4

### Simplify surd

So the centre is $(6, 8)$. Therefore

$$r = \sqrt{(6-1)^2 + (8-4)^2} = \sqrt{41}$$

---

**FINAL ANSWER**

$\sqrt{41}$.

4MA1/2HR JUNE 2025

# Question 24

6 marks | Linear Graphs

**24** $PQ$ is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

$P$ has coordinates $(-5,\ a)$ and $Q$ has coordinates $(7,\ 3a)$ where $a > 0$

The length of $PQ$ is $4\sqrt{10}$ cm

Find an equation of the perpendicular bisector of $PQ$
Give your answer in the form $y = mx + c$ where $m$ and $c$ are integers.

**(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2HR June 2025

### 1 Calculate value

$$P = (-5, a), \qquad Q = (7, 3a)$$

### 2 Simplify surd

The length of $PQ$ is $4\sqrt{10}$, so

$$\sqrt{(7-(-5))^2 + (3a-a)^2} = 4\sqrt{10}$$

$$\sqrt{12^2 + (2a)^2} = 4\sqrt{10}$$

### 3 Square both sides

Square both sides:

$$144 + 4a^2 = 160$$

$$4a^2 = 16$$

$$a^2 = 4$$

### 4 Solve inequality

Since $a > 0$,

$$a = 2$$

**5** **Calculate value**

So

$$P = (-5, 2), \qquad Q = (7, 6)$$

**6** **The midpoint of is**

The midpoint of $PQ$ is

$$\left(\frac{-5+7}{2}, \frac{2+6}{2}\right) = (1, 4)$$

**7** **The gradient of is**

The gradient of $PQ$ is

$$\frac{6-2}{7-(-5)} = \frac{4}{12} = \frac{1}{3}$$

**8** **Find the gradient**

So the gradient of the perpendicular bisector is

$$-3$$

**9** **Calculate value**

Using point $(1, 4)$:

$$y - 4 = -3(x - 1)$$

$$y = -3x + 7$$

✓ **FINAL ANSWER**

$$\boxed{y = -3x + 7}$$

4WM1/01 MAY 2025

# Question 24

5 marks | Linear Graphs

**24** $PQRS$ is a square.

$PR$ is a diagonal of the square.

$P$ is the point with coordinates $(4, 7)$
$R$ is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points $Q$ and $S$
Give your answer in the form $ay = bx + c$ where $a$, $b$ and $c$ are integers.

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Question 24 · 4WM1/01 May 2025

**1 The diagonals of a square**

The diagonals of a square bisect each other and are perpendicular.

**2 Find the midpoint**

Midpoint of $P(4, 7)$ and $R(8, -5)$:

$$\left(\frac{4+8}{2}, \frac{7-5}{2}\right) = (6, 1)$$

**3 Find the gradient**

Gradient of $PR$:

$$\frac{-5-7}{8-4} = -3$$

**4 Find the gradient**

So the perpendicular gradient is $\frac{1}{3}$.

**5** **The line through and passes**

The line through $Q$ and $S$ passes through $(6, 1)$, so

$$y - 1 = \frac{1}{3}(x - 6)$$

$$3y - 3 = x - 6$$

$$3y = x - 3$$

---

✓ **FINAL ANSWER**

$3y = x - 3$

4MA1/1H JUNE 2025

# Question 25

5 marks | Linear Graphs

**25** $PQRS$ is a square.

$PR$ is a diagonal of the square.

$P$ is the point with coordinates $(4, 7)$
$R$ is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points $Q$ and $S$
Give your answer in the form $ay = bx + c$ where $a$, $b$ and $c$ are integers.

**(Total for Question 25 is 5 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/1H June 2025

**1 Calculate value**

$$P = (4, 7), \qquad R = (8, -5)$$

**2 The gradient of diagonal is**

The gradient of diagonal $PR$ is

$$\frac{-5 - 7}{8 - 4} = -3$$

**3 The other diagonal is perpendicular**

The other diagonal $QS$ is perpendicular to $PR$, so its gradient is

$$\frac{1}{3}$$

**4 The diagonals of a square**

The diagonals of a square bisect each other.

**5 The midpoint of is**

The midpoint of $PR$ is

$$\left(\frac{4 + 8}{2}, \frac{7 + (-5)}{2}\right) = (6, 1)$$

6 **Find the gradient**

So $QS$ passes through $(6, 1)$ with gradient $\frac{1}{3}$.

$$y - 1 = \frac{1}{3}(x - 6)$$

$$y = \frac{1}{3}x - 1$$

---

7 **Multiply by 3**

Multiply by 3:

$$3y = x - 3$$

---

✓ **FINAL ANSWER**

$3y = x - 3$

4MA1/2H NOVEMBER 2025

# Question 1

2 marks | Linear Graphs

**1** The point $P$ has coordinates $(3, 4)$
The point $Q$ has coordinates $(9, 16)$

$M$ is the midpoint of the line $PQ$

Find the coordinates of $M$

(..... , .....)

**(Total for Question 1 is 2 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2H November 2025

**1 Calculate value**

$$P = (-3, 0), \qquad Q = (9, 16)$$

**2 The midpoint is**

The midpoint is

$$\left(\frac{-3+9}{2}, \frac{0+16}{2}\right)$$

$$= (3, 8)$$

**FINAL ANSWER**

$(3, 8)$.

4MA1/1H NOVEMBER 2025

# Question 20

3 marks | Linear Graphs

**20** The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H November 2025

**1 Line has equation**

Line $L$ has equation

$$y = 4x + 7$$

**2 Find the gradient**

So its gradient is

$$4$$

**3 Find the gradient**

Line $M$ is perpendicular to $L$, so its gradient is

$$-\frac{1}{4}$$

**4 Evaluate fraction**

Line $M$ passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

**FINAL ANSWER**

$$y = -\frac{1}{4}x + 3$$

4WM1/01 NOVEMBER 2025

# Question 27

5 marks | Linear Graphs

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

$A$ has coordinates $(16, -9)$

$C$ has coordinates $(24, 15)$

Find an equation of the line $BD$
Give your answer in the form $ax + by + c = 0$ where $a$, $b$ and $c$ are integers.
Show your working clearly.

**(Total for Question 27 is 5 marks)**

## WORKED SOLUTION

Question 27 · 4WM1/01 November 2025

**1 Diagonals of a rhombus bisect**

Diagonals of a rhombus bisect each other at right angles.

**2 Find the midpoint**

Midpoint of $AC$:

$$\left(\frac{16+24}{2}, \frac{-9+15}{2}\right) = (20, 3)$$

**3 Find the gradient**

Gradient of $AC$:

$$\frac{15-(-9)}{24-16} = 3$$

**4 Find the gradient**

So gradient of $BD$ is $-\frac{1}{3}$.

$$y - 3 = -\frac{1}{3}(x - 20)$$

$$x + 3y - 29 = 0$$

**FINAL ANSWER**

$x + 3y - 29 = 0$.

4MA1/1H NOVEMBER 2025

# Question 28

6 marks | Linear Graphs

**28** The curve **C** has equation $x^2 + y^2 = d - 11x$ where $d$ is an integer.
The line **L** has equation $y = 3x + e$ where $e$ is an integer.

**C** and **L** intersect at the point $A$ and at the point $B$

The coordinates of $A$ are $(0.2, 2.6)$

Work out the coordinates of $B$
Show your working clearly.

(..... , .....)

**(Total for Question 28 is 6 marks)**

## WORKED SOLUTION

Question 28 · 4MA1/1H November 2025

**1 The line is**

The line is

$$y = 3x + e$$

**2 Point lies on the line**

Point $A = (0.2, 2.6)$ lies on the line.

$$2.6 = 3(0.2) + e$$

$$2.6 = 0.6 + e$$

$$e = 2$$

**3 Calculate value**

So

$$y = 3x + 2$$

**4** **The curve is**

The curve is

$$x^2 + y^2 = d - 11x$$

---

**5** **Use point**

Use point $A = (0.2, 2.6)$:

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d = 9$$

---

**6** **Substitute into the curve**

Now substitute $y = 3x + 2$ into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

---

**7** **One root is**

One root is $0.2 = \frac{1}{5}$.

---

**8** **The product of the roots**

The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

---

**9** **Evaluate fraction**

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

---

**10** **Evaluate fraction**

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

---

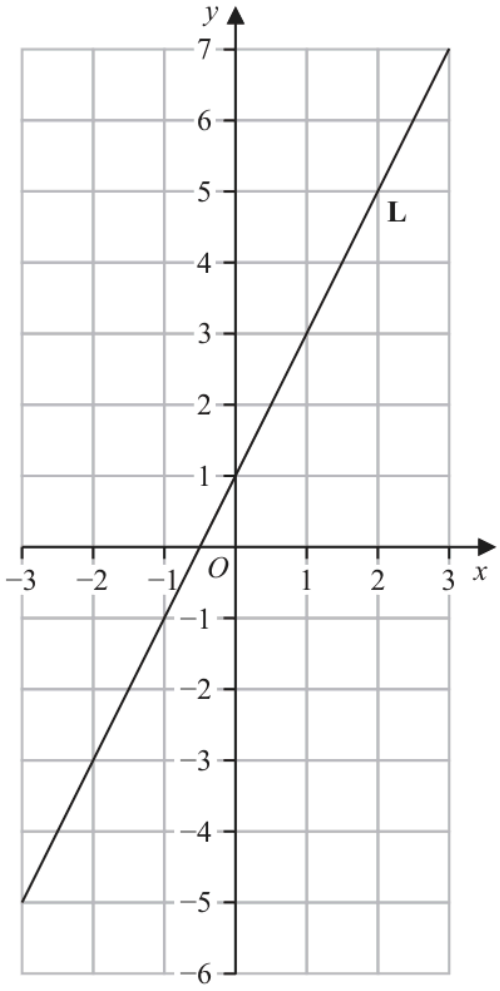
✓ **FINAL ANSWER**

$$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$$

# Question 10

3 marks | Linear Graphs

**10** The line **L** is drawn on a grid.




Find an equation for **L**

.....

**(Total for Question 10 is 3 marks)**

**1** **Read two points**

The line passes through $(0, 1)$ and $(2, 5)$.

**2** **Gradient and intercept**

$$m = \frac{5 - 1}{2 - 0} = 2, \qquad c = 1$$

**3** **Equation**

$$y = 2x + 1$$

✓ **FINAL ANSWER**

$y = 2x + 1$

4MA1/1HR JUNE 2026

# Question 13

3 marks | Linear Graphs

**13** The straight line **L** has equation $4y - 10x - 5 = 0$

(a) Find the gradient of **L**

.....

**(2)**

(b) Find the coordinates of the point where **L** crosses the $y$-axis.

(..... , .....)

**(1)**

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1HR June 2026

**1** **Part (a) | Step 1**

$$4y - 10x - 5 = 0 \implies 4y = 10x + 5 \implies y = \frac{5}{2}x + \frac{5}{4}.$$

Rearrange into the form y=mx+c.

**2** **Part (a) | Step 2**

The gradient is $\frac{5}{2}$.

The coefficient of x is the gradient.

**3** **Part (b) | Step 1**

$$x = 0 \implies 4y - 5 = 0 \implies y = \frac{5}{4}.$$

Every point on the y-axis has x-coordinate zero.

✓ **FINAL ANSWER**

(a) $\frac{5}{2}$ (b) $\left(0, \frac{5}{4}\right)$

4MA1/1H JUNE 2026

# Question 15

2 marks | Linear Graphs

**15** Point $A$ has coordinates $(-3, 4)$
Point $B$ has coordinates $(2, -6)$

Work out the gradient of the line $AB$

.....

**(Total for Question 15 is 2 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1H June 2026

### 1 Solution | Step 1

$$m = \frac{-6 - 4}{2 - (-3)} = \frac{-10}{5} = -2.$$

Use change in y divided by the corresponding change in x.

**FINAL ANSWER**

$-2$

# Question 17

4 marks | Linear Graphs

**17** The line **L** is perpendicular to the line with equation $y = 7 + 4x$

**L** passes through the point $A$ with coordinates $\left(0, \frac{19}{2}\right)$

**L** crosses the $x$-axis at the point $B$

Find the area of triangle $AOB$ where $O$ is the origin.
Show your working clearly.

.....

**(Total for Question 17 is 4 marks)**

**1** **Perpendicular gradient**

$y = 7 + 4x$ has gradient $4$, so $\mathbf{L}$ has gradient $-\frac{1}{4}$.

**2** **Equation of L**

$$y = -\tfrac{1}{4}x + \tfrac{19}{2}$$

**3** **Point B (y = 0)**

$$0 = -\tfrac{1}{4}x + \tfrac{19}{2} \Rightarrow x = 38,\ B = (38, 0)$$

**4** **Area of triangle AOB**

$$\tfrac{1}{2} \times 38 \times \tfrac{19}{2} = 180.5$$

✓ **FINAL ANSWER**

$180.5$



TOPIC

# Linear Equations

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 SPECIMEN

# Question 4

5 marks | Linear Equations

**4** (a) Expand and simplify $(n - 6)(n + 4)$

.....

**(2)**

(b) Solve $2x - 3 = \dfrac{3x - 5}{4}$

Show clear algebraic working.

$x =$ .....

**(3)**

**(Total for Question 4 is 5 marks)**

**1** **Expand the brackets**

$$(n-6)(n+4) = n^2 + 4n - 6n - 24$$

$$= n^2 - 2n - 24$$

**2** **Clear the denominator**

$$2x - 3 = \frac{3x - 5}{4}$$

$$4(2x - 3) = 3x - 5$$

**3** **Solve the equation**

$$8x - 12 = 3x - 5$$

$$5x = 7$$

$$x = \frac{7}{5}$$

✓ **FINAL ANSWER**

$n^2 - 2n - 24$, and $x = \frac{7}{5}$.

4MA1/1H JANUARY 2020

# Question 7

9 marks | Linear Equations

7 (a) Expand and simplify $(m - 8)(m + 5)$

.....

(2)

(b) Factorise fully $5y + 20y^2$

.....

(2)

(c) Simplify $(p^2 + 3)^0$

.....

(1)

(d) Solve $3(2x - 5) = \frac{9 - x}{2}$

Show clear algebraic working.

$x =$ .....

(4)

**(Total for Question 7 is 9 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1H January 2020

**1** **(a)**

(a) $(m-8)(m+5) = m^2 - 3m - 40$.

**2** **(b)**

(b) $5y + 20y^2 = 5y(1 + 4y)$.

**3** **(c)**

(c) $(p^2+3)^0 = 1$.

**4** **(d)**

(d)

$$3(2x-5)=\frac{9-x}{2}$$

$$12x-30=9-x$$

$$13x=39$$

---

**FINAL ANSWER**

(a) $m^2-3m-40$, (b) $5y(1+4y)$, (c) $1$, (d) $x=3$

4MA1/1HR MAY/NOVEMBER 2020

# Question 3

6 marks | Linear Equations

**3** (a) Simplify $h^7 \times h^2$

.....

(1)

$G = c^2 - 4c$

(b) Find the value of $G$ when $c = -5$

$G =$ .....

(2)

(c) Solve $\dfrac{5x - 3}{4} = 2x + 3$

Show clear algebraic working.

$x =$ .....

(3)

**(Total for Question 3 is 6 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1HR May/November 2020

**1** **(a)**

(a)

$$h^7 \times h^2 = h^{7+2} = h^9$$

**2** **(b)**

(b)

$$G = c^2 - 4c$$

**3** **Solve quadratic equation**

When $c = -5$:

$$G = (-5)^2 - 4(-5) = 25 + 20 = 45$$

**4** **(c)**

(c)

$$\frac{5x-3}{4}=2x+3$$

---

**5** **Multiply by 4**

Multiply by 4:

$$5x-3=8x+12$$

$$-3x=15$$

$$x=-5$$

---

**FINAL ANSWER**

$h^9$, $G=45$, and $x=-5$.

4MA1/1H JANUARY 2021

# Question 5

5 marks | Linear Equations

**5** (a) Factorise fully $15y^4 + 20uy^3$

.....

**(2)**

(b) Solve $4 - 3x = \dfrac{5 - 8x}{4}$

Show clear algebraic working.

$x =$ .....

**(3)**

**(Total for Question 5 is 5 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1H January 2021

**1** **(a)**

(a)

$$15y^4 + 20uy^3 = 5y^3(3y + 4u)$$

**2** **(b)**

(b)

$$4 - 3x = \frac{5 - 8x}{4}$$

$$16 - 12x = 5 - 8x$$

$$11 = 4x$$

**FINAL ANSWER**

(a) $5y^3(3y + 4u)$, (b) $x = \frac{11}{4}$

4MA1/2HR JANUARY 2022

# Question 3

5 marks | Linear Equations

3 (a) Solve $4y + 5 > 12$

.....

(2)

(b) Solve $6x - 5 = \frac{4x - 7}{2}$

Show clear algebraic working.

$x =$ .....

(3)

**(Total for Question 3 is 5 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2HR January 2022

**1** **(a)**

(a)

$$4y + 5 > 12$$

$$4y > 7$$

$$y > \frac{7}{4}$$

**2** **(b)**

(b)

$$6x - 5 = \frac{4x - 7}{2}$$

### 3 Multiply by 2

Multiply by 2:

$$12x - 10 = 4x - 7$$

$$8x = 3$$

$$x = \frac{3}{8}$$

---

**FINAL ANSWER**

$y > \frac{7}{4}$, and $x = \frac{3}{8}$.

4MA1/1H JUNE 2022

# Question 5

5 marks | Linear Equations

**5** (a) Expand and simplify $(n-6)(n+4)$

.....

(2)

(b) Solve $2x-3=\dfrac{3x-5}{4}$

Show clear algebraic working.

$x =$ .....

(3)

**(Total for Question 5 is 5 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1H June 2022

**1** **(a)**

(a)

$$(n-6)(n+4)=n^2-2n-24$$

**2** **(b)**

(b)

$$2x-3=\frac{3x-5}{4}$$

$$8x-12=3x-5$$

$$5x=7$$

✓ **FINAL ANSWER**

(a) $n^2-2n-24$, (b) $x=\frac{7}{5}$

4MA1/2HR JUNE 2022

# Question 6

8 marks | Linear Equations

**6** (a) Solve $p = \dfrac{3p - 5}{10}$

Show clear algebraic working.

$p =$ .....

(3)

(b) Simplify $a^0$ where $a > 0$

.....

(1)

(c) Simplify fully $\dfrac{3xy^3}{6x^2y}$

.....

(2)

(d) Factorise fully $10c^3d^2 + 15cd^4$

.....

(2)

**(Total for Question 6 is 8 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2HR June 2022

**1** **(a)**

(a)

$$p = \frac{3p - 5}{10}$$

$$10p = 3p - 5$$

$$7p = -5$$

$$p = -\frac{5}{7}$$

**2** **(b)**

(b) $a^0 = 1$.

---

**3** **(c)**

(c)

$$\frac{3xy^3}{6x^2y} = \frac{y^2}{2x}$$

---

**4** **(d)**

(d)

$$10c^3d^2 + 15cd^4 = 5cd^2(2c^2 + 3d^2)$$

---

✓ **FINAL ANSWER**

(a) $p = -\frac{5}{7}$, (b) $1$, (c) $\frac{y^2}{2x}$, (d) $5cd^2(2c^2 + 3d^2)$

4MA1/2HR JUNE 2024

# Question 6

8 marks | Linear Equations

6 (a) Solve $\frac{2f}{3} = 4f - 17$

Show clear algebraic working.

$f =$ .....

(3)

(b) Simplify $(e + 12)^0$ where $e > 0$

.....

(1)

(c) Simplify fully $\frac{12a^4h^6}{4ah^2}$

.....

(2)

(d) Factorise fully $20x^5y + 12x^3y^4$

.....

(2)

**(Total for Question 6 is 8 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2HR June 2024

**1 (a)**

(a)

$$\frac{2f}{3} = 4f - 17$$

$$2f = 12f - 51$$

$$10f = 51$$

$$f = \frac{51}{10}$$

**2** **(b)**

(b) $(e+12)^0 = 1$.

---

**3** **(c)**

(c)

$$\frac{12a^4h^6}{4ah^2} = 3a^3h^4$$

---

**4** **(d)**

(d)

$$20x^5y + 12x^3y^4 = 4x^3y(5x^2 + 3y^3)$$

---

✓ **FINAL ANSWER**

$f = \frac{51}{10}$, $1$, $3a^3h^4$, $4x^3y(5x^2 + 3y^3)$

4MA1/1H MAY 2024

# Question 6

5 marks | Linear Equations

**6** (a) Expand and simplify $(m + 5)(m - 8)$

.....

(2)

(b) Solve $3n - 4 = \dfrac{5n + 6}{3}$

Show clear algebraic working.

$n =$ .....

(3)

**(Total for Question 6 is 5 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1H May 2024

**1** **(a)**

(a)

$$(m+5)(m-8) = m^2 - 3m - 40$$

**2** **(b)**

(b)

$$3n - 4 = \frac{5n+6}{3}$$

$$9n - 12 = 5n + 6$$

$$4n = 18$$

**FINAL ANSWER**

(a) $m^2 - 3m - 40$, (b) $n = \frac{9}{2}$

4MA1/1H NOVEMBER 2024

# Question 3

6 marks | Linear Equations

3 (a) Simplify $\left(p^3\right)^5$

.....

(1)

(b) Expand and simplify $2n(4n+3)+n(n-4)$

.....

(2)

(c) Solve $\frac{2x+5}{3} = 4-x$

Show clear algebraic working.

$x =$ .....

(3)

**(Total for Question 3 is 6 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1H November 2024

**1** **(a)**

(a) $(p^3)^5 = p^{15}$.

**2** **(b)**

(b)

$$2n(4n+3)+n(n-4) = 8n^2+6n+n^2-4n = 9n^2+2n$$

**3** **(c)**

(c)

$$\frac{2x+5}{3} = 4-x$$

$$2x+5 = 12-3x$$

$$5x = 7$$

**FINAL ANSWER**

(a) $p^{15}$, (b) $9n^2 + 2n$, (c) $x = \frac{7}{5}$

4MA1/2HR JUNE 2025

# Question 1

5 marks | Linear Equations

**1** (a) Factorise fully $18c - 45cd$

.....

**(2)**

(b) Solve $\frac{5 - 2x}{6} = 3x - 4$

Show clear algebraic working.

$x =$ .....

**(3)**

**(Total for Question 1 is 5 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2HR June 2025

**1** **(a)**

(a)

$$18c - 45cd = 9c(2 - 5d)$$

**2** **(b)**

(b)

$$\frac{5 - 2x}{6} = 3x - 4$$

$$5 - 2x = 18x - 24$$

$$29 = 20x$$

$$x = \frac{29}{20}$$

**FINAL ANSWER**

(a) $9c(2 - 5d)$, (b) $x = \frac{29}{20}$

4WM1/01R MAY 2025

# Question 12

6 marks | Linear Equations

**12** (a) Solve $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$

Show clear algebraic working.

$a =$ .....

(4)

(b) Express $\left(\frac{\sqrt{y}}{3}\right)^{-1}$ in the form $cy^n$ where $c$ and $n$ are numbers to be found.

.....

(2)

**(Total for Question 12 is 6 marks)**

## WORKED SOLUTION

Question 12 · 4WM1/01R May 2025

### 1 Solve equation

For part (a),

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

### 2 Multiply by

Multiply by $12$:

$$4(5a+8) - 3(2a+5) = 276$$

$$20a + 32 - 6a - 15 = 276$$

$$14a + 17 = 276$$

$$14a = 259$$

$$a = \frac{259}{14} = \frac{37}{2}$$

### 3 Solve equation

For part (b),

$$\left(\frac{\sqrt{y}}{3}\right)^{-1} = \frac{3}{\sqrt{y}}$$

$$= 3y^{-1/2}$$

---

**FINAL ANSWER**

(a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

4MA1/1HR JUNE 2025

# Question 15

6 marks | Linear Equations

**15** (a) Solve $\frac{5a+8}{3} - \frac{2a+5}{4} = 23$

Show clear algebraic working.

$a =$ .....

(4)

(b) Express $\left(\frac{\sqrt{y}}{3}\right)^{-1}$ in the form $cy^n$ where $c$ and $n$ are numbers to be found.

.....

(2)

**(Total for Question 15 is 6 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1HR June 2025

### 1 Solve equation

For part (a),

$$\frac{5a+8}{3} - \frac{2a+5}{4} = 23$$

### 2 Multiply by

Multiply by $12$:

$$4(5a+8) - 3(2a+5) = 276$$

$$20a + 32 - 6a - 15 = 276$$

$$14a + 17 = 276$$

$$14a = 259$$

$$a = \frac{259}{14} = \frac{37}{2}$$

**3** **Solve equation**

For part (b),

$$\left(\frac{\sqrt{y}}{3}\right)^{-1} = \frac{3}{\sqrt{y}}$$

$$= 3y^{-1/2}$$

---

✓ **FINAL ANSWER**

(a) $a = \frac{37}{2}$, (b) $3y^{-1/2}$

4MA1/2HR JUNE 2026

# Question 4

6 marks | Linear Equations

**4** (a) Solve $3(x + 2) = 1 - 5x$
Show clear algebraic working.

$x =$ .....

**(3)**

(c) Simplify $m^0$

.....

**(1)**

(d) Simplify $(4r^2t)^3$

.....

**(2)**

## WORKED SOLUTION

Question 4 · 4MA1/2HR June 2026

**1** **Part (a) | Step 1**

$$3x + 6 = 1 - 5x.$$

Expand the bracket.

**2** **Part (a) | Step 2**

$$8x = -5 \quad \Rightarrow \quad x = -\frac{5}{8}.$$

Add 5x and subtract 6 on both sides, then divide by 8.

**3** **Part (c) | Step 1**

$m^0 = 1$ for $m \neq 0$.

Any nonzero base to the zero power equals 1.

**4** **Part (d) | Step 1**

$$(4r^2t)^3 = 4^3(r^2)^3t^3 = 64r^6t^3.$$

Cube each factor and multiply the exponents in the power of a power.

✓ **FINAL ANSWER**

(a) $x = -\frac{5}{8}$. (c) $1$. (d) $64r^6t^3$.

TOPIC

# Graphs of Functions

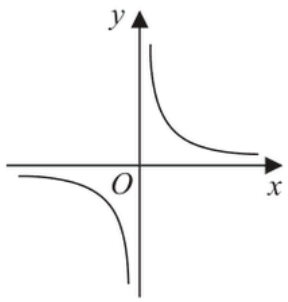
Every past-paper question on this topic, 2020 to 2026.

# Question 15

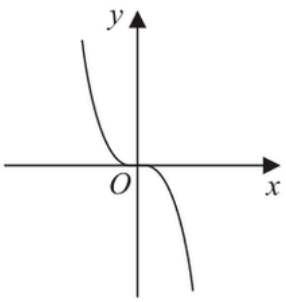
3 marks | Graphs of Functions

**15** Here are six graphs.

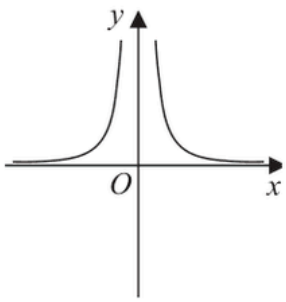
**Graph A**



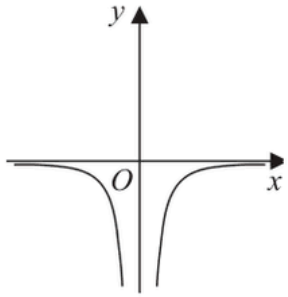

**Graph B**



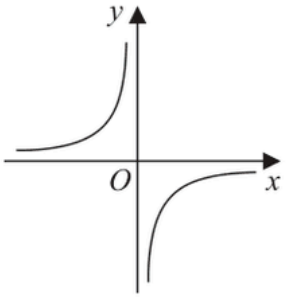

**Graph C**



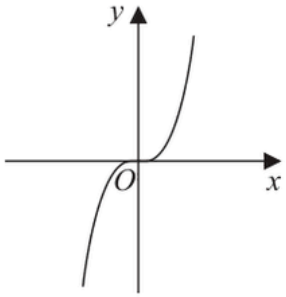

**Graph D**




**Graph E**




**Graph F**




Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

| Equation | Graph |
|---|---|
| $y = \frac{2}{x^2}$ | ..... |
| $y = -\frac{1}{2}x^3$ | ..... |
| $y = -\frac{5}{x}$ | ..... |

**(Total for Question 15 is 3 marks)**

**1** **Solve equation**

For

$$y = -\frac{2}{x^2}$$

the graph is negative on both sides of the $y$-axis and is symmetrical in the $y$-axis.

**2** **This is Graph D**

This is Graph D.

**3** **Solve equation**

For

$$y = -\frac{1}{2}x^3$$

the graph is a decreasing cubic through the origin.

**4** **This is Graph B**

This is Graph B.

**5** **Solve equation**

For

$$y = \frac{5}{x}$$

the graph is a positive reciprocal graph in quadrants 1 and 3.

**6** **This is Graph A**

This is Graph A.

✓ **FINAL ANSWER**

D, B, A.

# Question 15

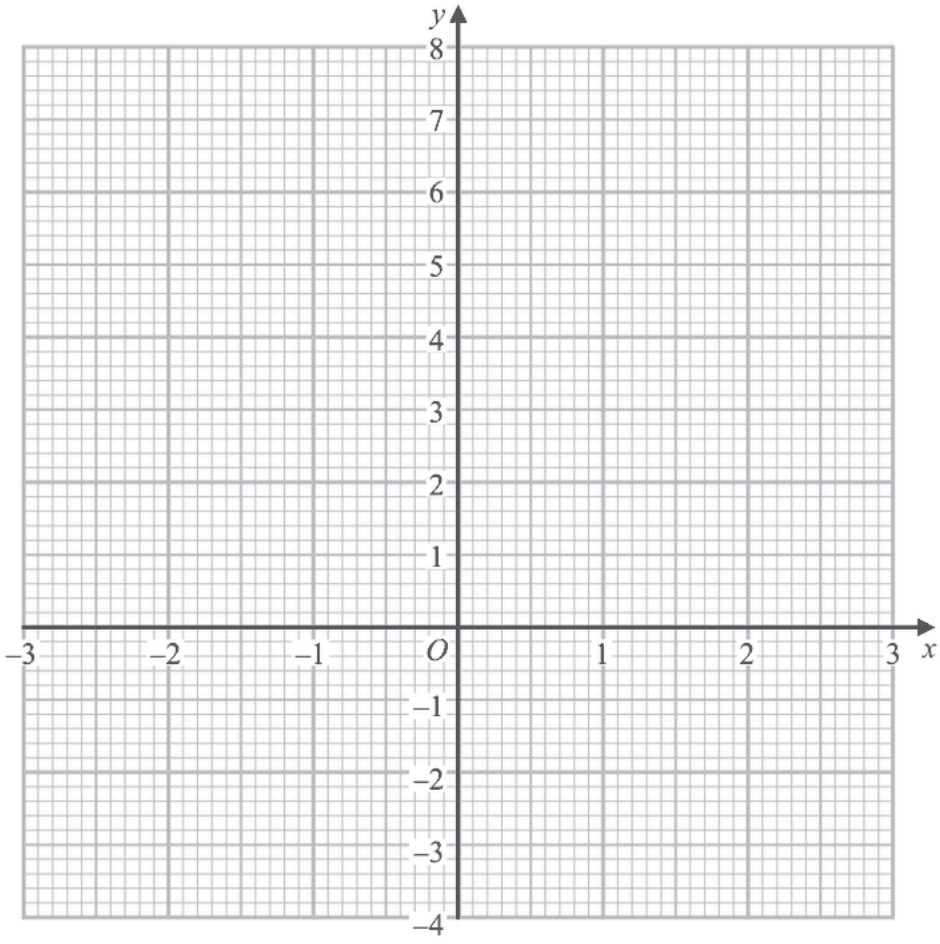
4 marks | Graphs of Functions

**15** (a) Complete the table of values for $y = x^2 - \frac{x}{2} - 3$

| $x$ | –3 | –2 | –1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | 7.5 | | | | –2.5 | | 4.5 |

**(2)**

(b) On the grid, draw the graph of $y = x^2 - \frac{x}{2} - 3$ for values of $x$ from –3 to 3




**(2)**

**(Total for Question 15 is 4 marks)**

## 1 Solve quadratic equation

$$y = x^2 - \frac{x}{2} - 3$$

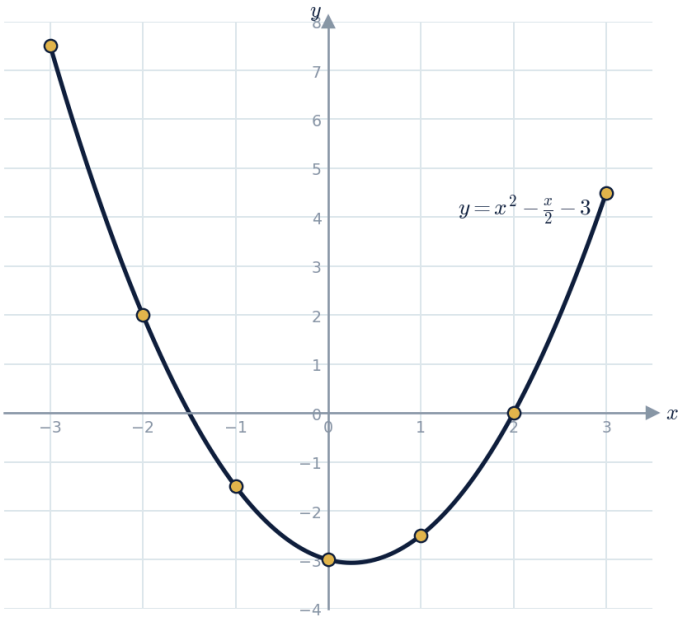
## 2 The completed table is

The completed table is:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $7.5$ | $2$ | $-1.5$ | $-3$ | $-2.5$ | $0$ | $4.5$ |

## 3 Plot these points and join

Plot these points and join them with a smooth U-shaped curve.




**FINAL ANSWER**

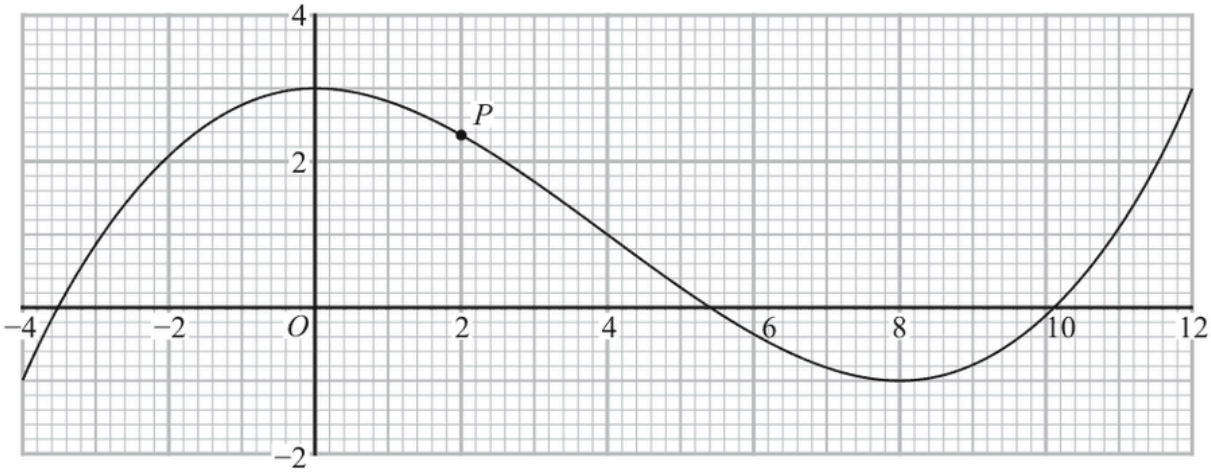
Missing values are $2, -1.5, -3, 0$.

4MA1/1HR JANUARY 2020

# Question 18

7 marks | Graphs of Functions

**18** The diagram shows the graph of $y = f(x)$ for $-4 \leqslant x \leqslant 12$




The point $P$ on the curve has $x$ coordinate 2

(a) (i) Use the graph to find an estimate for the gradient of the curve at $P$.

.....

(3)

(ii) Hence find an equation of the tangent to the curve at $P$.
Give your answer in the form $y = mx + c$

.....

(2)

The equation $f(x) = k$ has exactly two different solutions for $-4 \leqslant x \leqslant 12$

(b) Use the graph to find the two possible values of $k$.

..... , .....

(2)

**(Total for Question 18 is 7 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1HR January 2020

**1** **Draw a tangent to the**

Draw a tangent to the curve at $P$.

**2** **A reasonable tangent gives an**

A reasonable tangent gives an estimated gradient of about

$$m \approx -0.6$$

**3** **The point is approximately**

The point $P$ is approximately

$$(2, 2.4)$$

**4** **Find the gradient**

Using $y = mx + c$,

$$2.4 \approx -0.6(2) + c$$

$$c \approx 3.6$$

**5** **Find the gradient**

So an approximate tangent equation is

$$y \approx -0.6x + 3.6$$

**6** **The equation has exactly two**

The equation $f(x) = k$ has exactly two different solutions when the horizontal line touches one turning level.

**7** **Find the gradient**

From the graph, these levels are approximately

$$k = 3 \quad \text{and} \quad k = -1$$

✓ **FINAL ANSWER**

Gradient $\approx -0.6$, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

4MA1/1H JANUARY 2020

# Question 20

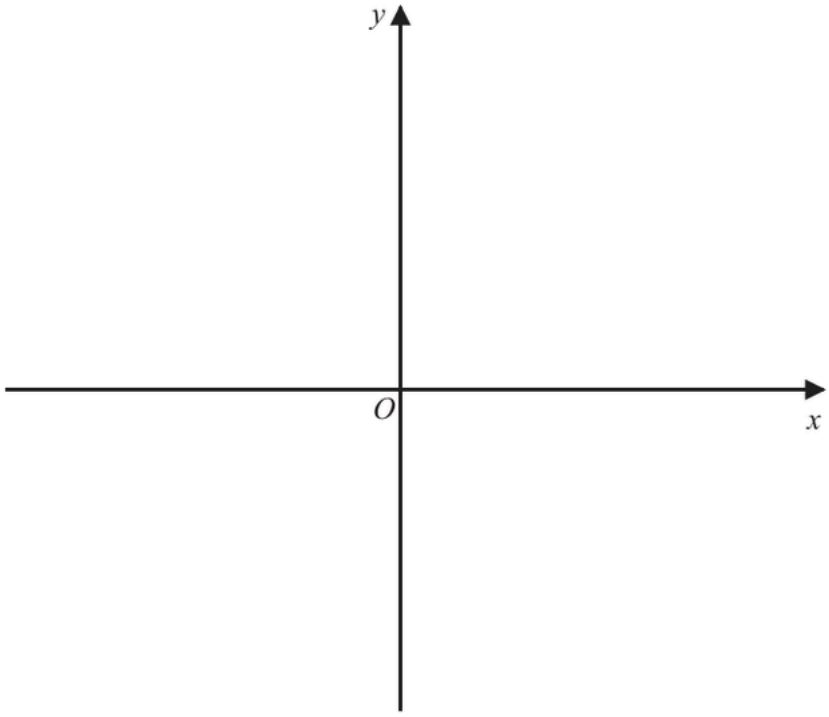
4 marks | Graphs of Functions

**20** The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of $a$,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.




**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H January 2020

**1** **Calculate value**

$$y = 4(x - 1)^2 - a$$

**2** **The turning point is read**

The turning point is read directly from the completed square form:

$$(1, -a)$$

3 **Find intercepts**

For the $y$-intercept, put $x = 0$:

$$y = 4(0-1)^2 - a$$

$$y = 4 - a$$

4 **Find intercepts**

So the $y$-intercept is

$$(0, 4-a)$$

5 **Find intercepts**

For the $x$-intercepts, put $y = 0$:

$$4(x-1)^2 - a = 0$$

$$4(x-1)^2 = a$$

$$(x-1)^2 = \frac{a}{4}$$

$$x - 1 = \pm\frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

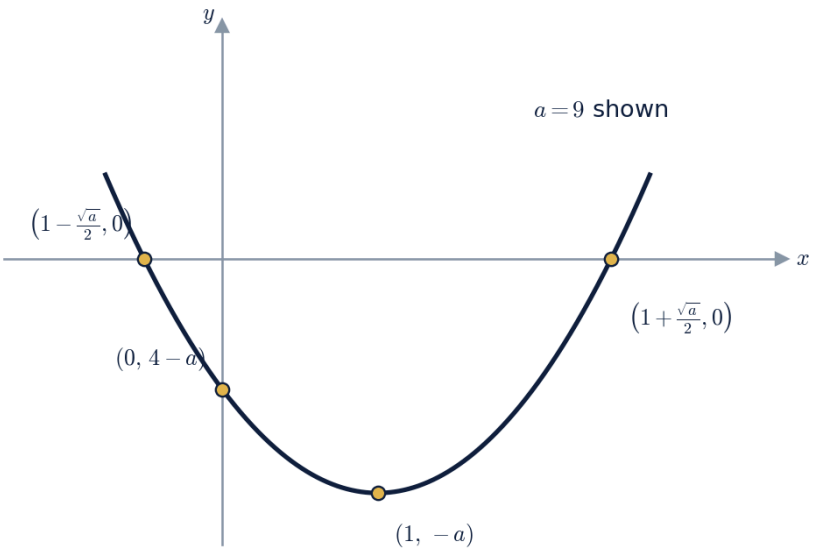
6 **Find intercepts**

So the $x$-intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

**7** **Solve inequality**

Since $a > 4$, the turning point is below the $x$-axis and the curve crosses the $x$-axis twice.




**FINAL ANSWER**

turning point $(1, -a)$, $y$-intercept $(0, 4-a)$, $x$-intercepts $\left(1-\frac{\sqrt{a}}{2}, 0\right)$ and $\left(1+\frac{\sqrt{a}}{2}, 0\right)$

# Question 15

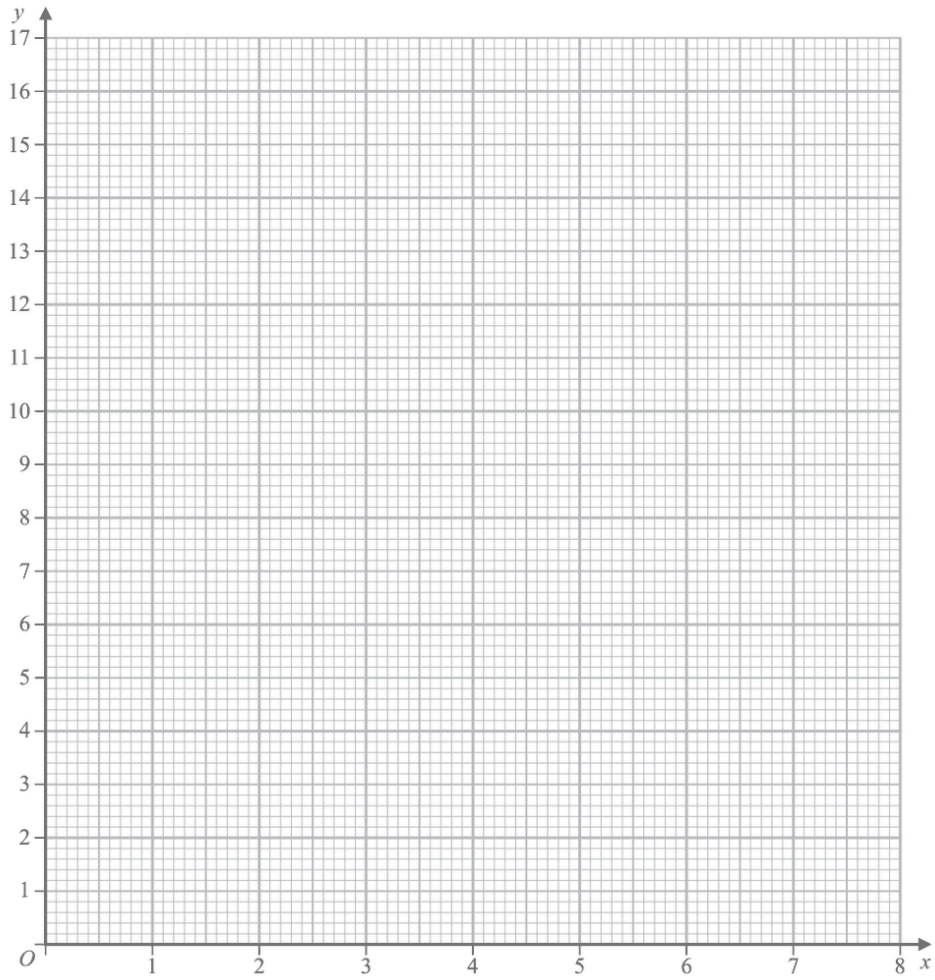
4 marks | Graphs of Functions

**15** (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

| $x$ | 0.25 | 0.5 | 1 | 2 | 4 | 8 |
|---|---|---|---|---|---|---|
| $y$ | 16.25 | | | | | 8.5 |

**(2)**

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leqslant x \leqslant 8$




**(2)**

**(Total for Question 15 is 4 marks)**

## 1 Read the graph

$$y = \frac{1}{x}(x^2 + 4) = x + \frac{4}{x}$$

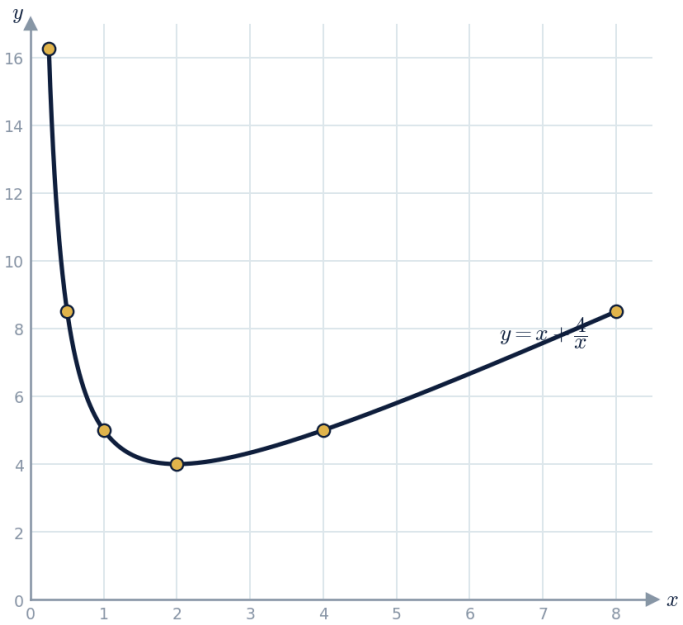
## 2 The completed table is

The completed table is:

| $x$ | 0.25 | 0.5 | 1 | 2 | 4 | 8 |
|---|---|---|---|---|---|---|
| $y$ | 16.25 | 8.5 | 5 | 4 | 5 | 8.5 |

## 3 Plot the points and join

Plot the points and join them with a smooth curve.




**FINAL ANSWER**

Missing values are $8.5, 5, 4, 5$.

# Question 4

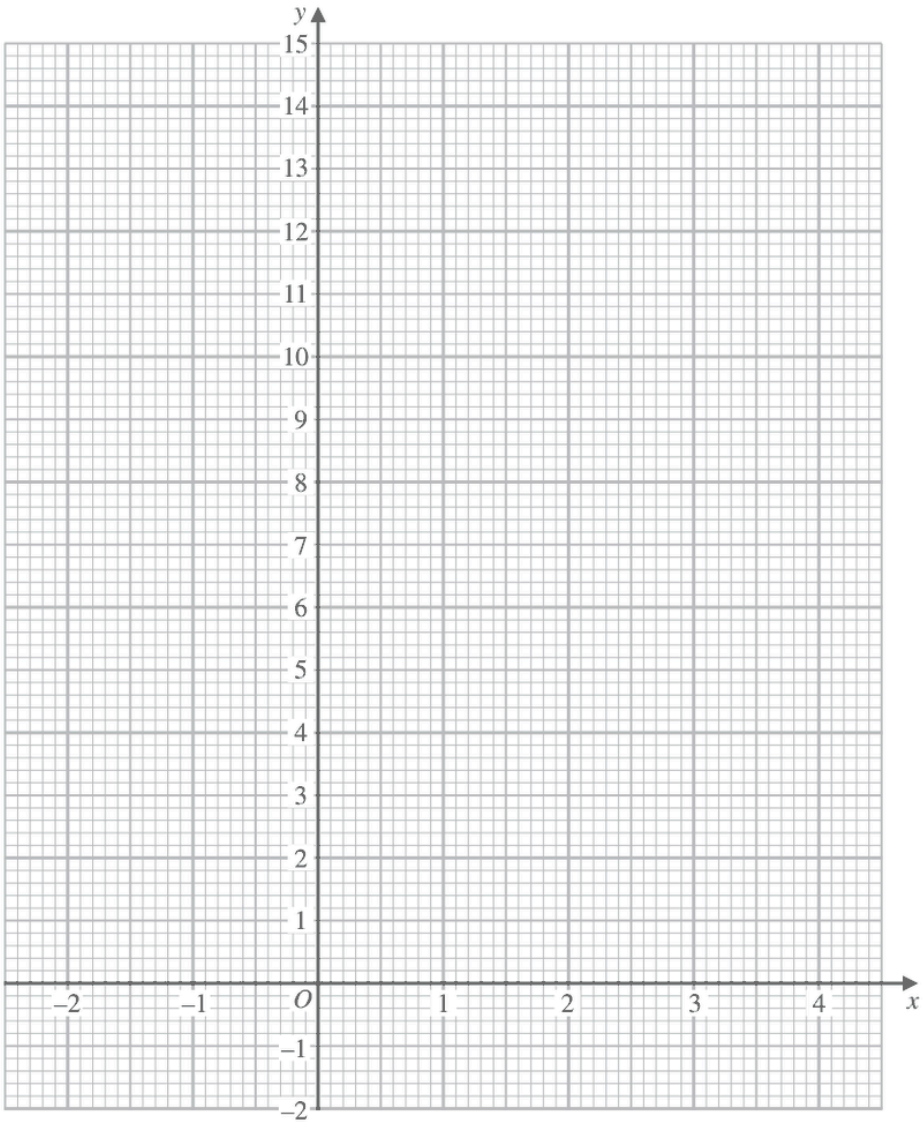
4 marks | Graphs of Functions

**4** (a) Complete the table of values for $y = x^2 - 4x + 3$

| $x$ | –2 | –1 | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|
| $y$ | | 8 | 3 | | | 0 | |

**(2)**

(b) On the grid, draw the graph of $y = x^2 - 4x + 3$ for values of $x$ from –2 to 4




**(2)**

**(Total for Question 4 is 4 marks)**

## 1 Solve quadratic equation

$$y = x^2 - 4x + 3$$

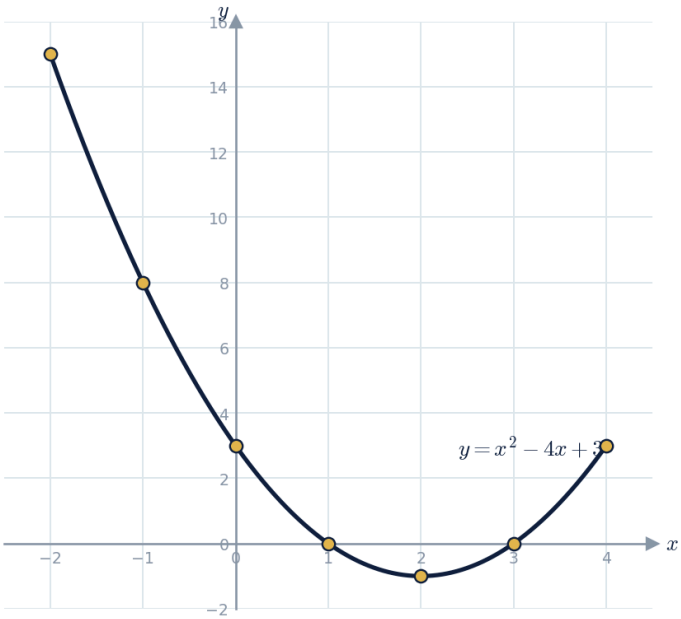
## 2 The completed table is

The completed table is:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|
| $y$ | $15$ | $8$ | $3$ | $0$ | $-1$ | $0$ | $3$ |

## 3 Plot the points and join

Plot the points and join them with a smooth U-shaped curve.




**FINAL ANSWER**

Missing values are $15, 0, -1, 3$.

# Question 11

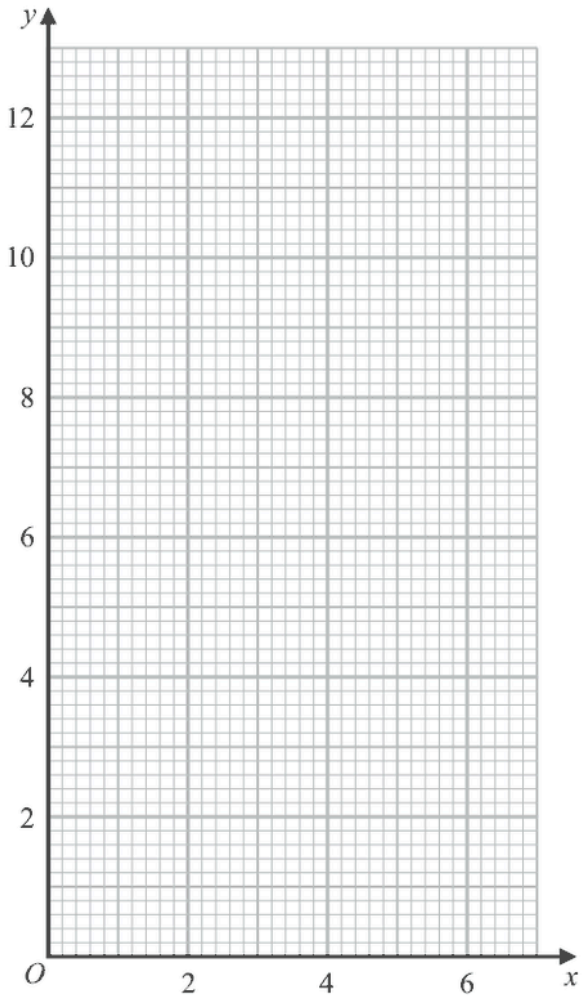
4 marks | Graphs of Functions

**11** (a) Complete the table of values for $y = \frac{6}{x}$

| $x$ | 0.5 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | | 6 | | 2 | | | 1 |

**(2)**

(b) On the grid, draw the graph of $y = \frac{6}{x}$ for $0.5 \leqslant x \leqslant 6$




**(2)**

**(Total for Question 11 is 4 marks)**

## 1 Read the graph

$$y = \frac{6}{x}$$

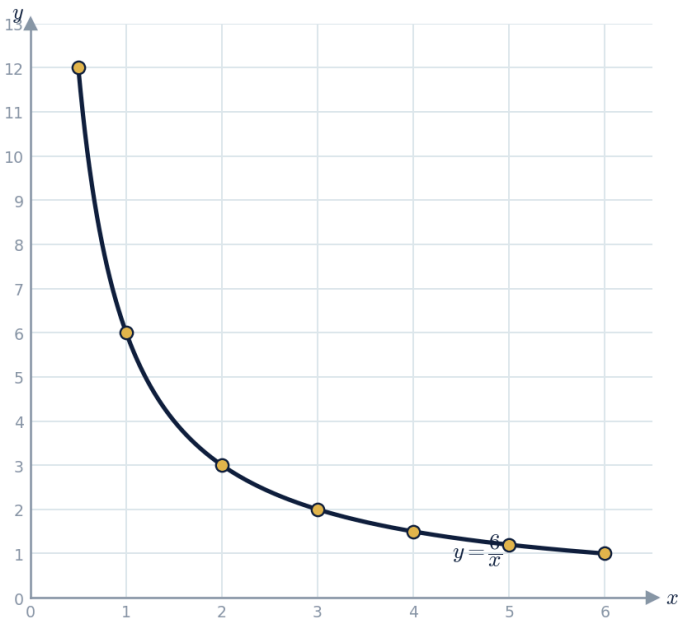
## 2 The completed table is

The completed table is:

| $x$ | 0.5 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | 12 | 6 | 3 | 2 | 1.5 | 1.2 | 1 |

## 3 Plot the points and join

Plot the points and join them with a smooth decreasing reciprocal curve.




**FINAL ANSWER**

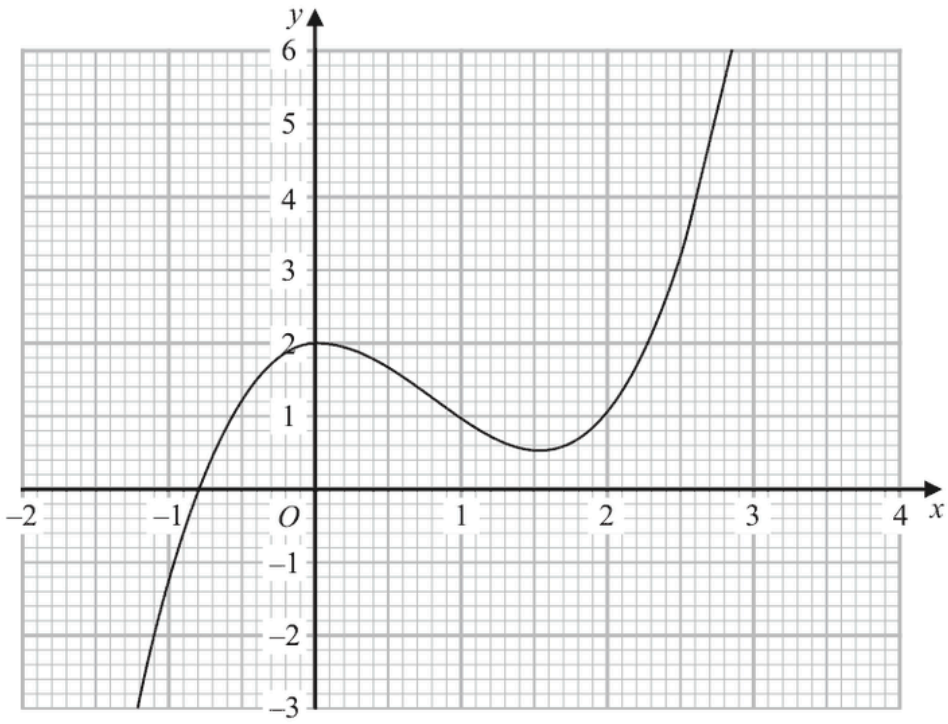
Missing values are $12, 3, 1.5, 1.2$.

4MA1/2HR JANUARY 2022

# Question 17

3 marks | Graphs of Functions

**17** Part of the curve with equation $y = f(x)$ is shown on the grid.




Find an estimate for the gradient of the curve at the point where $x = 2$
Show your working clearly.

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/2HR January 2022

**1 Draw a tangent to the**

Draw a tangent to the curve at the point where

$$x = 2$$

**2 Find the gradient**

Using two clear points on the tangent, for example approximately $(1.5, 0.6)$ and $(2.3, 2.2)$,

$$\text{gradient} \approx \frac{2.2 - 0.6}{2.3 - 1.5}$$

$$\text{gradient} \approx 2.0$$

**FINAL ANSWER**

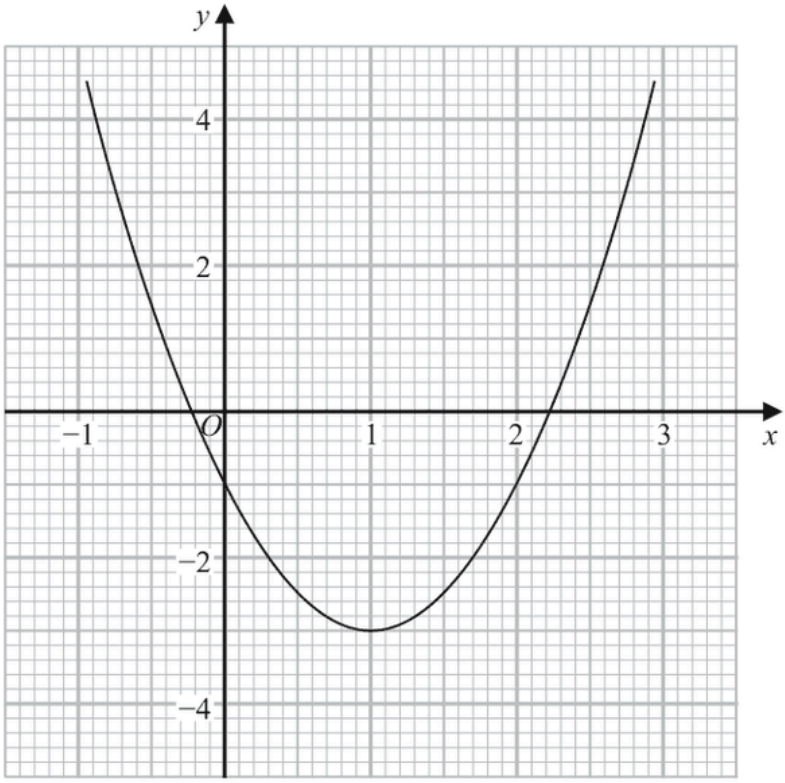
The gradient is approximately $\mathbf{2}$.

4MA1/2H JANUARY 2022

# Question 21

5 marks | Graphs of Functions

**21** Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.




(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

.....

**(2)**

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....

**(3)**

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2H January 2022

**1** **Read the graph**

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

**2** **Read the graph**

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

**3** **Read the graph**

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

**4** **Multiply by 2**

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

**5** **The given graph is**

The given graph is

$$y = 2x^2 - 4x - 1$$

**6** **Read the graph**

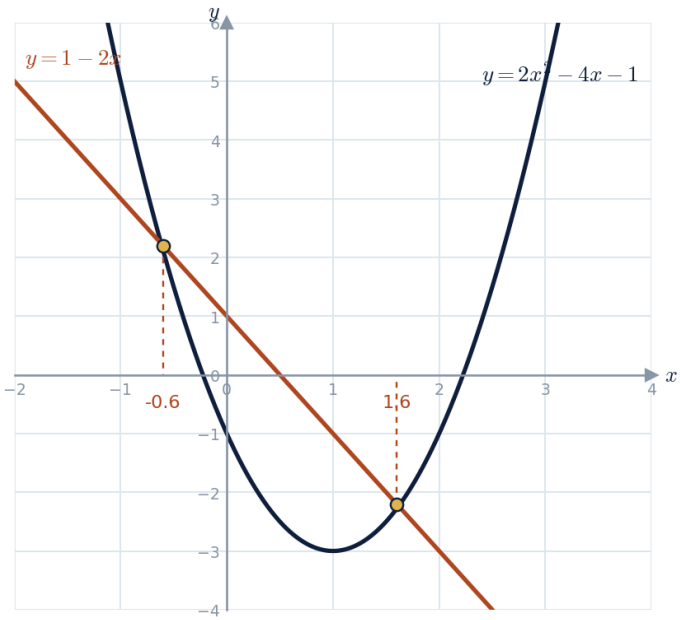
Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

**7** **Read the graph**

So draw the straight line

$$y = 1 - 2x$$



**8** **The intersections of this line**

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

---

✓ **FINAL ANSWER**

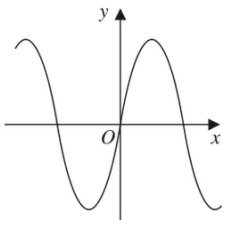
(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

# Question 16

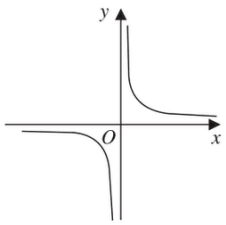
3 marks | Graphs of Functions

**16** Here are nine graphs.

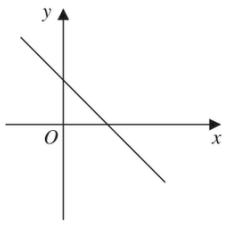
**Graph A**



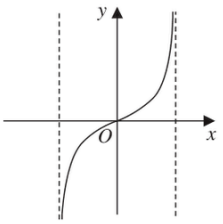

**Graph B**



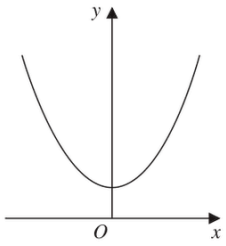

**Graph C**



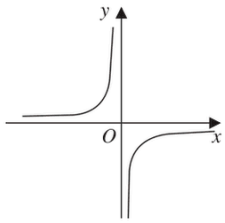

**Graph D**



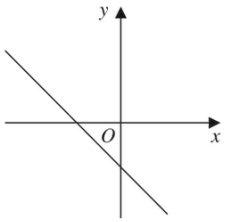

**Graph E**



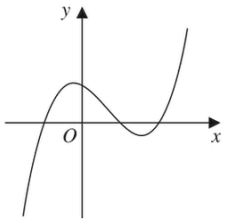

**Graph F**



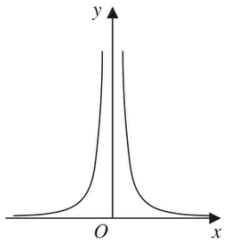

**Graph G**




**Graph H**




**Graph I**




Complete the table below with the letter of the graph that could represent each given equation.
Write each answer on the dotted line.

| Equation | Graph |
|---|---|
| $y = -2x + 3$ | ..... |
| $y = -\frac{1}{x}$ | ..... |
| $y = \tan x°$ | ..... |
| $y = (x + 1)(x - 1)(x - 2)$ | ..... |

**(Total for Question 16 is 3 marks)**

**1** **is a negative straight line**

$$y = -2x + 3$$

is a negative straight line with positive $y$-intercept, so it is **Graph C**.

**2** **Read the graph**

$$y = \frac{1}{x}$$

is the reciprocal graph in quadrants I and III, so it is **Graph B**.

**3** **has repeating branches with vertical**

$$y = \tan x$$

has repeating branches with vertical asymptotes and passes through the origin, so it is **Graph D**.

**4** **Read the graph**

$$y = (x + 1)(x - 1)(x - 2)$$

is a positive cubic with three roots, so it is **Graph H**.

✓ **FINAL ANSWER**

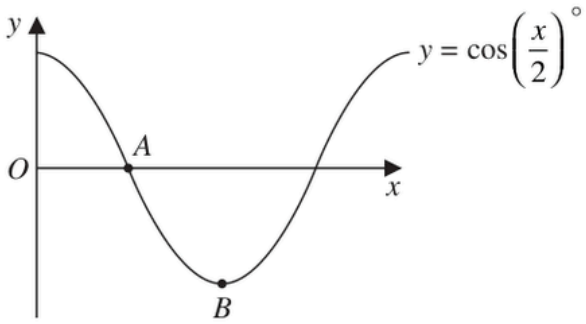
C, B, D, H.

4MA1/2H JUNE 2022

# Question 23

2 marks | Graphs of Functions

**23** The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$




(i) Find the coordinates of the point $A$

(.....,.....)

(1)

(ii) Find the coordinates of the point $B$

(.....,.....)

(1)

**(Total for Question 23 is 2 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2H June 2022

**1** **The graph is**

The graph is

$$y = \cos\left(\frac{x}{2}\right)^\circ$$

**2** **Point is the first positive**

Point $A$ is the first positive $x$-intercept, so $y = 0$.

$$\cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = 90^\circ$$

$$x = 180$$

**3** **Read the graph**

So

$$A = (180, 0)$$

---

**4** **Point is the minimum point**

Point $B$ is the minimum point.

---

**5** **The minimum value of cosine**

The minimum value of cosine is $-1$, which occurs when

$$\frac{x}{2} = 180°$$

$$x = 360$$

---

**6** **Read the graph**

So

$$B = (360, -1)$$

---

✓ **FINAL ANSWER**

$A = (180, 0),\ B = (360, -1)$

# Question 4

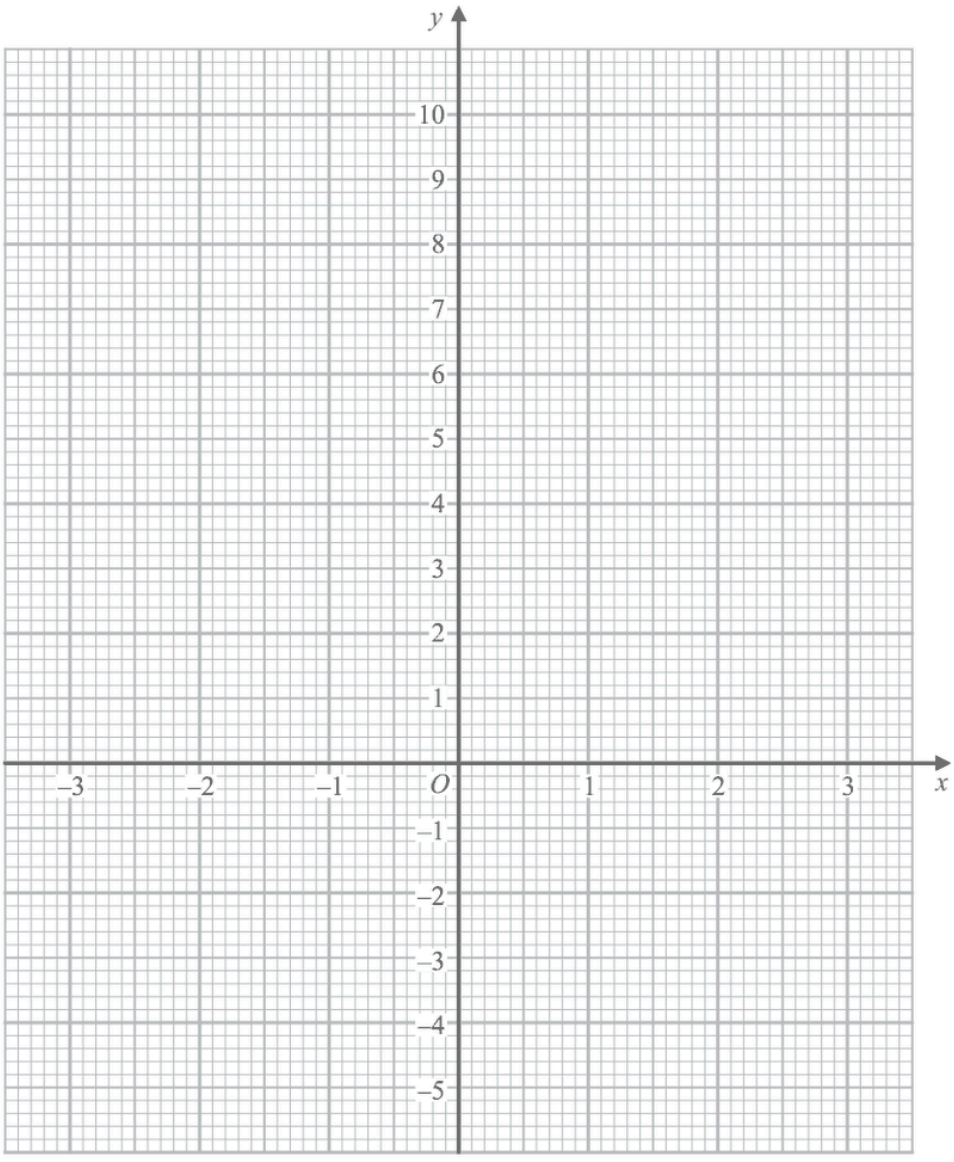
4 marks | Graphs of Functions

4 (a) Complete the table of values for $y = x^2 - x - 4$

| $x$ | –3 | –2 | –1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | | 2 | | | –4 | | |

(2)

(b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of $x$ from –3 to 3




(2)

**(Total for Question 4 is 4 marks)**

## 1 Solve quadratic equation

$$y = x^2 - x - 4$$

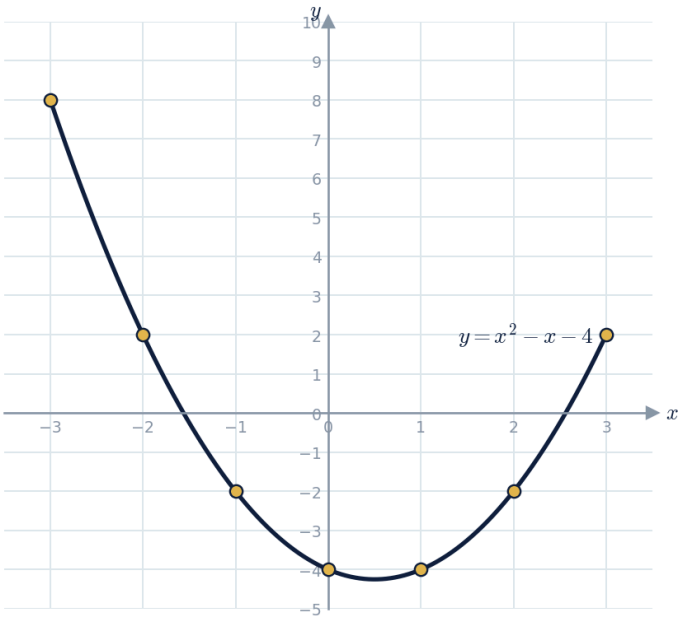
## 2 The completed table is

The completed table is:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $8$ | $2$ | $-2$ | $-4$ | $-4$ | $-2$ | $2$ |

## 3 Plot the points and join

Plot the points and join them with a smooth U-shaped curve.




**FINAL ANSWER**

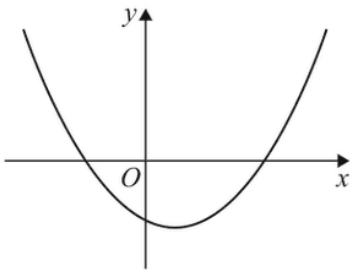
Missing values are $8, -2, -4, -2, 2$.

# Question 12

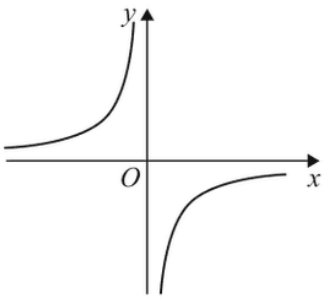
3 marks | Graphs of Functions

**12** Here are six graphs.

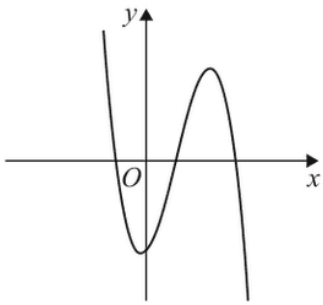
Graph **A**



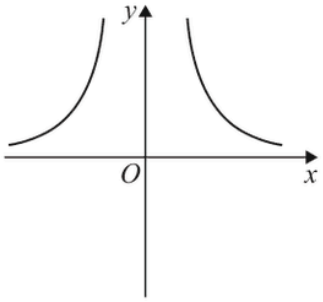

Graph **B**



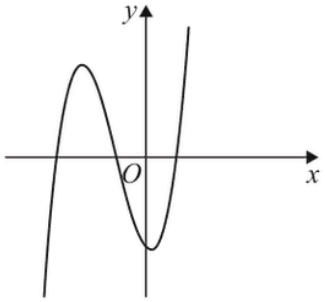

Graph **C**



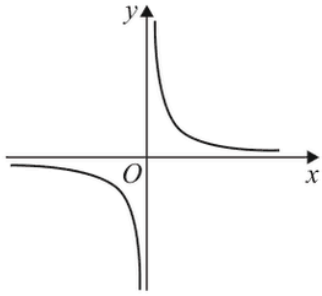

Graph **D**




Graph **E**




Graph **F**




Write down the letter of the graph of

(a) $y = \frac{10}{x^2}$

.....

**(1)**

(b) $y = x - 3 + 3x^2 - x^3$

.....

**(1)**

(c) $y = -\frac{3}{x}$

.....

**(1)**

**(Total for Question 12 is 3 marks)**

**1** **Solve equation**

For

$$y = \frac{10}{x^2}$$

the graph is positive on both sides and symmetrical in the $y$-axis.

**2** **This is Graph D**

This is Graph D.

**3** **Solve equation**

For

$$y = x - 3 + 3x^2 - x^3$$

the leading term is $-x^3$, so it is a cubic falling to the right with two turning points.

**4** **This is Graph C**

This is Graph C.

**5** **Solve equation**

For

$$y = -\frac{3}{x}$$

the graph is a negative reciprocal graph in quadrants 2 and 4.

**6** **This is Graph B**

This is Graph B.

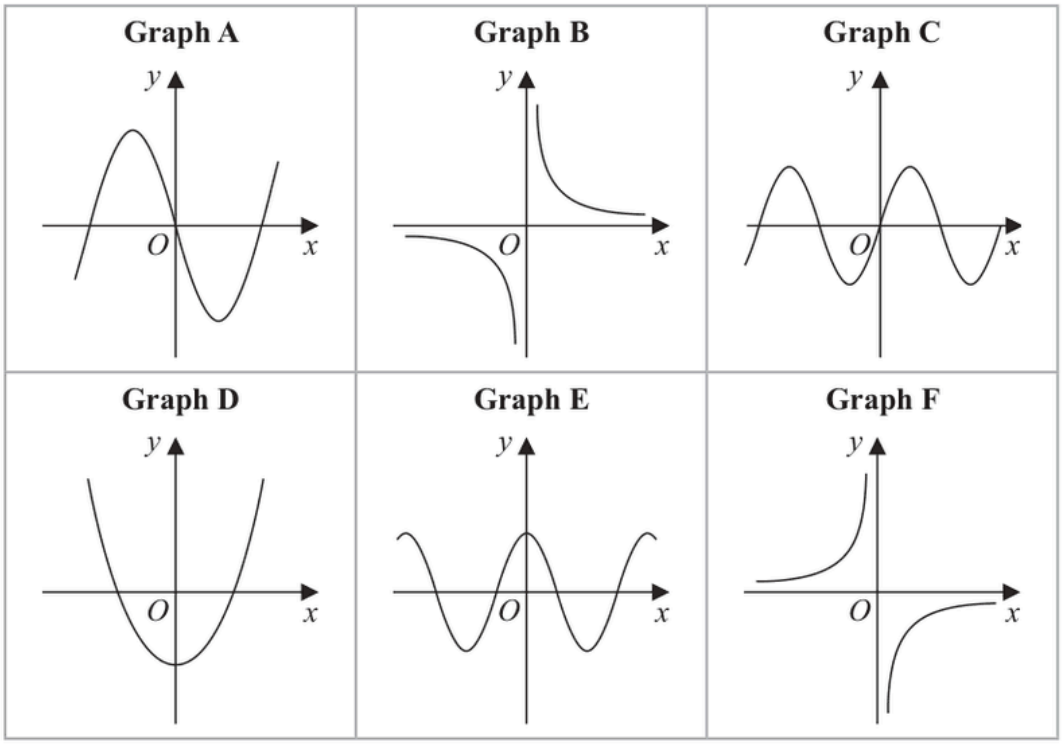
✓ **FINAL ANSWER**

D, C, B.

# Question 18

3 marks | Graphs of Functions

**18** Here are 6 graphs.




Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

| Equation | Graph |
|---|---|
| $y = \sin x$ | ..... |
| $y = -\frac{3}{x}$ | ..... |
| $y = 4x^3 - 5x$ | ..... |

**(Total for Question 18 is 3 marks)**

**1** **is the periodic sine shaped**

$$y = \sin x$$

is the periodic sine-shaped graph, so it is **Graph C**.

**2** **Read the graph**

$$y = \frac{3}{x}$$

is the positive reciprocal graph, so it is **Graph B**.

**3** **Find turning point**

$$y = 4x^3 - 5x$$

is a cubic graph with two turning points, so it is **Graph A**.

✓ **FINAL ANSWER**

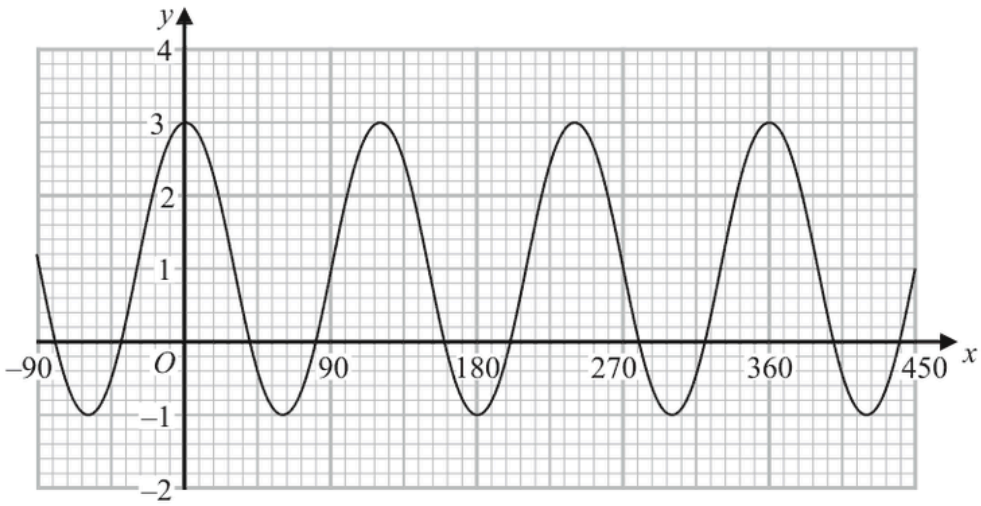
$C$, $B$, $A$.

4MA1/2HR JUNE 2023

# Question 26

3 marks | Graphs of Functions

**26** Here is a sketch of the curve with equation $y = a\cos bx° + c$ where $-90 \leqslant x \leqslant 450$




Find the value of $a$, the value of $b$ and the value of $c$

$a =$ .....

$b =$ .....

$c =$ .....

**(Total for Question 26 is 3 marks)**

## WORKED SOLUTION

Question 26 · 4MA1/2HR June 2023

**1 The curve is**

The curve is

$$y = a\cos bx° + c$$

**2 Use trigonometry**

From the graph, the maximum value is

$$3$$

**3 Use trigonometry**

and the minimum value is

$$-1$$

**4** **The amplitude is half the**

The amplitude is half the difference:

$$a = \frac{3 - (-1)}{2} = 2$$

**5** **The midline is the average**

The midline is the average:

$$c = \frac{3 + (-1)}{2} = 1$$

**6** **The distance between consecutive maxima**

The distance between consecutive maxima is $120°$, so the period is

$$120°$$

**7** **Use trigonometry**

For $y = a\cos bx° + c$, the period is

$$\frac{360°}{b}$$

**8** **Use trigonometry**

So

$$\frac{360}{b} = 120$$

$$b = 3$$

✓ **FINAL ANSWER**

$$\boxed{a = 2,\ b = 3,\ c = 1}$$

# Question 12

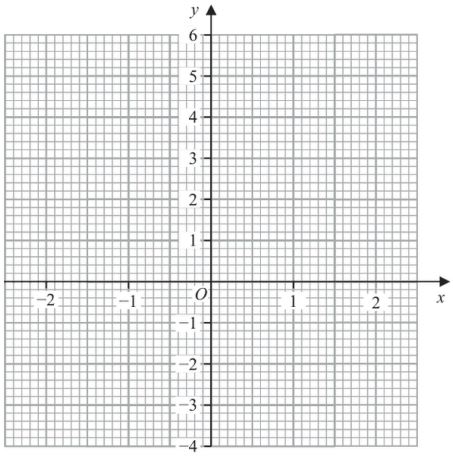
5 marks | Graphs of Functions

**12** (a) Complete the table of values for $y = x^3 - 3x + 1$

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | | | | −1 | |

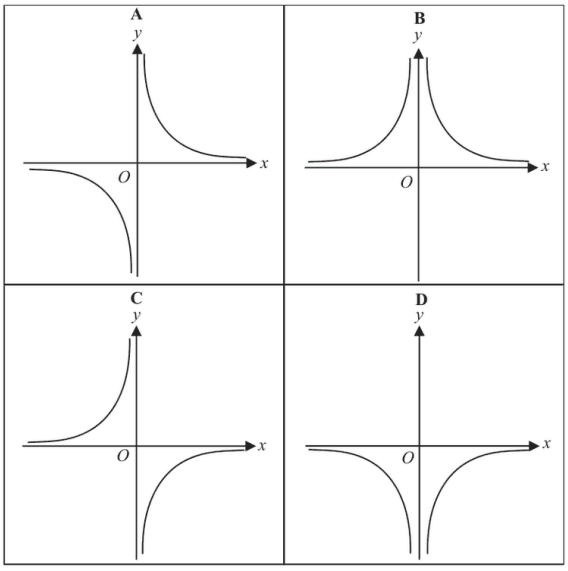
**(2)**

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of $x$ from − 2 to 2




**(2)**

Here are four graphs.




(c) Write down the letter of the graph that could have the equation $y = -\frac{1}{x^2}$

.....

**(1)**

**(Total for Question 12 is 5 marks)**

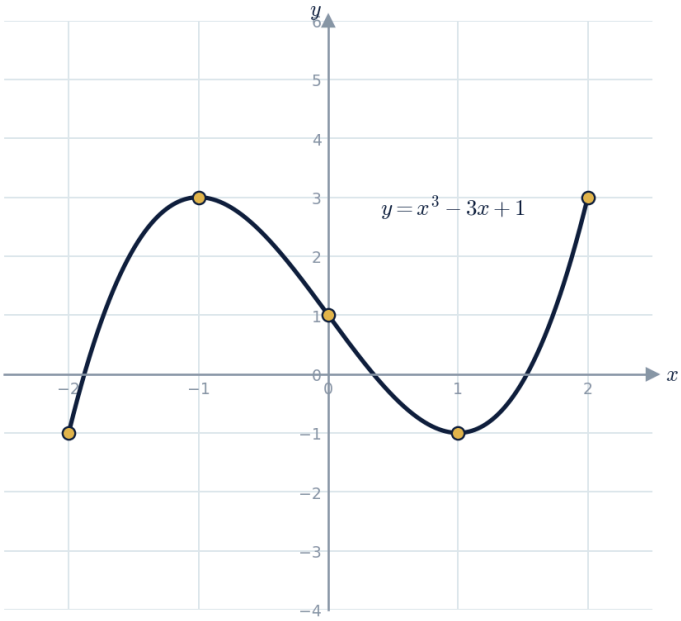
## 1 Read the graph

$$y = x^3 - 3x + 1$$

## 2 The completed table is

The completed table is:

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-1$ | $3$ | $1$ | $-1$ | $3$ |




## 3 The graph of is below

The graph of

$$y = -\frac{1}{x^2}$$

is below the $x$-axis on both sides and is symmetrical in the $y$-axis.

**FINAL ANSWER**

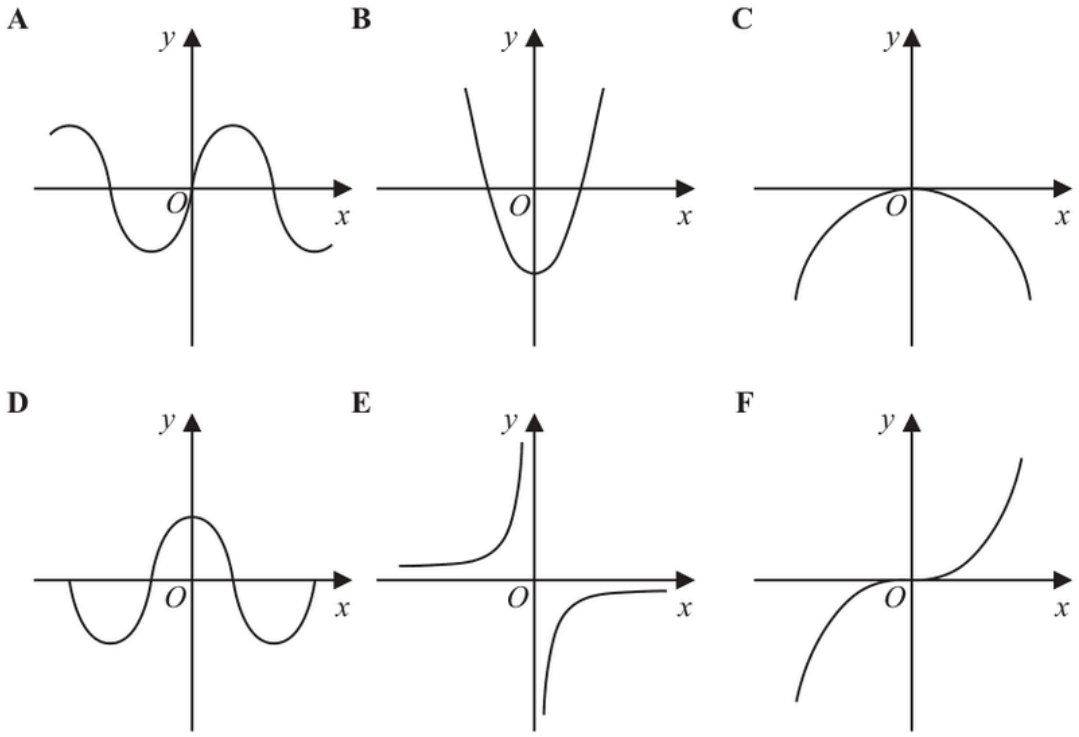
The graph letter is D.

4MA1/2HR JUNE 2024

# Question 16

2 marks | Graphs of Functions

**16** Here are six graphs.




Write down the letter of the graph that could have the equation

(i) $y = -\frac{1}{x}$

.....

(1)

(ii) $y = \sin x°$

.....

(1)

**(Total for Question 16 is 2 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/2HR June 2024

**1** **(i) has branches in quadrants**

(i)

$$y = -\frac{1}{x}$$

has branches in quadrants II and IV, so it is **Graph E**.

**2** **(ii) is the sine shaped**

(ii)

$$y = \sin x^\circ$$

is the sine-shaped graph through the origin, so it is **Graph A**.

---

✓ **FINAL ANSWER**

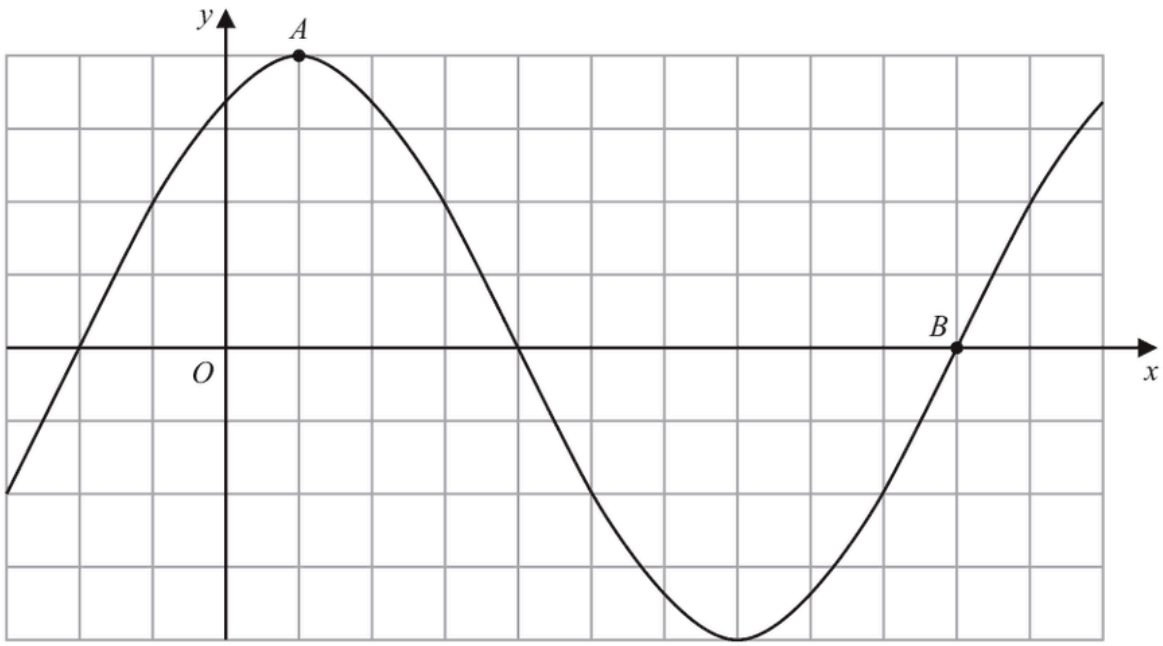
(i) $E$, (ii) $A$.

4MA1/2H JUNE 2024

# Question 25

2 marks | Graphs of Functions

**25** The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$




(i) Find the coordinates of the point $A$

(.....  ,  .....)

(1)

(ii) Find the coordinates of the point $B$

(.....  ,  .....)

(1)

**(Total for Question 25 is 2 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/2H June 2024

**1** **Use trigonometry**

$$y = 2\sin(x + 60)°$$

**2** **Point is the maximum point**

Point $A$ is the maximum point.

**3** **The maximum value is**

The maximum value is

$$y = 2$$

**4** **This happens when**

This happens when

$$x + 60 = 90$$

$$x = 30$$

**5** **Read the graph**

So

$$A = (30, 2)$$

**6** **Point is the positive intercept**

Point $B$ is the positive $x$-intercept after the minimum.

$$2\sin(x + 60)° = 0$$

$$\sin(x + 60)° = 0$$

**7** **Read the graph**

For this point,

$$x + 60 = 360$$

$$x = 300$$

**8** **Read the graph**

So

$$B = (300, 0)$$

✓ **FINAL ANSWER**

$A = (30, 2),\ B = (300, 0)$

# Question 11

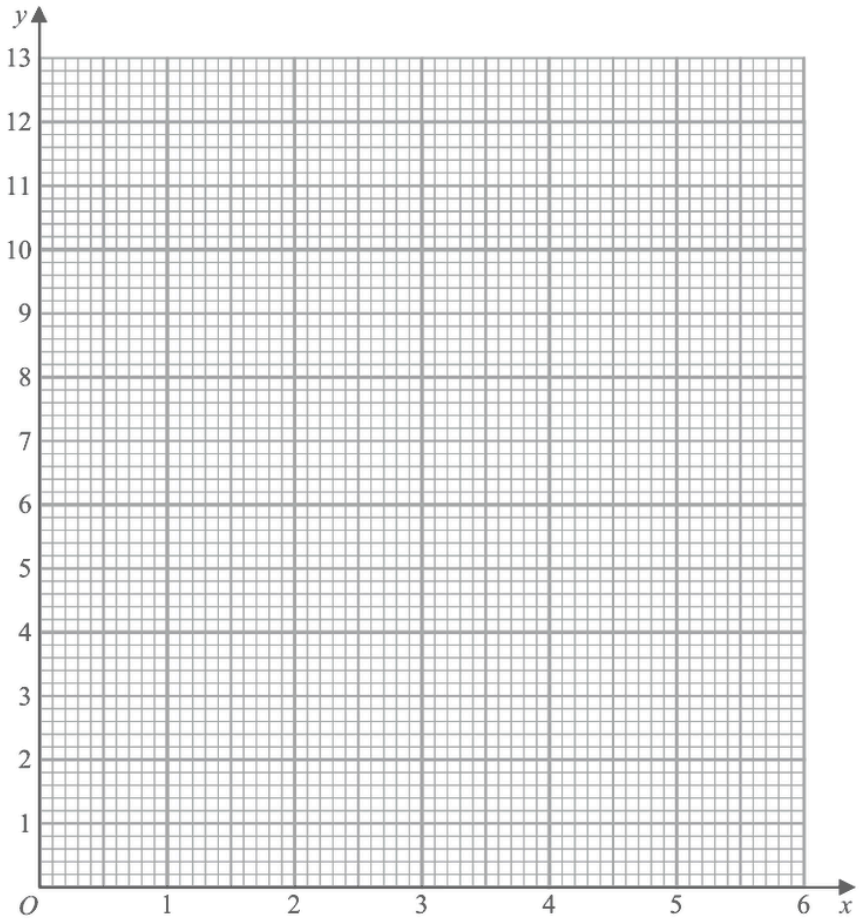
4 marks | Graphs of Functions

**11** (a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

| $x$ | 0.5 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | | | 5 | 6.7 | | | 12.3 |

**(2)**

(b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 6$




**(2)**

**(Total for Question 11 is 4 marks)**

## 1 Read the graph

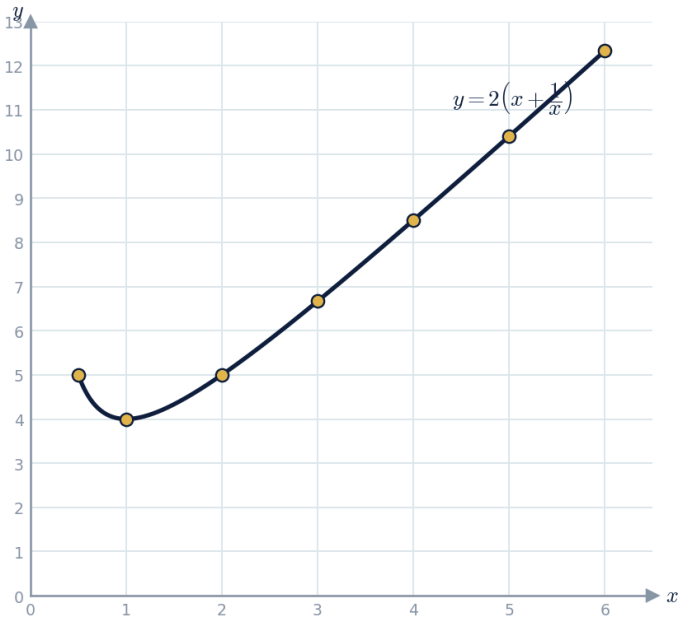
$$y = 2\left(x + \frac{1}{x}\right)$$

## 2 The completed table is

The completed table is:

| $x$ | 0.5 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | 5 | 4 | 5 | 6.7 | 8.5 | 10.4 | 12.3 |

## 3 Plot the points and join them with a smooth curve.




**FINAL ANSWER**

Missing values are $5, 4, 8.5, 10.4$.

4MA1/1H NOVEMBER 2024

# Question 23

5 marks | Graphs of Functions

**23** The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where $k$ is an integer.

**C** and **L** intersect at the points $A$ and $B$

The coordinates of point $A$ are $(p, k)$
The coordinates of point $B$ are $(q, k)$

Given that $p - q = 14$

find the value of $k$
Show clear algebraic working.

$k =$ .....

**(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/1H November 2024

**1** **The curve is**

The curve is

$$y = x^2 - 8x - 9$$

**2** **The line is**

The line is

$$y = k$$

**3** **Read the graph**

At the intersection points,

$$x^2 - 8x - 9 = k$$

$$x^2 - 8x - (9 + k) = 0$$

**4** **Read the graph**

The two roots are $p$ and $q$.

**5** **Solve quadratic equation**

For this quadratic,

$$p + q = 8$$

**6** **We are also given**

We are also given

$$p - q = 14$$

**7** **Solve the two equations**

Solve the two equations:

$$2p = 22$$

$$p = 11$$

$$q = -3$$

**8** **Read the graph**

Use one root in $y = x^2 - 8x - 9$:

$$k = 11^2 - 8(11) - 9$$

$$k = 121 - 88 - 9$$

$$k = 24$$

✓ **FINAL ANSWER**

$k = 24$

# Question 10

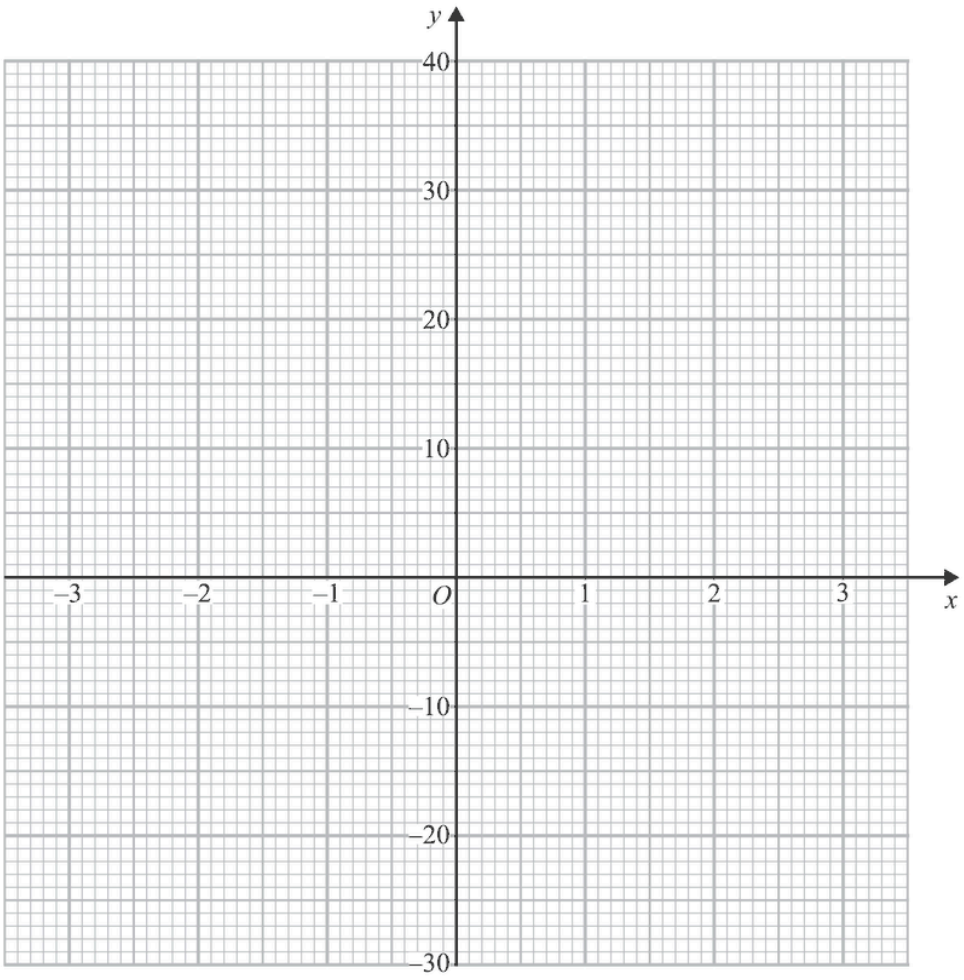
3 marks | Graphs of Functions

**10** (a) Complete the table of values for $y = x^3 + 2x + 3$

| $x$ | –3 | –2 | –1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | | –9 | 0 | 3 | 6 | | 36 |

**(1)**

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leqslant x \leqslant 3$




**(2)**

**(Total for Question 10 is 3 marks)**

## 1 Read the graph

$$y = x^3 + 2x + 3$$

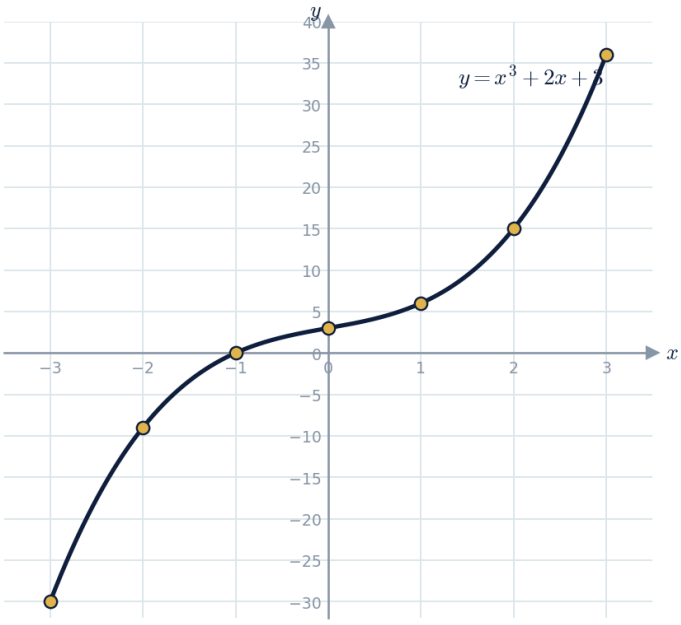
## 2 The completed table is

The completed table is:

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $-30$ | $-9$ | $0$ | $3$ | $6$ | $15$ | $36$ |

## 3 Plot the points and join

Plot the points and join them with a smooth cubic curve.




**FINAL ANSWER**

Missing values are $-30, 15$.

# Question 13

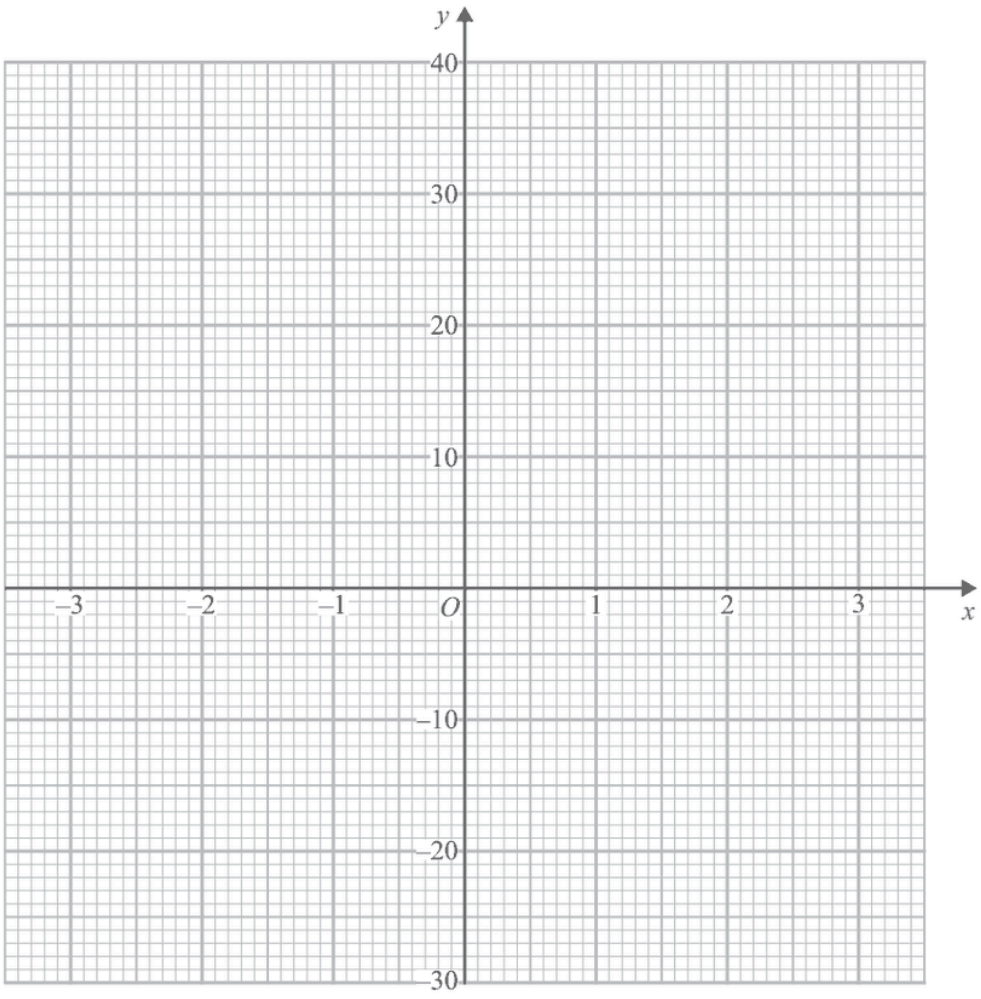
3 marks | Graphs of Functions

**13** (a) Complete the table of values for $y = x^3 + 2x + 3$

| $x$ | –3 | –2 | –1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | | –9 | 0 | 3 | 6 | | 36 |

**(1)**

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leqslant x \leqslant 3$




**(2)**

**(Total for Question 13 is 3 marks)**

**1** **Use**

Use $y = x^3 + 2x + 3$.

**2** **Read the graph**

For $x = -3$,

$$y = (-3)^3 + 2(-3) + 3 = -30$$

**3** **Read the graph**

For $x = 2$,

$$y = 2^3 + 2(2) + 3 = 15$$

**4** **Read the graph**

So the table is

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y$ | $-30$ | $-9$ | $0$ | $3$ | $6$ | $15$ | $36$ |

**5** **Plot these points and draw**

Plot these points and draw a smooth cubic curve through them.

✓ **FINAL ANSWER**

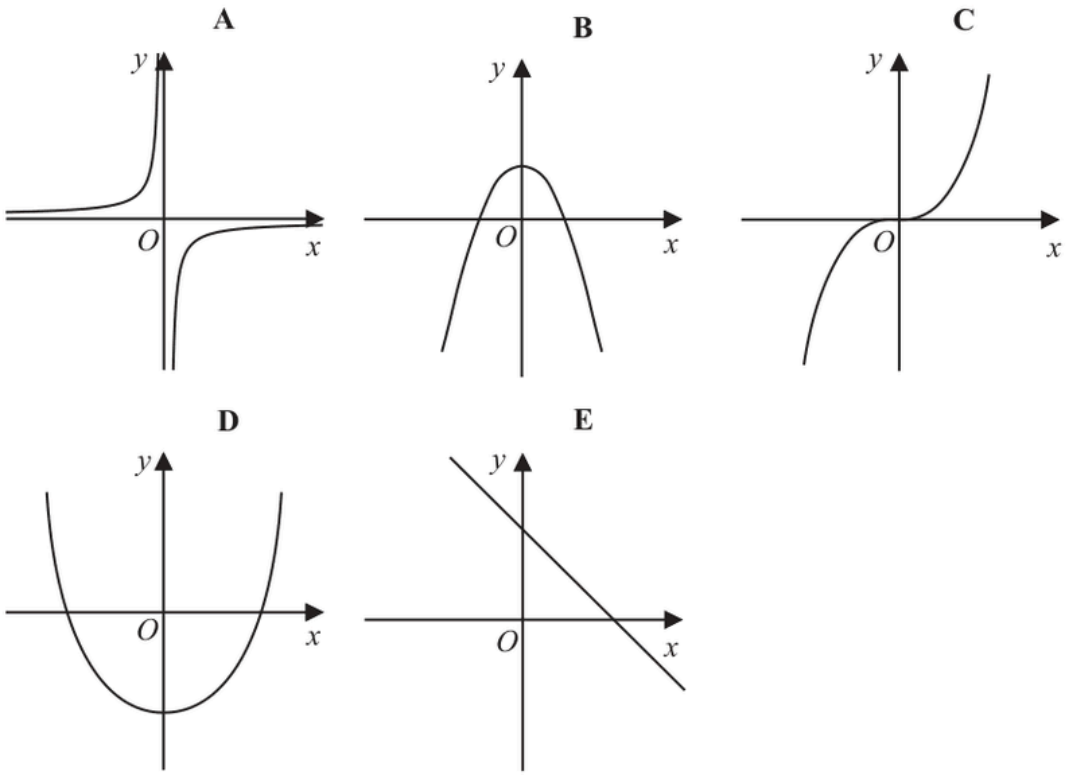
Missing values $-30$ and $15$; draw the smooth curve through the table points.

4MA1/2H JUNE 2025

# Question 14

2 marks | Graphs of Functions

**14** Here are five graphs.




(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$

.....

(1)

(b) Write down the letter of the graph that could have the equation $y = 2x^3$

.....

(1)

**(Total for Question 14 is 2 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2H June 2025

**1** **(a)**

(a)

$$y = 5 - x^2$$

2 **This is a quadratic graph**

This is a quadratic graph with a negative coefficient of $x^2$, so it is a downward-opening parabola.

3 **That is graph B**

That is graph **B**.

4 **(b)**

(b)

$$y = 2x^3$$

5 **This is an increasing cubic**

This is an increasing cubic graph passing through the origin.

6 **That is graph C**

That is graph **C**.

✓ **FINAL ANSWER**

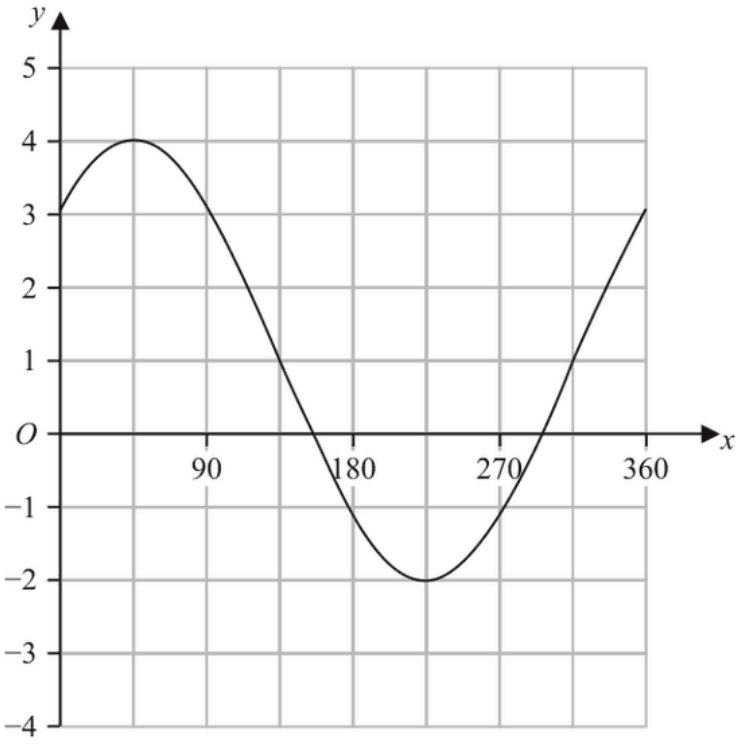
(a) B, (b) C.

4MA1/2HR JUNE 2025

# Question 25

3 marks | Graphs of Functions

**25** The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.




Find a suitable value for $a$, for $b$ and for $c$

$a =$ .....

$b =$ .....

$c =$ .....

**(Total for Question 25 is 3 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/2HR June 2025

**1** **Read the graph**

From the graph,

$$\text{maximum} = 4$$

$$\text{minimum} = -2$$

**2** **The amplitude is**

The amplitude is

$$a = \frac{4 - (-2)}{2} = 3$$

**3** **The centre line is**

The centre line is

$$c = \frac{4 + (-2)}{2} = 1$$

**4** **Use trigonometry**

The maximum is about $45°$. For a sine graph, a maximum happens when the angle inside the sine is $90°$, so

$$45 + b = 90$$

$$b = 45$$

✓ **FINAL ANSWER**

$a = 3,\ b = 45,\ c = 1$

# Question 12

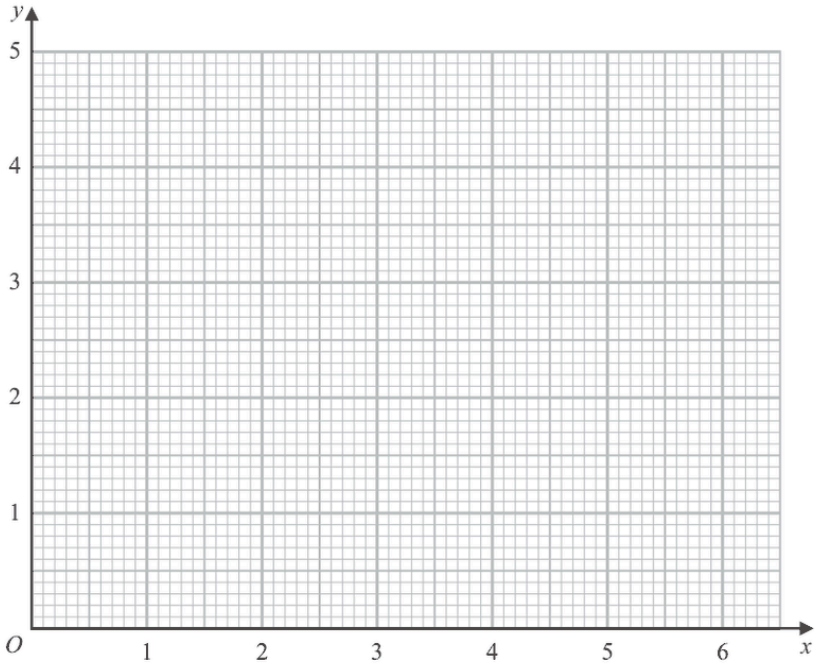
5 marks | Graphs of Functions

**12** (a) Complete the table of values for $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$

| $x$ | 0.5 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | 4.25 | | | 2.17 | 2.5 | 2.9 | 3.33 |

**(1)**

(b) Draw the graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ for $0.5 \leqslant x \leqslant 6$




**(2)**

(c) By drawing a suitable line on the grid, find estimates for the solutions of the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

.....

**(2)**

**(Total for Question 12 is 5 marks)**

**1** **Read the graph**

$$y = \frac{1}{2}\left(x + \frac{4}{x}\right)$$

**2** **The completed table is**

The completed table is:

| $x$ | 0.5 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | 4.25 | 2.5 | 2 | 2.17 | 2.5 | 2.9 | 3.33 |

**3** **Read the graph**

For

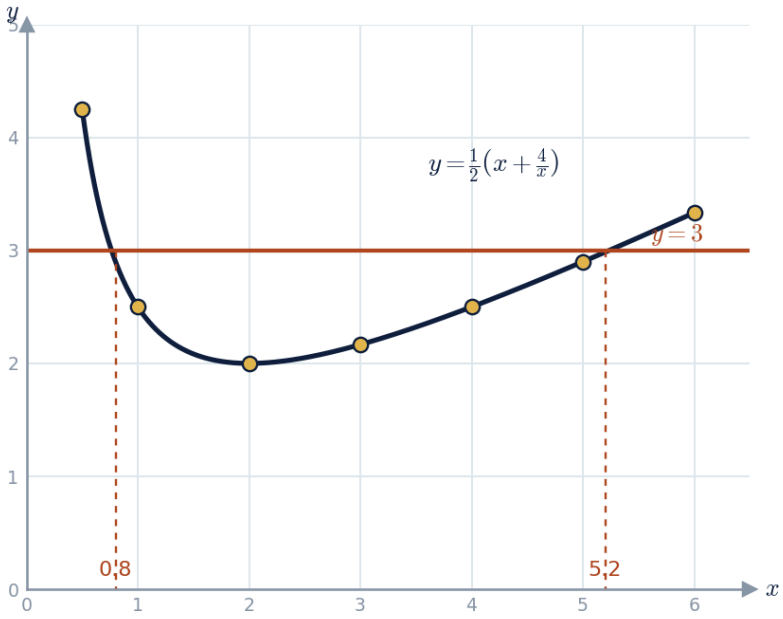
$$x + \frac{4}{x} = 6$$

we need

$$\frac{1}{2}\left(x + \frac{4}{x}\right) = 3$$

**4** **Read the graph**

So draw the horizontal line $y = 3$ on the graph.




**5** **The intersections give approximately**

The intersections give approximately

$$x = 0.8 \quad \text{and} \quad x = 5.2$$

**FINAL ANSWER**

Table missing values $2.5, 2$. Estimates $0.8$ and $5.2$.

# Question 13

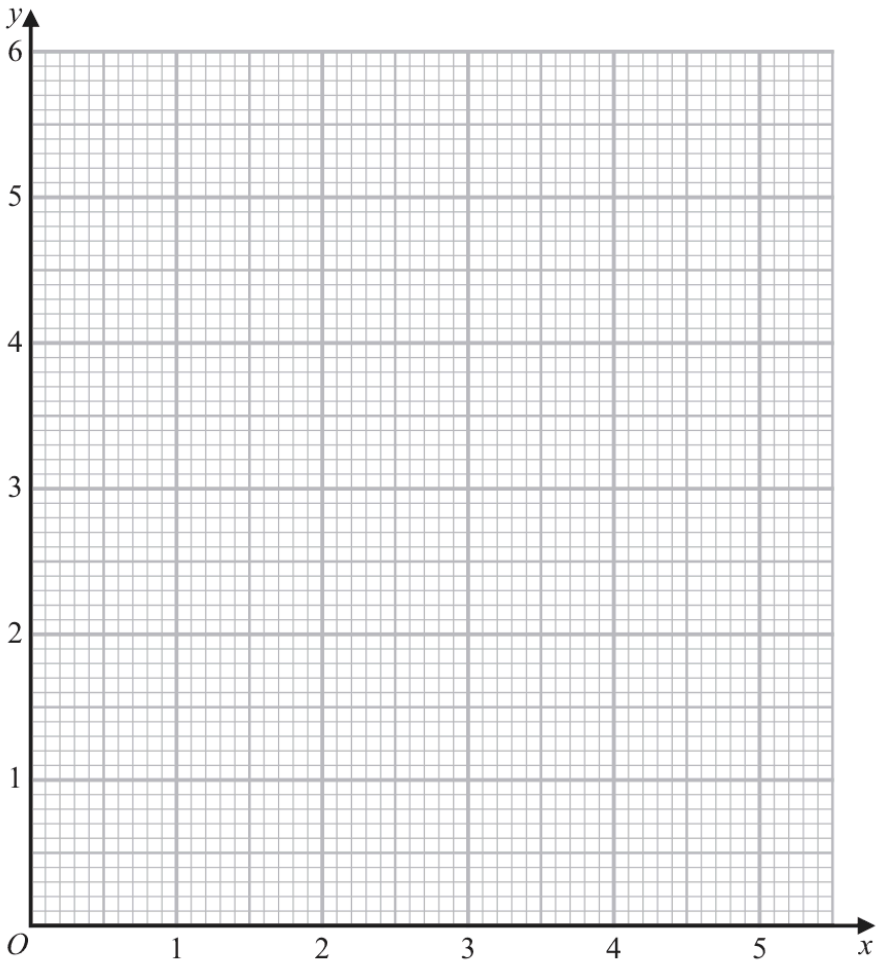
4 marks | Graphs of Functions

**13** (a) Complete the table of values for $y = x + \frac{1}{x}$

| $x$ | 0.25 | 0.5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|
| $y$ | 4.25 | | | 2.5 | 3.33 | | |

**(2)**

(b) On the grid, draw the graph of $y = x + \frac{1}{x}$ for $0.25 \leqslant x \leqslant 5$




**(2)**

**(Total for Question 13 is 4 marks)**

1 **Part (a) | Step 1**

Use $y = x + 1/x$:

$$y(0.5) = 0.5 + 2 = 2.5, \quad y(1) = 1 + 1 = 2, \quad y(4) = 4 + 0.25 = 4.25, \quad y(5) = 5 + 0.2 = 5.2.$$

Substitute each required x-value into the formula.

2 **Part (b) | Step 1**

Plot

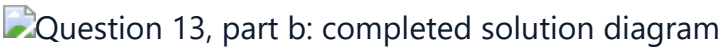
$$(0.25, 4.25),\ (0.5, 2.5),\ (1, 2),\ (2, 2.5),\ (3, 10/3),\ (4, 4.25),\ (5, 5.2).$$

Use x horizontally and y vertically on the supplied grid; 10/3 is approximately 3.33.

3 **Part (b) | Step 2**

Join the points with one smooth curve for $0.25 \le x \le 5$, descending to its minimum $(1, 2)$ and then ascending.

The graph is curved, not a set of straight segments, and includes both endpoints.



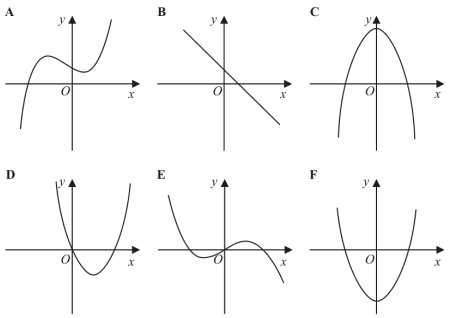

✓ **FINAL ANSWER**

(a) Table entries, in order: $2.5, 2, 4.25, 5.2$. (b) The smooth curve $y = x + 1/x$ through the stated coordinates on $[0.25, 5]$.

# Question 14

6 marks | Graphs of Functions

**14** Here are six graphs.




(a) Write down the letter of the graph that could have the equation

(i) $y = x^2 - 3x$

.....

(1)

(ii) $y = -x^3 + 9x$

.....

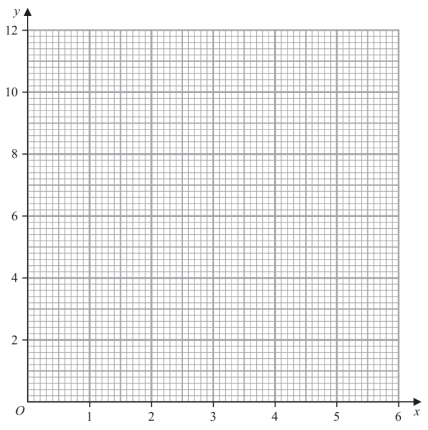
(1)

(b) Complete the table of values for $y = \frac{12}{x^2}$

| $x$ | 1 | 1.5 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $y$ | | 5.33 | | 1.33 | | | 0.33 |

(2)

(c) On the grid, draw the graph of $y = \frac{12}{x^2}$ for $1 \leqslant x \leqslant 6$




(2)

**(Total for Question 14 is 6 marks)**

**1** **(a)(i) Positive quadratic**

$y = x^2 - 3x = x(x-3)$: U-shape through $O$ and $x = 3$ → graph D.

**2** **(a)(ii) Negative cubic**

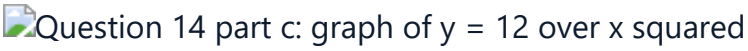
$y = -x^3 + 9x$: negative cubic through the origin → graph E.

**3** **(b) Substitute**

$$x = 1 : \ 12, \quad x = 2 : \ 3, \quad x = 4 : \ 0.75, \quad x = 5 : \ 0.48$$

**4** **(c) Plot and join with a smooth curve**

Plot all 7 points and draw a smooth curve (no straight segments).




✓ **FINAL ANSWER**

(a)(i) D (ii) E (b) $12,\ 3,\ 0.75,\ 0.48$ (c) smooth curve through the points



**TOPIC**

# Kinematic Graphs

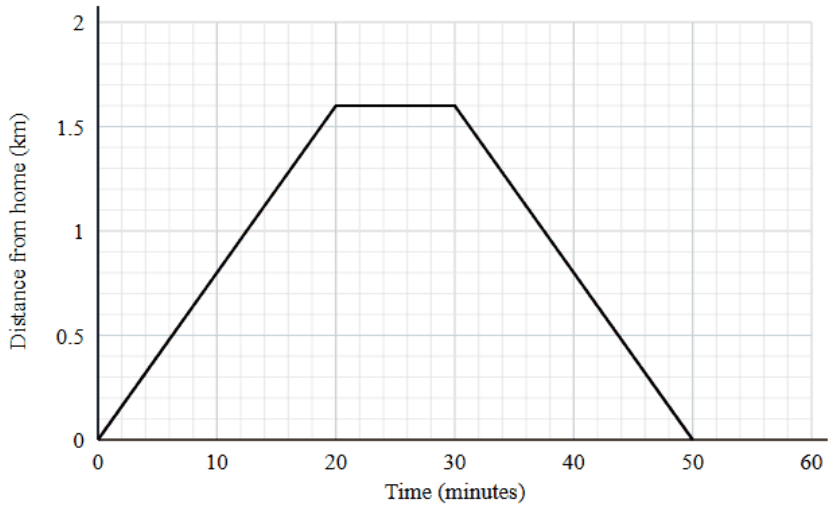
Every past-paper question on this topic, 2020 to 2026.

EP SPECIMEN

# Question 8

4 marks | Kinematic Graphs

**8** Ali walked from his home to a shop. He stayed at the shop and then walked back home.
Here is the distance-time graph for his journey.




(a) How far was Ali from home after 10 minutes?

..... km

**(1)**

(b) How many minutes did Ali stay at the shop?

..... minutes

**(1)**

(c) Work out Ali's speed, in kilometres per hour, on his walk back home.

..... km/h

**(2)**

**(Total for Question 8 is 4 marks)**

## WORKED SOLUTION

Question 8 · EP Specimen

**1** **(a) Read the graph at 10 minutes**

The line from $(0, 0)$ to $(20, 1.6)$ is halfway up at $10$ minutes: $0.8$ km.

**2** **(b) The flat section**

From $20$ to $30$ minutes: $10$ minutes.

**3** **(c) Speed = distance ÷ time**

He walks $1.6$ km in $50 - 30 = 20$ minutes $= \frac{1}{3}$ hour.

$$1.6 \div \frac{1}{3} = 4.8 \text{ km/h}$$

---

✓ **FINAL ANSWER**

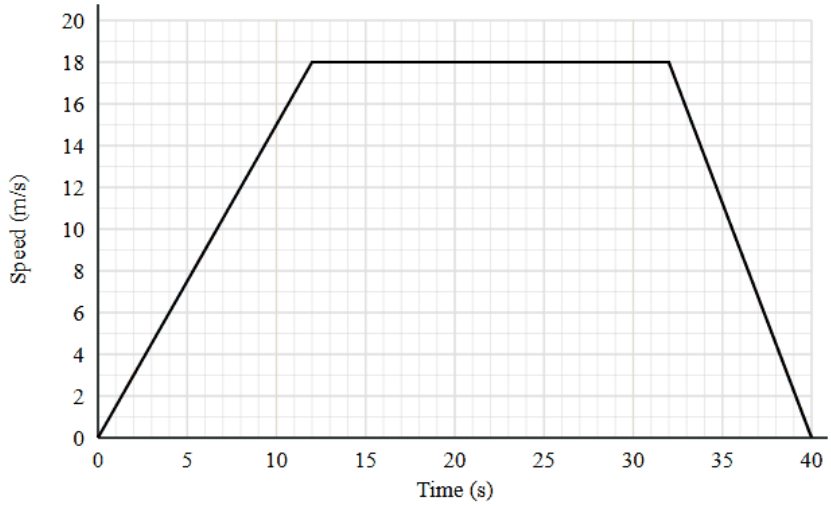
(a) $0.8$ km (b) $10$ minutes (c) $4.8$ km/h

EP SPECIMEN

# Question 9

5 marks | Kinematic Graphs

**9** Here is the speed-time graph for the first 40 seconds of a car's journey.




(a) Work out the acceleration of the car during the first 12 seconds.

..... m/s 2

**(2)**

(b) Work out the total distance the car travels in the 40 seconds.

..... m

**(3)**

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · EP Specimen

**1** **(a) Acceleration = gradient**

$$\frac{18-0}{12} = 1.5\ \text{m/s}^2$$

**2** **(b) Distance = area under the graph**

Split into a triangle, a rectangle and a triangle:

$$\tfrac{1}{2} \times 12 \times 18 + 20 \times 18 + \tfrac{1}{2} \times 8 \times 18 = 108 + 360 + 72 = 540\ \text{m}$$

✓ **FINAL ANSWER**

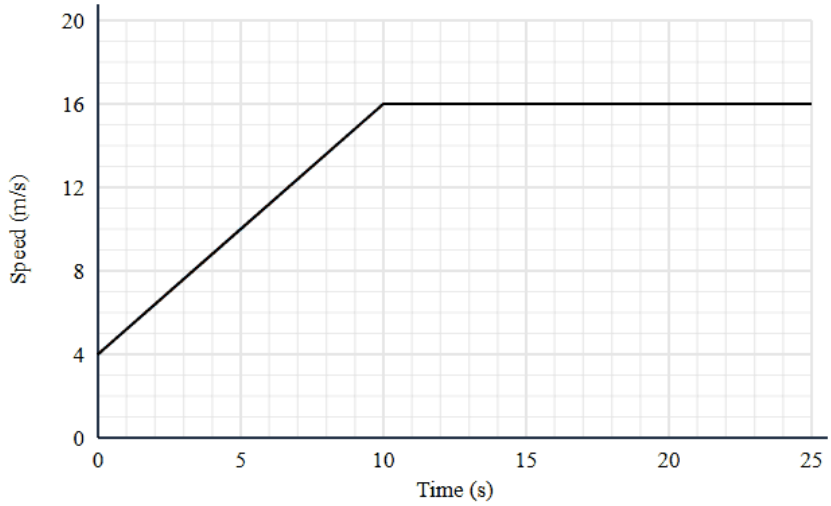
(a) $1.5$ m/s$^2$ (b) $540$ m

EP SPECIMEN

# Question 10

4 marks | Kinematic Graphs

**10** Here is the speed-time graph for 25 seconds of a cyclist's journey.




(a) Work out the distance the cyclist travels in the 25 seconds.

..... m

**(3)**

(b) Work out the average speed of the cyclist for the 25 seconds.

..... m/s

**(1)**

**(Total for Question 10 is 4 marks)**

## WORKED SOLUTION

Question 10 · EP Specimen

**1** **(a) Area of the trapezium (first 10 s)**

$$\frac{4+16}{2} \times 10 = 100 \text{ m}$$

**2** **(a) Area of the rectangle (last 15 s)**

$$16 \times 15 = 240 \text{ m}$$

Total: $100 + 240 = 340$ m.

**3** **(b) Average speed = total distance ÷ total time**

$$340 \div 25 = 13.6 \text{ m/s}$$

**FINAL ANSWER**

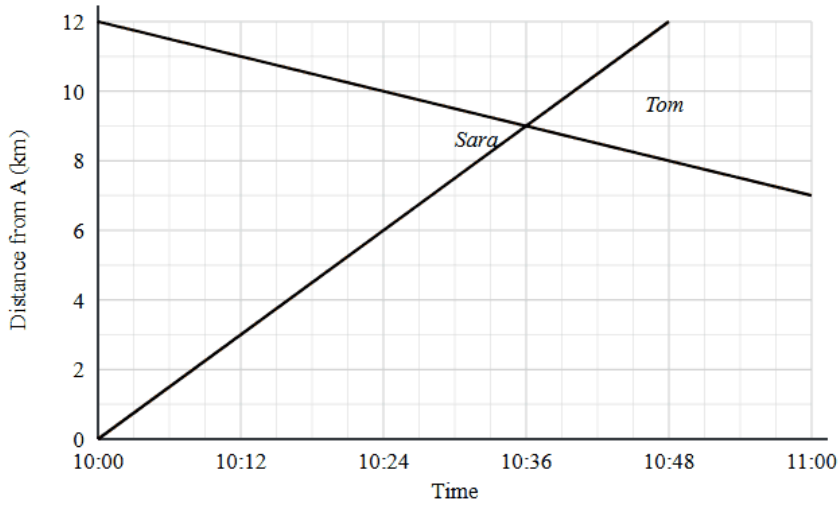
(a) $\mathbf{340}$ m (b) $\mathbf{13.6}$ m/s

EP SPECIMEN

# Question 11

4 marks | Kinematic Graphs

**11** At 10:00 Sara leaves town $A$ and cycles towards town $B$.
At 10:00 Tom leaves town $B$ and walks towards town $A$.
The graph shows their distances from town $A$.




(a) At what time do Sara and Tom pass each other?

.....

**(1)**

(b) How far from town $A$ are they when they pass each other?

..... km

**(1)**

(c) Work out Tom's speed in kilometres per hour.

..... km/h

**(2)**

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · EP Specimen

**1** **(a) Where the lines cross**

The lines cross at $36$ minutes after 10:00, so at $10{:}36$.

**2** **(b) Read across**

At the crossing point the distance from $A$ is $9$ km.

### 3 (c) Tom's speed

Tom goes from $12$ km to $7$ km from $A$ in $1$ hour:

$$\frac{12-7}{1} = 5\ \text{km/h}$$

---

**FINAL ANSWER**

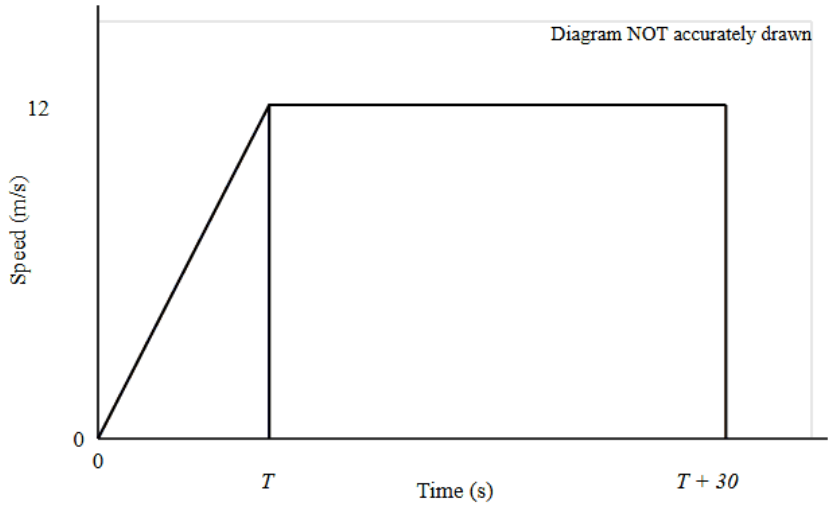
(a) $10{:}36$ (b) $9$ km (c) $5$ km/h

EP SPECIMEN

# Question 12

3 marks | Kinematic Graphs

**12** A particle starts from rest. It speeds up at a constant rate for $T$ seconds until its speed is 12 m/s.
It then travels at 12 m/s for 30 seconds.
The total distance it travels is 450 m.




Work out the value of $T$.

$T =$ .....

**(3)**

**(Total for Question 12 is 3 marks)**

## WORKED SOLUTION

Question 12 · EP Specimen

**1** **Distance = area under the graph**

Triangle: $\frac{1}{2} \times T \times 12 = 6T$. Rectangle: $30 \times 12 = 360$.

**2** **Form and solve the equation**

$$6T + 360 = 450 \Rightarrow 6T = 90 \Rightarrow T = 15$$

✓ **FINAL ANSWER**

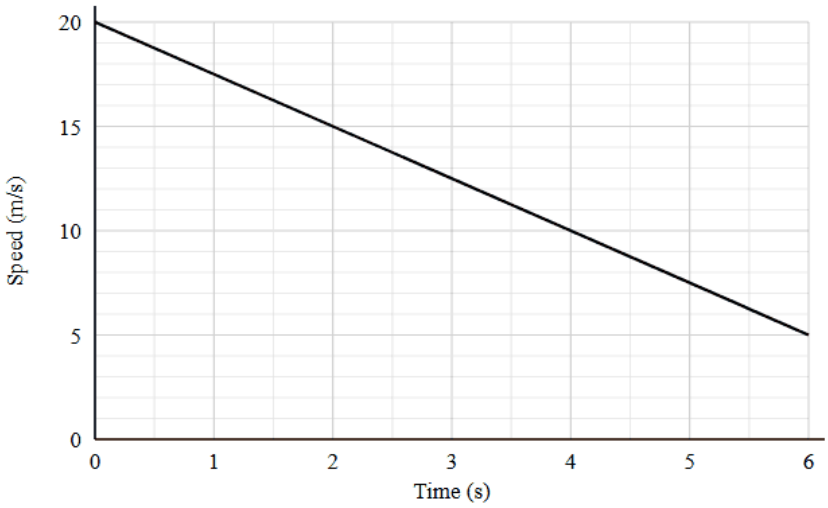
$T = 15$

EP SPECIMEN

# Question 13

4 marks | Kinematic Graphs

**13** A car is travelling at 20 m/s. The driver brakes and the car slows down at a constant rate. Here is the speed-time graph for the next 6 seconds.




(a) Work out the deceleration of the car.

..... m/s 2

**(2)**

(b) Work out the distance the car travels in the 6 seconds.

..... m

**(2)**

**(Total for Question 13 is 4 marks)**

## WORKED SOLUTION

Question 13 · EP Specimen

**1** **(a) Gradient of the line**

$$\frac{5-20}{6} = -2.5$$

So the deceleration is $2.5$ m/s$^2$.

**2** **(b) Area of the trapezium**

$$\frac{20+5}{2} \times 6 = 75 \text{ m}$$

**FINAL ANSWER**

(a) $2.5$ m/s$^2$ (b) $75$ m

EP SPECIMEN

# Question 14

4 marks | Kinematic Graphs

**14** A train starts from rest and speeds up at a constant rate for 50 seconds until its speed is 30 m/s.
It then travels at 30 m/s for $t$ seconds.
It then slows down at a constant rate and stops 40 seconds later.
The total distance the train travels is 6 km.

Work out the value of $t$.

$t =$ .....

**(4)**

**(Total for Question 14 is 4 marks)**

## WORKED SOLUTION

Question 14 · EP Specimen

**1 Sketch the speed-time graph**

A trapezium: up for $50$ s, flat at $30$ m/s for $t$ s, down for $40$ s.

**2 Distance = area (in metres)**

$$\tfrac{1}{2} \times 50 \times 30 + 30t + \tfrac{1}{2} \times 40 \times 30 = 750 + 30t + 600$$

**3 Form and solve the equation**

$6$ km $= 6000$ m:

$$1350 + 30t = 6000 \Rightarrow 30t = 4650 \Rightarrow t = 155$$

**FINAL ANSWER**

$t = 155$

TOPIC

# Perimeter & Area

Every past-paper question on this topic, 2020 to 2026.

# Question 6

4 marks | Perimeter & Area

**6** The diagram shows an 8-sided shape *ABCDEFGH*

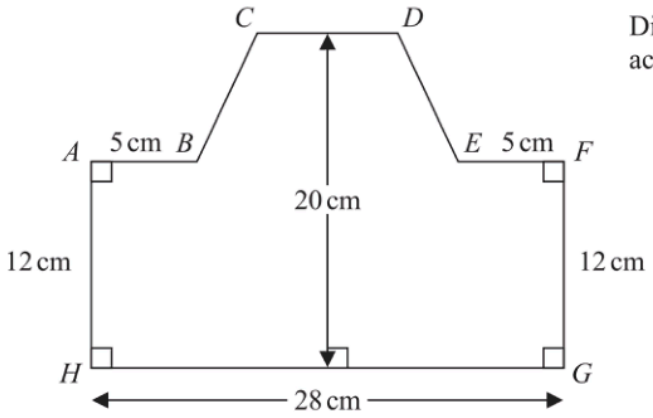



Diagram **NOT** accurately drawn

$HG = 28\,\text{cm}$ $AH = FG = 12\,\text{cm}$ $AB = EF = 5\,\text{cm}$
The height of the shape is 20 cm
*CD* is parallel to *HG*

The area of shape *ABCDEFGH* is $434\,\text{cm}^2$

Find the length of *CD*

## 1 Find the rectangular part

The rectangle at the bottom has width $28$ cm and height $12$ cm.

$$28 \times 12 = 336$$

## 2 Find the top area

$$434 - 336 = 98$$

## 3 Set up the trapezium

The top trapezium has height

$$20 - 12 = 8$$

Its lower parallel side is

$$28 - 5 - 5 = 18$$

## 4 Solve for the missing length

Let $CD = x$.

$$\frac{1}{2}(x + 18) \times 8 = 98$$

$$4(x + 18) = 98$$

$$x = 6.5$$

**FINAL ANSWER**

$CD = 6.5$ cm.

4MA1/1H JANUARY 2020

# Question 3

4 marks | Perimeter & Area

3 The diagram shows a shape.

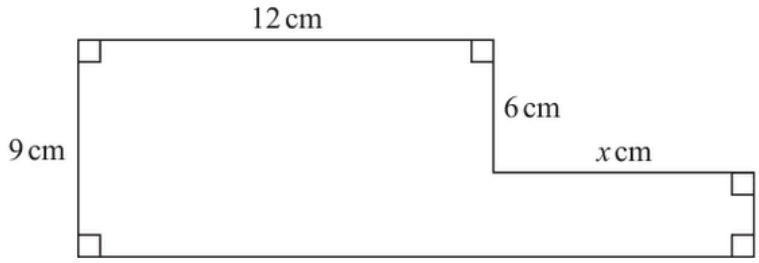



Diagram **NOT** accurately drawn

The shape has area 129 cm$^2$

Work out the value of $x$.

$x =$ .....

**(Total for Question 3 is 4 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1H January 2020

**1 Split the shape into two**

Split the shape into two rectangles.

**2 The left rectangle has area**

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

**3 The right rectangle has height**

The right rectangle has height

$$9 - 6 = 3 \text{ cm}$$

**4 Calculate area**

So its area is

$$3x$$

**5** **Total area**

Total area:

$$108 + 3x = 129$$

$$3x = 21$$

$$x = 7$$

---

✓ **FINAL ANSWER**

$7\ \text{cm}$.

4MA1/1HR JANUARY 2020

# Question 5

3 marks | Perimeter & Area

5 The diagram shows Yuen's garden.

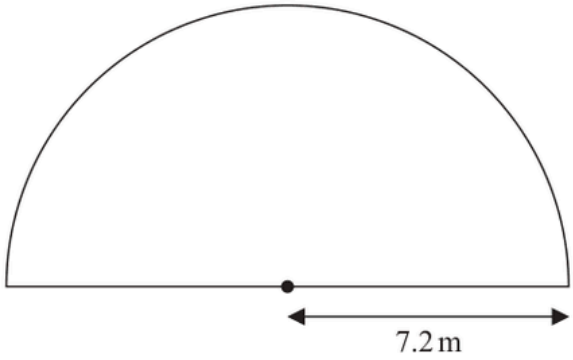



Diagram **NOT** accurately drawn

The garden is in the shape of a semicircle of radius 7.2 m.
Yuen is going to cover his garden with grass seed.

Yuen has 12 boxes of grass seed.
Each box of grass seed contains enough seed to cover 6 m² of the garden.

Has Yuen enough grass seed for his garden?
Show your working clearly.

**(Total for Question 5 is 3 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1HR January 2020

**1 The garden is a semicircle**

The garden is a semicircle of radius $7.2\ \text{m}$.

$$\text{area} = \frac{1}{2}\pi r^2$$

$$\text{area} = \frac{1}{2}\pi(7.2)^2 = 81.43\ldots\ \text{m}^2$$

**2 Calculate area**

Yuen has 12 boxes, and each box covers $6\ \text{m}^2$:

$$12 \times 6 = 72\ \text{m}^2$$

**3 Calculate area**

Since

$$72 < 81.43\ldots$$

4

### Yuen does not have enough

Yuen does not have enough grass seed.

---

✓ **FINAL ANSWER**

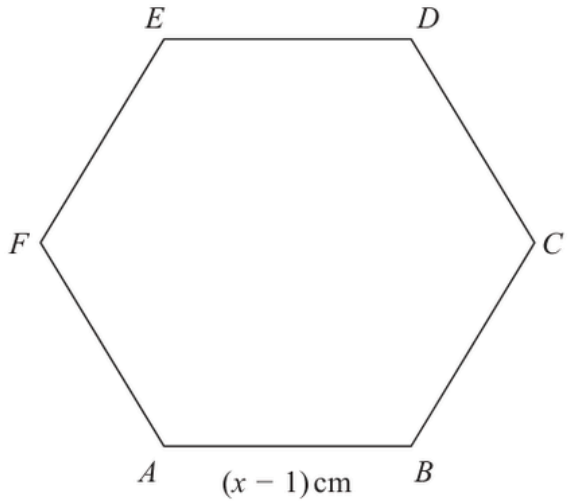
No, he does not have enough grass seed.

4MA1/1HR JANUARY 2020

# Question 10

5 marks | Perimeter & Area

**10** The diagram shows a regular hexagon, $ABCDEF$, and an isosceles triangle, $GHJ$.



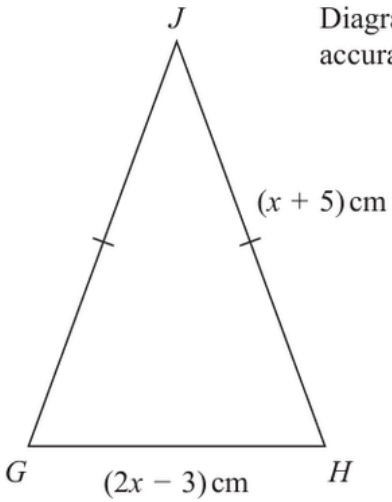




Diagram **NOT** accurately drawn

The perimeter of the hexagon is equal to the perimeter of the triangle.

Find the length of each side of the hexagon.
Show clear algebraic working.

..... cm

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1HR January 2020

**1** **The regular hexagon has side**

The regular hexagon has side length $(x - 1)$ cm, so its perimeter is

$$6(x - 1)$$

**2** **The isosceles triangle has perimeter**

The isosceles triangle has perimeter

$$(x + 5) + (x + 5) + (2x - 3) = 4x + 7$$

### 3 The perimeters are equal

The perimeters are equal:

$$6(x-1) = 4x + 7$$

$$6x - 6 = 4x + 7$$

$$2x = 13$$

$$x = 6.5$$

---

### 4 Each side of the hexagon

Each side of the hexagon is

$$x - 1 = 6.5 - 1 = 5.5$$

---

**FINAL ANSWER**

$5.5\ \text{cm}$.

4MA1/2H JANUARY 2020

# Question 11

5 marks | Perimeter & Area

**11** The diagram shows a shaded shape $ABCD$ made from a semicircle $ABC$ and a right-angled triangle $ACD$.

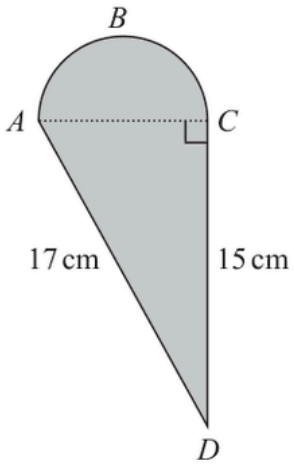



Diagram **NOT** accurately drawn

$AC$ is the diameter of the semicircle $ABC$.

Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 11 is 5 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2H January 2020

**1 Use trigonometry**

Triangle $ACD$ is right-angled, with

$$AD = 17 \text{ cm}, \qquad CD = 15 \text{ cm}$$

**2 Use Pythagoras to find**

Use Pythagoras to find $AC$:

$$AC^2 = 17^2 - 15^2 = 289 - 225 = 64$$

$$AC = 8 \text{ cm}$$

**3** **Calculate area**

So the semicircle has radius

$$r = 4 \text{ cm}$$

**4** **The arc length of the**

The arc length of the semicircle is

$$\pi r = 4\pi$$

**5** **The perimeter of the shaded**

The perimeter of the shaded shape is

$$AD + CD + \text{semicircle arc} = 17 + 15 + 4\pi = 44.566\ldots$$

✓ **FINAL ANSWER**

$44.6 \text{ cm}$.

4MA1/1H MAY/NOVEMBER 2020

# Question 11

6 marks | Perimeter & Area

**11** The diagram shows trapezium $ABCD$ in which $BC$ and $AD$ are parallel.

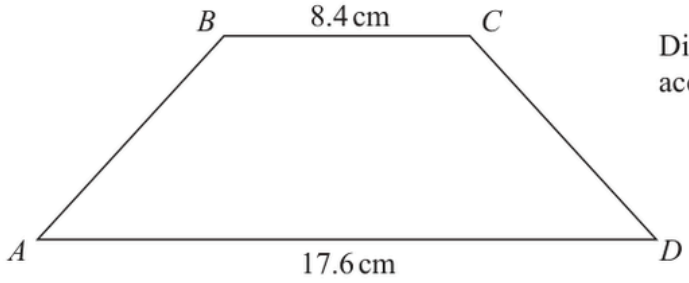



Diagram **NOT** accurately drawn

The trapezium has exactly one line of symmetry.

$BC = 8.4\,\text{cm}$
$AD = 17.6\,\text{cm}$

The trapezium has area $179.4\,\text{cm}^2$

Work out the size of angle $ABC$.
Give your answer correct to 1 decimal place.

.....°

**(Total for Question 11 is 6 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H May/November 2020

**1** **The average of the parallel**

The average of the parallel sides is

$$\frac{8.4 + 17.6}{2} = 13$$

**2** **Find the gradient**

So the perpendicular height is

$$179.4 \div 13 = 13.8$$

**3** **The horizontal offset on each**

The horizontal offset on each side is

$$\frac{17.6 - 8.4}{2} = 4.6$$

**4** **Use trigonometry**

The acute angle between $AB$ and the horizontal is

$$\tan^{-1}\left(\frac{13.8}{4.6}\right) = 71.565\ldots^\circ$$

**5** **Use trigonometry**

So the interior angle $ABC$ is

$$180^\circ - 71.565\ldots^\circ = 108.434\ldots^\circ$$

✓ **FINAL ANSWER**

$108.4^\circ$.

4MA1/1HR MAY/NOVEMBER 2020

# Question 22

6 marks | Perimeter & Area

**22** The diagram shows a regular octagon *ABCDEFGH*.

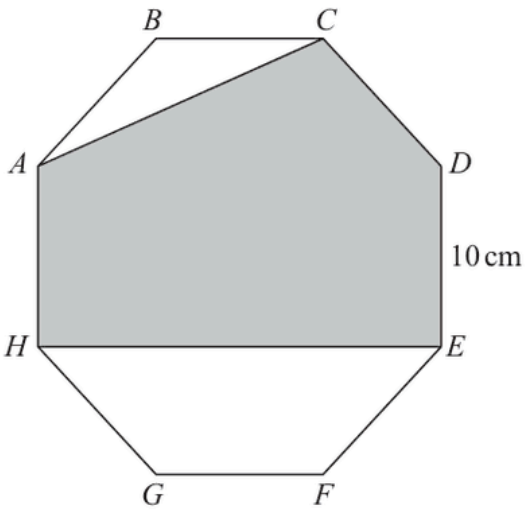



Diagram **NOT** accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region *ACDEH*.
Give your answer correct to the nearest $cm^2$

..... $cm^2$

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1HR May/November 2020

**1** **Each side of the regular**

Each side of the regular octagon is $10$ cm.

**2** **Calculate area**

For a regular octagon with side length $s$,

$$\text{area} = 2(1+\sqrt{2})s^2$$

**3** **Calculate area**

So the area of the whole octagon is

$$2(1+\sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

**4** **The unshaded triangle has**

The unshaded triangle $ABC$ has

$$AB = 10, \qquad BC = 10, \qquad \angle ABC = 135^\circ$$

**5** **Use trigonometry**

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10)\sin 135^\circ = 25\sqrt{2}$$

**6** **Find the gradient**

At the bottom, the unshaded trapezium $HEFG$ has parallel sides

$$HE = 10 + 10\sqrt{2}, \qquad GF = 10$$

**7** **Simplify surd**

and height

$$5\sqrt{2}$$

**8** **Calculate area**

So

$$\text{area of trapezium } HEFG = \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2})$$

$$= 50 + 50\sqrt{2}$$

**9** **The shaded area is**

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

$$= 326.7766\ldots$$

**FINAL ANSWER**

$\boxed{327\ \text{cm}^2}$

4MA1/2HR JANUARY 2021

# Question 2

3 marks | Perimeter & Area

**2** The diagram shows a rectangle $ABCD$ and a semicircle with diameter $AB$ where $AB = 12\,\text{cm}$. The point $E$ lies on $DC$ and also on the semicircle.

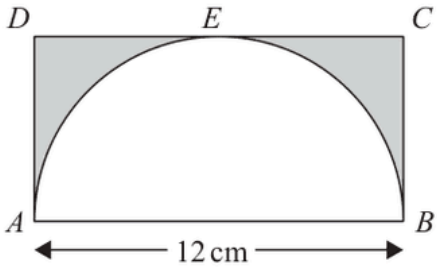



Diagram **NOT** accurately drawn

Work out the area of the shaded region.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2HR January 2021

**1** **Calculate area**

The semicircle has diameter $12\ \text{cm}$, so its radius is $6\ \text{cm}$.

**2** **The rectangle has area**

The rectangle has area

$$12 \times 6 = 72$$

**3** **Area of the semicircle**

Area of the semicircle:

$$\frac{1}{2}\pi(6)^2 = 18\pi$$

**4** **Shaded area**

Shaded area:

$$72 - 18\pi = 15.451\ldots$$

**FINAL ANSWER**

$15.5\ \text{cm}^2$.

4MA1/2H JANUARY 2021

# Question 8

5 marks | Perimeter & Area

**8** $A$, $B$ and $C$ are points on a circle with centre $O$.

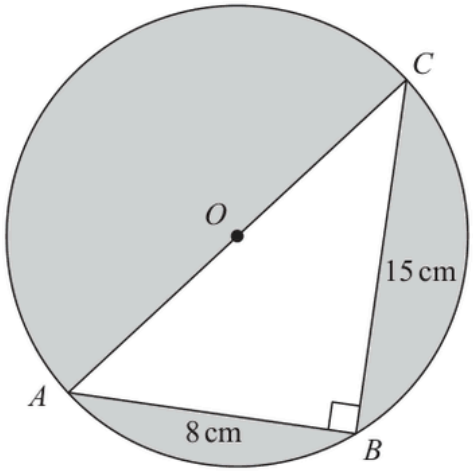



Diagram **NOT** accurately drawn

$AOC$ is a diameter of the circle.

$AB = 8\,\text{cm}$ $BC = 15\,\text{cm}$

Angle $ABC = 90°$

Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 8 is 5 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/2H January 2021

**1** **Triangle is right angled**

Triangle $ABC$ is right-angled:

$$AC^2 = 8^2 + 15^2 = 64 + 225 = 289$$

$$AC = 17\,\text{cm}$$

**2** **Calculate area**

Since $AOC$ is a diameter, the radius of the circle is

$$17 \div 2 = 8.5 \text{ cm}$$

**3** **Area of the circle**

Area of the circle:

$$\pi(8.5)^2 = 226.98\ldots \text{ cm}^2$$

**4** **Area of triangle**

Area of triangle $ABC$:

$$\frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$$

**5** **The shaded area is**

The shaded area is

$$226.98\ldots - 60 = 166.98\ldots$$

✓ **FINAL ANSWER**

$167 \text{ cm}^2$.

4MA1/1HR JANUARY 2021

# Question 10

5 marks | Perimeter & Area

**10** The diagram shows one face of a wall.
This face is in the shape of a pentagon with exactly one line of symmetry.

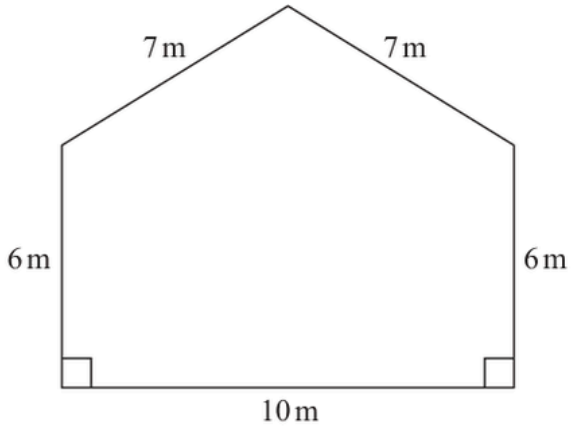



Diagram **NOT** accurately drawn

Omondi is going to paint this face of the wall once.
He has to buy all the paint that he needs to use.

The paint in each tin of paint Omondi is going to buy will cover $16\,m^2$ of the face of the wall.

Work out the least number of tins of paint Omondi will need to buy.
Show your working clearly.

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1HR January 2021

**1** **The wall is a rectangle**

The wall is a rectangle plus an isosceles triangle.

**2** **Rectangle area**

Rectangle area:

$$10 \times 6 = 60$$

**3** **Use trigonometry**

For the triangle, half the base is $5$ m.

$$h^2 = 7^2 - 5^2 = 24$$

$$h = \sqrt{24}$$

**4** **Triangle area**

Triangle area:

$$\frac{1}{2} \times 10 \times \sqrt{24} = 24.494\ldots$$

**5** **Total area**

Total area:

$$60 + 24.494\ldots = 84.494\ldots$$

**6** **Each tin covers**

Each tin covers $16\ \mathrm{m}^2$:

$$\frac{84.494\ldots}{16} = 5.28\ldots$$

**7** **Calculate area**

Omondi must buy whole tins, so he needs 6 tins.

✓ **FINAL ANSWER**

$6$ tins.

4MA1/2H JANUARY 2021

# Question 20

4 marks | Perimeter & Area

**20**

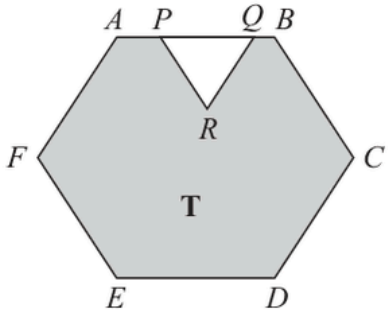



Diagram **NOT** accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle $PQR$ from a regular hexagon $ABCDEF$.

The points $P$ and $Q$ lie on $AB$ such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3}$ cm$^2$

work out the length of $PQ$.

..... cm

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H January 2021

**1 Let**

Let

$$PQ = p$$

**2 Given**

Given

$$AB = 1.5PQ$$

**3 Find the gradient**

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

**4** **Area of a regular hexagon**

Area of a regular hexagon with side length $s$ is

$$\frac{3\sqrt{3}}{2}s^2$$

**5** **Find the gradient**

So the area of the hexagon is

$$\frac{3\sqrt{3}}{2}\left(\frac{3}{2}p\right)^2$$

$$=\frac{27\sqrt{3}}{8}p^2$$

**6** **Find the gradient**

The removed triangle $PQR$ is equilateral, so its area is

$$\frac{\sqrt{3}}{4}p^2$$

**7** **The shaded area is**

The shaded area is

$$\frac{27\sqrt{3}}{8}p^2-\frac{\sqrt{3}}{4}p^2$$

$$=\frac{25\sqrt{3}}{8}p^2$$

**8** **Find the gradient**

This is $72\sqrt{3}$, so

$$\frac{25\sqrt{3}}{8}p^2=72\sqrt{3}$$

$$\frac{25}{8}p^2=72$$

$$p^2=\frac{576}{25}$$

$$p=4.8$$

**FINAL ANSWER**

$PQ = 4.8\ \text{cm}$

4MA1/1H MAY 2021

# Question 3

4 marks | Perimeter & Area

3

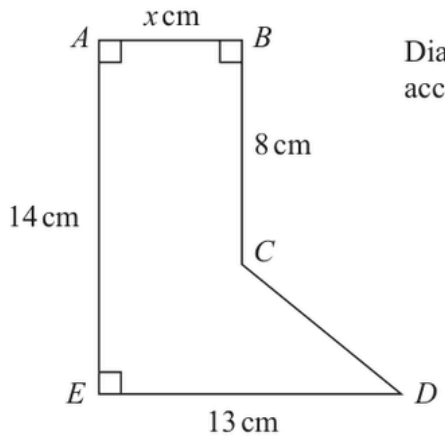



Diagram **NOT** accurately drawn

The diagram shows the shape $ABCDE$.

The area of the shape is $91.8\,\text{cm}^2$

Work out the value of $x$.

$x =$ .....

**(Total for Question 3 is 4 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1H May 2021

**1 Split the shape into a**

Split the shape into a rectangle and a triangle.

**2 Rectangle area**

Rectangle area:

$$14x$$

**3 Use trigonometry**

The triangle has base $13 - x$ and height $6$, so its area is

$$\frac{1}{2}(13 - x)(6) = 39 - 3x$$

**4** **Total area**

Total area:

$$14x + 39 - 3x = 91.5$$

$$11x = 52.5$$

$$x = 4.772\ldots$$

---

✓ **FINAL ANSWER**

$4.77\ \text{cm}$.

4MA1/1H NOVEMBER 2021

# Question 4

5 marks | Perimeter & Area

**4** The diagram shows the front of a wooden door with a semicircular glass window.

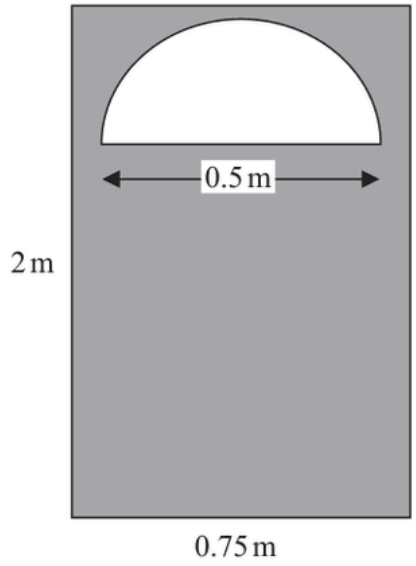



Diagram **NOT** accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers $4\,m^2$ of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

**(Total for Question 4 is 5 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1H November 2021

**1** **Area of the rectangular door**

Area of the rectangular door:

$$0.75 \times 2 = 1.5 \text{ m}^2$$

**2** **The semicircular window has diameter**

The semicircular window has diameter $0.5 \text{ m}$, so

$$r = 0.25 \text{ m}$$

### 3 Area of the semicircular window

Area of the semicircular window:

$$\frac{1}{2}\pi(0.25)^2 = 0.098174\ldots \text{ m}^2$$

### 4 Shaded area for one coat

Shaded area for one coat:

$$1.5 - 0.098174\ldots = 1.401825\ldots \text{ m}^2$$

### 5 Calculate area

For 2 coats:

$$2 \times 1.401825\ldots = 2.80365\ldots \text{ m}^2$$

### 6 Calculate area

Since 250 ml covers $4 \text{ m}^2$, varnish needed is

$$\frac{2.80365\ldots}{4} \times 250 = 175.22\ldots$$

**FINAL ANSWER**

$175$ millilitres.

4MA1/2H NOVEMBER 2021

# Question 10

4 marks | Perimeter & Area

**10** $ABC$ is an isosceles triangle with $BA = BC$.

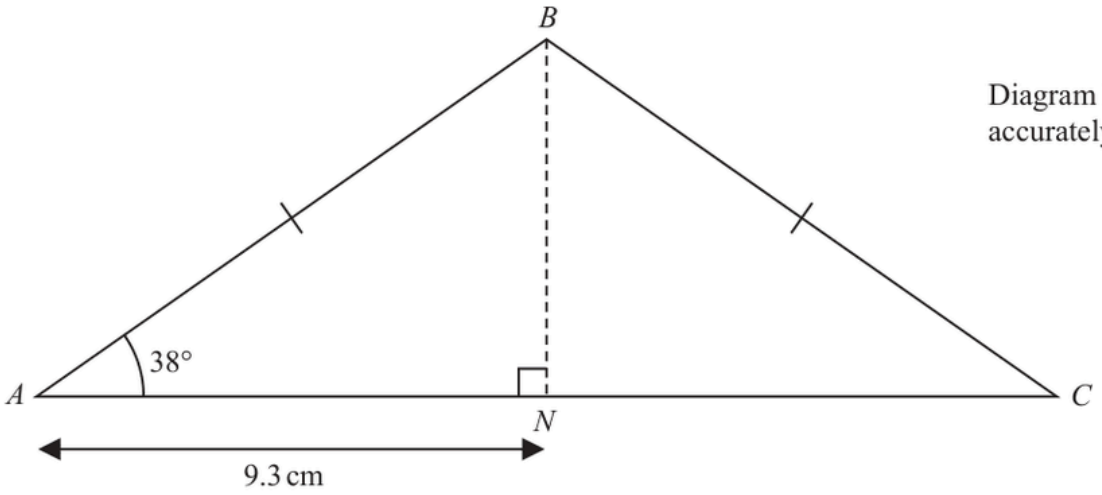



Diagram **NOT** accurately drawn

$N$ is the point on $AC$ such that $AN = 9.3$ cm and $BN$ is perpendicular to $AC$.

Work out the perimeter of triangle $ABC$.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 10 is 4 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2H November 2021

### 1 Use trigonometry

In right-angled triangle $ABN$,

$$\cos 38° = \frac{9.3}{AB}$$

$$AB = \frac{9.3}{\cos 38°} = 11.802\ldots$$

### 2 Calculate area

Since $ABC$ is isosceles,

$$BC = AB$$

**3** **Calculate area**

Also,

$$AC = 2 \times 9.3 = 18.6$$

---

**4** **Perimeter**

Perimeter:

$$11.802\ldots + 11.802\ldots + 18.6 = 42.204\ldots$$

---

✓ **FINAL ANSWER**

$42.2\ \text{cm}$.

4MA1/2H JUNE 2022

# Question 5

4 marks | Perimeter & Area

**5** The diagram shows an 8-sided shape *ABCDEFGH*.

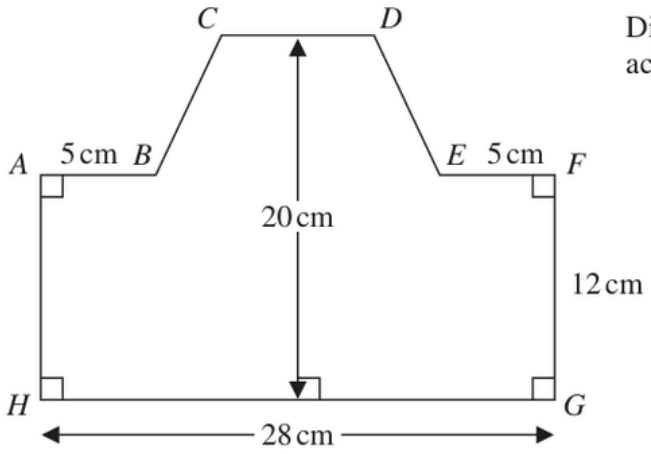



Diagram **NOT** accurately drawn

$HG = 28\,\text{cm}$ $FG = 12\,\text{cm}$ $AB = EF = 5\,\text{cm}$
The height of the shape is 20 cm
*CD* is parallel to *HG*

The area of shape *ABCDEFGH* is $434\,\text{cm}^2$

Find the length of *CD*.

..... cm

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2H June 2022

**1** **The lower rectangle has width**

The lower rectangle has width $28\ \text{cm}$ and height $12\ \text{cm}$:

$$28 \times 12 = 336\ \text{cm}^2$$

**2** **The top part is a**

The top part is a trapezium of height

$$20 - 12 = 8\ \text{cm}$$

**3** **Find the gradient**

The lower parallel side of this trapezium is

$$28 - 5 - 5 = 18\ \text{cm}$$

**4** **Area of the trapezium**

Area of the trapezium:

$$434 - 336 = 98 \text{ cm}^2$$

---

**5** **Use**

Use

$$\text{area} = \frac{1}{2}(a+b)h$$

$$98 = \frac{1}{2}(18+CD)(8)$$

$$98 = 4(18+CD)$$

$$18+CD = 24.5$$

$$CD = 6.5$$

---

✓ **FINAL ANSWER**

$6.5$ cm.

4MA1/2HR JUNE 2022

# Question 5

4 marks | Perimeter & Area

**5** The diagram shows a shape made from a square $ABCD$ and 4 identical semicircles.

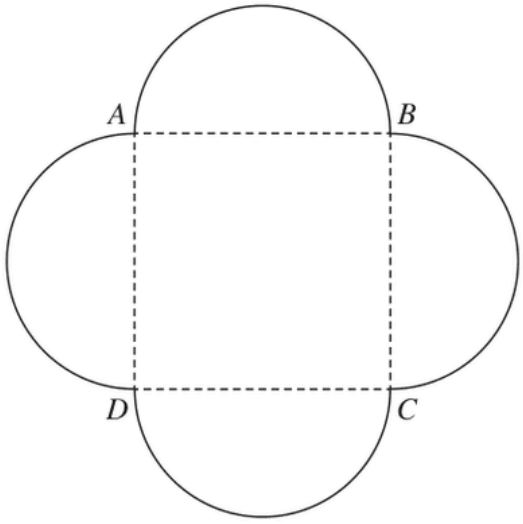



Diagram **NOT** accurately drawn

As shown in the diagram, the semicircles have $AB$, $BC$, $CD$ and $DA$ as diameters.

The area of the square is $36\,\text{cm}^2$

Calculate the total area of the shape.
Give your answer correct to one decimal place.

..... $\text{cm}^2$

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2HR June 2022

**1** **Calculate area**

The square has area $36\ \text{cm}^2$, so its side length is

$$\sqrt{36} = 6\ \text{cm}$$

**2** **Calculate area**

Each semicircle has diameter $6\ \text{cm}$, so radius

$$r = 3\ \text{cm}$$

### 3 Area of 4 semicircles

Area of 4 semicircles:

$$4 \times \frac{1}{2}\pi(3)^2 = 18\pi$$

### 4 Total area

Total area:

$$36 + 18\pi = 92.548\ldots$$

**FINAL ANSWER**

$92.5\ \text{cm}^2$.

# Question 11

6 marks | Perimeter & Area

**11** The diagram shows a quadrilateral $ABCD$

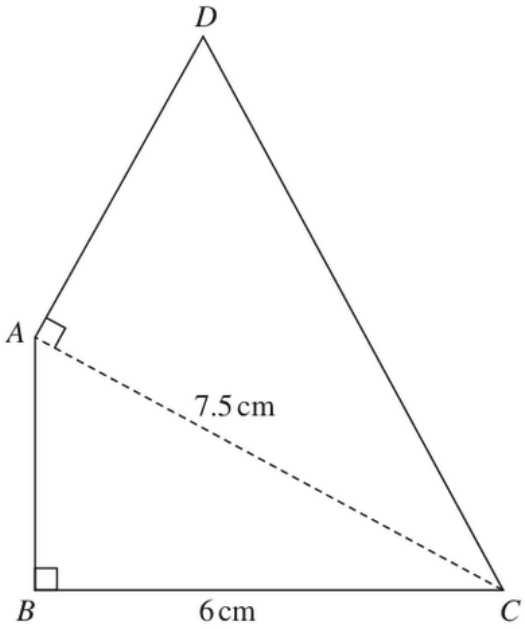



Diagram **NOT** accurately drawn

In the diagram, $ABC$ and $DAC$ are right-angled triangles.

$BC = 6\,\text{cm}$ $AC = 7.5\,\text{cm}$

The area of quadrilateral $ABCD$ is $31.5\,\text{cm}^2$

Work out the length of $AD$

.....cm

**(Total for Question 11 is 6 marks)**

**1** **Use trigonometry**

In right-angled triangle $ABC$,

$$AB^2 + BC^2 = AC^2$$

$$AB^2 + 6^2 = 7.5^2$$

$$AB^2 = 56.25 - 36 = 20.25$$

$$AB = 4.5 \text{ cm}$$

**2** **Area of triangle**

Area of triangle $ABC$:

$$\frac{1}{2} \times 6 \times 4.5 = 13.5 \text{ cm}^2$$

**3** **Area of triangle**

Area of triangle $ADC$:

$$31.5 - 13.5 = 18 \text{ cm}^2$$

**4** **Use trigonometry**

Triangle $ADC$ is right-angled, so

$$\frac{1}{2} \times AD \times 7.5 = 18$$

$$AD = \frac{36}{7.5} = 4.8$$

✓ **FINAL ANSWER**

$4.8 \text{ cm}$.

4MA1/2HR JUNE 2023

# Question 5

4 marks | Perimeter & Area

**5** A field is in the shape of a trapezium.

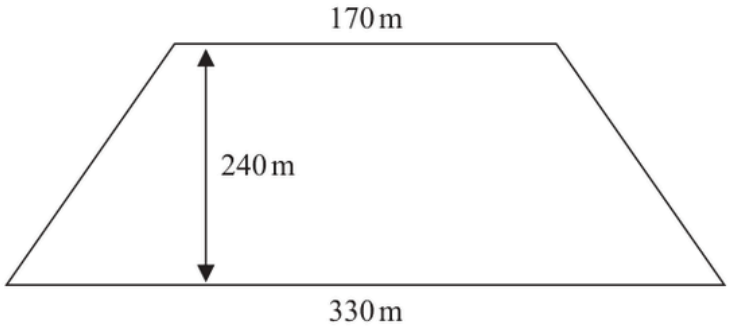



Diagram **NOT** accurately drawn

The field is sold for a price of \$49 650

Given that 1 hectare = 10 000 m²

work out the average price of the field per hectare.

\$.....

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2HR June 2023

### 1 Calculate area

$$\text{area} = \frac{1}{2}(170 + 330) \times 240 = 60000 \text{ m}^2$$

$$60000 \text{ m}^2 = 6 \text{ hectares}$$

$$\text{price per hectare} = 49650 \div 6 = 8275$$

**FINAL ANSWER**

\$8275.

4MA1/2H NOVEMBER 2023

# Question 10

5 marks | Perimeter & Area

**10** The diagram shows a hexagon $ABCDEF$

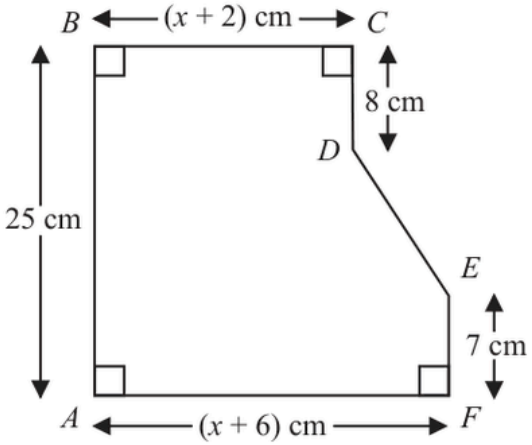



Diagram **NOT** accurately drawn

$AB = 25$ cm $\quad BC = (x + 2)$ cm $\quad CD = 8$ cm $\quad EF = 7$ cm $\quad AF = (x + 6)$ cm

The area of hexagon $ABCDEF$ is 258 cm²

Work out the value of $x$

$x =$ .....

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2H November 2023

**1 The area of the hexagon**

The area of the hexagon can be found from coordinates or by splitting the shape.

**2 Calculate area**

It simplifies to

$$\text{area} = 25x + 98$$

**3** **Calculate area**

Given the area is $258\ \text{cm}^2$,

$$25x + 98 = 258$$

$$25x = 160$$

$$x = 6.4$$

---

**FINAL ANSWER**

$x = 6.4$.

4MA1/2H JUNE 2024

# Question 7

4 marks | Perimeter & Area

7 The diagram shows a 6-sided shape $ABCDEF$

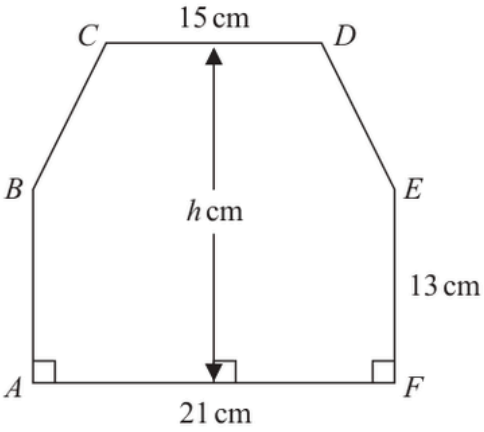



Diagram **NOT** accurately drawn

$AF = 21$ cm $\quad CD = 15$ cm $\quad AB = FE = 13$ cm

The perpendicular height of the shape is $h$ cm
$CD$ is parallel to $AF$

The area of the shape is $390\,\text{cm}^2$

Work out the value of $h$

$h =$ .....

**(Total for Question 7 is 4 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2H June 2024

**1 The bottom rectangle has area**

The bottom rectangle has area

$$21 \times 13 = 273$$

**2** **The top trapezium has height**

The top trapezium has height $h - 13$, with parallel sides $21$ and $15$.

$$273 + \frac{1}{2}(21 + 15)(h - 13) = 390$$

$$273 + 18(h - 13) = 390$$

$$18h + 39 = 390$$

$$h = 19.5$$

---

✓ **FINAL ANSWER**

$19.5\ \text{cm}$.

4MA1/1HR MAY 2024

# Question 8

6 marks | Perimeter & Area

**8** The diagram shows a trapezium, $ABCD$

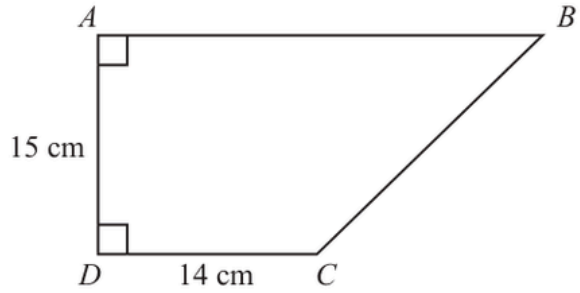



Diagram **NOT** accurately drawn

$DAB$ and $ADC$ are right angles.

$AD = 15$ cm  $DC = 14$ cm

The area of the trapezium is 360 cm²

Work out the perimeter of the trapezium.

..... cm

**(Total for Question 8 is 6 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1HR May 2024

**1** **Use the trapezium area formula**

Use the trapezium area formula:

$$360 = \frac{1}{2}(AB + 14)(15)$$

$$AB + 14 = 48$$

$$AB = 34$$

**2** **Use trigonometry**

Triangle $ABC$ is right-angled with horizontal length

$$34 - 14 = 20$$

### 3 Calculate area

So

$$BC = \sqrt{20^2 + 15^2} = 25$$

### 4 Perimeter

Perimeter:

$$15 + 14 + 25 + 34 = 88$$

**FINAL ANSWER**

$88\ \text{cm}$.

# Question 10

5 marks | Perimeter & Area

**10** The diagram shows a hexagon $ABCDEF$

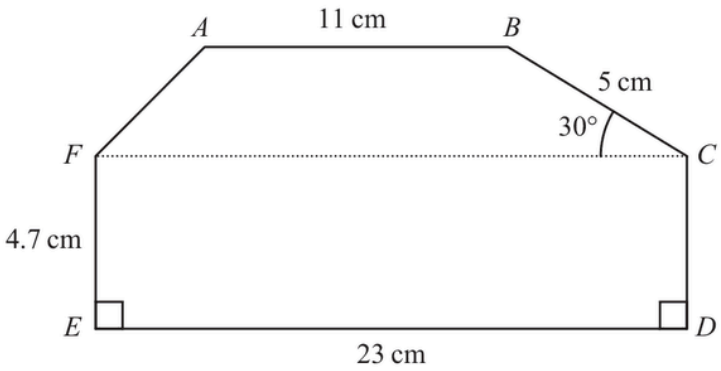



Diagram **NOT** accurately drawn

Angle $BCF = 30°$
$AB$, $FC$ and $ED$ are parallel.

Calculate the area of $ABCDEF$
Show your working clearly.

..... $cm^2$

**(Total for Question 10 is 5 marks)**

**1** **The rectangle has area**

The rectangle $EDCF$ has area

$$23 \times 4.7 = 108.1$$

**2** **The height of the top**

The height of the top trapezium is

$$5 \sin 30^\circ = 2.5$$

**3** **Area of the top trapezium**

Area of the top trapezium:

$$\frac{1}{2}(11 + 23) \times 2.5 = 42.5$$

**4** **Total area**

Total area:

$$108.1 + 42.5 = 150.6$$

✓ **FINAL ANSWER**

$150.6\ \text{cm}^2$.

4WM1/01 MAY 2025

# Question 4

3 marks | Perimeter & Area

**4** The diagram shows triangle $ABC$ and a semicircle.

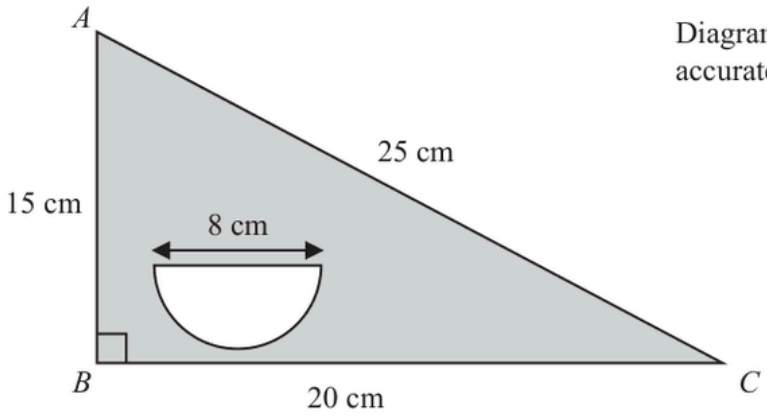



Diagram **NOT** accurately drawn

$AB = 15$ cm $\quad BC = 20$ cm $\quad AC = 25$ cm $\quad$ angle $ABC = 90°$

The diameter of the semicircle is 8 cm

Work out the area of the region shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4WM1/01 May 2025

**1** **The shaded area is the**

The shaded area is the triangle area minus the semicircle area.

$$\text{triangle area} = \frac{1}{2} \times 15 \times 20 = 150$$

**2** **Calculate area**

The semicircle has diameter $8$, so radius $4$.

$$\text{semicircle area} = \frac{1}{2}\pi(4)^2 = 8\pi$$

$$\text{shaded area} = 150 - 8\pi = 124.867\ldots$$

**3** **Calculate area**

To 3 significant figures,

$$125$$

---

**FINAL ANSWER**

$125\ \text{cm}^2$

4MA1/1H JUNE 2025

# Question 9

5 marks | Perimeter & Area

**9**

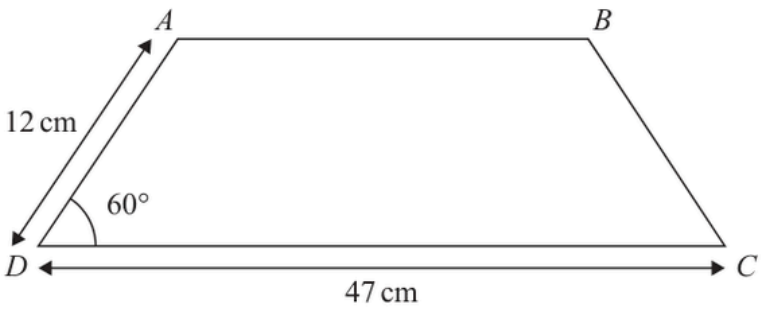



Diagram **NOT** accurately drawn

$ABCD$ is a trapezium with one line of symmetry.

angle $ADC = 60°$ $AD = 12\,\text{cm}$ $DC = 47\,\text{cm}$

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... $\text{cm}^2$

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1H June 2025

**1 Use trigonometry**

Because the trapezium has one line of symmetry, each sloping side makes the same $60°$ angle with the base.

**2 Height**

Height:

$$12\sin 60° = 10.392\ldots$$

**3 The top parallel side is**

The top parallel side is

$$47 - 2(12\cos 60°) = 35$$

**4** **Area**

Area:

$$\frac{1}{2}(47+35)(10.392\ldots) = 426.084\ldots$$

---

✓ **FINAL ANSWER**

$426\ \text{cm}^2$.

4WM1/01 MAY 2025

# Question 10

5 marks | Perimeter & Area

**10**

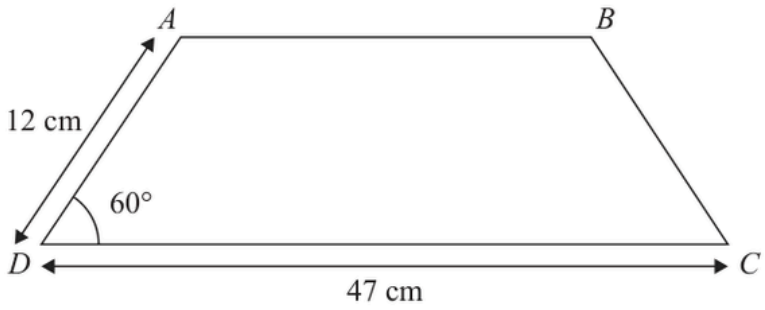



Diagram **NOT** accurately drawn

*ABCD* is a trapezium with one line of symmetry.

angle $ADC = 60°$ $AD = 12\,\text{cm}$ $DC = 47\,\text{cm}$

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... $\text{cm}^2$

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4WM1/01 May 2025

**1 The trapezium is symmetrical. The**

The trapezium is symmetrical. The horizontal shift from each sloping side is

$$12\cos 60° = 6$$

**2 Find the gradient**

So the top parallel side is

$$47 - 6 - 6 = 35$$

**3 The height is**

The height is

$$12\sin 60° = 10.392\ldots$$

### 4 Area of the trapezium

Area of the trapezium:

$$\frac{1}{2}(47+35)(12\sin 60°) = 426.084\ldots$$

### 5 Calculate area

To 3 significant figures,

$$426$$

**FINAL ANSWER**

$426\ \text{cm}^2$

4WM1/01R MAY 2025

# Question 19

5 marks | Perimeter & Area

**19** The diagram shows a pentagon $ABCDE$

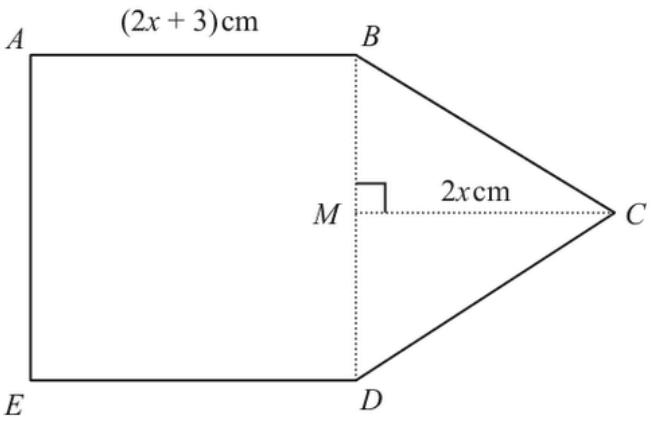



Diagram **NOT** accurately drawn

$ABCDE$ is made from a square $ABDE$ and a triangle $BCD$
$M$ is the midpoint of $BD$

$AB = (2x + 3)$ cm  $MC = 2x$ cm  angle $BMC = 90°$

The area of pentagon $ABCDE$ is $84$ cm$^2$

Work out the area of square $ABDE$
Show clear algebraic working.

..... cm$^2$

**(Total for Question 19 is 5 marks)**

## WORKED SOLUTION

Question 19 · 4WM1/01R May 2025

**1** **Square area**

Square area:

$$(2x + 3)^2$$

**2** **Triangle area**

Triangle area:

$$\frac{1}{2}(2x + 3)(2x)$$

### 3 Evaluate fraction

So

$$(2x+3)^2+\frac{1}{2}(2x+3)(2x)=84$$

$$6x^2+15x-75=0$$

$$2x^2+5x-25=0$$

$$(2x-5)(x+5)=0$$

### 4 Calculate value

The positive solution is $x=2.5$, so

$$(2x+3)^2=8^2=64$$

**FINAL ANSWER**

$64\ \text{cm}^2$.

4WM1/01 NOVEMBER 2025

# Question 1

2 marks | Perimeter & Area

**1** The diagram shows a semicircle.

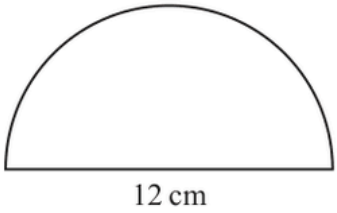



Diagram **NOT** accurately drawn

The semicircle has diameter 12 cm

Work out the perimeter of the semicircle.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 1 is 2 marks)**

## WORKED SOLUTION

Question 1 · 4WM1/01 November 2025

**1** **Radius cm**

Radius $= 6$ cm.

$$\text{perimeter} = \pi r + 12 = 6\pi + 12 \approx 30.8496$$

**FINAL ANSWER**

$30.8$ cm to 3 significant figures.

4WM1/01 NOVEMBER 2025

# Question 18

4 marks | Perimeter & Area

**18** The diagram shows a rectangle.

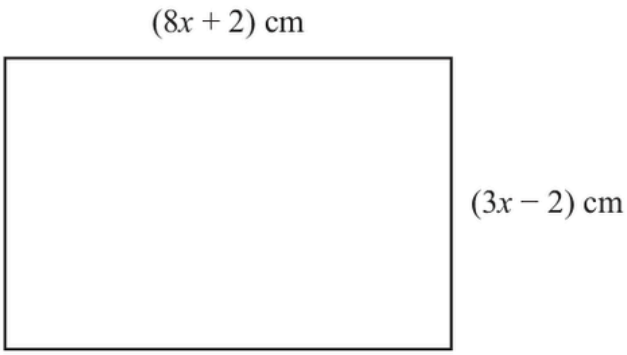



Diagram **NOT** accurately drawn

The rectangle has length $(8x + 2)$ cm and width $(3x - 2)$ cm
The area of the rectangle is $500\ \text{cm}^2$

(a) Show that $12x^2 - 5x - 252 = 0$

**(1)**

(b) Hence find the value of $x$
Give your answer correct to 3 significant figures.

$x =$ .....

**(3)**

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4WM1/01 November 2025

### 1 Calculate value

$$(8x + 2)(3x - 2) = 500$$

$$24x^2 - 10x - 4 = 500$$

$$12x^2 - 5x - 252 = 0$$

### 2 Solve quadratic equation

Using the quadratic formula,

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4(12)(-252)}}{24}$$

**3** **The positive solution is**

The positive solution is

$$x \approx 4.79564$$

---

✓ **FINAL ANSWER**

$x = 4.80$ to 3 significant figures.

4WM1/01 NOVEMBER 2025

# Question 21

4 marks | Perimeter & Area

**21** The diagram shows an equilateral triangle and a circle.

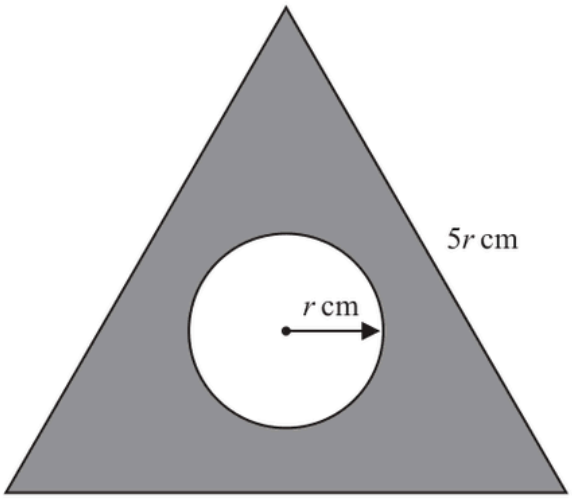



Diagram **NOT** accurately drawn

The equilateral triangle has sides of length $5r$ cm
The circle has radius $r$ cm

The area of the shaded region is $610\pi$ cm$^2$
Work out the value of $r$

Give your answer correct to 3 significant figures.

$r =$ .....

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Question 21 · 4WM1/01 November 2025

**1** **Triangle area**

Triangle area:

$$\frac{\sqrt{3}}{4}(5r)^2 = \frac{25\sqrt{3}}{4}r^2$$

**2** **Circle area**

Circle area:

$$\pi r^2$$

**3** **Simplify surd**

So

$$\left(\frac{25\sqrt{3}}{4} - \pi\right) r^2 = 610\pi$$

$$r \approx 15.7926$$

---

**FINAL ANSWER**

$r = 15.8$ to 3 significant figures.

# Question 4

5 marks | Perimeter & Area

**4** The diagram shows a shaded shape $ABCD$ made from a right-angled triangle $ABD$ and a semicircle $BCD$

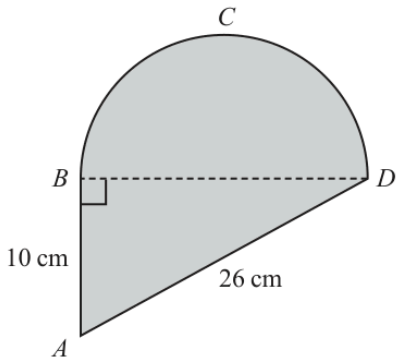



Diagram **NOT** accurately drawn

$AB = 10$ cm $AD = 26$ cm angle $ABD = 90°$

Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 4 is 5 marks)**

**1** **Find BD with Pythagoras**

$$BD = \sqrt{26^2 - 10^2} = \sqrt{576} = 24 \text{ cm}$$

**2** **Semicircle arc**

Diameter $BD = 24$, so the arc is

$$\tfrac{1}{2}\pi \times 24 = 12\pi = 37.699\ldots \text{ cm}$$

**3** **Perimeter (BD is inside, not counted)**

$$10 + 26 + 12\pi = 73.699\ldots$$

✓ **FINAL ANSWER**

$73.7$ cm

4MA1/1H JUNE 2026

# Question 11

5 marks | Perimeter & Area

**11** The diagram shows a shaded shape $ABCDEO$ made from a semicircle $BCD$ and two identical right-angled triangles $ABO$ and $EDO$

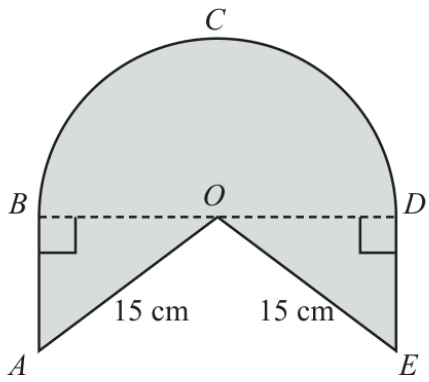



Diagram **NOT** accurately drawn

$BD$ is the diameter of the semicircle.
$O$ is the midpoint of $BD$

angle $ABO$ = angle $EDO$ = 90° $\quad AO = EO = 15$ cm

The area of semicircle $BCD$ is $227$ cm$^2$

Work out the perimeter of the shaded shape.
Show your working clearly.
Give your answer correct to one decimal place.

..... cm

**(Total for Question 11 is 5 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H June 2026

**1** **Solution | Step 1**

Let the radius be $r = OB = OD$. Then $\frac{1}{2}\pi r^2 = 227$, so $r = \sqrt{454/\pi} = 12.021343\ldots$ cm.
The given area belongs to the semicircle.

**2** **Solution | Step 2**

$$AB = DE = \sqrt{15^2 - r^2} = 8.971472\ldots \text{ cm.}$$

The identical triangles have hypotenuse 15 cm and horizontal leg r.

### 3 Solution | Step 3

$$P = \pi r + AB + DE + AO + EO = \pi r + 2\sqrt{225 - r^2} + 30 = 85.709107\ldots \text{ cm}.$$

The boundary contains the semicircular arc and the four outer triangle sides, but not the internal diameter.

---

### 4 Solution | Step 4

$$P = 85.7 \text{ cm} \quad (1 \text{ d.p.}).$$

Retain unrounded intermediate values.

---

**FINAL ANSWER**

$85.7 \text{ cm}$

TOPIC

# Circles, Arcs & Sectors

Every past-paper question on this topic, 2020 to 2026.

# Question 18

4 marks | Circles, Arcs & Sectors

**18** The diagram shows two identical circles drawn inside a rectangle.

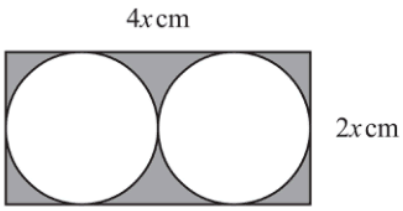



Diagram **NOT** accurately drawn

The length of the rectangle is $4x$ cm and the width of the rectangle is $2x$ cm
Each circle has a radius of $x$ cm

The circles touch each other and each circle touches three sides of the rectangle.

The region inside the rectangle that is outside the circles, shown shaded in the diagram, has a total area of $20\,\text{cm}^2$

Work out the perimeter of the rectangle.
Give your answer correct to 3 significant figures.

..... cm

## 1 Write the shaded area

Rectangle area:

$$4x \times 2x = 8x^2$$

Area of the two circles:

$$2\pi x^2$$

So

$$8x^2 - 2\pi x^2 = 20$$

## 2 Solve for x

$$x^2(8 - 2\pi) = 20$$

$$x = \sqrt{\frac{20}{8 - 2\pi}} = 3.413\ldots$$

## 3 Find the perimeter

$$P = 2(4x + 2x) = 12x$$

$$12(3.413\ldots) = 40.96\ldots$$

## 4 Round to 3 significant figures

$$40.96\ldots \approx 41.0$$

**FINAL ANSWER**

$41.0$ cm.

4MA1/1H JANUARY 2021

# Question 8

3 marks | Circles, Arcs & Sectors

**8** The region, shown shaded in the diagram, is a path.

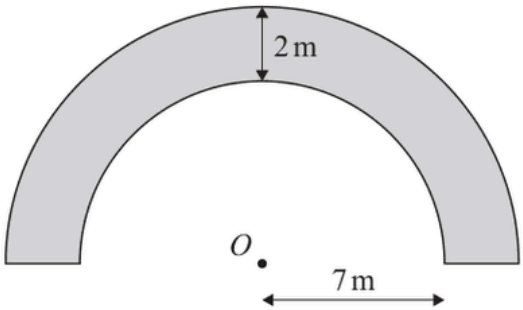



Diagram **NOT** accurately drawn

The boundary of the path is formed by two semicircles, with the same centre $O$, and two straight lines.

The inner semicircle has a radius of 7 metres.
The path has a width of 2 metres.

Work out the perimeter of the path.
Give your answer correct to one decimal place.

..... m

**(Total for Question 8 is 3 marks)**

## WORKED SOLUTION

Question 8 · 4MA1/1H January 2021

**1** **The inner radius is m**

The inner radius is $7$ m.

**2** **Calculate value**

The path is $2$ m wide, so the outer radius is

$$7 + 2 = 9 \text{ m}$$

### 3 Perimeter of the path

Perimeter of the path:

$$\text{inner semicircle} + \text{outer semicircle} + 2 \text{ straight ends}$$

$$= 7\pi + 9\pi + 2 + 2$$

$$= 16\pi + 4$$

$$= 54.265\ldots$$

---

**FINAL ANSWER**

$54.3\ \text{m}$.

4MA1/2HR JANUARY 2021

# Question 11

4 marks | Circles, Arcs & Sectors

**11** A circle centre $O$ has radius 9 cm.

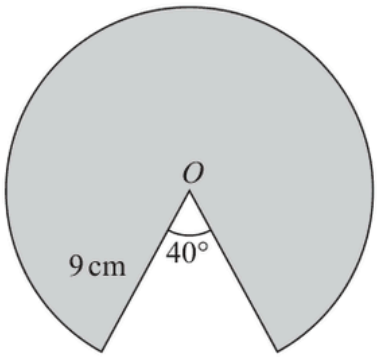



Diagram **NOT** accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2HR January 2021

**1** **The shaded sector is the**

The shaded sector is the major sector:

$$360° - 40° = 320°$$

**2** **Arc length**

Arc length:

$$\frac{320}{360} \times 2\pi(9) = 16\pi$$

**3** **Perimeter**

Perimeter:

$$16\pi + 2(9) = 68.265\ldots$$

✓ **FINAL ANSWER**

$68.3$ cm.

4MA1/2HR JANUARY 2021

# Question 20

5 marks | Circles, Arcs & Sectors

**20** $A$, $B$ and $C$ are points on a circle with centre $O$.

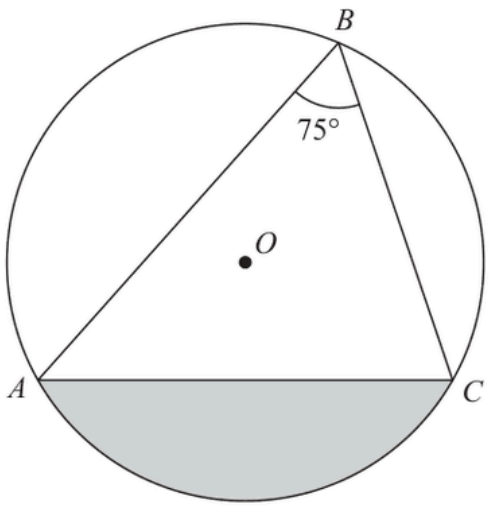



Diagram **NOT** accurately drawn

Angle $ABC = 75°$

The area of the shaded segment is $200\,\text{cm}^2$

Calculate the radius of the circle.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 5 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2HR January 2021

**1** **Use trigonometry**

$$\angle ABC = 75°$$

**2** **The angle at the centre**

The angle at the centre is twice the angle at the circumference, so

$$\angle AOC = 2(75°) = 150°$$

**3** **The shaded segment is**

The shaded segment is:

$$\text{sector } AOC - \text{triangle } AOC$$

**4** **Let the radius be**

Let the radius be $r$.

**5** **Sector area**

Sector area:

$$\frac{150}{360}\pi r^2$$

**6** **Triangle area**

Triangle area:

$$\frac{1}{2}r^2 \sin 150°$$

**7** **Use trigonometry**

So

$$\frac{150}{360}\pi r^2 - \frac{1}{2}r^2 \sin 150° = 200$$

$$r^2\left(\frac{5\pi}{12} - \frac{1}{4}\right) = 200$$

$$r^2 = \frac{200}{\frac{5\pi}{12} - \frac{1}{4}}$$

$$r = 13.7426\ldots$$

✓ **FINAL ANSWER**

$\boxed{13.7 \text{ cm}}$

4MA1/2H JANUARY 2021

# Question 22

6 marks | Circles, Arcs & Sectors

**22** The diagram shows a sector $OBC$ of a circle with centre $O$ and radius $(6 + x)$ cm.

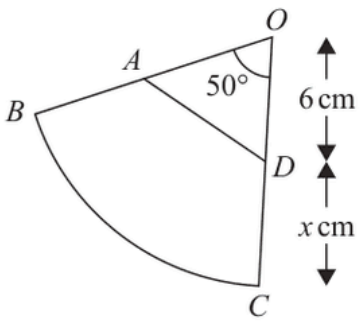



Diagram **NOT** accurately drawn

$A$ is the point on $OB$ and $D$ is the point on $OC$ such that $OA = OD = 6$ cm

Angle $BOC = 50°$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle $OAD$

find the value of $x$.
Give your answer correct to 3 significant figures.

$x =$ .....

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H January 2021

**1 The radius of sector is**

The radius of sector $OBC$ is

$$6 + x$$

**2 Use trigonometry**

The angle is $50°$, so the arc length of sector $OBC$ is

$$\frac{50}{360} \times 2\pi(6 + x)$$

**3** **Evaluate fraction**

Therefore the perimeter of sector $OBC$ is

$$2(6+x)+\frac{50}{360}\times 2\pi(6+x)$$

$$=(6+x)\left(2+\frac{5\pi}{18}\right)$$

**4** **Use trigonometry**

In triangle $OAD$,

$$OA=OD=6, \qquad \angle AOD=50^\circ$$

**5** **Use trigonometry**

So

$$AD=2(6)\sin 25^\circ=12\sin 25^\circ$$

**6** **The perimeter of triangle is**

The perimeter of triangle $OAD$ is

$$12+12\sin 25^\circ$$

**7** **Given that**

Given that

$$\text{perimeter of sector } OBC=2\times\text{perimeter of triangle } OAD$$

$$(6+x)\left(2+\frac{5\pi}{18}\right)=2(12+12\sin 25^\circ)$$

$$x=\frac{2(12+12\sin 25^\circ)}{2+\frac{5\pi}{18}}-6$$

$$x=5.8854\ldots$$

✓ **FINAL ANSWER**

$\boxed{5.89}$

4MA1/1H MAY 2021

# Question 12

2 marks | Circles, Arcs & Sectors

**12** The diagram shows a sector of a circle with radius 7 cm.

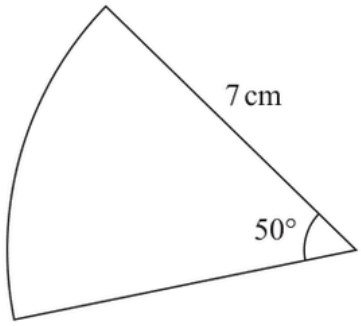



Diagram **NOT** accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

**(Total for Question 12 is 2 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H May 2021

**1** **Evaluate fraction**

$$\text{arc length} = \frac{50}{360} \times 2\pi(7)$$

$$= 6.108\ldots$$

**FINAL ANSWER**

$6.1$ cm.

4MA1/1H MAY 2021

# Question 25

6 marks | Circles, Arcs & Sectors

**25** The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

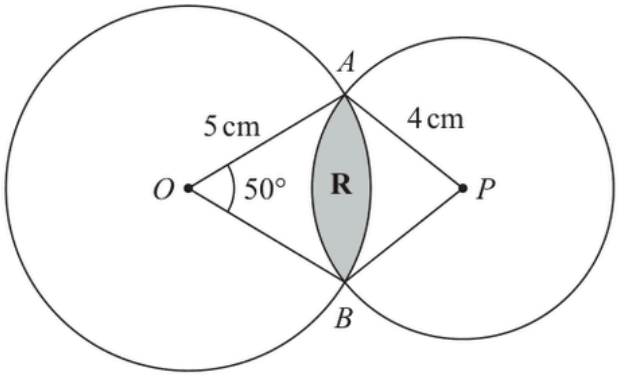



Diagram **NOT** accurately drawn

One of the circles has centre $O$ and radius 5 cm.
The other circle has centre $P$ and radius 4 cm.
Angle $AOB = 50°$

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.

..... cm$^2$

**(Total for Question 25 is 6 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/1H May 2021

**1** **Use trigonometry**

For the circle with centre $O$:

$$\angle AOB = 50°, \qquad OA = OB = 5$$

**2** **The half chord length is**

The half chord length is

$$\frac{AB}{2} = 5\sin 25°$$

**3** **Use trigonometry**

For the circle with centre $P$, let

$$\angle APB = \theta$$

---

**4** **Use trigonometry**

The radius is $4$, so

$$4\sin\left(\frac{\theta}{2}\right) = 5\sin 25°$$

$$\sin\left(\frac{\theta}{2}\right) = \frac{5\sin 25°}{4}$$

$$\theta = 63.7777\ldots°$$

---

**5** **The common region is made**

The common region is made from two minor segments.

---

**6** **Use trigonometry**

For the circle with centre $O$:

$$\text{segment area} = \frac{50}{360}\pi(5)^2 - \frac{1}{2}(5)^2\sin 50° = 1.3328\ldots$$

---

**7** **Use trigonometry**

For the circle with centre $P$:

$$\text{segment area} = \frac{\theta}{360}\pi(4)^2 - \frac{1}{2}(4)^2\sin\theta$$

$$= 1.7284\ldots$$

---

**8** **Total area of region**

Total area of region $R$:

$$1.3328\ldots + 1.7284\ldots = 3.0611\ldots$$

---

✓ **FINAL ANSWER**

$\boxed{3.06\ \text{cm}^2}$

4MA1/2HR JANUARY 2022

# Question 9

5 marks | Circles, Arcs & Sectors

**9** In the diagram, $ABC$ is a right-angled triangle and $DEF$ is a semicircular arc.

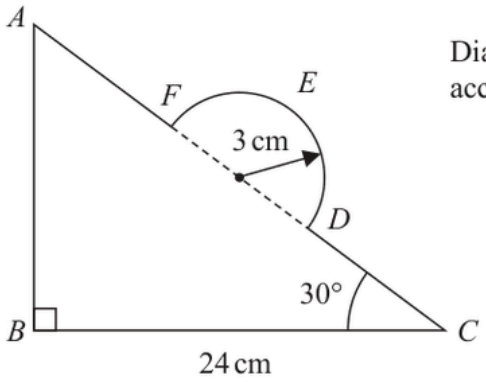



Diagram **NOT** accurately drawn

In triangle $ABC$

$BC = 24\,\text{cm}$ angle $ABC = 90°$ angle $BCA = 30°$

The points $D$ and $F$ lie on $AC$ so that $DF$ is the diameter of the semicircular arc $DEF$
The radius of the semicircular arc is 3 cm.

Work out the length of $AFEDC$
Give your answer correct to 2 significant figures.

..... cm

**(Total for Question 9 is 5 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/2HR January 2022

**1** **Use trigonometry**

In triangle $ABC$,

$$\cos 30° = \frac{24}{AC}$$

$$AC = \frac{24}{\cos 30°} = 27.712\ldots$$

**2** **Calculate value**

The semicircle has radius $3\ \text{cm}$, so its diameter is $6\ \text{cm}$.

$$AF + DC = AC - DF = 27.712\ldots - 6$$

**3** **Arc is a semicircle**

Arc $FED$ is a semicircle:

$$\text{arc length} = 3\pi$$

**4** **Calculate value**

Therefore

$$AFEDC = 27.712\ldots - 6 + 3\pi = 31.137\ldots$$

✓ **FINAL ANSWER**

$31\ \text{cm}$.

4MA1/1HR JANUARY 2022

# Question 11

2 marks | Circles, Arcs & Sectors

**11** The diagram shows sector $OPQ$ of a circle, centre $O$

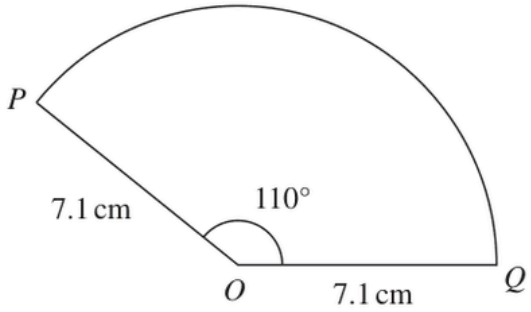



Diagram **NOT** accurately drawn

$OP = OQ = 7.1$ cm
Angle $POQ = 110°$

Calculate the area of sector $OPQ$
Give your answer correct to one decimal place.

..... cm²

**(Total for Question 11 is 2 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1HR January 2022

### 1 Calculate area

$$\text{area} = \frac{110}{360} \times \pi(7.1)^2$$

$$= 48.390\ldots$$

**FINAL ANSWER**

$48.4\ \text{cm}^2$.

4MA1/1H JANUARY 2022

# Question 16

4 marks | Circles, Arcs & Sectors

**16** The diagram shows a circle with centre $O$

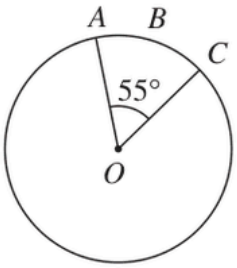



Diagram **NOT** accurately drawn

$A$, $B$ and $C$ are points on the circle so that the length of the arc $ABC$ is 5 cm.

Given that angle $AOC = 55°$

work out the area of the circle.
Give your answer correct to one decimal place.

..... $cm^2$

**(Total for Question 16 is 4 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H January 2022

### 1 Use trigonometry

The arc length is $5\ \text{cm}$, and the angle is $55°$.

$$5 = \frac{55}{360} \times 2\pi r$$

$$r = 5.208\ldots$$

### 2 Area of the circle

Area of the circle:

$$\pi r^2 = \pi(5.208\ldots)^2 = 85.233\ldots$$

**FINAL ANSWER**

$85.2\ \text{cm}^2$.

4MA1/2HR JUNE 2022

# Question 13

3 marks | Circles, Arcs & Sectors

**13** The diagram shows a sector $AOB$ of a circle with centre $O$

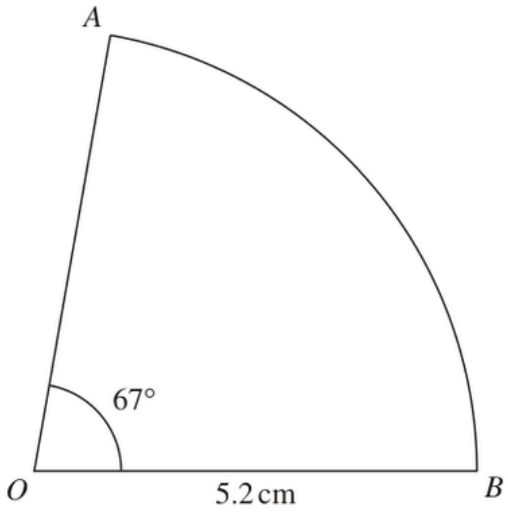



Diagram **NOT** accurately drawn

Angle $AOB = 67°$
$OA = OB = 5.2\,\text{cm}$

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/2HR June 2022

### 1 Arc length

Arc length:

$$\frac{67}{360} \times 2\pi(5.2) = 6.080\ldots$$

### 2 Perimeter

Perimeter:

$$5.2 + 5.2 + 6.080\ldots = 16.480\ldots$$

**FINAL ANSWER**

$16.5\ \text{cm}$.

4MA1/1H JUNE 2022

# Question 19

5 marks | Circles, Arcs & Sectors

**19**

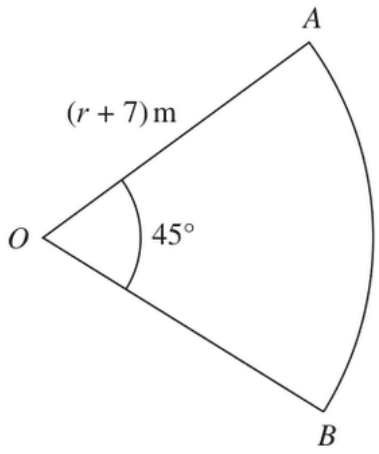



Diagram **NOT** accurately drawn

*OAB* is a sector **S** of a circle with centre *O* and radius $(r + 7)$ metres.
Angle $AOB = 45°$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of $r$
Show your working clearly.

**Question 19 continued.**

$r =$ .....

**(Total for Question 19 is 5 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1H June 2022

**1** **Area of sector**

Area of sector $S$:

$$\frac{45}{360}\pi(r+7)^2 = \frac{1}{8}\pi(r+7)^2$$

**2** **Area of circle**

Area of circle $C$:

$$\pi(r-2)^2$$

---

**3** **The sector area is twice**

The sector area is twice the circle area:

$$\frac{1}{8}\pi(r+7)^2 = 2\pi(r-2)^2$$

$$(r+7)^2 = 16(r-2)^2$$

---

**4** **Solve inequality**

Since $r > 2$,

$$r+7 = 4(r-2)$$

$$r+7 = 4r-8$$

$$r = 5$$

---

✓ **FINAL ANSWER**

$r = 5$.

4MA1/2H JUNE 2022

# Question 20

4 marks | Circles, Arcs & Sectors

**20** The diagram shows four identical circles drawn inside a square.

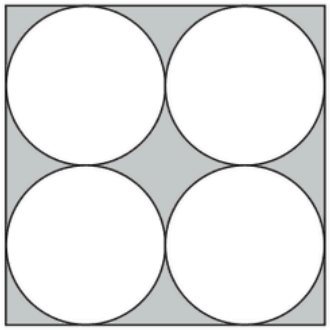


Diagram **NOT** accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of $40\,\text{cm}^2$

Work out the perimeter of the square.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H June 2022

**1 Let the radius of each**

Let the radius of each circle be $r$.

**2 There are two circles across**

There are two circles across the width of the square, so the side length of the square is

$$4r$$

**3 Area of the square**

Area of the square:

$$(4r)^2 = 16r^2$$

**4 Area of the four circles**

Area of the four circles:

$$4\pi r^2$$

**5** **Calculate area**

The shaded area is $40\ \text{cm}^2$, so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131\ldots$$

---

**6** **The perimeter of the square**

The perimeter of the square is

$$4(4r) = 16r$$

$$16r = 54.6101\ldots$$

---

✓ **FINAL ANSWER**

$\boxed{54.6\ \text{cm}}$

4MA1/1HR MAY 2023

# Question 4

5 marks | Circles, Arcs & Sectors

**4** The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

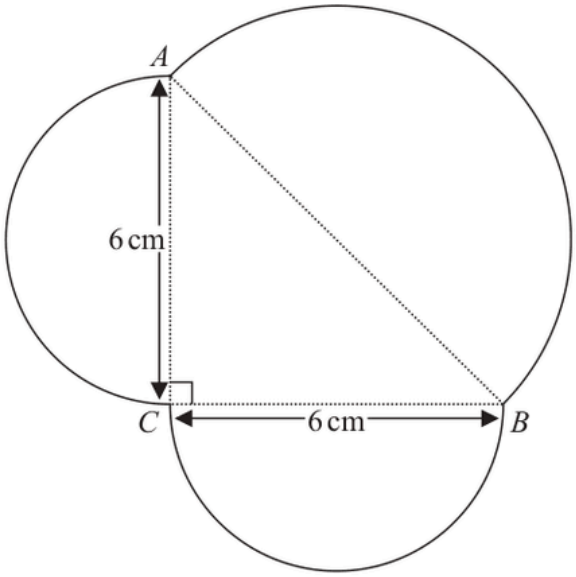



Diagram **NOT** accurately drawn

$AB$, $BC$ and $CA$ are each the diameter of a semicircle.

$BC = CA = 6$ cm.

Work out the perimeter of the shape.
Give your answer correct to one decimal place.

..... cm

**(Total for Question 4 is 5 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1HR May 2023

**1** **Simplify surd**

$$AB = \sqrt{6^2 + 6^2} = \sqrt{72} = 8.485\ldots$$

**2** **The perimeter is the sum**

The perimeter is the sum of three semicircular arcs:

$$\frac{\pi}{2}(AB + BC + CA) = \frac{\pi}{2}(8.485\ldots + 6 + 6) = 32.178\ldots$$

**FINAL ANSWER**

$32.2\ \text{cm}$.

4MA1/1H MAY 2023

# Question 22

6 marks | Circles, Arcs & Sectors

**22** The diagram shows two circles with centre $O$ and a regular pentagon $ABCDE$

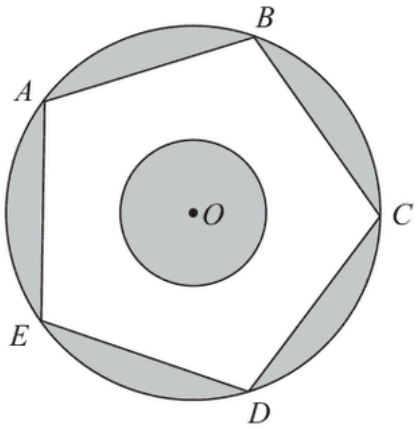



Diagram **NOT** accurately drawn

$A$, $B$, $C$, $D$ and $E$ are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H May 2023

**1** **The regular pentagon has side**

The regular pentagon has side length $8$ cm.

**2** **Use trigonometry**

For a regular pentagon,

$$R = \frac{s}{2\sin 36°}$$

**3** **Use trigonometry**

where $R$ is the radius of the larger circle.

$$R = \frac{8}{2\sin 36^\circ} = 6.8052\ldots$$

**4** **Area of the larger circle**

Area of the larger circle:

$$\pi R^2 = 145.4898\ldots$$

**5** **Area of the regular pentagon**

Area of the regular pentagon:

$$\frac{5s^2}{4\tan 36^\circ} = \frac{5(8)^2}{4\tan 36^\circ} = 110.1106\ldots$$

**6** **Let the radius of the**

Let the radius of the smaller circle be $r$.

**7** **The shaded area is**

The shaded area is:

$$\text{larger circle area} - \text{pentagon area} + \pi r^2$$

**8** **The unshaded area is**

The unshaded area is:

$$\text{pentagon area} - \pi r^2$$

**9** **Simplify surd**

These are equal, so

$$145.4898 - 110.1106 + \pi r^2 = 110.1106 - \pi r^2$$

$$2\pi r^2 = 2(110.1106) - 145.4898$$

$$\pi r^2 = 37.3657\ldots$$

$$r = \sqrt{\frac{37.3657\ldots}{\pi}} = 3.4487\ldots$$

✓ **FINAL ANSWER**

$$\boxed{3.45\text{ cm}}$$

4MA1/1H NOVEMBER 2023

# Question 5

3 marks | Circles, Arcs & Sectors

5 The diagram shows a shaded shape made from three identical semicircles, $X$, $Y$ and $Z$

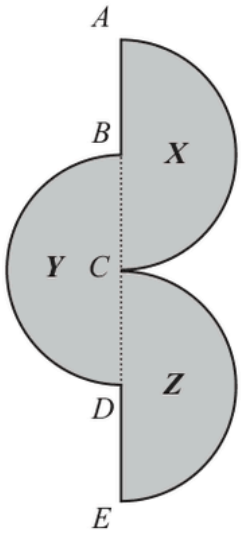



Diagram **NOT** accurately drawn

$ABCDE$ is a straight line.

$AC$ is the diameter of semicircle $X$ and $B$ is the centre of semicircle $X$
$BD$ is the diameter of semicircle $Y$ and $C$ is the centre of semicircle $Y$
$CE$ is the diameter of semicircle $Z$ and $D$ is the centre of semicircle $Z$

$AC = BD = CE = 20$ cm

Work out the perimeter of the shaded shape.
Give your answer correct to the nearest whole number.

..... cm

**(Total for Question 5 is 3 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1H November 2023

**1 Each semicircle has diameter**

Each semicircle has diameter $20$ cm, so each has radius $10$ cm.

**2 The total curved perimeter is**

The total curved perimeter is three semicircles:

$$3 \times \pi r = 3 \times 10\pi = 30\pi$$

### 3 The two exposed straight parts

The two exposed straight parts are

$$AB = 10, \qquad DE = 10$$

### 4 Calculate value

So the perimeter is

$$30\pi + 20 = 114.2\ldots$$

**FINAL ANSWER**

$114\ \text{cm}$ to the nearest whole number.

4MA1/2H NOVEMBER 2023

# Question 20

4 marks | Circles, Arcs & Sectors

**20** The diagram shows equilateral triangle $ABC$ with sides of length 10 cm.
A circle is drawn inside the triangle.

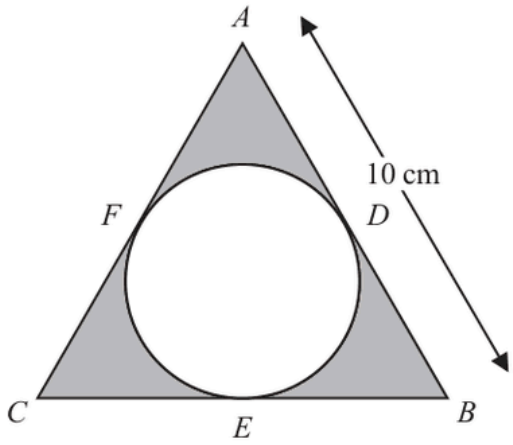



Diagram **NOT** accurately drawn

$D$, $E$ and $F$ are points on the circle.

$ADB$, $BEC$ and $CFA$ are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H November 2023

**1 Use trigonometry**

The triangle is equilateral with side length $10$ cm.

**2 Area of the triangle**

Area of the triangle:

$$\frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3}$$

**3 Use trigonometry**

For an equilateral triangle, the inradius is

$$r = \frac{s\sqrt{3}}{6}$$

**4** **Simplify surd**

So

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}$$

**5** **Area of the circle**

Area of the circle:

$$\pi r^2 = \pi\left(\frac{5\sqrt{3}}{3}\right)^2 = \frac{25\pi}{3}$$

**6** **The shaded area is**

The shaded area is

$$25\sqrt{3} - \frac{25\pi}{3} = 17.1213\ldots$$

✓ **FINAL ANSWER**

$\boxed{17.1\ \text{cm}^2}$

4MA1/2HR JUNE 2024

# Question 5

4 marks | Circles, Arcs & Sectors

**5** The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square $ABCD$

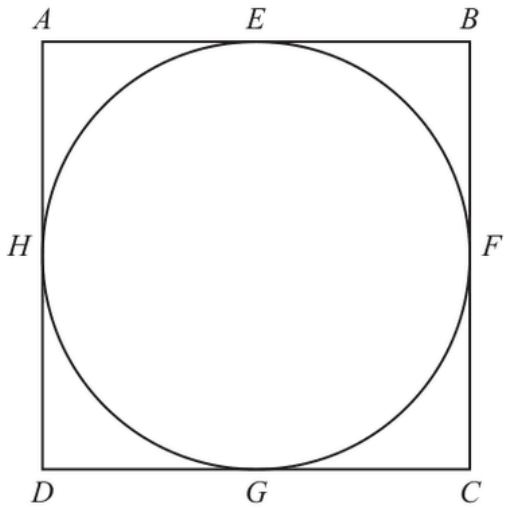



Diagram **NOT** accurately drawn

The circle touches the square at the points $E$, $F$, $G$ and $H$

The area of the square is $81\ \text{cm}^2$

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2HR June 2024

**1** **Calculate area**

The square has area $81\ \text{cm}^2$, so its side length is

$$\sqrt{81} = 9\ \text{cm}$$

**2** **Perimeter of the square**

Perimeter of the square:

$$4 \times 9 = 36$$

**3** **Calculate value**

The circle has diameter $9\ \text{cm}$, so its circumference is

$$\pi d = 9\pi$$

**4** **Total wire length**

Total wire length:

$$36 + 9\pi = 64.274\ldots$$

---

✓ **FINAL ANSWER**

$64.3\ \text{cm}$ to 3 significant figures.

4MA1/2HR JUNE 2024

# Question 13

3 marks | Circles, Arcs & Sectors

**13** $OAB$ is a sector of a circle with centre $O$ and radius $r$ cm.

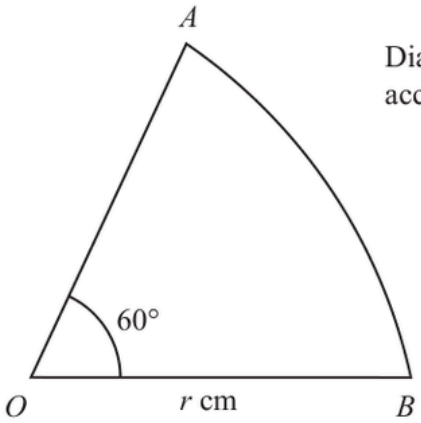



Diagram **NOT** accurately drawn

Angle $AOB = 60°$
The perimeter of the sector is $P$ cm.

Find a formula for $P$ in terms of $r$
Give your answer in the form $P = r(c\pi + k)$ where $c$ and $k$ are values to be found.

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/2HR June 2024

### 1 The arc length is

The arc length is

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

### 2 The perimeter is two radii

The perimeter is two radii plus the arc:

$$P = 2r + \frac{\pi r}{3}$$

$$P = r\left(\frac{\pi}{3} + 2\right)$$

**FINAL ANSWER**

$P = r\left(\frac{1}{3}\pi + 2\right)$.

4MA1/1H MAY 2024

# Question 20

4 marks | Circles, Arcs & Sectors

**20** The diagram shows a sector $OABC$ of a circle centre $O$

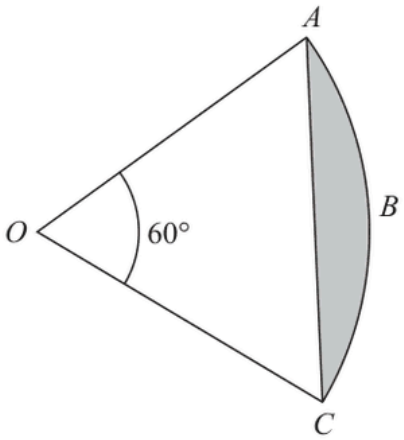



Diagram **NOT** accurately drawn

Angle $AOC = 60°$

The area of the shaded segment $ABC$ is $38\,\text{cm}^2$

Work out the perimeter of the shaded segment $ABC$
Give your answer correct to one decimal place.

..... cm

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H May 2024

**1** **Let the radius of the**

Let the radius of the sector be $r$.

**2** **The shaded segment is**

The shaded segment is

$$\text{sector area} - \text{triangle area}$$

**3** **Use trigonometry**

The angle is $60°$, so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60°$$

$$38 = r^2\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)$$

$$r = 20.4815\ldots$$

**4** **The perimeter of the shaded**

The perimeter of the shaded segment is the arc $AC$ plus the chord $AC$.

**5** **Arc length**

Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

**6** **Chord length**

Chord length:

$$2r \sin 30° = r$$

**7** **Evaluate fraction**

So the perimeter is

$$\frac{\pi r}{3} + r$$

$$= \frac{\pi(20.4815\ldots)}{3} + 20.4815\ldots$$

$$= 41.9296\ldots$$

✓ **FINAL ANSWER**

$\boxed{41.9\text{ cm}}$

4MA1/2H JUNE 2024

# Question 21

5 marks | Circles, Arcs & Sectors

**21** The diagram shows a square $ABCD$ and a circle.

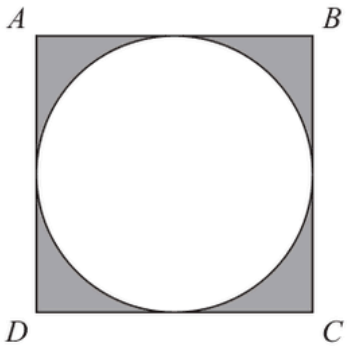



Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is $80\,\text{cm}^2$

Work out the length of $AC$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2H June 2024

**1** **Let the side length**

Let the side length of the square be $s$.

**2** **Square**

The circle is inscribed in the square, so its radius is

$$\frac{s}{2}$$

**3** **The shaded area is the**

The shaded area is the area of the square minus the area of the circle:

$$s^2 - \pi\left(\frac{s}{2}\right)^2 = 80$$

$$s^2 - \frac{\pi s^2}{4} = 80$$

$$s^2\left(1 - \frac{\pi}{4}\right) = 80$$

$$s = \sqrt{\frac{80}{1 - \frac{\pi}{4}}} = 19.3076\ldots$$

---

**4** **The diagonal of the square**

The diagonal of the square is

$$AC = s\sqrt{2}$$

$$AC = 19.3076\ldots\sqrt{2} = 27.3051\ldots$$

---

✓ **FINAL ANSWER**

$\boxed{27.3\text{ cm}}$

4MA1/1H NOVEMBER 2024

# Question 16

4 marks | Circles, Arcs & Sectors

**16** $OAPB$ is a sector of a circle, centre $O$

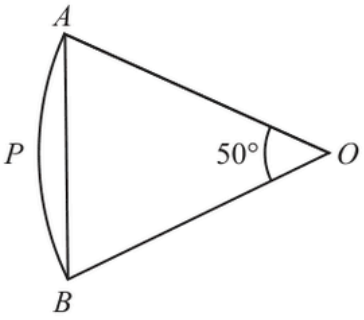



Diagram **NOT** accurately drawn

Angle $AOB = 50°$

Area of triangle $OAB = 120\ \text{cm}^2$

Work out the area of the sector $OAPB$
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 16 is 4 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H November 2024

### 1 Use trigonometry

For triangle $OAB$,

$$120 = \frac{1}{2} r^2 \sin 50°$$

$$r^2 = \frac{240}{\sin 50°}$$

### 2 Area of sector

Area of sector $OAPB$:

$$\frac{50}{360}\pi r^2 = \frac{50}{360}\pi \left(\frac{240}{\sin 50°}\right) = 136.701\ldots$$

**FINAL ANSWER**

$137\ \text{cm}^2$.

4MA1/2H NOVEMBER 2024

# Question 25

5 marks | Circles, Arcs & Sectors

**25** The diagram shows an equilateral triangle $ABC$ and a circle with centre $O$

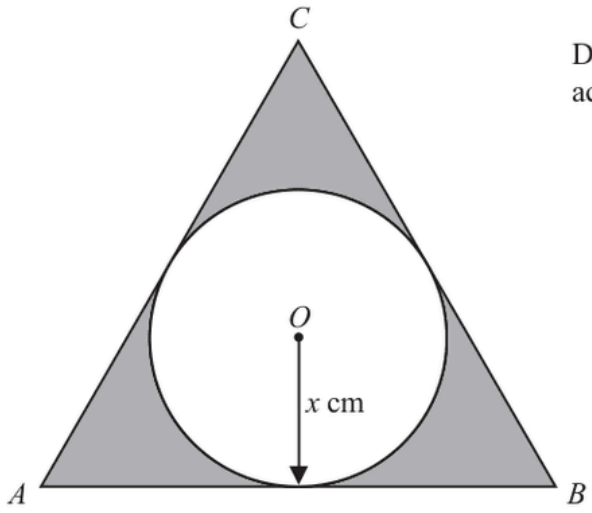



Diagram **NOT** accurately drawn

$AB$, $BC$ and $CA$ are tangents to the circle.

The radius of the circle is $x$ cm

The total area, in cm$^2$, of the regions shown shaded in the diagram is $nx^2$

Find the value of $n$
Give your answer correct to 3 significant figures.

$n =$ .....

**(Total for Question 25 is 5 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/2H November 2024

**1** **Use trigonometry**

The circle has radius $x$, so the inradius of the equilateral triangle is $x$.

**2** **Use trigonometry**

For an equilateral triangle with side length $a$,

$$\text{inradius} = \frac{a\sqrt{3}}{6}$$

**3** **Simplify surd**

So

$$x = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6x}{\sqrt{3}} = 2\sqrt{3}x$$

**4** **Area of the triangle**

Area of the triangle:

$$\frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(2\sqrt{3}x)^2$$

$$= \frac{\sqrt{3}}{4}(12x^2) = 3\sqrt{3}x^2$$

**5** **Area of the circle**

Area of the circle:

$$\pi x^2$$

**6** **The shaded area is**

The shaded area is

$$3\sqrt{3}x^2 - \pi x^2$$

$$= (3\sqrt{3} - \pi)x^2$$

**7** **Simplify surd**

Therefore

$$n = 3\sqrt{3} - \pi$$

✓ **FINAL ANSWER**

$$\boxed{3\sqrt{3} - \pi}$$

4MA1/1H JUNE 2025

# Question 11

2 marks | Circles, Arcs & Sectors

**11** $OABC$ is a sector of a circle, centre $O$

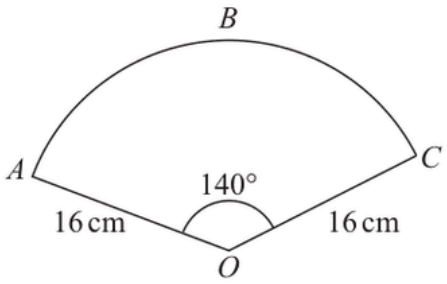



Diagram **NOT** accurately drawn

angle $AOC = 140°$
$OA = OC = 16\,\text{cm}$

Calculate the area of the sector.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 11 is 2 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H June 2025

### 1 Calculate area

$$\text{area} = \frac{140}{360} \times \pi(16)^2$$

$$= 312.763\ldots$$

**FINAL ANSWER**

$313\ \text{cm}^2$.

# Question 14

4 marks | Circles, Arcs & Sectors

**14** The diagram shows sector $OAB$ of a circle, centre $O$

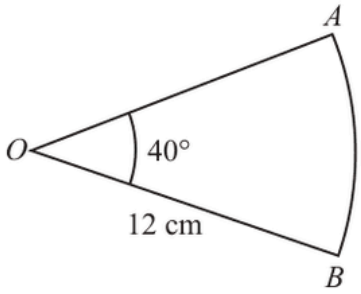



Diagram **NOT** accurately drawn

$OA = OB = 12$ cm  angle $AOB = 40°$

(a) Work out the area of the sector.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(2)**

The diagram shows triangle $XYZ$

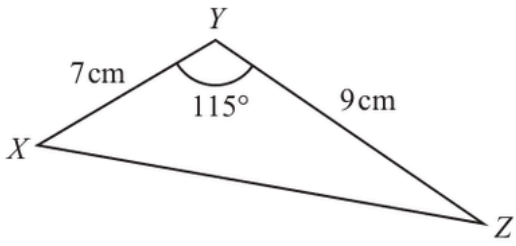



Diagram **NOT** accurately drawn

$XY = 7$ cm  $YZ = 9$ cm  angle $XYZ = 115°$

(b) Work out the area of the triangle.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(2)**

**(Total for Question 14 is 4 marks)**

**1** **Sector area**

Sector area:

$$\frac{40}{360}\pi(12)^2 = 16\pi = 50.265\ldots$$

**2** **To 3 significant figures**

To 3 significant figures,

$$50.3\ \text{cm}^2$$

**3** **Triangle area**

Triangle area:

$$\frac{1}{2}(7)(9)\sin 115^\circ = 28.548\ldots$$

**4** **To 3 significant figures**

To 3 significant figures,

$$28.5\ \text{cm}^2$$

✓ **FINAL ANSWER**

$50.3\ \text{cm}^2$ and $28.5\ \text{cm}^2$.

4WM1/01R MAY 2025

# Question 22

6 marks | Circles, Arcs & Sectors

**22**

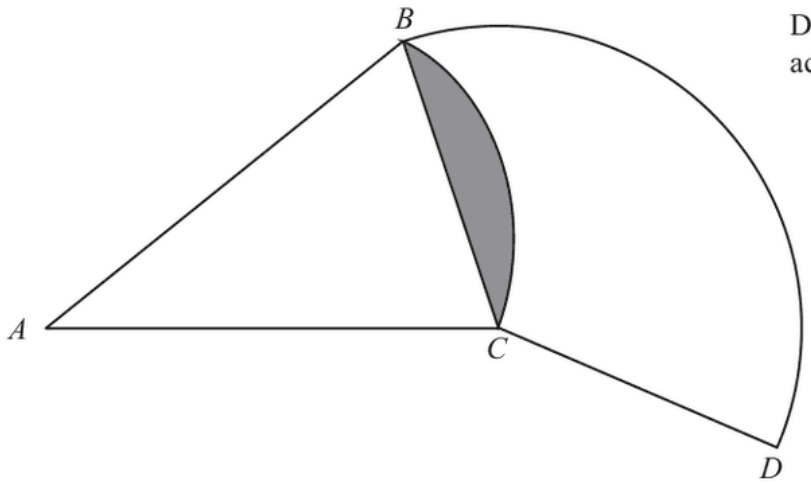



Diagram **NOT** accurately drawn

$BAC$ is a sector of a circle, centre $A$

$BCD$ is a sector of a circle, centre $C$

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc $BD$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4WM1/01R May 2025

**1 Let the radius of sector**

Let the radius of sector $BAC$ be $R$.

**2 The shaded segment is the**

The shaded segment is the segment cut off by chord $BC$ in the sector with centre $A$.

**3** **Its area is**

Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40°$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40°\right)$$

$$R = 31.8096\ldots$$

---

**4** **Use trigonometry**

Now find the chord $BC$:

$$BC = 2R\sin 20°$$

$$BC = 2(31.8096\ldots)\sin 20° = 21.7591\ldots$$

---

**5** **Calculate value**

In sector $BCD$, the centre is $C$, so the radius is

$$CB = CD = 21.7591\ldots$$

---

**6** **Use trigonometry**

The arc $BD$ has angle $130°$, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

---

✓ **FINAL ANSWER**

$\boxed{49.4 \text{ cm}}$

4MA1/1HR JUNE 2025

# Question 25

6 marks | Circles, Arcs & Sectors

**25**

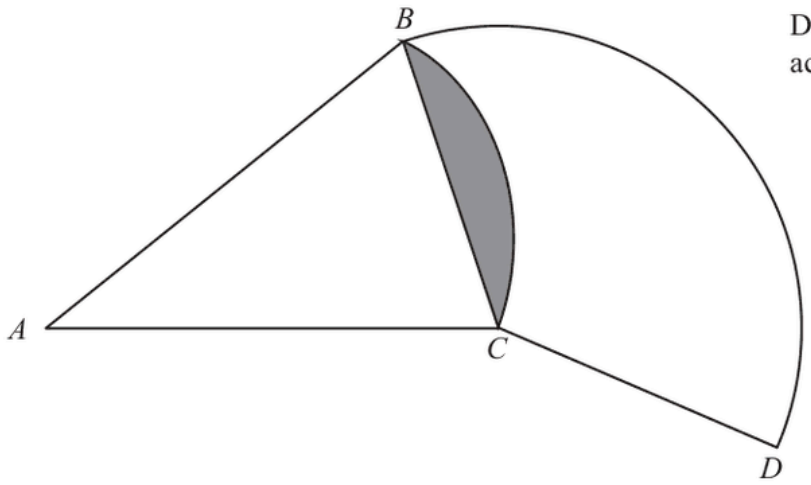



Diagram **NOT** accurately drawn

*BAC* is a sector of a circle, centre *A*

*BCD* is a sector of a circle, centre *C*

Angle *BAC* = 40°
Angle *BCD* = 130°
Area of shaded segment = 28 cm$^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 25 is 6 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/1HR June 2025

**1 Let the radius of sector**

Let the radius of sector $BAC$ be $R$.

**2 The shaded segment is the**

The shaded segment is the segment cut off by chord $BC$ in the sector with centre $A$.

**3** **Its area is**

Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40°$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40°\right)$$

$$R = 31.8096\ldots$$

---

**4** **Use trigonometry**

Now find the chord $BC$:

$$BC = 2R\sin 20°$$

$$BC = 2(31.8096\ldots)\sin 20° = 21.7591\ldots$$

---

**5** **Calculate value**

In sector $BCD$, the centre is $C$, so the radius is

$$CB = CD = 21.7591\ldots$$

---

**6** **Use trigonometry**

The arc $BD$ has angle $130°$, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

---

✓ **FINAL ANSWER**

$\boxed{49.4 \text{ cm}}$

4MA1/2H NOVEMBER 2025

# Question 6

5 marks | Circles, Arcs & Sectors

**6** A large circle with centre $O$ contains 3 identical small circles.
Each circle touches two other circles.

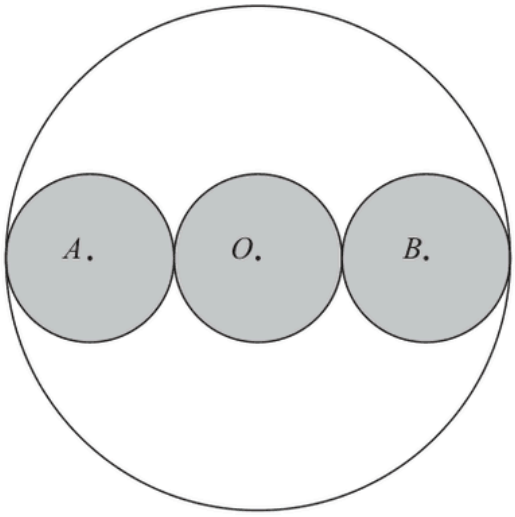



Diagram **NOT** accurately drawn

$A$, $O$ and $B$ are the centres of the small circles.
$AOB$ is a straight line.

The circumference of the large circle is 160 cm

Work out the area of one small circle.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(Total for Question 6 is 5 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2H November 2025

**1** **The circumference of the large**

The circumference of the large circle is $160\ \text{cm}$, so

$$2\pi R = 160$$

$$R = \frac{80}{\pi}$$

## 2 The three equal small circles

The three equal small circles fit along the diameter of the large circle, so

$$2R = 6r$$

$$r = \frac{R}{3} = \frac{80}{3\pi}$$

## 3 Area of one small circle

Area of one small circle is

$$\pi r^2 = \pi\left(\frac{80}{3\pi}\right)^2$$

$$= 226.35\ldots$$

**FINAL ANSWER**

$226\ \text{cm}^2$ to 3 significant figures.

4WM1/01 NOVEMBER 2025

# Question 14

2 marks | Circles, Arcs & Sectors

**14** $AOB$ is a sector of a circle with centre $O$ and radius 12 cm

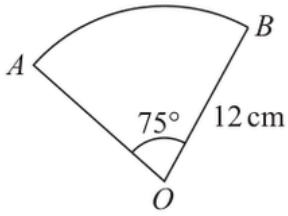



Diagram **NOT** accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(Total for Question 14 is 2 marks)**

## WORKED SOLUTION

Question 14 · 4WM1/01 November 2025

### 1 Calculate area

$$\text{area} = \frac{75}{360}\pi(12)^2$$

$$= 30\pi \approx 94.2478$$

**FINAL ANSWER**

$94.2\ \text{cm}^2$ to 3 significant figures.

4MA1/1H NOVEMBER 2025

# Question 15

2 marks | Circles, Arcs & Sectors

**15** $AOB$ is a sector of a circle with centre $O$ and radius 12 cm

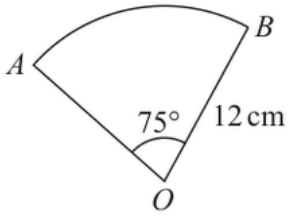



Diagram **NOT** accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 15 is 2 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1H November 2025

### 1 Calculate area

$$\text{area} = \frac{75}{360} \times \pi(12)^2$$

$$= 94.247\ldots$$

**FINAL ANSWER**

$94.2\ \text{cm}^2$.

4MA1/2H JUNE 2026

# Question 15

4 marks | Circles, Arcs & Sectors

**15** The diagram shows a sector $OABC$ of a circle, centre $O$

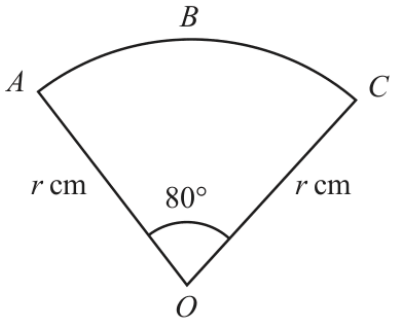



Diagram **NOT** accurately drawn

$OA = OC = r$ cm  angle $AOC = 80°$

The area of the sector $OABC$ is $32\pi$ cm$^2$

Work out the perimeter of the sector $OABC$
Give your answer correct to one decimal place.

..... cm

**(Total for Question 15 is 4 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2H June 2026

**1** **Solution | Step 1**

$$\frac{80}{360}\pi r^2 = 32\pi \quad \Rightarrow \quad r^2 = 32\frac{360}{80} = 144 \quad \Rightarrow \quad r = 12 \text{ cm}.$$

Use the fraction of the full circle and choose the positive radius.

**2** **Solution | Step 2**

$$\text{Arc length} = \frac{80}{360}(2\pi \times 12) = \frac{16\pi}{3} \text{ cm}.$$

The same central-angle fraction applies to the circumference.

**3** **Solution | Step 3**

$$P = 12 + 12 + \frac{16\pi}{3} = 40.7551608\ldots \text{ cm}.$$

The sector perimeter contains both radii as well as the arc.

**FINAL ANSWER**

$40.8\ \text{cm}$ (1 decimal place).

# Question 23

6 marks | Circles, Arcs & Sectors

**23** $OPQ$ and $OAB$ are sectors of a circle, centre $O$

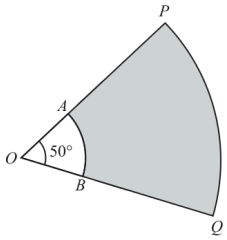



Diagram **NOT** accurately drawn

Angle $POQ = 50°$

$A$ is the point on $OP$ such that $OA = \frac{1}{4}OP$

The area of the shaded region is $\frac{243}{16}\pi\ \text{cm}^2$

Work out the perimeter of the shaded region.
Give your answer in the form $a\pi + b$ where $a$ and $b$ are constants.
Show your working clearly.

.....cm

**(Total for Question 23 is 6 marks)**

**1** **Let OP = r**

Then $OA = \frac{r}{4}$. Shaded area:

$$\frac{50}{360}\pi\left(r^2 - \frac{r^2}{16}\right) = \frac{243}{16}\pi$$

**2** **Solve for r**

$$\frac{75}{576}r^2 = \frac{243}{16} \Rightarrow r^2 = 116.64 \Rightarrow r = 10.8,\ OA = 2.7$$

**3** **Arcs**

$$\text{arc } PQ = \frac{50}{360} \times 2\pi(10.8) = 3\pi, \qquad \text{arc } AB = \frac{50}{360} \times 2\pi(2.7) = \tfrac{3}{4}\pi$$

**4** **Straight edges and total**

$AP = BQ = 10.8 - 2.7 = 8.1$:

$$3\pi + \tfrac{3}{4}\pi + 2(8.1) = \tfrac{15}{4}\pi + 16.2$$

**FINAL ANSWER**

$\frac{15}{4}\pi + 16.2$ cm



TOPIC

# Right-Angled Triangles - Pythagoras & Trigonometry

Every past-paper question on this topic, 2020 to 2026.

# Question 10

6 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**10** The diagram shows a quadrilateral $ABCD$

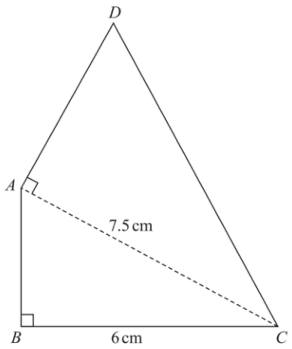



Diagram **NOT** accurately drawn

In the diagram, $ABC$ and $DAC$ are right-angled triangles.

$BC = 6$ cm $\quad AC = 7.5$ cm

The area of quadrilateral $ABCD$ is 31.5 cm$^2$

(a) Work out the length of $AB$

..... cm

(2)

(b) Work out the length of $AD$

..... cm

(4)

**(Total for Question 10 is 6 marks)**

**1** **Use Pythagoras for AB**

In triangle $ABC$,

$$AB^2 + 6^2 = 7.5^2$$

$$AB^2 = 56.25 - 36 = 20.25$$

$$AB = 4.5$$

**2** **Find the area of triangle ABC**

$$\frac{1}{2} \times 6 \times 4.5 = 13.5$$

**3** **Find the area of triangle DAC**

$$31.5 - 13.5 = 18$$

**4** **Use the triangle area formula**

In triangle $DAC$, $AD$ and $AC$ are perpendicular.

$$\frac{1}{2} \times AD \times 7.5 = 18$$

$$AD = \frac{36}{7.5} = 4.8$$

✓ **FINAL ANSWER**

$AB = 4.5$ cm, $AD = 4.8$ cm.

4MA1/2HR JANUARY 2020

# Question 4

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

4 The diagram shows a rectangle and a diagonal of the rectangle.

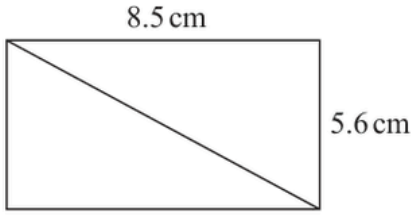



Diagram **NOT** accurately drawn

Work out the length of the diagonal of the rectangle.
Give your answer correct to 1 decimal place.

..... cm

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2HR January 2020

### 1 Use trigonometry

$$d^2 = 8.5^2 + 5.6^2$$

$$d = \sqrt{103.61} = 10.178\ldots$$

**FINAL ANSWER**

$10.2$ cm.

# Question 9

3 marks

Right-Angled Triangles - Pythagoras & Trigonometry

**9** Here is a right-angled triangle.

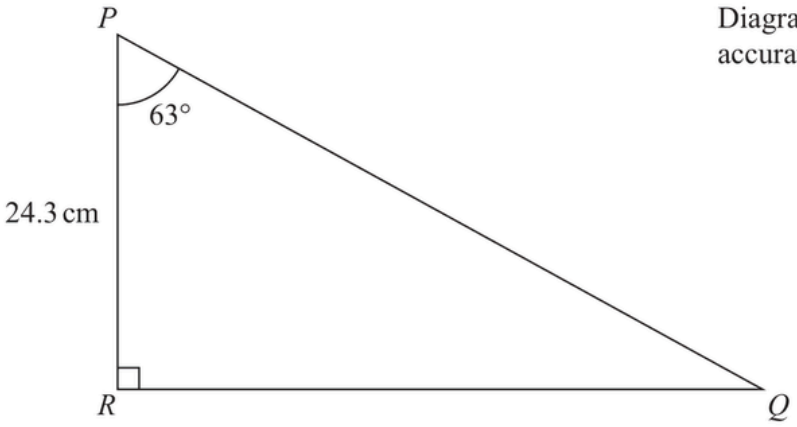



Diagram **NOT** accurately drawn

Calculate the length of $PQ$.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 9 is 3 marks)**

## 1 Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

$$PQ = \frac{24.3}{\cos 63^\circ} = 53.525\ldots$$

**FINAL ANSWER**

$53.5$ cm.

4MA1/1HR JANUARY 2020

# Question 15

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**15** The diagram shows two right-angled triangles, $DEF$ and $EFG$.

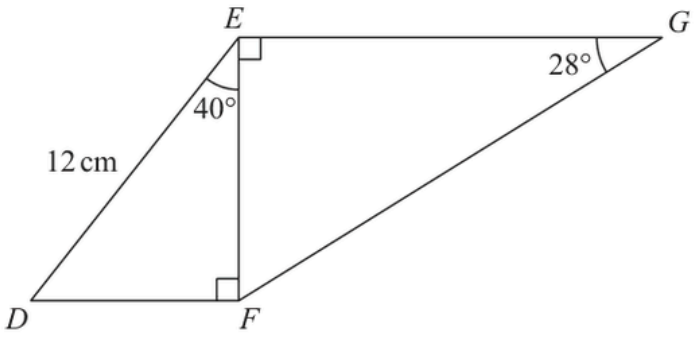



Diagram **NOT** accurately drawn

Work out the length of $EG$.
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 15 is 4 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1HR January 2020

**1 Use trigonometry**

In triangle $DEF$,

$$EF = 12\cos 40^\circ = 9.192\ldots$$

**2 Use trigonometry**

In triangle $EFG$,

$$\sin 28^\circ = \frac{EF}{EG}$$

$$EG = \frac{9.192\ldots}{\sin 28^\circ} = 19.580\ldots$$

**FINAL ANSWER**

$19.6$ cm.

4MA1/1H JANUARY 2021

# Question 10

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**10** Here is isosceles triangle $ABC$.

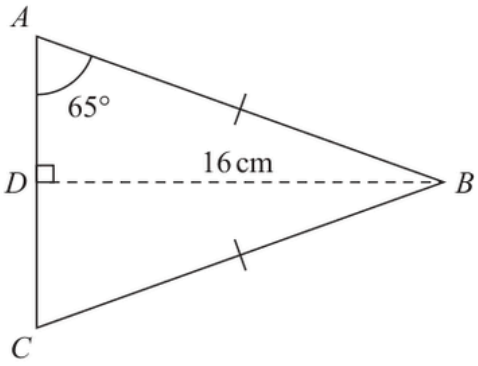



Diagram **NOT** accurately drawn

$D$ is the midpoint of $AC$ and $DB = 16$ cm.

Angle $DAB = 65°$

Work out the perimeter of triangle $ABC$.
Give your answer correct to one decimal place.

..... cm

**(Total for Question 10 is 4 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1H January 2021

**1 Find the gradient**

Since $ABC$ is isosceles and $D$ is the midpoint of $AC$, $BD$ is perpendicular to $AC$.

**2 Use trigonometry**

In right triangle $ABD$,

$$BD = 16, \qquad \angle DAB = 65°$$

$$\sin 65° = \frac{16}{AB}$$

$$AB = \frac{16}{\sin 65°}$$

**3** **Use trigonometry**

Also,

$$\tan 65^\circ = \frac{16}{AD}$$

$$AD = \frac{16}{\tan 65^\circ}$$

---

**4** **The perimeter is**

The perimeter is

$$AB + BC + AC = 2AB + 2AD$$

$$= 2\left(\frac{16}{\sin 65^\circ}\right) + 2\left(\frac{16}{\tan 65^\circ}\right)$$

$$= 50.2299\ldots$$

---

✓ **FINAL ANSWER**

$50.2\ \text{cm}$.

4MA1/2HR JANUARY 2021

# Question 13

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**13** The diagram shows two hot air balloons.
$A$ is a point on the base of one of the balloons and $B$ is a point on the base of the other balloon.

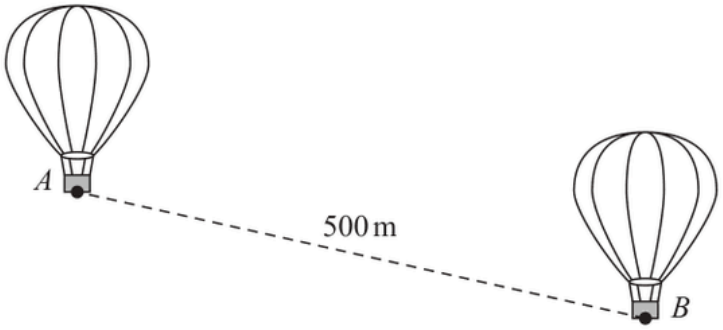



Diagram **NOT** accurately drawn

The distance between $A$ and $B$ is 500 metres.
The angle of depression of $B$ from $A$ is 23°

Calculate the vertical height of $A$ above $B$.
Give your answer correct to one decimal place.

..... metres

**(Total for Question 13 is 3 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/2HR January 2021

**1** **The vertical height is opposite**

The vertical height is opposite the $23°$ angle, and $AB = 500\ \text{m}$ is the hypotenuse.

$$\sin 23° = \frac{h}{500}$$

$$h = 500 \sin 23° = 195.365\ldots$$

**FINAL ANSWER**

$195.4\ \text{m}$.

4MA1/2H MAY 2021

# Question 7

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

7 Here is a right-angled triangle.

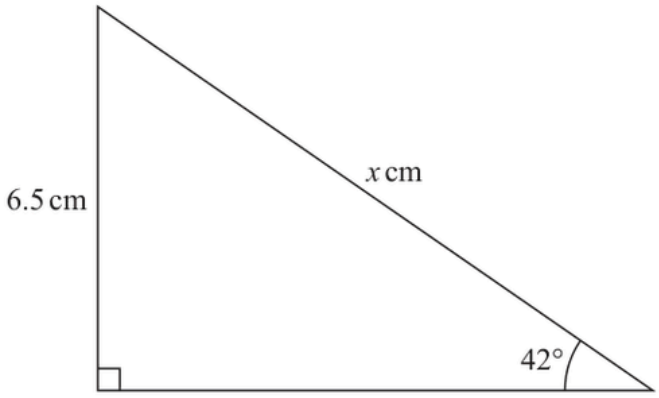



Diagram **NOT** accurately drawn

Work out the value of $x$.
Give your answer correct to one decimal place.

$x =$ .....

**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/2H May 2021

**1 The side cm is opposite**

The side $6.5$ cm is opposite the $42°$ angle and $x$ is the hypotenuse.

$$\sin 42° = \frac{6.5}{x}$$

$$x = \frac{6.5}{\sin 42°} = 9.714\ldots$$

**FINAL ANSWER**

$x = 9.7$ cm.

4MA1/1H NOVEMBER 2021

# Question 2

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**2** Here is a right-angled triangle.

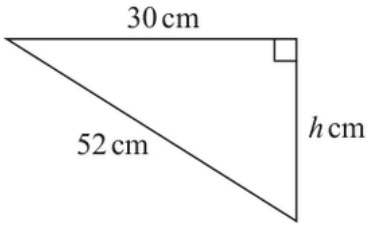



Diagram **NOT** accurately drawn

Calculate the value of $h$.
Give your answer correct to 3 significant figures.

$h$ = .....

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1H November 2021

### 1 Use trigonometry

$$h^2 + 30^2 = 52^2$$

$$h^2 = 52^2 - 30^2 = 1804$$

$$h = \sqrt{1804} = 42.473\ldots$$

**FINAL ANSWER**

$42.5$ cm to 3 significant figures.

# Question 12

6 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**12** The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.

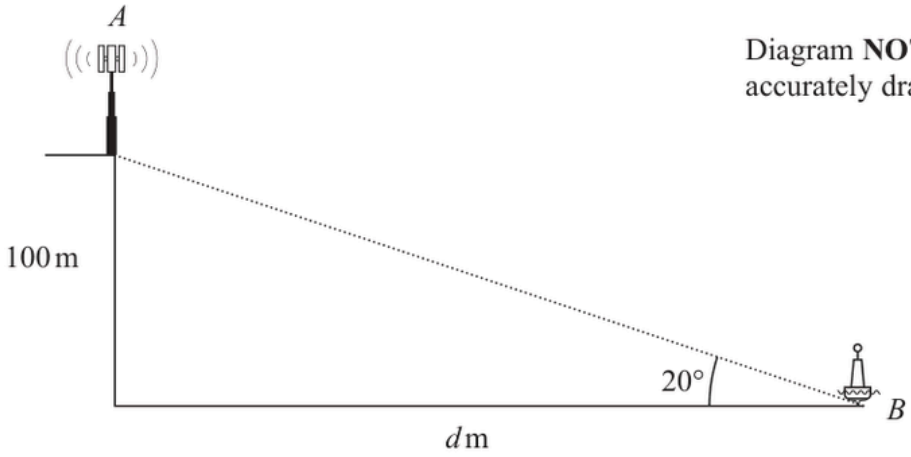



Diagram **NOT** accurately drawn

The height of the cliff is 100 metres.
The buoy is at the point $B$ that is $d$ metres from the base of the cliff.
The angle of elevation from $B$ to the top of the cliff is 20°

(a) Calculate the value of $d$.
Give your answer correct to 3 significant figures.

$d =$ .....

**(3)**

The point $A$ at the top of the radio mast is vertically above the top of the cliff.
The angle of elevation from $B$ to $A$ is 25°

(b) Calculate the height of the radio mast.
Give your answer correct to 3 significant figures.

..... m

**(3)**

**(Total for Question 12 is 6 marks)**

## 1 (a)

(a)

$$\tan 20° = \frac{100}{d}$$

$$d = \frac{100}{\tan 20°} = 274.747\ldots$$

## 2 Use trigonometry

So

$$d = 275 \text{ m}$$

## 3 (b) The angle of elevation

(b) The angle of elevation to the top of the radio mast is $25°$.

$$\tan 25° = \frac{100 + h}{274.747\ldots}$$

$$100 + h = 274.747\ldots \tan 25° = 128.116\ldots$$

$$h = 28.116\ldots$$

**FINAL ANSWER**

(a) $275$ m, (b) $28.1$ m.

4MA1/1H JANUARY 2022

# Question 3

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**3** The diagram shows triangle $ABC$

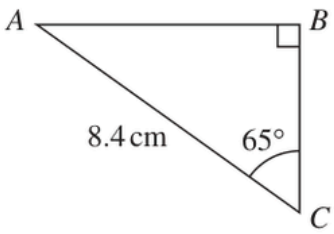



Diagram **NOT** accurately drawn

Work out the length of the side $AB$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1H January 2022

**1** **Use trigonometry**

Side $AB$ is opposite the $65°$ angle, and $AC = 8.4$ cm is the hypotenuse.

$$\sin 65° = \frac{AB}{8.4}$$

$$AB = 8.4 \sin 65° = 7.613\ldots$$

✓ **FINAL ANSWER**

$7.61$ cm.

4MA1/1HR JANUARY 2022

# Question 3

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**3** The shaded shape is made using three identical right-angled triangles and a square.

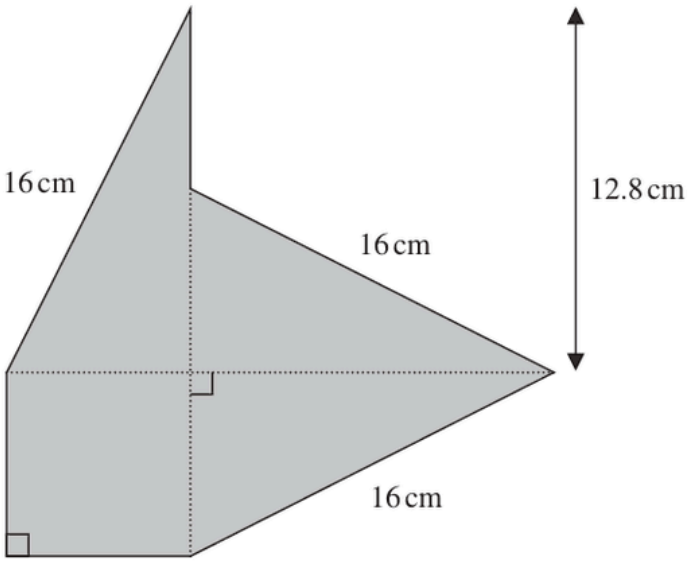



Diagram **NOT** accurately drawn

Work out the perimeter of the shaded shape.

..... cm

**(Total for Question 3 is 4 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1HR January 2022

**1** **Use trigonometry**

Each triangle has hypotenuse $16\ \text{cm}$ and one side $12.8\ \text{cm}$.

**2** **Find the other side**

Find the other side:

$$\sqrt{16^2 - 12.8^2} = 9.6\ \text{cm}$$

**3** **The exposed short vertical side**

The exposed short vertical side is

$$12.8 - 9.6 = 3.2\ \text{cm}$$

**4** **Perimeter**

Perimeter:

$$3(16) + 9.6 + 9.6 + 3.2 = 70.4$$

---

✓ **FINAL ANSWER**

$70.4\ \text{cm}$.

4MA1/2HR JANUARY 2022

# Question 12

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**12** The diagram shows two vertical phone masts, $AB$ and $CD$, on horizontal ground.

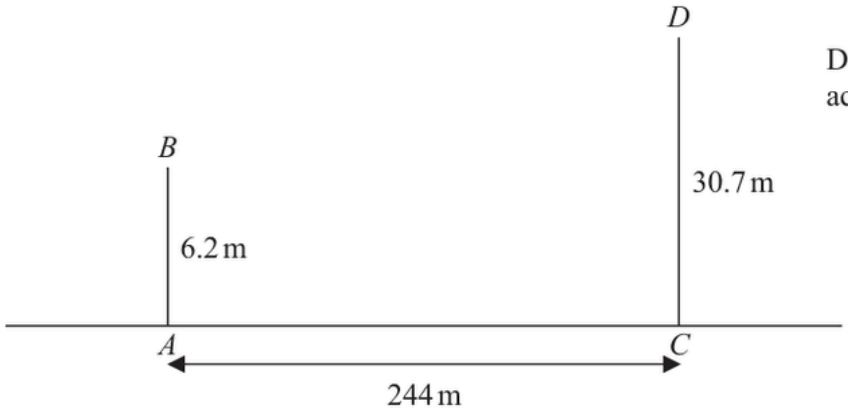



Diagram **NOT** accurately drawn

$AB = 6.2\,\text{m}$ $AC = 244\,\text{m}$ $CD = 30.7\,\text{m}$

Work out the size of the angle of depression of $B$ from $D$
Give your answer correct to one decimal place.

.....°

**(Total for Question 12 is 3 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/2HR January 2022

**1** **The vertical difference between the**

The vertical difference between the tops of the masts is

$$30.7 - 6.2 = 24.5\ \text{m}$$

**2** **The horizontal distance is**

The horizontal distance is $244\ \text{m}$.

$$\tan\theta = \frac{24.5}{244}$$

$$\theta = 5.733\ldots^\circ$$

✓ **FINAL ANSWER**

$5.7^\circ$.

4MA1/2H JANUARY 2022

# Question 14

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**14** A zip wire is shown as the dashed line $AC$ in the diagram.

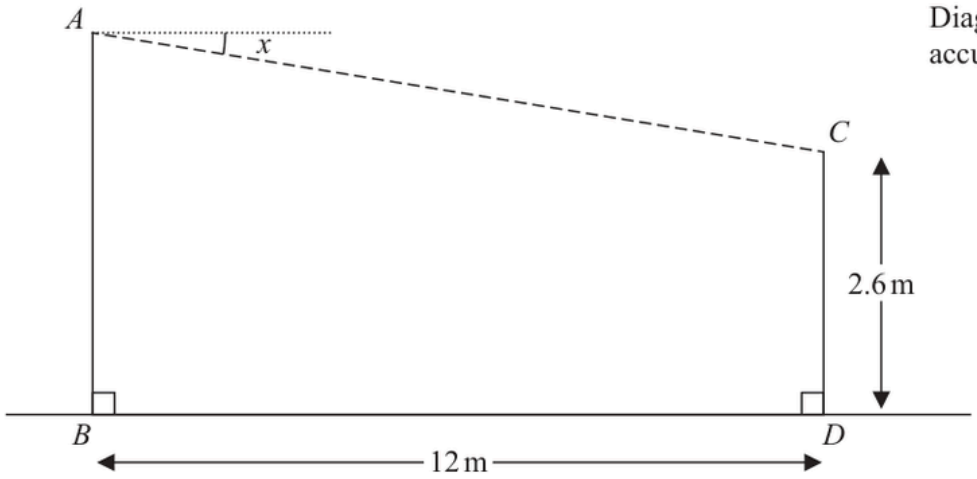



Diagram **NOT** accurately drawn

The zip wire is supported by two vertical posts $AB$ and $CD$ standing on horizontal ground.

$$CD = 2.6\,\text{m} \qquad BD = 12\,\text{m}$$

The zip wire makes an angle $x$ with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle $x$ to be at least 5°

Work out the least possible height of the post $AB$
Give your answer correct to 3 significant figures.

..... m

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2H January 2022

**1** **Use trigonometry**

The horizontal distance is $12\ \text{m}$ and the zip wire is $13\ \text{m}$.

**2** **Vertical difference**

Vertical difference:

$$\sqrt{13^2 - 12^2} = 5$$

**3** **Use trigonometry**

Since the shorter post is $2.6\ \text{m}$,

$$AB = 2.6 + 5 = 7.6$$

---

**FINAL ANSWER**

$7.60\ \text{m}$.

4MA1/2H JUNE 2022

# Question 6

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**6** The diagram shows triangle $PQR$.

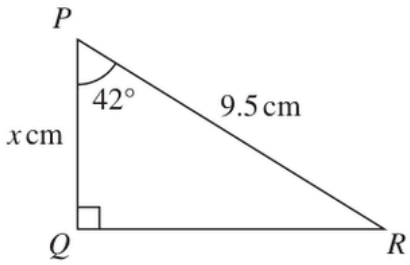



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to one decimal place.

$x =$ .....

**(Total for Question 6 is 3 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2H June 2022

**1** **Use trigonometry**

In the right-angled triangle, $x$ is adjacent to the $42°$ angle and $9.5$ cm is the hypotenuse.

$$\cos 42° = \frac{x}{9.5}$$

$$x = 9.5 \cos 42° = 7.059\ldots$$

✓ **FINAL ANSWER**

$7.1$ cm.

4MA1/2HR JUNE 2022

# Question 10

5 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**10** The diagram shows triangle $ABP$ inside the regular hexagon $ABCDEF$

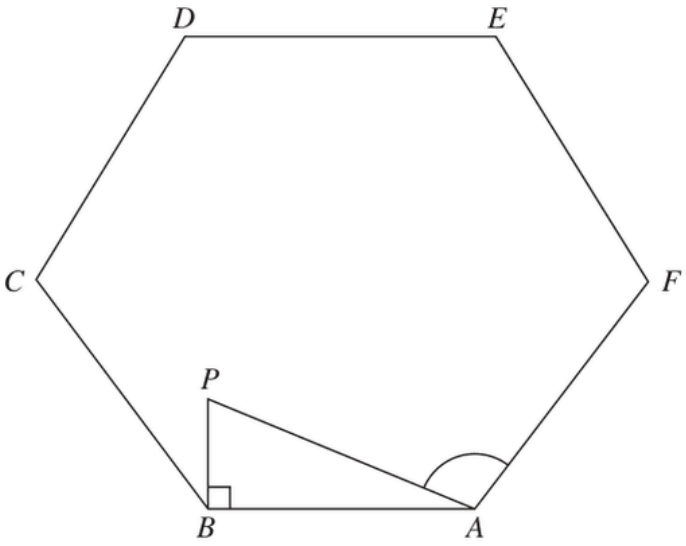



Diagram **NOT** accurately drawn

$AB = 5\,\text{cm}$ $BP = 2\,\text{cm}$ Angle $ABP = 90°$

Work out the size of angle $PAF$
Give your answer correct to 3 significant figures.

.....°

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2HR June 2022

**1** **A regular hexagon has interior**

A regular hexagon has interior angle

$$120°$$

**2** **Use trigonometry**

In right-angled triangle $ABP$,

$$\tan(\angle PAB) = \frac{BP}{AB} = \frac{2}{5}$$

$$\angle PAB = \tan^{-1}\left(\frac{2}{5}\right) = 21.801\ldots°$$

### 3 Use trigonometry

Therefore

$$\angle PAF = 120° - 21.801\ldots° = 98.198\ldots°$$

---

**FINAL ANSWER**

$98.2°$.

4MA1/1H MAY 2023

# Question 5

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**5** The diagram shows a shaded shape $AEBCD$ made by removing triangle $AEB$ from rectangle $ABCD$

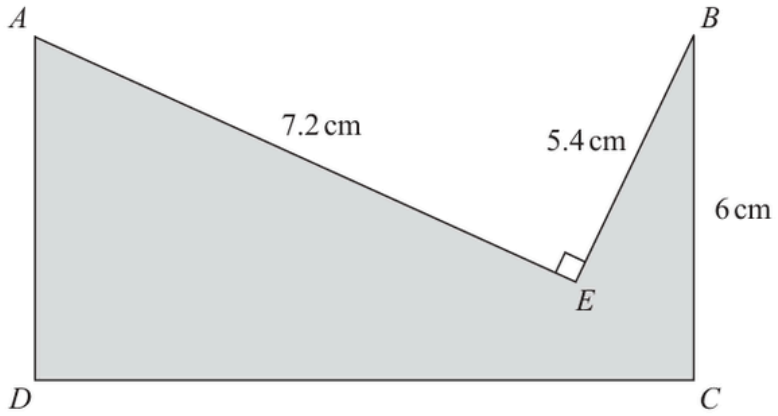



Diagram **NOT** accurately drawn

$AE = 7.2\,\text{cm}$ $BE = 5.4\,\text{cm}$ $BC = 6\,\text{cm}$ angle $AEB = 90°$

Work out the perimeter of the shaded shape.

..... cm

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1H May 2023

**1 Triangle is right angled**

Triangle $AEB$ is right-angled:

$$AB^2 = 7.2^2 + 5.4^2 = 81$$

$$AB = 9$$

**2 Use trigonometry**

For rectangle $ABCD$,

$$DC = 9, \qquad AD = BC = 6$$

**3 Perimeter of the shaded shape**

Perimeter of the shaded shape:

$$7.2 + 5.4 + 6 + 9 + 6 = 33.6$$

**FINAL ANSWER**

$33.6\ \text{cm}$.

4MA1/2H JUNE 2023

# Question 12

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**12** The diagram shows right-angled triangle $ABD$

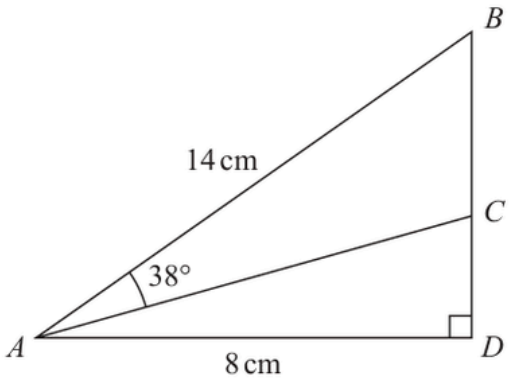



Diagram **NOT** accurately drawn

$AB = 14\,\text{cm}$ $AD = 8\,\text{cm}$

$C$ is the point on $BD$ such that angle $BAC = 38°$

Work out the length of $CD$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 12 is 4 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/2H June 2023

**1 Use trigonometry**

First find angle $DAB$:

$$\cos(\angle DAB) = \frac{8}{14}$$

$$\angle DAB = 55.150\ldots°$$

**2 Given**

Given

$$\angle BAC = 38°$$

**3 Use trigonometry**

So

$$\angle CAD = 55.150\ldots° - 38° = 17.150\ldots°$$

**4** **Use trigonometry**

In right-angled triangle $ACD$,

$$\tan(17.150\ldots^\circ) = \frac{CD}{8}$$

$$CD = 2.468\ldots$$

---

✓ **FINAL ANSWER**

$2.47\ \text{cm}$.

4MA1/2H NOVEMBER 2023

# Question 2

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**2** The diagram shows triangle $ABC$ and triangle $ECD$

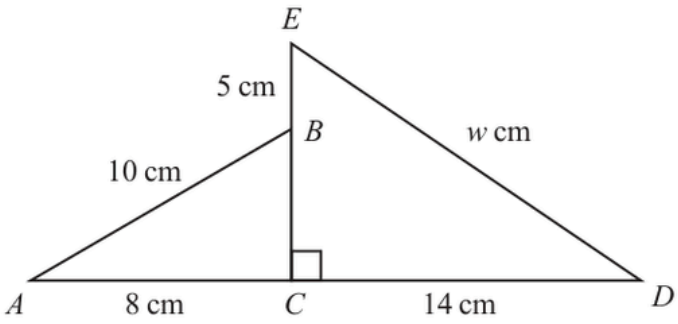



Diagram **NOT** accurately drawn

$ACD$ and $EBC$ are straight lines.

$AB = 10$ cm $\quad AC = 8$ cm $\quad EB = 5$ cm $\quad CD = 14$ cm $\quad ED = w$ cm

Work out the value of $w$
Give your answer correct to one decimal place.

$w =$ .....

**(Total for Question 2 is 4 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2H November 2023

**1** **Use trigonometry**

In triangle $ABC$,

$$BC^2 = 10^2 - 8^2$$

$$BC^2 = 36$$

$$BC = 6$$

**2** **Use trigonometry**

So

$$EC = EB + BC = 5 + 6 = 11$$

**3** **Use trigonometry**

In triangle $ECD$,

$$w^2 = 11^2 + 14^2$$

$$w^2 = 317$$

$$w = 17.804\ldots$$

---

✓ **FINAL ANSWER**

$w = 17.8\ \text{cm}$ to 1 decimal place.

4MA1/1H NOVEMBER 2023

# Question 10

5 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**10** Here is triangle $ABC$

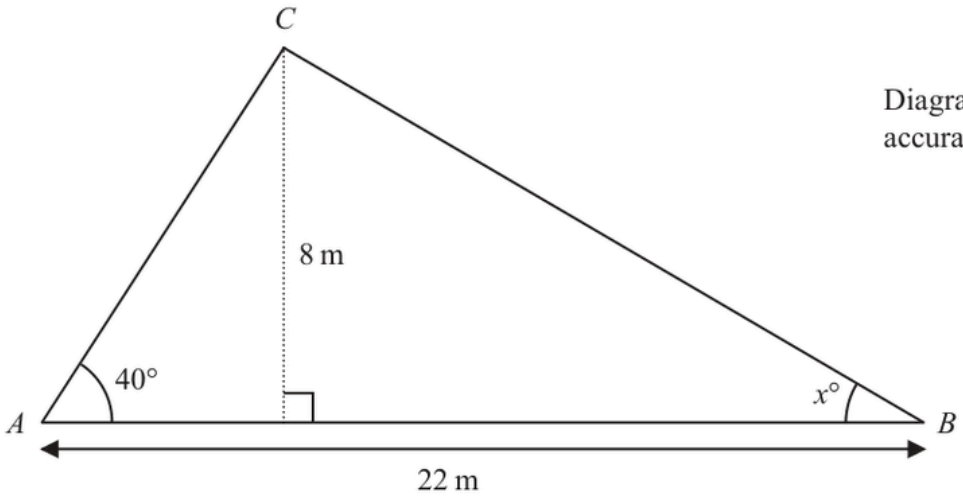



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to one decimal place.
Show your working clearly.

$x =$ .....

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/1H November 2023

**1** **Let the perpendicular from meet**

Let the perpendicular from $C$ meet $AB$ at $F$.

$$AF = \frac{8}{\tan 40^\circ} = 9.534\ldots$$

$$FB = 22 - 9.534\ldots = 12.465\ldots$$

**2** **Use trigonometry**

In right-angled triangle $CFB$,

$$\tan x = \frac{8}{12.465\ldots}$$

$$x = 32.7^\circ$$

**FINAL ANSWER**

$32.7°$.

4MA1/2H JUNE 2024

# Question 5

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**5** The diagram shows triangle $ABC$

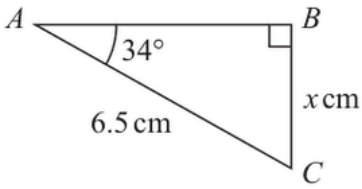



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to one decimal place.

$x =$ .....

**(Total for Question 5 is 3 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2H June 2024

### 1 Use trigonometry

The hypotenuse is $6.5$ cm, and $x$ is opposite the $34°$ angle.

$$\sin 34° = \frac{x}{6.5}$$

$$x = 6.5 \sin 34° = 3.634\ldots$$

**FINAL ANSWER**

$3.6$ cm.

4MA1/1H MAY 2024

# Question 11

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**11** The diagram shows a roof support.

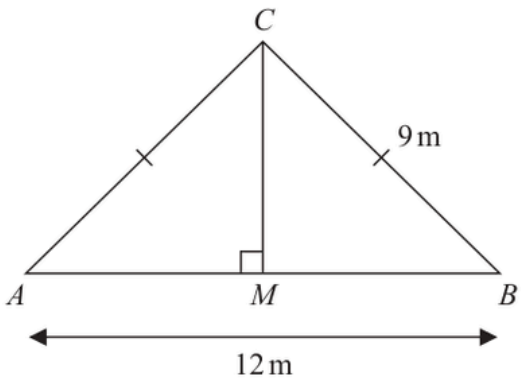



Diagram **NOT** accurately drawn

The roof support is made from four lengths of wood, $AB$, $AC$, $BC$ and $MC$

$AC = BC = 9\,\text{m}$ $\quad AB = 12\,\text{m}$

angle $AMC = 90°$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.
Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

..... euros

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H May 2024

**1** **Find the gradient**

Since $AC = BC$, the perpendicular support from $C$ meets $AB$ at its midpoint.

$$AM = \frac{12}{2} = 6$$

**2** **Use trigonometry**

In right-angled triangle $AMC$,

$$AC^2 = AM^2 + MC^2$$

$$9^2 = 6^2 + MC^2$$

$$MC^2 = 81 - 36 = 45$$

$$MC = \sqrt{45} = 6.708\ldots$$

---

**3** **A whole number of metres**

A whole number of metres must be bought, so buy $7$ m for $MC$.

---

**4** **Total length needed**

Total length needed:

$$12 + 9 + 9 + 7 = 37$$

---

**5** **Cost**

Cost:

$$37 \times 21.50 = 795.50$$

---

✓ **FINAL ANSWER**

$795.50$ euros.

4MA1/2H NOVEMBER 2024

# Question 3

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

3 The diagram shows triangle $PQR$

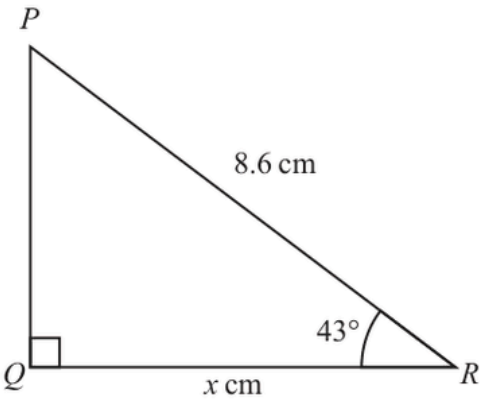



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to one decimal place.

$x =$ .....

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2H November 2024

### 1 Use trigonometry

In the right-angled triangle, $x$ is adjacent to the $43°$ angle.

$$\cos 43° = \frac{x}{8.6}$$

$$x = 8.6 \cos 43° = 6.289\ldots$$

**FINAL ANSWER**

$6.3$ cm.

4MA1/1H NOVEMBER 2024

# Question 12

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**12** $ABC$ is a right-angled triangle.
$D$ is a point on $BC$

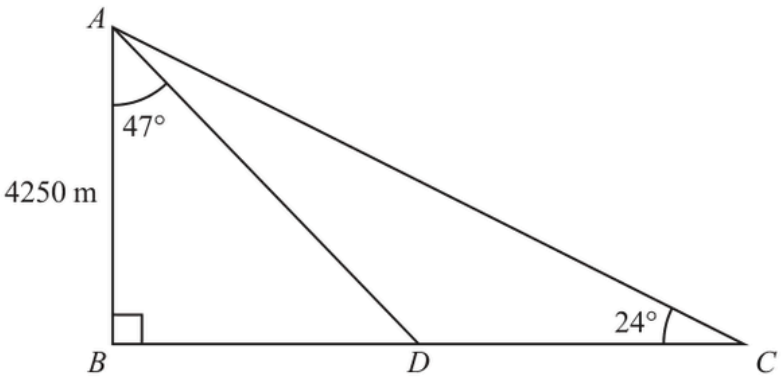



Diagram **NOT** accurately drawn

$AB = 4250$ m  angle $BAD = 47°$  angle $BCA = 24°$

Work out the length of $DC$
Give your answer correct to the nearest integer.

..... m

**(Total for Question 12 is 4 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H November 2024

**1** **Use trigonometry**

In triangle $ABD$,

$$\tan 47° = \frac{BD}{4250}$$

$$BD = 4250 \tan 47° = 4557.567\ldots$$

**2** **Use trigonometry**

In triangle $ABC$,

$$\tan 24^\circ = \frac{4250}{BC}$$

$$BC = \frac{4250}{\tan 24^\circ} = 9545.656\ldots$$

$$DC = BC - BD = 4988.089\ldots$$

---

**FINAL ANSWER**

$4988\ \text{m}$.

4WM1/01R MAY 2025

# Question 8

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**8**

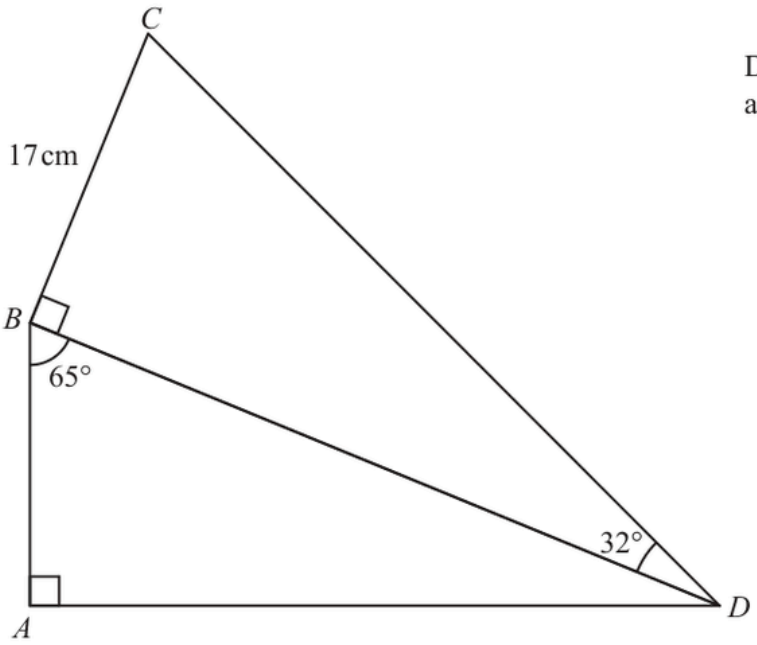



Diagram **NOT** accurately drawn

*ABD* and *BCD* are right-angled triangles.

angle $ABD = 65°$ angle $CDB = 32°$ $BC = 17$ cm

Work out the length of *AD*
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 8 is 4 marks)**

## WORKED SOLUTION

Question 8 · 4WM1/01R May 2025

**1** **Use trigonometry**

In right triangle $BCD$,

$$\tan 32° = \frac{17}{BD}$$

$$BD = \frac{17}{\tan 32°}$$

**2** **Use trigonometry**

In right triangle $ABD$,

$$\sin 65^\circ = \frac{AD}{BD}$$

$$AD = \frac{17}{\tan 32^\circ} \sin 65^\circ \approx 24.6567$$

---

✓ **FINAL ANSWER**

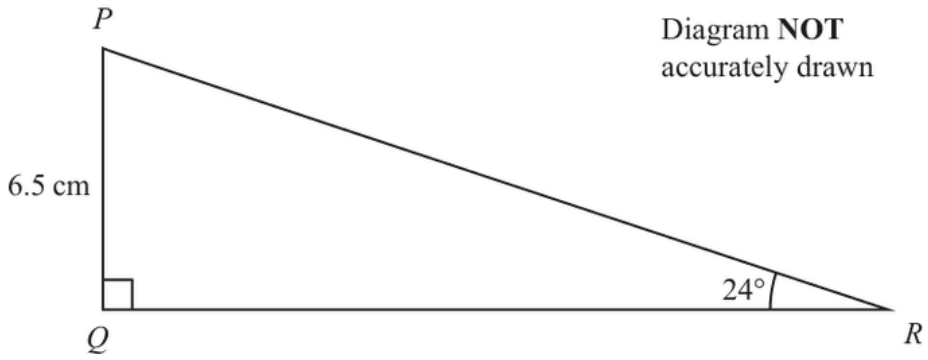
$AD = 24.7$ cm to 3 significant figures.

4MA1/1HR JUNE 2025

# Question 9

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**9** The diagram shows triangle $PQR$




Work out the length of $QR$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 9 is 3 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1HR June 2025

### 1 Use trigonometry

$$\tan 24^\circ = \frac{6.5}{QR}$$

$$QR = \frac{6.5}{\tan 24^\circ} = 14.599\ldots$$

**FINAL ANSWER**

$14.6$ cm.

4MA1/2H JUNE 2025

# Question 10

5 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**10** $ABD$ and $ABC$ are right-angled triangles.

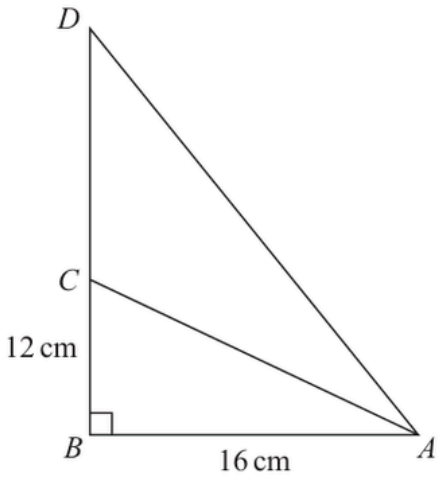



Diagram **NOT** accurately drawn

$AB = 16\,\text{cm}$ $BC = 12\,\text{cm}$ $AD = 1.5 \times AC$

Find the length of $CD$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 10 is 5 marks)**

## WORKED SOLUTION

Question 10 · 4MA1/2H June 2025

**1** **Use trigonometry**

In triangle $ABC$,

$$AC^2 = 16^2 + 12^2 = 400$$

$$AC = 20$$

**2** **Use trigonometry**

Given $AD = 1.5AC$,

$$AD = 30$$

**3** **Use trigonometry**

In triangle $ABD$,

$$BD^2 = 30^2 - 16^2 = 644$$

$$BD = 25.377\ldots$$

---

**4** **Use trigonometry**

So

$$CD = BD - BC = 25.377\ldots - 12 = 13.377\ldots$$

---

✓ **FINAL ANSWER**

$13.4\ \text{cm}$.

4MA1/2HR JUNE 2025

# Question 11

5 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**11** The diagram shows two triangles, $ADE$ and $CDB$

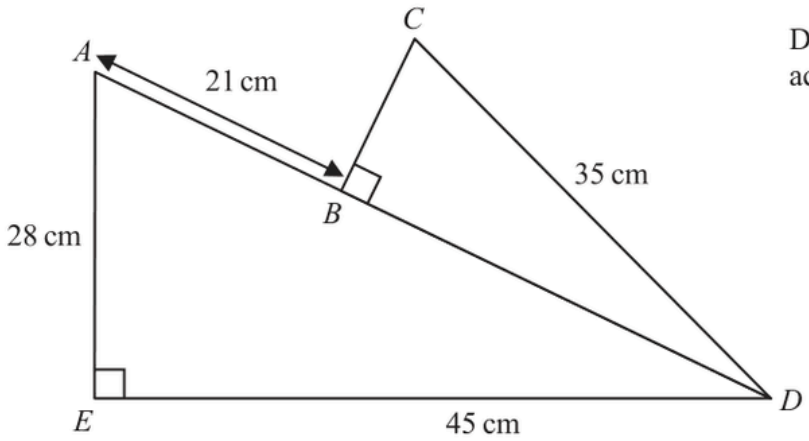



Diagram **NOT** accurately drawn

$ABD$ is a straight line.

$AE = 28\,\text{cm}$ $ED = 45\,\text{cm}$ $AB = 21\,\text{cm}$ $CD = 35\,\text{cm}$

angle $AED$ = angle $CBD = 90°$

Work out the area of triangle $CDB$
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 11 is 5 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2HR June 2025

**1** **Use trigonometry**

In right-angled triangle $AFD$,

$$AD^2 = 28^2 + 45^2$$

$$AD^2 = 2809$$

$$AD = 53$$

**2** **Use trigonometry**

Since $AB = 21$,

$$BD = 53 - 21 = 32$$

**3** **Use trigonometry**

In right-angled triangle $CBD$,

$$CB^2 = 35^2 - 32^2$$

$$CB^2 = 201$$

$$CB = \sqrt{201}$$

---

**4** **Area of triangle is**

Area of triangle $CDB$ is

$$\frac{1}{2} \times BD \times CB$$

$$\text{Area} = \frac{1}{2} \times 32 \times \sqrt{201} = 226.9\ldots$$

---

✓ **FINAL ANSWER**

$227\ \text{cm}^2$ to 3 significant figures.

# Question 13

5 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**13** In the diagram, $AB$ represents a vertical tower and $CD$ represents a vertical building.

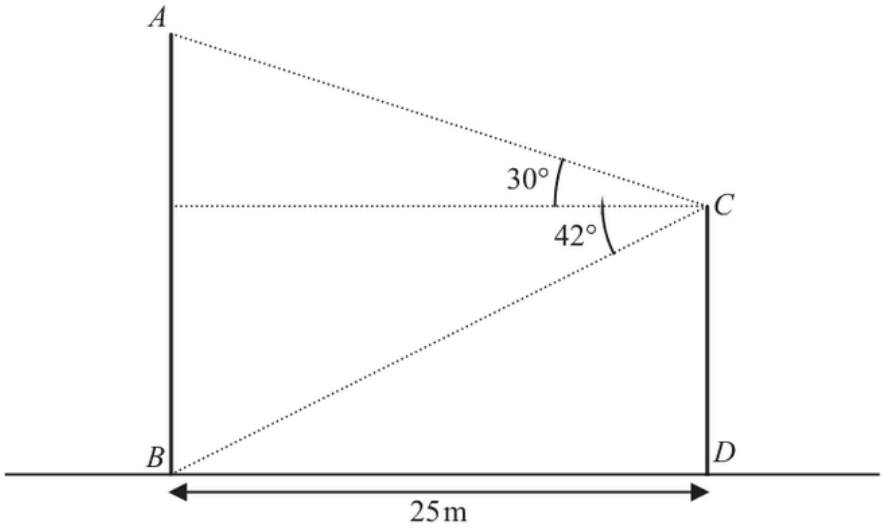



Diagram **NOT** accurately drawn

The bottom of the tower, $B$, and the bottom of the building, $D$, are on horizontal ground such that $BD = 25$ m

The angle of elevation of $A$ from $C$ is $30°$
The angle of depression of B from $C$ is $42°$

(a) Work out the height of the tower $AB$
Give your answer correct to one decimal place.

..... m

**(3)**

(b) Work out the angle of elevation of $A$ from $D$
Give your answer correct to one decimal place.

..... °

**(2)**

**(Total for Question 13 is 5 marks)**

**1** **The height of the building**

The height of the building is

$$CD = 25\tan 42°$$

**2** **Use trigonometry**

The extra height from $C$ to $A$ is

$$25\tan 30°$$

**3** **Use trigonometry**

So

$$AB = 25\tan 42° + 25\tan 30° \approx 36.9439$$

**4** **Use trigonometry**

For the angle of elevation $\theta$ from $D$,

$$\tan\theta = \frac{AB}{25}$$

$$\theta \approx 55.9137°$$

✓ **FINAL ANSWER**

$AB = 36.9$ m, angle of elevation $55.9°$.

4WM1/01 NOVEMBER 2025

# Question 9

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**9** The diagram shows a sketch of a right-angled triangle, $ABC$

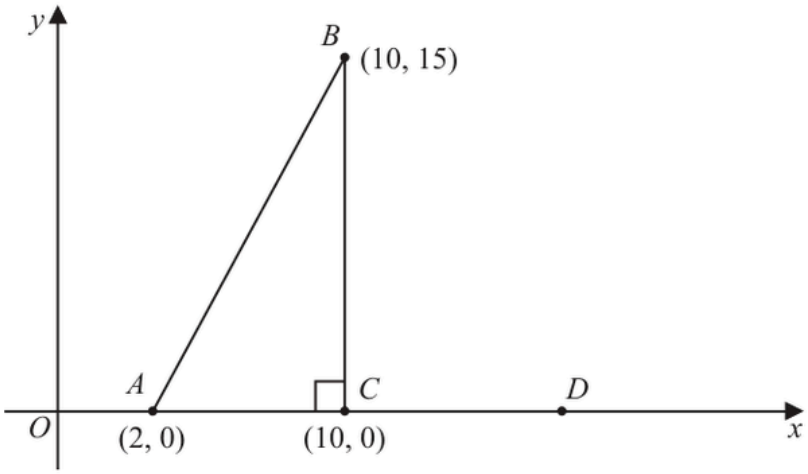



Diagram **NOT** accurately drawn

$A$ is the point with coordinates $(2, 0)$
$B$ is the point with coordinates $(10, 15)$
$C$ is the point with coordinates $(10, 0)$

$D$ is a point on the $x$-axis such that $AB = AD$

Find the coordinates of $D$

(..... , .....)

**(Total for Question 9 is 4 marks)**

## WORKED SOLUTION

Question 9 · 4WM1/01 November 2025

**1** **Simplify surd**

$$AB = \sqrt{(10-2)^2 + (15-0)^2}$$

$$= \sqrt{64 + 225} = 17$$

**2** **Calculate value**

Since $D$ is on the positive $x$-axis and $AD = 17$,

$$D = (2 + 17, 0) = (19, 0)$$

**FINAL ANSWER**

$(19, 0)$.

4WM1/01 NOVEMBER 2025

# Question 11

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**11** $ABC$ is a right-angled triangle.

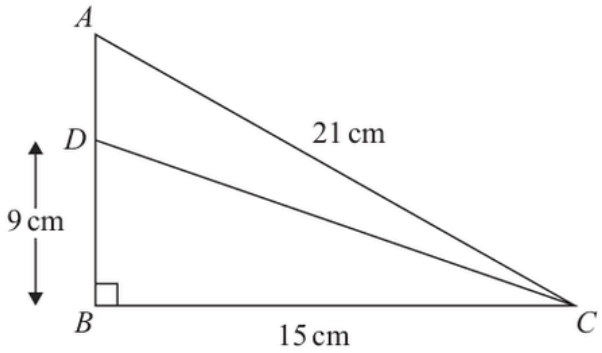



Diagram **NOT** accurately drawn

$AC = 21$ cm  $BC = 15$ cm  angle $ABC = 90°$

The point $D$ lies on $AB$ such that $DB = 9$ cm

Work out the size of angle $ACD$
Give your answer correct to one decimal place.

.....°

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4WM1/01 November 2025

**1 Use trigonometry**

First find $AB$:

$$AB^2 = 21^2 - 15^2 = 216$$

$$AB = 14.696\ldots$$

$$AD = AB - DB = 14.696\ldots - 9 = 5.696\ldots$$

**2 Angle**

Angle $ACB$:

$$\cos ACB = \frac{15}{21}$$

$$ACB = 44.415\ldots°$$

**3** **Angle**

Angle $DCB$:

$$\tan DCB = \frac{9}{15}$$

$$DCB = 30.963\ldots^\circ$$

$$ACD = 44.415\ldots - 30.963\ldots = 13.451\ldots^\circ$$

---

✓ **FINAL ANSWER**

$13.5^\circ$.

4MA1/1H NOVEMBER 2025

# Question 11

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**11** $ABC$ is a right-angled triangle.

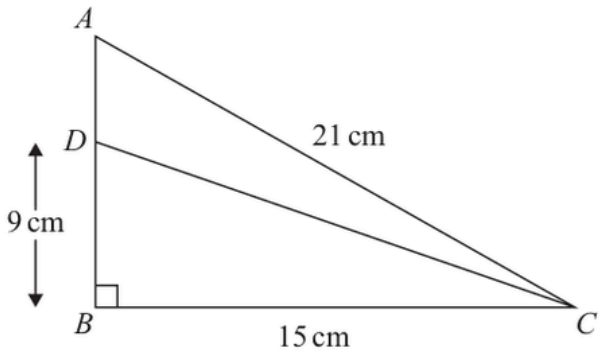



Diagram **NOT** accurately drawn

$AC = 21\,\text{cm}$ $BC = 15\,\text{cm}$ angle $ABC = 90°$

The point $D$ lies on $AB$ such that $DB = 9\,\text{cm}$

Work out the size of angle $ACD$
Give your answer correct to one decimal place.

.....°

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/1H November 2025

**1** **Use trigonometry**

First find $AB$:

$$AB^2 = 21^2 - 15^2 = 216$$

$$AB = 14.696\ldots$$

$$AD = AB - DB = 14.696\ldots - 9 = 5.696\ldots$$

**2** **Angle**

Angle $ACB$:

$$\cos ACB = \frac{15}{21}$$

$$ACB = 44.415\ldots^\circ$$

---

**3** **Angle**

Angle $DCB$:

$$\tan DCB = \frac{9}{15}$$

$$DCB = 30.963\ldots^\circ$$

$$ACD = 44.415\ldots - 30.963\ldots = 13.451\ldots^\circ$$

---

✓ **FINAL ANSWER**

$13.5^\circ$.

4MA1/2H NOVEMBER 2025

# Question 11

4 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**11** The diagram shows a right-angled triangle $ABC$

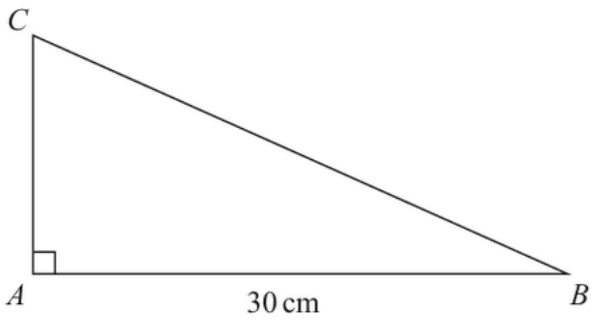



Diagram **NOT** accurately drawn

Area of triangle $ABC = 240\,\text{cm}^2$

Work out the perimeter of triangle $ABC$

..... cm

**(Total for Question 11 is 4 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2H November 2025

**1 Use the area to find**

Use the area to find $AC$:

$$\frac{1}{2} \times 30 \times AC = 240$$

$$15AC = 240$$

$$AC = 16$$

**2 Use trigonometry**

Then

$$BC^2 = 30^2 + 16^2 = 1156$$

$$BC = 34$$

**3** **Perimeter**

Perimeter:

$$30 + 16 + 34 = 80$$

---

✓ **FINAL ANSWER**

$80\ \text{cm}$.

4MA1/2HR JUNE 2026

# Question 1

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**1** The diagram shows a right-angled triangle.

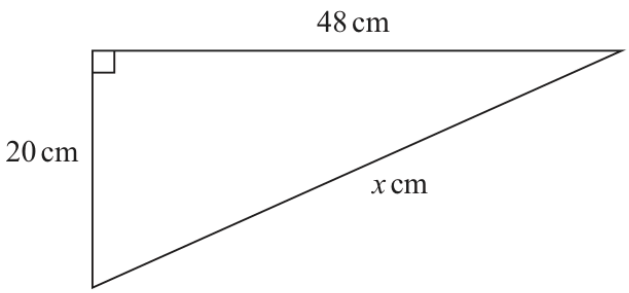



Diagram **NOT** accurately drawn

Work out the value of $x$

$x =$ .....

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2HR June 2026

### 1 Solution | Step 1

$$x^2 = 20^2 + 48^2 = 400 + 2304 = 2704.$$

Use Pythagoras because the 20 cm and 48 cm sides meet at the right angle.

### 2 Solution | Step 2

$$x = \sqrt{2704} = 52.$$

Take the positive square root because x is a length.

**FINAL ANSWER**

$x = 52$, so the hypotenuse is $52$ cm.

4MA1/1H JUNE 2026

# Question 2

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**2**

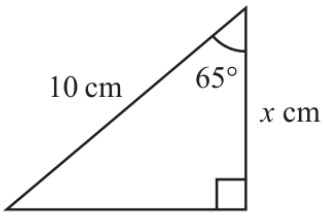



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to 3 significant figures.

$x =$ .....

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1H June 2026

### 1 Solution | Step 1

$$\cos 65^\circ = \frac{x}{10}.$$

The side of length 10 cm is the hypotenuse and x is adjacent to the given angle.

### 2 Solution | Step 2

$$x = 10\cos 65^\circ = 4.22618\ldots = 4.23 \text{ cm} \quad (3 \text{ s.f.}).$$

Multiply by 10 and round only the final value.

**✓ FINAL ANSWER**

$x = 4.23$ cm

4WM1/01 MAY 2026

# Question 2

3 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**2**

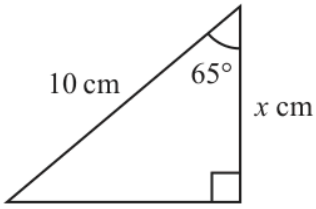



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to 3 significant figures.

$x =$ .....

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4WM1/01 May 2026

**1 Choose the ratio**

$x$ is next to the $65°$ angle and $10$ cm is the hypotenuse, so use cosine.

**2 Calculate**

$$x = 10\cos 65° = 4.2261\ldots$$

**FINAL ANSWER**

$x = 4.23$

4MA1/2HR JUNE 2026

# Question 11

5 marks | Right-Angled Triangles - Pythagoras & Trigonometry

**11** The diagram shows a square $ABCD$ and a right-angled triangle $EFG$

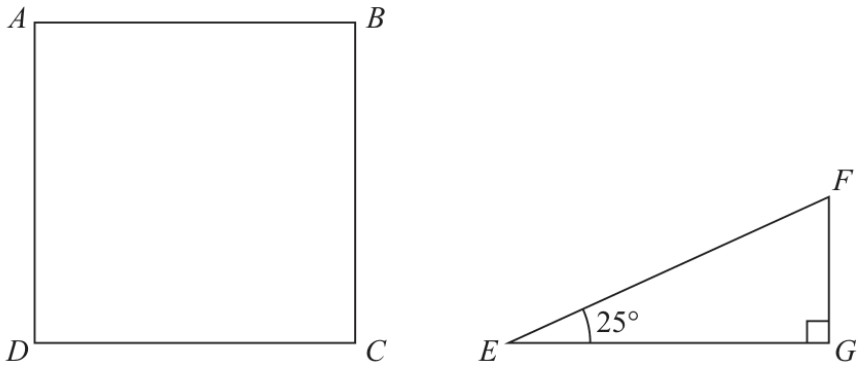



Diagram **NOT** accurately drawn

The area of square $ABCD$ is $36\,\text{cm}^2$
$DC = EG$

Work out the area of triangle $EFG$
Give your answer correct to one decimal place.
Show your working clearly.

..... $\text{cm}^2$

**(Total for Question 11 is 5 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2HR June 2026

**1** **Solution | Step 1**

$$DC = \sqrt{36} = 6\ \text{cm}, \qquad EG = 6\ \text{cm}.$$

All sides of the square have length equal to the square root of its area, and DC=EG.

**2** **Solution | Step 2**

$$\tan 25^\circ = \frac{FG}{EG} \quad \Rightarrow \quad FG = 6\tan 25^\circ = 2.7978459\ldots\ \text{cm}.$$

Relative to angle E, FG is opposite and EG is adjacent.

**3** **Solution | Step 3**

$$\text{Area} = \frac{1}{2}(EG)(FG) = \frac{1}{2}(6)(6\tan 25^\circ) = 8.3935378\ldots\ \text{cm}^2.$$

The base EG and height FG are perpendicular.

**FINAL ANSWER**

$8.4\ \text{cm}^2$ (1 decimal place).



TOPIC

# 3D Pythagoras & Trigonometry

Every past-paper question on this topic, 2020 to 2026.

# Question 19

4 marks | 3D Pythagoras & Trigonometry

**19** The diagram shows a triangular prism, $ABCDEF$, with a rectangular base $ABCD$

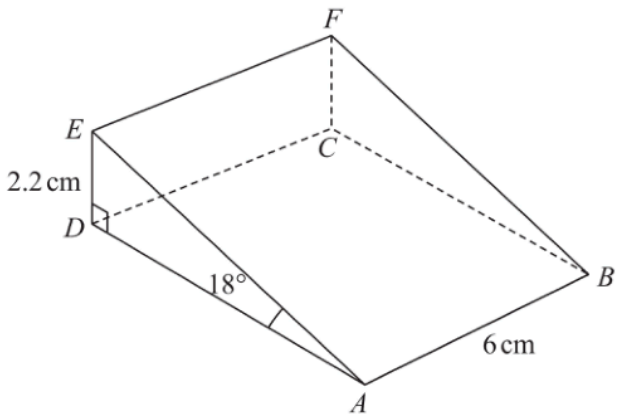



Diagram **NOT** accurately drawn

$AB = 6\,\text{cm}$ $DE = 2.2\,\text{cm}$ angle $DAE = 18°$ angle $ADE = 90°$

Work out the angle that $BE$ makes with the plane $ABCD$
Give your answer correct to one decimal place.

## 1 Find AD

In triangle $ADE$,

$$\tan 18^\circ = \frac{DE}{AD}$$

$$AD = \frac{2.2}{\tan 18^\circ} = 6.77\ldots$$

## 2 Find DB in the base

The base $ABCD$ is a rectangle, so

$$DB = \sqrt{6^2 + 6.77\ldots^2} = 9.04\ldots$$

## 3 Use the projection triangle

The projection of $BE$ on the plane is $BD$. Let the required angle be $\theta$.

$$\tan\theta = \frac{DE}{DB} = \frac{2.2}{9.04\ldots}$$

## 4 Calculate the angle

$$\theta = 13.7^\circ$$

**FINAL ANSWER**

$13.7^\circ$.

4MA1/1H JANUARY 2020

# Question 19

4 marks | 3D Pythagoras & Trigonometry

**19** The diagram shows a cuboid $ABCDEFGH$.

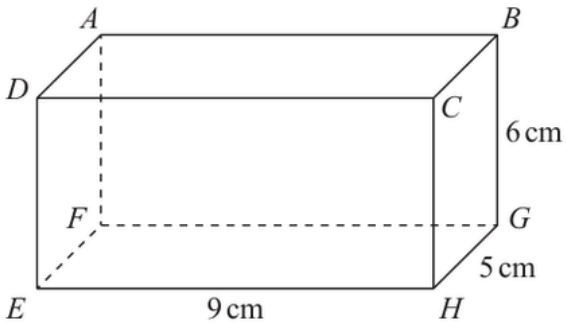



Diagram **NOT** accurately drawn

$EH = 9\,\text{cm}$, $HG = 5\,\text{cm}$ and $GB = 6\,\text{cm}$.

Work out the size of the angle between $AH$ and the plane $EFGH$.
Give your answer correct to 3 significant figures.

.....°

**(Total for Question 19 is 4 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1H January 2020

**1** **The projection of onto the**

The projection of $AH$ onto the base plane is the diagonal of the base rectangle:

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

**2** **Use trigonometry**

The vertical height is $6\ \text{cm}$, so

$$\tan\theta = \frac{6}{\sqrt{106}}$$

$$\theta = 30.232\ldots^\circ$$

**FINAL ANSWER**

$30.2^\circ$.

4MA1/1HR MAY/NOVEMBER 2020

# Question 17

6 marks | 3D Pythagoras & Trigonometry

**17** The diagram shows a prism $ABCDEFGH$ with a horizontal base.

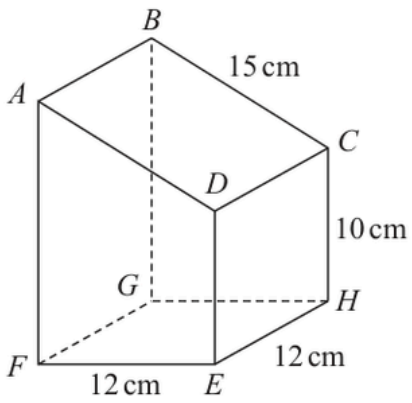



Diagram **NOT** accurately drawn

The base of the prism, $EFGH$, is a square of side 12 cm.
Trapezium $ADEF$ is a cross section of the prism where $AF$ and $DE$ are vertical edges.

$DE = CH = 10$ cm
$AD = BC = 15$ cm

(a) Work out the size of the angle between $CF$ and the base $EFGH$.
Give your answer correct to one decimal place.

..... °

(3)

(b) Work out the length of $BE$.
Give your answer correct to one decimal place.

..... cm

(3)

**(Total for Question 17 is 6 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1HR May/November 2020

**1** **Use trigonometry**

(a) $C$ is vertically above $H$, so the vertical height from $F$ to $C$ is $10$ cm.

2 **The projection of on the**

The projection of $CF$ on the base is $FH$:

$$FH = 12\sqrt{2}$$

$$\tan\theta = \frac{10}{12\sqrt{2}}$$

$$\theta = 30.508\ldots^\circ$$

---

3 **Answer**

**Answer:** $30.5^\circ$.

---

4 **(b) In the trapezium cross**

(b) In the trapezium cross-section, $AD = 15$ and the horizontal distance is $12$, so the difference in vertical heights is

$$\sqrt{15^2 - 12^2} = 9$$

---

5 **Use trigonometry**

Since $DE = 10$, the height $BG = 19\ \text{cm}$.

---

6 **The projection of on the**

The projection of $BE$ on the base is $GE = 12\sqrt{2}$, so

$$BE = \sqrt{19^2 + (12\sqrt{2})^2} = 25.475\ldots$$

---

✓ **FINAL ANSWER**

$25.5\ \text{cm}$.

4MA1/2HR JANUARY 2021

# Question 22

4 marks | 3D Pythagoras & Trigonometry

**22** $ABC$ is an isosceles triangle in a horizontal plane.
The point $T$ is vertically above $B$.

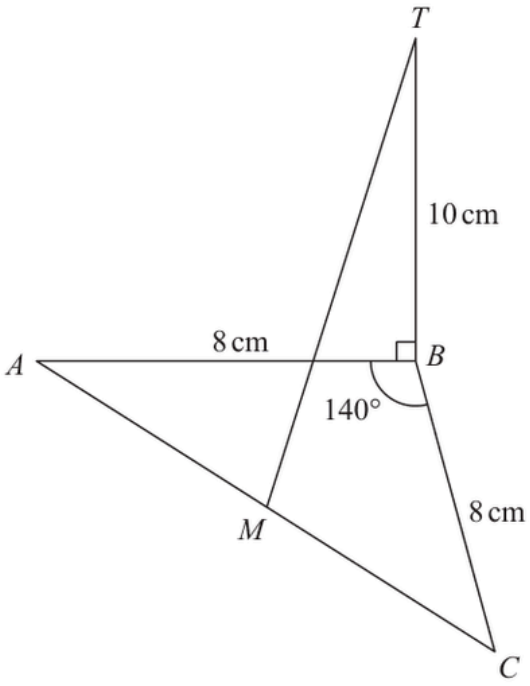



Diagram **NOT** accurately drawn

Angle $ABC = 140°$
$AB = BC = 8$ cm
$TB = 10$ cm
$M$ is the midpoint of $AC$.

Calculate the size of the angle between $MT$ and the horizontal plane $ABC$.
Give your answer correct to one decimal place.

.....°

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2HR January 2021

**1** **Use trigonometry**

Since $AB = BC = 8$, triangle $ABC$ is isosceles.

**2** **The line from to the**

The line from $B$ to the midpoint $M$ bisects angle $ABC$.

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

**3** **Use trigonometry**

In right triangle $ABM$,

$$\cos 70^\circ = \frac{BM}{8}$$

$$BM = 8\cos 70^\circ$$

**4** **Use trigonometry**

Since $T$ is vertically above $B$, the projection of $MT$ onto the horizontal plane is $MB$.

**5** **Let the required angle be**

Let the required angle be $\theta$.

$$\tan\theta = \frac{TB}{BM}$$

$$\tan\theta = \frac{10}{8\cos 70^\circ}$$

$$\theta = 74.7^\circ$$

**FINAL ANSWER**

$74.7^\circ$

4MA1/1H MAY 2021

# Question 22

4 marks | 3D Pythagoras & Trigonometry

**22** The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

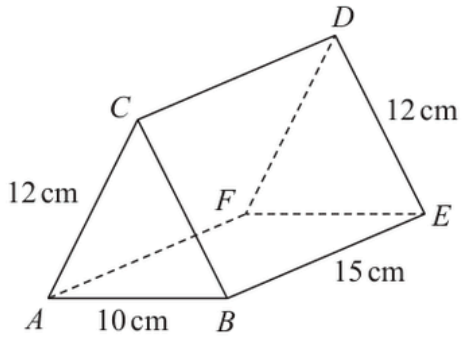



Diagram **NOT** accurately drawn

$AC = BC = FD = ED = 12\,\text{cm}$ $AB = 10\,\text{cm}$ $BE = 15\,\text{cm}$

Calculate the size of the angle between $AD$ and the base $ABEF$.
Give your answer correct to 3 significant figures.

.....°

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H May 2021

### 1 Calculate value

The triangular end is isosceles, with

$$AC = BC = 12, \qquad AB = 10$$

### 2 The perpendicular height of the

The perpendicular height of the triangle is found by halving $AB$:

$$AM = 5$$

$$CM^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$CM = \sqrt{119}$$

### 3 The projection of on the

The projection of $AD$ on the horizontal base goes from $A$ to the midpoint of $FE$. Its length is

$$\sqrt{15^2 + 5^2} = \sqrt{250}$$

### 4 Use trigonometry

So, for the angle $\theta$ between $AD$ and the base,

$$\tan\theta = \frac{\sqrt{119}}{\sqrt{250}}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{119}}{\sqrt{250}}\right) = 34.602\ldots$$

**FINAL ANSWER**

$34.6°$

4MA1/2HR JANUARY 2022

# Question 19

3 marks | 3D Pythagoras & Trigonometry

**19** $ABCD$ is a horizontal rectangular field.

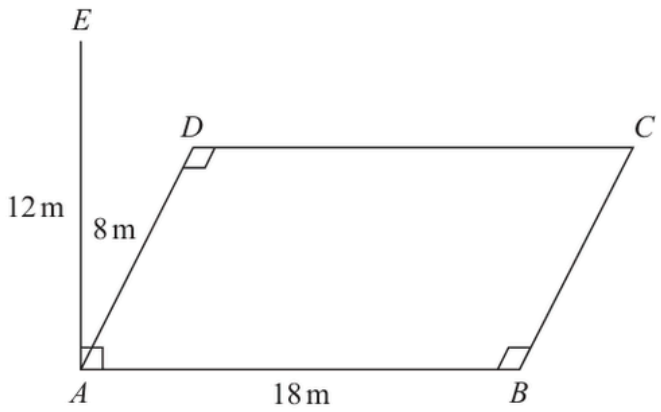



Diagram **NOT** accurately drawn

A vertical pole, $AE$, is placed at the corner $A$ of the field.

$AE = 12\,\text{m}$ $AB = 18\,\text{m}$ $AD = 8\,\text{m}$

Calculate the size of the angle between $EC$ and the plane $ABCD$
Give your answer correct to one decimal place.

.....°

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2HR January 2022

**1 The projection of onto the**

The projection of $EC$ onto the field is $AC$.

$$AC = \sqrt{18^2 + 8^2} = \sqrt{388}$$

**2 Use trigonometry**

The vertical height is $AE = 12\ \text{m}$, so

$$\tan\theta = \frac{12}{\sqrt{388}}$$

$$\theta = 31.350\ldots^\circ$$

**FINAL ANSWER**

$31.4°$.

4MA1/1H JANUARY 2022

# Question 21

5 marks | 3D Pythagoras & Trigonometry

**21** The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

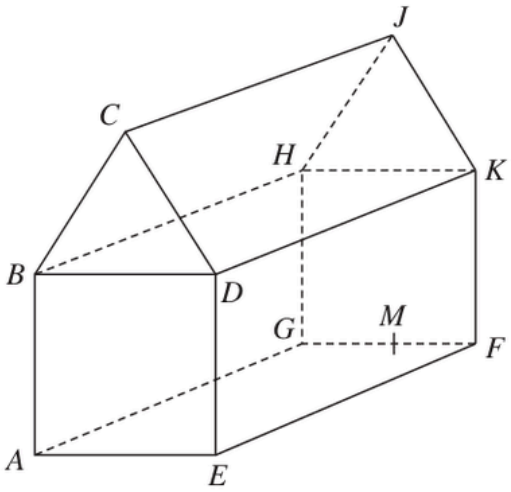



Diagram **NOT** accurately drawn

$ABCDE$ is a cross section of the prism where
$ABDE$ is a square
$BCD$ is an equilateral triangle

$EF = 2 \times AE$
$M$ is the midpoint of $GF$ so that $JM$ is vertical.

Angle $MAJ = y°$

Given that $\tan y° = T$

find the value of $T$, giving your answer in the form $\dfrac{\sqrt{p} + \sqrt{q}}{17}$ where $p$ and $q$ are integers.

$T =$ .....

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H January 2022

**1** **Let**

Let

$$AE = x$$

**2** **Use trigonometry**

Since $ABDE$ is a square,

$$AB = BD = DE = AE = x$$

**3** **Use trigonometry**

Since $BCD$ is an equilateral triangle, the height from $C$ to $BD$ is

$$\frac{\sqrt{3}}{2}x$$

**4** **Use trigonometry**

Therefore the vertical height from the base to $J$ is

$$JM = x + \frac{\sqrt{3}}{2}x$$

$$JM = \frac{x(2+\sqrt{3})}{2}$$

**5** **Use trigonometry**

Also,

$$EF = 2AE = 2x$$

**6** **Find the midpoint**

Since $M$ is the midpoint of $GF$, the horizontal distance from $A$ to $M$ is

$$AM = \sqrt{\left(\frac{x}{2}\right)^2 + (2x)^2}$$

$$AM = \sqrt{\frac{x^2}{4} + 4x^2}$$

$$AM = \frac{x\sqrt{17}}{2}$$

**7** **Use trigonometry**

In right triangle $AMJ$,

$$\tan y^\circ = \frac{JM}{AM}$$

$$T = \frac{\frac{x(2+\sqrt{3})}{2}}{\frac{x\sqrt{17}}{2}}$$

$$T = \frac{2+\sqrt{3}}{\sqrt{17}}$$

---

**8** **Rationalise the denominator**

Rationalise the denominator:

$$T = \frac{(2+\sqrt{3})\sqrt{17}}{17}$$

$$T = \frac{2\sqrt{17}+\sqrt{51}}{17}$$

$$T = \frac{\sqrt{68}+\sqrt{51}}{17}$$

---

✓ **FINAL ANSWER**

$$T = \frac{\sqrt{68}+\sqrt{51}}{17}$$

4MA1/1HR JUNE 2022

# Question 18

4 marks | 3D Pythagoras & Trigonometry

**18** The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

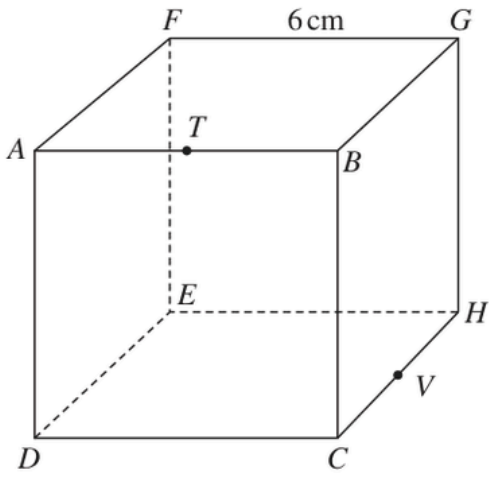



Diagram **NOT** accurately drawn

$T$ is the midpoint of $AB$ and $V$ is the midpoint of $CH$

Work out the distance from $T$ to $V$ in a straight line through the cube.
Give your answer in the form $\sqrt{a}$ cm where $a$ is an integer.

..... cm

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1HR June 2022

**1 The cube has side length**

The cube has side length $\mathbf{6}$ cm.

**2 Find the midpoint**

Let $T$ be the midpoint of $AB$ and $V$ be the midpoint of $CH$.

**3 The changes in the three**

The changes in the three perpendicular directions from $T$ to $V$ are

$$3, \quad 3, \quad 6$$

**4** **Use trigonometry**

So by 3D Pythagoras,

$$TV = \sqrt{3^2 + 3^2 + 6^2}$$

$$TV = \sqrt{9 + 9 + 36} = \sqrt{54}$$

---

✓ **FINAL ANSWER**

$\sqrt{54}\ \text{cm}$.

# Question 22

5 marks | 3D Pythagoras & Trigonometry

**22** Here is a cuboid *ABCDEFGH*

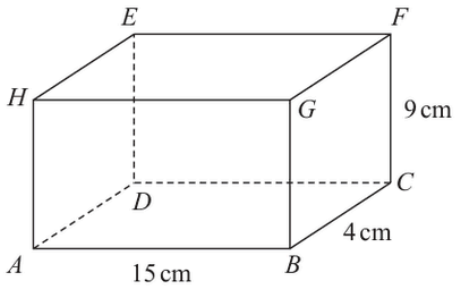



Diagram **NOT** accurately drawn

$AB = 15\,\text{cm}$ $BC = 4\,\text{cm}$ $CF = 9\,\text{cm}$

(a) Work out the length of *BE*
Give your answer correct to 3 significant figures.

..... cm

**(2)**

Here is a cuboid *PQRSTUVW*

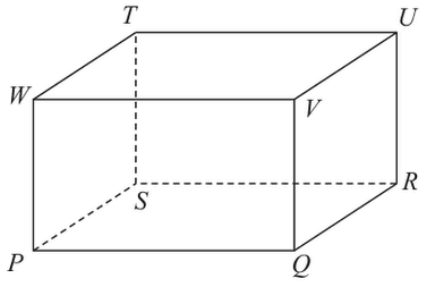



Diagram **NOT** accurately drawn

$PR = 42\,\text{cm}$

The size of the angle between *PU* and the plane *PQRS* is 30°

*M* is the midpoint of *PR*

(b) Work out the size of angle *UMR*
Give your answer correct to 3 significant figures.

.....°

**(3)**

**(Total for Question 22 is 5 marks)**

**1** **Find the gradient**

For part (a), $BE$ is the space diagonal using the three perpendicular lengths:

$$AB = 15, \qquad BC = 4, \qquad CF = 9$$

$$BE^2 = 15^2 + 4^2 + 9^2$$

$$BE^2 = 225 + 16 + 81 = 322$$

$$BE = \sqrt{322} = 17.9 \text{ cm}$$

**2** **Use trigonometry**

For part (b), the projection of $PU$ onto the plane $PQRS$ is $PR$.

$$PR = 42$$

**3** **The angle between and the**

The angle between $PU$ and the plane is $30°$, so

$$\tan 30° = \frac{RU}{PR}$$

$$RU = 42 \tan 30°$$

**4** **Find the midpoint**

Since $M$ is the midpoint of $PR$,

$$MR = 21$$

**5** **Use trigonometry**

In right triangle $MRU$,

$$\tan \angle UMR = \frac{RU}{MR}$$

$$\tan \angle UMR = \frac{42 \tan 30°}{21}$$

$$\tan \angle UMR = 2 \tan 30°$$

$$\angle UMR = 49.1°$$

**FINAL ANSWER**

$BE = 17.9\ \text{cm}$, $\angle UMR = 49.1°$

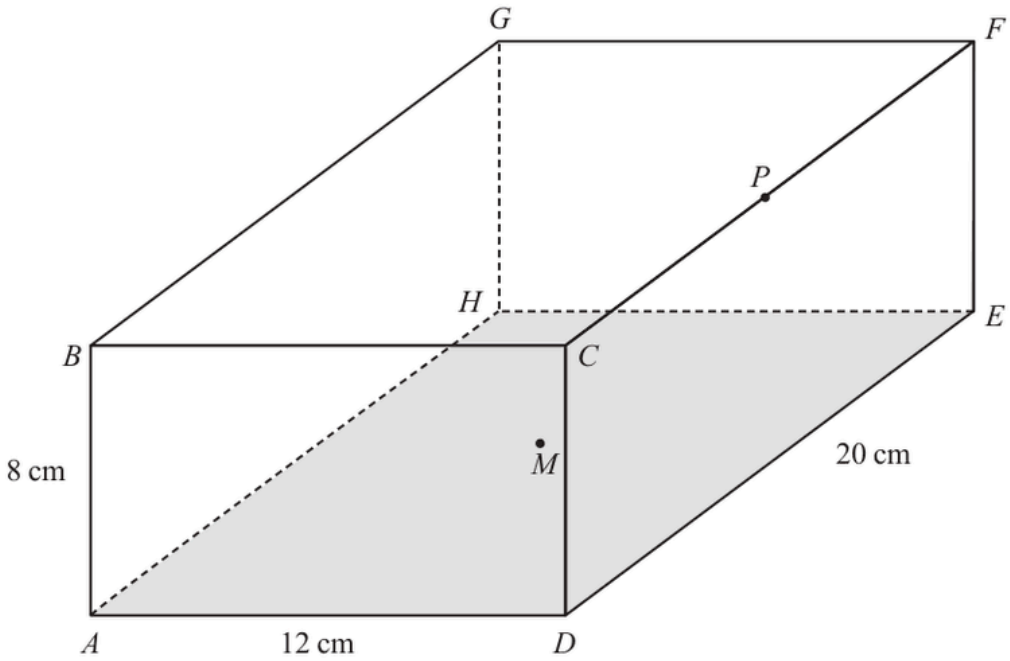
4MA1/1HR MAY 2024

# Question 22

6 marks | 3D Pythagoras & Trigonometry

**22** The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$

Diagram **NOT** accurately drawn




$AB = 8$ cm $\quad AD = 12$ cm $\quad DE = 20$ cm

$M$ is the midpoint of the base $ADEH$ and $P$ is the midpoint of the edge $CF$

Work out the size of angle $BMP$
Give your answer correct to one decimal place.

.....°

**(Total for Question 22 is 6 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1HR May 2024

**1** **Use trigonometry**

Use coordinates with

$$A = (0, 0, 0)$$

**2** **Use trigonometry**

Since

$$AD = 12, \qquad DE = 20, \qquad AB = 8$$

**3** **take**

take

$$B = (0, 0, 8)$$

**4** **The centre of the rectangular**

The centre of the rectangular base $ADEH$ is

$$M = (6, 10, 0)$$

**5** **Find the midpoint**

Point $P$ is the midpoint of $CF$.

$$C = (12, 0, 8), \qquad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

**6** **Use trigonometry**

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

### 7 Use the scalar product formula

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6°$$

---

**FINAL ANSWER**

$78.6°$

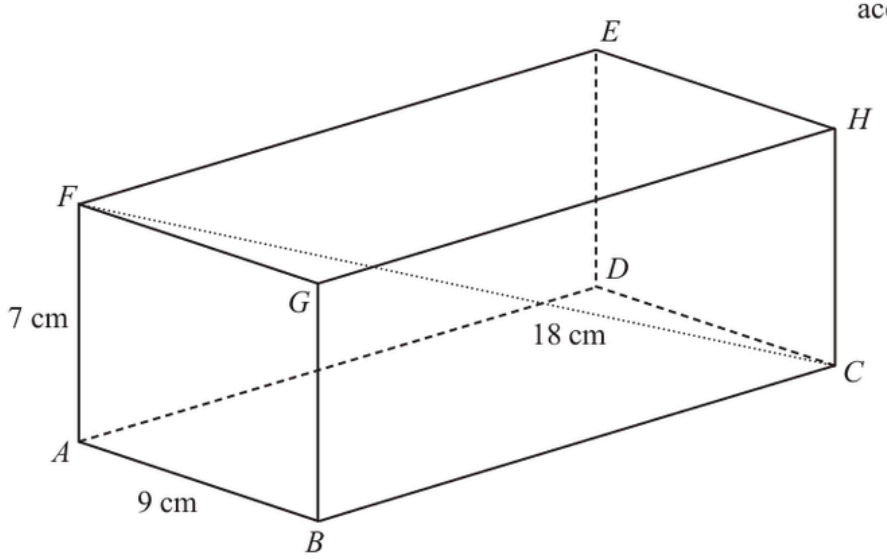
4MA1/1HR JUNE 2025

# Question 20

3 marks | 3D Pythagoras & Trigonometry

**20** The diagram shows cuboid $ABCDEFGH$

Diagram **NOT** accurately drawn




$AB = 9\,\text{cm}$ $AF = 7\,\text{cm}$ $FC = 18\,\text{cm}$

Calculate the length of $BC$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1HR June 2025

**1** **Use trigonometry**

In the rectangle $ABCF$, use Pythagoras with the diagonal $FC$:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

**2** **Use trigonometry**

Now use Pythagoras in triangle $ABC$:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

---

✓ **FINAL ANSWER**

$\mathbf{13.9\ cm}$ to 3 significant figures

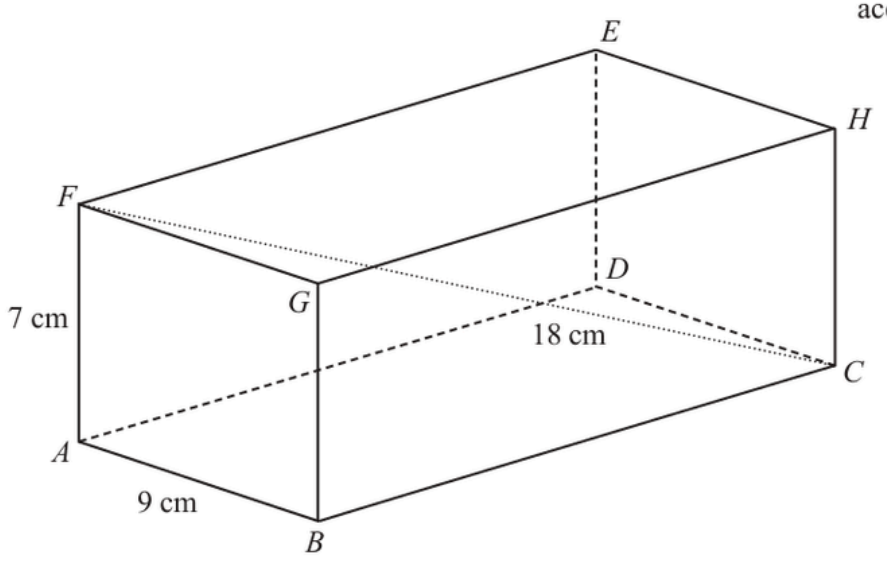
4WM1/01R MAY 2025

# Question 20

3 marks | 3D Pythagoras & Trigonometry

**20** The diagram shows cuboid $ABCDEFGH$

Diagram **NOT** accurately drawn




$AB = 9\,\text{cm}$ $AF = 7\,\text{cm}$ $FC = 18\,\text{cm}$

Calculate the length of $BC$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4WM1/01R May 2025

**1** **Use trigonometry**

In the rectangle $ABCF$, use Pythagoras with the diagonal $FC$:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

**2** **Use trigonometry**

Now use Pythagoras in triangle $ABC$:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

---

✓ **FINAL ANSWER**

$\mathbf{13.9\ cm}$ to 3 significant figures

4MA1/2HR JUNE 2025

# Question 23

3 marks | 3D Pythagoras & Trigonometry

**23** The diagram shows a triangular prism, $ABCDEF$, with a horizontal rectangular base $ABCD$

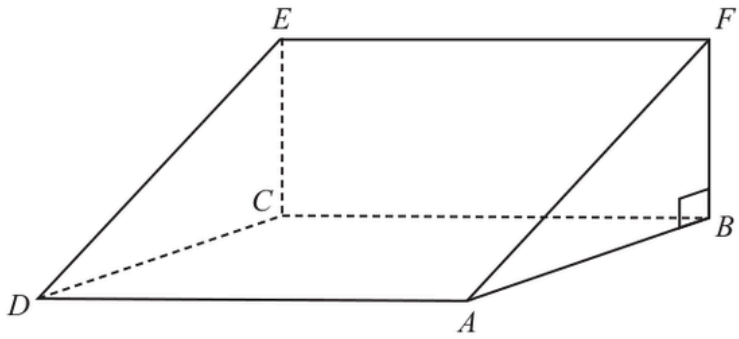



Diagram **NOT** accurately drawn

$M$ is the midpoint of the line $AC$

$AC = 40\,\text{cm}$ angle $CAE = 35°$ angle $ABF = 90°$

Work out the size of angle $CME$
Give your answer correct to 3 significant figures.

.....°

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2HR June 2025

### 1 Use trigonometry

$$AC = 40$$

### 2 The point is vertically above

The point $E$ is vertically above $C$, so triangle $ACE$ is right-angled at $C$.

$$\angle CAE = 35°$$

$$\tan 35° = \frac{CE}{AC}$$

$$CE = 40 \tan 35°$$

**3** **Find the midpoint**

Since $M$ is the midpoint of $AC$,

$$CM = 20$$

---

**4** **Use trigonometry**

In right triangle $CME$,

$$\tan \angle CME = \frac{CE}{CM}$$

$$\tan \angle CME = \frac{40 \tan 35°}{20}$$

$$\tan \angle CME = 2 \tan 35°$$

$$\angle CME = 54.5°$$

---

✓ **FINAL ANSWER**

$54.5°$

4WM1/01 MAY 2025

# Question 23

6 marks | 3D Pythagoras & Trigonometry

**23** The diagram shows cuboid $ABCDEFGH$

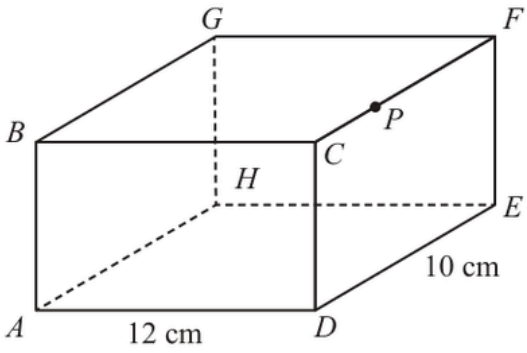



Diagram **NOT** accurately drawn

The point $P$ lies on $CF$

$AD = 12$ cm $\quad DE = 10$ cm $\quad CP = 3$ cm

The angle of elevation of $P$ from $A$ is $24°$

Calculate the size of angle $APG$
Give your answer correct to the nearest degree.
Show your working clearly.

.....°

**(Total for Question 23 is 6 marks)**

## WORKED SOLUTION

Question 23 · 4WM1/01 May 2025

**1** **Point is cm from along**

Point $P$ is $3$ cm from $C$ along $CF$. The horizontal distance from $A$ to the point directly below $P$ is

$$\sqrt{12^2 + 3^2} = \sqrt{153}$$

**2** **Use trigonometry**

Using the angle of elevation,

$$h = \sqrt{153}\tan 24° = 5.507\ldots$$

**3** **Use trigonometry**

So

$$AP = \frac{\sqrt{153}}{\cos 24^\circ} = 13.5399\ldots$$

**4** **Use trigonometry**

On the top face,

$$PG = \sqrt{12^2 + 7^2} = \sqrt{193} = 13.8924\ldots$$

**5** **Use trigonometry**

Also,

$$AG = \sqrt{12^2 + 10^2 + h^2} = 16.561\ldots$$

**6** **Use cosine rule**

Use the cosine rule in triangle $APG$.

$$\cos \angle APG = \frac{AP^2 + PG^2 - AG^2}{2(AP)(PG)}$$

$$\angle APG = 49.163\ldots^\circ$$

**7** **Use trigonometry**

To the nearest degree,

$$49^\circ$$

✓ **FINAL ANSWER**

$49^\circ$

4MA1/2H JUNE 2025

# Question 24

4 marks | 3D Pythagoras & Trigonometry

**24** The diagram shows a square-based pyramid $ABCDE$

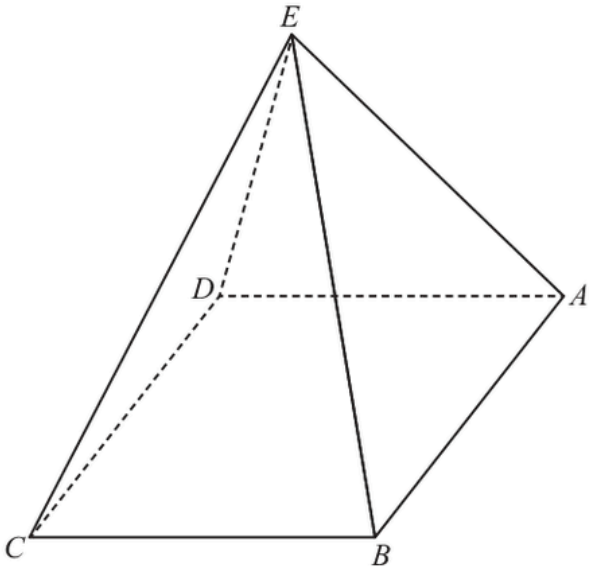



Diagram **NOT** accurately drawn

$EA = EB = EC = ED$

$M$ is the centre of the horizontal square base $ABCD$
$Q$ is the midpoint of $AB$
Angle $EQM = 80°$

$EA : AB = n : 1$

Find the value of $n$
Give your answer correct to 3 significant figures.

$n =$ .....

**(Total for Question 24 is 4 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/2H June 2025

**1 Let**

Let

$$AB = s$$

**2** **Find the midpoint**

Since $M$ is the centre of the square base and $Q$ is the midpoint of $AB$,

$$QM = \frac{s}{2}$$

**3** **Use trigonometry**

Since $EA = EB = EC = ED$, the point $E$ is vertically above $M$.

**4** **Use trigonometry**

In right triangle $EQM$,

$$\tan 80° = \frac{EM}{QM}$$

$$EM = \frac{s}{2}\tan 80°$$

**5** **The distance from the centre**

The distance from the centre of a square to a vertex is half the diagonal:

$$AM = \frac{s\sqrt{2}}{2}$$

**6** **Use trigonometry**

In right triangle $EMA$,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2}\tan 80°\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s\sqrt{\frac{\tan^2 80°}{4} + \frac{1}{2}}$$

**7** **Use trigonometry**

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80°}{4} + \frac{1}{2}}$$

$$n = 2.92$$

**FINAL ANSWER**

$n = 2.92$

# Question 26

5 marks | 3D Pythagoras & Trigonometry

**26** The diagram shows prism $ABCDEFGH$

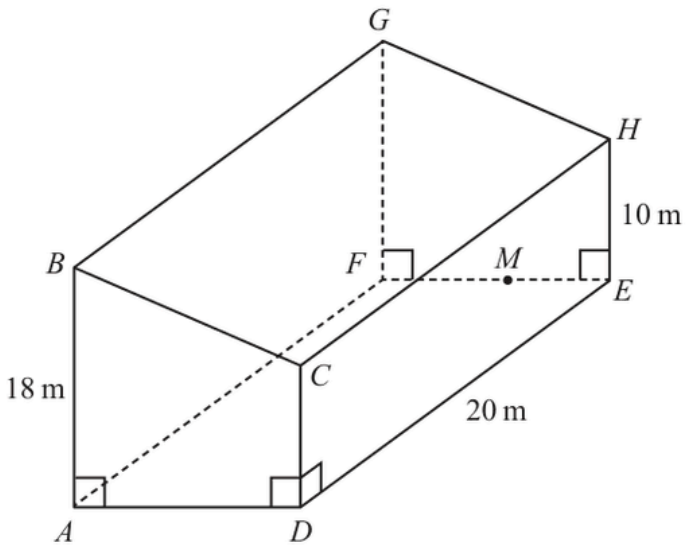



Diagram **NOT** accurately drawn

$$AB = FG = 18\,\text{m} \qquad DC = EH = 10\,\text{m} \qquad AD = FE$$
$$DE = CH = AF = BG = 20\,\text{m}$$
$$\text{angle } BAD = \text{angle } ADC = \text{angle } CDE = 90°$$

$M$ is the point on $FE$ such that $FM = \frac{3}{5}FE$

Given that angle $GMF = 60°$

calculate the angle of elevation of $G$ from $D$
Give your answer correct to one decimal place.
Show your working clearly.

.....°

**(Total for Question 26 is 5 marks)**

**1** **Use trigonometry**

From triangle $GMF$,

$$\tan 60^\circ = \frac{FG}{FM} = \frac{18}{FM}$$

$$FM = \frac{18}{\tan 60^\circ} = 6\sqrt{3}$$

**2** **Simplify surd**

Since $FM = \frac{3}{5}FE$,

$$FE = 10\sqrt{3}$$

**3** **Simplify surd**

The horizontal distance from $D$ to the point below $G$ is

$$\sqrt{20^2 + (10\sqrt{3})^2} = 10\sqrt{7}$$

**4** **Use trigonometry**

If $\theta$ is the angle of elevation,

$$\tan\theta = \frac{18}{10\sqrt{7}}$$

$$\theta \approx 34.2289^\circ$$

✓ **FINAL ANSWER**

$34.2^\circ$.

4MA1/1H NOVEMBER 2025

# Question 26

5 marks | 3D Pythagoras & Trigonometry

**26** Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

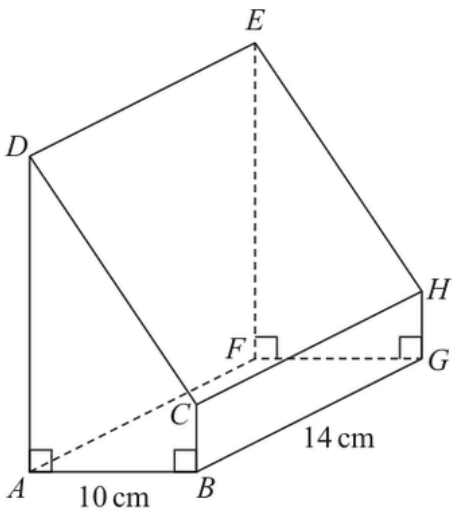



Diagram **NOT** accurately drawn

$AB = 10\,\text{cm}$ $BG = 14\,\text{cm}$

angle $DAB$ = angle $ABC$ = angle $EFG$ = angle $FGH = 90°$

$BC = \frac{1}{5}AD$

The angle of elevation of $E$ from $B$ is $50°$

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... $\text{cm}^3$

**(Total for Question 26 is 5 marks)**

## WORKED SOLUTION

Question 26 · 4MA1/1H November 2025

**1** **The horizontal base is a**

The horizontal base $ABGF$ is a rectangle.

$$AB = 10, \qquad BG = 14$$

**2** **Use trigonometry**

So the horizontal distance from $B$ to $F$ is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

**3** **Use trigonometry**

The angle of elevation of $E$ from $B$ is $50°$.

**4** **Use trigonometry**

Since $E$ is vertically above $F$,

$$\tan 50° = \frac{FE}{BF}$$

$$FE = \sqrt{296} \tan 50°$$

**5** **The prism has the same**

The prism has the same cross section throughout, so

$$AD = FE = \sqrt{296} \tan 50°$$

**6** **Use trigonometry**

Also

$$BC = \frac{1}{5} AD$$

**7** **Area of the trapezium cross**

Area of the trapezium cross section $ABCD$:

$$\frac{1}{2}(AD + BC) \times AB$$

$$= \frac{1}{2}\left(AD + \frac{1}{5}AD\right) \times 10$$

$$= 6AD$$

**8** **Volume of the prism**

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296}\tan 50°$$

$$= 1722.311\ldots$$

**9** **Correct to 3 significant figures**

Correct to 3 significant figures:

$$1720$$

✓ **FINAL ANSWER**

$$\boxed{1720\ \text{cm}^3}$$

# Question 19

3 marks | 3D Pythagoras & Trigonometry

**19** The diagram shows cuboid $PQRSTUVW$

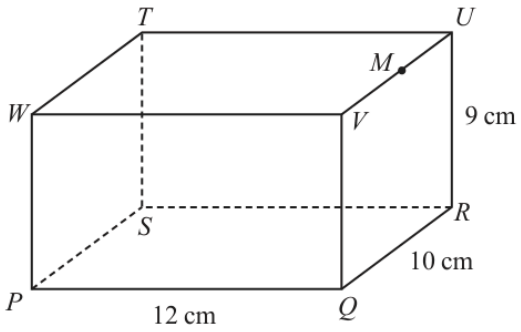



Diagram **NOT** accurately drawn

$PQ = 12$ cm $\quad QR = 10$ cm $\quad RU = 9$ cm

$M$ is the midpoint of $VU$

Work out the size of the angle between line $MP$ and plane $PQRS$
Give your answer correct to one decimal place.

..... °

**(Total for Question 19 is 3 marks)**

**1** **Drop M to the base**

$M$ is above the midpoint $N$ of $QR$, so $QN = 5$ and $MN = 9$.

**2** **Length PN**

$$PN = \sqrt{12^2 + 5^2} = 13$$

**3** **Angle**

$$\tan\theta = \frac{9}{13} \Rightarrow \theta = 34.69\ldots^\circ$$

✓ **FINAL ANSWER**

$34.7^\circ$

4MA1/1H JUNE 2026

# Question 25

5 marks | 3D Pythagoras & Trigonometry

**25** The diagram shows a triangular prism $ABCDEF$ with horizontal rectangular base $ABCD$
$F$ is vertically above $A$

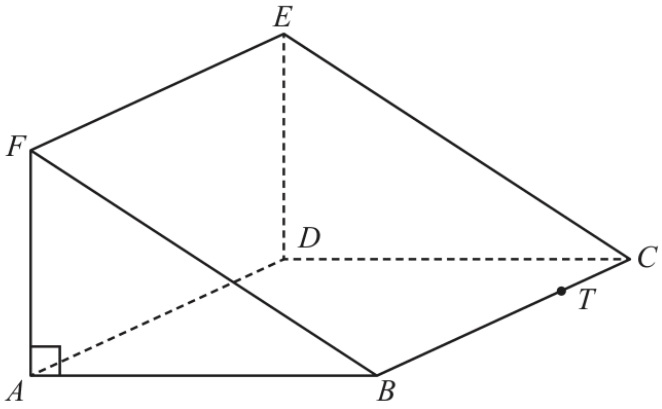



Diagram **NOT** accurately drawn

$AF = 12\,\text{cm}$ angle $ABF = 30°$ angle $BAF = 90°$

$T$ is the point on the line $BC$ such that $BF : BT = 3 : 2$

Work out the size of the angle between the line $FT$ and the plane $ABCD$
Give your answer correct to one decimal place.
Show your working clearly.

..... °

**(Total for Question 25 is 5 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/1H June 2026

### 1 Solution | Step 1

$$BF = \frac{12}{\sin 30°} = 24\,\text{cm}, \qquad AB = \frac{12}{\tan 30°} = 12\sqrt{3}\,\text{cm}.$$

Use right triangle ABF.

### 2 Solution | Step 2

$$BT = \frac{2}{3}BF = 16\,\text{cm}.$$

Apply the given BF:BT ratio.

3 **Solution | Step 3**

$$AT = \sqrt{AB^2 + BT^2} = \sqrt{432 + 256} = \sqrt{688} = 4\sqrt{43}\ \text{cm}.$$

AB is perpendicular to BC in the rectangular base, so triangle ABT is right-angled at B.

4 **Solution | Step 4**

The projection of FT onto plane ABCD is AT because F lies vertically above A. Therefore the required angle is $\theta = \angle FTA$.

A line-plane angle is the angle between the line and its perpendicular projection onto the plane.

5 **Solution | Step 5**

$$\tan\theta = \frac{AF}{AT} = \frac{12}{4\sqrt{43}} = \frac{3}{\sqrt{43}}, \quad \theta = 24.583891\ldots^\circ = 24.6^\circ.$$

Use right triangle FAT and round to one decimal place.

✓ **FINAL ANSWER**

$24.6^\circ$



TOPIC

# Sine & Cosine Rules

Every past-paper question on this topic, 2020 to 2026.

4MA1/1H JANUARY 2020

# Question 14

5 marks | Sine & Cosine Rules

**14** The diagram shows parallelogram $EFGH$.

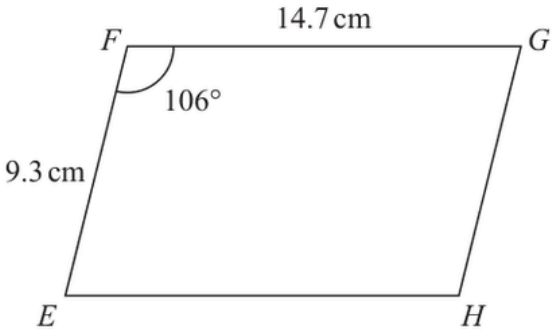



Diagram **NOT** accurately drawn

$EF = 9.3$ cm
$FG = 14.7$ cm
Angle $EFG = 106°$

(a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... $cm^2$
(2)

(b) Work out the length of the diagonal $EG$ of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

**(Total for Question 14 is 5 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1H January 2020

**1** **(a)**

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106° = 131.414\ldots$$

**2** **Answer**

**Answer:** $131\ cm^2$.

**3** **Use trigonometry**

(b) In triangle $EFG$,

$$EG^2 = 9.3^2 + 14.7^2 - 2(9.3)(14.7)\cos 106°$$

$$EG = 19.440\ldots$$

---

✓ **FINAL ANSWER**

$19.4\ \text{cm}$.

4MA1/1HR JANUARY 2020

# Question 21

6 marks | Sine & Cosine Rules

**21** The diagram shows the prism $ABCDEF$ with cross section triangle $ABC$.

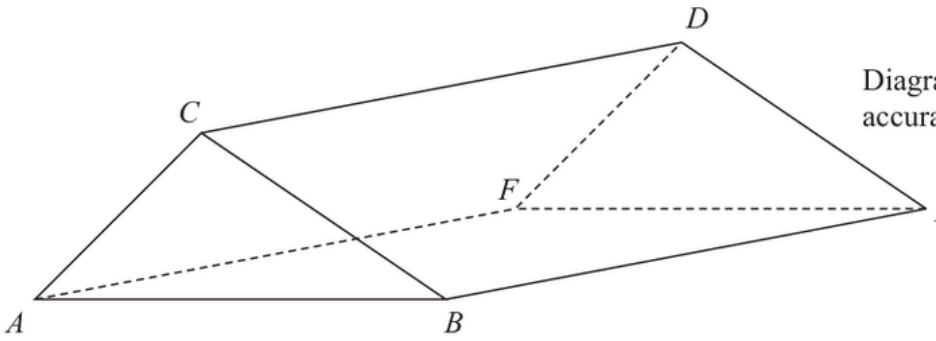



Diagram **NOT** accurately drawn

Angle $BEC = 40°$ and angle $ACB$ is obtuse.
$AC = 6\,\text{cm}$ and $CE = 13\,\text{cm}$

The area of triangle $ABC$ is $22\,\text{cm}^2$

Calculate the length of $AB$.
Give your answer correct to one decimal place.

..... cm

**(Total for Question 21 is 6 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1HR January 2020

**1 Use trigonometry**

In triangle $BEC$, $CE = 13$ and $\angle BEC = 40°$.

**2 Find the gradient**

Since this is a prism, $BE$ is perpendicular to the cross section, so triangle $BEC$ is right-angled at $B$.

$$BC = 13 \sin 40°$$

$$BC = 8.356238\ldots$$

**3 The area of triangle is**

The area of triangle $ABC$ is $22\ \text{cm}^2$.

**4** **Use trigonometry**

Using

$$\text{Area} = \frac{1}{2}ab\sin C$$

$$22 = \frac{1}{2}(AC)(BC)\sin(\angle ACB)$$

$$22 = \frac{1}{2}(6)(8.356238\ldots)\sin(\angle ACB)$$

$$\sin(\angle ACB) = 0.877587\ldots$$

---

**5** **Use trigonometry**

The angle $ACB$ is obtuse, so

$$\angle ACB = 180° - \sin^{-1}(0.877587\ldots)$$

$$\angle ACB = 118.647\ldots°$$

---

**6** **Use the cosine rule**

Use the cosine rule:

$$AB^2 = AC^2 + BC^2 - 2(AC)(BC)\cos(\angle ACB)$$

$$AB^2 = 6^2 + (8.356238\ldots)^2 - 2(6)(8.356238\ldots)\cos(118.647\ldots°)$$

$$AB = 12.4056\ldots$$

---

**7** **Correct to one decimal place**

Correct to one decimal place:

$$12.4$$

---

✓ **FINAL ANSWER**

$\boxed{12.4\text{ cm}}$

4MA1/2H NOVEMBER 2021

# Question 18

5 marks | Sine & Cosine Rules

**18** Here is a quadrilateral $ABCD$.

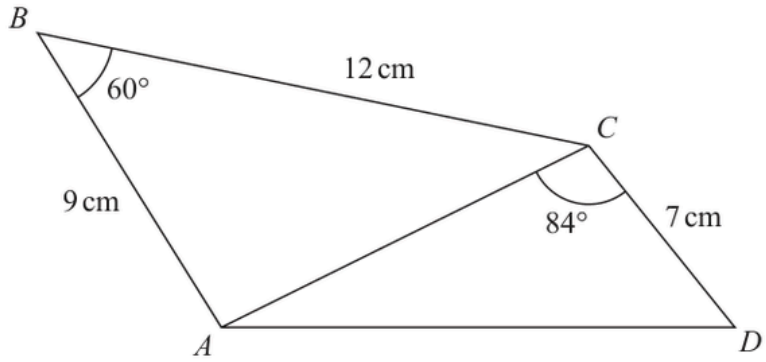



Diagram **NOT** accurately drawn

Calculate the area of quadrilateral $ABCD$.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... $cm^2$

**(Total for Question 18 is 5 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2H November 2021

**1 Area of triangle**

Area of triangle $ABC$:

$$\frac{1}{2}(9)(12)\sin 60° = 46.765\ldots$$

**2 Find using the cosine rule**

Find $AC$ using the cosine rule:

$$AC^2 = 9^2 + 12^2 - 2(9)(12)\cos 60°$$

$$AC = 10.816\ldots$$

**3 Area of triangle**

Area of triangle $ACD$:

$$\frac{1}{2}(10.816\ldots)(7)\sin 84° = 37.642\ldots$$

**4** **Total area**

Total area:

$$46.765\ldots + 37.642\ldots = 84.407\ldots$$

---

✓ **FINAL ANSWER**

$84.4\ \text{cm}^2$.

4MA1/1H JANUARY 2022

# Question 9

4 marks | Sine & Cosine Rules

9 The diagram shows isosceles triangle $ABC$

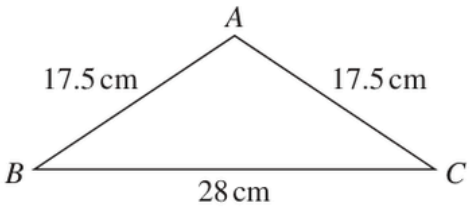



Diagram **NOT** accurately drawn

$AB = AC = 17.5$ cm $\quad\quad$ $BC = 28$ cm

Calculate the area of triangle $ABC$

..... cm$^2$

**(Total for Question 9 is 4 marks)**

## WORKED SOLUTION

Question 9 · 4MA1/1H January 2022

**1** **Use cosine rule**

Split the isosceles triangle into two right-angled triangles.

$$\frac{BC}{2} = 14$$

$$h^2 = 17.5^2 - 14^2 = 110.25$$

$$h = 10.5$$

$$\text{area} = \frac{1}{2} \times 28 \times 10.5 = 147$$

✓ **FINAL ANSWER**

$147\ \text{cm}^2$.

4MA1/2HR JANUARY 2022

# Question 16

5 marks | Sine & Cosine Rules

**16** The diagram shows quadrilateral $ABCD$

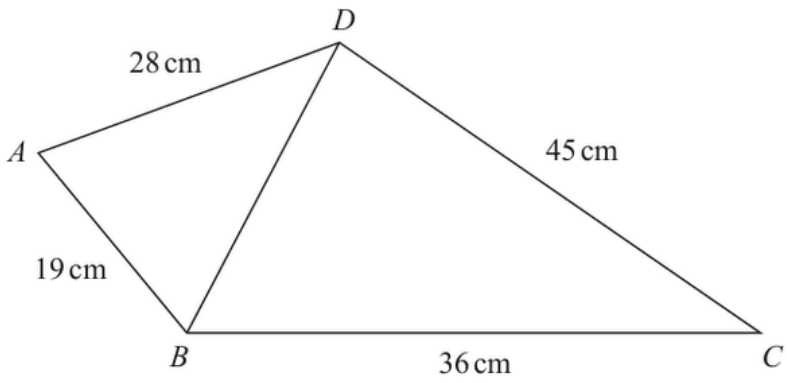



Diagram **NOT** accurately drawn

The angle $BCD$ is acute.

Given that the area of triangle $BCD = 405\,\text{cm}^2$

work out the size of angle $ABD$
Give your answer correct to one decimal place.

.....°

**(Total for Question 16 is 5 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/2HR January 2022

**1** **Use trigonometry**

For triangle $BCD$,

$$405 = \frac{1}{2}(36)(45)\sin BCD$$

$$\sin BCD = 0.5$$

**2** **Use trigonometry**

The angle $BCD$ is acute, so

$$BCD = 30°$$

**3** **Find**

Find $BD$:

$$BD^2 = 36^2 + 45^2 - 2(36)(45)\cos 30°$$

$$BD = 22.695\ldots$$

---

**4** **Use trigonometry**

Now use cosine rule in triangle $ABD$:

$$\cos ABD = \frac{19^2 + BD^2 - 28^2}{2(19)(BD)}$$

$$ABD = 83.871\ldots°$$

---

✓ **FINAL ANSWER**

$83.9°$.

4MA1/2H JANUARY 2022

# Question 23

6 marks | Sine & Cosine Rules

**23** The diagram shows triangle $PQR$

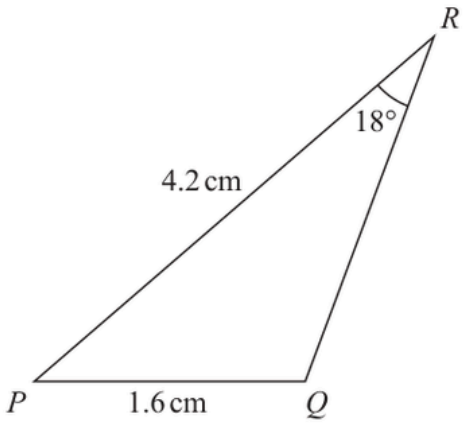



Diagram **NOT** accurately drawn

$PQ = 1.6\,\text{cm}$ $PR = 4.2\,\text{cm}$ Angle $PRQ = 18°$

Given that angle $PQR$ is obtuse,

work out the area of triangle $PQR$
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 23 is 6 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2H January 2022

**1** **Use trigonometry**

In triangle $PQR$,

$$PQ = 1.6, \quad PR = 4.2, \quad \angle PRQ = 18°$$

**2** **Use the sine rule**

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169\ldots$$

---

**3** **Use trigonometry**

Since angle $PQR$ is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

---

**4** **Use trigonometry**

Now find angle $P$:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

---

**5** **Area of triangle**

Area of triangle $PQR$:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

---

**6** **Correct to 3 significant figures**

Correct to 3 significant figures:

$$1.98$$

---

**FINAL ANSWER**

$\boxed{1.98\ \text{cm}^2}$

4MA1/2HR JUNE 2022

# Question 15

3 marks | Sine & Cosine Rules

**15** The diagram shows a kite $ABCD$

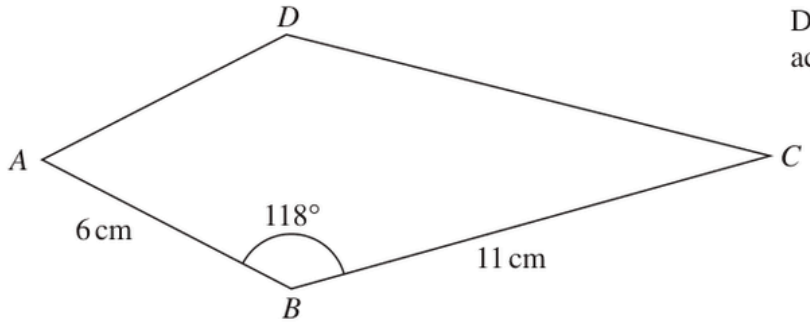



Diagram **NOT** accurately drawn

$AB = 6\,\text{cm}$ $BC = 11\,\text{cm}$ Angle $ABC = 118°$

Calculate the area of the kite.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2HR June 2022

**1** **A kite can be split**

A kite can be split into two congruent triangles.

**2** **Area of triangle**

Area of triangle $ABC$:

$$\frac{1}{2} \times 6 \times 11 \times \sin 118°$$

**3** **Use trigonometry**

So the area of the kite is

$$2 \times \frac{1}{2} \times 6 \times 11 \times \sin 118°$$

$$= 6 \times 11 \times \sin 118° = 58.274\ldots$$

**FINAL ANSWER**

$58.3\ \text{cm}^2$.

4MA1/1HR JUNE 2022

# Question 17

5 marks | Sine & Cosine Rules

**17** Here is a parallelogram $PQRS$, in which angle $SPQ$ is acute.

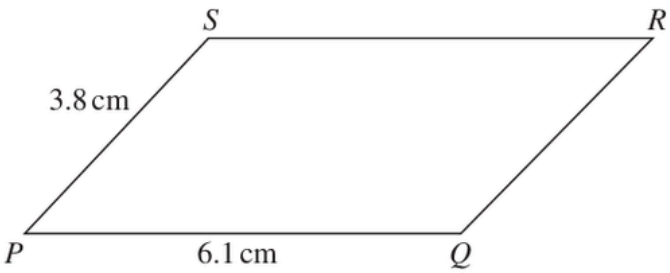



Diagram **NOT** accurately drawn

$PQ = 6.1\,\text{cm}$ $PS = 3.8\,\text{cm}$

The area of the parallelogram is $18\,\text{cm}^2$

Work out the length of $QS$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 17 is 5 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1HR June 2022

### 1 Height of the parallelogram

Height of the parallelogram:

$$h = \frac{18}{6.1} = 2.9508\ldots$$

### 2 Use cosine rule

The horizontal offset from $P$ to $S$ is

$$\sqrt{3.8^2 - h^2} = \sqrt{3.8^2 - 2.9508\ldots^2} = 2.394\ldots$$

### 3 Use trigonometry

So the horizontal distance from $S$ to $Q$ is

$$6.1 - 2.394\ldots = 3.705\ldots$$

$$QS = \sqrt{3.705\ldots^2 + 2.9508\ldots^2} = 4.737\ldots$$

**FINAL ANSWER**

4.74 cm.

4MA1/1H JUNE 2022

# Question 18

5 marks | Sine & Cosine Rules

**18** Here is triangle $ABC$

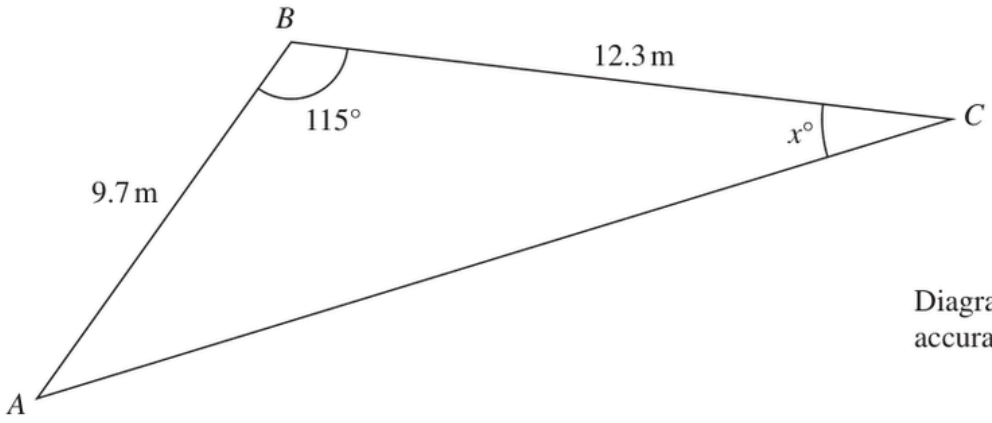



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to 3 significant figures.

$x =$ .....

**(Total for Question 18 is 5 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1H June 2022

### 1 Use trigonometry

First use the cosine rule to find $AC$:

$$AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos 115°$$

$$AC = 18.607\ldots$$

### 2 Use trigonometry

Now use the sine rule:

$$\frac{\sin x}{9.7} = \frac{\sin 115°}{18.607\ldots}$$

$$\sin x = \frac{9.7 \sin 115°}{18.607\ldots}$$

$$x = 28.194\ldots°$$

**FINAL ANSWER**

$28.2°$.

4MA1/2H JUNE 2023

# Question 22

5 marks | Sine & Cosine Rules

**22** The diagram shows a triangle $ABC$ and a flagpole $BF$

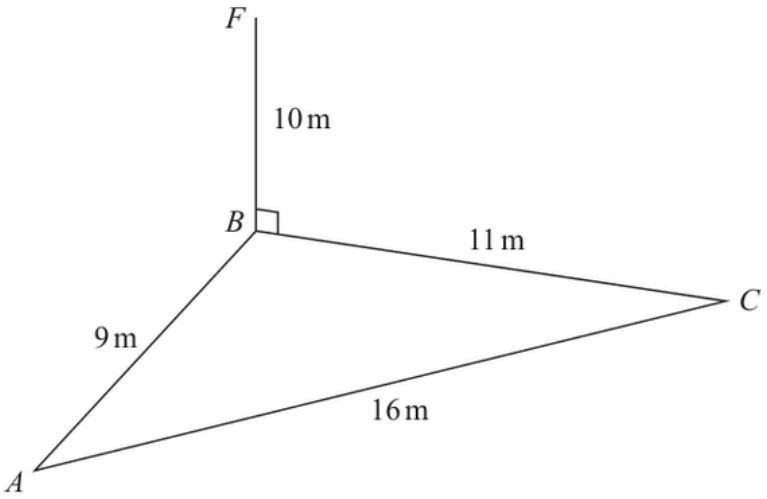



Diagram **NOT** accurately drawn

$A$, $B$ and $C$ are points on horizontal ground.

$BF$ is vertical.

$AB = 9\,\text{m}$ $BC = 11\,\text{m}$ $AC = 16\,\text{m}$ $BF = 10\,\text{m}$

$D$ is the point on $AC$ such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point $F$ from the point $D$
Give your answer correct to one decimal place.

.....°

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/2H June 2023

**1** **Use trigonometry**

In triangle $ABC$,

$$AB = 9, \qquad BC = 11, \qquad AC = 16$$

**2** **Use Heron s formula**

Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\text{area} = \sqrt{18(18-9)(18-11)(18-16)}$$

$$= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235\ldots$$

---

**3** **is the foot of the**

$D$ is the foot of the perpendicular from $B$ to $AC$, so $BD$ is the height of triangle $ABC$.

$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235\ldots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529\ldots$$

---

**4** **Use trigonometry**

In right-angled triangle $BDF$,

$$\tan\theta = \frac{BF}{BD}$$

$$\tan\theta = \frac{10}{5.9529\ldots}$$

$$\theta = \tan^{-1}\left(\frac{10}{5.9529\ldots}\right) = 59.2349\ldots^\circ$$

---

✓ **FINAL ANSWER**

$\boxed{59.2^\circ}$

4MA1/1HR MAY 2023

# Question 25

6 marks | Sine & Cosine Rules

**25** Here is a triangle $ABC$

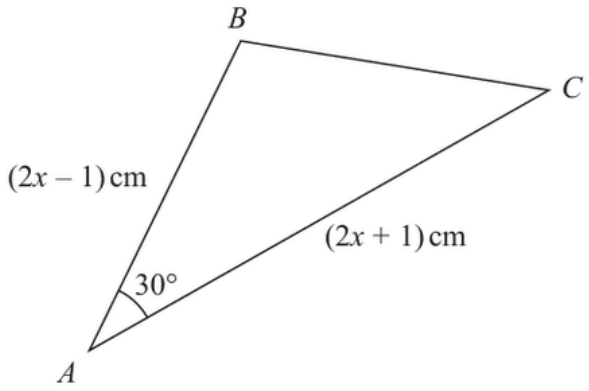



Diagram **NOT** accurately drawn

The area of the triangle is $(x^2 + x - 3.75)\,\text{cm}^2$

Find the size of the largest angle in triangle $ABC$
Give your answer correct to the nearest degree.

.....°

**(Total for Question 25 is 6 marks)**

## WORKED SOLUTION

Question 25 · 4MA1/1HR May 2023

**1** **The two sides around the**

The two sides around the $30°$ angle are

$$2x - 1 \quad \text{and} \quad 2x + 1$$

**2** **Use trigonometry**

Using the area formula,

$$\text{area} = \frac{1}{2}ab\sin C$$

$$x^2 + x - 3.75 = \frac{1}{2}(2x - 1)(2x + 1)\sin 30°$$

**3** **Use trigonometry**

Since $\sin 30° = \frac{1}{2}$,

$$x^2 + x - 3.75 = \frac{1}{4}(2x - 1)(2x + 1)$$

$$x^2 + x - 3.75 = \frac{1}{4}(4x^2 - 1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

---

**4** **Use trigonometry**

So

$$AB = 2x - 1 = 6, \qquad AC = 2x + 1 = 8$$

---

**5** **Use cosine rule**

Use the cosine rule to find $BC$:

$$BC^2 = 6^2 + 8^2 - 2(6)(8)\cos 30°$$

$$BC = 4.1063\ldots$$

---

**6** **Use trigonometry**

The largest side is $AC = 8$, so the largest angle is $\angle B$.

---

**7** **Use trigonometry**

Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643\ldots°$$

---

✓ **FINAL ANSWER**

$\boxed{103°}$

4MA1/2HR JUNE 2024

# Question 15

3 marks | Sine & Cosine Rules

**15**

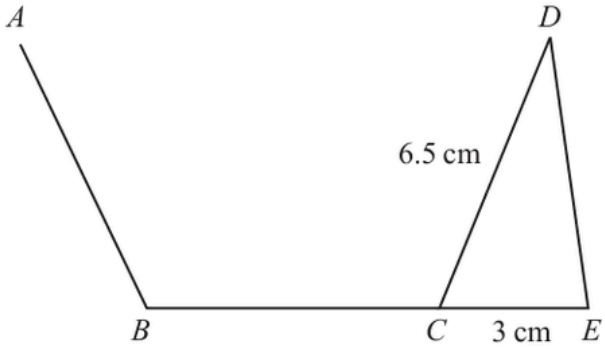



Diagram **NOT** accurately drawn

*AB*, *BC* and *CD* are three sides of a regular pentagon and *CDE* is a triangle.
*BCE* is a straight line.

$CD = 6.5\,\text{cm}$ $CE = 3\,\text{cm}$

Work out the area of triangle *CDE*
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2HR June 2024

**1 A regular pentagon has interior**

A regular pentagon has interior angle

$$\frac{(5-2)\times 180}{5} = 108^\circ$$

**2 Use trigonometry**

Since $BCE$ is a straight line,

$$\angle DCE = 180^\circ - 108^\circ = 72^\circ$$

**3** **Area of triangle is**

Area of triangle $CDE$ is

$$\frac{1}{2}ab\sin C$$

$$\text{Area} = \frac{1}{2}(6.5)(3)\sin 72°$$

$$\text{Area} = 9.27\ldots$$

---

✓ **FINAL ANSWER**

$9.27\ \text{cm}^2$ to 3 significant figures.

4MA1/1HR MAY 2024

# Question 15

3 marks | Sine & Cosine Rules

**15** The diagram shows isosceles triangle $EFG$

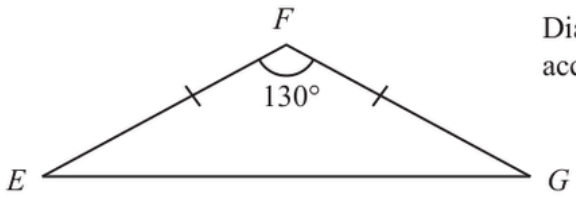



Diagram **NOT** accurately drawn

$EF = GF$
Angle $EFG = 130°$
The area of triangle $EFG$ is 74 cm²

Work out the length of $EF$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1HR May 2024

**1 Let**

Let

$$EF = FG = x$$

**2 Use the area formula**

Use the area formula:

$$74 = \frac{1}{2}x^2 \sin 130°$$

$$x^2 = \frac{148}{\sin 130°}$$

$$x = 13.899\ldots$$

**FINAL ANSWER**

$13.9$ cm.

4MA1/1H MAY 2024

# Question 19

5 marks | Sine & Cosine Rules

**19** Here is quadrilateral $ABCD$

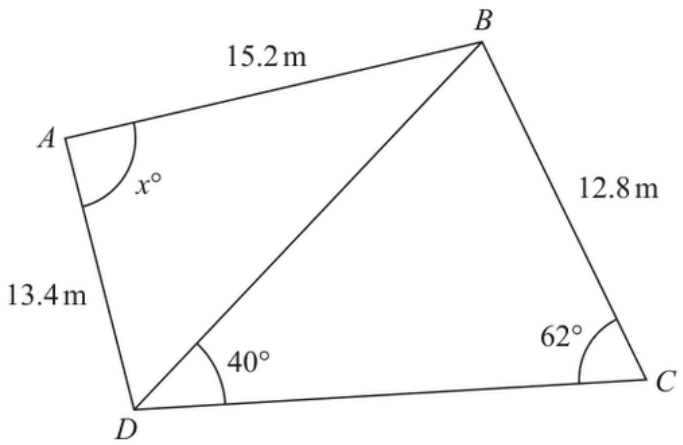



Diagram **NOT** accurately drawn

Work out the value of $x$
Give your answer correct to 3 significant figures.

$x =$ .....

**(Total for Question 19 is 5 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1H May 2024

**1** **Use trigonometry**

In triangle $BCD$,

$$\angle DBC = 180° - 40° - 62° = 78°$$

**2** **Use trigonometry**

By the sine rule,

$$\frac{DB}{\sin 62°} = \frac{12.8}{\sin 40°}$$

$$DB = 17.582\ldots$$

### 3 Use trigonometry

Now use cosine rule in triangle $ABD$:

$$\cos x = \frac{15.2^2 + 13.4^2 - 17.582\ldots^2}{2(15.2)(13.4)}$$

$$x = 75.577\ldots^\circ$$

---

**FINAL ANSWER**

$75.6^\circ$.

# Question 20

5 marks | Sine & Cosine Rules

**20**

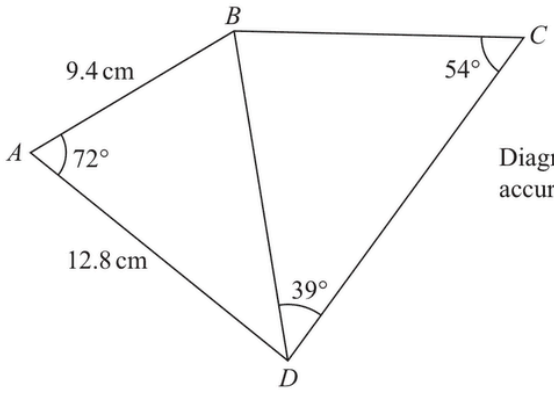



Diagram **NOT** accurately drawn

Work out the length of $BC$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 20 is 5 marks)**

**1** **Use trigonometry**

First use triangle $ABD$.

$$AB = 9.4, \quad AD = 12.8, \quad \angle BAD = 72°$$

**2** **Use cosine rule**

Use the cosine rule to find $BD$:

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72°$$

$$BD = 13.3356\ldots$$

**3** **Use trigonometry**

In triangle $BCD$,

$$\angle BDC = 39°, \quad \angle BCD = 54°$$

**4** **Use the sine rule**

Use the sine rule:

$$\frac{BC}{\sin 39°} = \frac{BD}{\sin 54°}$$

$$BC = \frac{13.3356\ldots \sin 39°}{\sin 54°}$$

$$BC = 10.3735\ldots$$

**5** **Use trigonometry**

Correct to 3 significant figures,

$$BC = 10.4$$

✓ **FINAL ANSWER**

$10.4$ cm

4WM1/01 MAY 2025

# Question 21

5 marks | Sine & Cosine Rules

**21**

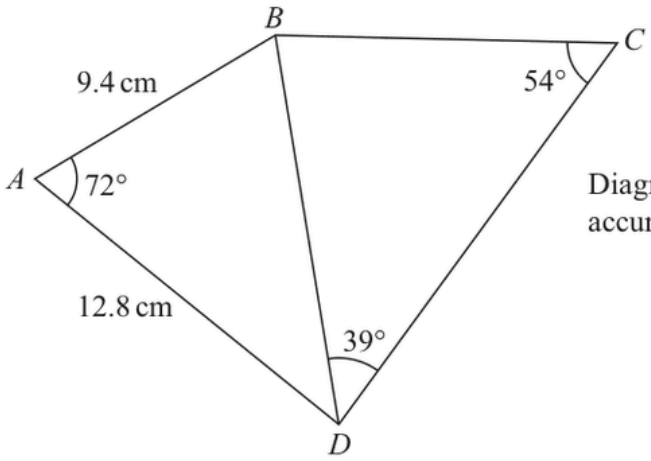



Diagram **NOT** accurately drawn

Work out the length of $BC$
Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4WM1/01 May 2025

**1** **Use trigonometry**

First find $BD$ in triangle $ABD$ using the cosine rule.

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72°$$

$$BD = 13.3356\ldots$$

**2** **Use trigonometry**

In triangle $BCD$, use the sine rule.

$$\frac{BC}{\sin 39°} = \frac{BD}{\sin 54°}$$

$$BC = 13.3356\ldots \times \frac{\sin 39°}{\sin 54°}$$

$$BC = 10.3735\ldots$$

**3** **Use trigonometry**

To 3 significant figures,

$$BC = 10.4$$

---

✓ **FINAL ANSWER**

$10.4\ \text{cm}$

4WM1/01 NOVEMBER 2025

# Question 19

3 marks | Sine & Cosine Rules

**19** The diagram shows triangle $ABC$

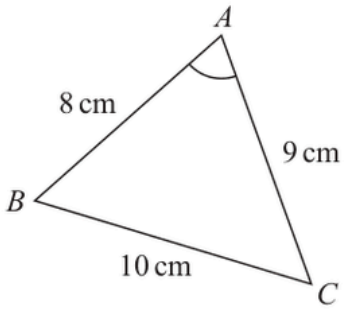



Diagram **NOT** accurately drawn

$AB = 8\,\text{cm}$ $BC = 10\,\text{cm}$ $CA = 9\,\text{cm}$

Work out the size of angle $BAC$
Give your answer correct to one decimal place.

.....°

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4WM1/01 November 2025

### 1 Use cosine rule

Use the cosine rule for angle $BAC$:

$$\cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)}$$

$$\cos A = \frac{45}{144}$$

$$A = 71.790\ldots^\circ$$

**FINAL ANSWER**

$71.8^\circ$.

4MA1/1H NOVEMBER 2025

# Question 19

3 marks | Sine & Cosine Rules

**19** The diagram shows triangle $ABC$

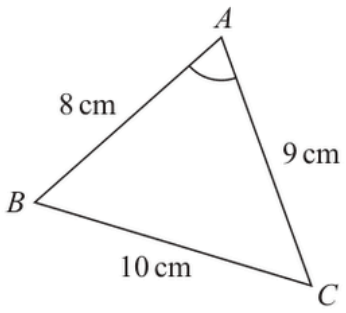



Diagram **NOT** accurately drawn

$AB = 8\,\text{cm}$ $BC = 10\,\text{cm}$ $CA = 9\,\text{cm}$

Work out the size of angle $BAC$
Give your answer correct to one decimal place.

.....°

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1H November 2025

### 1 Use cosine rule

Use the cosine rule for angle $BAC$:

$$\cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)}$$

$$\cos A = \frac{45}{144}$$

$$A = 71.790\ldots^\circ$$

**FINAL ANSWER**

$71.8^\circ$.

# Question 15

5 marks | Sine & Cosine Rules

**15** The diagram shows triangle $ABC$

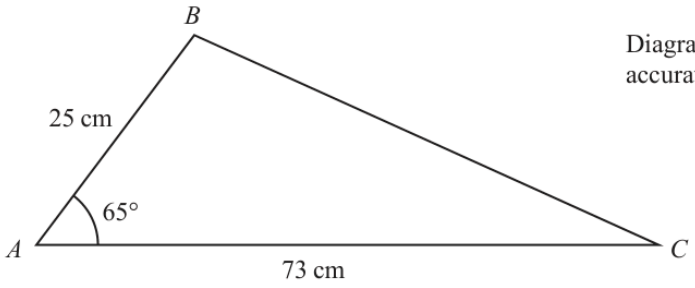



Diagram **NOT** accurately drawn

$AB = 25$ cm $\quad AC = 73$ cm $\quad$ angle $BAC = 65°$

(a) Work out the length of $BC$
Give your answer correct to 3 significant figures.

..... cm

**(3)**

(b) Work out the area of the triangle.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(2)**

**(Total for Question 15 is 5 marks)**

**1** **(a) Cosine rule**

$$BC^2 = 25^2 + 73^2 - 2(25)(73)\cos 65° = 4411.4\ldots$$

**2** **Square root**

$$BC = 66.418\ldots$$

**3** **(b) Area formula**

$$\tfrac{1}{2}(25)(73)\sin 65° = 827.005\ldots$$

✓ **FINAL ANSWER**

(a) $66.4$ cm (b) $827$ cm$^2$

4MA1/1H JUNE 2026

# Question 16

5 marks | Sine & Cosine Rules

**16** The diagram shows triangle $ABC$

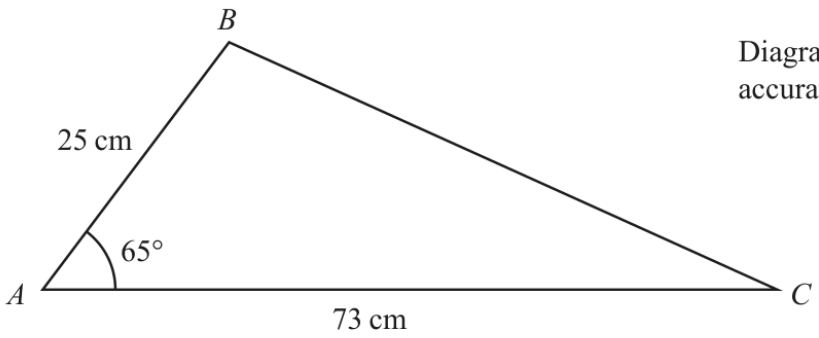



Diagram **NOT** accurately drawn

$AB = 25$ cm $\quad AC = 73$ cm $\quad$ angle $BAC = 65°$

(a) Work out the length of $BC$
Give your answer correct to 3 significant figures.

..... cm

**(3)**

(b) Work out the area of the triangle.
Give your answer correct to 3 significant figures.

..... $\text{cm}^2$

**(2)**

**(Total for Question 16 is 5 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H June 2026

**1** **Part (a) | Step 1**

$$BC^2 = 25^2 + 73^2 - 2(25)(73)\cos 65°.$$

The included angle is between the two given sides.

**2** **Part (a) | Step 2**

$$BC^2 = 4411.443344\ldots, \quad BC = \sqrt{4411.443344\ldots} = 66.418697\ldots = 66.4 \text{ cm} \quad (3 \text{ s.f.}).$$

Take the positive square root for a length.

**3** **Part (b) | Step 1**

$$\text{Area} = \frac{1}{2}(25)(73)\sin 65^\circ = 827.005855\ldots = 827\ \text{cm}^2 \quad (3\ \text{s.f.}).$$

Use half the product of the two sides and the sine of their included angle.

---

✓ **FINAL ANSWER**

(a) $66.4\ \text{cm}$ (b) $827\ \text{cm}^2$

4MA1/1HR JUNE 2026

# Question 22

4 marks | Sine & Cosine Rules

**22** The diagram shows triangle $ABC$

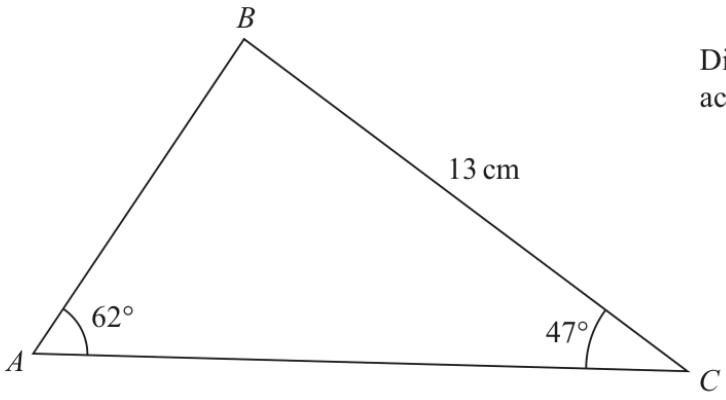



Diagram **NOT** accurately drawn

Calculate the area of the triangle.
Give your answer correct to 3 significant figures.

..... $cm^2$

**(Total for Question 22 is 4 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1HR June 2026

**1** **Solution | Step 1**

$$\angle ABC = 180^\circ - 62^\circ - 47^\circ = 71^\circ.$$

The three triangle angles sum to 180 degrees.

**2** **Solution | Step 2**

$$\frac{AB}{\sin 47^\circ} = \frac{13}{\sin 62^\circ} \quad \Longrightarrow \quad AB = \frac{13 \sin 47^\circ}{\sin 62^\circ} = 10.7680208859\ldots \text{ cm}.$$

Match each side to its opposite angle in the sine rule.

**3** **Solution | Step 3**

$$\text{Area} = \frac{1}{2}(13)\left(\frac{13 \sin 47^\circ}{\sin 62^\circ}\right)\sin 71^\circ = 66.1788645053\ldots \text{ cm}^2.$$

Use half the product of two sides times the sine of their included angle; retain unrounded values.

**4** **Solution | Step 4**

$$\text{Area} = 66.2\ \text{cm}^2 \quad (3\ \text{significant figures}).$$

Round only the final area.

---

✓ **FINAL ANSWER**

$66.2\ \text{cm}^2$



TOPIC

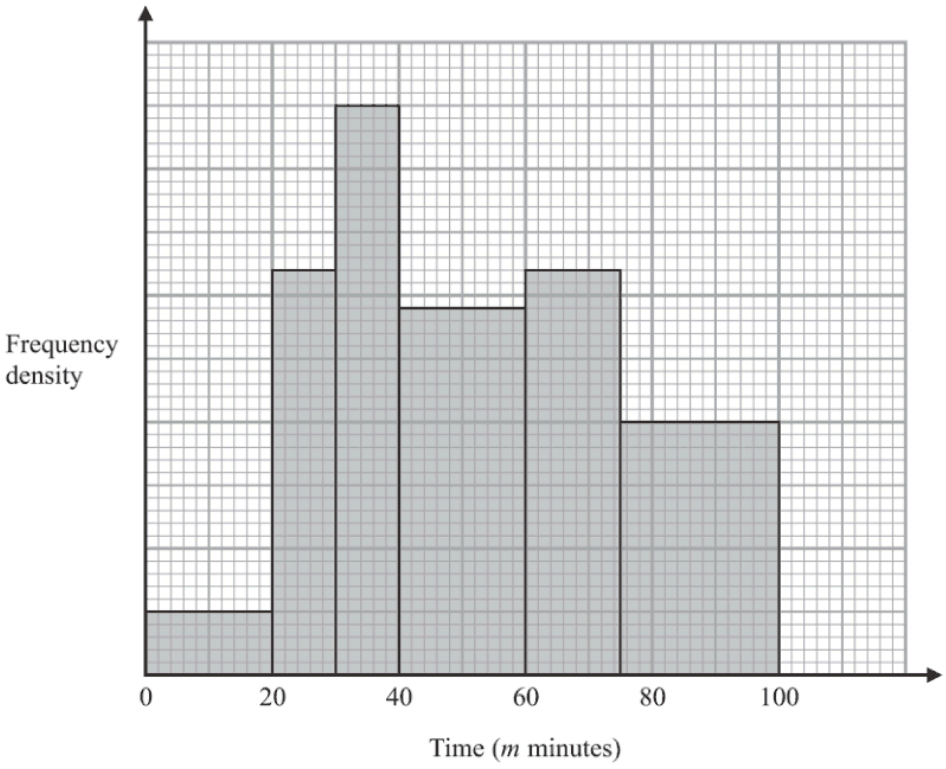
# Histograms

Every past-paper question on this topic, 2020 to 2026.

# Question 16

3 marks | Histograms

**16** The histogram shows information about the total time, $m$ minutes, taken by each child in a school to walk to school every day for one week.




There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$
Work out an estimate for the number of children for whom $50 < m \leqslant 80$

.....

**(Total for Question 16 is 3 marks)**

## 1 Set the frequency-density scale

For $0 < m \leq 20$, the frequency is $10$, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

## 2 Read the needed bar heights

Using the same scale from the histogram:

$$50 < m \leq 60 : \ 10 \times 2.9 = 29$$

$$60 < m \leq 75 : \ 15 \times 3.2 = 48$$

$$75 < m \leq 80 : \ 5 \times 2 = 10$$

## 3 Add the estimates

$$19 + 36 + 32 = 87$$

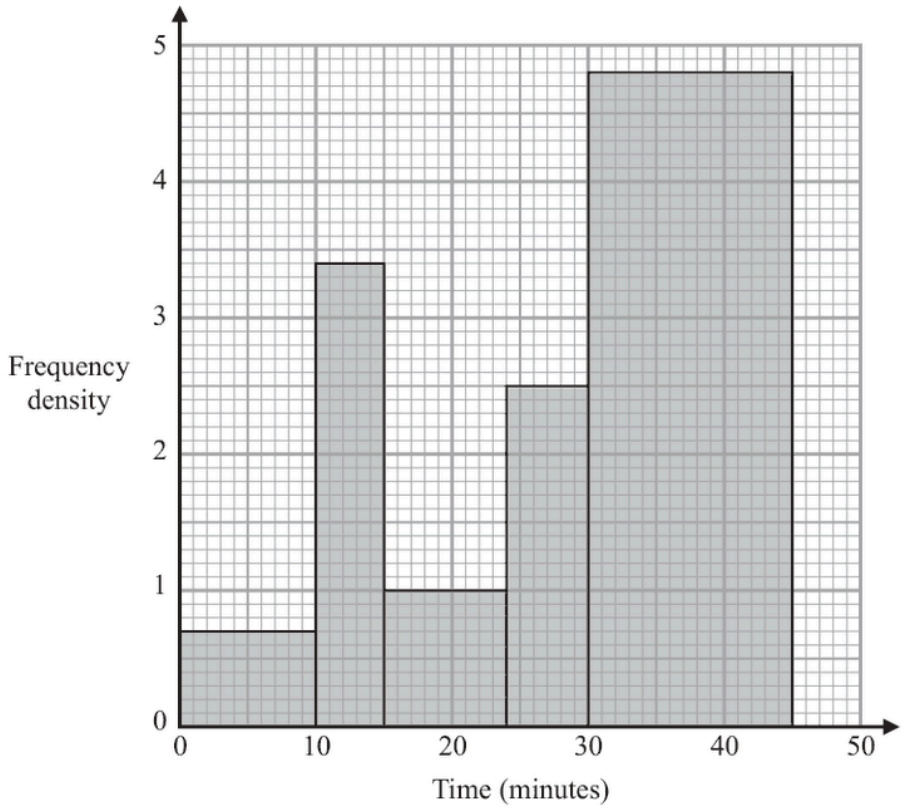
**FINAL ANSWER**

87.

# Question 18

5 marks | Histograms

**18** The histogram gives information about the times, in minutes, that some customers spent in a supermarket.




(a) Work out an estimate for the proportion of these customers who spent between 17 minutes and 35 minutes in the supermarket.

.....

**(3)**

One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

(b) find the probability that this customer spent more than 36 minutes in the supermarket.

.....

**(2)**

**(Total for Question 18 is 5 marks)**

## 1 Total frequency

Total frequency:

$$10(0.7) + 5(3.4) + 9(1) + 6(2.5) + 15(4.8) = 120$$

## 2 Between 17 and 35 minutes

Between 17 and 35 minutes:

$$7(1) + 6(2.5) + 5(4.8) = 46$$

$$\text{proportion} = \frac{46}{120} = \frac{23}{60}$$

## 3 Given the customer spent more

Given the customer spent more than 30 minutes, they are in the $30$ to $45$ minute class.

## 4 More than 36 minutes is

More than 36 minutes is $9$ minutes out of the $15$-minute class:

$$\frac{9}{15} = \frac{3}{5}$$

**FINAL ANSWER**

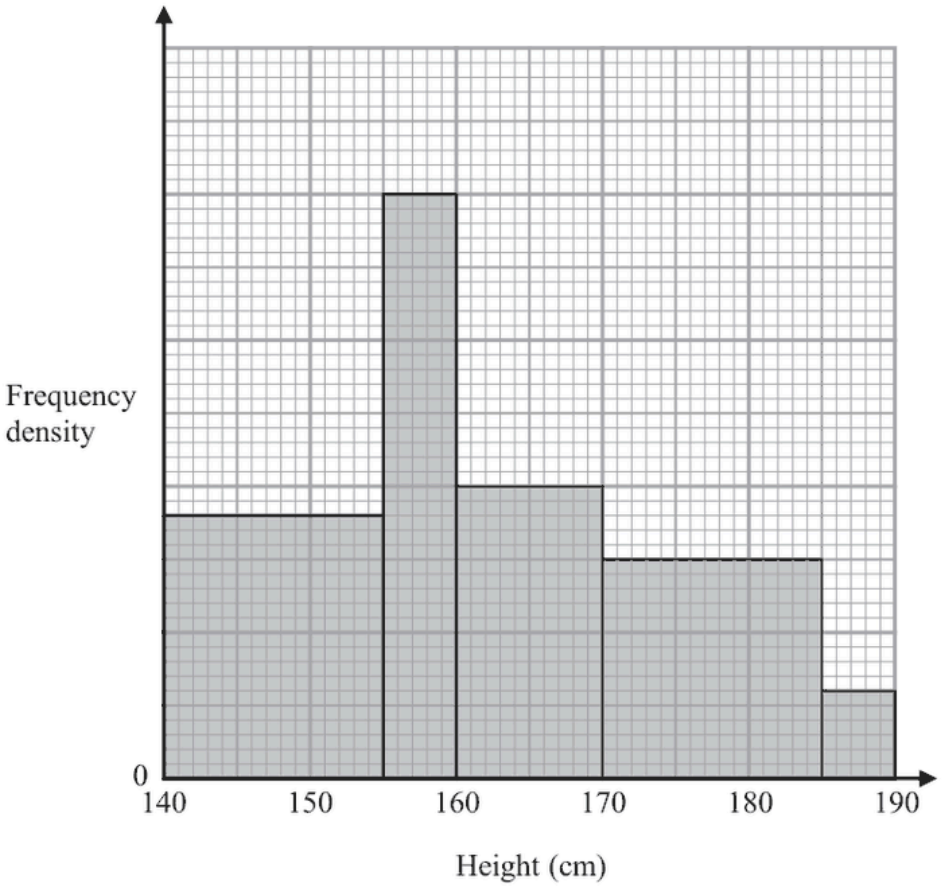
$\frac{23}{60}$, then $\frac{3}{5}$.

4MA1/1HR JANUARY 2020

# Question 19

4 marks | Histograms

19




The histogram gives information about the heights of all the Year 11 students at a school.

There are 160 students in Year 11 with a height between 155 cm and 170 cm.

Work out the total number of students in Year 11 at the school.

**(Total for Question 19 is 4 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1HR January 2020

**1 Use the relative histogram heights**

Use the relative histogram heights.

**2 Calculate area**

For $155 < h \leq 170$, the relative area is

$$5(40) + 10(20) = 400$$

### 3 Calculate area

This represents $160$ students, so each relative area unit represents

$$\frac{160}{400} = 0.4$$

### 4 Total relative area

Total relative area:

$$15(18) + 5(40) + 10(20) + 15(15) + 5(6) = 925$$

### 5 Estimated total

Estimated total:

$$925 \times 0.4 = 370$$

**FINAL ANSWER**

$370$ students.

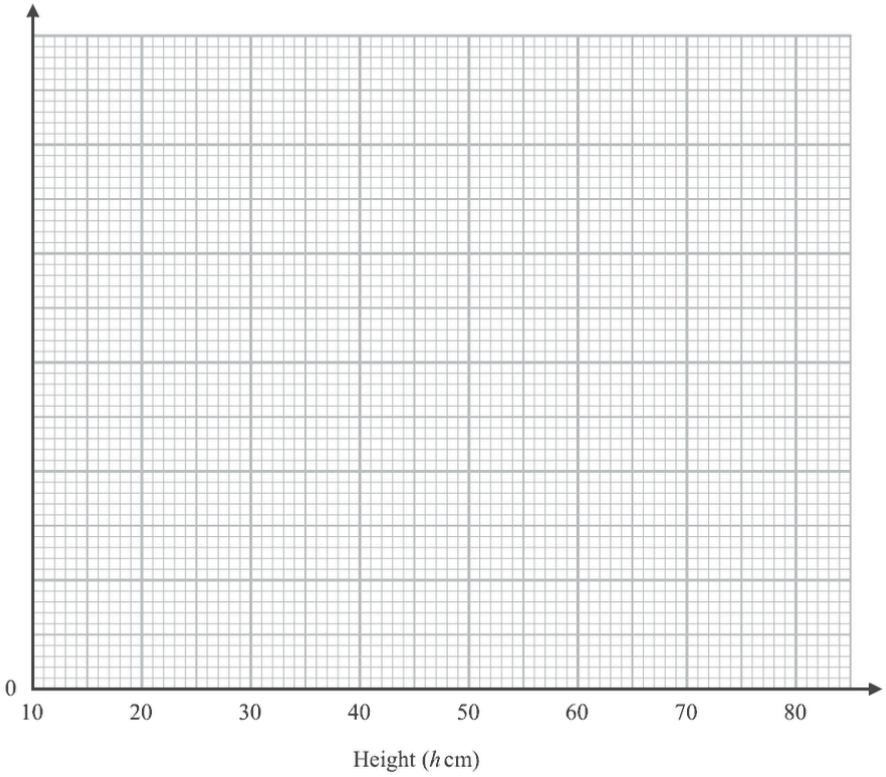
# Question 18

5 marks | Histograms

**18** The table gives information about the heights, in centimetres, of some plants.

| Height ($h$ cm) | Frequency |
|---|---|
| $10 < h \leqslant 20$ | 35 |
| $20 < h \leqslant 35$ | 45 |
| $35 < h \leqslant 50$ | 75 |
| $50 < h \leqslant 70$ | 40 |
| $70 < h \leqslant 80$ | 8 |

(a) On the grid, draw a histogram for this information.




**(3)**

(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

.....

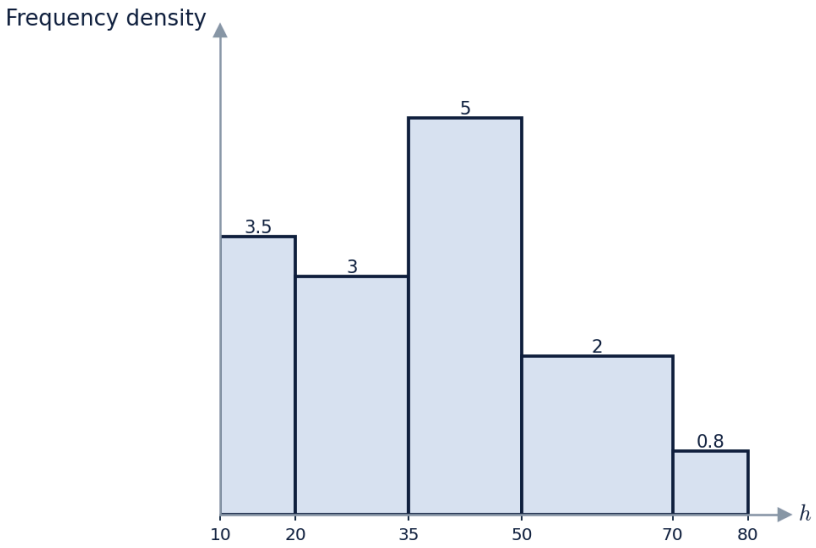
**(2)**

**(Total for Question 18 is 5 marks)**

**1** **The histogram densities are**

The histogram densities are

$$\frac{35}{10} = 3.5, \quad \frac{45}{15} = 3, \quad \frac{75}{15} = 5, \quad \frac{40}{20} = 2, \quad \frac{8}{10} = 0.8$$




**2** **Use histogram**

For plants taller than $40\ \text{cm}$:

**3** **Use histogram**

From $40 < h \leq 50$:

$$10(5) = 50$$

**4** **Use histogram**

From $50 < h \leq 70$:

$$40$$

**5** **Use histogram**

From $70 < h \leq 80$:

$$8$$

$$50 + 40 + 8 = 98$$

**FINAL ANSWER**

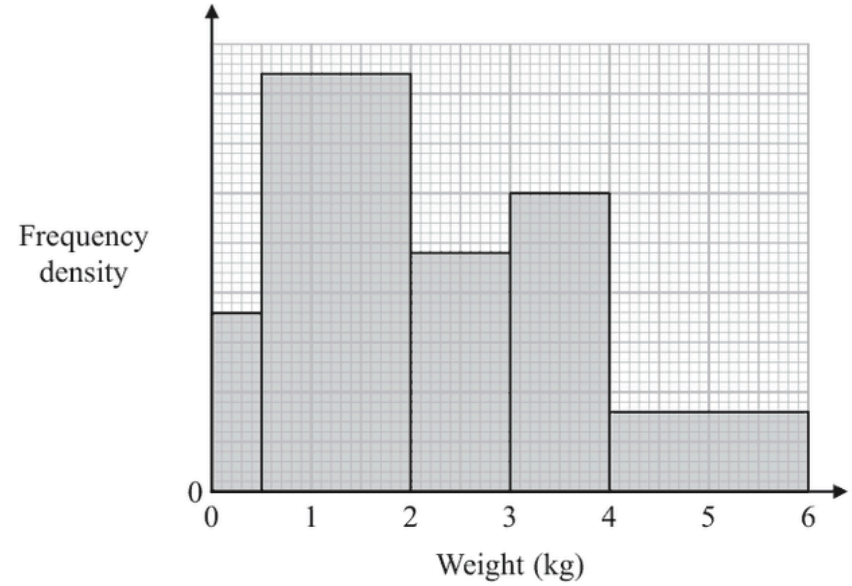
draw bars with densities $3.5, 3, 5, 2, 0.8$, and $98$ plants.

# Question 15

6 marks | Histograms

**15** A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.




63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

.....

**(3)**

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

.....

**(3)**

**(Total for Question 15 is 6 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2H January 2021

**1** **Use histogram**

For $0.5 < w \leq 2$, the frequency is $63$, so the density is

$$\frac{63}{1.5} = 42$$

**2** **Use histogram**

Using the histogram heights, the class frequencies are

$$9,\ 63,\ 24,\ 30,\ 16$$

---

**3** **Use histogram**

So the total number of parcels is

$$9 + 63 + 24 + 30 + 16 = 142$$

---

**4** **Use histogram**

For weight greater than $2.25\ \text{kg}$:

$$0.75(24) + 30 + 16 = 64$$

---

**5** **The probability that both randomly**

The probability that both randomly chosen records are greater than $2.25\ \text{kg}$ is

$$\frac{64}{142} \times \frac{63}{141} = 0.201\ldots$$

---

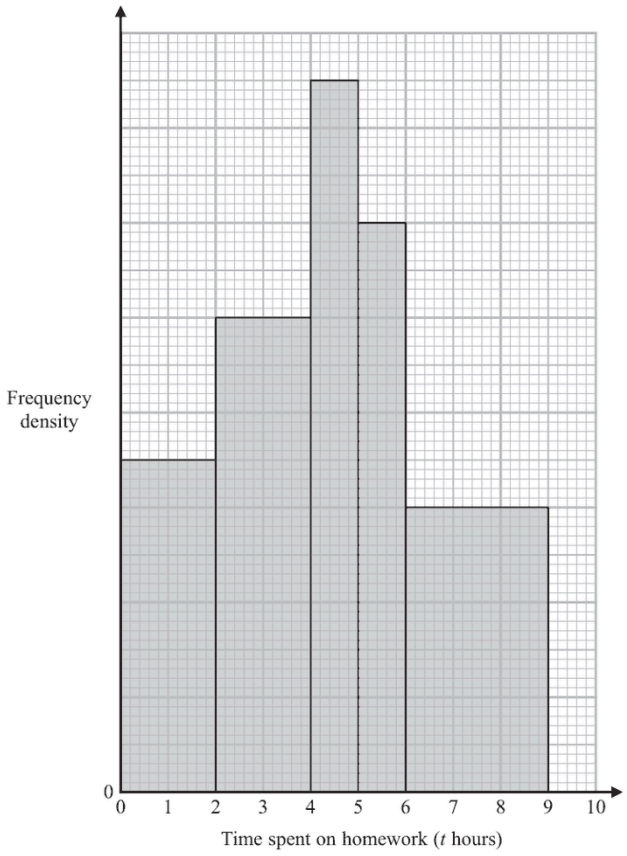
✓ **FINAL ANSWER**

$142$, then $0.201$ approximately.

# Question 18

5 marks | Histograms

**18** The histogram and the table give some information about the amounts of time, in hours, that Year 11 students at Bergdesh Academy spent, in total, on their homework last week.
No student in Year 11 spent longer than 9 hours on their homework.




| Time spent on homework ($t$ hours) | Frequency |
|---|---|
| $0 < t \leqslant 2$ | 28 |
| $2 < t \leqslant 4$ | |
| $4 < t \leqslant 5$ | |
| $5 < t \leqslant 6$ | |
| $6 < t \leqslant 9$ | |

Using the information in the histogram and in the table, work out an estimate for the mean amount of time the Year 11 students spent on their homework last week.
Give your answer in hours correct to 3 significant figures.

..... hours

**(Total for Question 18 is 5 marks)**

**1** **Use histogram**

The $0 < t \leq 2$ class has frequency $28$. From the histogram, the relative heights are

$$1,\ 4,\ 9,\ 6,\ 1$$

**2** **Use histogram**

So the frequencies are

$$28,\ 112,\ 126,\ 84,\ 42$$

**3** **Use midpoints**

Use midpoints $1, 3, 4.5, 5.5, 7.5$.

$$\text{total frequency} = 28 + 112 + 126 + 84 + 42 = 392$$

$$\text{estimated total} = 1(28) + 3(112) + 4.5(126) + 5.5(84) + 7.5(42) = 1708$$

$$\text{mean} = \frac{1708}{392} = 4.357\ldots$$

**FINAL ANSWER**

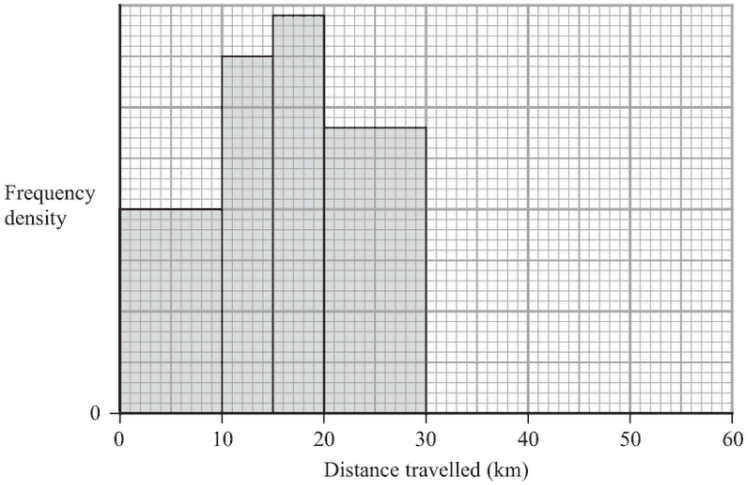
$4.36$ hours.

# Question 18

5 marks | Histograms

**18** The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

| Distance ($d$ km) | Frequency |
|---|---|
| $0 \leqslant d < 10$ | 40 |
| $10 \leqslant d < 15$ | |
| $15 \leqslant d < 20$ | |
| $20 \leqslant d < 30$ | |
| $30 \leqslant d < 60$ | 30 |




Using the information in the table and in the histogram,

(a) complete the table,

**(2)**

(b) complete the histogram.

**(1)**

One of the people working in the store is chosen at random.

(c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

.....

**(2)**

**(Total for Question 18 is 5 marks)**

**1** **Use histogram**

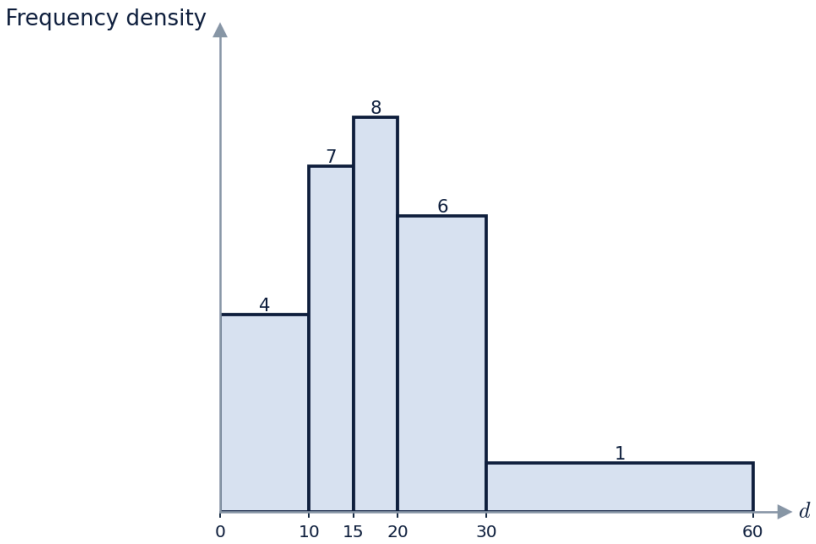
The $0 \leq d < 10$ class has frequency $40$, so its density is

$$\frac{40}{10} = 4$$

**2** **Use histogram**

From the histogram, the missing frequencies are

$$10 \leq d < 15 : \ 35, \qquad 15 \leq d < 20 : \ 40, \qquad 20 \leq d < 30 : \ 60$$




**3** **Together with the table**

Together with the table:

$$40 + 35 + 40 + 60 + 30 = 205$$

**4** **Use histogram**

For $d > 25$:

$$5(6) + 30 = 60$$

$$P(d > 25) = \frac{60}{205} = \frac{12}{41}$$

**FINAL ANSWER**

completed frequencies $35, 40, 60$, missing histogram density $1$, and probability $\frac{12}{41}$.

# Question 14

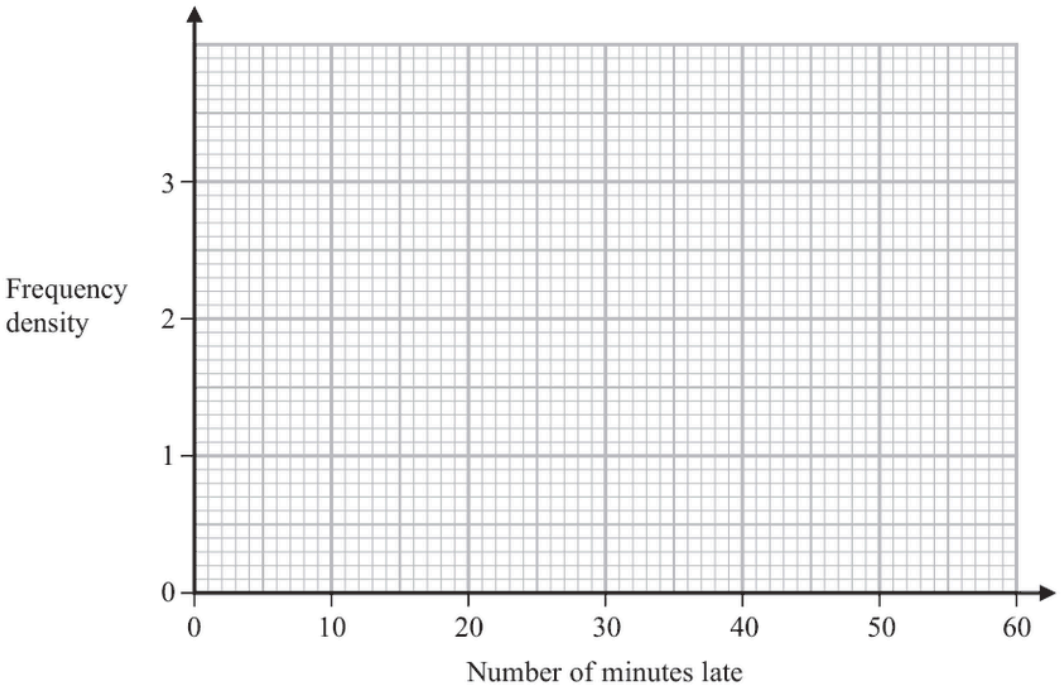
5 marks | Histograms

**14** Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

| Minutes late ($L$ minutes) | Frequency |
|---|---|
| $0 < L \leqslant 10$ | 8 |
| $10 < L \leqslant 15$ | 13 |
| $15 < L \leqslant 25$ | 19 |
| $25 < L \leqslant 40$ | 24 |
| $40 < L \leqslant 60$ | 6 |

(a) On the grid, draw a histogram for this information.




**(3)**

Manuel selected at random a flight that was late by 25 minutes or less from his results.

(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

.....

**(2)**

**(Total for Question 14 is 5 marks)**

**1** **The histogram densities are**

The histogram densities are

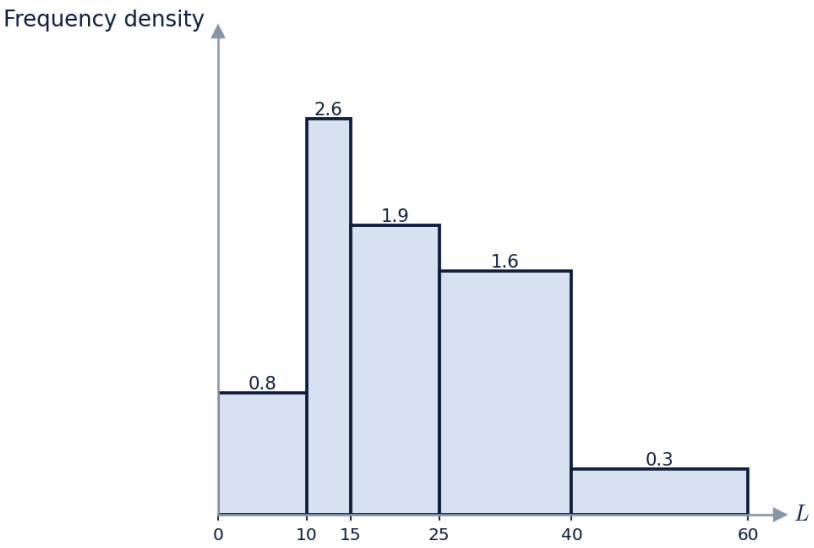
$$0 < L \leq 10 : \frac{8}{10} = 0.8$$

$$10 < L \leq 15 : \frac{13}{5} = 2.6$$

$$15 < L \leq 25 : \frac{19}{10} = 1.9$$

$$25 < L \leq 40 : \frac{24}{15} = 1.6$$

$$40 < L \leq 60 : \frac{6}{20} = 0.3$$




**2** **Use histogram**

Given $L \leq 25$, the total frequency is

$$8 + 13 + 19 = 40$$

**3** **Use histogram**

For $L \leq 5$, estimate half of the first class:

$$5(0.8) = 4$$

$$P(L \leq 5 \mid L \leq 25) = \frac{4}{40} = \frac{1}{10}$$

**FINAL ANSWER**

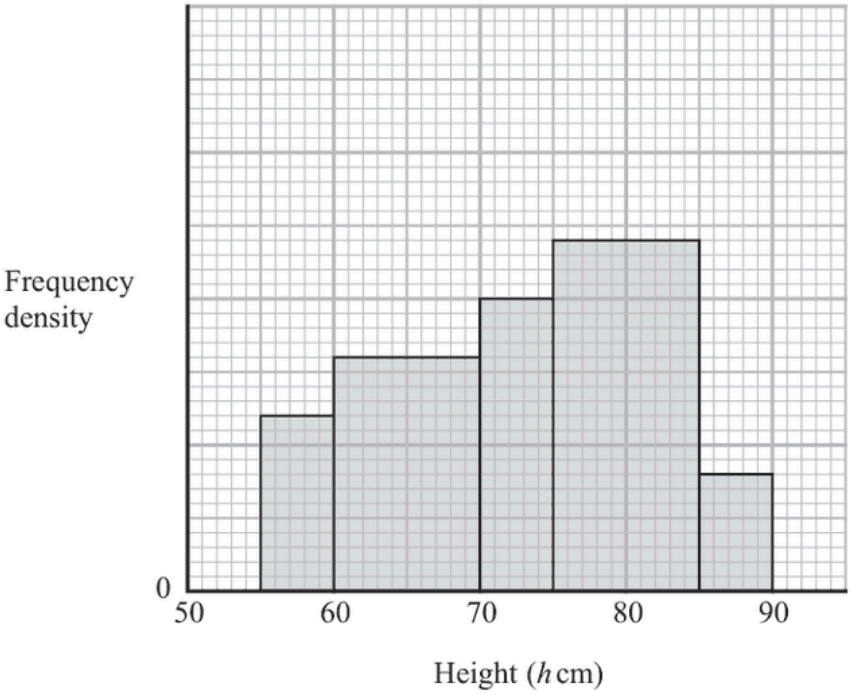
histogram densities $0.8, 2.6, 1.9, 1.6, 0.3$, and probability $\frac{1}{10}$.

4MA1/2H JANUARY 2022

# Question 20

4 marks | Histograms

**20** The histogram gives information about the heights, $h$ cm, of some tomato plants.




There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H January 2022

### 1 Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

### 2 Use frequency table

For $75 < h \leq 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

### 3 Use histogram

Using this scale from the histogram, the frequency densities are:

$$0.6,\ 0.8,\ 1.0,\ 1.2,\ 0.4$$

### 4 The total number of tomato

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

### 5 Solve inequality

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

### 6 Calculate probability

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

**FINAL ANSWER**

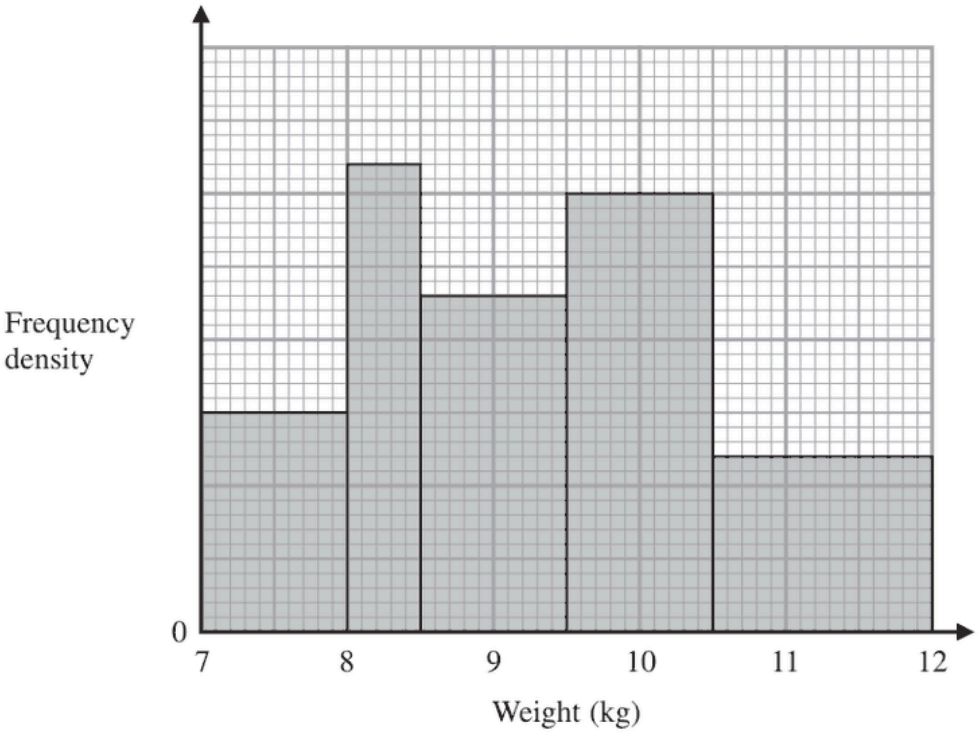
$\frac{1}{6}$

4MA1/1HR JANUARY 2022

# Question 21

3 marks | Histograms

21




The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1HR January 2022

**1 Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2 Use frequency table**

For $8 \leq w < 8.5$, the frequency is 16 and the class width is $0.5$, so

$$\text{frequency density} = \frac{16}{0.5} = 32$$

### 3 Use histogram

Using this scale from the histogram, the frequency densities are:

$$15,\ 32,\ 23,\ 30,\ 12$$

### 4 The total frequency is

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

**FINAL ANSWER**

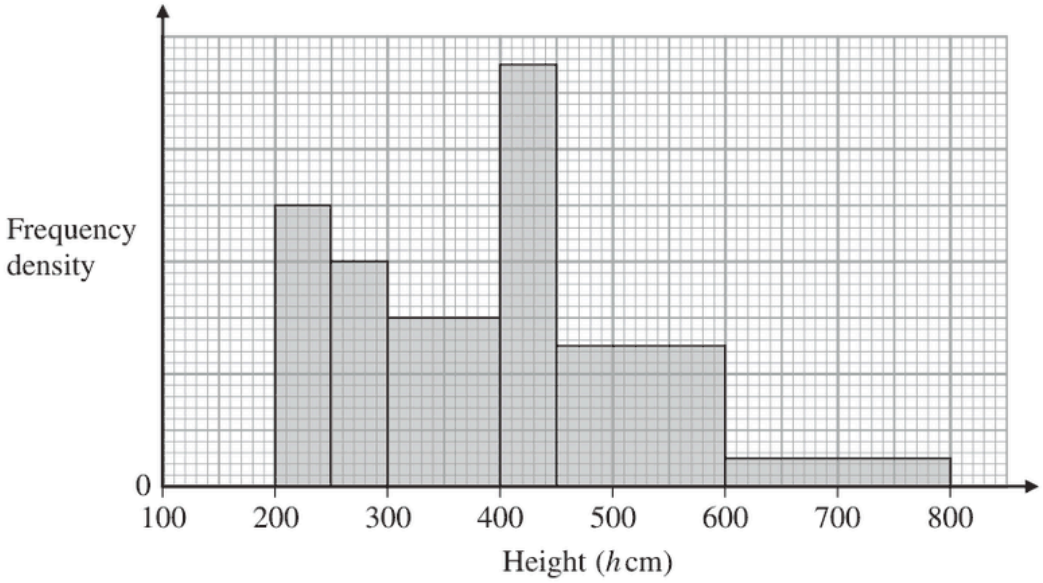
$102$ watermelons

4MA1/1HR JUNE 2022

# Question 19

3 marks | Histograms

**19** The histogram gives information about the height, $h$ cm, of each tree in part of a forest.




There are no trees for which $h \leqslant 200$ and for which $h > 800$

The number of trees for which $300 < h \leqslant 400$ is 8 fewer than the number of trees for which $400 < h \leqslant 500$

Work out an estimate for the number of trees in this part of the forest that have a height greater than 500 cm.

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1HR June 2022

**1** **Use relative densities from the**

Use relative densities from the histogram:

$$10,\ 8,\ 6,\ 15,\ 5,\ 1$$

**2** **Use histogram**

The number of trees with $300 < h \leq 400$ is

$$100(6s) = 600s$$

**3** **Use histogram**

The number of trees with $400 < h \leq 500$ is

$$50(15s) + 50(5s) = 1000s$$

**4** **The difference is**

The difference is $8$:

$$1000s - 600s = 8$$

$$400s = 8$$

$$s = 0.02$$

**5** **Use histogram**

For height greater than $500$:

$$100(5s) + 200(1s) = 700s$$

$$700(0.02) = 14$$

✓ **FINAL ANSWER**

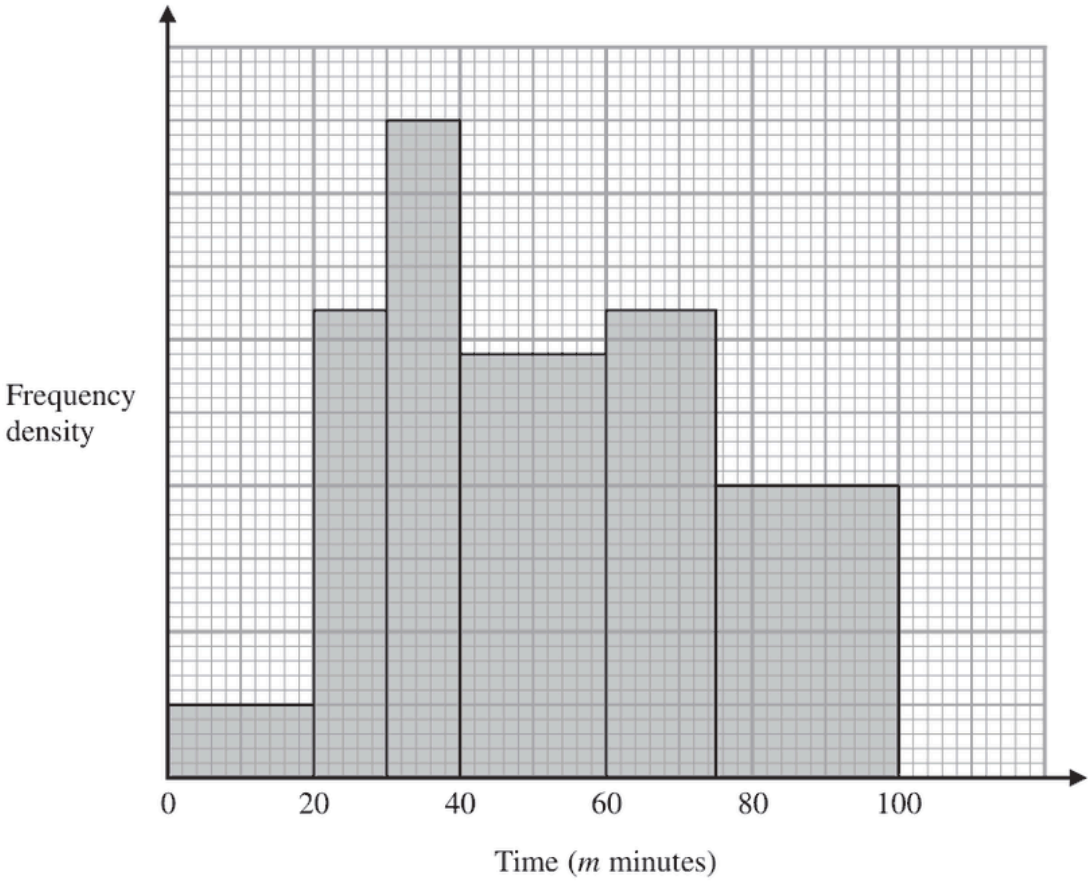
$14$ trees.

4MA1/1H JUNE 2022

# Question 21

3 marks | Histograms

**21** The histogram shows information about the total time, $m$ minutes, taken by each child in a school to walk to school every day for one week.




There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$

Work out an estimate for the number of children for whom $50 < m \leqslant 80$

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H June 2022

**1** **Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2** **Use frequency table**

For $m \leq 20$, the frequency is 10 and the class width is 20, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

---

**3** **Use histogram**

Using this scale from the histogram:

$$40 < m \leq 60 \quad \text{has frequency density } 2.9$$

$$60 < m \leq 75 \quad \text{has frequency density } 3.2$$

$$75 < m \leq 100 \quad \text{has frequency density } 2.0$$

---

**4** **Use part of each relevant**

For $50 < m \leq 80$, use part of each relevant bar:

$$(10)(2.9) + (15)(3.2) + (5)(2.0)$$

$$= 29 + 48 + 10$$

$$= 87$$

---

✓ **FINAL ANSWER**

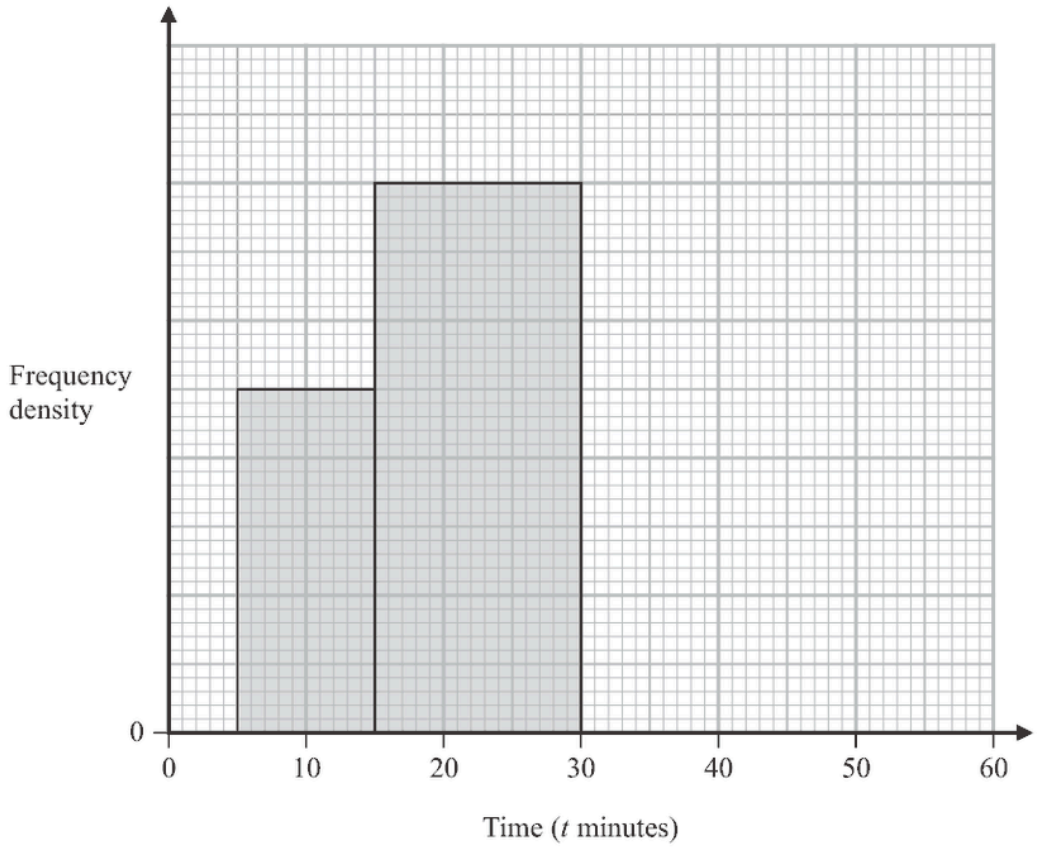
87 children

# Question 18

4 marks | Histograms

**18** The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

| Time ($t$ minutes) | Frequency |
|---|---|
| $0 < t \leqslant 5$ | 23 |
| $5 < t \leqslant 15$ | |
| $15 < t \leqslant 30$ | |
| $30 < t \leqslant 40$ | 18 |
| $40 < t \leqslant 60$ | 14 |




Complete the table and the histogram.

**(Total for Question 18 is 4 marks)**

**1** **The total frequency is**

The total frequency is $140$.

**2** **The known frequencies are**

The known frequencies are

$$23 + 18 + 14 = 55$$

**3** **Use histogram**

So the two missing frequencies have total

$$140 - 55 = 85$$

**4** **Use histogram**

From the histogram, the missing frequencies are

$$5 < t \leq 15 : \ 25, \qquad 15 < t \leq 30 : \ 60$$

**5** **The missing histogram densities are**

The missing histogram densities are

$$0 < t \leq 5 : \frac{23}{5} = 4.6$$

$$30 < t \leq 40 : \frac{18}{10} = 1.8$$

$$40 < t \leq 60 : \frac{14}{20} = 0.7$$

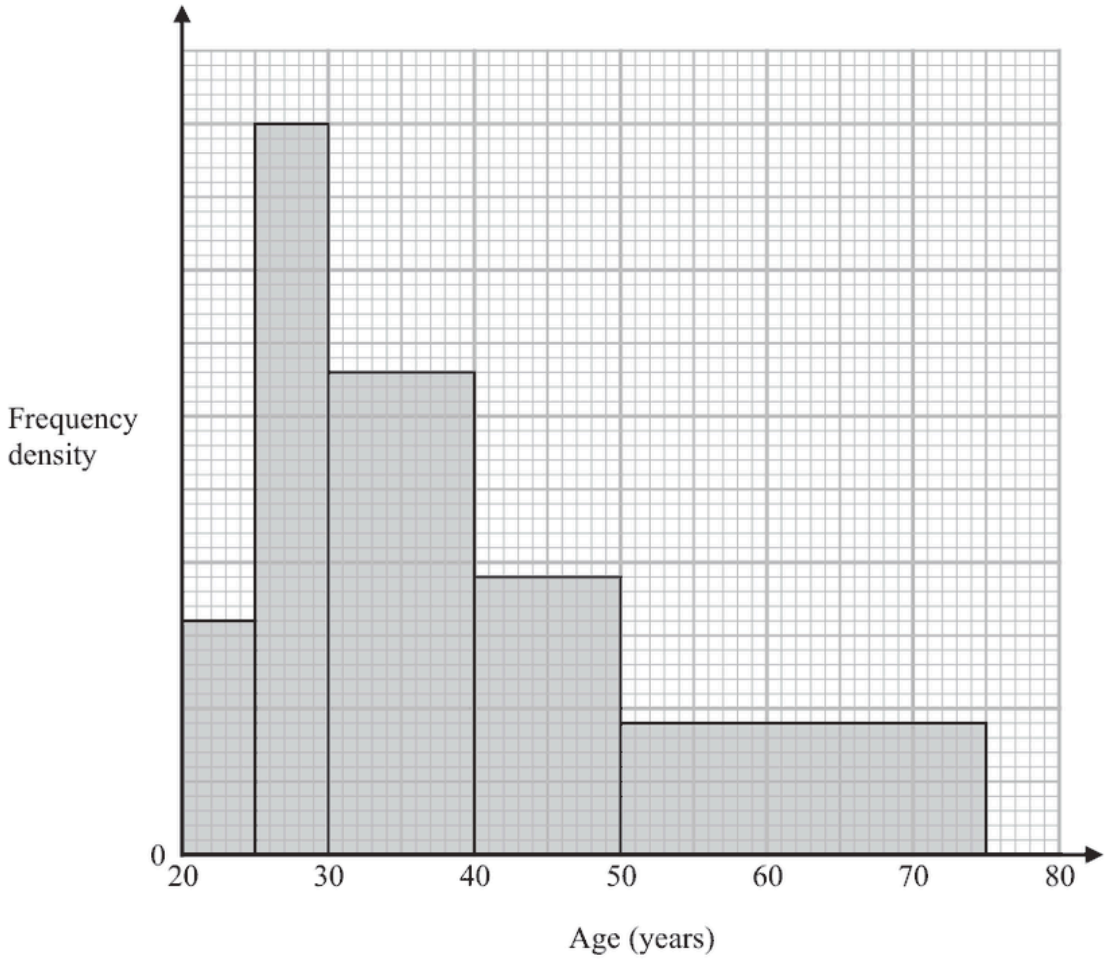
**FINAL ANSWER**

missing frequencies $25, 60$, and missing densities $4.6, 1.8, 0.7$.

# Question 22

4 marks | Histograms

**22** Some people attend a concert.
The histogram shows information about the ages of these people.




Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

.....%

**(Total for Question 22 is 4 marks)**

## 1 The scale of the frequency

The scale of the frequency density axis is not needed because the percentage is found from areas.

## 2 Calculate area

Using the bar heights from the histogram, the relative areas are:

$$(5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155)$$

$$= 1385 + 4330 + 5710 + 3290 + 3875$$

$$= 18590$$

## 3 Calculate value

For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

## 4 Calculate percentage

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675\ldots$$

**FINAL ANSWER**

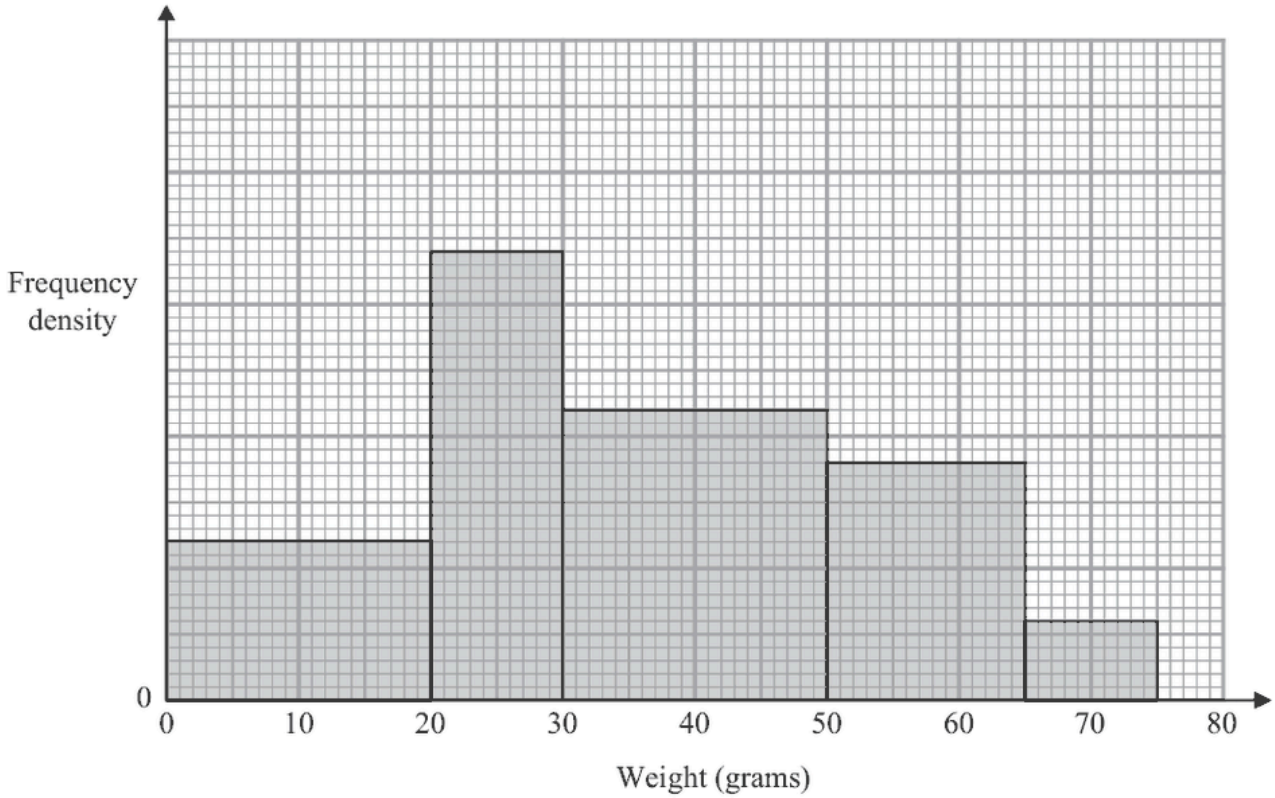
$16.7\%$

4MA1/1H NOVEMBER 2023

# Question 17

3 marks | Histograms

**17** The histogram gives information about the weights, in grams, of some oranges in a box.




24 of these oranges weigh less than 20 grams.

Medium oranges weigh between 35 grams and 55 grams.

Work out an estimate for the number of medium oranges in the box.

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1H November 2023

**1** **The to gram class has**

The $0$ to $20$ gram class has frequency $24$, so its density is

$$\frac{24}{20} = 1.2$$

**2** **Use histogram**

From the histogram scale, the density for $30 < w \leq 50$ is about $2.2$, and the density for $50 < w \leq 65$ is about $1.8$.

**3** **Medium oranges weigh between g**

Medium oranges weigh between $35$ g and $55$ g:

$$15(2.2) + 5(1.8) = 33 + 9 = 42$$

✓ **FINAL ANSWER**

about $42$ medium oranges.

4MA1/1HR MAY 2024

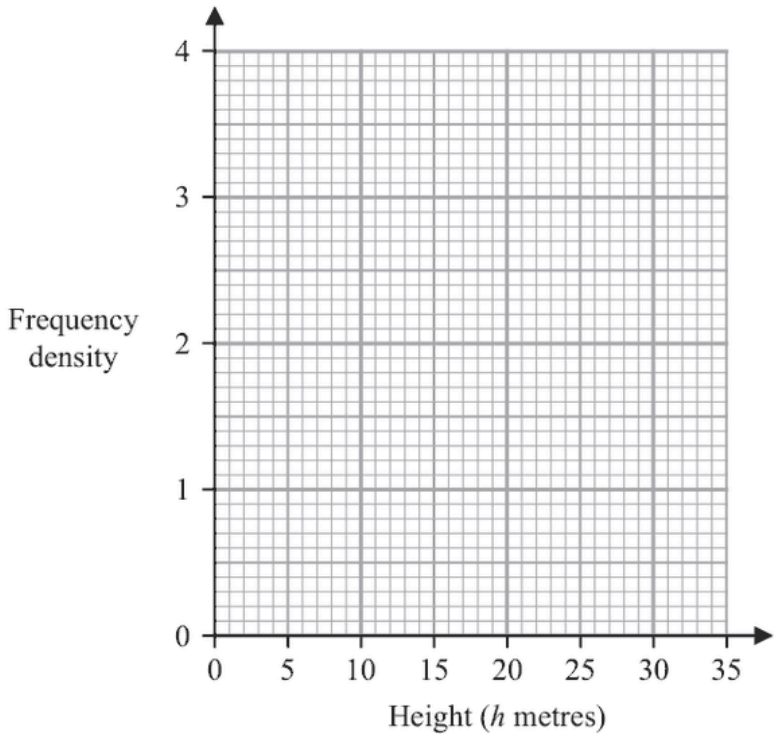
# Question 16

3 marks | Histograms

**16** The table gives information about the heights, in metres, of the trees in a park.

| Height ($h$ metres) | Frequency |
|---|---|
| $0 < h \leqslant 2$ | 5 |
| $2 < h \leqslant 5$ | 12 |
| $5 < h \leqslant 10$ | 18 |
| $10 < h \leqslant 20$ | 14 |
| $20 < h \leqslant 35$ | 9 |

On the grid, draw a histogram for this information.




**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1HR May 2024

**1** **Use histogram**

For a histogram,

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

**2** **The bar heights are**

The bar heights are:

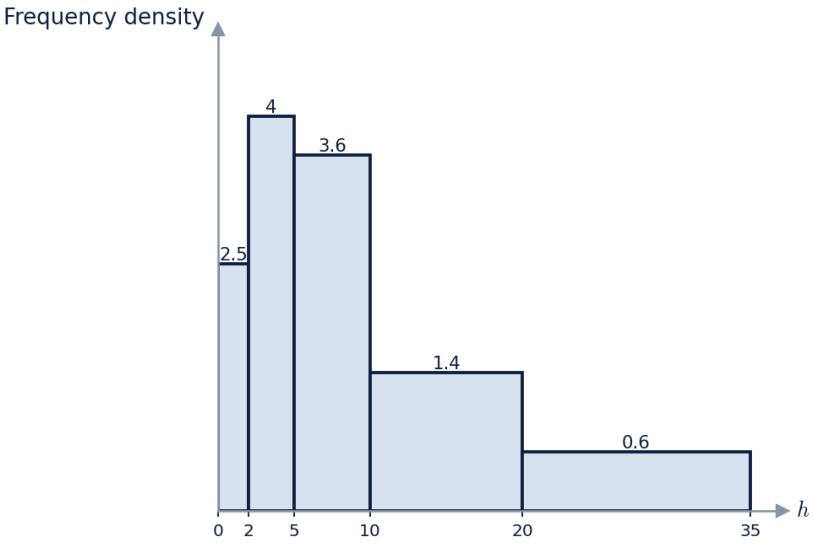
$$0 < h \leq 2 : \frac{5}{2} = 2.5$$

$$2 < h \leq 5 : \frac{12}{3} = 4$$

$$5 < h \leq 10 : \frac{18}{5} = 3.6$$

$$10 < h \leq 20 : \frac{14}{10} = 1.4$$

$$20 < h \leq 35 : \frac{9}{15} = 0.6$$




**FINAL ANSWER**

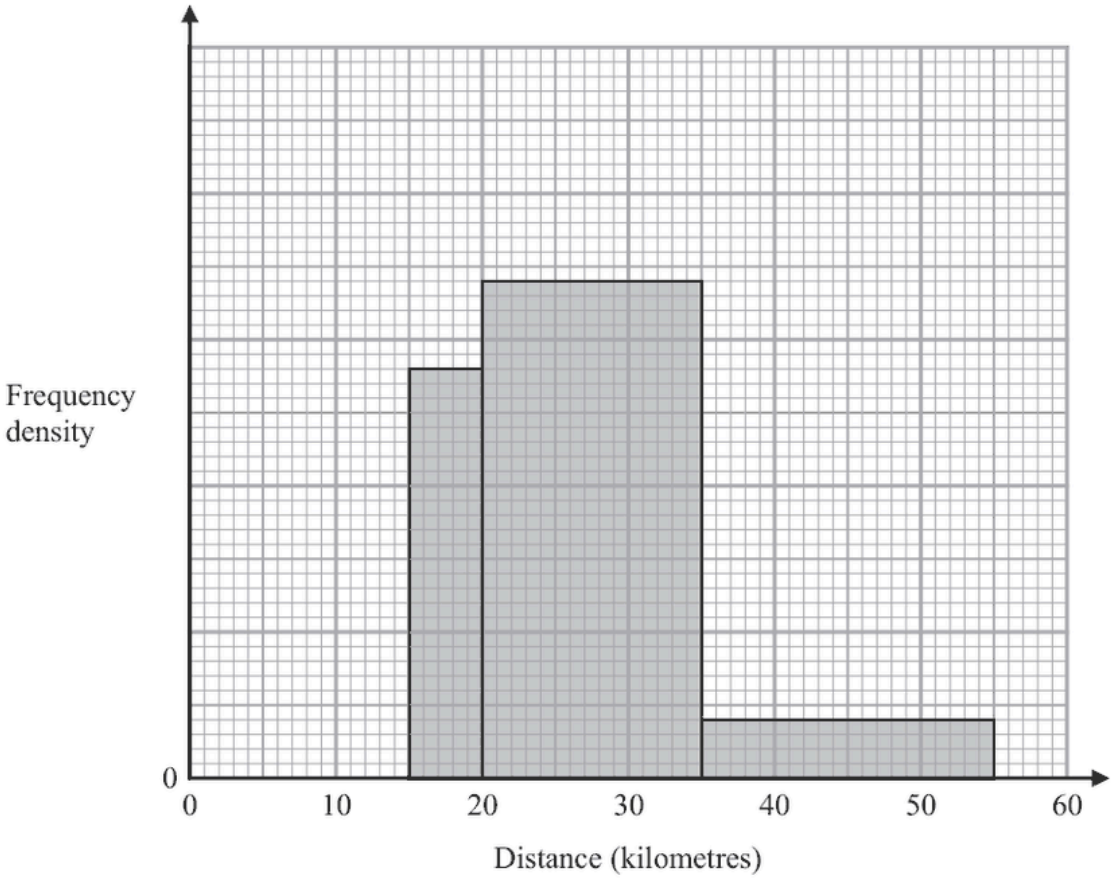
draw bars with heights $2.5, 4, 3.6, 1.4, 0.6$.

4MA1/1H MAY 2024

# Question 22

3 marks | Histograms

22 The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.




All of the adults ran at least 5 kilometres.
None of the adults ran more than 55 kilometres.
14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

**(Total for Question 22 is 3 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H May 2024

**1 Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2** **Use frequency table**

For $15 \leq d < 20$, the frequency is 14 and the class width is 5, so

$$\text{frequency density} = \frac{14}{5} = 2.8$$

**3** **Use histogram**

Using this scale from the histogram:

$$20 \leq d < 35 \quad \text{has frequency density } 3.4$$

$$35 \leq d \leq 55 \quad \text{has frequency density } 0.4$$

**4** **Calculate value**

So the known frequencies are

$$14 + (15)(3.4) + (20)(0.4)$$

$$= 14 + 51 + 8 = 73$$

**5** **Use frequency table**

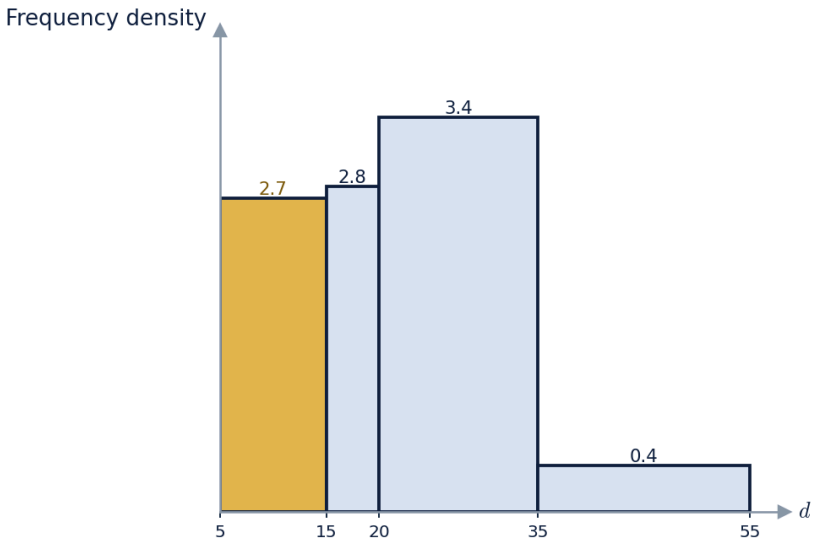
There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

**6** **The missing class width is**

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$



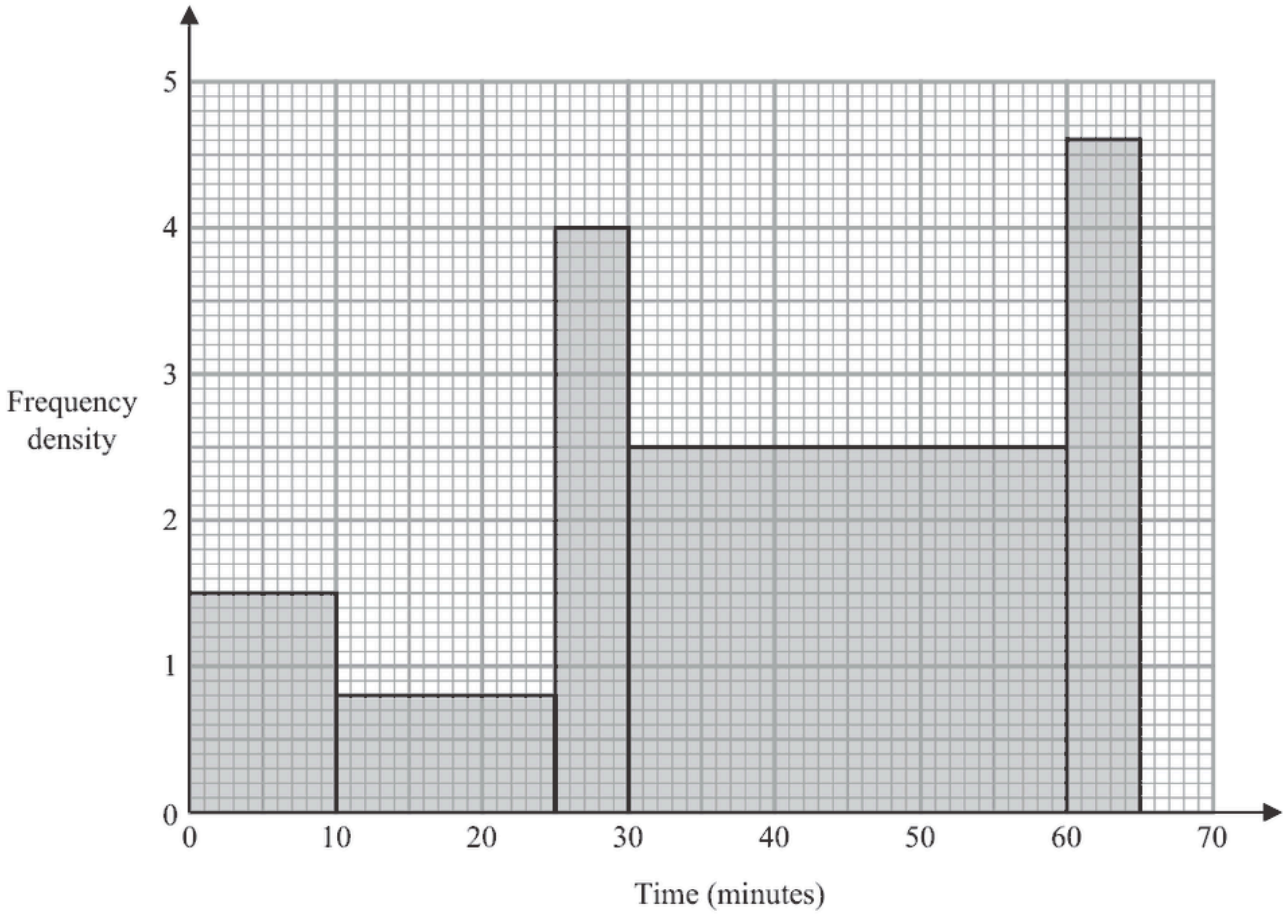
**FINAL ANSWER**

draw the missing bar from $5$ to $15$ with frequency density $2.7$

# Question 18

4 marks | Histograms

**18** The histogram shows information about the times some students took to complete a puzzle.




Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

**(Total for Question 18 is 4 marks)**

## 1 Estimate the frequencies from each

Estimate the frequencies from each bar:

$$0 \text{ to } 10: \ 10(1.5) = 15$$

$$10 \text{ to } 25: \ 15(0.8) = 12$$

$$25 \text{ to } 30: \ 5(4) = 20$$

$$30 \text{ to } 60: \ 30(2.5) = 75$$

$$60 \text{ to } 65: \ 5(4.6) = 23$$

## 2 Total frequency

Total frequency:

$$15 + 12 + 20 + 75 + 23 = 145$$

## 3 Between and minutes

Between $20$ and $60$ minutes:

$$5(0.8) + 20 + 75 = 99$$

$$\text{fraction} = \frac{99}{145}$$

**FINAL ANSWER**

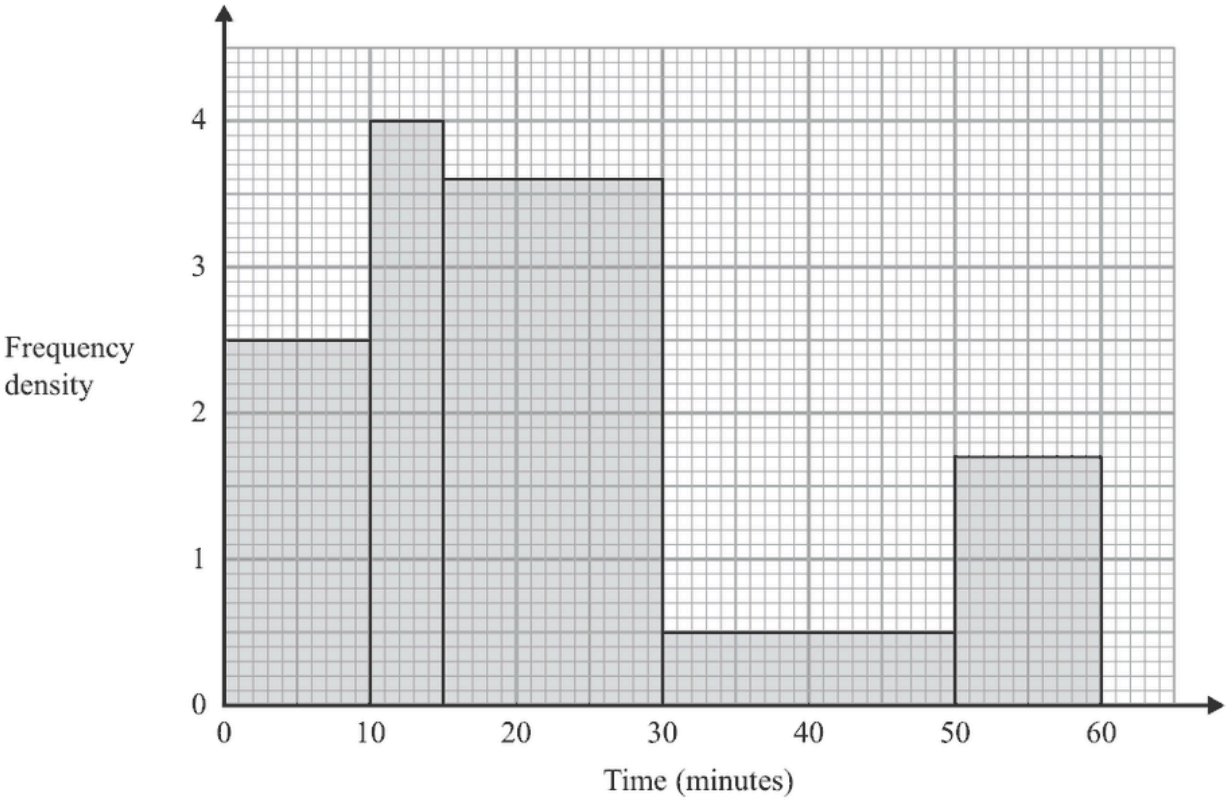
$\frac{99}{145}$.

4MA1/1H JUNE 2025

# Question 17

3 marks | Histograms

17 The histogram shows information about the times, in minutes, that some people were in a shop.




Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/1H June 2025

### 1 Total estimated frequency

Total estimated frequency:

$$10(2.5) + 5(4) + 15(3.6) + 20(0.5) + 10(1.7) = 126$$

### 2 More than minutes

More than $40$ minutes:

$$10(0.5) + 10(1.7) = 22$$

### 3 Use histogram

So the proportion is

$$\frac{22}{126} = \frac{11}{63}$$

---

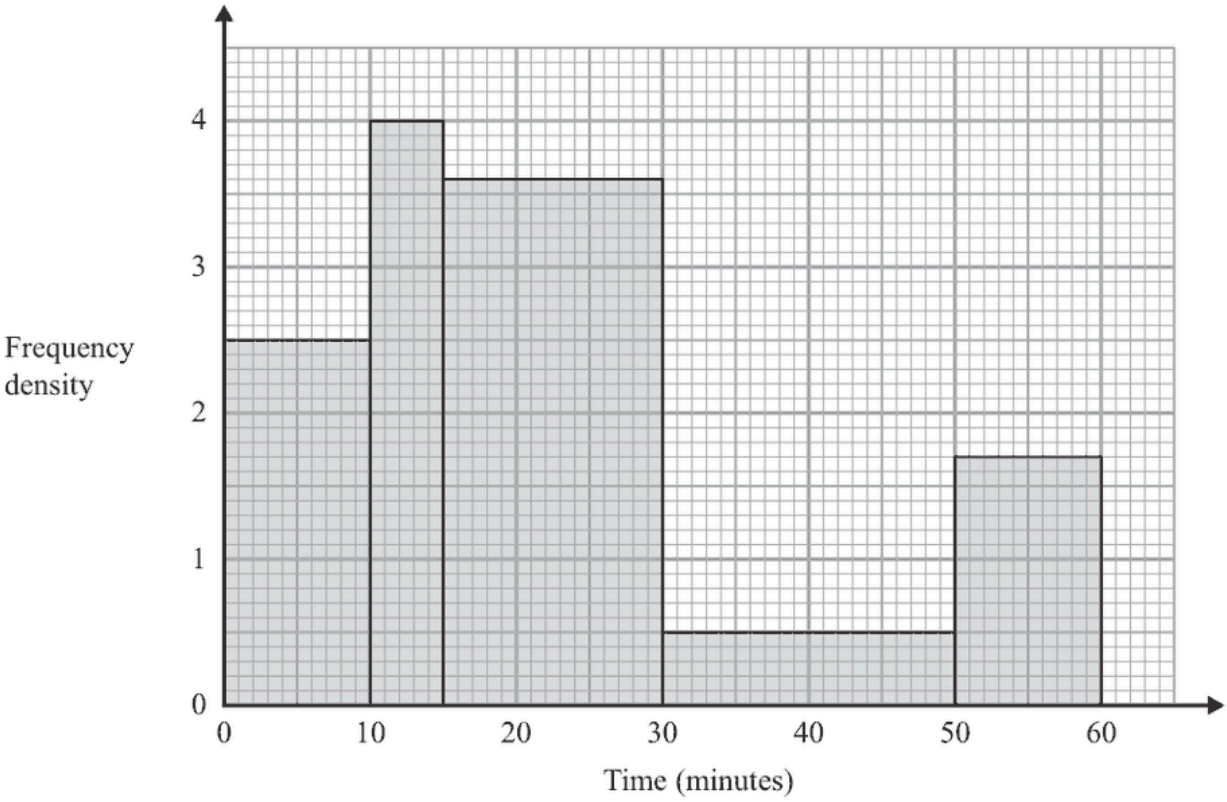
**FINAL ANSWER**

$\frac{11}{63}$.

# Question 18

3 marks | Histograms

**18** The histogram shows information about the times, in minutes, that some people were in a shop.




Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

**(Total for Question 18 is 3 marks)**

**1** **Frequency is class width multiplied**

Frequency is class width multiplied by frequency density.

$$0\text{--}10 : 10 \times 2.5 = 25$$

$$10\text{--}15 : 5 \times 4 = 20$$

$$15\text{--}30 : 15 \times 3.6 = 54$$

$$30\text{--}50 : 20 \times 0.5 = 10$$

$$50\text{--}60 : 10 \times 1.7 = 17$$

**2** **Total**

Total:

$$25 + 20 + 54 + 10 + 17 = 126$$

**3** **More than minutes is the**

More than $40$ minutes is the part from $40$ to $50$, plus $50$ to $60$.

$$10 \times 0.5 + 17 = 22$$

**4** **Use histogram**

So the proportion is

$$\frac{22}{126} = \frac{11}{63}$$

✓ **FINAL ANSWER**

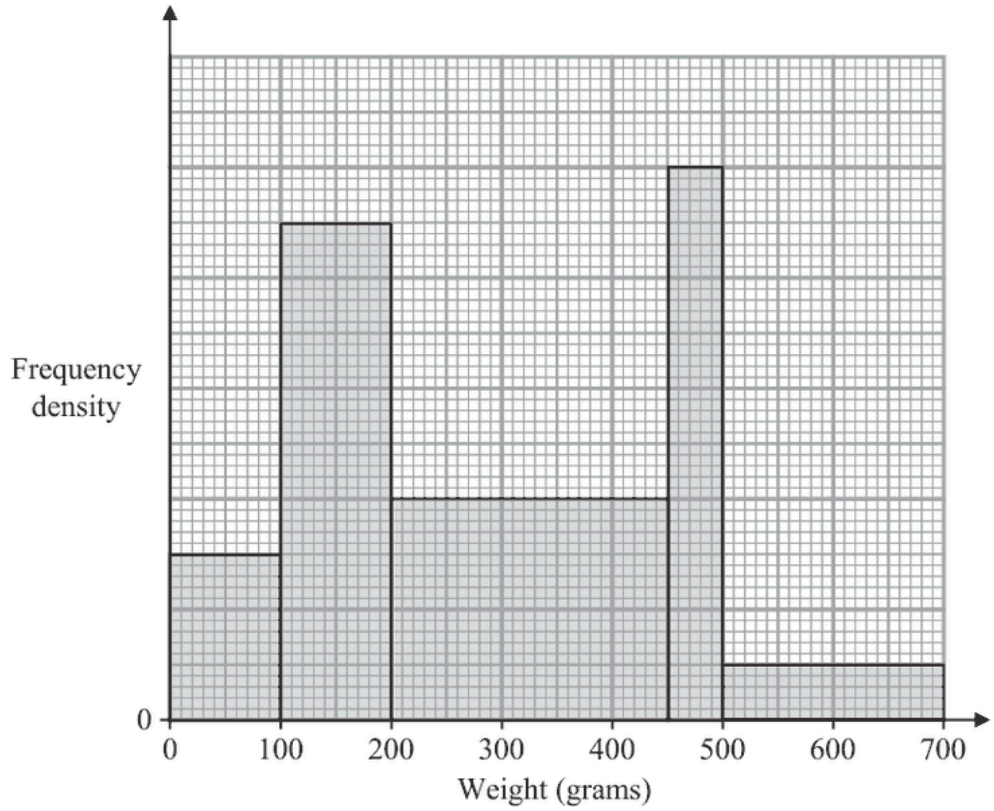
$\frac{11}{63}$

4MA1/2HR JUNE 2025

# Question 20

4 marks | Histograms

**20** The histogram gives some information about the weights, in grams, of some books.




75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2HR June 2025

**1** **Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2** **Use frequency table**

For books weighing less than 100 g, the frequency is 75 and the class width is 100, so

$$\text{frequency density} = \frac{75}{100} = 0.75$$

**3** **Use histogram**

Using this scale from the histogram, the frequency densities are:

$$0.75,\ 2.25,\ 1.0,\ 2.5,\ 0.25$$

**4** **The total number of books**

The total number of books is

$$(100)(0.75) + (100)(2.25) + (250)(1.0) + (50)(2.5) + (200)(0.25)$$

$$= 75 + 225 + 250 + 125 + 50 = 725$$

**5** **Calculate value**

For weights between 400 g and 600 g:

$$(50)(1.0) + (50)(2.5) + (100)(0.25)$$

$$= 50 + 125 + 25 = 200$$

**6** **Calculate probability**

So the probability is

$$\frac{200}{725} = \frac{8}{29}$$

✓ **FINAL ANSWER**

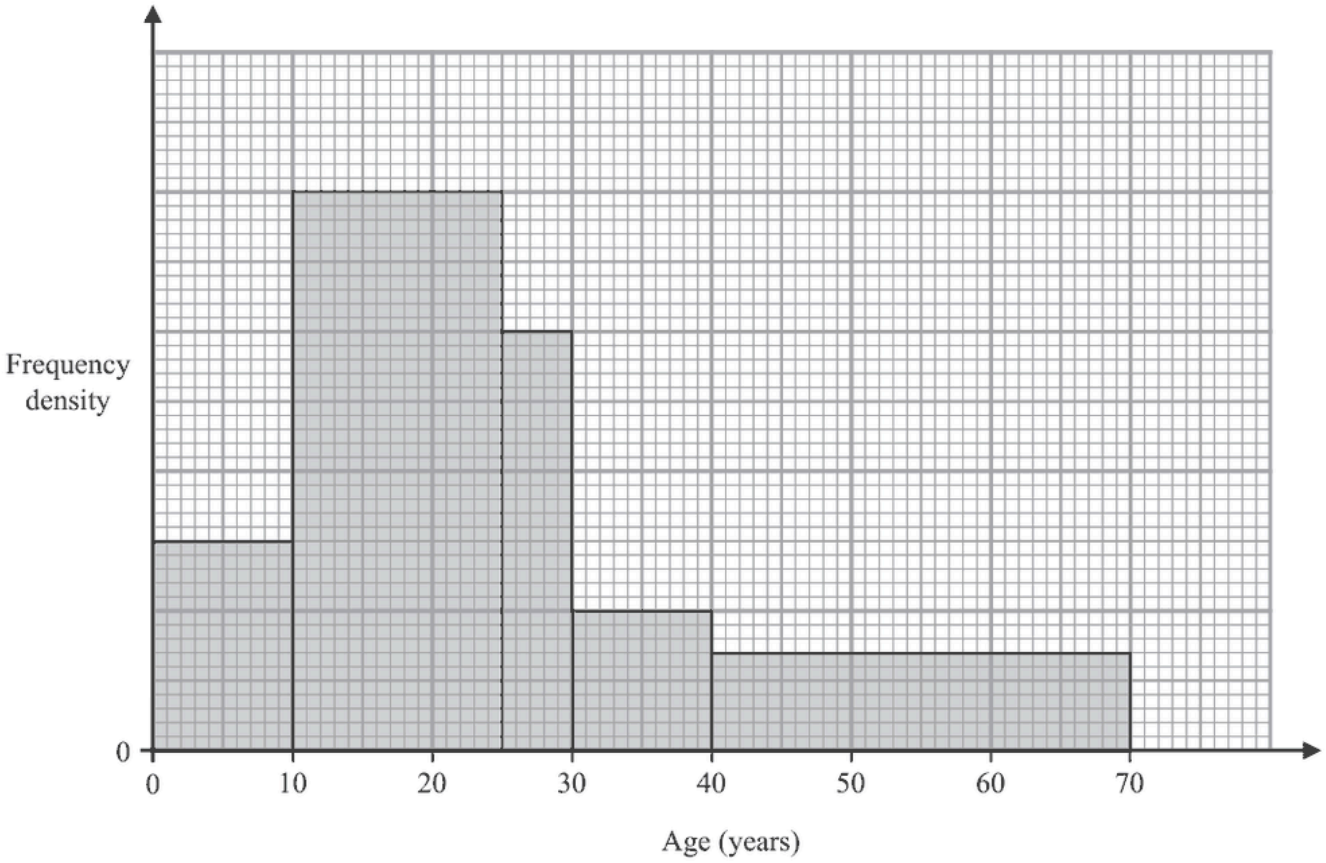
$\frac{8}{29}$ or $0.276$ to 3 significant figures

4WM1/01R MAY 2025

# Question 21

3 marks | Histograms

**21** The histogram shows information about the ages of the members of a chess club.




There are 60 members aged between 10 and 25
There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4WM1/01R May 2025

**1** **The interval to has width**

The interval $10$ to $25$ has width $15$ and frequency $60$, so its frequency density is

$$\frac{60}{15} = 4$$

**2** **Use histogram**

Using that scale from the histogram:

$$35 \text{ to } 40 : 5 \times 1 = 5$$

$$40 \text{ to } 70 : 30 \times 0.75 = 22.5$$

$$5 + 22.5 = 27.5$$

---

✓ **FINAL ANSWER**

about $28$ members.

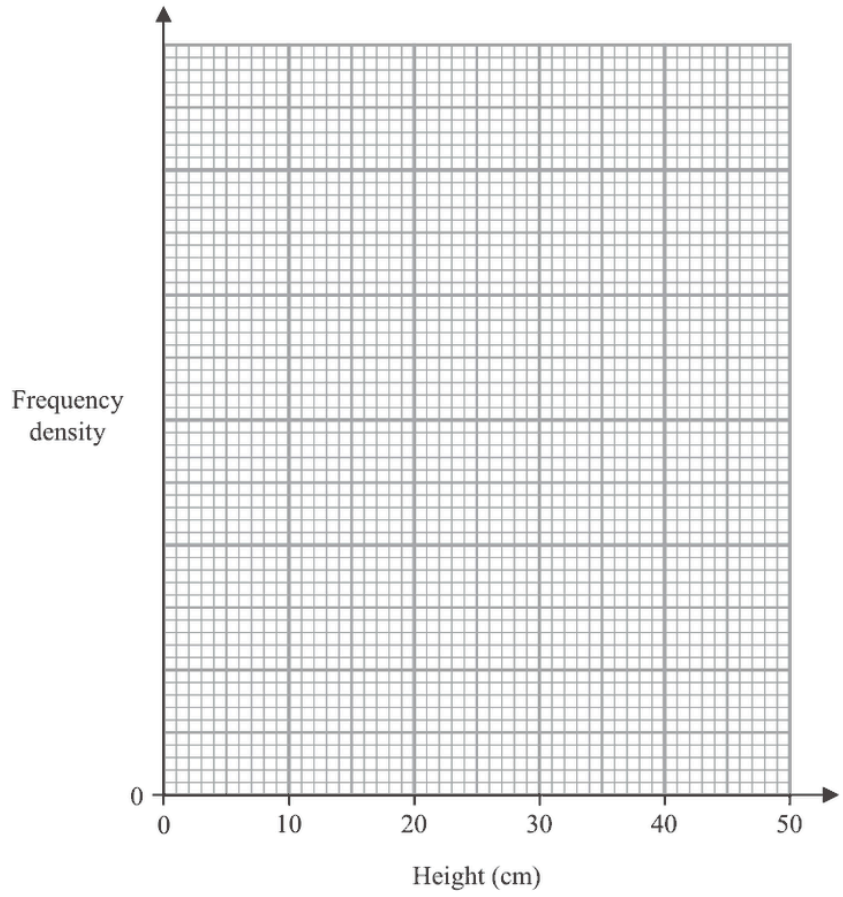
# Question 20

3 marks | Histograms

**20** The table gives information about the heights, in centimetres, of some plants.

| Height ($h$ cm) | Frequency |
|---|---|
| $0 < h \leqslant 5$ | 12 |
| $5 < h \leqslant 15$ | 34 |
| $15 < h \leqslant 30$ | 45 |
| $30 < h \leqslant 50$ | 28 |

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.




**(Total for Question 20 is 3 marks)**

## 1 Frequency density

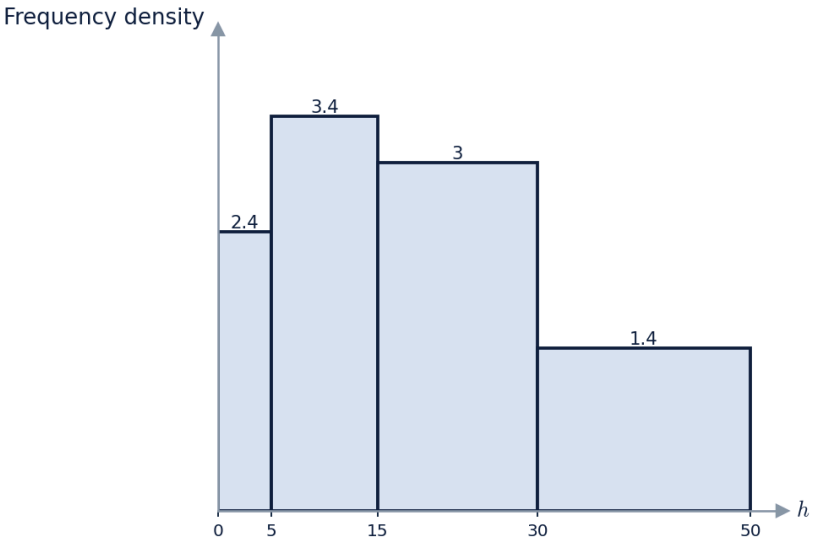
Frequency density $= \dfrac{\text{frequency}}{\text{class width}}$.

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

$$5 < h \leq 15 : \frac{34}{10} = 3.4$$

$$15 < h \leq 30 : \frac{45}{15} = 3$$

$$30 < h \leq 50 : \frac{28}{20} = 1.4$$




**FINAL ANSWER**

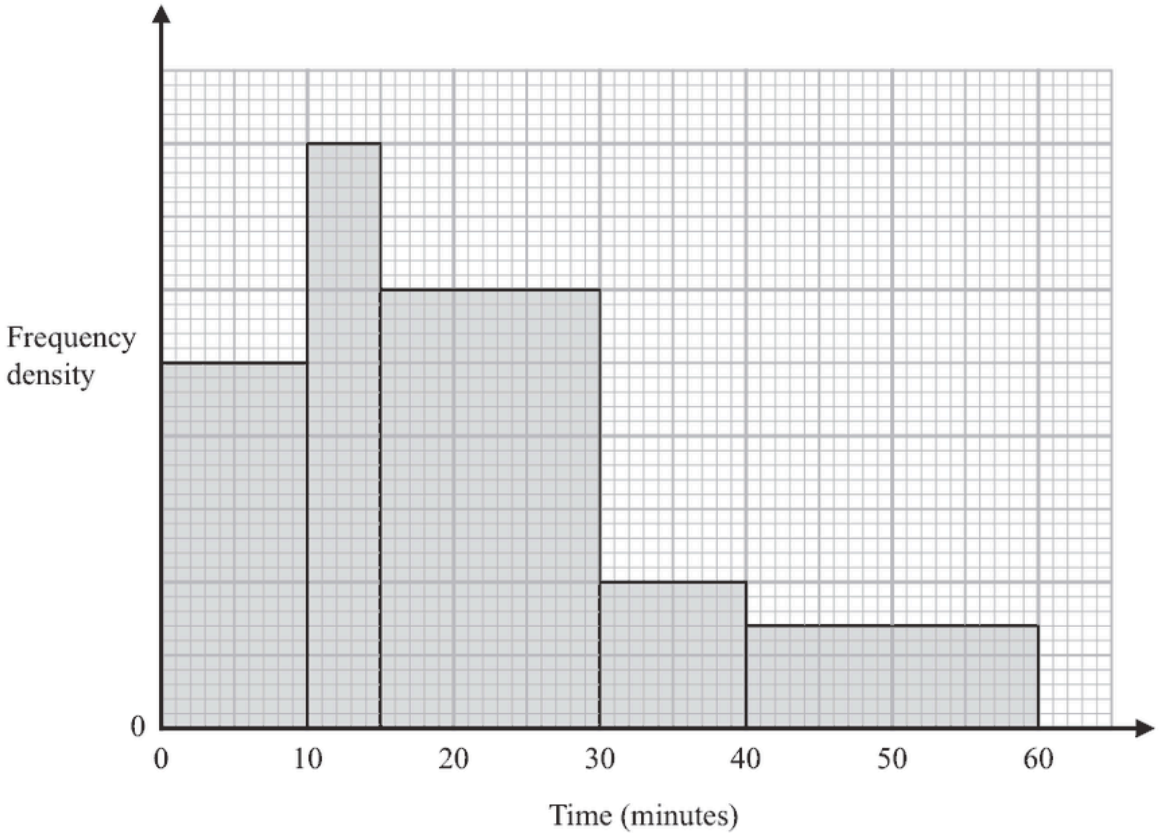
draw bars with densities $2.4, 3.4, 3, 1.4$.

4MA1/2H NOVEMBER 2025

# Question 21

3 marks | Histograms

**21** The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.




20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2H November 2025

**1** **Use histogram**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

**2** **Use frequency table**

For $10 < t \leq 15$, the frequency is 20 and the class width is 5, so

$$\text{frequency density} = \frac{20}{5} = 4$$

---

**3** **Use histogram**

Using this scale from the histogram:

$$15 < t \leq 30 \quad \text{has frequency density } 3$$

$$30 < t \leq 40 \quad \text{has frequency density } 1$$

$$40 < t \leq 60 \quad \text{has frequency density } 0.7$$

---

**4** **Calculate value**

For trains at least 25 minutes late:

$$(5)(3) + (10)(1) + (20)(0.7)$$

$$= 15 + 10 + 14$$

$$= 39$$

---

✓ **FINAL ANSWER**

$39$ trains

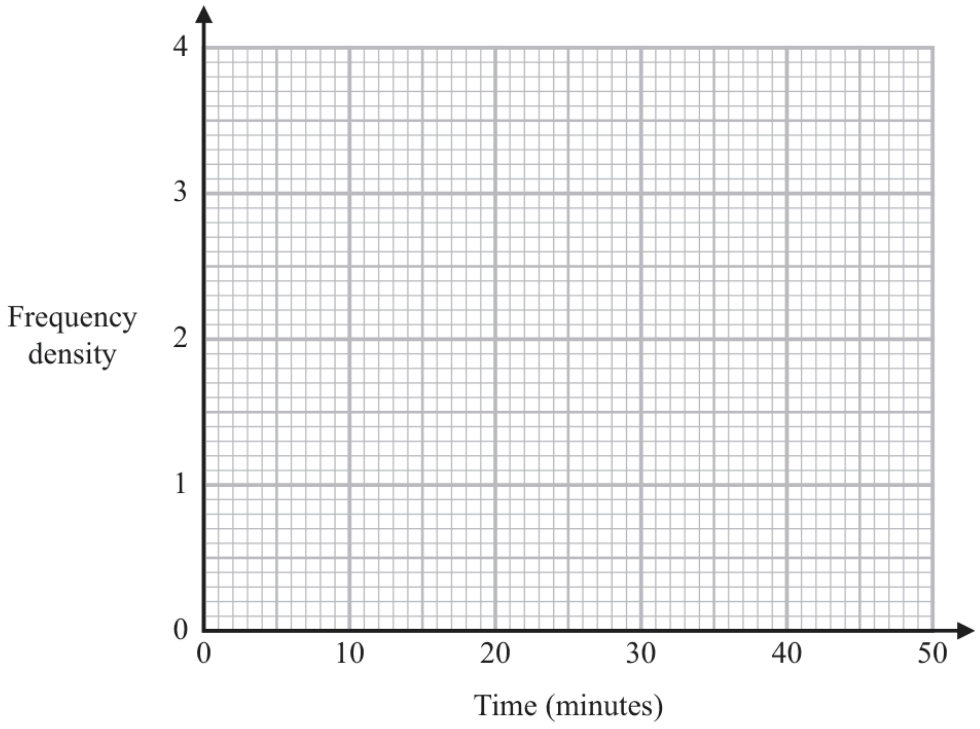
# Question 19

3 marks | Histograms

**19** The table gives information about the times, in minutes, some people spent waiting to go on a rollercoaster.

| Time ($t$ minutes) | Frequency |
|---|---|
| $0 < t \leqslant 20$ | 14 |
| $20 < t \leqslant 25$ | 12 |
| $25 < t \leqslant 35$ | 32 |
| $35 < t \leqslant 50$ | 30 |

On the grid, draw a histogram for this information.




**(Total for Question 19 is 3 marks)**

1 **Solution | Step 1**

Use $\textbf{frequency density} = \textbf{frequency}/\textbf{class width}$:

$$14/20 = 0.7, \quad 12/5 = 2.4, \quad 32/10 = 3.2, \quad 30/15 = 2.$$

Histogram areas, rather than heights alone, represent frequencies.

2 **Solution | Step 2**

Draw four adjoining rectangles on the supplied time and frequency-density axes: $[0, 20]$ at height $0.7$, $[20, 25]$ at height $2.4$, $[25, 35]$ at height $3.2$, and $[35, 50]$ at height $2$.

Each rectangle starts at density 0 and has the stated class boundaries as its vertical edges; leave no gaps between classes.

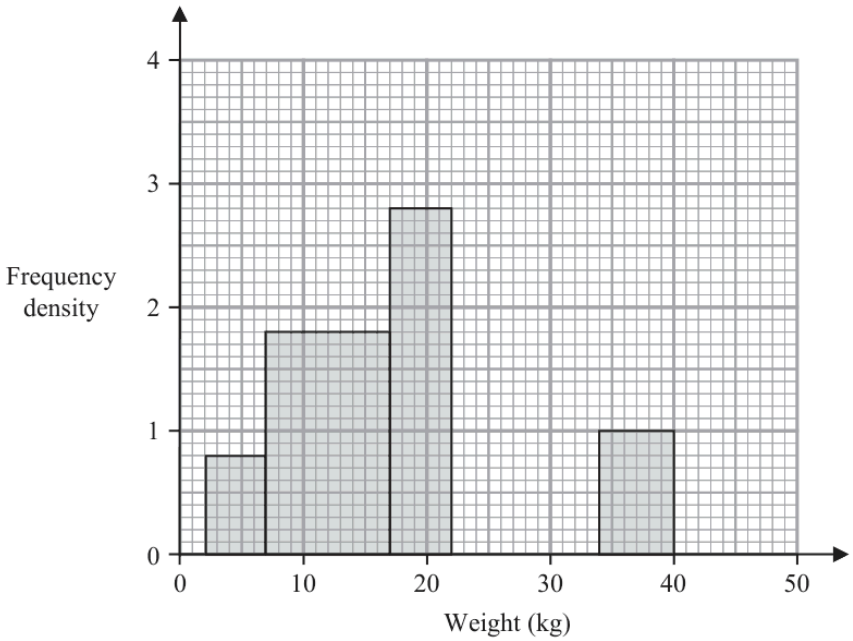
✓ **FINAL ANSWER**

Bar intervals/heights: $[0, 20] : 0.7;\ [20, 25] : 2.4;\ [25, 35] : 3.2;\ [35, 50] : 2$.

# Question 21

3 marks | Histograms

**21** The incomplete histogram shows information about the weights, in kg, of some parcels.




No parcels weigh less than 2 kg
No parcels weigh more than 40 kg
Half of the parcels weigh between 22 kg and 34 kg

Complete the histogram by drawing a bar to represent the parcels that weigh between 22 kg and 34 kg

**(Total for Question 21 is 3 marks)**

**1** **Frequency = width × height**

$$5(0.8) + 10(1.8) + 5(2.8) + 6(1) = 4 + 18 + 14 + 6 = 42$$

**2** **Half are in 22–34**

So the other half is also $42$ parcels.

**3** **Height of the new bar**

$$\text{f.d.} = \frac{42}{34 - 22} = 3.5$$

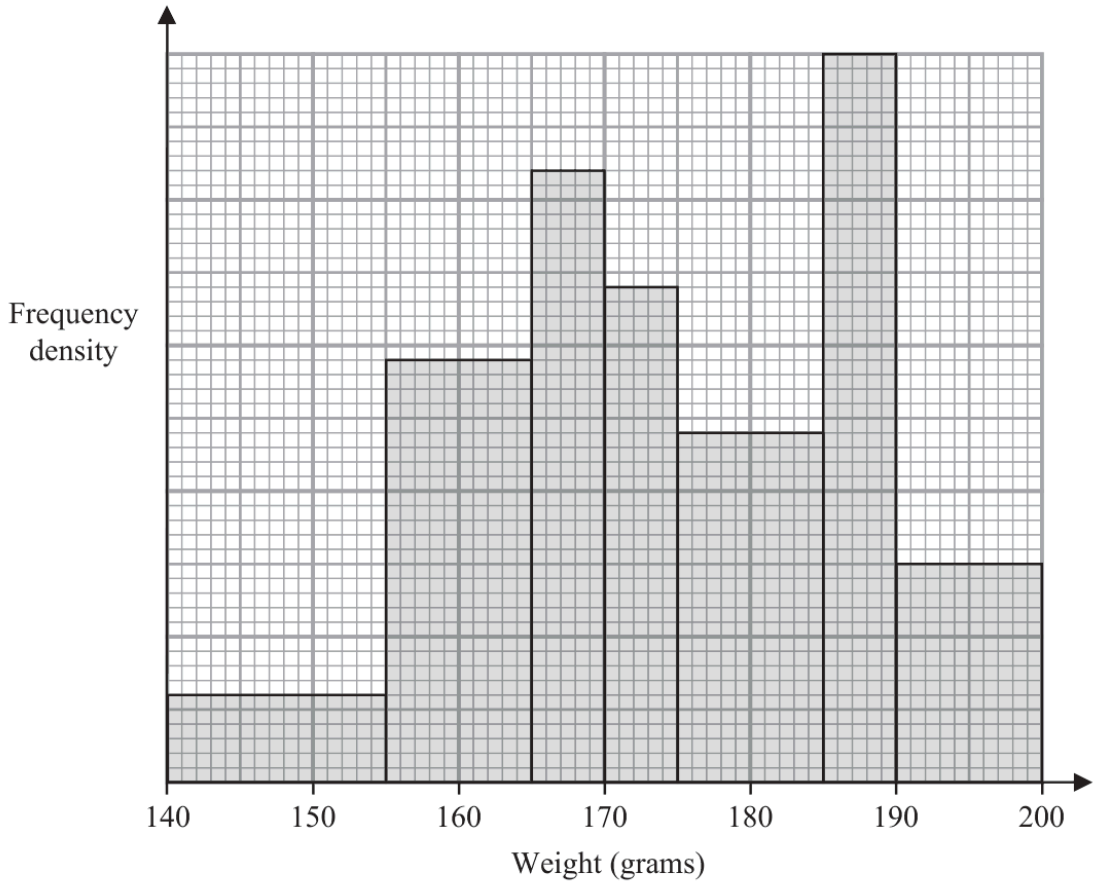
✓ **FINAL ANSWER**

Bar from $22$ to $34$ kg, height $3.5$

# Question 22

5 marks | Histograms

**22** The histogram gives information about the weights, in grams, of all the apples in a shop.




There are 9 apples with a weight between 140 grams and 155 grams.

(a) Work out the number of apples with a weight less than 175 grams.

.....

**(3)**

An apple is classified as a large apple if it has a weight of more than 180 grams.
One of the large apples is chosen at random.

(b) Work out the probability that it has a weight of more than 190 grams.

.....

**(2)**

**(Total for Question 22 is 5 marks)**

1 **Part (a) | Step 1**

The first class has width 15 g and frequency 9, so its density is $9/15 = 0.6$. Its height is 6 small grid squares, so each vertical square represents density $0.1$.

Frequency is class width multiplied by frequency density.

2 **Part (a) | Step 2**

The classes below 175 g have densities $0.6, 2.9, 4.2, 3.4$ and widths $15, 10, 5, 5$, respectively.

Read the bar heights using the calibrated vertical scale.

3 **Part (a) | Step 3**

$$N = 15(0.6) + 10(2.9) + 5(4.2) + 5(3.4) = 9 + 29 + 21 + 17 = 76.$$

Add the frequencies, not the heights.

4 **Part (b) | Step 1**

Above 180 g, the frequency from 180 to 185 is $5(2.4) = 12$, from 185 to 190 is $5(5) = 25$, and from 190 to 200 is $10(1.5) = 15$.

Use only half of the 175-185 bar and all of the two heavier classes.

5 **Part (b) | Step 2**

$$P(W > 190 \mid W > 180) = \frac{15}{12 + 25 + 15} = \frac{15}{52}.$$

Restrict the denominator to the large apples.

✓ **FINAL ANSWER**

(a) $76$ apples (b) $15/52$

TOPIC

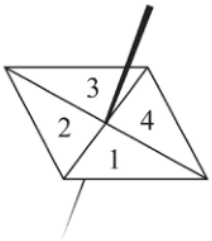
# Probability Toolkit

Every past-paper question on this topic, 2020 to 2026.

# Question 3

4 marks | Probability Toolkit

**3** Here is a biased 4-sided spinner.




The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

| **Number** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | 0.26 | | 0.18 | |

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Priya is going to spin the spinner 250 times.

Work out an estimate for the number of times the spinner will land on a 1 or a 2

.....

**(Total for Question 3 is 4 marks)**

### 1 Find the remaining probability

$$1 - 0.26 - 0.18 = 0.56$$

### 2 Split the equal probabilities

The probabilities for $2$ and $4$ are equal, so

$$P(2) = P(4) = \frac{0.56}{2} = 0.28$$

### 3 Find the probability of 1 or 2

$$P(1 \text{ or } 2) = 0.26 + 0.28 = 0.54$$

### 4 Calculate the estimate

$$0.54 \times 250 = 135$$

**FINAL ANSWER**

$135$.

4MA1/2H JANUARY 2020

# Question 6

4 marks | Probability Toolkit

6 A bag contains only red beads, blue beads, green beads and yellow beads.

The table gives the probabilities that, when a bead is taken at random from the bag, the bead will be blue or the bead will be yellow.

| Colour | red | blue | green | yellow |
|---|---|---|---|---|
| Probability | | 0.24 | | 0.31 |

The probability that the bead will be green is twice the probability that the bead will be red.

Sofia takes at random a bead from the bag.
She writes down the colour of the bead and puts the bead back into the bag.

She does this 180 times.

Work out an estimate for the number of times she takes a red bead from the bag.

**(Total for Question 6 is 4 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2H January 2020

### 1 Use the fact that probabilities add to 1

Let $P(\text{red}) = r$, so $P(\text{green}) = 2r$.

$$r + 0.24 + 2r + 0.31 = 1 \Rightarrow 3r = 0.45 \Rightarrow r = 0.15$$

### 2 Expected number of red beads

$$180 \times 0.15 = 27$$

**FINAL ANSWER**

27

4MA1/1H JANUARY 2020

# Question 18

4 marks | Probability Toolkit

**18** There are 16 sweets in a bowl.

4 of the sweets are blackcurrant.
5 of the sweets are lemon.
7 of the sweets are orange.

Anna, Ravi and Sam each take at random one sweet from the bowl.

Work out the probability that the 5 lemon sweets are still in the bowl.

**(Total for Question 18 is 4 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1H January 2020

### 1 Count the sweets that are not lemon

There are $16$ sweets and $5$ are lemon, so $11$ are not lemon.

### 2 Three picks without replacement, none lemon

All $5$ lemon sweets stay in the bowl only if Anna, Ravi and Sam each take a non-lemon sweet:

$$\frac{11}{16} \times \frac{10}{15} \times \frac{9}{14} = \frac{990}{3360} = \frac{33}{112}$$

**FINAL ANSWER**

$$\frac{33}{112}$$

4MA1/1H MAY/NOVEMBER 2020

# Question 2

4 marks | Probability Toolkit

2 Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

**(Total for Question 2 is 4 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1H May/November 2020

**1 Seconds in a working day**

$$15 \times 60 \times 60 = 54\,000 \text{ seconds}$$

**2 Toy cars made each day**

$$\frac{54\,000}{12} \times 5 = 22\,500$$

**3 Expected number of faulty cars**

$$22\,500 \times 0.002 = 45$$

✓ **FINAL ANSWER**

45

4MA1/1HR MAY/NOVEMBER 2020

# Question 23

4 marks | Probability Toolkit

**23** In a bag, there are only

3 blue beads
4 white beads
and $x$ orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.
Show clear algebraic working.

**(Total for Question 23 is 4 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/1HR May/November 2020

### 1 The total number of beads

The total number of beads is

$$x + 7$$

### 2 The probability of choosing two

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2} + \binom{4}{2} + \binom{x}{2}}{\binom{x+7}{2}}$$

**3** **Evaluate fraction**

This is equal to $\frac{3}{8}$, so

$$\frac{3+6+\frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}}=\frac{3}{8}$$

$$\frac{x^2-x+18}{(x+7)(x+6)}=\frac{3}{8}$$

$$8(x^2-x+18)=3(x+7)(x+6)$$

$$8x^2-8x+144=3x^2+39x+126$$

$$5x^2-47x+18=0$$

$$(x-9)(5x-2)=0$$

**4** **Calculate value**

Since $x$ must be a whole number,

$$x=9$$

**5** **Calculate value**

So the total number of beads is

$$9+7=16$$

**FINAL ANSWER**

$16$ beads

4MA1/2HR JANUARY 2021

# Question 21

6 marks | Probability Toolkit

**21** A bag contains $n$ beads.
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of $n$.

$n =$ .....

**(Total for Question 21 is 6 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2HR January 2021

**1 There are beads altogether**

There are $n$ beads altogether.

**2 Red beads**

Red beads:

$$6$$

**3 Blue beads**

Blue beads:

$$n - 6$$

**4 The probability of taking two**

The probability of taking two beads of the same colour is

$$P(RR) + P(BB) = \frac{9}{17}$$

**5** **Use trigonometry**

So

$$\frac{6}{n} \cdot \frac{5}{n-1} + \frac{n-6}{n} \cdot \frac{n-7}{n-1} = \frac{9}{17}$$

$$\frac{30 + (n-6)(n-7)}{n(n-1)} = \frac{9}{17}$$

**6** **Expand**

Expand:

$$(n-6)(n-7) = n^2 - 13n + 42$$

**7** **Use trigonometry**

So

$$\frac{n^2 - 13n + 72}{n(n-1)} = \frac{9}{17}$$

**8** **Cross multiply**

Cross multiply:

$$17(n^2 - 13n + 72) = 9n(n-1)$$

$$17n^2 - 221n + 1224 = 9n^2 - 9n$$

$$8n^2 - 212n + 1224 = 0$$

**9** **Divide by 4**

Divide by 4:

$$2n^2 - 53n + 306 = 0$$

**10** **Factor**

Factor:

$$(2n - 17)(n - 18) = 0$$

**11** **Use trigonometry**

So

$$n = \frac{17}{2} \quad \text{or} \quad n = 18$$

12 **The number of beads must**

The number of beads must be a whole number, so

$$n = 18$$

---

✓ **FINAL ANSWER**

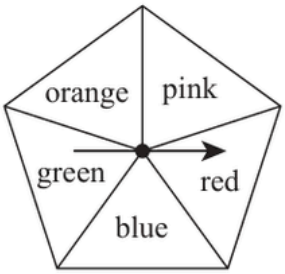
$\boxed{18}$

4MA1/1H MAY 2021

# Question 6

5 marks | Probability Toolkit

**6** Grace has a biased 5-sided spinner.




Grace is going to spin the arrow on the spinner once.

The table below gives the probabilities that the spinner will land on red or on blue or on green.

| **Colour** | Red | Blue | Green | Orange | Pink |
|---|---|---|---|---|---|
| **Probability** | 0.20 | 0.12 | 0.08 | | |

The probability that the spinner will land on orange is 3 times the probability that the spinner will land on pink.

(a) Work out the probability that the spinner will land on orange.

.....

(3)

Grace spins the arrow on the spinner 150 times.

(b) Work out an estimate for the number of times the spinner lands on blue.

.....

(2)

**(Total for Question 6 is 5 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/1H May 2021

**1** **(a) Set up with the missing probabilities**

Let $P(\text{pink}) = x$, so $P(\text{orange}) = 3x$.

$$0.20 + 0.12 + 0.08 + 3x + x = 1 \Rightarrow 4x = 0.60 \Rightarrow x = 0.15$$

**2** **(a) Probability of orange**

$$3x = 0.45$$

### 3 (b) Estimate for blue

$$150 \times 0.12 = 18$$

---

**FINAL ANSWER**

(a) $0.45$ (b) $18$

4MA1/2H MAY 2021

# Question 23

5 marks | Probability Toolkit

**23** Pippa has a box containing $N$ pens.

There are only black pens and red pens in the box.
The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of $N$.
Show clear algebraic working.

**(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/2H May 2021

**1 Let the number of red**

Let the number of red pens be $r$.

**2 List probability cases**

Then the number of black pens is

$$r + 3$$

**3 Calculate value**

and

$$N = 2r + 3$$

**4 Evaluate fraction**

So

$$r = \frac{N - 3}{2}$$

**5** **Evaluate fraction**

and

$$r+3=\frac{N+3}{2}$$

**6** **The probability of black followed**

The probability of black followed by red is

$$\frac{(N+3)/2}{N}\cdot\frac{(N-3)/2}{N-1}=\frac{9}{35}$$

$$\frac{N^2-9}{4N(N-1)}=\frac{9}{35}$$

$$35(N^2-9)=36N(N-1)$$

$$35N^2-315=36N^2-36N$$

$$N^2-36N+315=0$$

$$(N-15)(N-21)=0$$

✓ **FINAL ANSWER**

$N=15$ or $N=21$

4MA1/1H NOVEMBER 2021

# Question 3

4 marks | Probability Toolkit

3 There are 54 fish in a tank.
Some of the fish are white and the rest of the fish are red.

Jeevan takes at random a fish from the tank.
The probability that he takes a white fish is $\frac{4}{9}$

(a) Work out the number of white fish originally in the tank.

.....

(2)

Jeevan puts the fish he took out, back into the tank.
He puts some more white fish into the tank.

Jeevan takes at random a fish from the tank.
The probability that he takes a white fish is now $\frac{1}{2}$

(b) Work out the number of white fish Jeevan put into the tank.

.....

(2)

**(Total for Question 3 is 4 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1H November 2021

### 1 Calculate probability

(a) Let the number of white fish be $w$.

$$\frac{w}{54} = \frac{4}{9}$$

$$w = 54 \times \frac{4}{9} = 24$$

### 2 (b) There are still 54

(b) There are still 54 fish, and now

$$\frac{\text{white fish}}{54} = \frac{1}{2}$$

$$\text{white fish} = 27$$

**3** **Jeevan added**

Jeevan added

$$27 - 24 = 3$$

---

✓ **FINAL ANSWER**

(a) $24$ white fish, (b) $3$ white fish.

4MA1/2HR JANUARY 2022

# Question 1

4 marks | Probability Toolkit

**1** A tin contains tea bags with a choice of four different flavours of tea.
The four flavours of tea are Assam or Darjeeling or Nilgiri or Rize.

Sara takes at random a tea bag from the tin.

The table shows each of the probabilities that the flavour of the tea Sara takes is Assam or Darjeeling or Rize.

| **Flavour of tea** | Assam | Darjeeling | Nilgiri | Rize |
|---|---|---|---|---|
| **Probability** | 0.38 | 0.24 | | 0.16 |

(a) Work out the probability that the flavour of the tea Sara takes is Nilgiri.

.....

**(2)**

(b) Work out the probability that the flavour of the tea Sara takes is either Darjeeling or Rize.

.....

**(2)**

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/2HR January 2022

**1** **(a) Probabilities add to 1**

$$P(\text{Nilgiri}) = 1 - (0.38 + 0.24 + 0.16) = 1 - 0.78 = 0.22$$

**2** **(b) Darjeeling or Rize**

$$0.24 + 0.16 = 0.40$$

✓ **FINAL ANSWER**

(a) $0.22$ (b) $0.4$

4MA1/1HR JANUARY 2022

# Question 2

2 marks | Probability Toolkit

2 Each time John plays a game, the probability that he wins the game is 0.65

John is going to play the game 300 times.

Work out an estimate for the number of games that John wins.

**(Total for Question 2 is 2 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1HR January 2022

**1** **Expected number = probability × number of games**

$$0.65 \times 300 = 195$$

**FINAL ANSWER**

195

4MA1/2H JANUARY 2022

# Question 6

5 marks | Probability Toolkit

**6** A bag contains only pink sweets, white sweets, green sweets and red sweets.

The table gives each of the probabilities that, when a sweet is taken at random from the bag, the sweet will be green or the sweet will be red.

| **Sweet** | pink | white | green | red |
|---|---|---|---|---|
| **Probability** | | | 0.2 | 0.35 |

The ratio
number of pink sweets : number of white sweets = 2 : 1

There are 28 red sweets in the bag.

Work out the number of white sweets in the bag.

**(Total for Question 6 is 5 marks)**

## WORKED SOLUTION

Question 6 · 4MA1/2H January 2022

**1** **Split the ratio**

The probabilities for green and red are

$$0.20 + 0.35 = 0.55$$

**2** **Split the ratio**

So

$$P(\text{pink or white}) = 1 - 0.55 = 0.45$$

**3** **Split the ratio**

Pink : white $= 2 : 1$, so white is $\frac{1}{3}$ of $0.45$.

$$P(\text{white}) = \frac{1}{3} \times 0.45 = 0.15$$

**4** **Split the ratio**

The red probability is $0.35$, and there are $28$ red sweets.

$$\text{total sweets} = \frac{28}{0.35} = 80$$

**5** **Number of white sweets**

Number of white sweets:

$$0.15 \times 80 = 12$$

---

✓ **FINAL ANSWER**

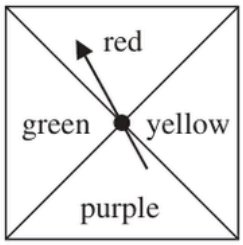
$12$

4MA1/1HR JUNE 2022

# Question 1

4 marks | Probability Toolkit

**1** Here is a biased spinner.




When the spinner is spun once, the probabilities that it lands on red or on yellow or on green are given in the table.

| **Colour** | red | yellow | purple | green |
|---|---|---|---|---|
| **Probability** | 0.25 | 0.2 | | 0.2 |

(a) Work out the probability that the spinner lands on red or on yellow.

.....

(1)

Yang is going to spin the spinner 300 times.

(b) Work out an estimate for the number of times the spinner will land on purple.

.....

(3)

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1HR June 2022

**1** **(a) Red or yellow**

$$0.25 + 0.2 = 0.45$$

**2** **(b) Probability of purple**

$$1 - (0.25 + 0.2 + 0.2) = 0.35$$

**3** **(b) Estimate for 300 spins**

$$300 \times 0.35 = 105$$

**FINAL ANSWER**

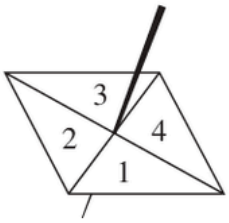
(a) $0.45$ (b) $105$

4MA1/1H JUNE 2022

# Question 2

4 marks | Probability Toolkit

**2** Here is a biased 4-sided spinner.




The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

| **Number** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | 0.26 | | 0.18 | |

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

**(Total for Question 2 is 4 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1H June 2022

**1** **Probabilities of landing on 2 and on 4**

The probabilities add to $1$, and $P(2) = P(4)$:

$$1 - (0.26 + 0.18) = 0.56, \qquad P(4) = 0.56 \div 2 = 0.28$$

**2** **Number of spins**

An estimate of $45$ for the number 3 means

$$0.18 \times n = 45 \Rightarrow n = 250$$

**3** **Estimate for landing on 4**

$$250 \times 0.28 = 70$$

**FINAL ANSWER**

70

4MA1/1H JUNE 2022

# Question 15

4 marks | Probability Toolkit

**15** Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.
When Abraham draws a game, he scores 0 points.
When Abraham loses a game, he scores −5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

**(Total for Question 15 is 4 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/1H June 2022

**1 Ways to score a total of 0**

Three draws ($0+0+0$), or one win and two losses ($10-5-5=0$) in any order.

**2 Three draws**

$$0.5^3 = 0.125$$

**3 One win and two losses (3 orders)**

$$3 \times 0.3 \times 0.2 \times 0.2 = 0.036$$

**4 Add the cases**

$$0.125 + 0.036 = 0.161$$

**FINAL ANSWER**

$0.161$

4MA1/1HR JUNE 2022

# Question 24

5 marks | Probability Toolkit

**24** Elliot has $x$ counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.

The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of $x$
Show clear algebraic working.

$x =$ .....

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Question 24 · 4MA1/1HR June 2022

**1 There are counters. Initially 5**

There are $x$ counters. Initially 5 show red, so $x - 5$ show green.

**2 List probability cases**

After two turns, there are still 5 red faces if:

**3 the same counter is picked**

- the same counter is picked twice, or
- one red counter and one green counter are picked.

**4 The probability of picking the**

The probability of picking the same counter twice is

$$\frac{1}{x}$$

**5** **The probability of picking one**

The probability of picking one red and one green in either order is

$$\frac{5}{x} \cdot \frac{x-5}{x} + \frac{x-5}{x} \cdot \frac{5}{x}$$

$$= \frac{10(x-5)}{x^2}$$

---

**6** **Evaluate fraction**

So

$$\frac{1}{x} + \frac{10(x-5)}{x^2} = \frac{19}{32}$$

$$\frac{11x-50}{x^2} = \frac{19}{32}$$

$$32(11x-50) = 19x^2$$

$$19x^2 - 352x + 1600 = 0$$

$$(x-8)(19x-200) = 0$$

---

**7** **Calculate value**

Since $x$ must be a whole number,

$$x = 8$$

---

✓ **FINAL ANSWER**

$x = 8$

4MA1/2H JUNE 2023

# Question 2

3 marks | Probability Toolkit

2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

| Colour | green | yellow | brown | pink |
|---|---|---|---|---|
| Probability | 0.32 | 0.13 | 0.28 | |

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/2H June 2023

### 1 Probability of pink

$$1 - (0.32 + 0.13 + 0.28) = 1 - 0.73 = 0.27$$

### 2 Estimate for 200 spins

$$200 \times 0.27 = 54$$

**FINAL ANSWER**

$54$

4MA1/1HR MAY 2023

# Question 5

2 marks | Probability Toolkit

5 Each time Evie plays a game against her computer, she will win or lose.

For each game, the probability that Evie will win is 0.74
Evie is going to play 300 games against her computer.

Work out an estimate for the number of games that Evie will lose.

**(Total for Question 5 is 2 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1HR May 2023

### 1 Probability of losing

$$1 - 0.74 = 0.26$$

### 2 Estimate for 300 games

$$300 \times 0.26 = 78$$

**FINAL ANSWER**

78

4MA1/2HR JUNE 2023

# Question 15

3 marks | Probability Toolkit

15 Osvaldo has a biased coin.
He spins the coin three times.

The probability that the coin lands on a head three times is $\frac{27}{64}$

Work out the probability that the coin will land on a tail three times.

**(Total for Question 15 is 3 marks)**

## WORKED SOLUTION

Question 15 · 4MA1/2HR June 2023

### 1 Find the probability of a head

$$P(H)^3 = \frac{27}{64} \Rightarrow P(H) = \sqrt[3]{\frac{27}{64}} = \frac{3}{4}$$

### 2 Three tails

$P(T) = 1 - \frac{3}{4} = \frac{1}{4}$, so

$$P(TTT) = \left(\frac{1}{4}\right)^3 = \frac{1}{64}$$

**FINAL ANSWER**

$$\frac{1}{64}$$

4MA1/2HR JUNE 2023

# Question 19

4 marks | Probability Toolkit

**19** 30 adults booked to stay in a hotel.

19 adults booked breakfast
15 adults booked dinner
4 adults did not book breakfast or dinner

Some adults booked breakfast **and** dinner.

Meihui chooses at random two of the 30 adults.

Work out the probability that these two adults each booked breakfast **and** dinner.

**(Total for Question 19 is 4 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/2HR June 2023

**1** **Adults who booked breakfast and dinner**

$30 - 4 = 26$ adults booked at least one meal.

$$n(B \cap D) = 19 + 15 - 26 = 8$$

**2** **Choose two of them (without replacement)**

$$\frac{8}{30} \times \frac{7}{29} = \frac{56}{870} = \frac{28}{435}$$

✓ **FINAL ANSWER**

$$\frac{28}{435}$$

4MA1/1H NOVEMBER 2023

# Question 4

4 marks | Probability Toolkit

**4** A biased spinner has three sections each of a different colour.

The table shows the probability that, when the spinner is spun once, it will land on blue or on orange or on white.

| **Colour** | blue | orange | white |
|---|---|---|---|
| **Probability** | 0.58 | $2x$ | $x$ |

(a) Work out the value of $x$

$x =$ .....

(2)

The spinner is spun 250 times.

(b) Work out an estimate for the number of times the spinner will land on blue.

.....

(2)

**(Total for Question 4 is 4 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1H November 2023

**1** **(a) Probabilities add to 1**

$$0.58 + 2x + x = 1 \Rightarrow 3x = 0.42 \Rightarrow x = 0.14$$

**2** **(b) Estimate for blue in 250 spins**

$$250 \times 0.58 = 145$$

✓ **FINAL ANSWER**

(a) $x = 0.14$ (b) $145$

4MA1/1H NOVEMBER 2023

# Question 20

4 marks | Probability Toolkit

**20** A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

**(Total for Question 20 is 4 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/1H November 2023

**1 Let the probability of taking**

Let the probability of taking a 10 cent coin be $p$.

**2 The mean is 10 cents**

The mean is 10 cents only when both coins are 10 cent coins.

$$p^2 = \frac{49}{121}$$

$$p = \frac{7}{11}$$

**3 Calculate probability**

So the probability of taking a 20 cent coin is

$$1 - \frac{7}{11} = \frac{4}{11}$$

**4 The mean is 15 cents**

The mean is 15 cents when one coin is 10 cents and the other is 20 cents.

**5** **There are two possible orders**

There are two possible orders:

$$P(10, 20) + P(20, 10)$$

$$= \frac{7}{11} \cdot \frac{4}{11} + \frac{4}{11} \cdot \frac{7}{11}$$

$$= \frac{56}{121}$$

---

✓ **FINAL ANSWER**

$\frac{56}{121}$

4MA1/1H MAY 2024

# Question 2

4 marks | Probability Toolkit

2 450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.
The table shows information about the probability of each method of travel.

| **Method of travel** | walk | bus | car | bicycle |
|---|---|---|---|---|
| **Probability** | 0.20 | $x$ | $2x$ | 0.26 |

Work out how many of the 450 students travelled by car.

**(Total for Question 2 is 4 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1H May 2024

**1 Find x**

$$0.20 + x + 2x + 0.26 = 1 \Rightarrow 3x = 0.54 \Rightarrow x = 0.18$$

**2 Probability of travelling by car**

$$2x = 0.36$$

**3 Number of students**

$$450 \times 0.36 = 162$$

✓ **FINAL ANSWER**

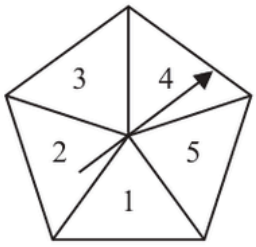
162

4MA1/1HR MAY 2024

# Question 2

4 marks | Probability Toolkit

**2** Here is a biased spinner.




The table gives information about the probability that, when the spinner is spun once, it will land on each number.

| **Number** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Probability** | $2x$ | 0.27 | 0.04 | $x$ | 0.12 |

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

**(Total for Question 2 is 4 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1HR May 2024

**1 Find x**

$$2x + 0.27 + 0.04 + x + 0.12 = 1 \Rightarrow 3x = 0.57 \Rightarrow x = 0.19$$

**2 Probability of an odd number (1, 3 or 5)**

$$2x + 0.04 + 0.12 = 0.38 + 0.04 + 0.12 = 0.54$$

**3 Estimate for 400 spins**

$$400 \times 0.54 = 216$$

**FINAL ANSWER**

$216$

4MA1/2HR JUNE 2024

# Question 14

3 marks | Probability Toolkit

**14** Adriana is going to roll a biased dice and spin a biased coin.

The probability that the coin will land on Heads is 0.8
The probability that the dice will land on 6 and the coin will land on Heads is 0.24

Work out the probability that the dice will land on 6 and the coin will land on Tails.

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2HR June 2024

### 1 Find the probability of a 6

The dice and the coin are independent:

$$P(6) \times 0.8 = 0.24 \Rightarrow P(6) = 0.3$$

### 2 6 and Tails

$P(\text{Tails}) = 1 - 0.8 = 0.2$, so

$$0.3 \times 0.2 = 0.06$$

**FINAL ANSWER**

$0.06$

4MA1/1HR MAY 2024

# Question 21

5 marks | Probability Toolkit

**21** There are 25 counters in a bag such that

6 counters are blue
$x$ counters are orange, where $x > 9$
the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.
Show clear algebraic working.

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1HR May 2024

**1** **There are 25 counters**

There are 25 counters.

**2** **Blue counters**

Blue counters:

$$6$$

**3** **Orange counters**

Orange counters:

$$x$$

**4** **Pink counters**

Pink counters:

$$25 - 6 - x = 19 - x$$

**5** **The probability of taking one**

The probability of taking one orange and one pink in either order is

$$\frac{x}{25} \cdot \frac{19-x}{24} + \frac{19-x}{25} \cdot \frac{x}{24}$$

$$= \frac{2x(19-x)}{25 \cdot 24}$$

**6** **Evaluate fraction**

This is equal to $\frac{22}{75}$, so

$$\frac{2x(19-x)}{600} = \frac{22}{75}$$

$$\frac{x(19-x)}{300} = \frac{22}{75}$$

$$x(19-x) = 88$$

$$19x - x^2 = 88$$

$$x^2 - 19x + 88 = 0$$

$$(x-8)(x-11) = 0$$

**7** **Solve inequality**

Since $x > 9$,

$$x = 11$$

**8** **Calculate value**

So the number of pink counters is

$$19 - 11 = 8$$

**9** **The probability of taking 2**

The probability of taking 2 pink counters is

$$\frac{8}{25} \cdot \frac{7}{24} = \frac{56}{600}$$

$$= \frac{7}{75}$$

**FINAL ANSWER**

$$\frac{7}{75}$$

4MA1/2H NOVEMBER 2024

# Question 5

4 marks | Probability Toolkit

5 Cody has a biased 5-sided spinner, numbered 1, 2, 3, 4, 5

The table gives the probabilities that when the spinner is spun it will land on 2 or on 3 or on 5

| Number | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Probability | | 0.14 | 0.17 | | 0.21 |

The probability that the spinner will land on 1 is the same as the probability that the spinner will land on 4

Cody is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on 4

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/2H November 2024

**1 Probabilities of 1 and 4 together**

$$1 - (0.14 + 0.17 + 0.21) = 1 - 0.52 = 0.48$$

**2 They are equal**

$$P(4) = 0.48 \div 2 = 0.24$$

**3 Estimate for 400 spins**

$$400 \times 0.24 = 96$$

✓ **FINAL ANSWER**

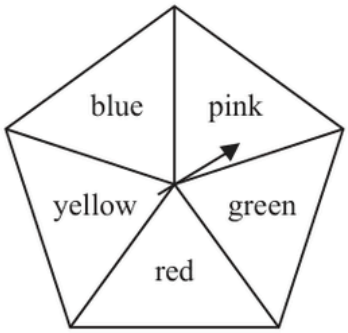
96

4WM1/01 MAY 2025

# Question 3

4 marks | Probability Toolkit

3 Here is a biased spinner.
The spinner can land on pink or on green or on red or on yellow or on blue.




The table gives information about the probability that, when the spinner is spun, it will land on each colour.

| **Colour** | pink | green | red | yellow | blue |
|---|---|---|---|---|---|
| **Probability** | 0.14 | 0.23 | $2x$ | 0.18 | $7x$ |

Jim is going to spin the spinner 500 times.

Work out the expected number of times the spinner will land on blue.

**(Total for Question 3 is 4 marks)**

## WORKED SOLUTION

Question 3 · 4WM1/01 May 2025

**1 Calculate probability**

The probabilities add to $1$.

$$0.14 + 0.23 + 2x + 0.18 + 7x = 1$$

$$0.55 + 9x = 1$$

$$x = 0.05$$

**2 The probability of blue is**

The probability of blue is

$$7x = 7(0.05) = 0.35$$

### 3 Expected number of blue results

Expected number of blue results:

$$500 \times 0.35 = 175$$

---

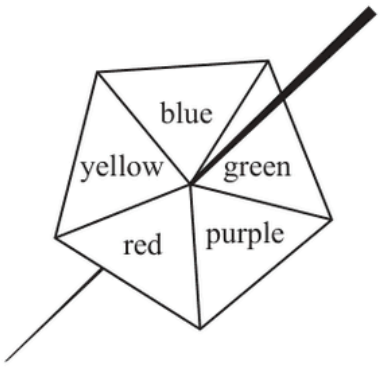
**FINAL ANSWER**

$175$

# Question 3

5 marks | Probability Toolkit

**3** Here is a biased 5-sided spinner.
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.




The table gives information about the probability of the spinner landing on each colour.

| **Colour** | blue | green | purple | red | yellow |
|---|---|---|---|---|---|
| **Probability** | 0.12 | 0.20 | 0.38 | $4x$ | $x$ |

Sophie spins the spinner once.

(a) Work out the probability that the spinner lands on blue or on green or on purple.

.....

**(1)**

Max spins the spinner 350 times.

(b) Work out an estimate for the number of times the spinner lands on red.

.....

**(4)**

**(Total for Question 3 is 5 marks)**

## 1 Let the yellow probability be

Let the yellow probability be $x$. Then the red probability is $4x$.

$$0.12 + 0.20 + 0.30 + 4x + x = 1$$

$$5x = 0.38$$

$$x = 0.076$$

## 2 Calculate probability

So

$$P(\text{red}) = 4x = 0.304$$

$$P(\text{blue, green or purple}) = 0.12 + 0.20 + 0.30 = 0.62$$

## 3 Expected red

Expected red:

$$350 \times 0.304 = 106.4$$

**FINAL ANSWER**

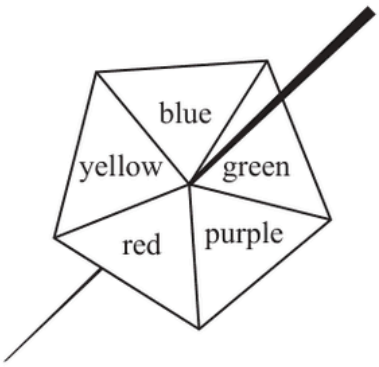
$0.62$, and about $106$.

4MA1/1HR JUNE 2025

# Question 4

5 marks | Probability Toolkit

**4** Here is a biased 5-sided spinner.
When the spinner is spun, it can land on blue or on green or on purple or on red or on yellow.




The table gives information about the probability of the spinner landing on each colour.

| **Colour** | blue | green | purple | red | yellow |
|---|---|---|---|---|---|
| **Probability** | 0.12 | 0.20 | 0.38 | $4x$ | $x$ |

Sophie spins the spinner once.

(a) Work out the probability that the spinner lands on blue or on green or on purple.

.....

**(1)**

Max spins the spinner 350 times.

(b) Work out an estimate for the number of times the spinner lands on red.

.....

**(4)**

**(Total for Question 4 is 5 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1HR June 2025

**1** **(a) Blue, green or purple**

$$0.12 + 0.20 + 0.38 = 0.70$$

**2** **(b) Find x**

Red and yellow share the rest: $4x + x = 1 - 0.70 = 0.30$, so $5x = 0.30$ and $x = 0.06$.

### 3 (b) Probability of red

$$4x = 4 \times 0.06 = 0.24$$

### 4 (b) Estimate for 350 spins

$$350 \times 0.24 = 84$$

**FINAL ANSWER**

(a) $0.7$ (b) $84$

4MA1/2H JUNE 2025

# Question 11

3 marks | Probability Toolkit

**11** Khalid has a box of counters.

17 of the counters are red
28 of the counters are blue
the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

**(Total for Question 11 is 3 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2H June 2025

**1 Counters that are not orange**

$17 + 28 = 45$ counters are not orange, and $P(\text{not orange}) = 1 - \frac{4}{9} = \frac{5}{9}$.

**2 Total number of counters**

$$\frac{5}{9} \times T = 45 \Rightarrow T = 81$$

**3 Orange counters**

$$81 - 45 = 36$$

**FINAL ANSWER**

$36$

4WM1/01R MAY 2025

# Question 17

3 marks | Probability Toolkit

**17** Kannika has 9 counters.
There is a number on each counter.

Kannika puts the 9 counters in a bag.
She takes at random a counter from the bag and does not replace the counter.
She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

**(Total for Question 17 is 3 marks)**

## WORKED SOLUTION

Question 17 · 4WM1/01R May 2025

**1 The counters are**

The counters are

$$1,\ 1,\ 2,\ 3,\ 3,\ 3,\ 5,\ 6,\ 6$$

**2 There are**

There are

$$\binom{9}{2} = 36$$

**3 ways to choose two counters**

ways to choose two counters.

**4 Calculate probability**

For a sum less than $5$, the pairs are:

$$(1,1),\quad (1,2),\quad (1,3)$$

**5 Counting the repeated counters gives**

Counting the repeated counters gives

$$1+2+6=9$$

### 6 Calculate probability

So

$$P(\text{sum} < 5) = \frac{9}{36} = \frac{1}{4}$$

---

**FINAL ANSWER**

$\frac{1}{4}$.

4MA1/1HR JUNE 2025

# Question 19

3 marks | Probability Toolkit

**19** Kannika has 9 counters.
There is a number on each counter.

Kannika puts the 9 counters in a bag.
She takes at random a counter from the bag and does not replace the counter.
She then takes at random a second counter from the bag.

Work out the probability that the sum of the numbers on the two counters is less than 5

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1HR June 2025

### 1 Pairs with a sum less than 5

The counters are $1, 1, 2, 3, 3, 3, 5, 6, 6$. The only possible sums under $5$ are $1+1$, $1+2$ and $1+3$.

### 2 First counter, then second (no replacement)

$$P(1,1) = \frac{2}{9} \times \frac{1}{8} = \frac{2}{72}$$

$$P(1 \text{ and } 2) = 2 \times \frac{2}{9} \times \frac{1}{8} = \frac{4}{72}, \qquad P(1 \text{ and } 3) = 2 \times \frac{2}{9} \times \frac{3}{8} = \frac{12}{72}$$

### 3 Add the cases

$$\frac{2+4+12}{72} = \frac{18}{72} = \frac{1}{4}$$

✓ **FINAL ANSWER**

$$\frac{1}{4}$$

4MA1/1H JUNE 2025

# Question 21

3 marks | Probability Toolkit

**21** A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

**(Total for Question 21 is 3 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1H June 2025

**1 There are**

There are

$$\binom{20}{3} = 1140$$

**2 ways to choose 3 counters**

ways to choose 3 counters.

**3 Exactly two counters are the**

Exactly two counters are the same colour:

$$\binom{9}{2}(11) + \binom{7}{2}(13) + \binom{4}{2}(16)$$

$$= 36(11) + 21(13) + 6(16)$$

$$= 396 + 273 + 96 = 765$$

**4 Calculate probability**

So the probability is

$$\frac{765}{1140} = \frac{51}{76}$$

**FINAL ANSWER**

$$\frac{51}{76}$$

4WM1/01 MAY 2025

# Question 22

3 marks | Probability Toolkit

**22** A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

**(Total for Question 22 is 3 marks)**

## WORKED SOLUTION

Question 22 · 4WM1/01 May 2025

### 1 Total ways to choose counters

Total ways to choose $3$ counters:

$$\binom{20}{3} = 1140$$

### 2 Exactly two red

Exactly two red:

$$\binom{9}{2}\binom{11}{1} = 396$$

### 3 Exactly two yellow

Exactly two yellow:

$$\binom{7}{2}\binom{13}{1} = 273$$

### 4 Exactly two green

Exactly two green:

$$\binom{4}{2}\binom{16}{1} = 96$$

### 5 Favourable outcomes

Favourable outcomes:

$$396 + 273 + 96 = 765$$

$$P = \frac{765}{1140} = \frac{51}{76}$$

---

**FINAL ANSWER**

$\frac{51}{76}$

4WM1/01 NOVEMBER 2025

# Question 5

5 marks | Probability Toolkit

5 All students at a school are asked which sport they prefer from football, tennis, rugby and cricket.

The table gives information about the probabilities that, when a student is chosen at random, the student prefers football or tennis or rugby or cricket.

| **Favourite sport** | football | tennis | rugby | cricket |
|---|---|---|---|---|
| **Probability** | $2x$ | 0.2 | $x$ | 0.35 |

One of the students at the school is chosen at random.

(a) Work out the probability that this student prefers tennis or cricket.

.....

(1)

There are 700 students at the school.

(b) Work out how many students prefer football.

.....

(4)

**(Total for Question 5 is 5 marks)**

## WORKED SOLUTION

Question 5 · 4WM1/01 November 2025

**1** **Calculate value**

$$P(\text{tennis or cricket}) = 0.2 + 0.35 = 0.55$$

**2** **Calculate value**

Also,

$$2x + 0.2 + x + 0.35 = 1$$

$$3x = 0.45$$

$$x = 0.15$$

### 3 Calculate value

So

$$P(\text{football}) = 2x = 0.30$$

$$700 \times 0.30 = 210$$

---

**FINAL ANSWER**

$0.55$, $210$ students.

# Question 9

3 marks | Probability Toolkit

**9** Mark has three bags, bag A, bag B and bag C

Bag A contains 120 counters.
The probability of taking at random a red counter from bag A is 0.45

Bag B contains 80 counters.
The probability of taking at random a red counter from bag B is 0.3

Bag C is empty.

Mark puts all the counters from bag A and bag B into bag C

Mark takes at random a counter from bag C

Work out the probability that he takes a red counter.

.....

**(Total for Question 9 is 3 marks)**

## 1 Red counters in each bag

Bag A: $120 \times 0.45 = 54$. Bag B: $80 \times 0.3 = 24$.

## 2 Probability of red from bag C

Bag C has $54 + 24 = 78$ red counters out of $120 + 80 = 200$.

$$\frac{78}{200} = 0.39$$

**FINAL ANSWER**

$0.39$

4MA1/1HR JUNE 2026

# Question 5

4 marks | Probability Toolkit

**5** A bag contains only blue counters, green counters, red counters and yellow counters.

Danny takes a counter at random from the bag.

The table gives information about the probabilities that he takes a blue counter or a green counter or a red counter or a yellow counter.

| **Colour** | blue | green | red | yellow |
|---|---|---|---|---|
| **Probability** | 0.45 | 0.31 | $5x$ | $x$ |

There are 400 counters in the bag.

Work out the number of red counters in the bag.

.....

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4MA1/1HR June 2026

### 1 Solution | Step 1

$$0.45 + 0.31 + 5x + x = 1.$$

The four colours are mutually exclusive and exhaust all counters, so their probabilities sum to 1.

### 2 Solution | Step 2

$$6x = 1 - 0.76 = 0.24, \qquad x = 0.04.$$

Collect the unknown probabilities and solve the linear equation.

### 3 Solution | Step 3

$$P(\text{red}) = 5(0.04) = 0.20, \qquad N_{\text{red}} = 400(0.20) = 80.$$

Multiply the red proportion by the total number of counters.

**FINAL ANSWER**

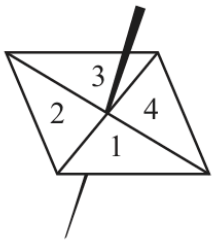
80 red counters

4MA1/1H JUNE 2026

# Question 7

4 marks | Probability Toolkit

7 Here is a biased 4-sided spinner.




The table gives information about the probability that, when the spinner is spun once, it will land on each number.

| **Number** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | 0.12 | $x$ | $2x$ | 0.16 |

Nevan spins the spinner 500 times.

Work out the expected number of times that the spinner lands on an even number.

.....

**(Total for Question 7 is 4 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1H June 2026

**1 Solution | Step 1**

$$0.12 + x + 2x + 0.16 = 1 \implies 3x = 0.72 \implies x = 0.24.$$

The four exhaustive outcome probabilities sum to 1.

**2 Solution | Step 2**

$$P(\text{even}) = P(2) + P(4) = 0.24 + 0.16 = 0.40.$$

The events 2 and 4 are mutually exclusive.

**3 Solution | Step 3**

$$\text{Expected number} = 500 \times 0.40 = 200.$$

Multiply the number of spins by the probability of an even result.

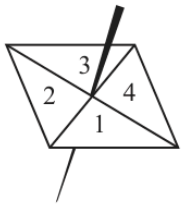
**FINAL ANSWER**

200

# Question 7

4 marks | Probability Toolkit

**7** Here is a biased 4-sided spinner.




The table gives information about the probability that, when the spinner is spun once, it will land on each number.

| **Number** | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **Probability** | 0.12 | $x$ | $2x$ | 0.16 |

Nevan spins the spinner 500 times.

Work out the expected number of times that the spinner lands on an even number.

.....

**(Total for Question 7 is 4 marks)**

**1** **Probabilities add to 1**

$$0.12 + x + 2x + 0.16 = 1 \Rightarrow 3x = 0.72 \Rightarrow x = 0.24$$

**2** **Even numbers are 2 and 4**

$$P(\text{even}) = 0.24 + 0.16 = 0.40$$

**3** **Expected number**

$$500 \times 0.40 = 200$$

✓ **FINAL ANSWER**

200

# Question 11

3 marks | Probability Toolkit

**11** There are 40 counters in a bag.

25 of the counters are green
7 of the counters are blue
8 of the counters are red

Ben puts some extra blue counters and some extra red counters into the bag.
The number of blue counters added to the bag is twice as many as the number of red counters added to the bag.

Ben takes at random a counter from the bag.

The probability that he takes a green counter is $\frac{5}{11}$

How many of the extra counters that Ben put into the bag were blue?

.....

**(Total for Question 11 is 3 marks)**

**1** **Let r red be added**

Then $2r$ blue are added and the total is $40 + 3r$.

**2** **Use P(green)**

$$\frac{25}{40+3r} = \frac{5}{11} \Rightarrow 275 = 200 + 15r \Rightarrow r = 5$$

**3** **Blue added**

$$2r = 10$$

✓ **FINAL ANSWER**

10

4MA1/2H JUNE 2026

# Question 20

3 marks | Probability Toolkit

**20** A box only contains green counters and red counters.

Nuno chooses at random a counter from the box.
He puts the counter back into the box.

Nuno then chooses at random a second counter from the box.

The probability that Nuno chooses at least one red counter is $\frac{39}{64}$

Work out the probability that Nuno chooses exactly one red counter.

.....

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H June 2026

### 1 Solution | Step 1

Let $g = P(G)$. With replacement, the draws are independent.

$$P(GG) = 1 - P(\text{at least one red}) = 1 - \frac{39}{64} = \frac{25}{64}.$$

The complementary event to at least one red is two greens.

### 2 Solution | Step 2

$$g^2 = \frac{25}{64} \quad \Rightarrow \quad g = \frac{5}{8}, \qquad P(R) = 1 - \frac{5}{8} = \frac{3}{8}.$$

Multiply identical independent green probabilities and take the nonnegative root.

### 3 Solution | Step 3

$$P(\text{exactly one red}) = P(RG) + P(GR) = \frac{3}{8}\frac{5}{8} + \frac{5}{8}\frac{3}{8} = \frac{30}{64} = \frac{15}{32}.$$

The two possible orders are mutually exclusive and must both be included.

✓ **FINAL ANSWER**

$$\frac{15}{32}$$

TOPIC

# Tree Diagrams & Conditional Probability

Every past-paper question on this topic, 2020 to 2026.

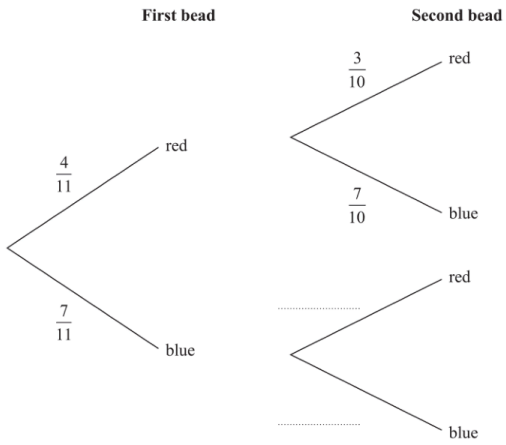
# Question 11

6 marks | Tree Diagrams & Conditional Probability

**11** Tess has a bag containing 11 beads.
4 of the beads are red.
7 of the beads are blue.

Tess takes at random a bead from the bag and keeps it.
Tess takes at random another bead from the bag and keeps it.

(a) Complete the probability tree diagram.




(1)

(b) Work out the probability that both of the beads are red.

.....

(2)

(c) Work out the probability that the two beads are of different colours.

.....

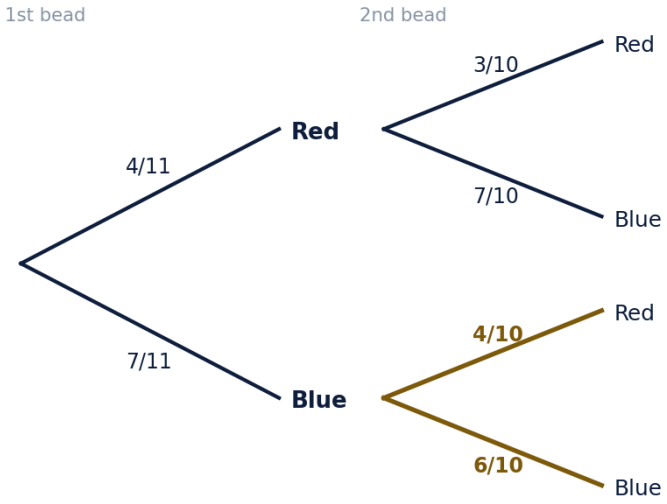
(3)

**(Total for Question 11 is 6 marks)**

## 1 Complete the blue-first branch

After a blue bead is taken first, there are $4$ red and $6$ blue beads left out of $10$.

$$P(\text{red second after blue}) = \frac{4}{10}, \qquad P(\text{blue second after blue}) = \frac{6}{10}$$




## 2 Find both red

$$P(RR) = \frac{4}{11} \times \frac{3}{10} = \frac{12}{110} = \frac{6}{55}$$

## 3 Find different colours

$$P(RB) = \frac{4}{11} \times \frac{7}{10}$$

$$P(BR) = \frac{7}{11} \times \frac{4}{10}$$

$$P(\text{different}) = \frac{28}{110} + \frac{28}{110} = \frac{28}{55}$$

**FINAL ANSWER**

Missing probabilities $\frac{4}{10}$ and $\frac{6}{10}$; $P(RR) = \frac{6}{55}$; $P(\text{different}) = \frac{28}{55}$.

4MA1/2H JANUARY 2020

# Question 14

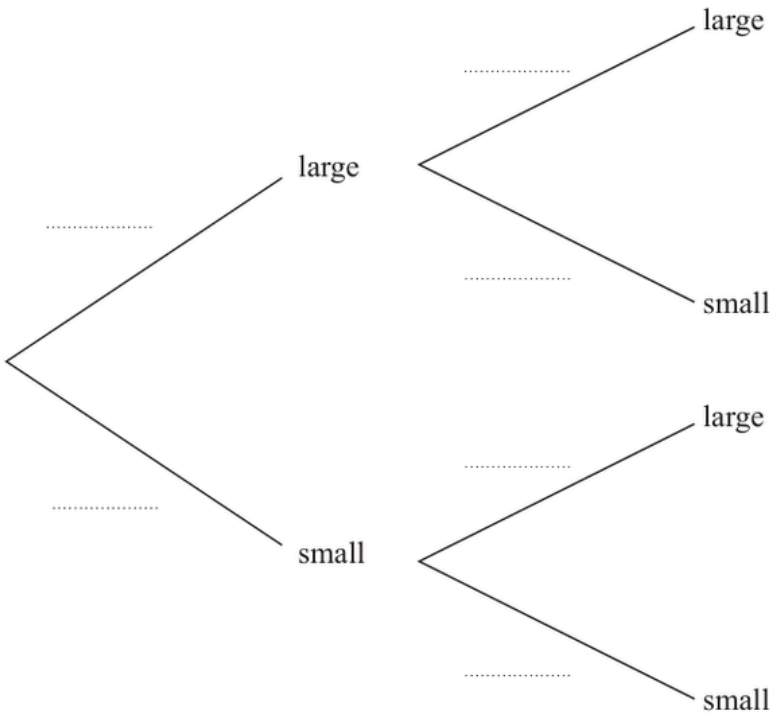
4 marks | Tree Diagrams & Conditional Probability

**14** There are 20 glasses in a cupboard.

13 of the glasses are large
7 of the glasses are small

Roberto takes at random two glasses from the cupboard.

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Roberto takes two small glasses.

.....

(2)

**(Total for Question 14 is 4 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2H January 2020

**1** **(a) First glass**

$$P(\text{large}) = \frac{13}{20}, \qquad P(\text{small}) = \frac{7}{20}$$

**2** **(a) Second glass (one fewer glass left)**

After a large glass: large $\frac{12}{19}$, small $\frac{7}{19}$.
After a small glass: large $\frac{13}{19}$, small $\frac{6}{19}$.

---

**3** **(b) Two small glasses**

$$\frac{7}{20} \times \frac{6}{19} = \frac{42}{380} = \frac{21}{190}$$

---

✓ **FINAL ANSWER**

(a) $\frac{13}{20}, \frac{7}{20}$; $\frac{12}{19}, \frac{7}{19}$; $\frac{13}{19}, \frac{6}{19}$ (b) $\frac{21}{190}$

4MA1/1HR JANUARY 2020

# Question 16

4 marks | Tree Diagrams & Conditional Probability

**16** Steffi is going to play one game of tennis and one game of chess.

The probability that she will win the game of tennis is 0.6
The probability that she will win **both** games is 0.42

Work out the probability that she will **not** win either game.

**(Total for Question 16 is 4 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1HR January 2020

### 1 Find the probability of winning chess

Treat the two games as independent:

$$0.6 \times P(\text{win chess}) = 0.42 \Rightarrow P(\text{win chess}) = 0.42 \div 0.6 = 0.7$$

### 2 Probabilities of losing

$$P(\text{not win tennis}) = 1 - 0.6 = 0.4, \qquad P(\text{not win chess}) = 1 - 0.7 = 0.3$$

### 3 Win neither game

$$0.4 \times 0.3 = 0.12$$

**FINAL ANSWER**

$0.12$

# Question 16

8 marks | Tree Diagrams & Conditional Probability

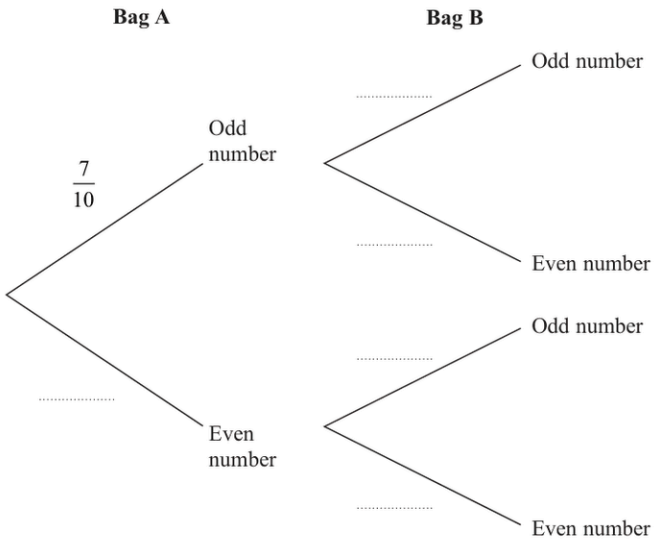
**16** Cody has two bags of counters, bag **A** and bag **B**.

Each of the counters has either an odd number or an even number written on it.

There are 10 counters in bag **A** and 7 of these counters have an **odd** number written on them.
There are 12 counters in bag **B** and 7 of these counters have an **odd** number written on them.

Cody is going to take at random a counter from bag **A** and a counter from bag **B**.

(a) Complete the probability tree diagram.




**(2)**

(b) Calculate the probability that the total of the numbers on the two counters will be an odd number.

.....

**(3)**

Harriet also has a bag of counters.
Each of her counters also has either an odd number or an even number written on it.

Harriet is going to take at random a counter from her bag of counters.

The probability that the number on each of Cody's two counters **and** the number on Harriet's counter will all be even is $\frac{3}{100}$

(c) Find the least number of counters that Harriet has in her bag.
Show your working clearly.

.....

**(3)**

**(Total for Question 16 is 8 marks)**

1 **(a) Complete the tree**

Bag A: odd $\frac{7}{10}$, even $\frac{3}{10}$.

Bag B (on both branches): odd $\frac{7}{12}$, even $\frac{5}{12}$.

2 **(b) Odd total = odd + even or even + odd**

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{35}{120} + \frac{21}{120} = \frac{56}{120} = \frac{7}{15}$$

3 **(c) Probability Harriet takes an even counter**

Cody takes two even counters with probability $\frac{3}{10} \times \frac{5}{12} = \frac{1}{8}$.

$$\frac{1}{8} \times p = \frac{3}{100} \Rightarrow p = \frac{24}{100} = \frac{6}{25}$$

4 **(c) Least number of counters**

$\frac{6}{25}$ is in its simplest form, so the number of counters must be a multiple of $25$. The least number is $25$.

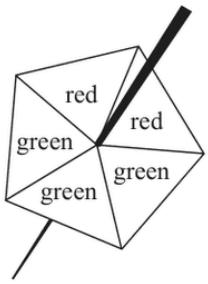
✓ **FINAL ANSWER**

(a) $\frac{3}{10}$; $\frac{7}{12}$, $\frac{5}{12}$ on both branches (b) $\frac{7}{15}$ (c) $25$

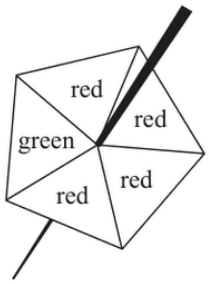
# Question 15

5 marks | Tree Diagrams & Conditional Probability

**15** Harry has two fair 5-sided spinners.



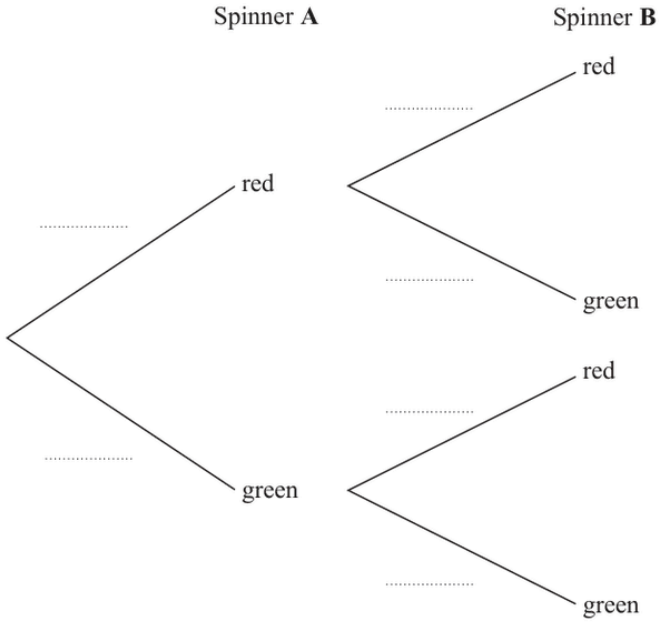

Spinner **A**




Spinner **B**

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.




**(2)**

(b) Work out the probability that at least one of the spinners will land on green.

.....

**(3)**

**(Total for Question 15 is 5 marks)**

1

## (a) Complete the tree

Spinner A has $2$ red and $3$ green sections: red $\frac{2}{5}$, green $\frac{3}{5}$.

Spinner B has $4$ red and $1$ green (on both branches): red $\frac{4}{5}$, green $\frac{1}{5}$.

2

## (b) Use the complement

"At least one green" is the opposite of "both red":

$$1 - \frac{2}{5} \times \frac{4}{5} = 1 - \frac{8}{25} = \frac{17}{25}$$

**FINAL ANSWER**

(a) $\frac{2}{5}, \frac{3}{5}$; $\frac{4}{5}, \frac{1}{5}$ on both branches (b) $\frac{17}{25}$

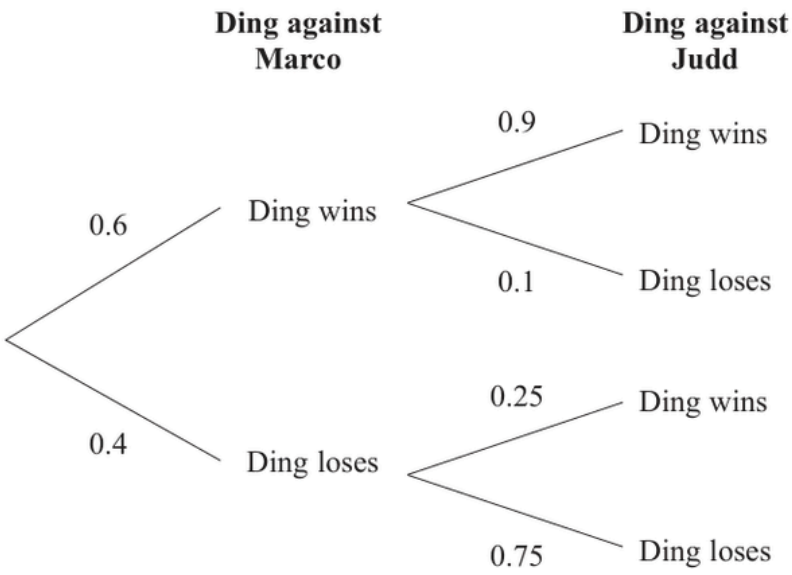
4MA1/1H JANUARY 2021

# Question 14

5 marks | Tree Diagrams & Conditional Probability

**14** Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.




(a) Work out the probability that Ding will win both games.

.....

(2)

(b) Work out the probability that Ding will win exactly one of the games.

.....

(3)

**(Total for Question 14 is 5 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1H January 2021

**1** **(a) Win both games**

Multiply along the "win, win" branch:

$$0.6 \times 0.9 = 0.54$$

### 2 (b) Win exactly one game

Win then lose, or lose then win:

$$0.6 \times 0.1 + 0.4 \times 0.25 = 0.06 + 0.10 = 0.16$$

---

**FINAL ANSWER**

(a) $0.54$ (b) $0.16$

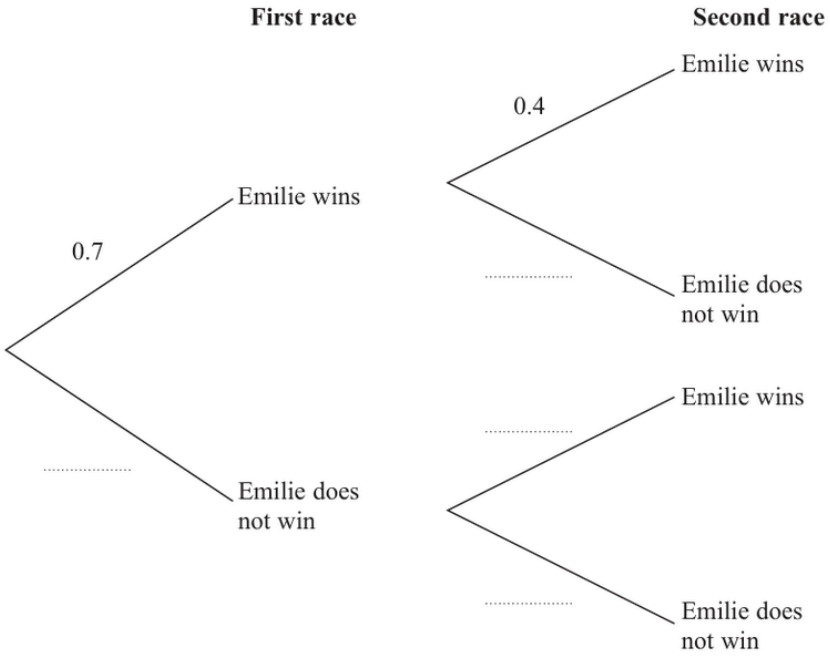
# Question 13

8 marks | Tree Diagrams & Conditional Probability

**13** Emilie takes part in two races.

The probability that she wins the first race is 0.7
The probability that she wins the second race is 0.4
The outcomes of the two races are independent.

(a) Complete the probability tree diagram.




**(2)**

(b) Work out the probability that Emilie wins exactly one of the two races.

.....

**(3)**

Emilie is going to take part in a third race.

If she wins both of the first two races, the probability that she will win the third race is 0.6

If she wins exactly one of the first two races, the probability that she will win the third race is 0.3

(c) Work out the probability that Emilie will win exactly two of the three races.

.....

**(3)**

**(Total for Question 13 is 8 marks)**

**1** **(a) Complete the tree**

First race: does not win $0.3$.

Second race: after a win, does not win $0.6$; after not winning, wins $0.4$ and does not win $0.6$.

**2** **(b) Exactly one win**

$$0.7 \times 0.6 + 0.3 \times 0.4 = 0.42 + 0.12 = 0.54$$

**3** **(c) Win the first two, lose the third**

$P(\text{win both}) = 0.7 \times 0.4 = 0.28$, then lose the third with $1 - 0.6 = 0.4$:

$$0.28 \times 0.4 = 0.112$$

**4** **(c) Exactly one of the first two, then win the third**

$$0.54 \times 0.3 = 0.162$$

**5** **(c) Add the cases**

$$0.112 + 0.162 = 0.274$$

✓ **FINAL ANSWER**

(a) $0.3$; $0.6$; $0.4, 0.6$ (b) $0.54$ (c) $0.274$

# Question 15

8 marks | Tree Diagrams & Conditional Probability

**15** Magnus and Garry play 2 games of chess against each other.

The probability that Magnus beats Garry in any game is $\frac{2}{9}$

The probability that any game between Magnus and Garry is drawn is $\frac{4}{9}$

The result of any game is independent of the result of any other game.

(a) Complete the probability tree diagram.

**(2)**

For each game of chess,

the winner gets 2 points and the loser gets 0 points,
when the game is drawn, each player gets 1 point.

(b) Work out the probability that, after 2 games, Magnus and Garry have the same number of points.

.....

**(3)**

Magnus and Garry now play a third game of chess.

(c) Work out the probability that, after 3 games, Magnus and Garry have the same number of points.

.....

**(3)**

**(Total for Question 15 is 8 marks)**

1 **(a) Complete the tree**

Magnus loses: $1 - \frac{2}{9} - \frac{4}{9} = \frac{3}{9}$.

Every second-game branch repeats $\frac{2}{9}$, $\frac{4}{9}$, $\frac{3}{9}$ (the games are independent).

2 **(b) Equal points after 2 games**

Two draws, or one win and one loss (either order):

$$\left(\frac{4}{9}\right)^2 + 2 \times \frac{2}{9} \times \frac{3}{9} = \frac{16}{81} + \frac{12}{81} = \frac{28}{81}$$

3 **(c) Equal points after 3 games**

Each player needs $3$ of the $6$ points: three draws, or one win, one draw and one loss ($3! = 6$ orders).

4 **(c) Add the cases**

$$\left(\frac{4}{9}\right)^3 + 6 \times \frac{2}{9} \times \frac{4}{9} \times \frac{3}{9} = \frac{64}{729} + \frac{144}{729} = \frac{208}{729}$$

✓ **FINAL ANSWER**

(a) $\frac{3}{9}$; $\frac{2}{9}$, $\frac{4}{9}$, $\frac{3}{9}$ on every branch (b) $\frac{28}{81}$ (c) $\frac{208}{729}$

4MA1/2H NOVEMBER 2021

# Question 16

3 marks | Tree Diagrams & Conditional Probability

**16** A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/2H November 2021

### 1 Count the colours

$4$ red, $5$ green and $15 - 9 = 6$ yellow. The counter is replaced, so the second pick has the same probabilities.

### 2 Both the same colour

$$\left(\frac{4}{15}\right)^2 + \left(\frac{5}{15}\right)^2 + \left(\frac{6}{15}\right)^2 = \frac{16 + 25 + 36}{225} = \frac{77}{225}$$

**FINAL ANSWER**

$$\frac{77}{225}$$

# Question 13

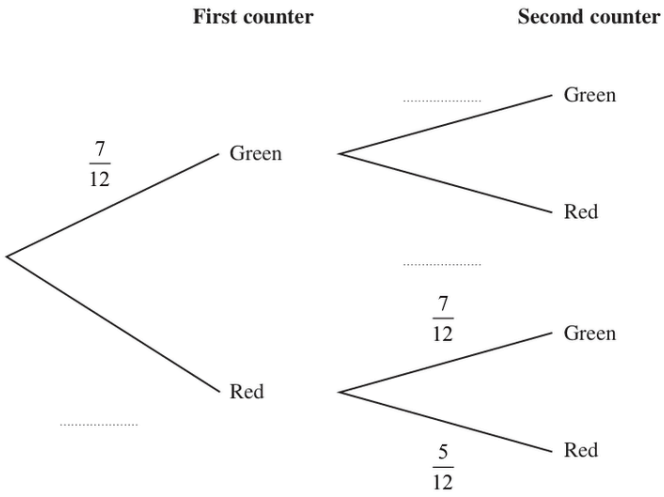
7 marks | Tree Diagrams & Conditional Probability

**13** Hector has a bag that contains 12 counters.
There are 7 green counters and 5 red counters in the bag.

Hector takes at random a counter from the bag.
He looks at the counter and puts the counter back into the bag.

Hector then takes at random a second counter from the bag.
He looks at the counter and puts the counter back into the bag.

(a) Complete the probability tree diagram.




**(2)**

(b) Work out the probability that both counters are red.

.....

**(2)**

Meghan has a jar containing 15 counters.
There are only blue counters, green counters and red counters in the jar.

Hector is going to take at random one of the counters from his bag of 12 counters.
He will look at the counter and put the counter back into the bag.

Hector is then going to take at random a second counter from his bag.
He will look at the counter and put the counter back into the bag.

Meghan is then going to take at random one of the counters from her jar of counters.
She will look at the counter and put the counter back into the jar.

The probability that the 3 counters each have a different colour is $\frac{7}{24}$

(c) Work out how many blue counters there are in the jar.

.....

**(3)**

**(Total for Question 13 is 7 marks)**

### 1 (a) Complete the tree (counter is replaced)

First counter red: $\frac{5}{12}$.

After green, second counter: green $\frac{7}{12}$, red $\frac{5}{12}$.

### 2 (b) Both red

$$\frac{5}{12} \times \frac{5}{12} = \frac{25}{144}$$

### 3 (c) What "three different colours" needs

Hector only has green and red, so he must take one green and one red (either order), and Meghan must take blue.

$$P(\text{one green, one red}) = 2 \times \frac{7}{12} \times \frac{5}{12} = \frac{35}{72}$$

### 4 (c) Solve for the number of blue counters

Let $b$ be the number of blue counters out of $15$.

$$\frac{35}{72} \times \frac{b}{15} = \frac{7}{24} \Rightarrow \frac{b}{15} = \frac{7}{24} \times \frac{72}{35} = \frac{3}{5} \Rightarrow b = 9$$

**FINAL ANSWER**

(a) $\frac{5}{12}$; $\frac{7}{12}$, $\frac{5}{12}$ (b) $\frac{25}{144}$ (c) $9$

4MA1/1H JANUARY 2022

# Question 14

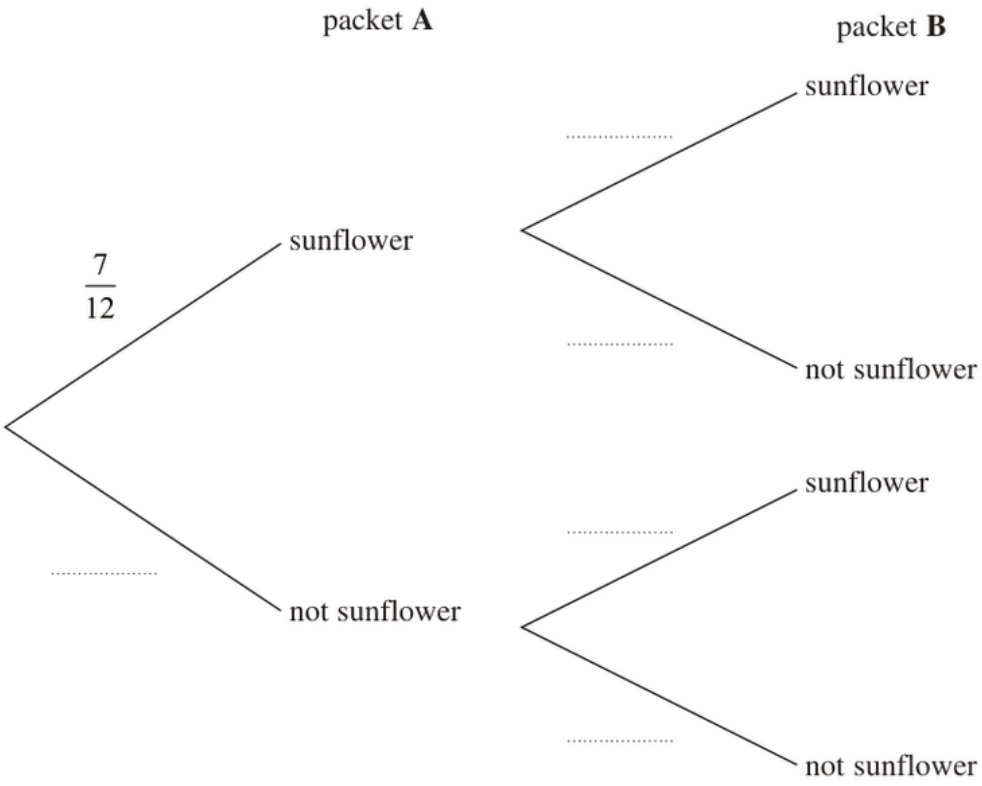
4 marks | Tree Diagrams & Conditional Probability

**14** Aika has 2 packets of seeds, packet **A** and packet **B**

There are 12 seeds in packet **A** and 7 of these are sunflower seeds.
There are 15 seeds in packet **B** and 8 of these are sunflower seeds.

Aika is going to take at random a seed from packet **A** and a seed from packet **B**

(a) Complete the probability tree diagram.




**(2)**

(b) Calculate the probability that Aika will take two sunflower seeds.

.....

**(2)**

**(Total for Question 14 is 4 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1H January 2022

**1** **(a) Complete the tree**

Packet A: not sunflower $1 - \frac{7}{12} = \frac{5}{12}$.

Packet B (on both branches): sunflower $\frac{8}{15}$, not sunflower $\frac{7}{15}$.

**2** **(b) Two sunflower seeds**

$$\frac{7}{12} \times \frac{8}{15} = \frac{56}{180} = \frac{14}{45}$$

---

✓ **FINAL ANSWER**

(a) $\frac{5}{12}$; $\frac{8}{15}$, $\frac{7}{15}$ on both branches (b) $\frac{14}{45}$

4MA1/1H JUNE 2022

# Question 12

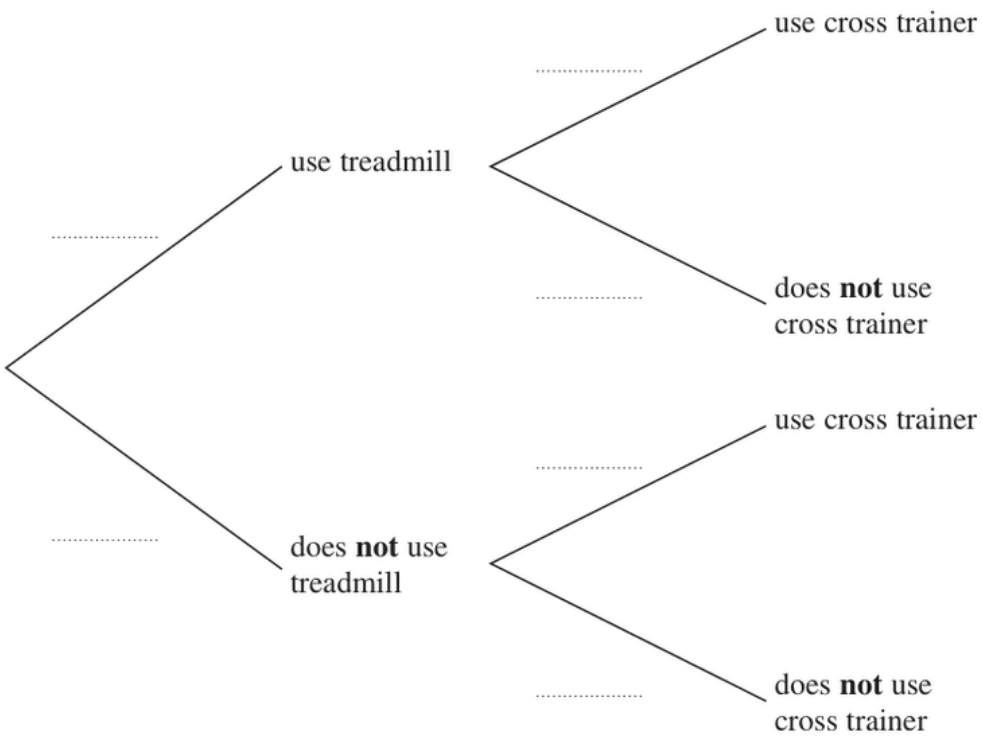
4 marks | Tree Diagrams & Conditional Probability

**12** Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3
When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.




(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

.....

(2)

**(Total for Question 12 is 4 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/1H June 2022

**1** **(a) Complete the tree**

Treadmill: use $0.8$, not use $0.2$.

After using the treadmill: cross trainer $0.3$, not $0.7$.

After not using the treadmill: cross trainer $0.6$, not $0.4$.

## 2 (b) Treadmill and cross trainer

$$0.8 \times 0.3 = 0.24$$

---

**FINAL ANSWER**

(a) $0.8, 0.2$; $0.3, 0.7$; $0.6, 0.4$ (b) $0.24$

4MA1/2HR JUNE 2022

# Question 14

3 marks | Tree Diagrams & Conditional Probability

**14** Ciara throws **four** fair six-sided dice.

The faces of each dice are labelled with the numbers 1, 2, 3, 4, 5, 6

Work out the probability that at least one of the dice lands on an even number.

**(Total for Question 14 is 3 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/2HR June 2022

**1 Use the complement**

"At least one even" is the opposite of "all four dice are odd". Each die is odd with probability $\frac{1}{2}$.

**2 All four dice odd**

$$\left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

**3 At least one even**

$$1 - \frac{1}{16} = \frac{15}{16}$$

**FINAL ANSWER**

$$\frac{15}{16}$$

# Question 12

5 marks | Tree Diagrams & Conditional Probability

**12** Moeen has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

| **Number** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Probability** | $x$ | 0.15 | 0.5 | $y$ | 0.13 | 0.03 |

(a) Show that $x + y = 0.19$

**(2)**

Given that $3x - y = 0.09$

and $x + y = 0.19$

(b) work out the value of $x$ and the value of $y$
Show clear algebraic working.

$x =$ .....

$y =$ .....

**(3)**

**(Total for Question 12 is 5 marks)**

**1** **(a) Probabilities add to 1**

$$x + 0.15 + 0.5 + y + 0.13 + 0.03 = 1 \Rightarrow x + y + 0.81 = 1 \Rightarrow x + y = 0.19$$

**2** **(b) Add the two equations**

$$(3x - y) + (x + y) = 0.09 + 0.19 \Rightarrow 4x = 0.28 \Rightarrow x = 0.07$$

**3** **(b) Find y**

$$y = 0.19 - 0.07 = 0.12$$

✓ **FINAL ANSWER**

(b) $x = 0.07,\ y = 0.12$

4MA1/2H JUNE 2023

# Question 13

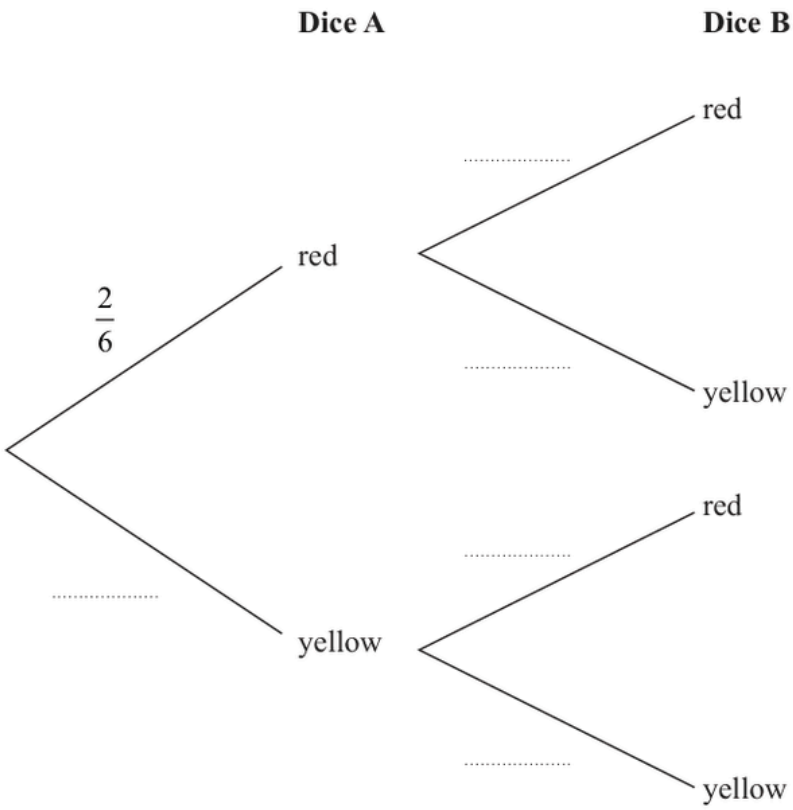
4 marks | Tree Diagrams & Conditional Probability

**13** Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.
Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that both dice land on yellow.

.....

(2)

**(Total for Question 13 is 4 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/2H June 2023

**1** **(a) Complete the tree**

Dice A: yellow $\frac{4}{6}$.

Dice B (on both branches): red $\frac{1}{6}$, yellow $\frac{5}{6}$.

**2** **(b) Both yellow**

$$\frac{4}{6} \times \frac{5}{6} = \frac{20}{36} = \frac{5}{9}$$

---

✓ **FINAL ANSWER**

(a) $\frac{4}{6}$; $\frac{1}{6}, \frac{5}{6}$ on both branches (b) $\frac{5}{9}$

# Question 13

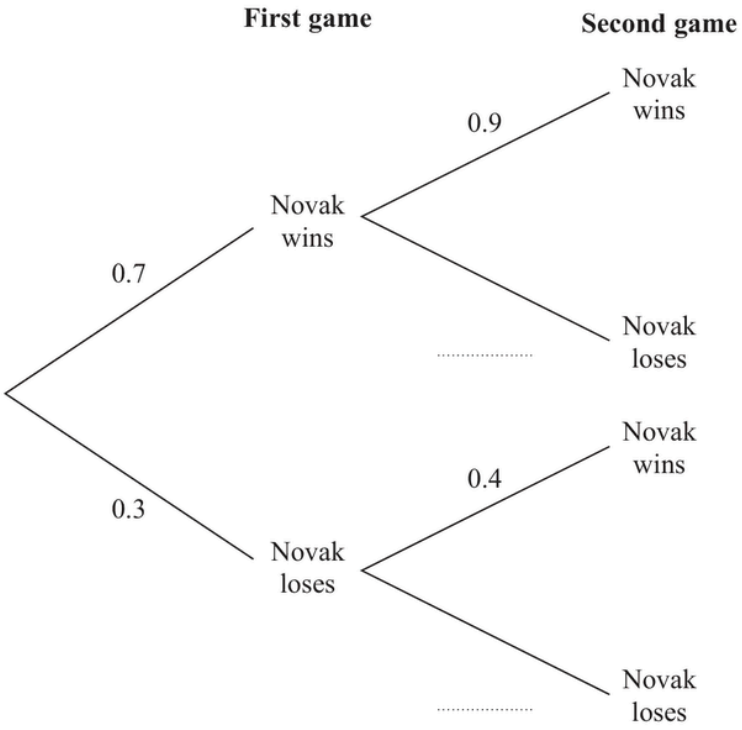
3 marks | Tree Diagrams & Conditional Probability

**13** Novak is going to play two games of tennis.

The probability that he wins the first game is 0.7

If he wins the first game, the probability that he wins the second game is 0.9
If he loses the first game, the probability that he wins the second game is 0.4

(a) Complete the probability tree diagram.




**(1)**

(b) Work out the probability that Novak wins both games of tennis.

.....

**(2)**

**(Total for Question 13 is 3 marks)**

## 1 (a) Complete the tree

After a win: loses $1 - 0.9 = 0.1$. After a loss: loses $1 - 0.4 = 0.6$.

## 2 (b) Wins both games

$$0.7 \times 0.9 = 0.63$$

**FINAL ANSWER**

(a) $0.1$, $0.6$ (b) $0.63$

4MA1/2H JUNE 2023

# Question 20

3 marks | Tree Diagrams & Conditional Probability

**20** There are 12 counters in a bag.

3 of the counters are red
9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Question 20 · 4MA1/2H June 2023

**1 There are 3 red counters**

There are 3 red counters.

**2 At least one red counter**

At least one red counter is still in the bag unless all 3 red counters are taken.

**3 Use the complement**

So use the complement:

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

$$= \frac{6}{1320}$$

$$= \frac{1}{220}$$

**4 Evaluate fraction**

Therefore,

$$P(\text{at least one red is still in the bag}) = 1 - \frac{1}{220}$$

$$= \frac{219}{220}$$

**FINAL ANSWER**

$$\frac{219}{220}$$

4MA1/2H NOVEMBER 2023

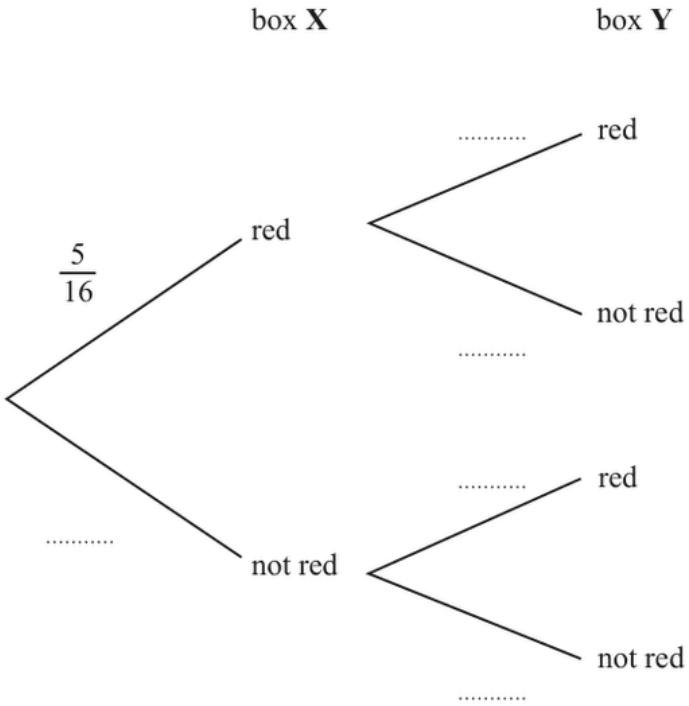
# Question 11

5 marks | Tree Diagrams & Conditional Probability

**11** Sid has 2 boxes of crayons, box **X** and box **Y**

5 of the 16 crayons in box **X** are red.
7 of the 20 crayons in box **Y** are red.
Sid takes at random one crayon from box **X** and one crayon from box **Y**

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Sid takes two crayons that are red or two crayons that are not red.

.....

(3)

**(Total for Question 11 is 5 marks)**

## WORKED SOLUTION

Question 11 · 4MA1/2H November 2023

**1** **(a) Complete the tree**

Box X: not red $\frac{11}{16}$.

Box Y (on both branches): red $\frac{7}{20}$, not red $\frac{13}{20}$.

**2** **(b) Both red or both not red**

$$\frac{5}{16} \times \frac{7}{20} + \frac{11}{16} \times \frac{13}{20} = \frac{35}{320} + \frac{143}{320} = \frac{178}{320} = \frac{89}{160}$$

---

✓ **FINAL ANSWER**

(a) $\frac{11}{16}$; $\frac{7}{20}$, $\frac{13}{20}$ on both branches (b) $\frac{89}{160}$

4MA1/1H MAY 2024

# Question 13

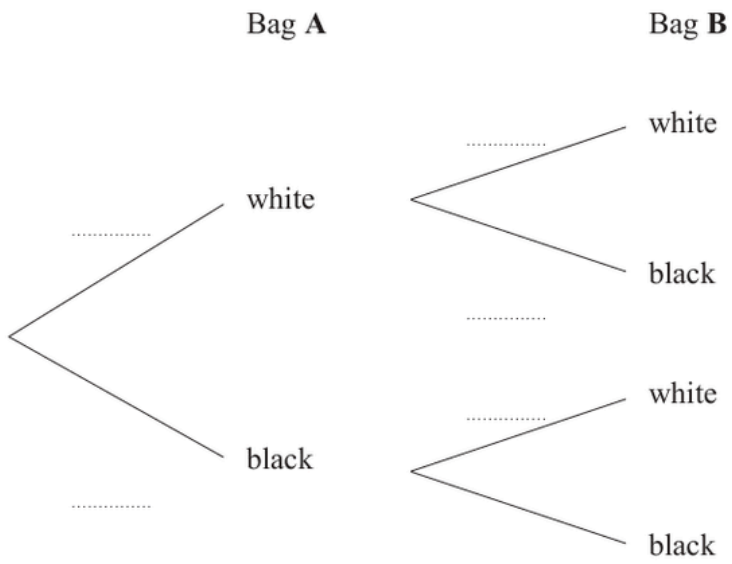
5 marks | Tree Diagrams & Conditional Probability

**13** Harman has two bags of beads.

In bag **A**, there are 3 white beads and 7 black beads.
In bag **B**, there are 5 white beads and 4 black beads.

Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Harman takes two beads of the same colour.

.....

(3)

**(Total for Question 13 is 5 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H May 2024

**1** **(a) Complete the tree**

Bag A: white $\frac{3}{10}$, black $\frac{7}{10}$.
Bag B (on both branches): white $\frac{5}{9}$, black $\frac{4}{9}$.

**2** **(b) Same colour = both white or both black**

$$\frac{3}{10} \times \frac{5}{9} + \frac{7}{10} \times \frac{4}{9} = \frac{15}{90} + \frac{28}{90} = \frac{43}{90}$$

**FINAL ANSWER**

(a) $\frac{3}{10}, \frac{7}{10}$; $\frac{5}{9}, \frac{4}{9}$ on both branches (b) $\frac{43}{90}$

4MA1/1H MAY 2024

# Question 16

4 marks | Tree Diagrams & Conditional Probability

**16** There are 20 sweets in a box.

15 of the sweets are red
5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

**(Total for Question 16 is 4 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H May 2024

**1 Use the complement**

"At least one of each colour" fails only if all three sweets are red or all three are yellow.

**2 All red, all yellow (no replacement)**

$$P(RRR) = \frac{15}{20} \times \frac{14}{19} \times \frac{13}{18} = \frac{2730}{6840}, \qquad P(YYY) = \frac{5}{20} \times \frac{4}{19} \times \frac{3}{18} = \frac{60}{6840}$$

**3 At least one of each**

$$1 - \frac{2730 + 60}{6840} = \frac{4050}{6840} = \frac{45}{76}$$

**FINAL ANSWER**

$$\frac{45}{76}$$

4MA1/1H NOVEMBER 2024

# Question 13

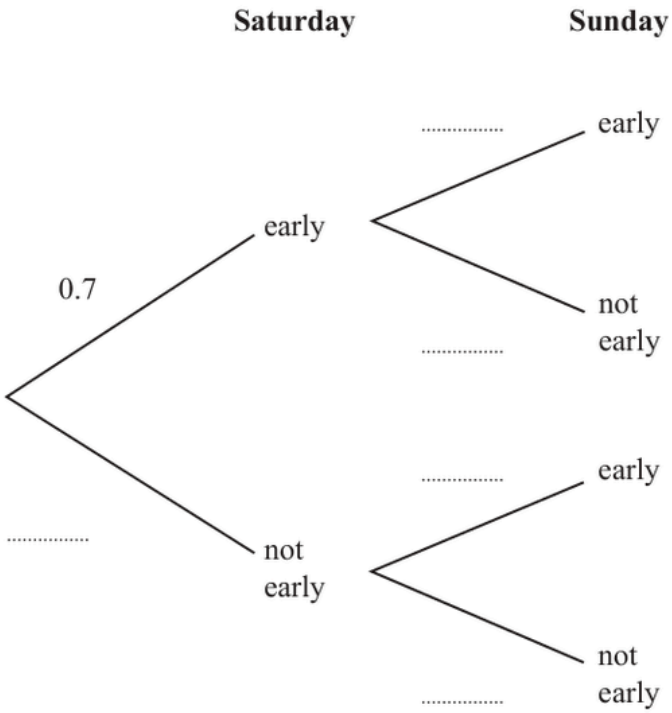
4 marks | Tree Diagrams & Conditional Probability

**13** The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

(a) Use this information to complete the probability tree diagram.




(2)

(b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

.....

(2)

**(Total for Question 13 is 4 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H November 2024

**1** **(a) Complete the tree**

Saturday: not early $0.3$.

After early: Sunday early $0.9$, not early $0.1$. After not early: Sunday early $0.6$, not early $0.4$.

## 2 (b) Early on both days

$$0.7 \times 0.9 = 0.63$$

---

**FINAL ANSWER**

(a) $0.3$; $0.9, 0.1$; $0.6, 0.4$ (b) $0.63$

# Question 11

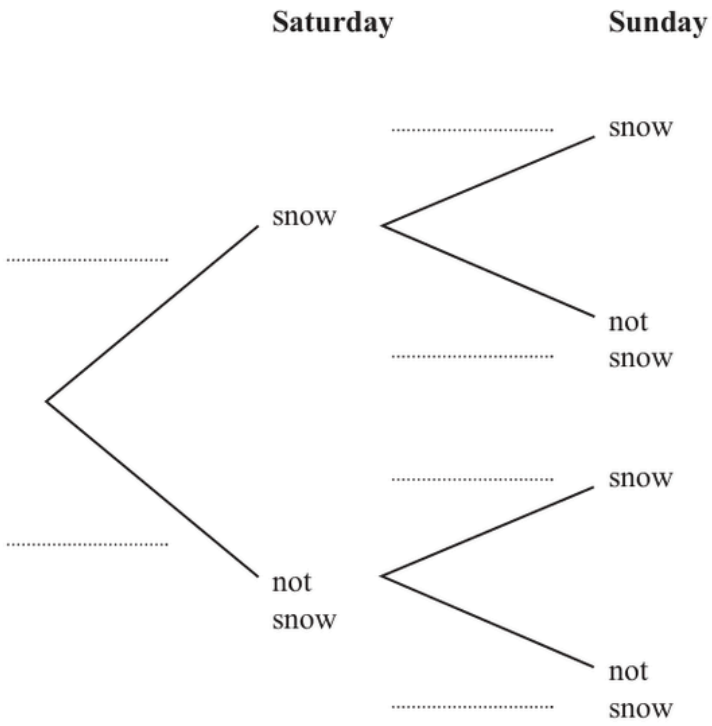
4 marks | Tree Diagrams & Conditional Probability

**11** The probability that it will snow on Saturday is 0.2

If it snows on Saturday, the probability that it will snow on Sunday is 0.35

If it does not snow on Saturday, the probability that it will snow on Sunday is 0.4

(a) Use this information to complete the probability tree diagram.




**(2)**

(b) Work out the probability that it will snow on both Saturday and Sunday.

.....

**(2)**

**(Total for Question 11 is 4 marks)**

**1** **The missing branch probabilities are**

The missing branch probabilities are complements.

$$P(\text{not snow on Saturday}) = 0.8$$

$$P(\text{not snow on Sunday} \mid \text{snow Saturday}) = 0.65$$

$$P(\text{not snow on Sunday} \mid \text{not snow Saturday}) = 0.6$$

**2** **Calculate value**

For snow on both days,

$$0.2 \times 0.35 = 0.07$$

✓ **FINAL ANSWER**

tree labels $0.2, 0.8, 0.35, 0.65, 0.4, 0.6$; probability $0.07$.

4WM1/01 MAY 2025

# Question 12

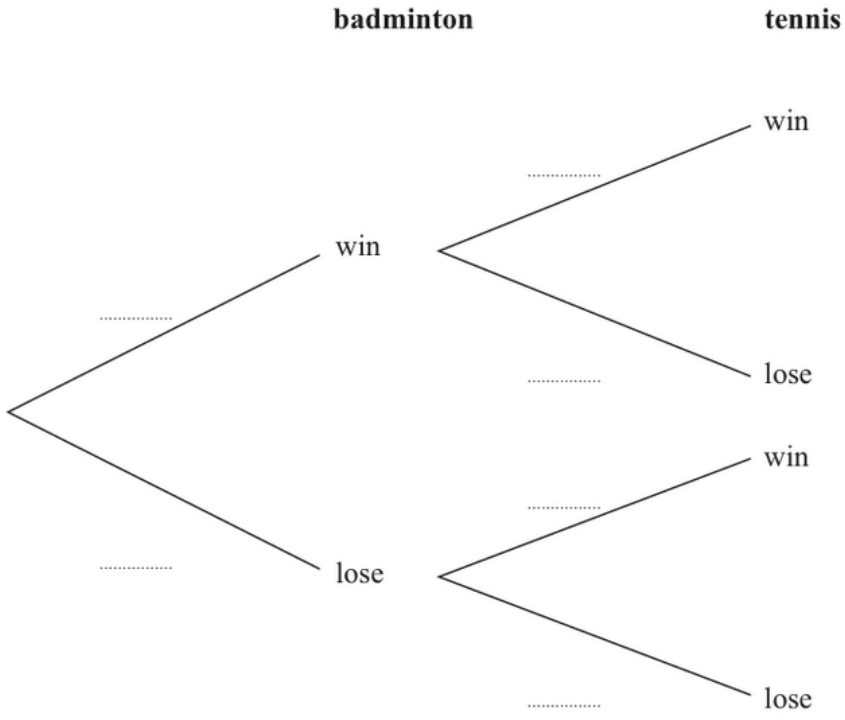
4 marks | Tree Diagrams & Conditional Probability

**12** Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7
The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Ricardo loses both games.

.....

(2)

**(Total for Question 12 is 4 marks)**

## WORKED SOLUTION

Question 12 · 4WM1/01 May 2025

**1** **Calculate probability**

For badminton,

$$P(\text{win}) = 0.7, \qquad P(\text{lose}) = 0.3$$

**2** **Calculate probability**

For tennis,

$$P(\text{win}) = 0.4, \qquad P(\text{lose}) = 0.6$$

**3** **The probability of losing both**

The probability of losing both games is

$$0.3 \times 0.6 = 0.18$$

✓ **FINAL ANSWER**

Tennis branches are $0.4$ and $0.6$ on each branch; probability of losing both is $0.18$.

4MA1/1H JUNE 2025

# Question 14

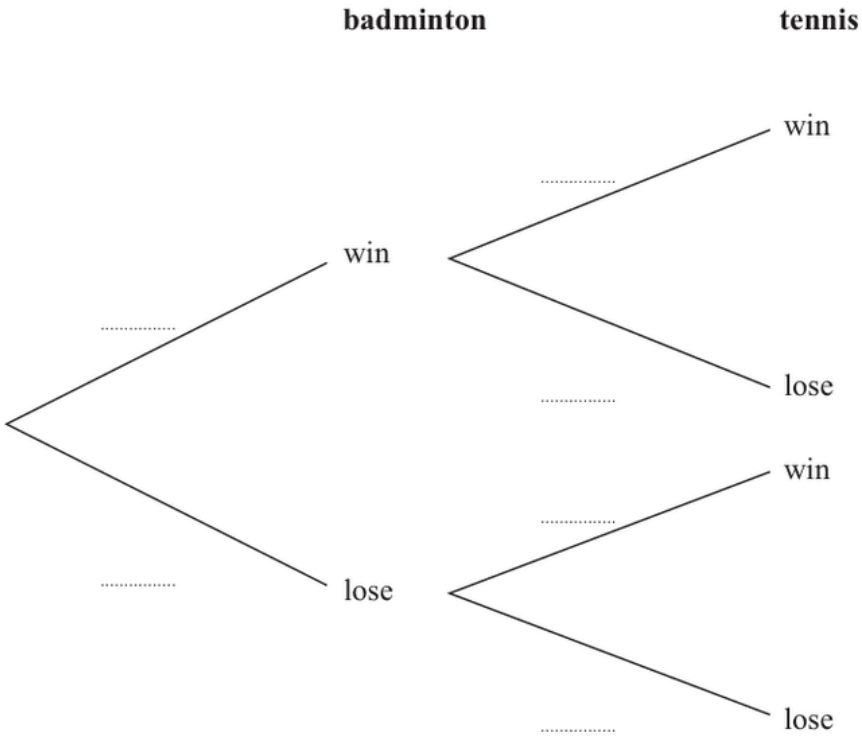
4 marks | Tree Diagrams & Conditional Probability

**14** Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7
The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Ricardo loses both games.

.....

(2)

**(Total for Question 14 is 4 marks)**

## WORKED SOLUTION

Question 14 · 4MA1/1H June 2025

**1** **(a) Complete the tree**

Badminton: win $0.7$, lose $0.3$.

Tennis (on both branches): win $0.4$, lose $0.6$.

**2** **(b) Lose both**

$$0.3 \times 0.6 = 0.18$$

---

✓ **FINAL ANSWER**

(a) $0.7, 0.3$; $0.4, 0.6$ on both branches (b) $0.18$

# Question 14

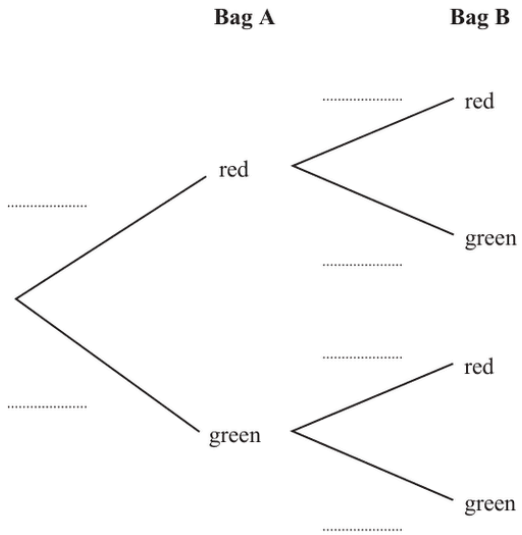
7 marks | Tree Diagrams & Conditional Probability

**14** Sara has two bags, **A** and **B**

In bag **A**, there are only 5 red beads and 4 green beads.
In bag **B**, there are only 7 red beads and 3 green beads.

Sara takes at random a bead from bag **A**
She then takes at random a bead from bag **B**

(a) Use this information to complete the probability tree diagram.




(2)

(b) Work out the probability that Sara takes two red beads.

.....

(2)

Sara puts the beads back into their original bags.

Sara also has a box of beads.
In the box, there are only red beads and green beads.

When a bead is taken at random from the box, the probability that it is

a green bead is $\frac{2}{11}$

Sara takes at random a bead from bag **A**
She then takes at random a bead from bag **B**
She then takes at random a bead from the box.

(c) Work out the probability that Sara takes more red beads than green beads.

.....

(3)

**(Total for Question 14 is 7 marks)**

1 **(a) Complete the tree**

Bag A: red $\frac{5}{9}$, green $\frac{4}{9}$.

Bag B (on both branches): red $\frac{7}{10}$, green $\frac{3}{10}$.

2 **(b) Two red beads**

$$\frac{5}{9} \times \frac{7}{10} = \frac{35}{90} = \frac{7}{18}$$

3 **(c) More red than green with three beads**

She needs at least two reds: RRR, RRG, RGR or GRR. The box gives red $\frac{9}{11}$, green $\frac{2}{11}$.

4 **(c) Add the four cases**

$$\frac{5\cdot7\cdot9 + 5\cdot7\cdot2 + 5\cdot3\cdot9 + 4\cdot7\cdot9}{9\cdot10\cdot11} = \frac{315+70+135+252}{990} = \frac{772}{990} = \frac{386}{495}$$

✓ **FINAL ANSWER**

(a) $\frac{5}{9}, \frac{4}{9}; \frac{7}{10}, \frac{3}{10}$ on both branches (b) $\frac{7}{18}$ (c) $\frac{386}{495}$

4WM1/01 NOVEMBER 2025

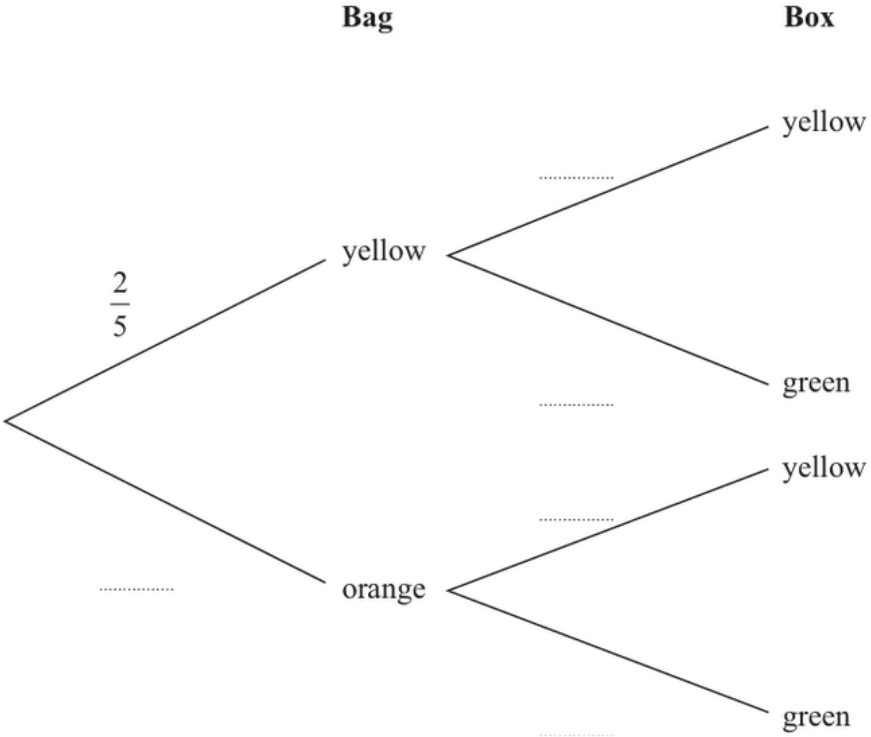
# Question 12

5 marks | Tree Diagrams & Conditional Probability

**12** In a bag, there are only 2 yellow beads and 3 orange beads.
In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Chi takes two beads of different colours.

.....

(3)

**(Total for Question 12 is 5 marks)**

## WORKED SOLUTION

Question 12 · 4WM1/01 November 2025

**1** **Bag 1**

Bag 1:

$$P(Y) = \frac{2}{5}, \qquad P(O) = \frac{3}{5}$$

**2** **Bag 2**

Bag 2:

$$P(Y) = \frac{3}{8}, \qquad P(G) = \frac{5}{8}$$

---

**3** **The only way to get**

The only way to get the same colour is yellow then yellow.

$$P(\text{different}) = 1 - \frac{2}{5} \times \frac{3}{8}$$

$$= 1 - \frac{3}{20} = \frac{17}{20}$$

---

✓ **FINAL ANSWER**

$\frac{17}{20}$.

4MA1/1H NOVEMBER 2025

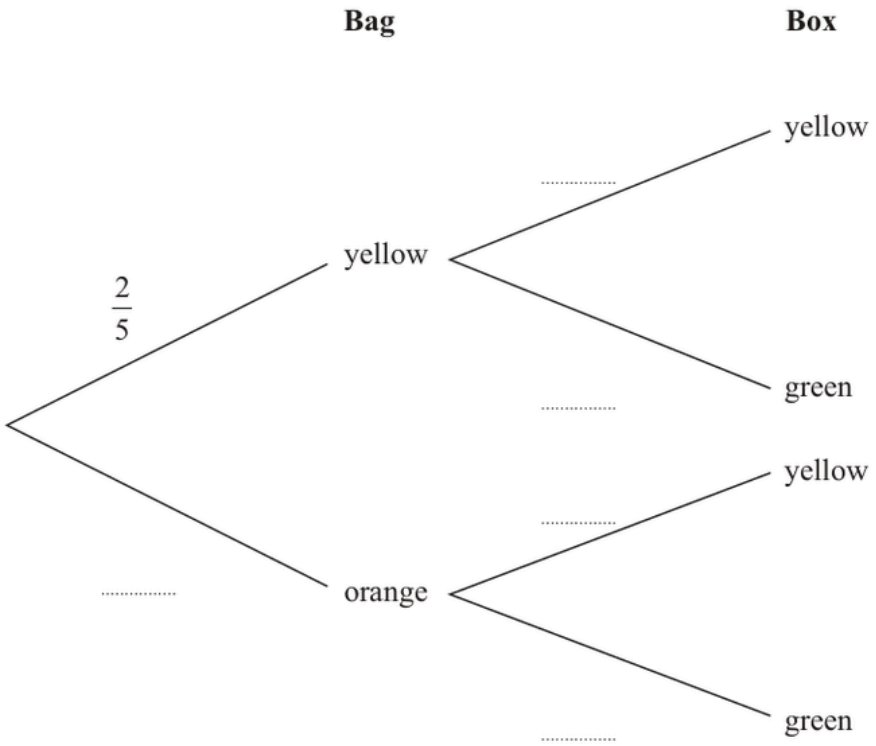
# Question 13

5 marks | Tree Diagrams & Conditional Probability

**13** In a bag, there are only 2 yellow beads and 3 orange beads.
In a box, there are only 3 yellow beads and 5 green beads.

Chi takes at random a bead from the bag and a bead from the box.

(a) Complete the probability tree diagram.




(2)

(b) Work out the probability that Chi takes two beads of different colours.

.....

(3)

**(Total for Question 13 is 5 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H November 2025

**1** **(a) Complete the tree**

Bag: orange $\frac{3}{5}$.

Box (on both branches): yellow $\frac{3}{8}$, green $\frac{5}{8}$.

**2** **(b) Use the complement**

The only way to get the same colour is yellow from both, so

$$P(\text{different}) = 1 - \frac{2}{5} \times \frac{3}{8} = 1 - \frac{6}{40} = \frac{17}{20}$$

---

✓ **FINAL ANSWER**

(a) $\frac{3}{5}$; $\frac{3}{8}, \frac{5}{8}$ on both branches (b) $\frac{17}{20}$

4MA1/1H NOVEMBER 2025

# Question 22

3 marks | Tree Diagrams & Conditional Probability

**22** There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

**(Total for Question 22 is 3 marks)**

## WORKED SOLUTION

Question 22 · 4MA1/1H November 2025

**1 There are 2 pink buttons**

There are 2 pink buttons.

**2 At least one pink button**

At least one pink button is still in the box unless both pink buttons are taken.

**3 Use the complement**

So use the complement.

**4 The probability that both pink**

The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

**5** **Evaluate fraction**

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

---

✓ **FINAL ANSWER**

$\frac{34}{35}$

4WM1/01 NOVEMBER 2025

# Question 23

3 marks | Tree Diagrams & Conditional Probability

**23** There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Question 23 · 4WM1/01 November 2025

**1 There are 2 pink buttons**

There are 2 pink buttons.

**2 At least one pink button**

At least one pink button is still in the box unless both pink buttons are taken.

**3 Use the complement**

So use the complement.

**4 The probability that both pink**

The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

**5** **Evaluate fraction**

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

---

✓ **FINAL ANSWER**

$\frac{34}{35}$

4MA1/1H JUNE 2026

# Question 13

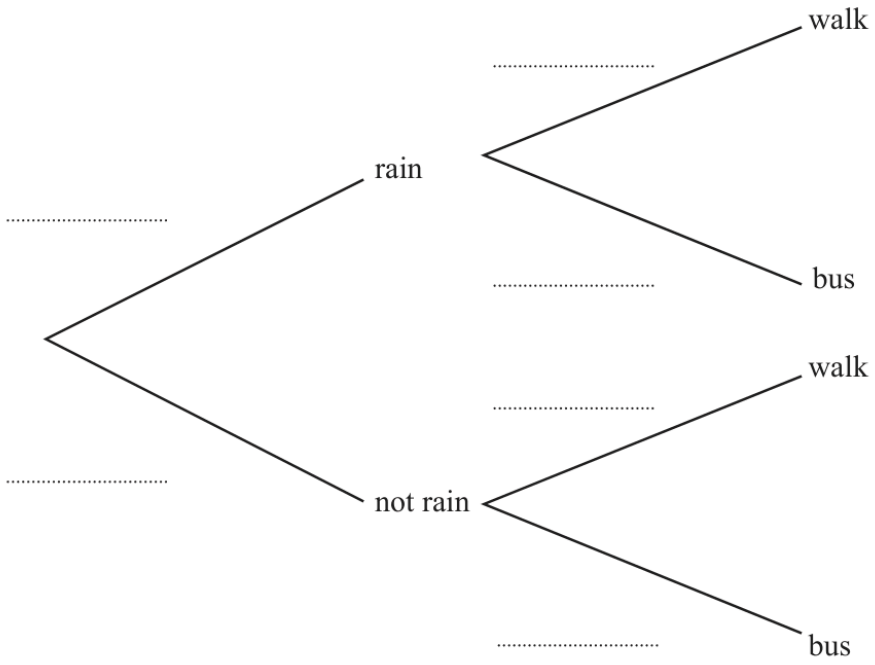
4 marks | Tree Diagrams & Conditional Probability

**13** On any school day, Paulo either walks to school or travels by bus to school.

The probability that it rains on any school day is 0.15

If it rains on a school day, the probability that Paulo walks to school is 0.2
If it does not rain on a school day, the probability that Paulo walks to school is 0.75

(a) Complete the probability tree diagram for this information.




**(2)**

(b) Work out the probability that on a school day, it rains and Paulo walks to school.

.....

**(2)**

**(Total for Question 13 is 4 marks)**

## WORKED SOLUTION

Question 13 · 4MA1/1H June 2026

**1** **Part (a) | Step 1**

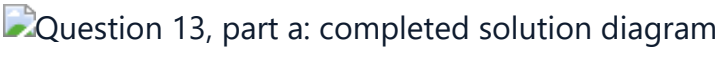
First split: rain $0.15$, not rain $0.85$.

Complementary branches sum to 1.

**2** **Part (a) | Step 2**

After rain: walk $0.20$, bus $0.80$. After not rain: walk $0.75$, bus $0.25$.

Use the stated conditional probabilities and their complements on the corresponding branches.



3 **Part (b) | Step 1**

$$P(\text{rain and walk}) = P(\text{rain})P(\text{walk} \mid \text{rain}) = 0.15 \times 0.2 = 0.03.$$

Multiply the probabilities along the rain-then-walk path; independence is not assumed.

---

✓ **FINAL ANSWER**

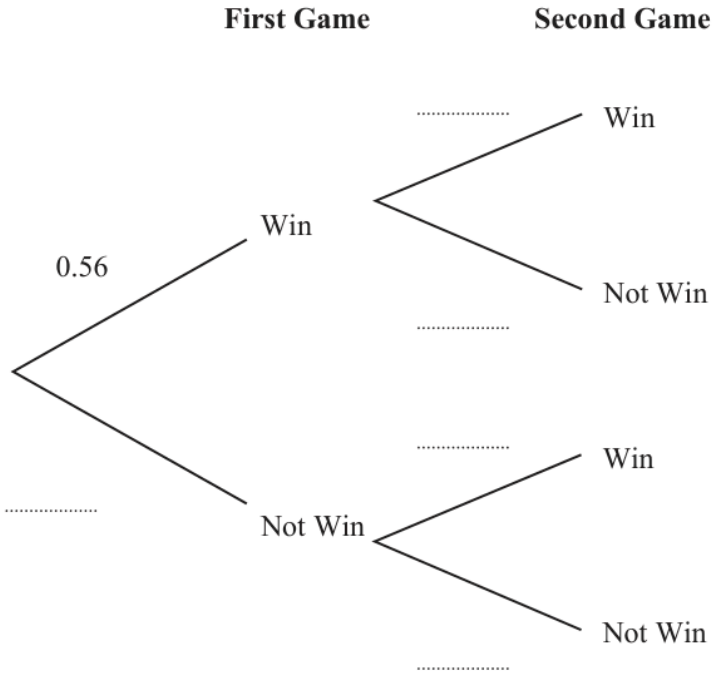
(a) Rain/not rain: $0.15, 0.85$; walk/bus after rain: $0.2, 0.8$; walk/bus after no rain: $0.75, 0.25$. (b) $0.03$

# Question 13

4 marks | Tree Diagrams & Conditional Probability

**13** A team plays two games of football.
For each game, the probability that the team wins is 0.56

(a) Complete the probability tree diagram.




**(2)**

(b) Work out the probability that the team wins both games.

.....

**(2)**

**(Total for Question 13 is 4 marks)**

## 1 (a) Branches add to 1

Not win $= 1 - 0.56 = 0.44$ on every pair of branches.

## 2 (b) Win and win

$$0.56 \times 0.56 = 0.3136$$

**FINAL ANSWER**

(a) see diagram (b) $0.3136$

# Question 14

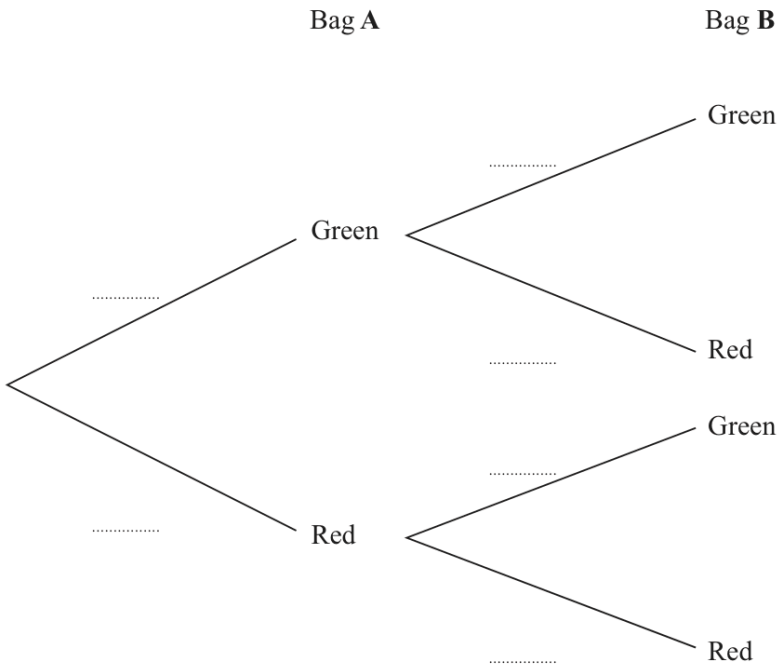
8 marks | Tree Diagrams & Conditional Probability

**14** Pras has 2 bags of counters, bag **A** and bag **B**

In bag **A**, there are 4 green counters and 3 red counters.
In bag **B**, there are 5 green counters and 1 red counter.

Pras is going to take at random a counter from bag **A** and a counter from bag **B**

(a) Complete the probability tree diagram.




**(2)**

(b) Work out the probability that Pras will take two green counters.

.....

**(2)**

Pras also has a third bag of counters, bag **C**
In bag **C**, there are only green counters and red counters.
All the original counters are back in bag **A** and bag **B**

Pras is going to take at random a counter from bag **A**, a counter from bag **B** and a counter from bag **C**

The probability that Pras will take three red counters is $\frac{3}{70}$

(c) Work out the probability that Pras will take 1 green counter and 2 red counters.

.....

**(4)**

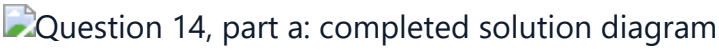
**(Total for Question 14 is 8 marks)**

1 **Part (a) | Step 1**

From the start, label the Bag A branches $P(G_A) = 4/7$ and $P(R_A) = 3/7$.
Bag A has 7 counters in total.

2 **Part (a) | Step 2**

From each Bag A outcome, label the Bag B green branch $5/6$ and red branch $1/6$.
Bag B has 6 counters and is unaffected by the result from bag A.


Question 14, part a: completed solution diagram

3 **Part (b) | Step 1**

$$P(G_A G_B) = \frac{4}{7} \times \frac{5}{6} = \frac{20}{42} = \frac{10}{21}.$$

Both events must occur, so multiply along the green-green path.

4 **Part (c) | Step 1**

Let $r = P(R_C)$. Then

$$\frac{3}{7} \times \frac{1}{6} \times r = \frac{3}{70} \quad \Rightarrow \quad r = \frac{3}{5}, \qquad P(G_C) = \frac{2}{5}.$$

Use the given probability of three reds to find the bag C probabilities.

5 **Part (c) | Step 2**

Exactly one green means the disjoint outcomes $G_A R_B R_C$, $R_A G_B R_C$, or $R_A R_B G_C$.
The single green counter can come from A, B or C.

6 **Part (c) | Step 3**

$$P = \frac{4}{7}\frac{1}{6}\frac{3}{5} + \frac{3}{7}\frac{5}{6}\frac{3}{5} + \frac{3}{7}\frac{1}{6}\frac{2}{5} = \frac{2}{35} + \frac{3}{14} + \frac{1}{35} = \frac{3}{10}.$$

Multiply within each outcome and add the three mutually exclusive probabilities.

✓ **FINAL ANSWER**

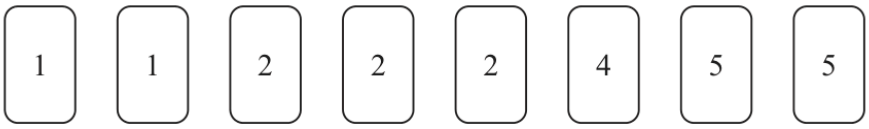
(a) Branch labels from top to bottom: $4/7, 3/7$ for A; $5/6, 1/6$ for B after either A result. (b) $\frac{10}{21}$. (c) $\frac{3}{10}$.

4MA1/1HR JUNE 2026

# Question 18

3 marks | Tree Diagrams & Conditional Probability

**18** Here are 8 cards.
There is a number on each card.




Gordon puts the 8 cards in a bag.
He takes at random a card from the bag and does not replace it.
Gordon then takes at random a second card from the bag.

Calculate the probability that the number on the second card is 3 more than the number on the first card.

.....

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/1HR June 2026

### 1 Solution | Step 1

The only possible ordered number pairs are $(1, 4)$ and $(2, 5)$.
The second number must be exactly 3 more than the first, not merely differ by 3.

### 2 Solution | Step 2

$$P(1 \text{ then } 4) = \frac{2}{8} \times \frac{1}{7}, \qquad P(2 \text{ then } 5) = \frac{3}{8} \times \frac{2}{7}.$$

There are 7 cards remaining for the second draw; there are two 1s, three 2s, one 4 and two 5s.

### 3 Solution | Step 3

$$P = \frac{2}{56} + \frac{6}{56} = \frac{8}{56} = \frac{1}{7}.$$

The two ordered events are mutually exclusive, so add their probabilities.

**FINAL ANSWER**

$\frac{1}{7}$

# Question 22

5 marks | Tree Diagrams & Conditional Probability

**22** A bag contains $x$ marbles.
12 of the marbles are pink and the rest are yellow.

Greta takes at random 2 marbles from the bag.

The probability that Greta takes a pink marble and a yellow marble is $\frac{18}{47}$

Work out the value of $x$
Show clear algebraic working.

$x =$ .....

**(Total for Question 22 is 5 marks)**

**1** **Pink then yellow, or yellow then pink**

$$2 \times \frac{12}{x} \times \frac{x-12}{x-1} = \frac{18}{47}$$

**2** **Cross-multiply**

$$1128(x-12) = 18x(x-1) \Rightarrow 18x^2 - 1146x + 13536 = 0$$

**3** **Simplify (÷6) and solve**

$$3x^2 - 191x + 2256 = 0 \Rightarrow (x-48)(3x-47) = 0$$

**4** **x must be a whole number**

$$x = 48$$

✓ **FINAL ANSWER**

$x = 48$

4MA1/1H JUNE 2026

# Question 23

5 marks | Tree Diagrams & Conditional Probability

**23** A bag contains $x$ marbles.
12 of the marbles are pink and the rest are yellow.

Greta takes at random 2 marbles from the bag.

The probability that Greta takes a pink marble and a yellow marble is $\frac{18}{47}$

Work out the value of $x$
Show clear algebraic working.

$x =$ .....

**(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Question 23 · 4MA1/1H June 2026

**1 Solution | Step 1**

$$P(\text{one pink, one yellow}) = \frac{12}{x}\frac{x-12}{x-1} + \frac{x-12}{x}\frac{12}{x-1} = \frac{24(x-12)}{x(x-1)}.$$

Include pink-then-yellow and yellow-then-pink, without replacement.

**2 Solution | Step 2**

$$\frac{24(x-12)}{x(x-1)} = \frac{18}{47} \implies 1128(x-12) = 18x(x-1).$$

Equate to the given probability and cross-multiply.

**3 Solution | Step 3**

$$18x^2 - 1146x + 13536 = 0 \implies 3x^2 - 191x + 2256 = 0.$$

Rearrange and divide by 6.

**4 Solution | Step 4**

$$(x-48)(3x-47) = 0 \implies x = 48 \text{ or } x = \frac{47}{3}.$$

Factorise the quadratic.

5

**Solution | Step 5**

The number of marbles must be an integer, so $x = 48$.

Reject the non-integer root.

---

✓ **FINAL ANSWER**

$x = 48$



TOPIC

# Set Notation & Venn Diagrams

Every past-paper question on this topic, 2020 to 2026.

# Question 14

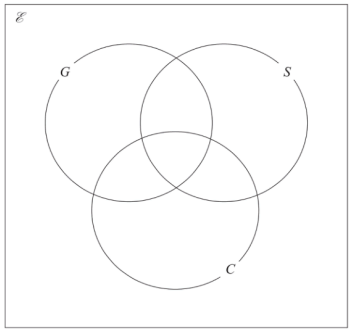
8 marks | Set Notation & Venn Diagrams

**14** 100 farmers are asked if they have goats ($G$), sheep ($S$) or chickens ($C$) on their farms.

Of these farmers

31 have sheep
53 have chickens
6 have goats, sheep and chickens
11 have sheep and goats
17 have sheep and chickens
18 have goats and chickens
20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.




**(3)**

(b) Find

(i) $n(G)$

.....

**(1)**

(ii) $n([G \cup S]')$

.....

**(1)**

(iii) $n(G' \cap C)$

.....

**(1)**

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

.....

**(2)**

**(Total for Question 14 is 8 marks)**

**1** **Place the centre value**

The number with goats, sheep and chickens is $6$.

**2** **Fill the pair-only regions**

$$G \cap S \text{ only} = 11 - 6 = 5$$

$$S \cap C \text{ only} = 17 - 6 = 11$$

$$G \cap C \text{ only} = 18 - 6 = 12$$

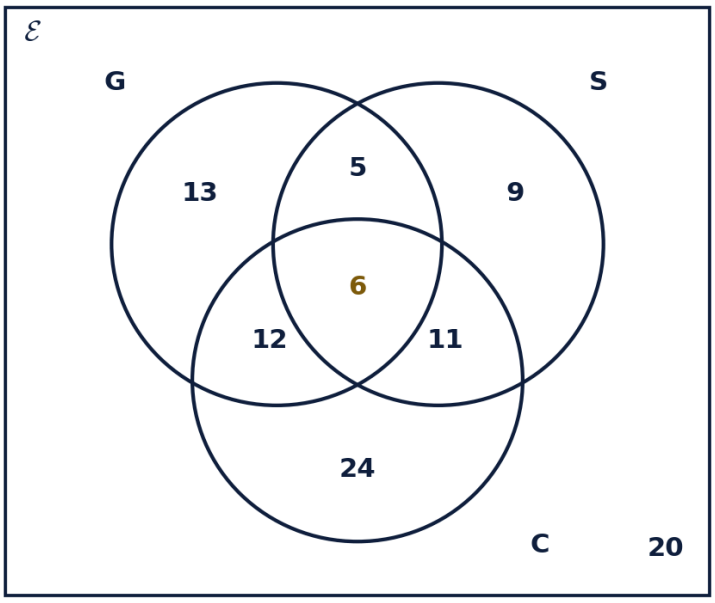
**3** **Fill the single-set regions**

$$S \text{ only} = 31 - 5 - 11 - 6 = 9$$

$$C \text{ only} = 53 - 11 - 12 - 6 = 24$$

Since $20$ have none, the total inside the circles is $80$.

$$G \text{ only} = 80 - (5 + 11 + 12 + 6 + 9 + 24) = 13$$



### 4 Read the set values

$$n(G) = 13 + 5 + 12 + 6 = 36$$

$$n((G \cup S)') = 24 + 20 = 44$$

$$n(G' \cap C) = 24 + 11 = 35$$

---

### 5 Find the conditional probability

Out of the $53$ farmers with chickens, $18$ also have goats.

$$P(G \mid C) = \frac{18}{53}$$

---

**FINAL ANSWER**

Venn regions: $G$ only $13$, $S$ only $9$, $C$ only $24$, $G \cap S$ only $5$, $S \cap C$ only $11$, $G \cap C$ only $12$, all three $6$, none $20$. Also $36$, $44$, $35$, and $\frac{18}{53}$.

4MA1/1HR JANUARY 2020

# Question 4

3 marks | Set Notation & Venn Diagrams

4 $B = \{b, l, u, e\}$

$G = \{g, r, e, y\}$

$W = \{w, h, i, t, e\}$

(a) List all the members of the set

(i) $B \cup G$

(ii) $W \cap G'$

.....

(2)

Serena writes down the statement $B \cap G \cap W = \varnothing$

(b) Is Serena's statement correct?
You must give a reason for your answer.

.....

(1)

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1HR January 2020

**1 Calculate value**

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

**2 means all the letters that**

$B \cup G$ means all the letters that are in $B$ or in $G$.

$$B \cup G = \{b, l, u, e, g, r, y\}$$

**3 Calculate statistic**

$W \cap G'$ means letters that are in $W$, but not in $G$. Since $e$ is in $G$, it is not included.

$$W \cap G' = \{w, h, i, t\}$$

**4** **Serena is not correct because**

Serena is not correct because $e$ is in all three sets.

$$B \cap G \cap W = \{e\}$$

---

✓ **FINAL ANSWER**

(a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

# Question 17

6 marks | Set Notation & Venn Diagrams

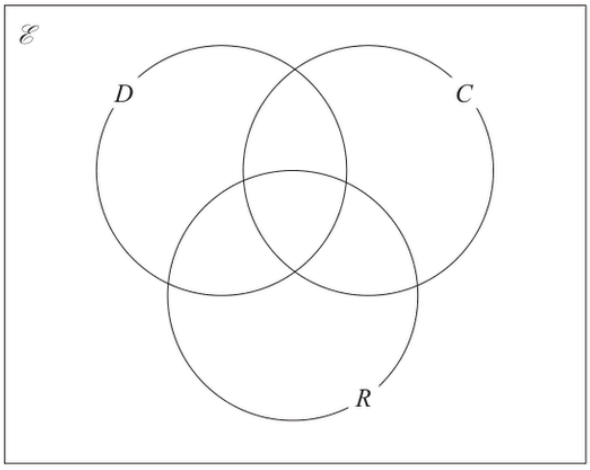
**17** Some students in a school were asked the following question.

"Do you have a dog ($D$), a cat ($C$) or a rabbit ($R$)?"

Of these students

28 have a dog
18 have a cat
20 have a rabbit
8 have both a cat and a rabbit
9 have both a dog and a rabbit
$x$ have both a dog and a cat
6 have a dog, a cat and a rabbit
5 have not got a dog or a cat or a rabbit

(a) Using this information, complete the Venn diagram to show the number of students in each appropriate subset.
Give the numbers in terms of $x$ where necessary.




**(3)**

Given that a total of 50 students answered the question,

(b) work out the value of $x$.

$x =$ .....

**(2)**

(c) Find $n(C' \cap D')$

.....

**(1)**

**(Total for Question 17 is 6 marks)**

**1** **Let the sets dogs**

Let the sets be $D$ for dogs, $C$ for cats, and $R$ for rabbits.

**2** **The number in all three**

The number in all three sets is $6$.

$$D \cap R \text{ only} = 9 - 6 = 3$$

$$C \cap R \text{ only} = 8 - 6 = 2$$

$$D \cap C \text{ only} = x - 6$$

**3** **Calculate value**

So the remaining one-set regions are

$$D \text{ only} = 25 - x, \quad C \text{ only} = 16 - x, \quad R \text{ only} = 9$$

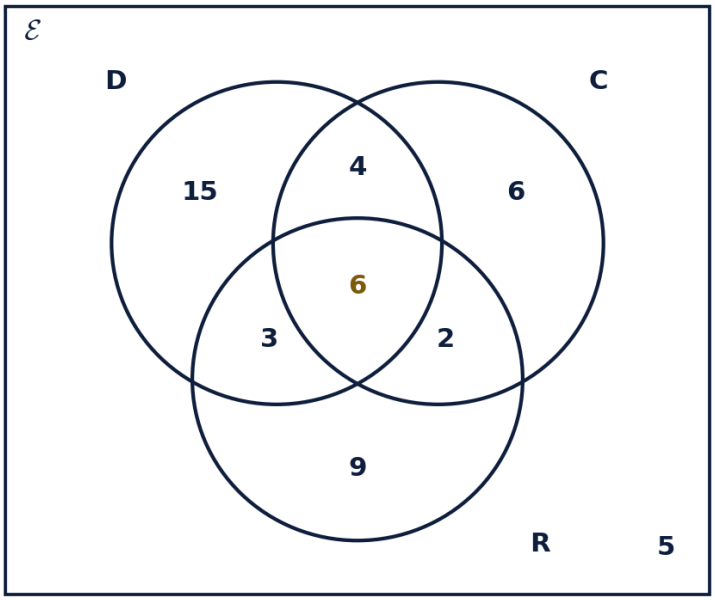
**4** **There are students outside all**

There are $5$ students outside all three sets. The total number of students is $50$, so

$$(25 - x) + (16 - x) + 9 + (x - 6) + 3 + 2 + 6 + 5 = 50$$

$$60 - x = 50$$

$$x = 10$$



**5** **Calculate value**

For $C' \cap D'$, we need students who are not in $C$ and not in $D$. These are the rabbit-only students and the students outside all three sets.

$$9 + 5 = 14$$

---

✓ **FINAL ANSWER**

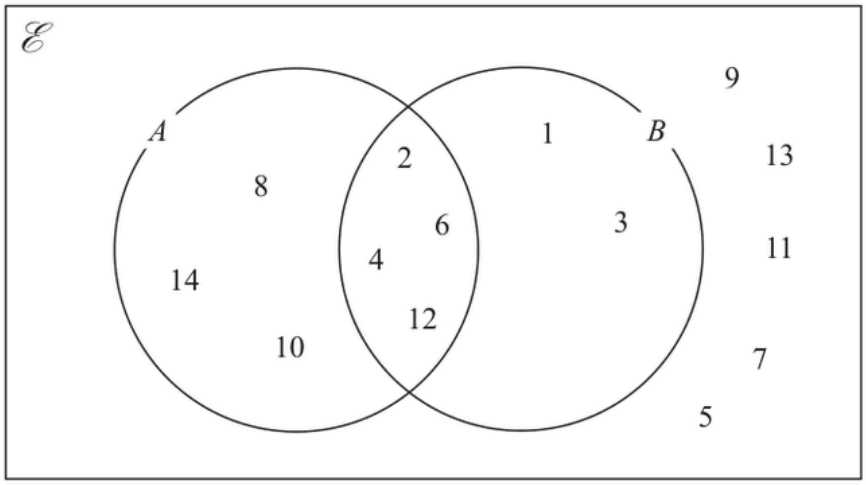
$x = 10$ and $n(C' \cap D') = 14$

4MA1/1H MAY/NOVEMBER 2020

# Question 1

4 marks | Set Notation & Venn Diagrams

**1** The numbers from 1 to 14 are shown in the Venn diagram.




(a) List the members of the set $A \cap B$

.....

**(1)**

(b) List the members of the set $B'$

.....

**(1)**

A number is picked at random from the numbers in the Venn diagram.

(c) Find the probability that this number is in set $A$ but is **not** in set $B$.

.....

**(2)**

**(Total for Question 1 is 4 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H May/November 2020

**1** **(a) In both A and B**

$A \cap B = \{2, 4, 6, 12\}$

**2** **(b) Not in B**

Everything outside circle $B$: $B' = \{5, 7, 8, 9, 10, 11, 13, 14\}$

**3** **(c) In A but not in B**

$\{8, 10, 14\}$: $3$ of the $14$ numbers.

$$\frac{3}{14}$$

---

✓ **FINAL ANSWER**

(a) $2, 4, 6, 12$ (b) $5, 7, 8, 9, 10, 11, 13, 14$ (c) $\frac{3}{14}$

4MA1/2HR JANUARY 2021

# Question 3

3 marks | Set Notation & Venn Diagrams

3 $\mathscr{E} = \{21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$
$A = \{22, 24, 26, 28, 30\}$
$B = \{21, 24, 27, 30\}$

(a) List the members of the set

(i) $A \cap B$

(ii) $A'$

.....

(2)

$C = \{23, 25, 29\}$

(b) Using set notation, find an expression for $C$ in terms of $A$ and $B$.

.....

(1)

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/2HR January 2021

**1 Calculate value**

$$A \cap B = \{24, 30\}$$

**2 Set is**

The complement of $A$, using the given universal set, is

$$A' = \{21, 23, 25, 27, 29\}$$

**3 The set**

The set

$$C = \{23, 25, 29\}$$

**4 Calculate value**

is everything not in $A$ and not in $B$, so

$$C = A' \cap B'$$

**5** **Calculate value**

Equivalently,

$$C = (A \cup B)'$$

---

**FINAL ANSWER**

$A \cap B = \{24, 30\}$, $A' = \{21, 23, 25, 27, 29\}$, and $C = A' \cap B'$.

4MA1/1H JANUARY 2021

# Question 4

2 marks | Set Notation & Venn Diagrams

4 $\mathscr{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$A = \{\text{odd numbers}\}$
$B = \{\text{multiples of 3}\}$

List the members of the set

(i) $A \cap B$

.....

(1)

(ii) $A \cup B$

.....

(1)

**(Total for Question 4 is 2 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1H January 2021

**1 Calculate value**

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

**2 The odd numbers are**

The odd numbers are

$$A = \{21, 23, 25, 27, 29\}$$

**3 The multiples of are**

The multiples of $3$ are

$$B = \{21, 24, 27\}$$

**4 The numbers in both sets**

The numbers in both sets are

$$A \cap B = \{21, 27\}$$

**5 The numbers in either set**

The numbers in either set are

$$A \cup B = \{21, 23, 24, 25, 27, 29\}$$

**FINAL ANSWER**

(i) $\{21, 27\}$, (ii) $\{21, 23, 24, 25, 27, 29\}$

# Question 16

6 marks | Set Notation & Venn Diagrams

**16** Some students were asked the following question.

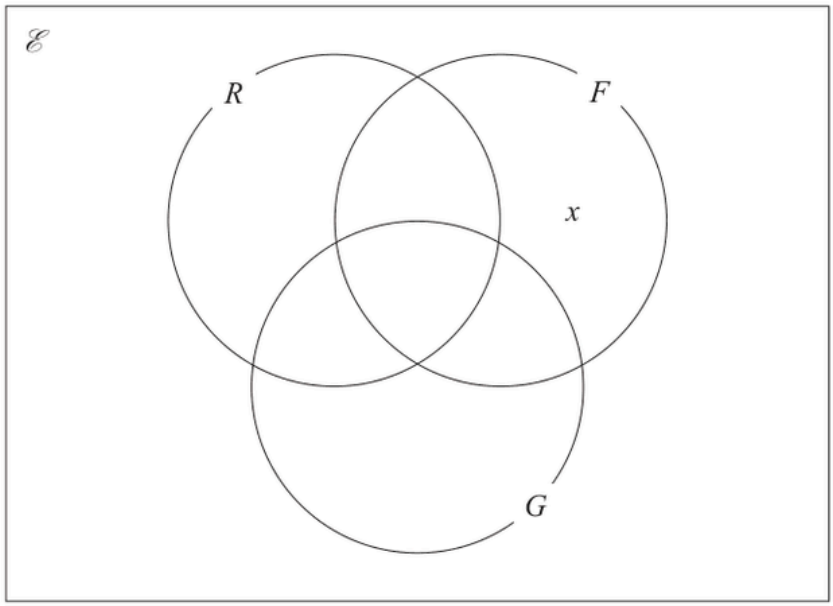
"Which of the subjects Russian ($R$), French ($F$) and German ($G$) do you study?"

Of these students

4 study all three of Russian, French and German
10 study Russian and French
13 study French and German
6 study Russian and German
24 study German
11 study none of the three subjects
the number who study Russian only is twice the number who study French only.

Let $x$ be the number of students who study French only.

(a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of $x$ where necessary.




**(3)**

Given that the number of students who were asked the question was 80

(b) work out the number of these students that study Russian.

.....

**(3)**

**(Total for Question 16 is 6 marks)**

**1** **The number studying all three**

The number studying all three languages is $4$.

**2** **Calculate value**

So the two-language-only regions are

$$R \cap F \text{ only} = 10 - 4 = 6$$

$$F \cap G \text{ only} = 13 - 4 = 9$$

$$R \cap G \text{ only} = 6 - 4 = 2$$

**3** **The German only region is**

The German-only region is

$$24 - 9 - 2 - 4 = 9$$

**4** **The French only region is**

The French-only region is $x$ and the Russian-only region is $2x$. There are $11$ students outside all three sets.

**5** **Calculate value**

Using the total $80$,

$$2x + x + 9 + 6 + 9 + 2 + 4 + 11 = 80$$

$$3x + 41 = 80$$

$$x = 13$$

**6** **The number studying Russian is**

The number studying Russian is

$$2x + 6 + 2 + 4 = 26 + 6 + 2 + 4 = 38$$

✓ **FINAL ANSWER**

$38$

4MA1/2HR JANUARY 2021

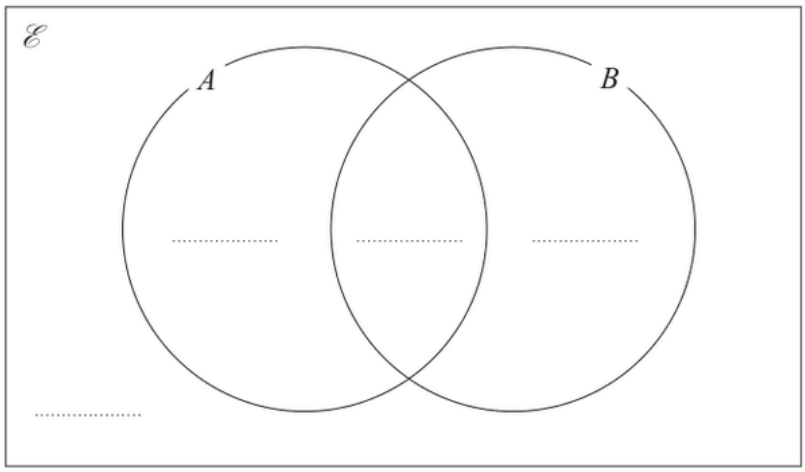
# Question 16

5 marks | Set Notation & Venn Diagrams

**16** Two events $A$ and $B$ are such that $n(A) = 62$ $n(B) = 30$ and $n(A \cup B) = 68$

Given that $n(\mathscr{E}) = 80$

(a) complete the Venn diagram to show the number of elements in each region.




**(2)**

An element is chosen at random from $\mathscr{E}$.

(b) Using the Venn diagram, find the probability that this element is in

(i) $A \cap B$

.....

**(1)**

(ii) $A \cup B'$

.....

**(2)**

**(Total for Question 16 is 5 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/2HR January 2021

**1** **(a) Find the intersection**

$$n(A \cap B) = n(A) + n(B) - n(A \cup B) = 62 + 30 - 68 = 24$$

**2** **(a) Fill every region**

$A$ only $= 62 - 24 = 38$, $B$ only $= 30 - 24 = 6$, outside $= 80 - 68 = 12$.

**3** **(b)(i) In both A and B**

$$P(A \cap B) = \frac{24}{80} = \frac{3}{10}$$

---

**4** **(b)(ii) In A or not in B**

$A \cup B'$ is everything except "$B$ only": $80 - 6 = 74$.

$$P(A \cup B') = \frac{74}{80} = \frac{37}{40}$$

---

✓ **FINAL ANSWER**

(a) $38, 24, 6$, outside $12$ (b)(i) $\frac{3}{10}$ (ii) $\frac{37}{40}$

4MA1/2H MAY 2021

# Question 4

3 marks | Set Notation & Venn Diagrams

4 $\mathscr{E}$ = {letters of the alphabet}
$B$ = {b, r, a, z, i, l}
$I$ = {i, r, e, l, a, n, d}

(a) List the members of the set

(i) $B \cup I$

(ii) $B \cap I'$

.....

(2)

$K$ = {k, e, n, y, a}

Cody writes down the statement $B \cap K = \varnothing$
Cody's statement is wrong.

(b) Explain why.

.....

(1)

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2H May 2021

**1 Calculate value**

$$B = \{b, r, a, z, i, l\}$$

$$I = \{i, r, e, l, a, n, d\}$$

**2 The union contains every letter**

The union contains every letter that is in $B$ or $I$:

$$B \cup I = \{a, b, d, e, i, l, n, r, z\}$$

**3 Calculate statistic**

$B \cap I'$ means letters in $B$, but not in $I$:

$$B \cap I' = \{b, z\}$$

**4** **Calculate value**

For $K = \{k, e, n, y, a\}$, Cody is not correct because the letter $a$ is in both $B$ and $K$.

$$B \cap K = \{a\}$$

---

✓ **FINAL ANSWER**

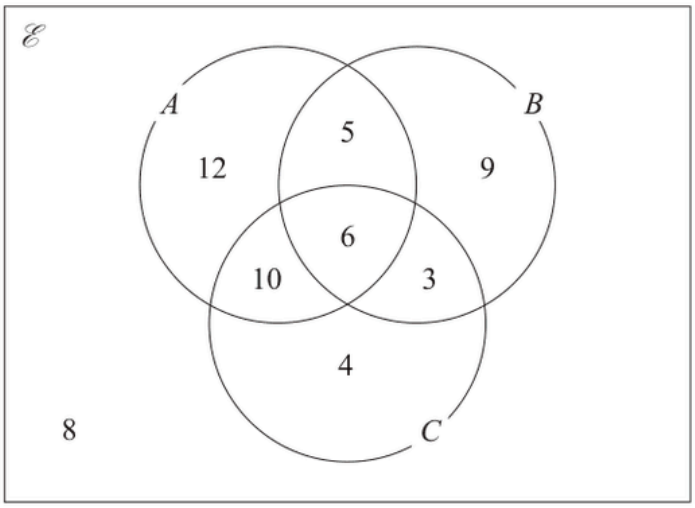
(a)(i) $\{a, b, d, e, i, l, n, r, z\}$, (a)(ii) $\{b, z\}$. Cody is incorrect.

4MA1/1H MAY 2021

# Question 19

3 marks | Set Notation & Venn Diagrams

**19** The Venn diagram shows a universal set, $\mathscr{E}$ and sets $A$, $B$ and $C$.




12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

**(1)**

(ii) $n(A' \cap B')$

.....

**(1)**

(iii) $n([A \cap B] \cup C)$

.....

**(1)**

**(Total for Question 19 is 3 marks)**

## WORKED SOLUTION

Question 19 · 4MA1/1H May 2021

**1** **From the Venn diagram**

From the Venn diagram:

**2** **only is only is only**

- $A$ only is $12$
- $A \cap B$ only is $5$
- $B$ only is $9$
- $A \cap C$ only is $10$
- all three is $6$
- $B \cap C$ only is $3$
- $C$ only is $4$
- outside is $8$

---

**3** **Add every region inside**

For $A \cup B$, add every region inside $A$ or $B$:

$$12 + 5 + 9 + 10 + 6 + 3 = 45$$

---

**4** **Add the regions outside both**

For $A' \cap B'$, add the regions outside both $A$ and $B$:

$$4 + 8 = 12$$

---

**5** **Add and all**

For $(A \cap B) \cup C$, add $A \cap B$ and all of $C$:

$$5 + 6 + 10 + 3 + 4 = 28$$

---

✓ **FINAL ANSWER**

(i) $45$, (ii) $12$, (iii) $28$

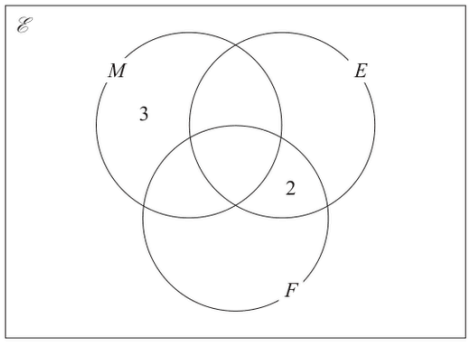
# Question 16

5 marks | Set Notation & Venn Diagrams

**16** There are 32 students in a class.
In one term these 32 students each took a test in Maths ($M$), in English ($E$) and in French ($F$).

25 students passed the test in Maths.
20 students passed the test in English.
14 students passed the test in French.
18 students passed the tests in Maths and English.
11 students passed the tests in Maths and French.
4 students failed all three tests.
$x$ students passed all three tests.

The incomplete Venn diagram gives some more information about the results of the 32 students.

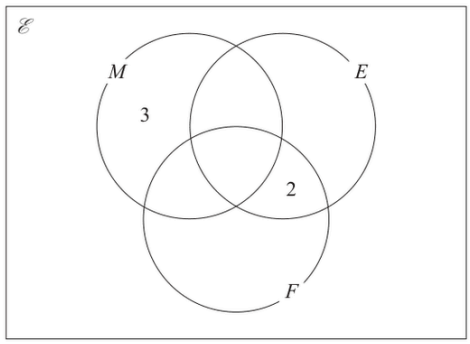



(a) Use all the given information about the results of students who passed the test in Maths to find the value of $x$.

$x =$ .....

**(2)**

(b) Use your value of $x$ to complete the Venn diagram to show the number of students in each subset.




**(2)**

A student who passed the test in Maths is chosen at random.

(c) Find the probability that this student failed the test in French.

.....

**(1)**

**(Total for Question 16 is 5 marks)**

1 **(a) Use the Maths circle**

Maths only $= 3$, Maths and English only $= 18 - x$, Maths and French only $= 11 - x$, all three $= x$.

$$3 + (18 - x) + (11 - x) + x = 25 \;\Rightarrow\; 32 - x = 25 \;\Rightarrow\; x = 7$$

2 **(b) Fill every region**

$M \cap E$ only $= 11$, $M \cap F$ only $= 4$, centre $= 7$, $E \cap F$ only $= 2$.

$$E \text{ only} = 20 - 11 - 7 - 2 = 0, \qquad F \text{ only} = 14 - 4 - 7 - 2 = 1$$

Outside: $4$. (Check: $3 + 11 + 4 + 7 + 0 + 2 + 1 + 4 = 32$.)

3 **(c) Passed Maths, failed French**

Maths only and Maths-and-English only: $3 + 11 = 14$ of the $25$.

$$\frac{14}{25}$$

✓ **FINAL ANSWER**

(a) $x = 7$ (b) $M$ only 3, $M \cap E$ only 11, $M \cap F$ only 4, centre 7, $E \cap F$ only 2, $E$ only 0, $F$ only 1, outside 4 (c) $\frac{14}{25}$

4MA1/1H JANUARY 2022

# Question 7

3 marks | Set Notation & Venn Diagrams

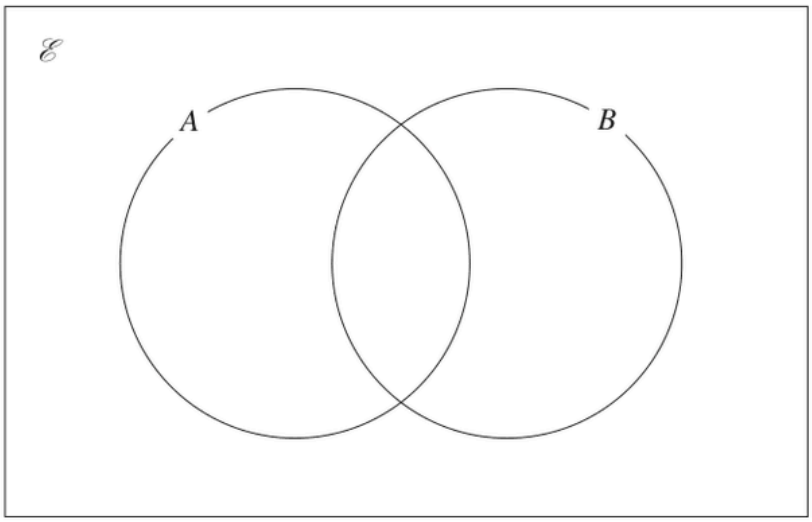
7 $\mathscr{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$A \cap B = \{5, 10, 15\}$

$B' = \{7, 8, 9, 11, 12, 13, 14\}$

$A' = \{4, 6, 7, 8, 14\}$

Complete the Venn diagram for this information.




**(Total for Question 7 is 3 marks)**

## WORKED SOLUTION

Question 7 · 4MA1/1H January 2022

**1 Calculate value**

$$\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

**2 We are given**

We are given

$$A \cap B = \{5, 10, 15\}$$

**3 Calculate value**

Also,

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

**4** **Calculate value**

so

$$B = \{4, 5, 6, 10, 15\}$$

**5** **Calculate value**

And

$$A' = \{4, 6, 7, 8, 14\}$$

**6** **Calculate value**

so

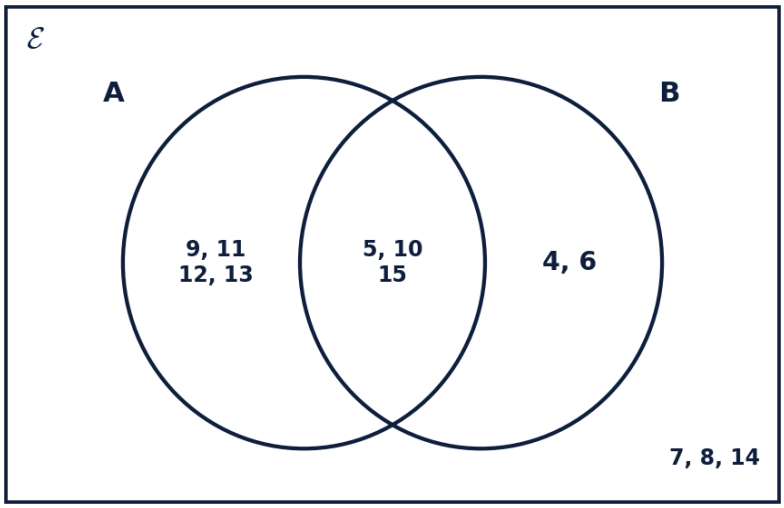
$$A = \{5, 9, 10, 11, 12, 13, 15\}$$

**7** **Use Set notation Venn diagrams**

Therefore the Venn diagram regions are:

**8** **only only outside both sets**

- $A \cap B$ : $\{5, 10, 15\}$
- $A$ only: $\{9, 11, 12, 13\}$
- $B$ only: $\{4, 6\}$
- outside both sets: $\{7, 8, 14\}$



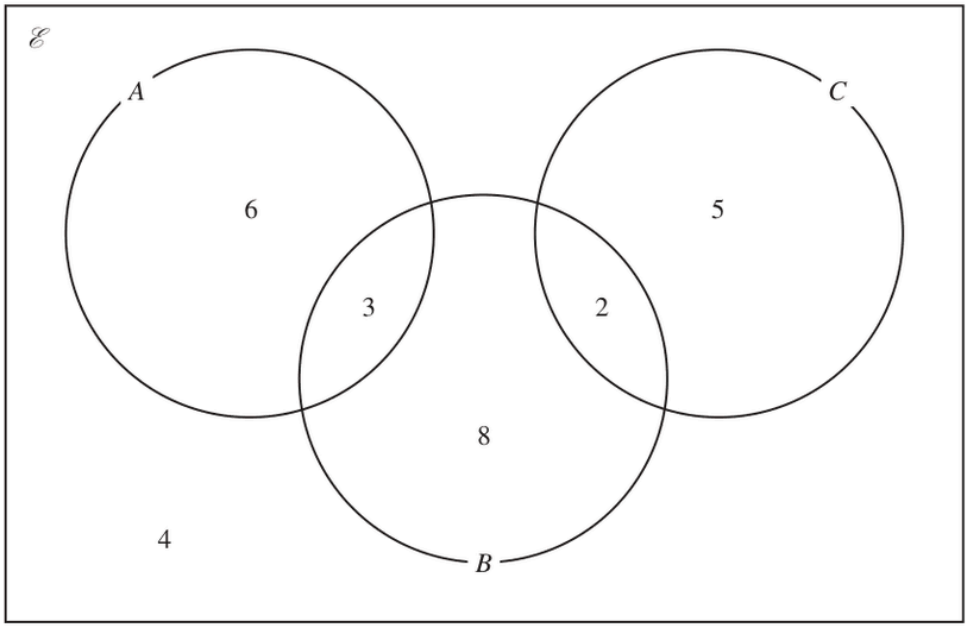

**FINAL ANSWER**

Put $\{5, 10, 15\}$ in the overlap, $\{9, 11, 12, 13\}$ in $A$ only, $\{4, 6\}$ in $B$ only, and $\{7, 8, 14\}$ outside.

# Question 17

4 marks | Set Notation & Venn Diagrams

**17** The Venn diagram shows a universal set $\mathscr{E}$ and three sets $A$, $B$ and $C$.




6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

**(1)**

(ii) $n(A \cap C)$

.....

**(1)**

(iii) $n(B \cap C')$

.....

**(1)**

(iv) $n(A' \cup B' \cup C')$

.....

**(1)**

**(Total for Question 17 is 4 marks)**

**1** **From the Venn diagram**

From the Venn diagram:

$$A \text{ only} = 6, \quad A \cap B = 3, \quad B \text{ only} = 8, \quad B \cap C = 2, \quad C \text{ only} = 5, \quad \text{outside} = 4$$

**2** **Calculate value**

There is no region in both $A$ and $C$, and there is no triple intersection.

$$n(A \cup B) = 6 + 3 + 8 + 2 = 19$$

$$n(A \cap C) = 0$$

$$n(B \cap C') = 3 + 8 = 11$$

**3** **Use**

For $A' \cup B' \cup C'$, use

$$A' \cup B' \cup C' = (A \cap B \cap C)'$$

**4** **Set**

Since $A \cap B \cap C$ is empty, this includes everyone in the universal set.

$$6 + 3 + 8 + 2 + 5 + 4 = 28$$

✓ **FINAL ANSWER**

(i) $19$, (ii) $0$, (iii) $11$, (iv) $28$

4MA1/2HR JUNE 2022

# Question 4

5 marks | Set Notation & Venn Diagrams

**4** $\mathscr{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$
$A = \{\text{multiples of } 3\}$
$B = \{\text{odd numbers}\}$

(a) List the members of the set

(i) $A \cap B$

.....

**(1)**

(ii) $A \cup B$

.....

**(1)**

(b) Is it true that $24 \in A$?

Tick one of the boxes below.

| Yes | No |
|---|---|
| ☐ | ☐ |

Give a reason for your answer.

.....

**(1)**

Set $C$ has 4 members such that $C \cap B' = \{10, 18\}$

(c) List the members of one possible set $C$

.....

**(2)**

**(Total for Question 4 is 5 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/2HR June 2022

**1** **Calculate value**

$$\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$$

**2** **The multiples of in are**

The multiples of $3$ in $\mathcal{E}$ are

$$A = \{9, 12, 15, 18\}$$

**3** **The odd numbers in are**

The odd numbers in $\mathcal{E}$ are

$$B = \{9, 11, 13, 15, 17, 19\}$$

**4** **Calculate value**

So

$$A \cap B = \{9, 15\}$$

**5** **Calculate value**

and

$$A \cup B = \{9, 11, 12, 13, 15, 17, 18, 19\}$$

**6** **The statement is false because**

The statement $24 \in A$ is false because $24$ is not in the universal set $\mathcal{E}$.

**7** **Calculate value**

For $C$, we need $C \cap B' = \{10, 18\}$ and $C$ has $4$ members. One valid choice is

$$C = \{9, 10, 11, 18\}$$

✓ **FINAL ANSWER**

(a)(i) $\{9, 15\}$, (a)(ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$, (b) No, (c) one possible set is $\{9, 10, 11, 18\}$

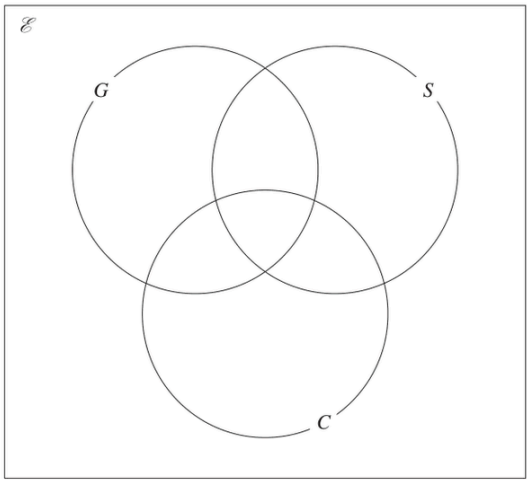
# Question 16

8 marks | Set Notation & Venn Diagrams

**16** 100 farmers are asked if they have goats ($G$), sheep ($S$) or chickens ($C$) on their farms.

Of these farmers
- 31 have sheep
- 53 have chickens
- 6 have goats, sheep and chickens
- 11 have sheep and goats
- 17 have sheep and chickens
- 18 have goats and chickens
- 20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.




**(3)**

(b) Find

(i) n($G$)

.....

**(1)**

(ii) n($[G \cup S]'$)

.....

**(1)**

(iii) n($G' \cap C$)

.....

**(1)**

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

.....

**(2)**

**(Total for Question 16 is 8 marks)**

**1** **(a) Start from the centre**

All three: $6$. Then the "two only" regions:

$$G \cap S \text{ only} = 11 - 6 = 5, \quad S \cap C \text{ only} = 17 - 6 = 11, \quad G \cap C \text{ only} = 18 - 6 = 12$$

**2** **(a) Single-animal regions**

$$S \text{ only} = 31 - 5 - 11 - 6 = 9, \qquad C \text{ only} = 53 - 11 - 12 - 6 = 24$$

$80$ farmers are inside the circles ($100 - 20$), so

$$G \text{ only} = 80 - (9 + 24 + 5 + 11 + 12 + 6) = 13$$

Outside the circles: $20$.

**3** **(b)(i) n(G)**

$$13 + 5 + 12 + 6 = 36$$

**4** **(b)(ii) Not in G or S**

Only "$C$ only" and the outside region: $24 + 20 = 44$.

**5** **(b)(iii) In C but not in G**

"$C$ only" and "$S \cap C$ only": $24 + 11 = 35$.

**6** **(c) Chickens farmers who also have goats**

$53$ farmers have chickens; $12 + 6 = 18$ of them also have goats.

$$\frac{18}{53}$$

**FINAL ANSWER**

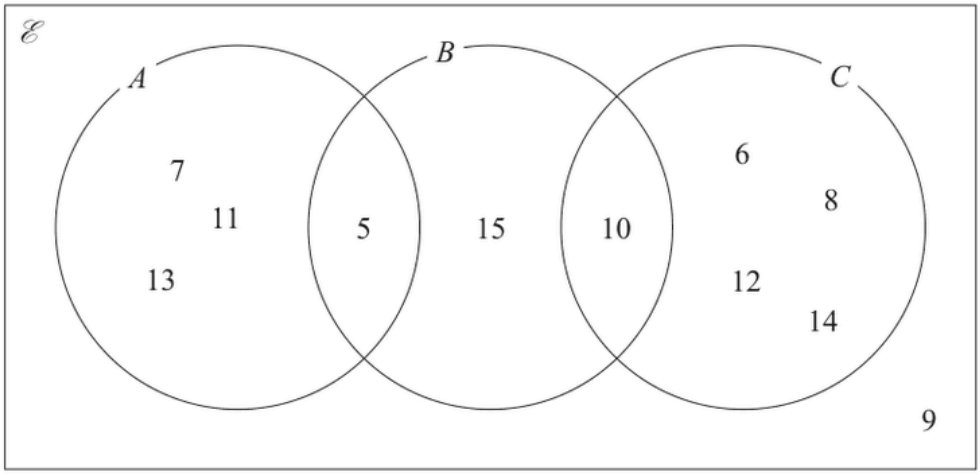
(a) $G$ only 13, $S$ only 9, $C$ only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, centre 6, outside 20
(b)(i) $36$ (ii) $44$ (iii) $35$ (c) $\frac{18}{53}$

4MA1/1HR MAY 2023

# Question 2

3 marks | Set Notation & Venn Diagrams

**2** Here is a Venn diagram.




(a) Write down the numbers that are in the set

(i) $A$

.....

(1)

(ii) $B \cup C$

.....

(1)

Dominic writes down $9 \notin C$

(b) Explain why Dominic is correct.

.....

.....

(1)

**(Total for Question 2 is 3 marks)**

## WORKED SOLUTION

Question 2 · 4MA1/1HR May 2023

**1** **The numbers inside are the**

The numbers inside $A$ are the numbers in the $A$-only region and the overlap with $B$:

$$A = \{5, 7, 11, 13\}$$

**2** **Calculate statistic**

$B \cup C$ means every number inside $B$ or $C$:

$$B \cup C = \{5, 6, 8, 10, 12, 14, 15\}$$

---

**3** **Dominic writes . This is**

Dominic writes $9 \notin C$. This is correct because $9$ is outside circle $C$ in the Venn diagram.

---

✓ **FINAL ANSWER**

(a)(i) $\{5, 7, 11, 13\}$, (a)(ii) $\{5, 6, 8, 10, 12, 14, 15\}$. Dominic is correct because $9$ is not in $C$.

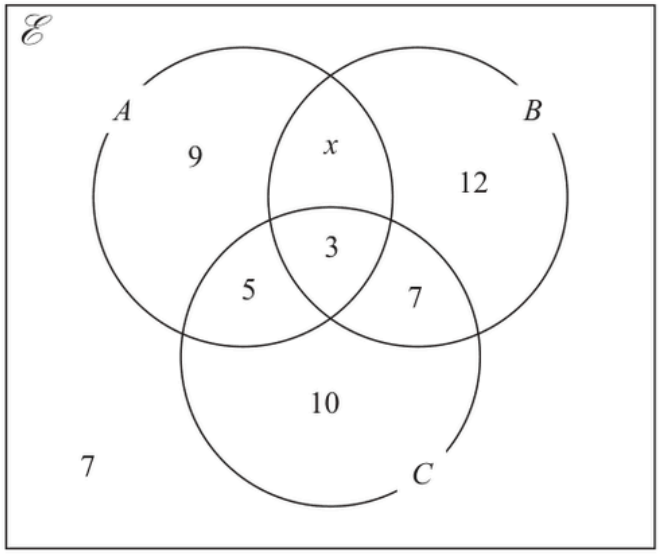
4MA1/1H MAY 2023

# Question 16

3 marks | Set Notation & Venn Diagrams

**16** The Venn diagram shows a universal set $\mathscr{E}$ and sets $A$, $B$ and $C$

The numbers and the letter $x$ represent **numbers** of elements.




Given that $n(A \cup B) = 42$

(a) find the value of $x$

$x =$ .....

(1)

(b) Find $n(A')$

.....

(1)

(c) Find $n(B' \cap C)$

.....

(1)

**(Total for Question 16 is 3 marks)**

## WORKED SOLUTION

Question 16 · 4MA1/1H May 2023

**1** **The regions inside are**

The regions inside $A \cup B$ are

$$9 + x + 12 + 5 + 3 + 7$$

**2** **We are told that**

We are told that

$$n(A \cup B) = 42$$

**3** **Calculate value**

So

$$36 + x = 42$$

$$x = 6$$

**4** **Add every region not inside**

For $A'$, add every region not inside $A$:

$$12 + 7 + 10 + 7 = 36$$

**5** **Add the regions**

For $B' \cap C$, add the regions that are in $C$ but not in $B$:

$$5 + 10 = 15$$

✓ **FINAL ANSWER**

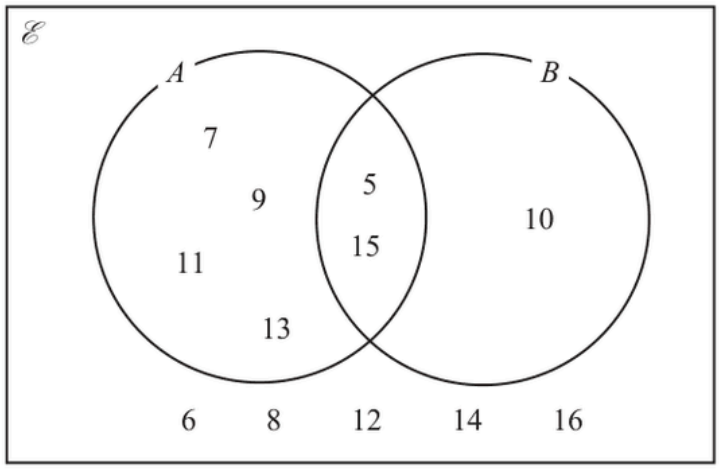
(a) $x = 6$, (b) $36$, (c) $15$

4MA1/1H NOVEMBER 2023

# Question 1

3 marks | Set Notation & Venn Diagrams

**1** Here is a Venn diagram.




List the members of the set

(a) $A$

.....

(1)

(b) $A \cap B$

.....

(1)

(c) $(A \cup B)'$

.....

(1)

**(Total for Question 1 is 3 marks)**

## WORKED SOLUTION

Question 1 · 4MA1/1H November 2023

**1** **The members of are all**

The members of $A$ are all the numbers inside circle $A$, including the overlap:

$$A = \{5, 7, 9, 11, 13, 15\}$$

**2** **The members of are the**

The members of $A \cap B$ are the numbers in both circles:

$$A \cap B = \{5, 15\}$$

**3** **means outside both circles**

$(A \cup B)'$ means outside both circles:

$$(A \cup B)' = \{6, 8, 12, 14, 16\}$$

---

✓ **FINAL ANSWER**

(a) $\{5, 7, 9, 11, 13, 15\}$, (b) $\{5, 15\}$, (c) $\{6, 8, 12, 14, 16\}$

# Question 4

5 marks | Set Notation & Venn Diagrams

**4** $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{\text{factors of } 6\}$
$B = \{\text{prime numbers}\}$

(a) List the members of the set

(i) $A \cup B$

.....

**(1)**

(ii) $A'$

.....

**(1)**

Harpreet states that $A \cap B = \varnothing$

Harpreet is incorrect.

(b) Explain why.

.....

**(1)**

$C$ is a set with 4 members such that

the set $A \cap C$ has 2 members
the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.

(c) List the 4 members of set $C$

.....

**(2)**

**(Total for Question 4 is 5 marks)**

**1** **Calculate value**

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 6\}$$

$$B = \{2, 3, 5, 7\}$$

**2** **Calculate value**

So

$$A \cup B = \{1, 2, 3, 5, 6, 7\}$$

**3** **Calculate value**

and

$$A' = \{4, 5, 7, 8, 9, 10\}$$

**4** **Harpreet is not correct because**

Harpreet is not correct because $2$ and $3$ are in both $A$ and $B$.

$$A \cap B = \{2, 3\}$$

**5** **Calculate value**

For $C$, it has $4$ members, $n(A \cap C) = 2$, $n(B \cap C) = 2$, and no member is in both of those overlaps. So choose the two elements of $A$ only and the two elements of $B$ only:

$$C = \{1, 5, 6, 7\}$$

✓ **FINAL ANSWER**

(a)(i) $\{1, 2, 3, 5, 6, 7\}$, (a)(ii) $\{4, 5, 7, 8, 9, 10\}$, (b) Harpreet is incorrect, (c) $C = \{1, 5, 6, 7\}$

# Question 7

5 marks | Set Notation & Venn Diagrams

7 $\mathscr{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$
$A = \{\text{even numbers}\}$
$B = \{23, 29, 31\}$
$C = \{\text{multiples of } 3\}$

(a) List the members of the set

(i) $B \cup C$

.....

**(1)**

(ii) $A' \cap C$

.....

**(1)**

(b) Is it true that $B \cap C = \varnothing$ ?

Tick (✓) one of the boxes below.

| Yes | No |
|---|---|
| ☐ | ☐ |

Give a reason for your answer.

.....

**(1)**

The set $D$ has 4 members and is such that $D \cap (A \cup C) = \varnothing$

(c) List the members of set $D$

.....

**(2)**

**(Total for Question 7 is 5 marks)**

**1** **Calculate value**

$$\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$$

$$A = \{24, 26, 28, 30, 32, 34\}$$

$$B = \{23, 29, 31\}$$

$$C = \{24, 27, 30, 33\}$$

**2** **Calculate value**

So

$$B \cup C = \{23, 24, 27, 29, 30, 31, 33\}$$

**3** **Set of odd numbers**

Also $A'$ is the set of odd numbers in $\mathcal{E}$, so

$$A' \cap C = \{27, 33\}$$

**4** **The statement is true because**

The statement $B \cap C = \emptyset$ is true because none of $23, 29, 31$ is a multiple of $3$.

**5** **Calculate value**

For $D$, we need $D \cap (A \cup C) = \emptyset$. So $D$ must contain the four numbers outside $A \cup C$:

$$D = \{23, 25, 29, 31\}$$

✓ **FINAL ANSWER**

(a)(i) $\{23, 24, 27, 29, 30, 31, 33\}$, (a)(ii) $\{27, 33\}$, (b) Yes, (c) $\{23, 25, 29, 31\}$

# Question 16

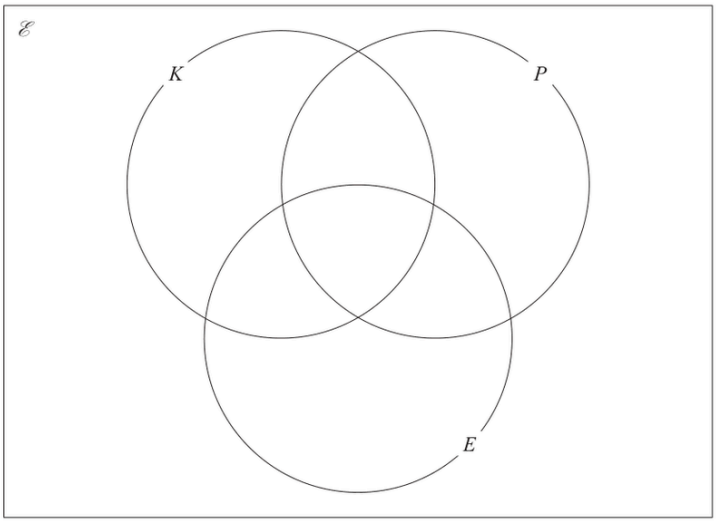
7 marks | Set Notation & Venn Diagrams

**16** 60 art students were asked if they would like to attend workshops for knitting ($K$), for photography ($P$) or for embroidery ($E$)

Of these students

9 chose knitting, photography and embroidery
17 chose knitting and photography
16 chose photography and embroidery
20 chose knitting and embroidery
28 chose photography
39 chose embroidery
2 chose none of the workshops

(a) Using this information, complete the Venn diagram to show the numbers of students in each subset.




**(3)**

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

**(2)**

(c) Find $n(P \cap K')$

**(1)**

(d) Find $n([P \cup E] \cap K)$

**(1)**

**(Total for Question 16 is 7 marks)**

**1** **(a) Start from the centre**

All three: $9$.

$$K \cap P \text{ only} = 17 - 9 = 8, \quad P \cap E \text{ only} = 16 - 9 = 7, \quad K \cap E \text{ only} = 20 - 9 = 11$$

**2** **(a) Single regions**

$$P \text{ only} = 28 - 8 - 7 - 9 = 4, \qquad E \text{ only} = 39 - 11 - 7 - 9 = 12$$

$58$ students are inside the circles ($60 - 2$):

$$K \text{ only} = 58 - (8 + 7 + 11 + 9 + 4 + 12) = 7$$

Outside: $2$.

**3** **(b) Knitting, given photography**

$28$ chose photography; $8 + 9 = 17$ of them also chose knitting.

$$\frac{17}{28}$$

**4** **(c) Photography but not knitting**

$P$ only and $P \cap E$ only: $4 + 7 = 11$.

**5** **(d) Knitting and (photography or embroidery)**

$8 + 9 + 11 = 28$

**FINAL ANSWER**

(a) $K$ only 7, $P$ only 4, $E$ only 12, $K \cap P$ only 8, $P \cap E$ only 7, $K \cap E$ only 11, centre 9, outside 2

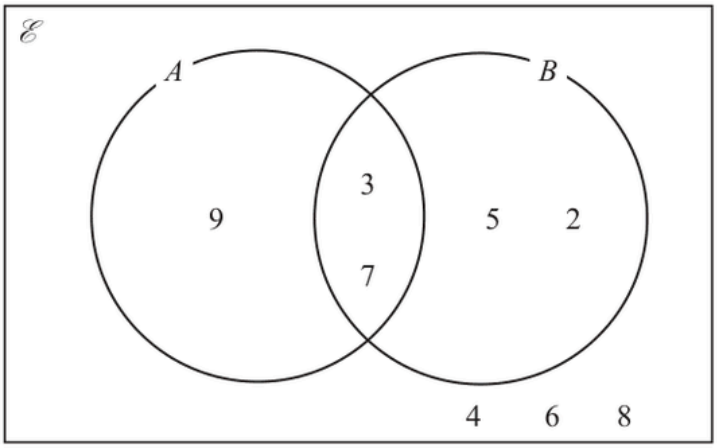
(b) $\frac{17}{28}$ (c) $11$ (d) $28$

4MA1/1H NOVEMBER 2024

# Question 4

3 marks | Set Notation & Venn Diagrams

**4** Here is a Venn diagram.




(a) List the members of the set $B$

.....

(1)

(b) List the members of the set $A \cap B$

.....

(1)

(c) List the members of the set $A'$

.....

(1)

**(Total for Question 4 is 3 marks)**

## WORKED SOLUTION

Question 4 · 4MA1/1H November 2024

**1** **The members of are all**

The members of $B$ are all the numbers inside circle $B$, including the overlap:

$$B = \{2, 3, 5, 7\}$$

**2** **The members of are the**

The members of $A \cap B$ are the numbers in both circles:

$$A \cap B = \{3, 7\}$$

### 3 Calculate statistic

$A'$ means all the numbers in the universal set that are not in $A$:

$$A' = \{2, 4, 5, 6, 8\}$$

---

**FINAL ANSWER**

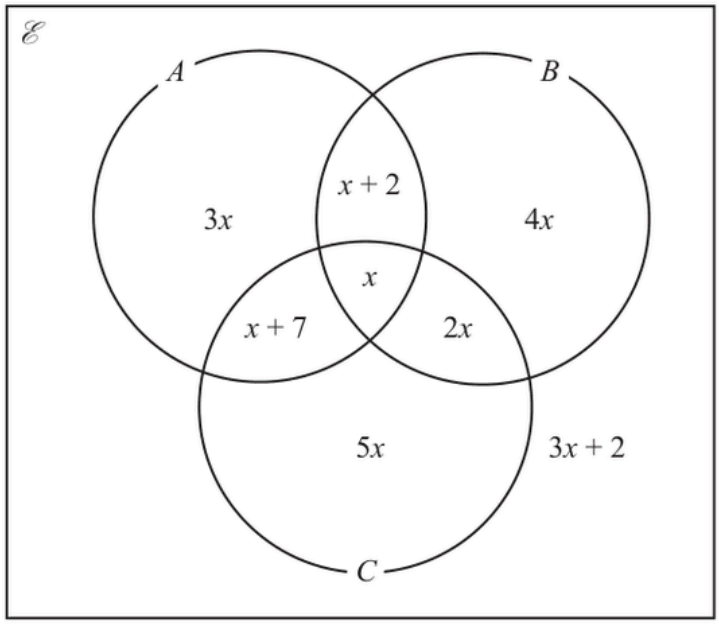
(a) $\{2, 3, 5, 7\}$, (b) $\{3, 7\}$, (c) $\{2, 4, 5, 6, 8\}$

4MA1/2H NOVEMBER 2024

# Question 21

4 marks | Set Notation & Venn Diagrams

**21** The Venn diagram shows information about the numbers of items in set $A$, set $B$ and set $C$, where $x$ is an integer.




Given that $\text{n}(A \cup B)' = 26$

find $\text{n}(A' \cap C)$

$\text{n}(A' \cap C) =$ .....

**(Total for Question 21 is 4 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/2H November 2024

**1 The complement of contains the**

The complement of $A \cup B$ contains the regions outside both $A$ and $B$. These are $5x$ and $3x + 2$.

**2 Calculate value**

So $5x + (3x + 2) = 26$, hence $8x + 2 = 26$ and $x = 3$.

**3 Calculate statistic**

Now $A' \cap C$ means inside $C$, but not inside $A$. These are the regions $5x$ and $2x$.

**4 Calculate value**

So $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$.

**FINAL ANSWER**

21

# Question 4

4 marks | Set Notation & Venn Diagrams

**4** $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

.....

(ii) $B'$

.....

**(2)**

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

| $\mathscr{E}$ | $\cap$ | $\cup$ | $\varnothing$ | $\in$ | $\notin$ |
|---|---|---|---|---|---|

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C =$ .....

(ii) 13 ..... $\mathscr{E}$

**(2)**

**(Total for Question 4 is 4 marks)**

**1** **Calculate value**

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

**2** **Calculate value**

Therefore

$$A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$$

**3** **Calculate value**

and

$$B' = \{1, 2, 4, 5, 7, 8, 10, 11\}$$

**4** **No number is both even**

No number is both even and odd, so

$$A \cap C = \emptyset$$

**5** **Set**

Also $13 \notin \mathcal{E}$ because $13$ is not in the universal set.

✓ **FINAL ANSWER**

(a)(i) $\{2, 3, 4, 6, 8, 9, 10, 12\}$, (a)(ii) $\{1, 2, 4, 5, 7, 8, 10, 11\}$, (b)(i) $\emptyset$, (b)(ii) $13 \notin \mathcal{E}$

# Question 5

4 marks | Set Notation & Venn Diagrams

**5** $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

(a) List the members of the set

(i) $A \cup B$

.....

(ii) $B'$

.....

**(2)**

$\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 4, 6, 8, 10, 12\}$

$B = \{3, 6, 9, 12\}$

$C = \{1, 3, 5, 7, 9, 11\}$

| | | | | | |
|---|---|---|---|---|---|
| $\mathscr{E}$ | $\cap$ | $\cup$ | $\varnothing$ | $\in$ | $\notin$ |

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C =$ .....

(ii) 13 ..... $\mathscr{E}$

**(2)**

**(Total for Question 5 is 4 marks)**

**1** **Calculate value**

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

**2** **Calculate value**

Therefore

$$A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$$

**3** **Calculate value**

and

$$B' = \{1, 2, 4, 5, 7, 8, 10, 11\}$$

**4** **No number is both even**

No number is both even and odd, so

$$A \cap C = \emptyset$$

**5** **Set**

Also $13 \notin \mathcal{E}$ because $13$ is not in the universal set.

✓ **FINAL ANSWER**

(a)(i) $\{2, 3, 4, 6, 8, 9, 10, 12\}$, (a)(ii) $\{1, 2, 4, 5, 7, 8, 10, 11\}$, (b)(i) $\emptyset$, (b)(ii) $13 \notin \mathcal{E}$

4WM1/01 MAY 2025

# Question 8

5 marks | Set Notation & Venn Diagrams

**8** $\mathscr{E} = \{18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$

$A = \{\text{even numbers}\}$

$B = \{\text{multiples of 3}\}$

$C = \{\text{multiples of 5}\}$

(a) List the members of the set $(A \cup B)'$

.....

(2)

Sophie writes down the statement $B \cap C = \emptyset$

(b) Explain why Sophie's statement is wrong.

.....

(1)

$D$ is a set such that $A \cap D = \{18, 26\}$
The set $D$ has exactly 5 members.

(c) List the members of one possible set $D$

.....

(2)

**(Total for Question 8 is 5 marks)**

## WORKED SOLUTION

Question 8 · 4WM1/01 May 2025

**1** **Calculate value**

$$A = \{18, 20, 22, 24, 26, 28, 30\}$$

$$B = \{18, 21, 24, 27, 30\}$$

$$A \cup B = \{18, 20, 21, 22, 24, 26, 27, 28, 30\}$$

**2** **Calculate value**

So

$$(A \cup B)' = \{19, 23, 25, 29\}$$

**3** **Sophie is wrong because is**

Sophie is wrong because $30$ is both a multiple of $3$ and a multiple of $5$, so $30 \in B \cap C$.

---

**4** **Set is**

For $D$, include $18$ and $26$, but no other even numbers. One possible set is

$$D = \{18, 19, 23, 25, 26\}$$

---

✓ **FINAL ANSWER**

$\{19, 23, 25, 29\}$; Sophie is wrong because $30 \in B \cap C$; one possible $D$ is $\{18, 19, 23, 25, 26\}$.

# Question 20

6 marks | Set Notation & Venn Diagrams

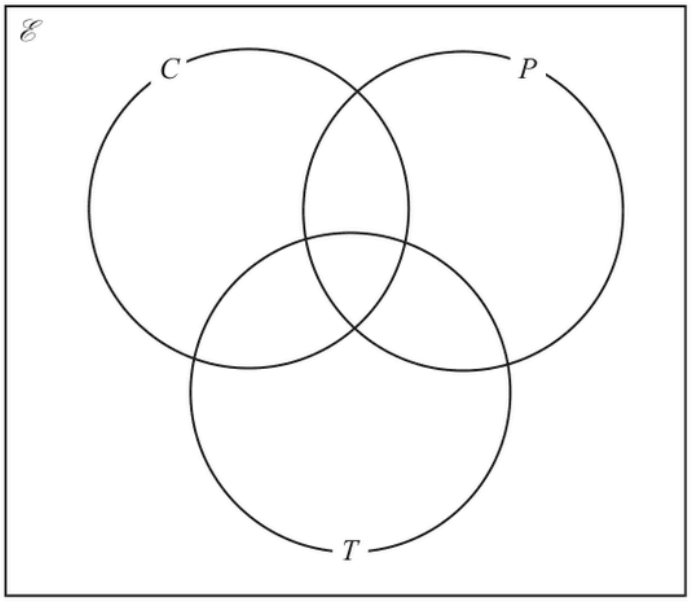
**20** 120 gardeners were asked if they grow carrots ($C$) or potatoes ($P$) or tomatoes ($T$)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.




**(3)**

(b) Find $n(T' \cap C)$

.....

**(1)**

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

.....

**(2)**

**(Total for Question 20 is 6 marks)**

**1** **Calculate value**

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

**2** **The triple intersection is**

The triple intersection is

$$n(C \cap P \cap T) = 12$$

**3** **Calculate value**

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

**4** **Calculate value**

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

**5** **Let the number who grow**

Let the number who grow only potatoes be $y$. The number who grow only tomatoes is also $y$.

**6** **Add the regions union**

Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

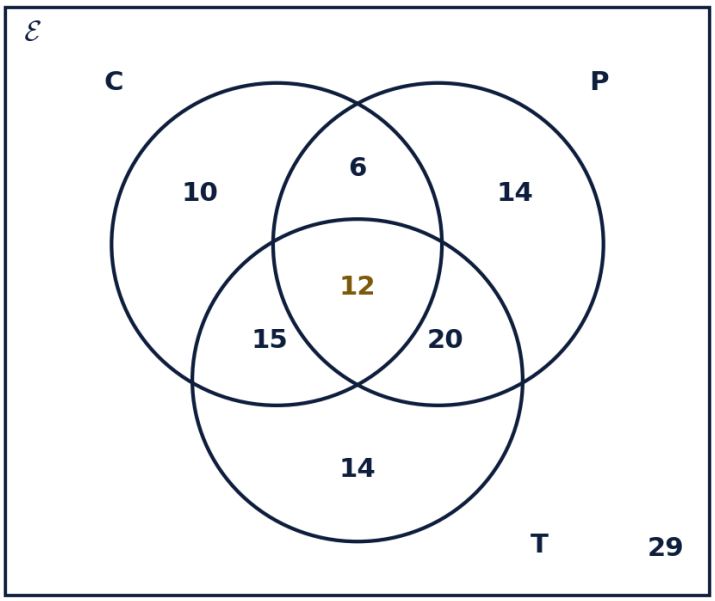
**7** **Calculate value**

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$




**8** **Calculate value**

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

**9** **Evaluate fraction**

For part (c), one gardener who grows carrots is chosen.

$$P(\text{also grows potatoes}) = \frac{n(C \cap P)}{n(C)}$$

$$= \frac{18}{43}$$

✓ **FINAL ANSWER**

(b) $16$, (c) $\frac{18}{43}$

# Question 4

4 marks | Set Notation & Venn Diagrams

**4** $\mathscr{E} = \{8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18\}$
$A = \{\text{odd numbers}\}$
$B = \{\text{multiples of 3}\}$
$C = \{\text{prime numbers}\}$

(a) List the members of the set $A \cup C$

.....

**(1)**

(b) List the members of the set $(A \cap B)'$

.....

**(1)**

(c) Is it true that $15 \in (B \cap C)$?

Tick (✓) the correct box.

| Yes | No |
|---|---|
| ☐ | ☐ |

Give a reason for your answer.

.....

**(1)**

$D$ is a set such that $D \cup C = A$
$D$ has 2 members.

(d) List the members of $D$

.....

**(1)**

**(Total for Question 4 is 4 marks)**

**1** **Calculate value**

$$A = \{9, 11, 13, 15, 17\}, \quad B = \{9, 12, 15, 18\}, \quad C = \{11, 13, 17\}$$

$$A \cup C = \{9, 11, 13, 15, 17\}$$

$$A \cap B' = \{11, 13, 17\}$$

**2** **Check prime condition**

$15 \notin B \cap C$ because $15$ is not prime.

**3** **Calculate value**

Since $D \cup C = A$ and $D$ has two members,

$$D = \{9, 15\}$$

**FINAL ANSWER**

$\{9, 11, 13, 15, 17\}$, $\{11, 13, 17\}$, No, $D = \{9, 15\}$.

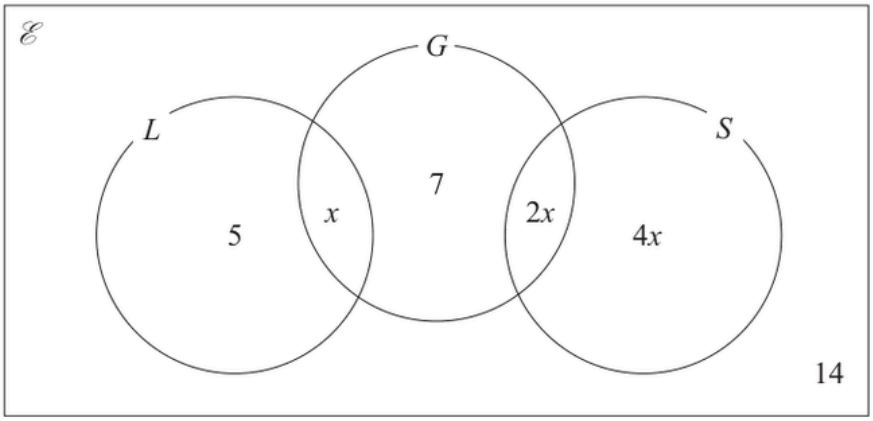
4MA1/2H NOVEMBER 2025

# Question 18

3 marks | Set Notation & Venn Diagrams

**18** Bianca asked 40 students which of the languages Latin ($L$), Greek ($G$) and Sanskrit ($S$) they study.

The Venn diagram gives some information about her results.
It shows the number of students in each subset.




One of these students is selected at random.
Work out the probability that this student studies Latin or Sanskrit.

**(Total for Question 18 is 3 marks)**

## WORKED SOLUTION

Question 18 · 4MA1/2H November 2025

**1 Find x (all regions add to 40)**

$$5 + x + 7 + 2x + 4x + 14 = 40 \Rightarrow 26 + 7x = 40 \Rightarrow x = 2$$

**2 Latin or Sanskrit**

Regions in $L$ or $S$: $5 + x + 2x + 4x = 5 + 2 + 4 + 8 = 19$.

$$\frac{19}{40}$$

✓ **FINAL ANSWER**

$$\frac{19}{40}$$

4MA1/1H JUNE 2026

# Question 3

3 marks | Set Notation & Venn Diagrams

**3** $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19\}$

$A = \{\text{factors of } 20\}$

$B = \{\text{multiples of } 5\}$

(a) List the members of the set

(i) $A \cup B$

.....

**(1)**

(ii) $A \cap B'$

.....

**(1)**

$C = \{\text{prime numbers}\}$

Ana writes down the statement $A \cap C = \varnothing$
Ana's statement is wrong.

(b) Explain why.

.....

.....

**(1)**

**(Total for Question 3 is 3 marks)**

## WORKED SOLUTION

Question 3 · 4MA1/1H June 2026

**1** **Part (a(i)) | Step 1**

$A = \{1, 2, 4, 5, 10\}$ and $B = \{5, 10, 15\}$, so $A \cup B = \{1, 2, 4, 5, 10, 15\}$.

Include every member of either set once; 20 lies outside the universal set.

**2** **Part (a(ii)) | Step 1**

$A \cap B' = \{1, 2, 4\}$.

Retain the factors of 20 in the universal set which are not multiples of 5.

**3** **Part (b) | Step 1**

Both 2 and 5 are prime factors of 20, so $A \cap C = \{2, 5\} \neq \varnothing$.

A shared member is enough to disprove the claim that the intersection is empty.

**FINAL ANSWER**

(a(i)) $\{1, 2, 4, 5, 10, 15\}$ (a(ii)) $\{1, 2, 4\}$ (b) $A \cap C = \{2, 5\}$, not the empty set.

4MA1/2HR JUNE 2026

# Question 3

1 marks | Set Notation & Venn Diagrams

3 $\mathscr{E}$ = {integers from 1 to 12}
$A$ = {factors of 12}

List the members of the set $A'$

.....

**(Total for Question 3 is 1 mark)**

## WORKED SOLUTION

Question 3 · 4MA1/2HR June 2026

**1** **Solution | Step 1**

$A = \{1, 2, 3, 4, 6, 12\}$. Remove these factors from the integers 1 to 12.
The complement contains precisely the universal-set elements not in A.

**FINAL ANSWER**

$A' = \{5, 7, 8, 9, 10, 11\}$.

4WM1/01 MAY 2026

# Question 5

4 marks | Set Notation & Venn Diagrams

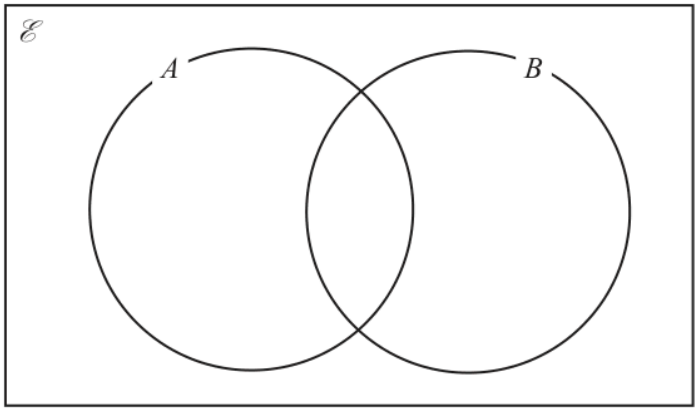
**5** $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$A = \{2, 3, 5, 7, 11\}$

$A \cap B = \{2, 5\}$

$B' = \{3, 4, 6, 7, 9, 11\}$

(a) Complete the Venn diagram for this information.




**(3)**

A number is chosen at random from the universal set $\mathscr{E}$

(b) Write down the probability that this number is in the set $(A \cup B)'$

.....

**(1)**

**(Total for Question 5 is 4 marks)**

## WORKED SOLUTION

Question 5 · 4WM1/01 May 2026

**1** **Find set B**

$B' = \{3, 4, 6, 7, 9, 11\}$, so $B = \{1, 2, 5, 8, 10, 12\}$.

**2** **Fill each region**

$A \cap B = \{2, 5\}$; $A$ only $= \{3, 7, 11\}$; $B$ only $= \{1, 8, 10, 12\}$; outside both $= \{4, 6, 9\}$.

Question 5 part a: completed Venn diagram

**3** **(b) Outside both circles**

$$P\left((A \cup B)'\right) = \frac{3}{12} = \frac{1}{4}$$

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**FINAL ANSWER**

(a) see diagram (b) $\frac{3}{12} = \frac{1}{4}$

4MA1/2H JUNE 2026

# Question 12

2 marks | Set Notation & Venn Diagrams

**12** $\mathscr{E} = \{\text{integers } x \text{ such that } 20 \leqslant x \leqslant 32\}$

$A = \{x \text{ such that } x < 27\}$

$B = \{x \text{ such that } 25 \leqslant x \leqslant 30\}$

(a) List the members of the set $A \cap B$

.....

**(1)**

(b) List the members of the set $(A \cup B)'$

.....

**(1)**

**(Total for Question 12 is 2 marks)**

## WORKED SOLUTION

Question 12 · 4MA1/2H June 2026

**1** **Part (a) | Step 1**

Within the universal set, $A = \{20, 21, 22, 23, 24, 25, 26\}$ and $B = \{25, 26, 27, 28, 29, 30\}$. Their common members are $25$ and $26$.

Intersection consists of elements in both sets.

**2** **Part (b) | Step 1**

$A \cup B = \{20, 21, \ldots, 30\}$. The members of $\mathcal{E} = \{20, 21, \ldots, 32\}$ outside this union are $31$ and $32$.

Take the complement relative to the stated universal set, not all integers.

✓ **FINAL ANSWER**

(a) $A \cap B = \{25, 26\}$ (b) $(A \cup B)' = \{31, 32\}$

4MA1/1HR JUNE 2026

# Question 21

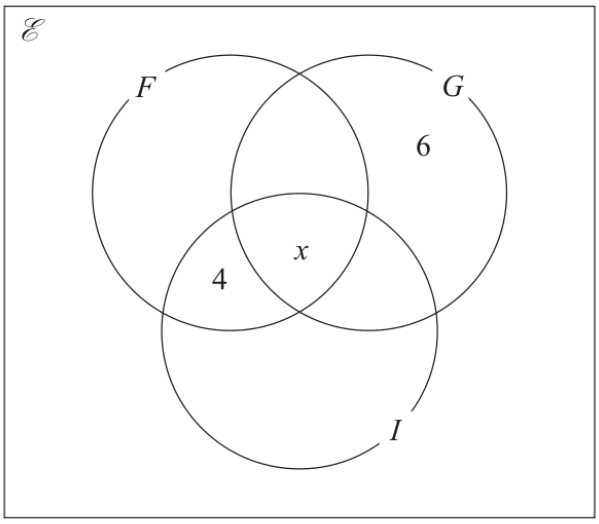
5 marks | Set Notation & Venn Diagrams

**21** 60 students were asked if they study French ($F$), German ($G$) or Italian ($I$)

Of these students

32 students study French
24 students study German
27 students study Italian
16 students study French and German
15 students study German and Italian

The incomplete Venn diagram gives some more information about the languages the 60 students study.




(a) Work out the value of $x$

$x =$ .....

**(3)**

(b) Work out how many of the 60 students study none of French or German or Italian.

.....

**(2)**

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Question 21 · 4MA1/1HR June 2026

**1** **Part (a) | Step 1**

The French-and-German-only region is $16 - x$, and the German-and-Italian-only region is $15 - x$. The totals 16 and 15 each include the triple intersection x.

**2** **Part (a) | Step 2**

$$6 + (16 - x) + (15 - x) + x = 24.$$

The four disjoint regions in the German set must total 24.

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**3** **Part (a) | Step 3**

$$37 - x = 24 \quad \Longrightarrow \quad x = 13.$$

Solve for the common triple-intersection count.

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**4** **Part (b) | Step 1**

$$FG \text{ only} = 16 - 13 = 3, \qquad GI \text{ only} = 15 - 13 = 2.$$

Use the triple-intersection value from part (a).

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**5** **Part (b) | Step 2**

$$F \text{ only} = 32 - (3 + 4 + 13) = 12, \qquad I \text{ only} = 27 - (4 + 2 + 13) = 8.$$

The French-and-Italian-only count is the given 4.

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**6** **Part (b) | Step 3**

$$n(F \cup G \cup I) = 12 + 6 + 8 + 3 + 4 + 2 + 13 = 48.$$

Sum all seven mutually exclusive interior regions.

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**7** **Part (b) | Step 4**

$$n(\text{none}) = 60 - 48 = 12.$$

Subtract the union from the total number of students.

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✓ **FINAL ANSWER**

(a) $x = 13$ (b) $12$ students.

TOPIC

# Forming & Solving Equations

Every past-paper question on this topic, 2020 to 2026.

# Question 6

3 marks | Forming & Solving Equations

$\left(6^2\right)^{10} = 6^p$

(c) Find the value of $p$

$p =$ .....

(1)

$5^r = 1$

(d) Write down the value of $r$

$r =$ .....

(1)

**(Total for Question 5 is 4 marks)**

**6** The diagram shows triangle $ABC$

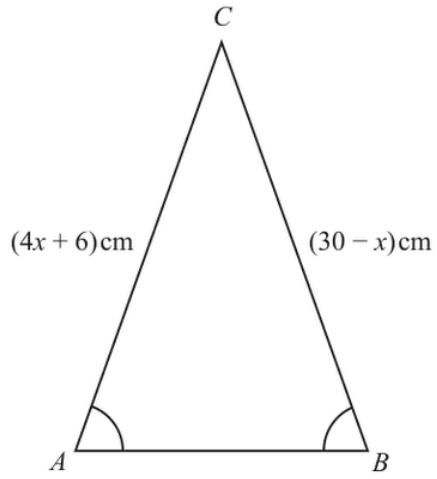


Diagram **NOT** accurately drawn

$CA = (4x + 6)$ cm $\quad CB = (30 - x)$ cm $\quad$ angle $CAB$ = angle $CBA$

Work out the value of $x$

$x =$ .....

**(Total for Question 6 is 3 marks)**

**1** **Use trigonometry**

The base angles are equal, so the opposite sides are equal.

$$4x + 6 = 30 - x$$

$$5x = 24$$

$$x = 4.8$$

**FINAL ANSWER**

$x = 4.8$.



TOPIC

# Algebraic Proof

Every past-paper question on this topic, 2020 to 2026.

4WM1/01 NOVEMBER 2025

# Question 17

2 marks | Algebraic Proof

**17** Use algebra to show that $0.95\dot{4}\dot{5} = \frac{21}{22}$

**(Total for Question 17 is 2 marks)**

## WORKED SOLUTION

Question 17 · 4WM1/01 November 2025

**1 Let**

Let

$$x = 0.95454545\ldots$$

**2 Calculate value**

Then

$$10x = 9.5454545\ldots$$

**3 Calculate value**

and

$$1000x = 954.5454545\ldots$$

**4 Subtract**

Subtract:

$$990x = 945$$

$$x = \frac{945}{990} = \frac{21}{22}$$

**FINAL ANSWER**

$0.954545\ldots = \frac{21}{22}$.

4MA1/2H JUNE 2026

# Question 17

2 marks | Algebraic Proof

**17** Use algebra to show that $0.5\dot{3}\dot{9} = \frac{89}{165}$

**(Total for Question 17 is 2 marks)**

## WORKED SOLUTION

Question 17 · 4MA1/2H June 2026

### 1 Solution | Step 1

Let $x = 0.5\overline{39} = 0.5393939\ldots$. Then $10x = 5.393939\ldots$ and $1000x = 539.393939\ldots$.

Shift past the non-recurring digit and then one complete two-digit recurring block.

### 2 Solution | Step 2

Subtract:

$$1000x - 10x = 539.393939\ldots - 5.393939\ldots = 534.$$

$$990x = 534 \quad \Rightarrow \quad x = \frac{534}{990} = \frac{89}{165}.$$

Identical infinite decimal tails cancel exactly; reduce the resulting fraction by 6.

**FINAL ANSWER**

$0.5\overline{39} = \frac{89}{165}$.